

Stellaris® LM3S6432 Microcontroller

DATA SHEET

DS-LM3S6432-7007

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Register 21: ADC Sample Sequence Input Multiplexer Select 1 (ADCSSMUX1), offset 0x060 284
Register 22: ADC Sample Sequence Input Multiplexer Select 2 (ADCSSMUX2), offset 0x080 284
Register 23: ADC Sample Sequence Control 1 (ADCSSCTL1), offset 0x064 285
Register 24: ADC Sample Sequence Control 2 (ADCSSCTL2), offset 0x084 285
Register 25: ADC Sample Sequence Input Multiplexer Select 3 (ADCSSMUX3), offset 0x0A0 287
Register 26: ADC Sample Sequence Control 3 (ADCSSCTL3), offset 0x0A4 288
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Register 2: UART Receive Status/Error Clear (UARTRSR/UARTECR), offset 0x004 300
Register 3: UART Flag (UARTFR), offset 0x018 302
Register 4: UART IrDA Low-Power Register (UARTILPR), offset 0x020 304
Register 5: UART Integer Baud-Rate Divisor (UARTIBRD), offset 0x024 305
Register 6: UART Fractional Baud-Rate Divisor (UARTFBRD), offset 0x028 306
Register 7: UART Line Control (UARTLCRH), offset 0x02C 307
Register 8: UART Control (UARTCTL), offset 0x030 309
Register 9: UART Interrupt FIFO Level Select (UARTIFLS), offset 0x034 311
Register 10: UART Interrupt Mask (UARTIM), offset 0x038 313
Register 11: UART Raw Interrupt Status (UARTRIS), offset 0x03C 315
Register 12: UART Masked Interrupt Status (UARTMIS), offset 0x040 316
Register 13: UART Interrupt Clear (UARTICR), offset 0x044 317
Register 14: UART Peripheral Identification 4 (UARTPeriphID4), offset 0xFD0 319
Register 15: UART Peripheral Identification 5 (UARTPeriphID5), offset 0xFD4 320
Register 16: UART Peripheral Identification 6 (UARTPeriphID6), offset 0xFD8 321
Register 17: UART Peripheral Identification 7 (UARTPeriphID7), offset 0xFDC 322
Register 18: UART Peripheral Identification 0 (UARTPeriphID0), offset 0xFE0 323
Register 19: UART Peripheral Identification 1 (UARTPeriphID1), offset 0xFE4 324
Register 20: UART Peripheral Identification 2 (UARTPeriphID2), offset 0xFE8 325
Register 21: UART Peripheral Identification 3 (UARTPeriphID3), offset 0xFEC 326
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Register 6: SSI Interrupt Mask (SSIIM), offset 0x014 351
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Register 10: SSI Peripheral Identification 4 (SSIPeriphID4), offset 0xFD0 356
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Register 18: SSI PrimeCell Identification 0 (SSIPCellID0), offset 0xFF0 364
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Register 4: I^2C Master Timer Period (I2CMTPR), offset 0x00C 389
Register 5: I^2C Master Interrupt Mask (I2CMIMR), offset 0x010 390
Register 6: I^2C Master Raw Interrupt Status (I2CMRIS), offset 0x014 391
Register 7: I^2C Master Masked Interrupt Status (I2CMMIS), offset 0x018 392
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Register 13: I^2C Slave Interrupt Mask (I2CSIMR), offset 0x00C 400
Register 14: I^2C Slave Raw Interrupt Status (I2CSRIS), offset 0x010 401
Register 15: I^2C Slave Masked Interrupt Status (I2CSMIS), offset 0x014 402
Register 16: I^2C Slave Interrupt Clear (I2CSICR), offset 0x018 403

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Register 2: Ethernet MAC Interrupt Mask (MACIM), offset 0x004 418
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Register 4: Ethernet MAC Transmit Control (MACTCTL), offset 0x00C 420
Register 5: Ethernet MAC Data (MACDATA), offset 0x010 421
Register 6: Ethernet MAC Individual Address 0 (MACIA0), offset 0x014 423
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Register 17: Ethernet PHY Management Register 2 – PHY Identifier 1 (MR2), address 0x02 437
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Register 19: Ethernet PHY Management Register 4 – Auto-Negotiation Advertisement (MR4), address 0x04 439

Register 20: Ethernet PHY Management Register 5 – Auto-Negotiation Link Partner Base Page Ability (MR5), address 0x05 441
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Register 24: Ethernet PHY Management Register 18 – Diagnostic (MR18), address 0x12 447
Register 25: Ethernet PHY Management Register 19 – Transceiver Control (MR19), address 0x13 448
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Register 4: Analog Comparator Reference Voltage Control (ACREFCTL), offset 0x010 459
Register 5: Analog Comparator Status 0 (ACSTAT0), offset 0x020 460
Register 6: Analog Comparator Status 1 (ACSTAT1), offset 0x040 460
Register 7: Analog Comparator Control 0 (ACCTL0), offset 0x024 461
Register 8: Analog Comparator Control 1 (ACCTL1), offset 0x044 461

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Register 5: PWM Output Fault (PWMFAULT), offset 0x010 475
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Register 7: PWM Raw Interrupt Status (PWMRIS), offset 0x018 477
Register 8: PWM Interrupt Status and Clear (PWMISC), offset 0x01C 478
Register 9: PWM Status (PWMSTATUS), offset 0x020 479
Register 10: PWM0 Control (PWM0CTL), offset 0x040 480
Register 11: PWM0 Interrupt and Trigger Enable (PWM0INTEN), offset 0x044 482
Register 12: PWM0 Raw Interrupt Status (PWM0RIS), offset 0x048 485
Register 13: PWM0 Interrupt Status and Clear (PWM0ISC), offset 0x04C 486
Register 14: PWM0 Load (PWM0LOAD), offset 0x050 487
Register 15: PWM0 Counter (PWM0COUNT), offset 0x054 488
Register 16: PWM0 Compare A (PWM0CMPA), offset 0x058 489
Register 17: PWM0 Compare B (PWM0CMPB), offset 0x05C 490
Register 18: PWM0 Generator A Control (PWM0GENA), offset 0x060 491
Register 19: PWM0 Generator B Control (PWM0GENB), offset 0x064 494
Register 20: PWM0 Dead-Band Control (PWM0DBCTL), offset 0x068 497
Register 21: PWM0 Dead-Band Rising-Edge Delay (PWM0DBRISE), offset 0x06C 498
Register 22: PWM0 Dead-Band Falling-Edge-Delay (PWM0DBFALL), offset 0x070 499

Revision History

The revision history table notes changes made between the indicated revisions of the LM3S6432 data sheet.

Table 1. Revision History

Date	Revision	Description
March 2008	2550	Started tracking revision history.
April 2008	2881	■ The Θ_{JA} value was changed from 55.3 to 34 in the "Thermal Characteristics" table in the Operating Characteristics chapter. ■ Bit 31 of the **DC3** register was incorrectly described in prior versions of the data sheet. A reset of 1 indicates that an even CCP pin is present and can be used as a 32-KHz input clock. ■ Values for $I_{DD_HIBERNATE}$ were added to the "Detailed Power Specifications" table in the "Electrical Characteristics" chapter. ■ The "Hibernation Module DC Electricals" table was added to the "Electrical Characteristics" chapter. ■ The maximum value on Core supply voltage (V_{DD25}) in the "Maximum Ratings" table in the "Electrical Characteristics" chapter was changed from 4 to 3. ■ The operational frequency of the internal 30-kHz oscillator clock source is 30 kHz ± 50% (prior data sheets incorrectly noted it as 30 kHz ± 30%). ■ A value of 0x3 in bits 5:4 of the **MISC** register (OSCSRC) indicates the 30-KHz internal oscillator is the input source for the oscillator. Prior data sheets incorrectly noted 0x3 as a reserved value. ■ The reset for bits 6:4 of the **RCC2** register (OSCSRC2) is 0x1 (IOSC). Prior data sheets incorrectly noted the reset was 0x0 (MOSC). ■ A note on high-current applications was added to the GPIO chapter: For special high-current applications, the GPIO output buffers may be used with the following restrictions. With the GPIO pins configured as 8-mA output drivers, a total of four GPIO outputs may be used to sink current loads up to 18 mA each. At 18-mA sink current loading, the VOL value is specified as 1.2 V. The high-current GPIO package pins must be selected such that there are only a maximum of two per side of the physical package or BGA pin group with the total number of high-current GPIO outputs not exceeding four for the entire package. ■ A note on Schmitt inputs was added to the GPIO chapter: Pins configured as digital inputs are Schmitt-triggered. ■ The Buffer type on the WAKE pin changed from OD to - in the Signal Tables. ■ The "Differential Sampling Range" figures in the ADC chapter were clarified. ■ The last revision of the data sheet (revision 2550) introduced two errors that have now been corrected: – The LQFP pin diagrams and pin tables were missing the comparator positive and negative input pins. – The base address was listed incorrectly in the **FMPRE0** and **FMPPE0** register bit diagrams. ■ Additional minor data sheet clarifications and corrections.

Table 1. Revision History *(continued)*

Date	Revision	Description
May 2008	2972	■ The 108-Ball BGA pin diagram and pin tables had an error. The following signals were erroneously indicated as available and have now been changed to a No Connect (NC): – Ball C1: Changed PE7 to NC – Ball C2: Changed PE6 to NC – Ball D2: Changed PE5 to NC ■ As noted in the PCN, three of the nine Ethernet LED configuration options are no longer supported: TX Activity (0x2), RX Activity (0x3), and Collision (0x4). These values for the LED0 and LED1 bit fields in the **MR23** register are now marked as reserved. ■ As noted in the PCN, the option to provide VDD25 power from external sources was removed. Use the LDO output as the source of VDD25 input. ■ As noted in the PCN, pin 41 (ball K3 on the BGA package) was renamed from GNDPHY to ERBIAS. A 12.4-kΩ resistor should be connected between ERBIAS and ground to accommodate future device revisions (see “Functional Description” on page 405). ■ Additional minor data sheet clarifications and corrections.
July 2008	3108	■ Corrected resistor value in ERBIAS signal description. ■ Additional minor data sheet clarifications and corrections.
August 2008	3447	■ Added note on clearing interrupts to Interrupts chapter. ■ Added Power Architecture diagram to System Control chapter. ■ Additional minor data sheet clarifications and corrections.
October 2008	4149	■ Corrected values for DSOSCSRC bit field in **Deep Sleep Clock Configuration (DSLPCLKCFG)** register. ■ The FMA value for the **FMPRE3** register was incorrect in the Flash Resident Registers table in the Internal Memory chapter. The correct value is 0x0000.0006. ■ In the Ethernet chapter, major improvements were made including a rewrite of the conceptual information and the addition of new figures to clarify how to use the Ethernet Controller interface. ■ Incorrect Comparator Operating Modes tables were removed from the Analog Comparators chapter.
November 2008	4283	■ Revised High-Level Block Diagram. ■ Additional minor data sheet clarifications and corrections were made.
January 2009	4660	■ Corrected bit type for RELOAD bit field in SysTick Reload Value register; changed to R/W. ■ Clarification added as to what happens when the SSI in slave mode is required to transmit but there is no data in the TX FIFO. ■ Added "Hardware Configuration" section to Ethernet Controller chapter. ■ Additional minor data sheet clarifications and corrections.

Table 1. Revision History *(continued)*

Date	Revision	Description
April 2009	5367	■ Added JTAG/SWD clarification (see “Communication with JTAG/SWD” on page 58). ■ Added clarification that the PLL operates at 400 MHz, but is divided by two prior to the application of the output divisor. ■ Added "GPIO Module DC Characteristics" table (see Table 21-4 on page 528). ■ Additional minor data sheet clarifications and corrections.
July 2009	5902	■ Clarified Power-on reset and $\overline{\text{RST}}$ pin operation; added new diagrams. ■ Clarified explanation of nonvolatile register programming in Internal Memory chapter. ■ Added explanation of reset value to **FMPRE0/1/2/3**, **FMPPE0/1/2/3**, **USER_DBG**, and **USER_REG0/1** registers. ■ Added description for Ethernet PHY power-saving modes. ■ Corrected the reset values for bits 6 and 7 in the Ethernet **MR24** register. ■ Changed buffer type for $\overline{\text{WAKE}}$ pin to TTL and HIB pin to OD. ■ In ADC characteristics table, changed Max value for GAIN parameter from ±1 to ±3 and added E_{IR} (Internal voltage reference error) parameter. ■ Additional minor data sheet clarifications and corrections.
July 2009	5920	Corrected ordering numbers.
October 2009	6462	■ Deleted MAXADCSPD bit field from **DCGC0** register as it is not applicable in Deep-Sleep mode. ■ Deleted reset value for 16-bit mode from **GPTMTAILR**, **GPTMTAMATCHR**, and **GPTMTAR** registers because the module resets in 32-bit mode. ■ Clarified PWM source for ADC triggering. ■ Made these changes to the Electrical Characteristics chapter: – Removed V_{SIH} and V_{SIL} parameters from Operating Conditions table. – Added table showing actual PLL frequency depending on input crystal. – Changed the name of the $t_{HIB_REG_WRITE}$ parameter to $t_{HIB_REG_ACCESS}$. – Revised ADC electrical specifications to clarify, including reorganizing and adding new data. – Changed SSI set up and hold times to be expressed in system clocks, not ns.

Table 1. Revision History *(continued)*

Date	Revision	Description
January 2010	6712	■ In "System Control" section, clarified Debug Access Port operation after Sleep modes. ■ Clarified wording on Flash memory access errors. ■ Added section on Flash interrupts. ■ Changed the reset value of the **ADC Sample Sequence Result FIFO n (ADCSSFIFOn)** registers to be indeterminate. ■ Clarified operation of SSI transmit FIFO. ■ Made these changes to the Operating Characteristics chapter: – Added storage temperature ratings to "Temperature Characteristics" table – Added "ESD Absolute Maximum Ratings" table ■ Made these changes to the Electrical Characteristics chapter: – In "Flash Memory Characteristics" table, corrected Mass erase time – Added sleep and deep-sleep wake-up times ("Sleep Modes AC Characteristics" table) – In "Reset Characteristics" table, corrected units for supply voltage (VDD) rise time
April 2010	7007	■ Added caution note to the **I^2C Master Timer Period (I2CMTPR)** register description and changed field width to 7 bits. ■ Removed erroneous text about restoring the Flash Protection registers. ■ Added note about $\overline{\text{RST}}$ signal routing. ■ Clarified the function of the TnSTALL bit in the **GPTMCTL** register. ■ Corrected XTALNPHY pin description. ■ Additional minor data sheet clarifications and corrections.

About This Document

This data sheet provides reference information for the LM3S6432 microcontroller, describing the functional blocks of the system-on-chip (SoC) device designed around the ARM® Cortex™-M3 core.

Audience

This manual is intended for system software developers, hardware designers, and application developers.

About This Manual

This document is organized into sections that correspond to each major feature.

Related Documents

The following related documents are available on the documentation CD or from the Stellaris® web site at www.ti.com/stellaris:

- *ARM® CoreSight Technical Reference Manual*
- *ARM® Cortex™-M3 Errata*
- *ARM® Cortex™-M3 Technical Reference Manual*
- *ARM® v7-M Architecture Application Level Reference Manual*
- *Stellaris® Graphics Library User's Guide*
- *Stellaris® Peripheral Driver Library User's Guide*
- *Stellaris® Errata*

The following related documents are also referenced:

- *IEEE Standard 1149.1-Test Access Port and Boundary-Scan Architecture*

This documentation list was current as of publication date. Please check the web site for additional documentation, including application notes and white papers.

Documentation Conventions

This document uses the conventions shown in Table 2 on page 22.

Table 2. Documentation Conventions

Notation	Meaning
General Register Notation	
REGISTER	APB registers are indicated in uppercase bold. For example, **PBORCTL** is the Power-On and Brown-Out Reset Control register. If a register name contains a lowercase n, it represents more than one register. For example, **SRCRn** represents any (or all) of the three Software Reset Control registers: **SRCR0, SRCR1** , and **SRCR2**.
bit	A single bit in a register.

Table 2. Documentation Conventions *(continued)*

Notation	Meaning
bit field	Two or more consecutive and related bits.
offset 0x*nnn*	A hexadecimal increment to a register's address, relative to that module's base address as specified in “Memory Map” on page 47.
Register *N*	Registers are numbered consecutively throughout the document to aid in referencing them. The register number has no meaning to software.
reserved	Register bits marked *reserved* are reserved for future use. In most cases, reserved bits are set to 0; however, user software should not rely on the value of a reserved bit. To provide software compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
yy:xx	The range of register bits inclusive from xx to yy. For example, 31:15 means bits 15 through 31 in that register.
Register Bit/Field Types	This value in the register bit diagram indicates whether software running on the controller can change the value of the bit field.
RC	Software can read this field. The bit or field is cleared by hardware after reading the bit/field.
RO	Software can read this field. Always write the chip reset value.
R/W	Software can read or write this field.
R/W1C	Software can read or write this field. A write of a 0 to a W1C bit does not affect the bit value in the register. A write of a 1 clears the value of the bit in the register; the remaining bits remain unchanged. This register type is primarily used for clearing interrupt status bits where the read operation provides the interrupt status and the write of the read value clears only the interrupts being reported at the time the register was read.
R/W1S	Software can read or write a 1 to this field. A write of a 0 to a R/W1S bit does not affect the bit value in the register.
W1C	Software can write this field. A write of a 0 to a W1C bit does not affect the bit value in the register. A write of a 1 clears the value of the bit in the register; the remaining bits remain unchanged. A read of the register returns no meaningful data. This register is typically used to clear the corresponding bit in an interrupt register.
WO	Only a write by software is valid; a read of the register returns no meaningful data.
Register Bit/Field Reset Value	This value in the register bit diagram shows the bit/field value after any reset, unless noted.
0	Bit cleared to 0 on chip reset.
1	Bit set to 1 on chip reset.
-	Nondeterministic.
Pin/Signal Notation	
[ ]	Pin alternate function; a pin defaults to the signal without the brackets.
pin	Refers to the physical connection on the package.
signal	Refers to the electrical signal encoding of a pin.
assert a signal	Change the value of the signal from the logically False state to the logically True state. For active High signals, the asserted signal value is 1 (High); for active Low signals, the asserted signal value is 0 (Low). The active polarity (High or Low) is defined by the signal name (see SIGNAL and $\overline{\text{SIGNAL}}$ below).
deassert a signal	Change the value of the signal from the logically True state to the logically False state.
$\overline{\text{SIGNAL}}$	Signal names are in uppercase and in the Courier font. An overbar on a signal name indicates that it is active Low. To assert $\overline{\text{SIGNAL}}$ is to drive it Low; to deassert $\overline{\text{SIGNAL}}$ is to drive it High.
SIGNAL	Signal names are in uppercase and in the Courier font. An active High signal has no overbar. To assert SIGNAL is to drive it High; to deassert SIGNAL is to drive it Low.
Numbers	

Table 2. Documentation Conventions *(continued)*

Notation	Meaning
X	An uppercase X indicates any of several values is allowed, where X can be any legal pattern. For example, a binary value of 0X00 can be either 0100 or 0000, a hex value of 0xX is 0x0 or 0x1, and so on.
0x	Hexadecimal numbers have a prefix of 0x. For example, 0x00FF is the hexadecimal number FF. All other numbers within register tables are assumed to be binary. Within conceptual information, binary numbers are indicated with a b suffix, for example, 1011b, and decimal numbers are written without a prefix or suffix.

1 Architectural Overview

The Stellaris® family of microcontrollers—the first ARM® Cortex™-M3 based controllers—brings high-performance 32-bit computing to cost-sensitive embedded microcontroller applications. These pioneering parts deliver customers 32-bit performance at a cost equivalent to legacy 8- and 16-bit devices, all in a package with a small footprint.

The Stellaris® family offers efficient performance and extensive integration, favorably positioning the device into cost-conscious applications requiring significant control-processing and connectivity capabilities. The Stellaris® LM3S6000 series combines both a 10/100 Ethernet Media Access Control (MAC) and Physical (PHY) layer, marking the first time that integrated connectivity is available with an ARM Cortex-M3 MCU and the only integrated 10/100 Ethernet MAC and PHY available in an ARM architecture MCU.

The LM3S6432 microcontroller is targeted for industrial applications, including remote monitoring, electronic point-of-sale machines, test and measurement equipment, network appliances and switches, factory automation, HVAC and building control, gaming equipment, motion control, medical instrumentation, and fire and security.

In addition, the LM3S6432 microcontroller offers the advantages of ARM's widely available development tools, System-on-Chip (SoC) infrastructure IP applications, and a large user community. Additionally, the microcontroller uses ARM's Thumb®-compatible Thumb-2 instruction set to reduce memory requirements and, thereby, cost. Finally, the LM3S6432 microcontroller is code-compatible to all members of the extensive Stellaris® family; providing flexibility to fit our customers' precise needs.

Texas Instruments offers a complete solution to get to market quickly, with evaluation and development boards, white papers and application notes, an easy-to-use peripheral driver library, and a strong support, sales, and distributor network. See “Ordering and Contact Information” on page 565 for ordering information for Stellaris® family devices.

1.1 Product Features

The LM3S6432 microcontroller includes the following product features:

- 32-Bit RISC Performance
  - 32-bit ARM® Cortex™-M3 v7M architecture optimized for small-footprint embedded applications
  - System timer (SysTick), providing a simple, 24-bit clear-on-write, decrementing, wrap-on-zero counter with a flexible control mechanism
  - Thumb®-compatible Thumb-2-only instruction set processor core for high code density
  - 50-MHz operation
  - Hardware-division and single-cycle-multiplication
  - Integrated Nested Vectored Interrupt Controller (NVIC) providing deterministic interrupt handling
  - 29 interrupts with eight priority levels

  - Memory protection unit (MPU), providing a privileged mode for protected operating system functionality
  - Unaligned data access, enabling data to be efficiently packed into memory
  - Atomic bit manipulation (bit-banding), delivering maximum memory utilization and streamlined peripheral control

- ARM® Cortex™-M3 Processor Core
  - Compact core.
  - Thumb-2 instruction set, delivering the high-performance expected of an ARM core in the memory size usually associated with 8- and 16-bit devices; typically in the range of a few kilobytes of memory for microcontroller class applications.
  - Rapid application execution through Harvard architecture characterized by separate buses for instruction and data.
  - Exceptional interrupt handling, by implementing the register manipulations required for handling an interrupt in hardware.
  - Deterministic, fast interrupt processing: always 12 cycles, or just 6 cycles with tail-chaining
  - Memory protection unit (MPU) to provide a privileged mode of operation for complex applications.
  - Migration from the ARM7™ processor family for better performance and power efficiency.
  - Full-featured debug solution
    - Serial Wire JTAG Debug Port (SWJ-DP)
    - Flash Patch and Breakpoint (FPB) unit for implementing breakpoints
    - Data Watchpoint and Trigger (DWT) unit for implementing watchpoints, trigger resources, and system profiling
    - Instrumentation Trace Macrocell (ITM) for support of printf style debugging
    - Trace Port Interface Unit (TPIU) for bridging to a Trace Port Analyzer
  - Optimized for single-cycle flash usage
  - Three sleep modes with clock gating for low power
  - Single-cycle multiply instruction and hardware divide
  - Atomic operations
  - ARM Thumb2 mixed 16-/32-bit instruction set
  - 1.25 DMIPS/MHz
- JTAG

  - IEEE 1149.1-1990 compatible Test Access Port (TAP) controller
  - Four-bit Instruction Register (IR) chain for storing JTAG instructions
  - IEEE standard instructions: BYPASS, IDCODE, SAMPLE/PRELOAD, EXTEST and INTEST
  - ARM additional instructions: APACC, DPACC and ABORT
  - Integrated ARM Serial Wire Debug (SWD)
- Internal Memory
  - 96 KB single-cycle flash
    - User-managed flash block protection on a 2-KB block basis
    - User-managed flash data programming
    - User-defined and managed flash-protection block
  - 32 KB single-cycle SRAM
- GPIOs
  - 14-43 GPIOs, depending on configuration
  - 5-V-tolerant input/outputs
  - Programmable control for GPIO interrupts
    - Interrupt generation masking
    - Edge-triggered on rising, falling, or both
    - Level-sensitive on High or Low values
  - Bit masking in both read and write operations through address lines
  - Can initiate an ADC sample sequence
  - Pins configured as digital inputs are Schmitt-triggered.
  - Programmable control for GPIO pad configuration
    - Weak pull-up or pull-down resistors
    - 2-mA, 4-mA, and 8-mA pad drive for digital communication; up to four pads can be configured with an 18-mA pad drive for high-current applications
    - Slew rate control for the 8-mA drive
    - Open drain enables
    - Digital input enables
- General-Purpose Timers

- Three General-Purpose Timer Modules (GPTM), each of which provides two 16-bit timers/counters. Each GPTM can be configured to operate independently:
  - As a single 32-bit timer
  - As one 32-bit Real-Time Clock (RTC) to event capture
  - For Pulse Width Modulation (PWM)
  - To trigger analog-to-digital conversions
- 32-bit Timer modes
  - Programmable one-shot timer
  - Programmable periodic timer
  - Real-Time Clock when using an external 32.768-KHz clock as the input
  - User-enabled stalling when the controller asserts CPU Halt flag during debug
  - ADC event trigger
- 16-bit Timer modes
  - General-purpose timer function with an 8-bit prescaler (for one-shot and periodic modes only)
  - Programmable one-shot timer
  - Programmable periodic timer
  - User-enabled stalling when the controller asserts CPU Halt flag during debug
  - ADC event trigger
- 16-bit Input Capture modes
  - Input edge count capture
  - Input edge time capture
- 16-bit PWM mode
  - Simple PWM mode with software-programmable output inversion of the PWM signal

■ ARM FiRM-compliant Watchdog Timer

- 32-bit down counter with a programmable load register
- Separate watchdog clock with an enable
- Programmable interrupt generation logic with interrupt masking
- Lock register protection from runaway software
- Reset generation logic with an enable/disable

  - User-enabled stalling when the controller asserts the CPU Halt flag during debug

- ADC
  - Three analog input channels
  - Single-ended and differential-input configurations
  - On-chip internal temperature sensor
  - Sample rate of 250 thousand samples/second
  - Flexible, configurable analog-to-digital conversion
  - Four programmable sample conversion sequences from one to eight entries long, with corresponding conversion result FIFOs
  - Flexible trigger control
    - Controller (software)
    - Timers
    - Analog Comparators
    - PWM
    - GPIO
  - Hardware averaging of up to 64 samples for improved accuracy
  - Converter uses an internal 3-V reference
  - Power and ground for the analog circuitry is separate from the digital power and ground
- UART
  - Two fully programmable 16C550-type UARTs with IrDA support
  - Separate 16x8 transmit (TX) and receive (RX) FIFOs to reduce CPU interrupt service loading
  - Programmable baud-rate generator allowing speeds up to 3.125 Mbps
  - Programmable FIFO length, including 1-byte deep operation providing conventional double-buffered interface
  - FIFO trigger levels of 1/8, 1/4, 1/2, 3/4, and 7/8
  - Standard asynchronous communication bits for start, stop, and parity
  - False-start bit detection
  - Line-break generation and detection
  - Fully programmable serial interface characteristics
    - 5, 6, 7, or 8 data bits

- Even, odd, stick, or no-parity bit generation/detection
- 1 or 2 stop bit generation

– IrDA serial-IR (SIR) encoder/decoder providing

- Programmable use of IrDA Serial Infrared (SIR) or UART input/output
- Support of IrDA SIR encoder/decoder functions for data rates up to 115.2 Kbps half-duplex
- Support of normal 3/16 and low-power (1.41-2.23 µs) bit durations
- Programmable internal clock generator enabling division of reference clock by 1 to 256 for low-power mode bit duration

■ Synchronous Serial Interface (SSI)

– Master or slave operation

– Programmable clock bit rate and prescale

– Separate transmit and receive FIFOs, 16 bits wide, 8 locations deep

– Programmable interface operation for Freescale SPI, MICROWIRE, or Texas Instruments synchronous serial interfaces

– Programmable data frame size from 4 to 16 bits

– Internal loopback test mode for diagnostic/debug testing

■ I^2C

– Devices on the I^2C bus can be designated as either a master or a slave

- Supports both sending and receiving data as either a master or a slave
- Supports simultaneous master and slave operation

– Four I^2C modes

- Master transmit
- Master receive
- Slave transmit
- Slave receive

– Two transmission speeds: Standard (100 Kbps) and Fast (400 Kbps)

– Master and slave interrupt generation

- Master generates interrupts when a transmit or receive operation completes (or aborts due to an error)
- Slave generates interrupts when data has been sent or requested by a master

- Master with arbitration and clock synchronization, multimaster support, and 7-bit addressing mode

■ 10/100 Ethernet Controller

- Conforms to the *IEEE 802.3-2002 specification*
  - 10BASE-T/100BASE-TX IEEE-802.3 compliant. Requires only a dual 1:1 isolation transformer interface to the line
  - 10BASE-T/100BASE-TX ENDEC, 100BASE-TX scrambler/descrambler
  - Full-featured auto-negotiation
- Multiple operational modes
  - Full- and half-duplex 100 Mbps
  - Full- and half-duplex 10 Mbps
  - Power-saving and power-down modes
- Highly configurable
  - Programmable MAC address
  - LED activity selection
  - Promiscuous mode support
  - CRC error-rejection control
  - User-configurable interrupts
- Physical media manipulation
  - Automatic MDI/MDI-X cross-over correction
  - Register-programmable transmit amplitude
  - Automatic polarity correction and 10BASE-T signal reception

■ Analog Comparators

- Two independent integrated analog comparators
- Configurable for output to drive an output pin, generate an interrupt, or initiate an ADC sample sequence
- Compare external pin input to external pin input or to internal programmable voltage reference
- Compare a test voltage against any one of these voltages
  - An individual external reference voltage
  - A shared single external reference voltage

- A shared internal reference voltage

■ PWM

– One PWM generator blocks, each with one 16-bit counter, two PWM comparators, a PWM signal generator, a dead-band generator, and an interrupt/ADC-trigger selector

– One fault input in hardware to promote low-latency shutdown

– One 16-bit counter

- Runs in Down or Up/Down mode
- Output frequency controlled by a 16-bit load value
- Load value updates can be synchronized
- Produces output signals at zero and load value

– Two PWM comparators

- Comparator value updates can be synchronized
- Produces output signals on match

– PWM generator

- Output PWM signal is constructed based on actions taken as a result of the counter and PWM comparator output signals
- Produces two independent PWM signals

– Dead-band generator

- Produces two PWM signals with programmable dead-band delays suitable for driving a half-H bridge
- Can be bypassed, leaving input PWM signals unmodified

– Flexible output control block with PWM output enable of each PWM signal

- PWM output enable of each PWM signal
- Optional output inversion of each PWM signal (polarity control)
- Optional fault handling for each PWM signal
- Synchronization of timers in the PWM generator blocks
- Synchronization of timer/comparator updates across the PWM generator blocks
- Interrupt status summary of the PWM generator blocks

– Can initiate an ADC sample sequence

■ Power

  - On-chip Low Drop-Out (LDO) voltage regulator, with programmable output user-adjustable from 2.25 V to 2.75 V
  - Low-power options on controller: Sleep and Deep-sleep modes
  - Low-power options for peripherals: software controls shutdown of individual peripherals
  - 3.3-V supply brown-out detection and reporting via interrupt or reset
- Flexible Reset Sources
  - Power-on reset (POR)
  - Reset pin assertion
  - Brown-out (BOR) detector alerts to system power drops
  - Software reset
  - Watchdog timer reset
  - Internal low drop-out (LDO) regulator output goes unregulated
- Industrial and extended temperature 100-pin RoHS-compliant LQFP package
- Industrial-range 108-ball RoHS-compliant BGA package

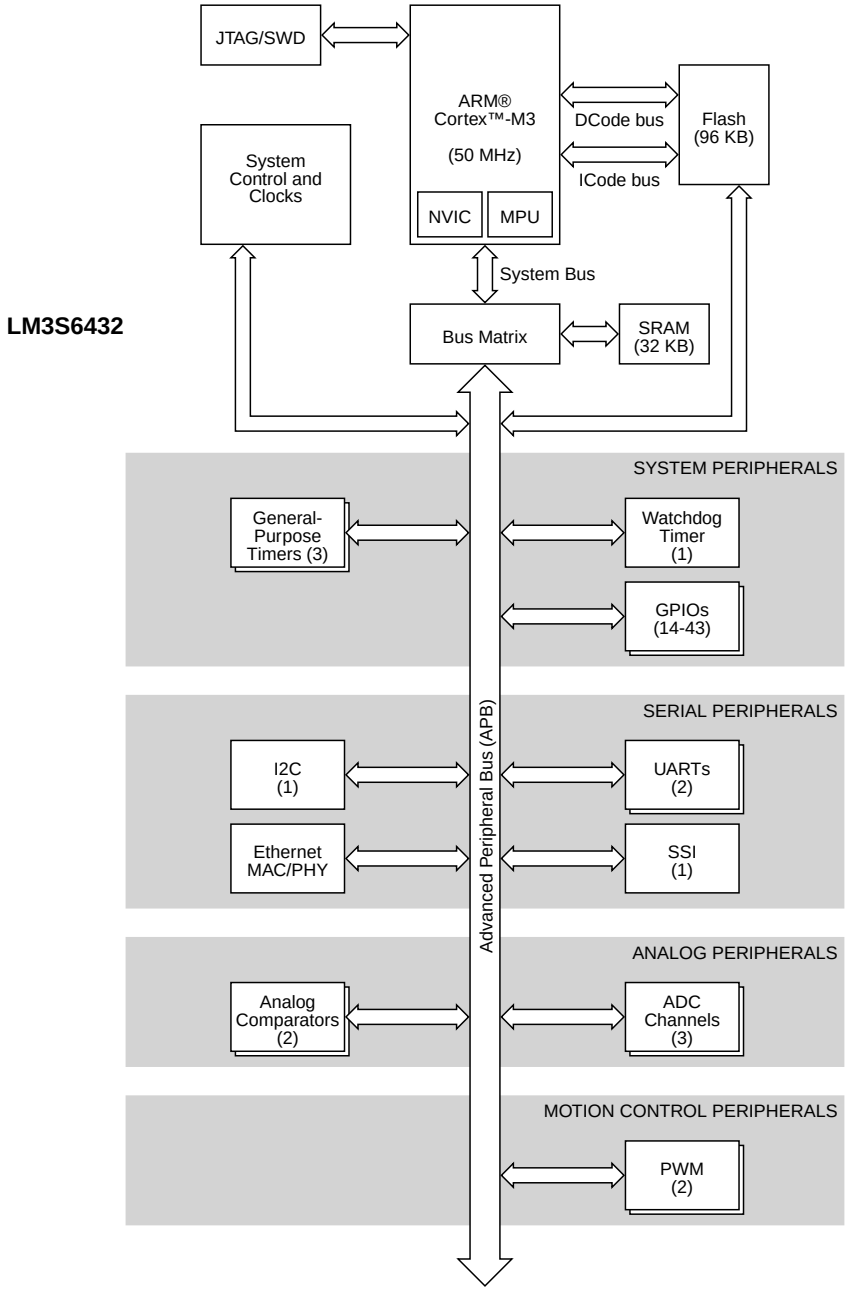
1.2 Target Applications

- Remote monitoring
- Electronic point-of-sale (POS) machines
- Test and measurement equipment
- Network appliances and switches
- Factory automation
- HVAC and building control
- Gaming equipment
- Motion control
- Medical instrumentation
- Fire and security
- Power and energy
- Transportation

1.3 High-Level Block Diagram

Figure 1-1 on page 34 depicts the features on the Stellaris® LM3S6432 microcontroller.

Figure 1-1. Stellaris® LM3S6432 Microcontroller High-Level Block Diagram



1.4 Functional Overview

The following sections provide an overview of the features of the LM3S6432 microcontroller. The page number in parenthesis indicates where that feature is discussed in detail. Ordering and support information can be found in “Ordering and Contact Information” on page 565.

1.4.1 ARM Cortex™-M3

1.4.1.1 Processor Core (see page 41)

All members of the Stellaris® product family, including the LM3S6432 microcontroller, are designed around an ARM Cortex™-M3 processor core. The ARM Cortex-M3 processor provides the core for a high-performance, low-cost platform that meets the needs of minimal memory implementation, reduced pin count, and low-power consumption, while delivering outstanding computational performance and exceptional system response to interrupts.

“ARM Cortex-M3 Processor Core” on page 41 provides an overview of the ARM core; the core is detailed in the *ARM® Cortex™-M3 Technical Reference Manual*.

1.4.1.2 System Timer (SysTick) (see page 44)

Cortex-M3 includes an integrated system timer, SysTick. SysTick provides a simple, 24-bit clear-on-write, decrementing, wrap-on-zero counter with a flexible control mechanism. The counter can be used in several different ways, for example:

- An RTOS tick timer which fires at a programmable rate (for example, 100 Hz) and invokes a SysTick routine.
- A high-speed alarm timer using the system clock.
- A variable rate alarm or signal timer—the duration is range-dependent on the reference clock used and the dynamic range of the counter.
- A simple counter. Software can use this to measure time to completion and time used.
- An internal clock source control based on missing/meeting durations. The COUNTFLAG bit-field in the control and status register can be used to determine if an action completed within a set duration, as part of a dynamic clock management control loop.

1.4.1.3 Nested Vectored Interrupt Controller (NVIC) (see page 49)

The LM3S6432 controller includes the ARM Nested Vectored Interrupt Controller (NVIC) on the ARM® Cortex™-M3 core. The NVIC and Cortex-M3 prioritize and handle all exceptions. All exceptions are handled in Handler Mode. The processor state is automatically stored to the stack on an exception, and automatically restored from the stack at the end of the Interrupt Service Routine (ISR). The vector is fetched in parallel to the state saving, which enables efficient interrupt entry. The processor supports tail-chaining, which enables back-to-back interrupts to be performed without the overhead of state saving and restoration. Software can set eight priority levels on 7 exceptions (system handlers) and 29 interrupts.

“Interrupts” on page 49 provides an overview of the NVIC controller and the interrupt map. Exceptions and interrupts are detailed in the *ARM® Cortex™-M3 Technical Reference Manual*.

1.4.2 Motor Control Peripherals

To enhance motor control, the LM3S6432 controller features Pulse Width Modulation (PWM) outputs.

1.4.2.1 PWM

Pulse width modulation (PWM) is a powerful technique for digitally encoding analog signal levels. High-resolution counters are used to generate a square wave, and the duty cycle of the square wave is modulated to encode an analog signal. Typical applications include switching power supplies and motor control.

On the LM3S6432, PWM motion control functionality can be achieved through:

- Dedicated, flexible motion control hardware using the PWM pins
- The motion control features of the general-purpose timers using the CCP pins

PWM Pins (see page 463)

The LM3S6432 PWM module consists of one PWM generator blocks and a control block. Each PWM generator block contains one timer (16-bit down or up/down counter), two comparators, a PWM signal generator, a dead-band generator, and an interrupt/ADC-trigger selector. The control block determines the polarity of the PWM signals, and which signals are passed through to the pins.

Each PWM generator block produces two PWM signals that can either be independent signals or a single pair of complementary signals with dead-band delays inserted. The output of the PWM generation blocks are managed by the output control block before being passed to the device pins.

CCP Pins (see page 199)

The General-Purpose Timer Module's CCP (Capture Compare PWM) pins are software programmable to support a simple PWM mode with a software-programmable output inversion of the PWM signal.

Fault Pin (see page 468)

The LM3S6432 PWM module includes one fault-condition handling input to quickly provide low-latency shutdown and prevent damage to the motor being controlled.

1.4.3 Analog Peripherals

To handle analog signals, the LM3S6432 microcontroller offers an Analog-to-Digital Converter (ADC).

For support of analog signals, the LM3S6432 microcontroller offers two analog comparators.

1.4.3.1 ADC (see page 253)

An analog-to-digital converter (ADC) is a peripheral that converts a continuous analog voltage to a discrete digital number.

The LM3S6432 ADC module features 10-bit conversion resolution and supports three input channels, plus an internal temperature sensor. Four buffered sample sequences allow rapid sampling of up to eight analog input sources without controller intervention. Each sample sequence provides flexible programming with fully configurable input source, trigger events, interrupt generation, and sequence priority.

1.4.3.2 Analog Comparators (see page 451)

An analog comparator is a peripheral that compares two analog voltages, and provides a logical output that signals the comparison result.

The LM3S6432 microcontroller provides two independent integrated analog comparators that can be configured to drive an output or generate an interrupt or ADC event.

A comparator can compare a test voltage against any one of these voltages:

- An individual external reference voltage
- A shared single external reference voltage
- A shared internal reference voltage

The comparator can provide its output to a device pin, acting as a replacement for an analog comparator on the board, or it can be used to signal the application via interrupts or triggers to the ADC to cause it to start capturing a sample sequence. The interrupt generation and ADC triggering logic is separate. This means, for example, that an interrupt can be generated on a rising edge and the ADC triggered on a falling edge.

1.4.4 Serial Communications Peripherals

The LM3S6432 controller supports both asynchronous and synchronous serial communications with:

- Two fully programmable 16C550-type UARTs
- One SSI module
- One I^2C module
- Ethernet controller

1.4.4.1 UART (see page 290)

A Universal Asynchronous Receiver/Transmitter (UART) is an integrated circuit used for RS-232C serial communications, containing a transmitter (parallel-to-serial converter) and a receiver (serial-to-parallel converter), each clocked separately.

The LM3S6432 controller includes two fully programmable 16C550-type UARTs that support data transfer speeds up to 3.125 Mbps. (Although similar in functionality to a 16C550 UART, it is not register-compatible.) In addition, each UART is capable of supporting IrDA.

Separate 16x8 transmit (TX) and receive (RX) FIFOs reduce CPU interrupt service loading. The UART can generate individually masked interrupts from the RX, TX, modem status, and error conditions. The module provides a single combined interrupt when any of the interrupts are asserted and are unmasked.

1.4.4.2 SSI (see page 331)

Synchronous Serial Interface (SSI) is a four-wire bi-directional full and low-speed communications interface.

The LM3S6432 controller includes one SSI module that provides the functionality for synchronous serial communications with peripheral devices, and can be configured to use the Freescale SPI, MICROWIRE, or TI synchronous serial interface frame formats. The size of the data frame is also configurable, and can be set between 4 and 16 bits, inclusive.

The SSI module performs serial-to-parallel conversion on data received from a peripheral device, and parallel-to-serial conversion on data transmitted to a peripheral device. The TX and RX paths are buffered with internal FIFOs, allowing up to eight 16-bit values to be stored independently.

The SSI module can be configured as either a master or slave device. As a slave device, the SSI module can also be configured to disable its output, which allows a master device to be coupled with multiple slave devices.

The SSI module also includes a programmable bit rate clock divider and prescaler to generate the output serial clock derived from the SSI module's input clock. Bit rates are generated based on the input clock and the maximum bit rate is determined by the connected peripheral.

1.4.4.3 I^2C (see page 368)

The Inter-Integrated Circuit (I^2C) bus provides bi-directional data transfer through a two-wire design (a serial data line SDA and a serial clock line SCL).

The I^2C bus interfaces to external I^2C devices such as serial memory (RAMs and ROMs), networking devices, LCDs, tone generators, and so on. The I^2C bus may also be used for system testing and diagnostic purposes in product development and manufacture.

The LM3S6432 controller includes one I^2C module that provides the ability to communicate to other IC devices over an I^2C bus. The I^2C bus supports devices that can both transmit and receive (write and read) data.

Devices on the I^2C bus can be designated as either a master or a slave. The I^2C module supports both sending and receiving data as either a master or a slave, and also supports the simultaneous operation as both a master and a slave. The four I^2C modes are: Master Transmit, Master Receive, Slave Transmit, and Slave Receive.

A Stellaris® I^2C module can operate at two speeds: Standard (100 Kbps) and Fast (400 Kbps).

Both the I^2C master and slave can generate interrupts. The I^2C master generates interrupts when a transmit or receive operation completes (or aborts due to an error). The I^2C slave generates interrupts when data has been sent or requested by a master.

1.4.4.4 Ethernet Controller (see page 404)

Ethernet is a frame-based computer networking technology for local area networks (LANs). Ethernet has been standardized as IEEE 802.3. It defines a number of wiring and signaling standards for the physical layer, two means of network access at the Media Access Control (MAC)/Data Link Layer, and a common addressing format.

The Stellaris® Ethernet Controller consists of a fully integrated media access controller (MAC) and network physical (PHY) interface device. The Ethernet Controller conforms to IEEE 802.3 specifications and fully supports 10BASE-T and 100BASE-TX standards. In addition, the Ethernet Controller supports automatic MDI/MDI-X cross-over correction.

1.4.5 System Peripherals

1.4.5.1 Programmable GPIOs (see page 151)

General-purpose input/output (GPIO) pins offer flexibility for a variety of connections.

The Stellaris® GPIO module is comprised of seven physical GPIO blocks, each corresponding to an individual GPIO port. The GPIO module is FiRM-compliant (compliant to the ARM Foundation IP for Real-Time Microcontrollers specification) and supports 14-43 programmable input/output pins. The number of GPIOs available depends on the peripherals being used (see “Signal Tables” on page 502 for the signals available to each GPIO pin).

The GPIO module features programmable interrupt generation as either edge-triggered or level-sensitive on all pins, programmable control for GPIO pad configuration, and bit masking in

both read and write operations through address lines. Pins configured as digital inputs are Schmitt-triggered.

1.4.5.2 Three Programmable Timers (see page 193)

Programmable timers can be used to count or time external events that drive the Timer input pins.

The Stellaris® General-Purpose Timer Module (GPTM) contains three GPTM blocks. Each GPTM block provides two 16-bit timers/counters that can be configured to operate independently as timers or event counters, or configured to operate as one 32-bit timer or one 32-bit Real-Time Clock (RTC). Timers can also be used to trigger analog-to-digital (ADC) conversions.

When configured in 32-bit mode, a timer can run as a Real-Time Clock (RTC), one-shot timer or periodic timer. When in 16-bit mode, a timer can run as a one-shot timer or periodic timer, and can extend its precision by using an 8-bit prescaler. A 16-bit timer can also be configured for event capture or Pulse Width Modulation (PWM) generation.

1.4.5.3 Watchdog Timer (see page 229)

A watchdog timer can generate an interrupt or a reset when a time-out value is reached. The watchdog timer is used to regain control when a system has failed due to a software error or to the failure of an external device to respond in the expected way.

The Stellaris® Watchdog Timer module consists of a 32-bit down counter, a programmable load register, interrupt generation logic, and a locking register.

The Watchdog Timer can be configured to generate an interrupt to the controller on its first time-out, and to generate a reset signal on its second time-out. Once the Watchdog Timer has been configured, the lock register can be written to prevent the timer configuration from being inadvertently altered.

1.4.6 Memory Peripherals

The LM3S6432 controller offers both single-cycle SRAM and single-cycle Flash memory.

1.4.6.1 SRAM (see page 126)

The LM3S6432 static random access memory (SRAM) controller supports 32 KB SRAM. The internal SRAM of the Stellaris® devices is located at offset 0x0000.0000 of the device memory map. To reduce the number of time-consuming read-modify-write (RMW) operations, ARM has introduced *bit-banding* technology in the new Cortex-M3 processor. With a bit-band-enabled processor, certain regions in the memory map (SRAM and peripheral space) can use address aliases to access individual bits in a single, atomic operation.

1.4.6.2 Flash (see page 127)

The LM3S6432 Flash controller supports 96 KB of flash memory. The flash is organized as a set of 1-KB blocks that can be individually erased. Erasing a block causes the entire contents of the block to be reset to all 1s. These blocks are paired into a set of 2-KB blocks that can be individually protected. The blocks can be marked as read-only or execute-only, providing different levels of code protection. Read-only blocks cannot be erased or programmed, protecting the contents of those blocks from being modified. Execute-only blocks cannot be erased or programmed, and can only be read by the controller instruction fetch mechanism, protecting the contents of those blocks from being read by either the controller or by a debugger.

1.4.7 Additional Features

1.4.7.1 Memory Map (see page 47)

A memory map lists the location of instructions and data in memory. The memory map for the LM3S6432 controller can be found in “Memory Map” on page 47. Register addresses are given as a hexadecimal increment, relative to the module's base address as shown in the memory map.

The *ARM® Cortex™-M3 Technical Reference Manual* provides further information on the memory map.

1.4.7.2 JTAG TAP Controller (see page 52)

The Joint Test Action Group (JTAG) port is an IEEE standard that defines a Test Access Port and Boundary Scan Architecture for digital integrated circuits and provides a standardized serial interface for controlling the associated test logic. The TAP, Instruction Register (IR), and Data Registers (DR) can be used to test the interconnections of assembled printed circuit boards and obtain manufacturing information on the components. The JTAG Port also provides a means of accessing and controlling design-for-test features such as I/O pin observation and control, scan testing, and debugging.

The JTAG port is composed of the standard five pins: TRST, TCK, TMS, TDI, and TDO. Data is transmitted serially into the controller on TDI and out of the controller on TDO. The interpretation of this data is dependent on the current state of the TAP controller. For detailed information on the operation of the JTAG port and TAP controller, please refer to the *IEEE Standard 1149.1-Test Access Port and Boundary-Scan Architecture*.

The Stellaris® JTAG controller works with the ARM JTAG controller built into the Cortex-M3 core. This is implemented by multiplexing the TDO outputs from both JTAG controllers. ARM JTAG instructions select the ARM TDO output while Stellaris® JTAG instructions select the Stellaris® TDO outputs. The multiplexer is controlled by the Stellaris® JTAG controller, which has comprehensive programming for the ARM, Stellaris®, and unimplemented JTAG instructions.

1.4.7.3 System Control and Clocks (see page 64)

System control determines the overall operation of the device. It provides information about the device, controls the clocking of the device and individual peripherals, and handles reset detection and reporting.

1.4.8 Hardware Details

Details on the pins and package can be found in the following sections:

- “Pin Diagram” on page 500
- “Signal Tables” on page 502
- “Operating Characteristics” on page 526
- “Electrical Characteristics” on page 527
- “Package Information” on page 567

2 ARM Cortex-M3 Processor Core

The ARM Cortex-M3 processor provides the core for a high-performance, low-cost platform that meets the needs of minimal memory implementation, reduced pin count, and low power consumption, while delivering outstanding computational performance and exceptional system response to interrupts. Features include:

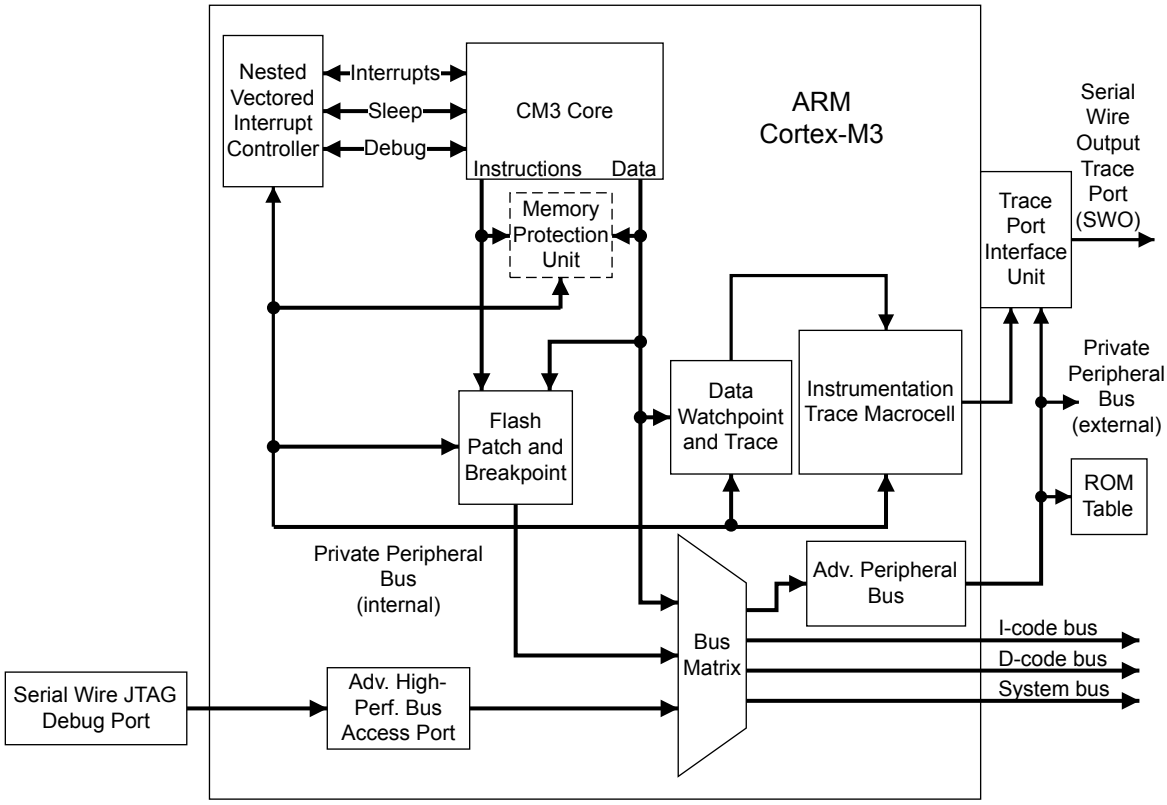
- Compact core.
- Thumb-2 instruction set, delivering the high-performance expected of an ARM core in the memory size usually associated with 8- and 16-bit devices; typically in the range of a few kilobytes of memory for microcontroller class applications.
- Rapid application execution through Harvard architecture characterized by separate buses for instruction and data.
- Exceptional interrupt handling, by implementing the register manipulations required for handling an interrupt in hardware.
- Deterministic, fast interrupt processing: always 12 cycles, or just 6 cycles with tail-chaining
- Memory protection unit (MPU) to provide a privileged mode of operation for complex applications.
- Migration from the ARM7™ processor family for better performance and power efficiency.
- Full-featured debug solution
  - Serial Wire JTAG Debug Port (SWJ-DP)
  - Flash Patch and Breakpoint (FPB) unit for implementing breakpoints
  - Data Watchpoint and Trigger (DWT) unit for implementing watchpoints, trigger resources, and system profiling
  - Instrumentation Trace Macrocell (ITM) for support of printf style debugging
  - Trace Port Interface Unit (TPIU) for bridging to a Trace Port Analyzer
- Optimized for single-cycle flash usage
- Three sleep modes with clock gating for low power
- Single-cycle multiply instruction and hardware divide
- Atomic operations
- ARM Thumb2 mixed 16-/32-bit instruction set
- 1.25 DMIPS/MHz

The Stellaris® family of microcontrollers builds on this core to bring high-performance 32-bit computing to cost-sensitive embedded microcontroller applications, such as factory automation and control, industrial control power devices, building and home automation, and stepper motors.

For more information on the ARM Cortex-M3 processor core, see the *ARM® Cortex™-M3 Technical Reference Manual*. For information on SWJ-DP, see the *ARM® CoreSight Technical Reference Manual*.

2.1 Block Diagram

Figure 2-1. CPU Block Diagram



2.2 Functional Description

Important: The *ARM® Cortex™-M3 Technical Reference Manual* describes all the features of an ARM Cortex-M3 in detail. However, these features differ based on the implementation. This section describes the Stellaris® implementation.

Texas Instruments has implemented the ARM Cortex-M3 core as shown in Figure 2-1 on page 42. As noted in the *ARM® Cortex™-M3 Technical Reference Manual*, several Cortex-M3 components are flexible in their implementation: SW/JTAG-DP, ETM, TPIU, the ROM table, the MPU, and the Nested Vectored Interrupt Controller (NVIC). Each of these is addressed in the sections that follow.

2.2.1 Serial Wire and JTAG Debug

Texas Instruments has replaced the ARM SW-DP and JTAG-DP with the ARM CoreSight™-compliant Serial Wire JTAG Debug Port (SWJ-DP) interface. The SWJ-DP interface combines the SWD and JTAG debug ports into one module. See the *CoreSight™ Design Kit Technical Reference Manual* for details on SWJ-DP.

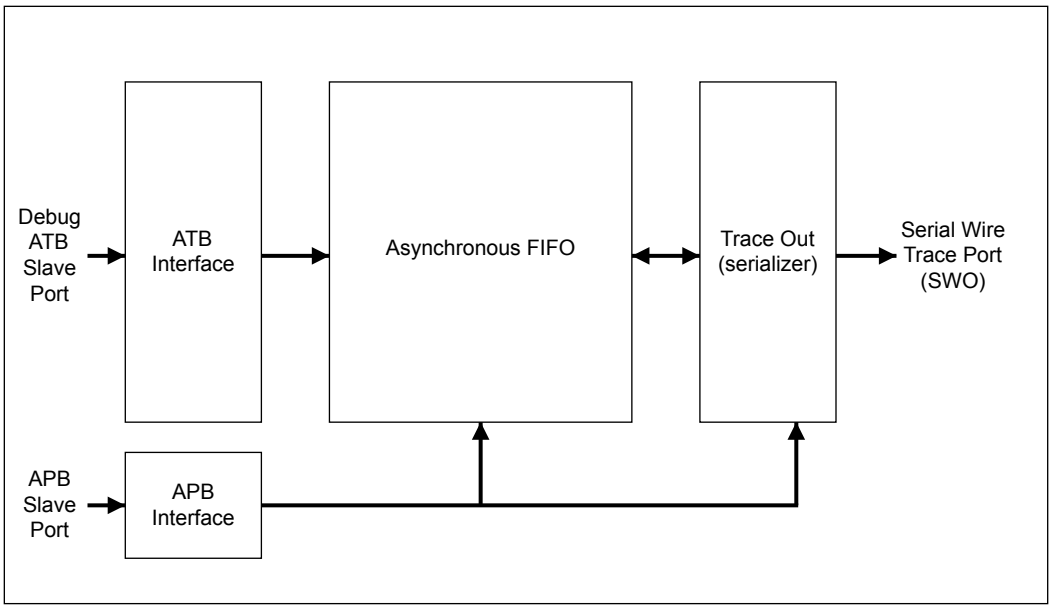
2.2.2 Embedded Trace Macrocell (ETM)

ETM was not implemented in the Stellaris® devices. This means Chapters 15 and 16 of the *ARM® Cortex™-M3 Technical Reference Manual* can be ignored.

2.2.3 Trace Port Interface Unit (TPIU)

The TPIU acts as a bridge between the Cortex-M3 trace data from the ITM, and an off-chip Trace Port Analyzer. The Stellaris® devices have implemented TPIU as shown in Figure 2-2 on page 43. This is similar to the non-ETM version described in the *ARM® Cortex™-M3 Technical Reference Manual*, however, SWJ-DP only provides SWV output for the TPIU.

Figure 2-2. TPIU Block Diagram



2.2.4 ROM Table

The default ROM table was implemented as described in the *ARM® Cortex™-M3 Technical Reference Manual*.

2.2.5 Memory Protection Unit (MPU)

The Memory Protection Unit (MPU) is included on the LM3S6432 controller and supports the standard ARMv7 Protected Memory System Architecture (PMSA) model. The MPU provides full support for protection regions, overlapping protection regions, access permissions, and exporting memory attributes to the system.

2.2.6 Nested Vectored Interrupt Controller (NVIC)

The Nested Vectored Interrupt Controller (NVIC):

- Facilitates low-latency exception and interrupt handling
- Controls power management
- Implements system control registers

The NVIC supports up to 240 dynamically reprioritizable interrupts each with up to 256 levels of priority. The NVIC and the processor core interface are closely coupled, which enables low latency interrupt processing and efficient processing of late arriving interrupts. The NVIC maintains knowledge of the stacked (nested) interrupts to enable tail-chaining of interrupts.

You can only fully access the NVIC from privileged mode, but you can pend interrupts in user-mode if you enable the Configuration Control Register (see the ARM® Cortex™-M3 Technical Reference Manual). Any other user-mode access causes a bus fault.

All NVIC registers are accessible using byte, halfword, and word unless otherwise stated.

2.2.6.1 Interrupts

The *ARM® Cortex™-M3 Technical Reference Manual* describes the maximum number of interrupts and interrupt priorities. The LM3S6432 microcontroller supports 29 interrupts with eight priority levels.

2.2.6.2 System Timer (SysTick)

Cortex-M3 includes an integrated system timer, SysTick. SysTick provides a simple, 24-bit clear-on-write, decrementing, wrap-on-zero counter with a flexible control mechanism. The counter can be used in several different ways, for example:

- An RTOS tick timer which fires at a programmable rate (for example, 100 Hz) and invokes a SysTick routine.
- A high-speed alarm timer using the system clock.
- A variable rate alarm or signal timer—the duration is range-dependent on the reference clock used and the dynamic range of the counter.
- A simple counter. Software can use this to measure time to completion and time used.
- An internal clock source control based on missing/meeting durations. The COUNTFLAG bit-field in the control and status register can be used to determine if an action completed within a set duration, as part of a dynamic clock management control loop.

Functional Description

The timer consists of three registers:

- A control and status counter to configure its clock, enable the counter, enable the SysTick interrupt, and determine counter status.
- The reload value for the counter, used to provide the counter's wrap value.
- The current value of the counter.

A fourth register, the SysTick Calibration Value Register, is not implemented in the Stellaris® devices.

When enabled, the timer counts down from the reload value to zero, reloads (wraps) to the value in the SysTick Reload Value register on the next clock edge, then decrements on subsequent clocks. Writing a value of zero to the Reload Value register disables the counter on the next wrap. When the counter reaches zero, the COUNTFLAG status bit is set. The COUNTFLAG bit clears on reads.

Writing to the Current Value register clears the register and the COUNTFLAG status bit. The write does not trigger the SysTick exception logic. On a read, the current value is the value of the register at the time the register is accessed.

If the core is in debug state (halted), the counter will not decrement. The timer is clocked with respect to a reference clock. The reference clock can be the core clock or an external clock source.

SysTick Control and Status Register

Use the SysTick Control and Status Register to enable the SysTick features. The reset is 0x0000.0000.

Bit/Field	Name	Type	Reset	Description
31:17	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
16	COUNTFLAG	R/W	0	Count Flag Returns 1 if timer counted to 0 since last time this was read. Clears on read by application. If read by the debugger using the DAP, this bit is cleared on read-only if the MasterType bit in the AHB-AP Control Register is set to 0. Otherwise, the COUNTFLAG bit is not changed by the debugger read.
15:3	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
2	CLKSOURCE	R/W	0	Clock Source Value Description 0 External reference clock. (Not implemented for Stellaris microcontrollers.) 1 Core clock If no reference clock is provided, it is held at 1 and so gives the same time as the core clock. The core clock must be at least 2.5 times faster than the reference clock. If it is not, the count values are unpredictable.
1	TICKINT	R/W	0	Tick Interrupt Value Description 0 Counting down to 0 does not generate the interrupt request to the NVIC. Software can use the COUNTFLAG to determine if ever counted to 0. 1 Counting down to 0 pends the SysTick handler.
0	ENABLE	R/W	0	Enable Value Description 0 Counter disabled. 1 Counter operates in a multi-shot way. That is, counter loads with the Reload value and then begins counting down. On reaching 0, it sets the COUNTFLAG to 1 and optionally pends the SysTick handler, based on TICKINT. It then loads the Reload value again, and begins counting.

SysTick Reload Value Register

Use the SysTick Reload Value Register to specify the start value to load into the current value register when the counter reaches 0. It can be any value between 1 and 0x00FF.FFFF. A start value of 0 is possible, but has no effect because the SysTick interrupt and COUNTFLAG are activated when counting from 1 to 0.

Therefore, as a multi-shot timer, repeated over and over, it fires every N+1 clock pulse, where N is any value from 1 to 0x00FF.FFFF. So, if the tick interrupt is required every 100 clock pulses, 99 must be written into the RELOAD. If a new value is written on each tick interrupt, so treated as single shot, then the actual count down must be written. For example, if a tick is next required after 400 clock pulses, 400 must be written into the RELOAD.

Bit/Field	Name	Type	Reset	Description
31:24	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
23:0	RELOAD	R/W	-	Reload Value to load into the SysTick Current Value Register when the counter reaches 0.

SysTick Current Value Register

Use the SysTick Current Value Register to find the current value in the register.

Bit/Field	Name	Type	Reset	Description
31:24	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
23:0	CURRENT	W1C	-	Current Value Current value at the time the register is accessed. No read-modify-write protection is provided, so change with care. This register is write-clear. Writing to it with any value clears the register to 0. Clearing this register also clears the COUNTFLAG bit of the SysTick Control and Status Register.

SysTick Calibration Value Register

The SysTick Calibration Value register is not implemented.

3 Memory Map

The memory map for the LM3S6432 controller is provided in Table 3-1 on page 47.

In this manual, register addresses are given as a hexadecimal increment, relative to the module's base address as shown in the memory map. See also Chapter 4, "Memory Map" in the *ARM® Cortex™-M3 Technical Reference Manual*.

Table 3-1. Memory Map[a]

Start	End	Description	For details on registers, see page ...
Memory			
0x0000.0000	0x0001.7FFF	On-chip flash [b]	130
0x0001.8000	0x1FFF.FFFF	Reserved	-
0x2000.0000	0x2000.7FFF	Bit-banded on-chip SRAM[c]	130
0x2000.8000	0x21FF.FFFF	Reserved	-
0x2200.0000	0x220F.FFFF	Bit-band alias of 0x2000.0000 through 0x200F.FFFF	126
0x2210.0000	0x3FFF.FFFF	Reserved	-
FiRM Peripherals			
0x4000.0000	0x4000.0FFF	Watchdog timer	232
0x4000.1000	0x4000.3FFF	Reserved	-
0x4000.4000	0x4000.4FFF	GPIO Port A	158
0x4000.5000	0x4000.5FFF	GPIO Port B	158
0x4000.6000	0x4000.6FFF	GPIO Port C	158
0x4000.7000	0x4000.7FFF	GPIO Port D	158
0x4000.8000	0x4000.8FFF	SSI0	342
0x4000.9000	0x4000.BFFF	Reserved	-
0x4000.C000	0x4000.CFFF	UART0	297
0x4000.D000	0x4000.DFFF	UART1	297
0x4000.E000	0x4001.FFFF	Reserved	-
Peripherals			
0x4002.0000	0x4002.07FF	I2C Master 0	382
0x4002.0800	0x4002.0FFF	I2C Slave 0	395
0x4002.1000	0x4002.3FFF	Reserved	-
0x4002.4000	0x4002.4FFF	GPIO Port E	158
0x4002.5000	0x4002.5FFF	GPIO Port F	158
0x4002.6000	0x4002.6FFF	GPIO Port G	158
0x4002.7000	0x4002.7FFF	Reserved	-
0x4002.8000	0x4002.8FFF	PWM	470
0x4002.9000	0x4002.FFFF	Reserved	-
0x4003.0000	0x4003.0FFF	Timer0	204
0x4003.1000	0x4003.1FFF	Timer1	204
0x4003.2000	0x4003.2FFF	Timer2	204
0x4003.3000	0x4003.7FFF	Reserved	-

Table 3-1. Memory Map *(continued)*

Start	End	Description	For details on registers, see page ...
0x4003.8000	0x4003.8FFF	ADC	261
0x4003.9000	0x4003.BFFF	Reserved	-
0x4003.C000	0x4003.CFFF	Analog Comparators	451
0x4003.D000	0x4004.7FFF	Reserved	-
0x4004.8000	0x4004.8FFF	Ethernet Controller	414
0x4004.9000	0x400F.CFFF	Reserved	-
0x400F.D000	0x400F.DFFF	Flash control	130
0x400F.E000	0x400F.EFFF	System control	74
0x400F.F000	0x41FF.FFFF	Reserved	-
0x4200.0000	0x43FF.FFFF	Bit-banded alias of 0x4000.0000 through 0x400F.FFFF	-
0x4400.0000	0xDFFF.FFFF	Reserved	-
Private Peripheral Bus			
0xE000.0000	0xE000.0FFF	Instrumentation Trace Macrocell (ITM)	*ARM® Cortex™-M3 Technical Reference Manual*
0xE000.1000	0xE000.1FFF	Data Watchpoint and Trace (DWT)	*ARM® Cortex™-M3 Technical Reference Manual*
0xE000.2000	0xE000.2FFF	Flash Patch and Breakpoint (FPB)	*ARM® Cortex™-M3 Technical Reference Manual*
0xE000.3000	0xE000.DFFF	Reserved	-
0xE000.E000	0xE000.EFFF	Nested Vectored Interrupt Controller (NVIC)	*ARM® Cortex™-M3 Technical Reference Manual*
0xE000.F000	0xE003.FFFF	Reserved	-
0xE004.0000	0xE004.0FFF	Trace Port Interface Unit (TPIU)	*ARM® Cortex™-M3 Technical Reference Manual*
0xE004.1000	0xFFFF.FFFF	Reserved	-

a. All reserved space returns a bus fault when read or written.

b. The unavailable flash will bus fault throughout this range.

c. The unavailable SRAM will bus fault throughout this range.

4 Interrupts

The ARM Cortex-M3 processor and the Nested Vectored Interrupt Controller (NVIC) prioritize and handle all exceptions. All exceptions are handled in Handler Mode. The processor state is automatically stored to the stack on an exception, and automatically restored from the stack at the end of the Interrupt Service Routine (ISR). The vector is fetched in parallel to the state saving, which enables efficient interrupt entry. The processor supports tail-chaining, which enables back-to-back interrupts to be performed without the overhead of state saving and restoration.

Table 4-1 on page 49 lists all exception types. Software can set eight priority levels on seven of these exceptions (system handlers) as well as on 29 interrupts (listed in Table 4-2 on page 50).

Priorities on the system handlers are set with the NVIC System Handler Priority registers. Interrupts are enabled through the NVIC Interrupt Set Enable register and prioritized with the NVIC Interrupt Priority registers. You also can group priorities by splitting priority levels into pre-emption priorities and subpriorities. All of the interrupt registers are described in Chapter 8, “Nested Vectored Interrupt Controller” in the *ARM® Cortex™-M3 Technical Reference Manual*.

Internally, the highest user-settable priority (0) is treated as fourth priority, after a Reset, NMI, and a Hard Fault. Note that 0 is the default priority for all the settable priorities.

If you assign the same priority level to two or more interrupts, their hardware priority (the lower position number) determines the order in which the processor activates them. For example, if both GPIO Port A and GPIO Port B are priority level 1, then GPIO Port A has higher priority.

Important: It may take several processor cycles after a write to clear an interrupt source in order for NVIC to see the interrupt source de-assert. This means if the interrupt clear is done as the last action in an interrupt handler, it is possible for the interrupt handler to complete while NVIC sees the interrupt as still asserted, causing the interrupt handler to be re-entered errantly. This can be avoided by either clearing the interrupt source at the beginning of the interrupt handler or by performing a read or write after the write to clear the interrupt source (and flush the write buffer).

See Chapter 5, “Exceptions” and Chapter 8, “Nested Vectored Interrupt Controller” in the *ARM® Cortex™-M3 Technical Reference Manual* for more information on exceptions and interrupts.

Table 4-1. Exception Types

Exception Type	Vector Number	Priority[a]	Description
-	0	-	Stack top is loaded from first entry of vector table on reset.
Reset	1	-3 (highest)	Invoked on power up and warm reset. On first instruction, drops to lowest priority (and then is called the base level of activation). This is asynchronous.
Non-Maskable Interrupt (NMI)	2	-2	Cannot be stopped or preempted by any exception but reset. This is asynchronous. An NMI is only producible by software, using the NVIC **Interrupt Control State** register.
Hard Fault	3	-1	All classes of Fault, when the fault cannot activate due to priority or the configurable fault handler has been disabled. This is synchronous.
Memory Management	4	settable	MPU mismatch, including access violation and no match. This is synchronous. The priority of this exception can be changed.

Table 4-1. Exception Types *(continued)*

Exception Type	Vector Number	Priority[a]	Description
Bus Fault	5	settable	Pre-fetch fault, memory access fault, and other address/memory related faults. This is synchronous when precise and asynchronous when imprecise. You can enable or disable this fault.
Usage Fault	6	settable	Usage fault, such as undefined instruction executed or illegal state transition attempt. This is synchronous.
-	7-10	-	Reserved.
SVCall	11	settable	System service call with SVC instruction. This is synchronous.
Debug Monitor	12	settable	Debug monitor (when not halting). This is synchronous, but only active when enabled. It does not activate if lower priority than the current activation.
-	13	-	Reserved.
PendSV	14	settable	Pendable request for system service. This is asynchronous and only pended by software.
SysTick	15	settable	System tick timer has fired. This is asynchronous.
Interrupts	16 and above	settable	Asserted from outside the ARM Cortex-M3 core and fed through the NVIC (prioritized). These are all asynchronous. Table 4-2 on page 50 lists the interrupts on the LM3S6432 controller.

a. 0 is the default priority for all the settable priorities.

Table 4-2. Interrupts

Vector Number	Interrupt Number (Bit in Interrupt Registers)	Description
0-15	-	Processor exceptions
16	0	GPIO Port A
17	1	GPIO Port B
18	2	GPIO Port C
19	3	GPIO Port D
20	4	GPIO Port E
21	5	UART0
22	6	UART1
23	7	SSI0
24	8	I2C0
25	9	PWM Fault
26	10	PWM Generator 0
27-29	11-13	Reserved
30	14	ADC Sequence 0
31	15	ADC Sequence 1
32	16	ADC Sequence 2
33	17	ADC Sequence 3
34	18	Watchdog timer
35	19	Timer0 A
36	20	Timer0 B
37	21	Timer1 A

Table 4-2. Interrupts *(continued)*

Vector Number	Interrupt Number (Bit in Interrupt Registers)	Description
38	22	Timer1 B
39	23	Timer2 A
40	24	Timer2 B
41	25	Analog Comparator 0
42	26	Analog Comparator 1
43	27	Reserved
44	28	System Control
45	29	Flash Control
46	30	GPIO Port F
47	31	GPIO Port G
48-57	32-41	Reserved
58	42	Ethernet Controller
59-70	43-54	Reserved

5 JTAG Interface

The Joint Test Action Group (JTAG) port is an IEEE standard that defines a Test Access Port and Boundary Scan Architecture for digital integrated circuits and provides a standardized serial interface for controlling the associated test logic. The TAP, Instruction Register (IR), and Data Registers (DR) can be used to test the interconnections of assembled printed circuit boards and obtain manufacturing information on the components. The JTAG Port also provides a means of accessing and controlling design-for-test features such as I/O pin observation and control, scan testing, and debugging.

The JTAG port is comprised of five pins: $\overline{\text{TRST}}$, TCK, TMS, TDI, and TDO. Data is transmitted serially into the controller on TDI and out of the controller on TDO. The interpretation of this data is dependent on the current state of the TAP controller. For detailed information on the operation of the JTAG port and TAP controller, please refer to the *IEEE Standard 1149.1-Test Access Port and Boundary-Scan Architecture*.

The Stellaris® JTAG controller works with the ARM JTAG controller built into the Cortex-M3 core. This is implemented by multiplexing the TDO outputs from both JTAG controllers. ARM JTAG instructions select the ARM TDO output while Stellaris® JTAG instructions select the Stellaris® TDO outputs. The multiplexer is controlled by the Stellaris® JTAG controller, which has comprehensive programming for the ARM, Stellaris®, and unimplemented JTAG instructions.

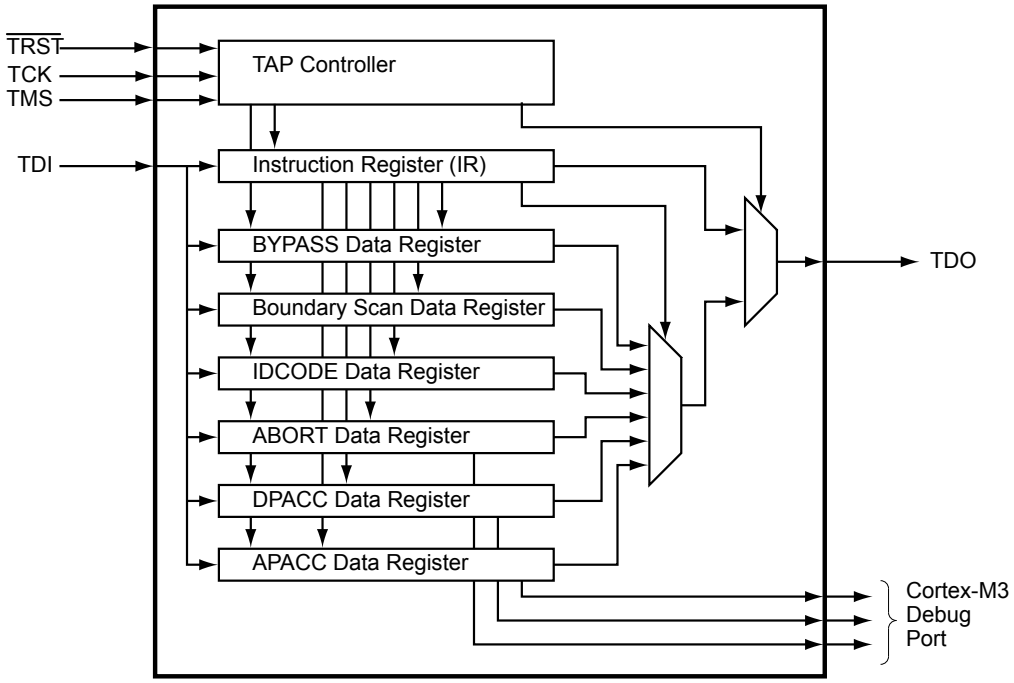
The Stellaris® JTAG module has the following features:

- IEEE 1149.1-1990 compatible Test Access Port (TAP) controller
- Four-bit Instruction Register (IR) chain for storing JTAG instructions
- IEEE standard instructions: BYPASS, IDCODE, SAMPLE/PRELOAD, EXTEST and INTEST
- ARM additional instructions: APACC, DPACC and ABORT
- Integrated ARM Serial Wire Debug (SWD)

See the *ARM® Cortex™-M3 Technical Reference Manual* for more information on the ARM JTAG controller.

5.1 Block Diagram

Figure 5-1. JTAG Module Block Diagram



5.2 Functional Description

A high-level conceptual drawing of the JTAG module is shown in Figure 5-1 on page 53. The JTAG module is composed of the Test Access Port (TAP) controller and serial shift chains with parallel update registers. The TAP controller is a simple state machine controlled by the TRST, TCK and TMS inputs. The current state of the TAP controller depends on the current value of TRST and the sequence of values captured on TMS at the rising edge of TCK. The TAP controller determines when the serial shift chains capture new data, shift data from TDI towards TDO, and update the parallel load registers. The current state of the TAP controller also determines whether the Instruction Register (IR) chain or one of the Data Register (DR) chains is being accessed.

The serial shift chains with parallel load registers are comprised of a single Instruction Register (IR) chain and multiple Data Register (DR) chains. The current instruction loaded in the parallel load register determines which DR chain is captured, shifted, or updated during the sequencing of the TAP controller.

Some instructions, like EXTEST and INTEST, operate on data currently in a DR chain and do not capture, shift, or update any of the chains. Instructions that are not implemented decode to the BYPASS instruction to ensure that the serial path between TDI and TDO is always connected (see Table 5-2 on page 59 for a list of implemented instructions).

See “JTAG and Boundary Scan” on page 532 for JTAG timing diagrams.

5.2.1 JTAG Interface Pins

The JTAG interface consists of five standard pins: TRST,TCK, TMS, TDI, and TDO. These pins and their associated reset state are given in Table 5-1 on page 54. Detailed information on each pin follows.

Table 5-1. JTAG Port Pins Reset State

Pin Name	Data Direction	Internal Pull-Up	Internal Pull-Down	Drive Strength	Drive Value
TRST	Input	Enabled	Disabled	N/A	N/A
TCK	Input	Enabled	Disabled	N/A	N/A
TMS	Input	Enabled	Disabled	N/A	N/A
TDI	Input	Enabled	Disabled	N/A	N/A
TDO	Output	Enabled	Disabled	2-mA driver	High-Z

5.2.1.1 Test Reset Input ($\overline{\text{TRST}}$)

The $\overline{\text{TRST}}$ pin is an asynchronous active Low input signal for initializing and resetting the JTAG TAP controller and associated JTAG circuitry. When $\overline{\text{TRST}}$ is asserted, the TAP controller resets to the Test-Logic-Reset state and remains there while $\overline{\text{TRST}}$ is asserted. When the TAP controller enters the Test-Logic-Reset state, the JTAG Instruction Register (IR) resets to the default instruction, IDCODE.

By default, the internal pull-up resistor on the $\overline{\text{TRST}}$ pin is enabled after reset. Changes to the pull-up resistor settings on GPIO Port B should ensure that the internal pull-up resistor remains enabled on PB7/$\overline{\text{TRST}}$; otherwise JTAG communication could be lost.

5.2.1.2 Test Clock Input (TCK)

The TCK pin is the clock for the JTAG module. This clock is provided so the test logic can operate independently of any other system clocks. In addition, it ensures that multiple JTAG TAP controllers that are daisy-chained together can synchronously communicate serial test data between components. During normal operation, TCK is driven by a free-running clock with a nominal 50% duty cycle. When necessary, TCK can be stopped at 0 or 1 for extended periods of time. While TCK is stopped at 0 or 1, the state of the TAP controller does not change and data in the JTAG Instruction and Data Registers is not lost.

By default, the internal pull-up resistor on the TCK pin is enabled after reset. This assures that no clocking occurs if the pin is not driven from an external source. The internal pull-up and pull-down resistors can be turned off to save internal power as long as the TCK pin is constantly being driven by an external source.

5.2.1.3 Test Mode Select (TMS)

The TMS pin selects the next state of the JTAG TAP controller. TMS is sampled on the rising edge of TCK. Depending on the current TAP state and the sampled value of TMS, the next state is entered. Because the TMS pin is sampled on the rising edge of TCK, the *IEEE Standard 1149.1* expects the value on TMS to change on the falling edge of TCK.

Holding TMS high for five consecutive TCK cycles drives the TAP controller state machine to the Test-Logic-Reset state. When the TAP controller enters the Test-Logic-Reset state, the JTAG Instruction Register (IR) resets to the default instruction, IDCODE. Therefore, this sequence can be used as a reset mechanism, similar to asserting $\overline{\text{TRST}}$. The JTAG Test Access Port state machine can be seen in its entirety in Figure 5-2 on page 56.

By default, the internal pull-up resistor on the TMS pin is enabled after reset. Changes to the pull-up resistor settings on GPIO Port C should ensure that the internal pull-up resistor remains enabled on PC1/TMS; otherwise JTAG communication could be lost.

5.2.1.4 Test Data Input (TDI)

The TDI pin provides a stream of serial information to the IR chain and the DR chains. TDI is sampled on the rising edge of TCK and, depending on the current TAP state and the current instruction, presents this data to the proper shift register chain. Because the TDI pin is sampled on the rising edge of TCK, the *IEEE Standard 1149.1* expects the value on TDI to change on the falling edge of TCK.

By default, the internal pull-up resistor on the TDI pin is enabled after reset. Changes to the pull-up resistor settings on GPIO Port C should ensure that the internal pull-up resistor remains enabled on PC2/TDI; otherwise JTAG communication could be lost.

5.2.1.5 Test Data Output (TDO)

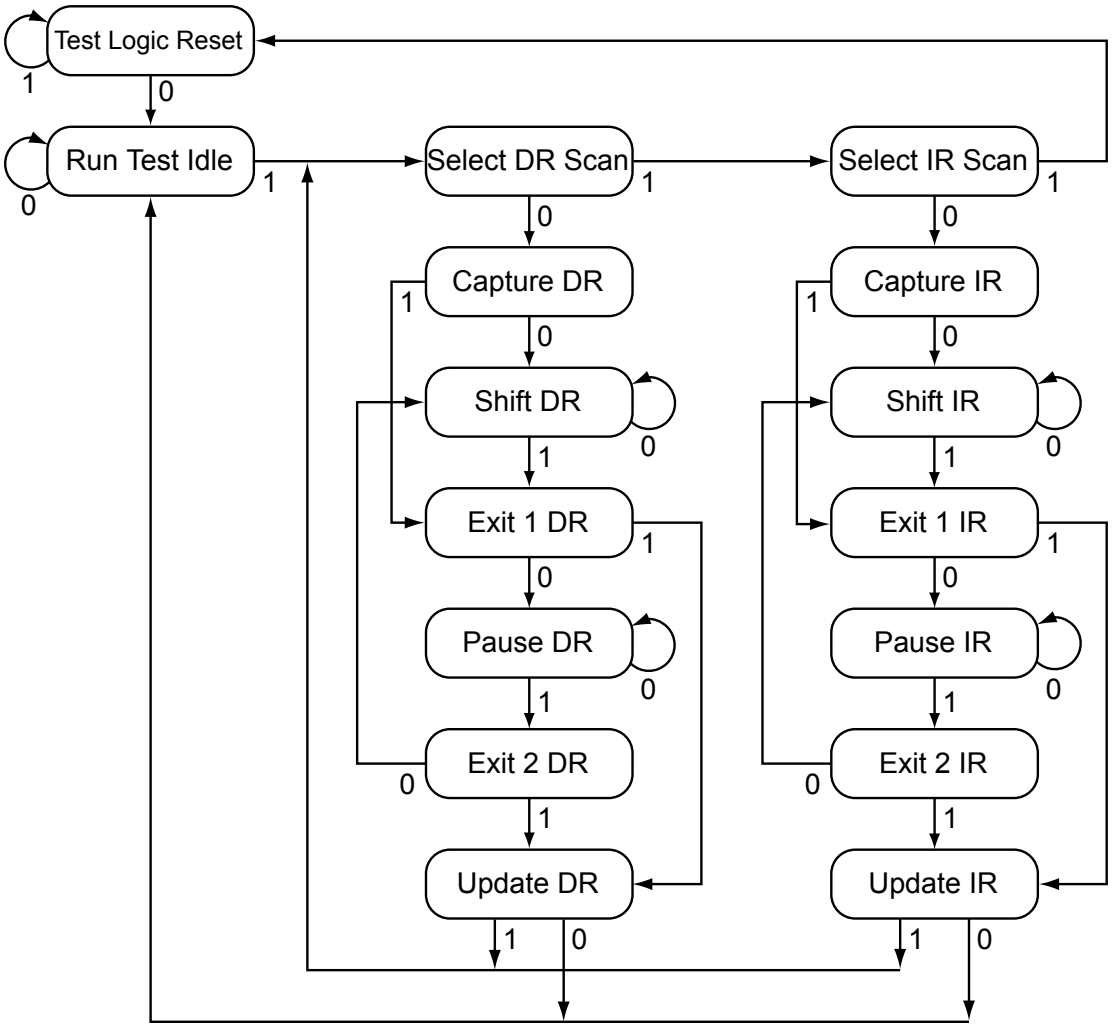
The TDO pin provides an output stream of serial information from the IR chain or the DR chains. The value of TDO depends on the current TAP state, the current instruction, and the data in the chain being accessed. In order to save power when the JTAG port is not being used, the TDO pin is placed in an inactive drive state when not actively shifting out data. Because TDO can be connected to the TDI of another controller in a daisy-chain configuration, the *IEEE Standard 1149.1* expects the value on TDO to change on the falling edge of TCK.

By default, the internal pull-up resistor on the TDO pin is enabled after reset. This assures that the pin remains at a constant logic level when the JTAG port is not being used. The internal pull-up and pull-down resistors can be turned off to save internal power if a High-Z output value is acceptable during certain TAP controller states.

5.2.2 JTAG TAP Controller

The JTAG TAP controller state machine is shown in Figure 5-2 on page 56. The TAP controller state machine is reset to the Test-Logic-Reset state on the assertion of a Power-On-Reset (POR) or the assertion of TRST. Asserting the correct sequence on the TMS pin allows the JTAG module to shift in new instructions, shift in data, or idle during extended testing sequences. For detailed information on the function of the TAP controller and the operations that occur in each state, please refer to *IEEE Standard 1149.1*.

Figure 5-2. Test Access Port State Machine



5.2.3 Shift Registers

The Shift Registers consist of a serial shift register chain and a parallel load register. The serial shift register chain samples specific information during the TAP controller's CAPTURE states and allows this information to be shifted out of TDO during the TAP controller's SHIFT states. While the sampled data is being shifted out of the chain on TDO, new data is being shifted into the serial shift register on TDI. This new data is stored in the parallel load register during the TAP controller's UPDATE states. Each of the shift registers is discussed in detail in "Register Descriptions" on page 59.

5.2.4 Operational Considerations

There are certain operational considerations when using the JTAG module. Because the JTAG pins can be programmed to be GPIOs, board configuration and reset conditions on these pins must be considered. In addition, because the JTAG module has integrated ARM Serial Wire Debug, the method for switching between these two operational modes is described below.

5.2.4.1 GPIO Functionality

When the controller is reset with either a POR or RST, the JTAG/SWD port pins default to their JTAG/SWD configurations. The default configuration includes enabling digital functionality (setting **GPIODEN** to 1), enabling the pull-up resistors (setting **GPIOPUR** to 1), and enabling the alternate hardware function (setting **GPIOAFSEL** to 1) for the PB7 and PC[3:0] JTAG/SWD pins.

It is possible for software to configure these pins as GPIOs after reset by writing 0s to PB7 and PC[3:0] in the **GPIOAFSEL** register. If the user does not require the JTAG/SWD port for debugging or board-level testing, this provides five more GPIOs for use in the design.

Caution – It is possible to create a software sequence that prevents the debugger from connecting to the Stellaris® microcontroller. If the program code loaded into flash immediately changes the JTAG pins to their GPIO functionality, the debugger may not have enough time to connect and halt the controller before the JTAG pin functionality switches. This may lock the debugger out of the part. This can be avoided with a software routine that restores JTAG functionality based on an external or software trigger.

The GPIO commit control registers provide a layer of protection against accidental programming of critical hardware peripherals. Protection is currently provided for the five JTAG/SWD pins (PB7 and PC[3:0]). Writes to protected bits of the **GPIO Alternate Function Select (GPIOAFSEL)** register (see page 168) are not committed to storage unless the **GPIO Lock (GPIOLOCK)** register (see page 178) has been unlocked and the appropriate bits of the **GPIO Commit (GPIOCR)** register (see page 179) have been set to 1.

Recovering a "Locked" Device

Note: Performing the sequence below causes the nonvolatile registers discussed in “Nonvolatile Register Programming” on page 129 to be restored to their factory default values. The mass erase of the flash memory caused by the below sequence occurs prior to the nonvolatile registers being restored.

If software configures any of the JTAG/SWD pins as GPIO and loses the ability to communicate with the debugger, there is a debug sequence that can be used to recover the device. Performing a total of ten JTAG-to-SWD and SWD-to-JTAG switch sequences while holding the device in reset mass erases the flash memory. The sequence to recover the device is:

1. Assert and hold the RST signal.
2. Perform the JTAG-to-SWD switch sequence.
3. Perform the SWD-to-JTAG switch sequence.
4. Perform the JTAG-to-SWD switch sequence.
5. Perform the SWD-to-JTAG switch sequence.
6. Perform the JTAG-to-SWD switch sequence.
7. Perform the SWD-to-JTAG switch sequence.
8. Perform the JTAG-to-SWD switch sequence.
9. Perform the SWD-to-JTAG switch sequence.
10. Perform the JTAG-to-SWD switch sequence.

11. Perform the SWD-to-JTAG switch sequence.

12. Release the $\overline{\text{RST}}$ signal.

13. Wait 400 ms.

14. Power-cycle the device.

The JTAG-to-SWD and SWD-to-JTAG switch sequences are described in “ARM Serial Wire Debug (SWD)” on page 58. When performing switch sequences for the purpose of recovering the debug capabilities of the device, only steps 1 and 2 of the switch sequence in the section called “JTAG-to-SWD Switching” on page 58 must be performed.

5.2.4.2 Communication with JTAG/SWD

Because the debug clock and the system clock can be running at different frequencies, care must be taken to maintain reliable communication with the JTAG/SWD interface. In the Capture-DR state, the result of the previous transaction, if any, is returned, together with a 3-bit ACK response. Software should check the ACK response to see if the previous operation has completed before initiating a new transaction. Alternatively, if the system clock is at least 8 times faster than the debug clock (TCK or SWCLK), the previous operation has enough time to complete and the ACK bits do not have to be checked.

5.2.4.3 ARM Serial Wire Debug (SWD)

In order to seamlessly integrate the ARM Serial Wire Debug (SWD) functionality, a serial-wire debugger must be able to connect to the Cortex-M3 core without having to perform, or have any knowledge of, JTAG cycles. This is accomplished with a SWD preamble that is issued before the SWD session begins.

The switching preamble used to enable the SWD interface of the SWJ-DP module starts with the TAP controller in the Test-Logic-Reset state. From here, the preamble sequences the TAP controller through the following states: Run Test Idle, Select DR, Select IR, Test Logic Reset, Test Logic Reset, Run Test Idle, Run Test Idle, Select DR, Select IR, Test Logic Reset, Test Logic Reset, Run Test Idle, Run Test Idle, Select DR, Select IR, and Test Logic Reset states.

Stepping through this sequences of the TAP state machine enables the SWD interface and disables the JTAG interface. For more information on this operation and the SWD interface, see the *ARM® Cortex™-M3 Technical Reference Manual* and the *ARM® CoreSight Technical Reference Manual*.

Because this sequence is a valid series of JTAG operations that could be issued, the ARM JTAG TAP controller is not fully compliant to the *IEEE Standard 1149.1*. This is the only instance where the ARM JTAG TAP controller does not meet full compliance with the specification. Due to the low probability of this sequence occurring during normal operation of the TAP controller, it should not affect normal performance of the JTAG interface.

JTAG-to-SWD Switching

To switch the operating mode of the Debug Access Port (DAP) from JTAG to SWD mode, the external debug hardware must send the switching preamble to the device. The 16-bit switch sequence for switching to SWD mode is defined as b1110011110011110, transmitted LSB first. This can also be represented as 16'hE79E when transmitted LSB first. The complete switch sequence should consist of the following transactions on the TCK/SWCLK and TMS/SWDIO signals:

1. Send at least 50 TCK/SWCLK cycles with TMS/SWDIO set to 1. This ensures that both JTAG and SWD are in their reset/idle states.

2. Send the 16-bit JTAG-to-SWD switch sequence, 16'hE79E.

3. Send at least 50 TCK/SWCLK cycles with TMS/SWDIO set to 1. This ensures that if SWJ-DP was already in SWD mode, before sending the switch sequence, the SWD goes into the line reset state.

SWD-to-JTAG Switching

To switch the operating mode of the Debug Access Port (DAP) from SWD to JTAG mode, the external debug hardware must send a switch sequence to the device. The 16-bit switch sequence for switching to JTAG mode is defined as b1110011100111100, transmitted LSB first. This can also be represented as 16'hE73C when transmitted LSB first. The complete switch sequence should consist of the following transactions on the TCK/SWCLK and TMS/SWDIO signals:

1. Send at least 50 TCK/SWCLK cycles with TMS/SWDIO set to 1. This ensures that both JTAG and SWD are in their reset/idle states.

2. Send the 16-bit SWD-to-JTAG switch sequence, 16'hE73C.

3. Send at least 5 TCK/SWCLK cycles with TMS/SWDIO set to 1. This ensures that if SWJ-DP was already in JTAG mode, before sending the switch sequence, the JTAG goes into the Test Logic Reset state.

5.3 Initialization and Configuration

After a Power-On-Reset or an external reset ($\overline{\text{RST}}$), the JTAG pins are automatically configured for JTAG communication. No user-defined initialization or configuration is needed. However, if the user application changes these pins to their GPIO function, they must be configured back to their JTAG functionality before JTAG communication can be restored. This is done by enabling the five JTAG pins (PB7 and PC[3:0]) for their alternate function using the **GPIOAFSEL** register. In addition to enabling the alternate functions, any other changes to the GPIO pad configurations on the five JTAG pins (PB7 andPC[3:0]) should be reverted to their default settings.

5.4 Register Descriptions

There are no APB-accessible registers in the JTAG TAP Controller or Shift Register chains. The registers within the JTAG controller are all accessed serially through the TAP Controller. The registers can be broken down into two main categories: Instruction Registers and Data Registers.

5.4.1 Instruction Register (IR)

The JTAG TAP Instruction Register (IR) is a four-bit serial scan chain connected between the JTAG TDI and TDO pins with a parallel load register. When the TAP Controller is placed in the correct states, bits can be shifted into the Instruction Register. Once these bits have been shifted into the chain and updated, they are interpreted as the current instruction. The decode of the Instruction Register bits is shown in Table 5-2 on page 59. A detailed explanation of each instruction, along with its associated Data Register, follows.

Table 5-2. JTAG Instruction Register Commands

IR[3:0]	Instruction	Description
0000	EXTEST	Drives the values preloaded into the Boundary Scan Chain by the SAMPLE/PRELOAD instruction onto the pads.
0001	INTEST	Drives the values preloaded into the Boundary Scan Chain by the SAMPLE/PRELOAD instruction into the controller.

Table 5-2. JTAG Instruction Register Commands *(continued)*

IR[3:0]	Instruction	Description
0010	SAMPLE / PRELOAD	Captures the current I/O values and shifts the sampled values out of the Boundary Scan Chain while new preload data is shifted in.
1000	ABORT	Shifts data into the ARM Debug Port Abort Register.
1010	DPACC	Shifts data into and out of the ARM DP Access Register.
1011	APACC	Shifts data into and out of the ARM AC Access Register.
1110	IDCODE	Loads manufacturing information defined by the *IEEE Standard 1149.1* into the IDCODE chain and shifts it out.
1111	BYPASS	Connects TDI to TDO through a single Shift Register chain.
All Others	Reserved	Defaults to the BYPASS instruction to ensure that TDI is always connected to TDO.

5.4.1.1 EXTEST Instruction

The EXTEST instruction is not associated with its own Data Register chain. The EXTEST instruction uses the data that has been preloaded into the Boundary Scan Data Register using the SAMPLE/PRELOAD instruction. When the EXTEST instruction is present in the Instruction Register, the preloaded data in the Boundary Scan Data Register associated with the outputs and output enables are used to drive the GPIO pads rather than the signals coming from the core. This allows tests to be developed that drive known values out of the controller, which can be used to verify connectivity. While the EXTEST instruction is present in the Instruction Register, the Boundary Scan Data Register can be accessed to sample and shift out the current data and load new data into the Boundary Scan Data Register.

5.4.1.2 INTEST Instruction

The INTEST instruction is not associated with its own Data Register chain. The INTEST instruction uses the data that has been preloaded into the Boundary Scan Data Register using the SAMPLE/PRELOAD instruction. When the INTEST instruction is present in the Instruction Register, the preloaded data in the Boundary Scan Data Register associated with the inputs are used to drive the signals going into the core rather than the signals coming from the GPIO pads. This allows tests to be developed that drive known values into the controller, which can be used for testing. It is important to note that although the $\overline{\text{RST}}$ input pin is on the Boundary Scan Data Register chain, it is only observable. While the INTEXT instruction is present in the Instruction Register, the Boundary Scan Data Register can be accessed to sample and shift out the current data and load new data into the Boundary Scan Data Register.

5.4.1.3 SAMPLE/PRELOAD Instruction

The SAMPLE/PRELOAD instruction connects the Boundary Scan Data Register chain between TDI and TDO. This instruction samples the current state of the pad pins for observation and preloads new test data. Each GPIO pad has an associated input, output, and output enable signal. When the TAP controller enters the Capture DR state during this instruction, the input, output, and output-enable signals to each of the GPIO pads are captured. These samples are serially shifted out of TDO while the TAP controller is in the Shift DR state and can be used for observation or comparison in various tests.

While these samples of the inputs, outputs, and output enables are being shifted out of the Boundary Scan Data Register, new data is being shifted into the Boundary Scan Data Register from TDI. Once the new data has been shifted into the Boundary Scan Data Register, the data is saved in the parallel load registers when the TAP controller enters the Update DR state. This update of the parallel load register preloads data into the Boundary Scan Data Register that is associated with

each input, output, and output enable. This preloaded data can be used with the EXTEST and INTEST instructions to drive data into or out of the controller. Please see “Boundary Scan Data Register” on page 62 for more information.

5.4.1.4 ABORT Instruction

The ABORT instruction connects the associated ABORT Data Register chain between TDI and TDO. This instruction provides read and write access to the ABORT Register of the ARM Debug Access Port (DAP). Shifting the proper data into this Data Register clears various error bits or initiates a DAP abort of a previous request. Please see the “ABORT Data Register” on page 63 for more information.

5.4.1.5 DPACC Instruction

The DPACC instruction connects the associated DPACC Data Register chain between TDI and TDO. This instruction provides read and write access to the DPACC Register of the ARM Debug Access Port (DAP). Shifting the proper data into this register and reading the data output from this register allows read and write access to the ARM debug and status registers. Please see “DPACC Data Register” on page 63 for more information.

5.4.1.6 APACC Instruction

The APACC instruction connects the associated APACC Data Register chain between TDI and TDO. This instruction provides read and write access to the APACC Register of the ARM Debug Access Port (DAP). Shifting the proper data into this register and reading the data output from this register allows read and write access to internal components and buses through the Debug Port. Please see “APACC Data Register” on page 63 for more information.

5.4.1.7 IDCODE Instruction

The IDCODE instruction connects the associated IDCODE Data Register chain between TDI and TDO. This instruction provides information on the manufacturer, part number, and version of the ARM core. This information can be used by testing equipment and debuggers to automatically configure their input and output data streams. IDCODE is the default instruction that is loaded into the JTAG Instruction Register when a Power-On-Reset (POR) is asserted, $\overline{\text{TRST}}$ is asserted, or the Test-Logic-Reset state is entered. Please see “IDCODE Data Register” on page 62 for more information.

5.4.1.8 BYPASS Instruction

The BYPASS instruction connects the associated BYPASS Data Register chain between TDI and TDO. This instruction is used to create a minimum length serial path between the TDI and TDO ports. The BYPASS Data Register is a single-bit shift register. This instruction improves test efficiency by allowing components that are not needed for a specific test to be bypassed in the JTAG scan chain by loading them with the BYPASS instruction. Please see “BYPASS Data Register” on page 62 for more information.

5.4.2 Data Registers

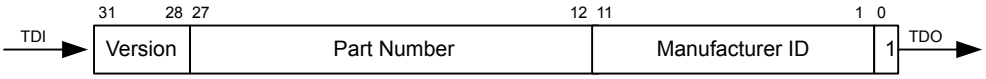
The JTAG module contains six Data Registers. These include: IDCODE, BYPASS, Boundary Scan, APACC, DPACC, and ABORT serial Data Register chains. Each of these Data Registers is discussed in the following sections.

5.4.2.1 IDCODE Data Register

The format for the 32-bit IDCODE Data Register defined by the *IEEE Standard 1149.1* is shown in Figure 5-3 on page 62. The standard requires that every JTAG-compliant device implement either the IDCODE instruction or the BYPASS instruction as the default instruction. The LSB of the IDCODE Data Register is defined to be a 1 to distinguish it from the BYPASS instruction, which has an LSB of 0. This allows auto configuration test tools to determine which instruction is the default instruction.

The major uses of the JTAG port are for manufacturer testing of component assembly, and program development and debug. To facilitate the use of auto-configuration debug tools, the IDCODE instruction outputs a value of 0x3BA0.0477. This allows the debuggers to automatically configure themselves to work correctly with the Cortex-M3 during debug.

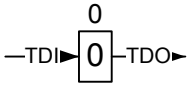
Figure 5-3. IDCODE Register Format



5.4.2.2 BYPASS Data Register

The format for the 1-bit BYPASS Data Register defined by the *IEEE Standard 1149.1* is shown in Figure 5-4 on page 62. The standard requires that every JTAG-compliant device implement either the BYPASS instruction or the IDCODE instruction as the default instruction. The LSB of the BYPASS Data Register is defined to be a 0 to distinguish it from the IDCODE instruction, which has an LSB of 1. This allows auto configuration test tools to determine which instruction is the default instruction.

Figure 5-4. BYPASS Register Format

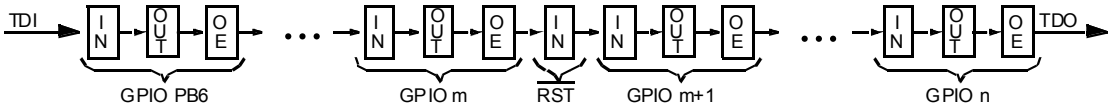


5.4.2.3 Boundary Scan Data Register

The format of the Boundary Scan Data Register is shown in Figure 5-5 on page 63. Each GPIO pin, starting with a GPIO pin next to the JTAG port pins, is included in the Boundary Scan Data Register. Each GPIO pin has three associated digital signals that are included in the chain. These signals are input, output, and output enable, and are arranged in that order as can be seen in the figure.

When the Boundary Scan Data Register is accessed with the SAMPLE/PRELOAD instruction, the input, output, and output enable from each digital pad are sampled and then shifted out of the chain to be verified. The sampling of these values occurs on the rising edge of TCK in the Capture DR state of the TAP controller. While the sampled data is being shifted out of the Boundary Scan chain in the Shift DR state of the TAP controller, new data can be preloaded into the chain for use with the EXTEST and INTEST instructions. These instructions either force data out of the controller, with the EXTEST instruction, or into the controller, with the INTEST instruction.

Figure 5-5. Boundary Scan Register Format



5.4.2.4 APACC Data Register

The format for the 35-bit APACC Data Register defined by ARM is described in the *ARM® Cortex™-M3 Technical Reference Manual*.

5.4.2.5 DPACC Data Register

The format for the 35-bit DPACC Data Register defined by ARM is described in the *ARM® Cortex™-M3 Technical Reference Manual*.

5.4.2.6 ABORT Data Register

The format for the 35-bit ABORT Data Register defined by ARM is described in the *ARM® Cortex™-M3 Technical Reference Manual*.

6 System Control

System control determines the overall operation of the device. It provides information about the device, controls the clocking to the core and individual peripherals, and handles reset detection and reporting.

6.1 Functional Description

The System Control module provides the following capabilities:

- Device identification, see “Device Identification” on page 64
- Local control, such as reset (see “Reset Control” on page 64), power (see “Power Control” on page 67) and clock control (see “Clock Control” on page 69)
- System control (Run, Sleep, and Deep-Sleep modes), see “System Control” on page 72

6.1.1 Device Identification

Several read-only registers provide software with information on the microcontroller, such as version, part number, SRAM size, flash size, and other features. See the **DID0**, **DID1**, and **DC0**-**DC4** registers.

6.1.2 Reset Control

This section discusses aspects of hardware functions during reset as well as system software requirements following the reset sequence.

6.1.2.1 CMOD0 and CMOD1 Test-Mode Control Pins

Two pins, CMOD0 and CMOD1, are defined for internal use for testing the microcontroller during manufacture. They have no end-user function and should not be used. The CMOD pins should be connected to ground.

6.1.2.2 Reset Sources

The controller has five sources of reset:

1. External reset input pin ($\overline{\text{RST}}$) assertion, see “External $\overline{\text{RST}}$ Pin” on page 65.
2. Power-on reset (POR), see “Power-On Reset (POR)” on page 64.
3. Internal brown-out (BOR) detector, see “Brown-Out Reset (BOR)” on page 66.
4. Software-initiated reset (with the software reset registers), see “Software Reset” on page 67.
5. A watchdog timer reset condition violation, see “Watchdog Timer Reset” on page 67.

After a reset, the **Reset Cause (RESC)** register is set with the reset cause. The bits in this register are sticky and maintain their state across multiple reset sequences, except when an internal POR is the cause, and then all the other bits in the **RESC** register are cleared except for the POR indicator.

6.1.2.3 Power-On Reset (POR)

Note: The power-on reset also resets the JTAG controller. An external reset does not.

The internal Power-On Reset (POR) circuit monitors the power supply voltage (V_{DD}) and generates a reset signal to all of the internal logic including JTAG when the power supply ramp reaches a threshold value (V_{TH}). The microcontroller must be operating within the specified operating parameters when the on-chip power-on reset pulse is complete. The 3.3-V power supply to the microcontroller must reach 3.0 V within 10 msec of V_{DD} crossing 2.0 V to guarantee proper operation. For applications that require the use of an external reset signal to hold the microcontroller in reset longer than the internal POR, the $\overline{RST}$ input may be used as discussed in "External $\overline{RST}$ Pin" on page 65.

The Power-On Reset sequence is as follows:

1. The microcontroller waits for internal POR to go inactive.
2. The internal reset is released and the core loads from memory the initial stack pointer, the initial program counter, and the first instruction designated by the program counter, and then begins execution.

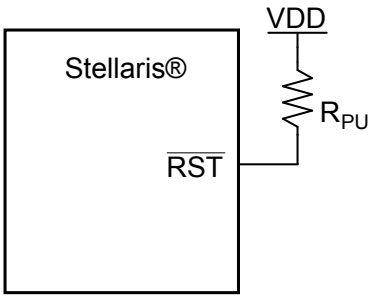
The internal POR is only active on the initial power-up of the microcontroller. The Power-On Reset timing is shown in Figure 21-6 on page 534.

6.1.2.4 External $\overline{RST}$ Pin

Note: It is recommended that the trace for the $\overline{RST}$ signal must be kept as short as possible. Be sure to place any components connected to the $\overline{RST}$ signal as close to the microcontroller as possible.

If the application only uses the internal POR circuit, the $\overline{RST}$ input must be connected to the power supply (V_{DD}) through an optional pull-up resistor (0 to 100K Ω) as shown in Figure 6-1 on page 65.

Figure 6-1. Basic $\overline{RST}$ Configuration



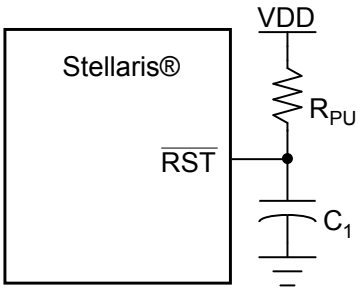
R_{PU} = 0 to 100 kΩ

The external reset pin ($\overline{RST}$) resets the microcontroller including the core and all the on-chip peripherals except the JTAG TAP controller (see "JTAG Interface" on page 52). The external reset sequence is as follows:

1. The external reset pin ($\overline{RST}$) is asserted for the duration specified by T_{MIN} and then de-asserted (see "Reset" on page 533).
2. The internal reset is released and the core loads from memory the initial stack pointer, the initial program counter, and the first instruction designated by the program counter, and then begins execution.

To improve noise immunity and/or to delay reset at power up, the $\overline{RST}$ input may be connected to an RC network as shown in Figure 6-2 on page 66.

Figure 6-2. External Circuitry to Extend Power-On Reset

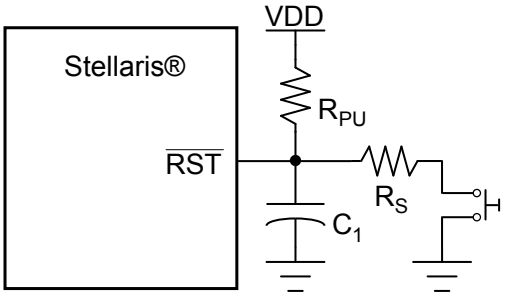


R_{PU} = 1 kΩ to 100 kΩ

C_1 = 1 nF to 10 µF

If the application requires the use of an external reset switch, Figure 6-3 on page 66 shows the proper circuitry to use.

Figure 6-3. Reset Circuit Controlled by Switch



Typical R_{PU} = 10 kΩ

Typical R_S = 470 Ω

C_1 = 10 nF

The R_{PU} and C_1 components define the power-on delay.

The external reset timing is shown in Figure 21-5 on page 534.

6.1.2.5 Brown-Out Reset (BOR)

A drop in the input voltage resulting in the assertion of the internal brown-out detector can be used to reset the controller. This is initially disabled and may be enabled by software.

The system provides a brown-out detection circuit that triggers if the power supply (V_{DD}) drops below a brown-out threshold voltage (V_{BTH}). If a brown-out condition is detected, the system may generate a controller interrupt or a system reset.

Brown-out resets are controlled with the **Power-On and Brown-Out Reset Control (PBORCTL)** register. The `BORIOR` bit in the **PBORCTL** register must be set for a brown-out condition to trigger a reset.

The brown-out reset is equivalent to an assertion of the external $\overline{RST}$ input and the reset is held active until the proper V_{DD} level is restored. The **RESC** register can be examined in the reset interrupt

handler to determine if a Brown-Out condition was the cause of the reset, thus allowing software to determine what actions are required to recover.

The internal Brown-Out Reset timing is shown in Figure 21-7 on page 534.

6.1.2.6 Software Reset

Software can reset a specific peripheral or generate a reset to the entire system .

Peripherals can be individually reset by software via three registers that control reset signals to each peripheral (see the **SRCRn** registers). If the bit position corresponding to a peripheral is set and subsequently cleared, the peripheral is reset. The encoding of the reset registers is consistent with the encoding of the clock gating control for peripherals and on-chip functions (see “System Control” on page 72). Note that all reset signals for all clocks of the specified unit are asserted as a result of a software-initiated reset.

The entire system can be reset by software by setting the SYSRESETREQ bit in the Cortex-M3 Application Interrupt and Reset Control register resets the entire system including the core. The software-initiated system reset sequence is as follows:

1. A software system reset is initiated by writing the SYSRESETREQ bit in the ARM Cortex-M3 Application Interrupt and Reset Control register.
2. An internal reset is asserted.
3. The internal reset is deasserted and the controller loads from memory the initial stack pointer, the initial program counter, and the first instruction designated by the program counter, and then begins execution.

The software-initiated system reset timing is shown in Figure 21-8 on page 534.

6.1.2.7 Watchdog Timer Reset

The watchdog timer module's function is to prevent system hangs. The watchdog timer can be configured to generate an interrupt to the controller on its first time-out, and to generate a reset signal on its second time-out.

After the first time-out event, the 32-bit counter is reloaded with the value of the **Watchdog Timer Load (WDTLOAD)** register, and the timer resumes counting down from that value. If the timer counts down to its zero state again before the first time-out interrupt is cleared, and the reset signal has been enabled, the watchdog timer asserts its reset signal to the system. The watchdog timer reset sequence is as follows:

1. The watchdog timer times out for the second time without being serviced.
2. An internal reset is asserted.
3. The internal reset is released and the controller loads from memory the initial stack pointer, the initial program counter, the first instruction designated by the program counter, and begins execution.

The watchdog reset timing is shown in Figure 21-9 on page 535.

6.1.3 Power Control

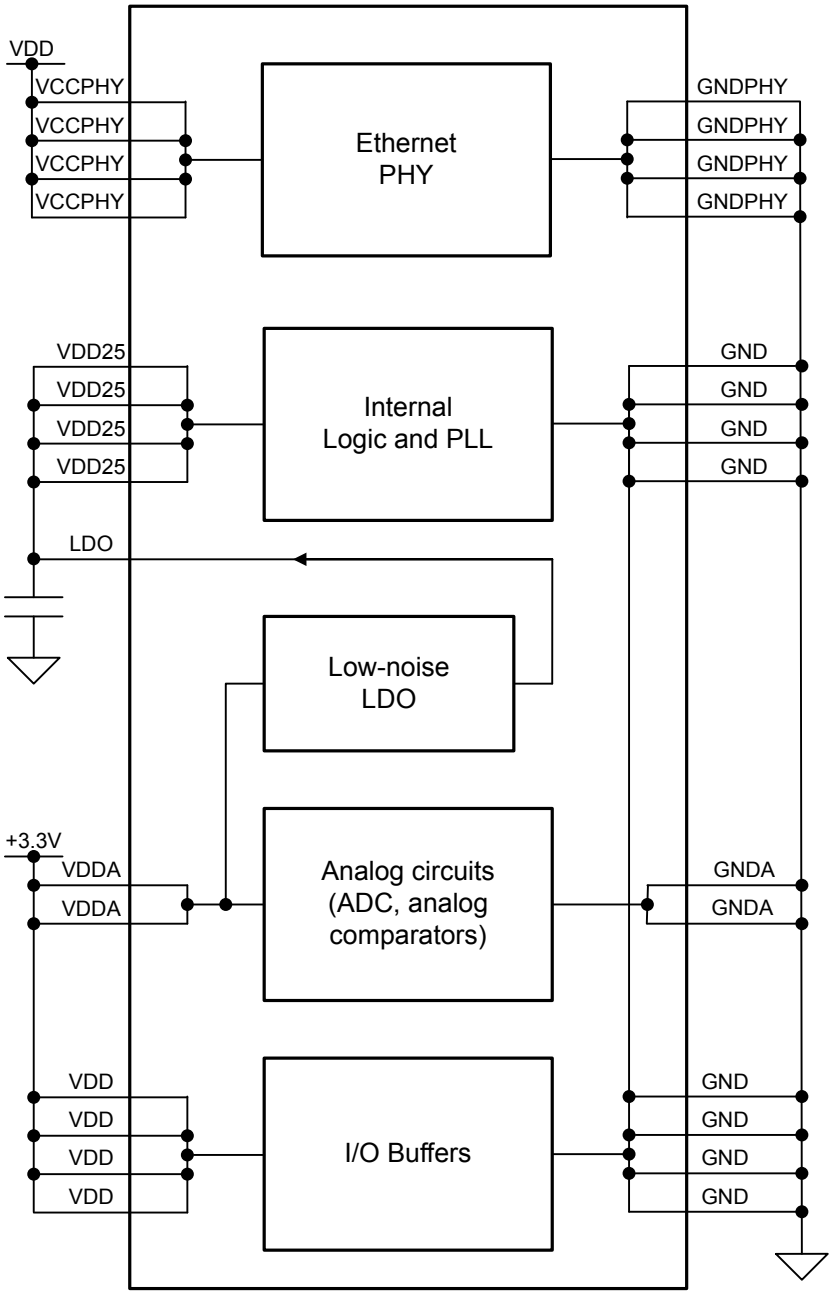
The Stellaris® microcontroller provides an integrated LDO regulator that may be used to provide power to the majority of the controller's internal logic. For power reduction, the LDO regulator provides

software a mechanism to adjust the regulated value, in small increments (VSTEP), over the range of 2.25 V to 2.75 V (inclusive)—or 2.5 V ± 10%. The adjustment is made by changing the value of the VADJ field in the **LDO Power Control (LDOPCTL)** register.

Figure 6-4 on page 68 shows the power architecture.

Note: On the printed circuit board, use the LDO output as the source of VDD25 input. In addition, the LDO requires decoupling capacitors. See “On-Chip Low Drop-Out (LDO) Regulator Characteristics” on page 528.

Figure 6-4. Power Architecture



6.1.4 Clock Control

System control determines the control of clocks in this part.

6.1.4.1 Fundamental Clock Sources

There are multiple clock sources for use in the device:

- **Internal Oscillator (IOSC).** The internal oscillator is an on-chip clock source. It does not require the use of any external components. The frequency of the internal oscillator is 12 MHz ± 30%. Applications that do not depend on accurate clock sources may use this clock source to reduce system cost. The internal oscillator is the clock source the device uses during and following POR. If the main oscillator is required, software must enable the main oscillator following reset and allow the main oscillator to stabilize before changing the clock reference.

- **Main Oscillator (MOSC).** The main oscillator provides a frequency-accurate clock source by one of two means: an external single-ended clock source is connected to the OSC0 input pin, or an external crystal is connected across the OSC0 input and OSC1 output pins. If the PLL is being used, the crystal value must be one of the supported frequencies between 3.579545 MHz through 8.192 MHz (inclusive). If the PLL is not being used, the crystal may be any one of the supported frequencies between 1 MHz and 8.192 MHz. The single-ended clock source range is from DC through the specified speed of the device. The supported crystals are listed in the XTAL bit field in the **RCC** register (see page 83).

- **Internal 30-kHz Oscillator.** The internal 30-kHz oscillator is similar to the internal oscillator, except that it provides an operational frequency of 30 kHz ± 50%. It is intended for use during Deep-Sleep power-saving modes. This power-savings mode benefits from reduced internal switching and also allows the main oscillator to be powered down.

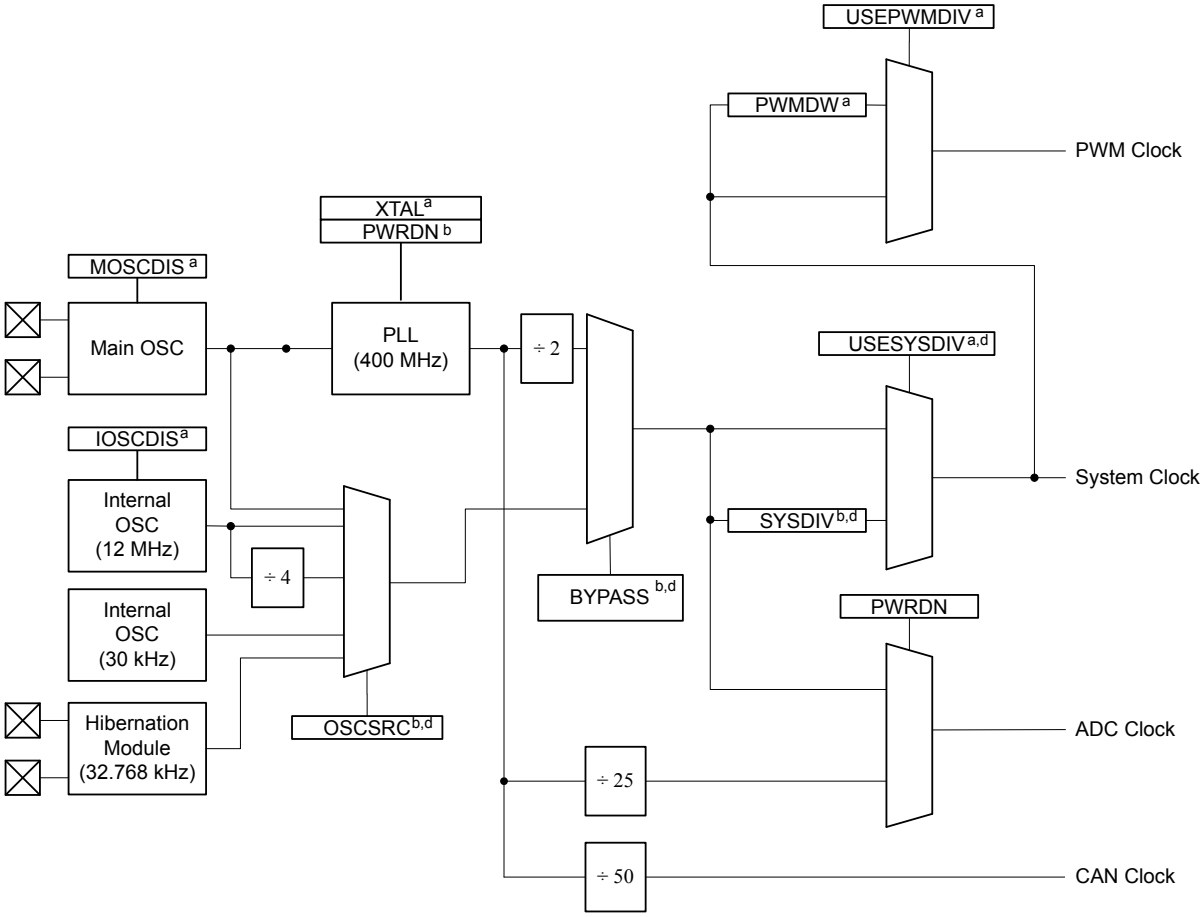
The internal system clock (SysClk), is derived from any of the above sources plus two others: the output of the main internal PLL, and the internal oscillator divided by four (3 MHz ± 30%). The frequency of the PLL clock reference must be in the range of 3.579545 MHz to 8.192 MHz (inclusive).

The **Run-Mode Clock Configuration (RCC)** and **Run-Mode Clock Configuration 2 (RCC2)** registers provide control for the system clock. The **RCC2** register is provided to extend fields that offer additional encodings over the **RCC** register. When used, the **RCC2** register field values are used by the logic over the corresponding field in the **RCC** register. In particular, **RCC2** provides for a larger assortment of clock configuration options.

Figure 6-5 on page 70 shows the logic for the main clock tree. The peripheral blocks are driven by the system clock signal and can be individually enabled/disabled. The ADC clock signal is automatically divided down to 16 MHz for proper ADC operation. The PWM clock signal is a synchronous divide of the system clock to provide the PWM circuit with more range (set with PWMDIV in **RCC**).

Note: When the ADC module is in operation, the system clock must be at least 16 MHz.

Figure 6-5. Main Clock Tree



a. Control provided by RCC register bit/field.
b. Control provided by RCC register bit/field or RCC2 register bit/field, if overridden with RCC2 register bit USERCC2.
c. Control provided by RCC2 register bit/field.
d. Also may be controlled by DSLPCLKCFG when in deep sleep mode.

Note: The figure above shows all features available on all Stellaris® Fury-class devices.

6.1.4.2 Crystal Configuration for the Main Oscillator (MOSC)

The main oscillator supports the use of a select number of crystals. If the main oscillator is used by the PLL as a reference clock, the supported range of crystals is 3.579545 to 8.192 MHz, otherwise, the range of supported crystals is 1 to 8.192 MHz.

The XTAL bit in the **RCC** register (see page 83) describes the available crystal choices and default programming values.

Software configures the **RCC** register XTAL field with the crystal number. If the PLL is used in the design, the XTAL field value is internally translated to the PLL settings.

6.1.4.3 Main PLL Frequency Configuration

The main PLL is disabled by default during power-on reset and is enabled later by software if required. Software specifies the output divisor to set the system clock frequency, and enables the main PLL to drive the output. The PLL operates at 400 MHz, but is divided by two prior to the application of the output divisor.

If the main oscillator provides the clock reference to the main PLL, the translation provided by hardware and used to program the PLL is available for software in the **XTAL to PLL Translation (PLLCFG)** register (see page 88). The internal translation provides a translation within ± 1% of the targeted PLL VCO frequency. Table 21-8 on page 530 shows the actual PLL frequency and error for a given crystal choice.

The Crystal Value field (XTAL) on page 83 describes the available crystal choices and default programming of the **PLLCFG** register. The crystal number is written into the XTAL field of the **Run-Mode Clock Configuration (RCC)** register. Any time the XTAL field changes, the new settings are translated and the internal PLL settings are updated.

6.1.4.4 PLL Modes

The PLL has two modes of operation: Normal and Power-Down

- Normal: The PLL multiplies the input clock reference and drives the output.
- Power-Down: Most of the PLL internal circuitry is disabled and the PLL does not drive the output.

The modes are programmed using the **RCC**/**RCC2** register fields (see page 83 and page 89).

6.1.4.5 PLL Operation

If a PLL configuration is changed, the PLL output frequency is unstable until it reconverges (relocks) to the new setting. The time between the configuration change and relock is T_{READY} (see Table 21-7 on page 530). During the relock time, the affected PLL is not usable as a clock reference.

The PLL is changed by one of the following:

- Change to the XTAL value in the **RCC** register—writes of the same value do not cause a relock.
- Change in the PLL from Power-Down to Normal mode.

A counter is defined to measure the T_{READY} requirement. The counter is clocked by the main oscillator. The range of the main oscillator has been taken into account and the down counter is set to 0x1200 (that is, ~600 µs at an 8.192 MHz external oscillator clock). Hardware is provided to keep the PLL from being used as a system clock until the T_{READY} condition is met after one of the two changes above. It is the user's responsibility to have a stable clock source (like the main oscillator) before the **RCC**/**RCC2** register is switched to use the PLL.

If the main PLL is enabled and the system clock is switched to use the PLL in one step, the system control hardware continues to clock the controller from the oscillator selected by the **RCC**/**RCC2** register until the main PLL is stable (T_{READY} time met), after which it changes to the PLL. Software can use many methods to ensure that the system is clocked from the main PLL, including periodically polling the PLLLRIS bit in the **Raw Interrupt Status (RIS)** register, and enabling the PLL Lock interrupt.

6.1.5 System Control

For power-savings purposes, the **RCGCn** , **SCGCn** , and **DCGCn** registers control the clock gating logic for each peripheral or block in the system while the controller is in Run, Sleep, and Deep-Sleep mode, respectively.

There are four levels of operation for the device defined as:

- **Run Mode.** In Run mode, the controller actively executes code. Run mode provides normal operation of the processor and all of the peripherals that are currently enabled by the **RCGCn** registers. The system clock can be any of the available clock sources including the PLL.

- **Sleep Mode.** In Sleep mode, the clock frequency of the active peripherals is unchanged, but the processor and the memory subsystem are not clocked and therefore no longer execute code. Sleep mode is entered by the Cortex-M3 core executing a `WFI(Wait for Interrupt)` instruction. Any properly configured interrupt event in the system will bring the processor back into Run mode. See the system control NVIC section of the *ARM® Cortex™-M3 Technical Reference Manual* for more details.

  Peripherals are clocked that are enabled in the **SCGCn** register when auto-clock gating is enabled (see the **RCC** register) or the **RCGCn** register when the auto-clock gating is disabled. The system clock has the same source and frequency as that during Run mode.

- **Deep-Sleep Mode.** In Deep-Sleep mode, the clock frequency of the active peripherals may change (depending on the Run mode clock configuration) in addition to the processor clock being stopped. An interrupt returns the device to Run mode from one of the sleep modes; the sleep modes are entered on request from the code. Deep-Sleep mode is entered by first writing the Deep Sleep Enable bit in the ARM Cortex-M3 NVIC system control register and then executing a `WFI` instruction. Any properly configured interrupt event in the system will bring the processor back into Run mode. See the system control NVIC section of the *ARM® Cortex™-M3 Technical Reference Manual* for more details.

  The Cortex-M3 processor core and the memory subsystem are not clocked. Peripherals are clocked that are enabled in the **DCGCn** register when auto-clock gating is enabled (see the **RCC** register) or the **RCGCn** register when auto-clock gating is disabled. The system clock source is the main oscillator by default or the internal oscillator specified in the **DSLPCLKCFG** register if one is enabled. When the **DSLPCLKCFG** register is used, the internal oscillator is powered up, if necessary, and the main oscillator is powered down. If the PLL is running at the time of the WFI instruction, hardware will power the PLL down and override the `SYSDIV` field of the active **RCC**/**RCC2** register, to be determined by the `DSDIVORIDE` setting in the **DSLPCLKCFG** register, up to /16 or /64 respectively. When the Deep-Sleep exit event occurs, hardware brings the system clock back to the source and frequency it had at the onset of Deep-Sleep mode before enabling the clocks that had been stopped during the Deep-Sleep duration.

Caution – If the Cortex-M3 Debug Access Port (DAP) has been enabled, and the device wakes from a low power sleep or deep-sleep mode, the core may start executing code before all clocks to peripherals have been restored to their run mode configuration. The DAP is usually enabled by software tools accessing the JTAG or SWD interface when debugging or flash programming. If this condition occurs, a Hard Fault is triggered when software accesses a peripheral with an invalid clock.

A software delay loop can be used at the beginning of the interrupt routine that is used to wake up a system from a WFI (Wait For Interrupt) instruction. This stalls the execution of any code that accesses a peripheral register that might cause a fault. This loop can be removed for production software as the DAP is most likely not enabled during normal execution.

Because the DAP is disabled by default (power on reset), the user can also power-cycle the device. The DAP is not enabled unless it is enabled through the JTAG or SWD interface.

6.2 Initialization and Configuration

The PLL is configured using direct register writes to the **RCC**/**RCC2** register. If the **RCC2** register is being used, the USERCC2 bit must be set and the appropriate **RCC2** bit/field is used. The steps required to successfully change the PLL-based system clock are:

1. Bypass the PLL and system clock divider by setting the BYPASS bit and clearing the USESYS bit in the **RCC** register. This configures the system to run off a “raw” clock source and allows for the new PLL configuration to be validated before switching the system clock to the PLL.
2. Select the crystal value (XTAL) and oscillator source (OSCSRC), and clear the PWRDN bit in **RCC**/**RCC2**. Setting the XTAL field automatically pulls valid PLL configuration data for the appropriate crystal, and clearing the PWRDN bit powers and enables the PLL and its output.
3. Select the desired system divider (SYSDIV) in **RCC**/**RCC2** and set the USESYS bit in **RCC**. The SYSDIV field determines the system frequency for the microcontroller.
4. Wait for the PLL to lock by polling the PLLLRIS bit in the **Raw Interrupt Status (RIS**) register.
5. Enable use of the PLL by clearing the BYPASS bit in **RCC**/**RCC2**.

6.3 Register Map

Table 6-1 on page 73 lists the System Control registers, grouped by function. The offset listed is a hexadecimal increment to the register's address, relative to the System Control base address of 0x400F.E000.

Note: Spaces in the System Control register space that are not used are reserved for future or internal use. Software should not modify any reserved memory address.

Table 6-1. System Control Register Map

Offset	Name	Type	Reset	Description	See page
0x000	DID0	RO	-	Device Identification 0	75
0x004	DID1	RO	-	Device Identification 1	92
0x008	DC0	RO	0x007F.002F	Device Capabilities 0	94
0x010	DC1	RO	0x0011.31BF	Device Capabilities 1	95

Table 6-1. System Control Register Map *(continued)*

Offset	Name	Type	Reset	Description	See page
0x014	DC2	RO	0x0307.1013	Device Capabilities 2	97
0x018	DC3	RO	0x8F07.87C3	Device Capabilities 3	99
0x01C	DC4	RO	0x5000.007F	Device Capabilities 4	101
0x030	PBORCTL	R/W	0x0000.7FFD	Brown-Out Reset Control	77
0x034	LDOPCTL	R/W	0x0000.0000	LDO Power Control	78
0x040	SRCR0	R/W	0x00000000	Software Reset Control 0	121
0x044	SRCR1	R/W	0x00000000	Software Reset Control 1	122
0x048	SRCR2	R/W	0x00000000	Software Reset Control 2	124
0x050	RIS	RO	0x0000.0000	Raw Interrupt Status	79
0x054	IMC	R/W	0x0000.0000	Interrupt Mask Control	80
0x058	MISC	R/W1C	0x0000.0000	Masked Interrupt Status and Clear	81
0x05C	RESC	R/W	-	Reset Cause	82
0x060	RCC	R/W	0x078E.3AD1	Run-Mode Clock Configuration	83
0x064	PLLCFG	RO	-	XTAL to PLL Translation	88
0x070	RCC2	R/W	0x0780.2810	Run-Mode Clock Configuration 2	89
0x100	RCGC0	R/W	0x00000040	Run Mode Clock Gating Control Register 0	103
0x104	RCGC1	R/W	0x00000000	Run Mode Clock Gating Control Register 1	109
0x108	RCGC2	R/W	0x00000000	Run Mode Clock Gating Control Register 2	115
0x110	SCGC0	R/W	0x00000040	Sleep Mode Clock Gating Control Register 0	105
0x114	SCGC1	R/W	0x00000000	Sleep Mode Clock Gating Control Register 1	111
0x118	SCGC2	R/W	0x00000000	Sleep Mode Clock Gating Control Register 2	117
0x120	DCGC0	R/W	0x00000040	Deep Sleep Mode Clock Gating Control Register 0	107
0x124	DCGC1	R/W	0x00000000	Deep Sleep Mode Clock Gating Control Register 1	113
0x128	DCGC2	R/W	0x00000000	Deep Sleep Mode Clock Gating Control Register 2	119
0x144	DSLPCLKCFG	R/W	0x0780.0000	Deep Sleep Clock Configuration	91

6.4 Register Descriptions

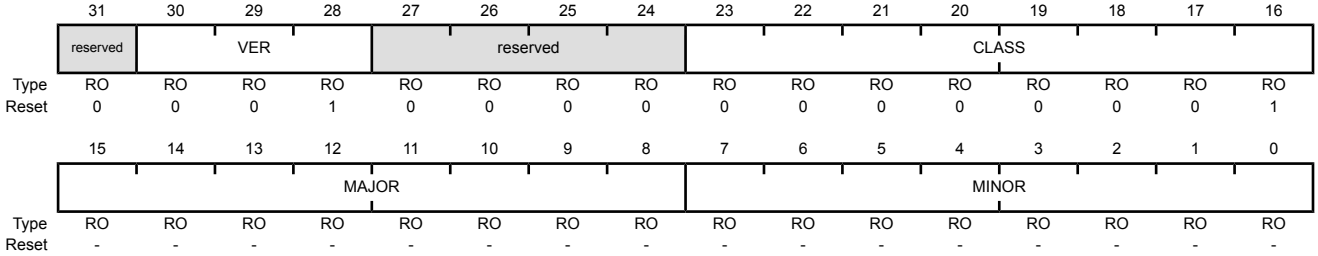
All addresses given are relative to the System Control base address of 0x400F.E000.

Register 1: Device Identification 0 (DID0), offset 0x000

This register identifies the version of the device.

Device Identification 0 (DID0)

Base 0x400F.E000
Offset 0x000
Type RO, reset -



Bit/Field	Name	Type	Reset	Description
31	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
30:28	VER	RO	0x1	DID0 Version This field defines the **DID0** register format version. The version number is numeric. The value of the VER field is encoded as follows: Value Description 0x1 Second version of the **DID0** register format.
27:24	reserved	RO	0x0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
23:16	CLASS	RO	0x1	Device Class The CLASS field value identifies the internal design from which all mask sets are generated for all devices in a particular product line. The CLASS field value is changed for new product lines, for changes in fab process (for example, a remap or shrink), or any case where the MAJOR or MINOR fields require differentiation from prior devices. The value of the CLASS field is encoded as follows (all other encodings are reserved): Value Description 0x1 Stellaris® Fury-class devices.

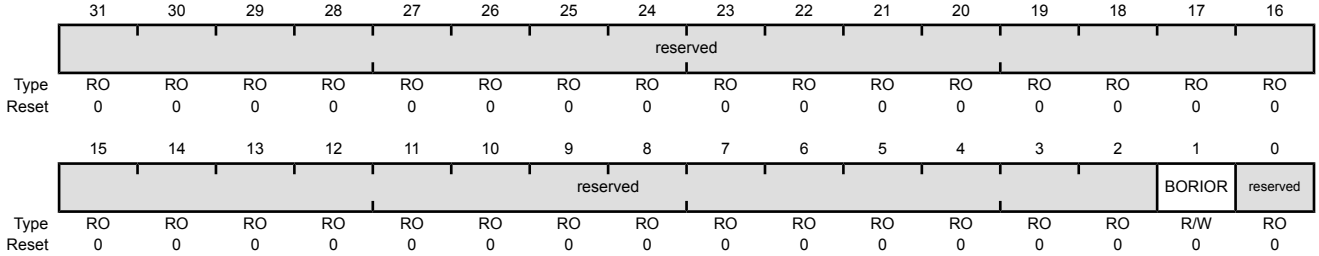
Bit/Field	Name	Type	Reset	Description
15:8	MAJOR	RO	-	Major Revision This field specifies the major revision number of the device. The major revision reflects changes to base layers of the design. The major revision number is indicated in the part number as a letter (A for first revision, B for second, and so on). This field is encoded as follows: Value Description 0x0 Revision A (initial device) 0x1 Revision B (first base layer revision) 0x2 Revision C (second base layer revision) and so on.
7:0	MINOR	RO	-	Minor Revision This field specifies the minor revision number of the device. The minor revision reflects changes to the metal layers of the design. The MINOR field value is reset when the MAJOR field is changed. This field is numeric and is encoded as follows: Value Description 0x0 Initial device, or a major revision update. 0x1 First metal layer change. 0x2 Second metal layer change. and so on.

Register 2: Brown-Out Reset Control (PBORCTL), offset 0x030

This register is responsible for controlling reset conditions after initial power-on reset.

Brown-Out Reset Control (PBORCTL)

Base 0x400F.E000
Offset 0x030
Type R/W, reset 0x0000.7FFD



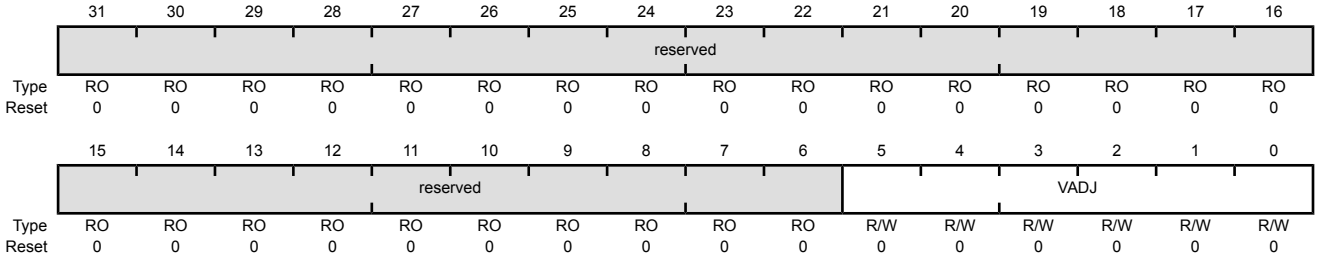
Bit/Field	Name	Type	Reset	Description
31:2	reserved	RO	0x0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1	BORIOR	R/W	0	BOR Interrupt or Reset This bit controls how a BOR event is signaled to the controller. If set, a reset is signaled. Otherwise, an interrupt is signaled.
0	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Register 3: LDO Power Control (LDOPCTL), offset 0x034

The VADJ field in this register adjusts the on-chip output voltage (V_{OUT}).

LDO Power Control (LDOPCTL)

Base 0x400F.E000
Offset 0x034
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:6	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
5:0	VADJ	R/W	0x0	LDO Output Voltage This field sets the on-chip output voltage. The programming values for the VADJ field are provided below.

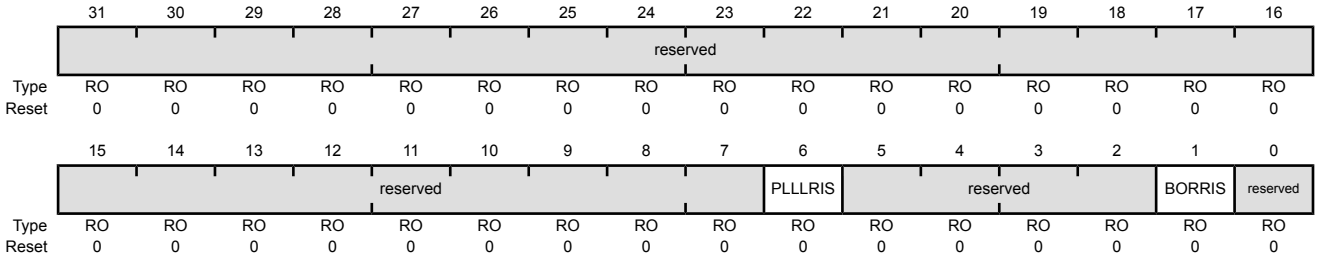
Value	V_{OUT} (V)
0x00	2.50
0x01	2.45
0x02	2.40
0x03	2.35
0x04	2.30
0x05	2.25
0x06-0x3F	Reserved
0x1B	2.75
0x1C	2.70
0x1D	2.65
0x1E	2.60
0x1F	2.55

Register 4: Raw Interrupt Status (RIS), offset 0x050

Central location for system control raw interrupts. These are set and cleared by hardware.

Raw Interrupt Status (RIS)

Base 0x400F.E000
Offset 0x050
Type RO, reset 0x0000.0000



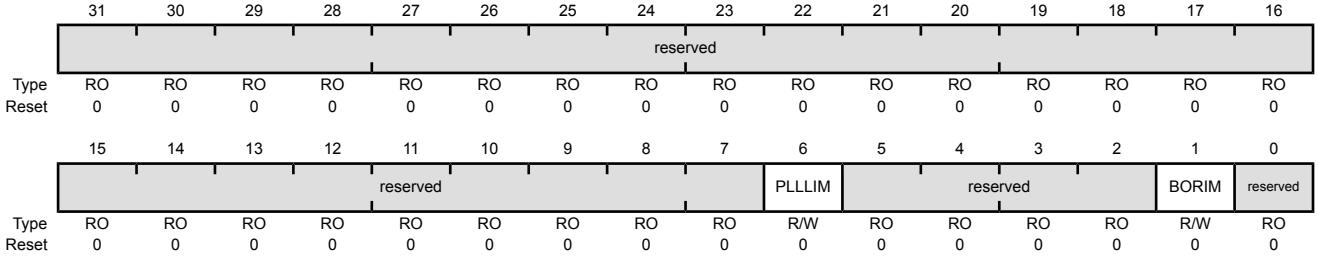
Bit/Field	Name	Type	Reset	Description
31:7	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
6	PLLLRIS	RO	0	PLL Lock Raw Interrupt Status This bit is set when the PLL T_{READY} Timer asserts.
5:2	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1	BORRIS	RO	0	Brown-Out Reset Raw Interrupt Status This bit is the raw interrupt status for any brown-out conditions. If set, a brown-out condition is currently active. This is an unregistered signal from the brown-out detection circuit. An interrupt is reported if the BORIM bit in the **IMC** register is set and the BORIOR bit in the **PBORCTL** register is cleared.
0	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Register 5: Interrupt Mask Control (IMC), offset 0x054

Central location for system control interrupt masks.

Interrupt Mask Control (IMC)

Base 0x400F.E000
Offset 0x054
Type R/W, reset 0x0000.0000



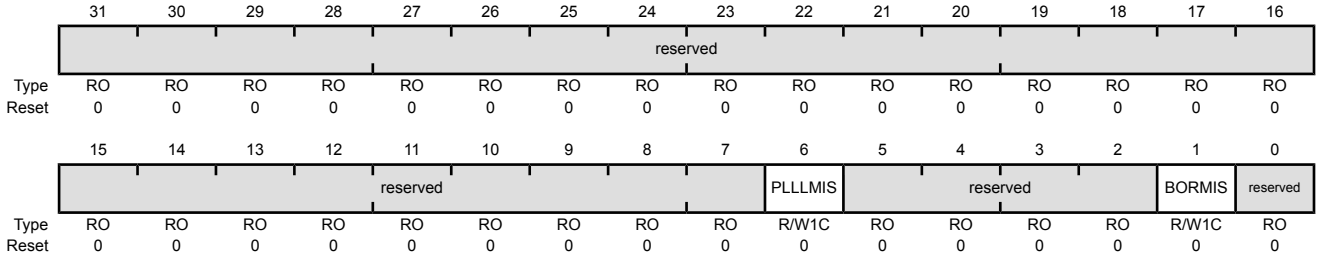
Bit/Field	Name	Type	Reset	Description
31:7	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
6	PLLLIM	R/W	0	PLL Lock Interrupt Mask This bit specifies whether a PLL Lock interrupt is promoted to a controller interrupt. If set, an interrupt is generated if PLLLRIS in **RIS** is set; otherwise, an interrupt is not generated.
5:2	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1	BORIM	R/W	0	Brown-Out Reset Interrupt Mask This bit specifies whether a brown-out condition is promoted to a controller interrupt. If set, an interrupt is generated if BORRIS is set; otherwise, an interrupt is not generated.
0	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Register 6: Masked Interrupt Status and Clear (MISC), offset 0x058

On a read, this register gives the current masked status value of the corresponding interrupt. All of the bits are R/W1C and this action also clears the corresponding raw interrupt bit in the **RIS** register (see page 79).

Masked Interrupt Status and Clear (MISC)

Base 0x400F.E000
Offset 0x058
Type R/W1C, reset 0x0000.0000



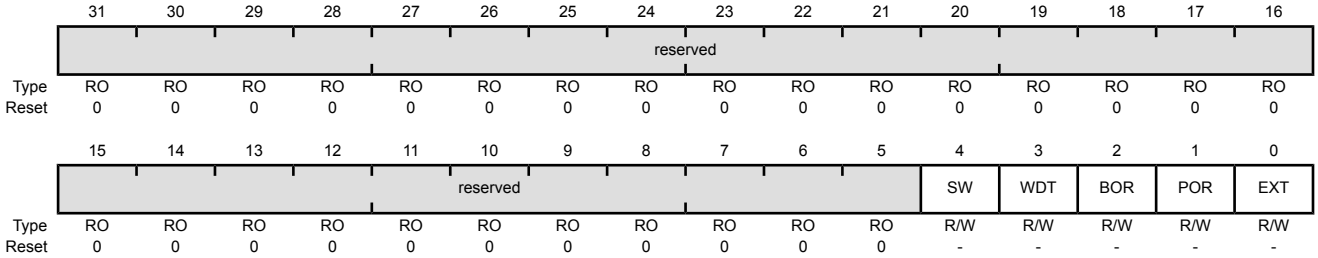
Bit/Field	Name	Type	Reset	Description
31:7	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
6	PLLLMIS	R/W1C	0	PLL Lock Masked Interrupt Status This bit is set when the PLL T_{READY} timer asserts. The interrupt is cleared by writing a 1 to this bit.
5:2	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1	BORMIS	R/W1C	0	BOR Masked Interrupt Status The BORMIS is simply the BORRIS ANDed with the mask value, BORIM.
0	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Register 7: Reset Cause (RESC), offset 0x05C

This register is set with the reset cause after reset. The bits in this register are sticky and maintain their state across multiple reset sequences, except when an power-on reset is the cause, in which case, all bits other than POR in the **RESC** register are cleared.

Reset Cause (RESC)

Base 0x400F.E000
Offset 0x05C
Type R/W, reset -



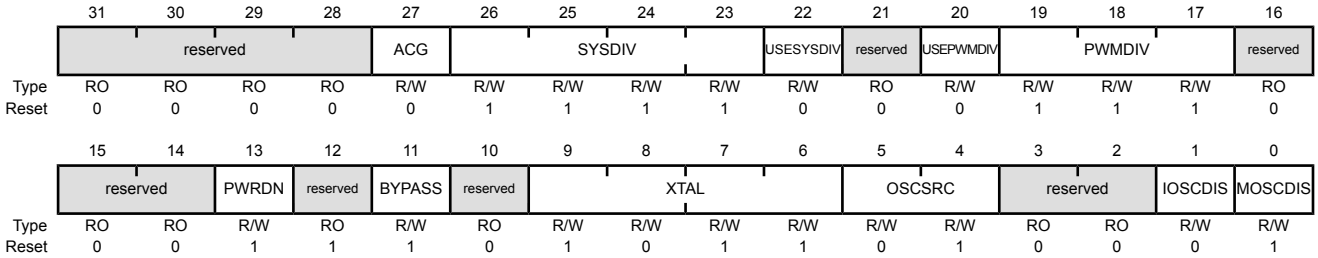
Bit/Field	Name	Type	Reset	Description
31:5	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
4	SW	R/W	-	Software Reset When set, indicates a software reset is the cause of the reset event.
3	WDT	R/W	-	Watchdog Timer Reset When set, indicates a watchdog reset is the cause of the reset event.
2	BOR	R/W	-	Brown-Out Reset When set, indicates a brown-out reset is the cause of the reset event.
1	POR	R/W	-	Power-On Reset When set, indicates a power-on reset is the cause of the reset event.
0	EXT	R/W	-	External Reset When set, indicates an external reset ($\overline{\text{RST}}$ assertion) is the cause of the reset event.

Register 8: Run-Mode Clock Configuration (RCC), offset 0x060

This register is defined to provide source control and frequency speed.

Run-Mode Clock Configuration (RCC)

Base 0x400F.E000
Offset 0x060
Type R/W, reset 0x078E.3AD1



Bit/Field	Name	Type	Reset	Description
31:28	reserved	RO	0x0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
27	ACG	R/W	0	Auto Clock Gating This bit specifies whether the system uses the **Sleep-Mode Clock Gating Control (SCGCn)** registers and **Deep-Sleep-Mode Clock Gating Control (DCGCn)** registers if the controller enters a Sleep or Deep-Sleep mode (respectively). If set, the **SCGCn** or **DCGCn** registers are used to control the clocks distributed to the peripherals when the controller is in a sleep mode. Otherwise, the **Run-Mode Clock Gating Control (RCGCn)** registers are used when the controller enters a sleep mode. The **RCGCn** registers are always used to control the clocks in Run mode. This allows peripherals to consume less power when the controller is in a sleep mode and the peripheral is unused.

Bit/Field	Name	Type	Reset	Description
26:23	SYSDIV	R/W	0xF	System Clock Divisor Specifies which divisor is used to generate the system clock from the PLL output. Although the PLL VCO frequency is 400 MHz, it is predivided by 2 before the divisor is applied. Value Divisor (BYPASS=1) Frequency (BYPASS=0) 0x0 reserved reserved 0x1 /2 reserved 0x2 /3 reserved 0x3 /4 50 MHz 0x4 /5 40 MHz 0x5 /6 33.33 MHz 0x6 /7 28.57 MHz 0x7 /8 25 MHz 0x8 /9 22.22 MHz 0x9 /10 20 MHz 0xA /11 18.18 MHz 0xB /12 16.67 MHz 0xC /13 15.38 MHz 0xD /14 14.29 MHz 0xE /15 13.33 MHz 0xF /16 12.5 MHz (default) When reading the **Run-Mode Clock Configuration (RCC)** register (see page 83), the SYSDIV value is MINSYSDIV if a lower divider was requested and the PLL is being used. This lower value is allowed to divide a non-PLL source.
22	USESYSDIV	R/W	0	Enable System Clock Divider Use the system clock divider as the source for the system clock. The system clock divider is forced to be used when the PLL is selected as the source.
21	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
20	USEPWMDIV	R/W	0	Enable PWM Clock Divisor Use the PWM clock divider as the source for the PWM clock.

Bit/Field	Name	Type	Reset	Description
19:17	PWMDIV	R/W	0x7	PWM Unit Clock Divisor This field specifies the binary divisor used to predivide the system clock down for use as the timing reference for the PWM module. This clock is only power 2 divide and rising edge is synchronous without phase shift from the system clock. Value Divisor 0x0 /2 0x1 /4 0x2 /8 0x3 /16 0x4 /32 0x5 /64 0x6 /64 0x7 /64 (default)
16:14	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
13	PWRDN	R/W	1	PLL Power Down This bit connects to the PLL PWRDN input. The reset value of 1 powers down the PLL.
12	reserved	RO	1	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
11	BYPASS	R/W	1	PLL Bypass Chooses whether the system clock is derived from the PLL output or the OSC source. If set, the clock that drives the system is the OSC source. Otherwise, the clock that drives the system is the PLL output clock divided by the system divider. **Note:** The ADC must be clocked from the PLL or directly from a 14-MHz to 18-MHz clock source to operate properly. While the ADC works in a 14-18 MHz range, to maintain a 1 M sample/second rate, the ADC must be provided a 16-MHz clock source.
10	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Bit/Field	Name	Type	Reset	Description
9:6	XTAL	R/W	0xB	Crystal Value This field specifies the crystal value attached to the main oscillator. The encoding for this field is provided below.

Value	Crystal Frequency (MHz) Not Using the PLL	Crystal Frequency (MHz) Using the PLL
0x0	1.000	reserved
0x1	1.8432	reserved
0x2	2.000	reserved
0x3	2.4576	reserved
0x4	3.579545 MHz	
0x5	3.6864 MHz	
0x6	4 MHz	
0x7	4.096 MHz	
0x8	4.9152 MHz	
0x9	5 MHz	
0xA	5.12 MHz	
0xB	6 MHz (reset value)	
0xC	6.144 MHz	
0xD	7.3728 MHz	
0xE	8 MHz	
0xF	8.192 MHz	

5:4	OSCSRC	R/W	0x1	Oscillator Source Selects the input source for the OSC. The values are:

Value	Input Source
0x0	MOSC
	Main oscillator
0x1	IOSC
	Internal oscillator (default)
0x2	IOSC/4
	Internal oscillator / 4 (this is necessary if used as input to PLL)
0x3	30 kHz
	30-KHz internal oscillator

For additional oscillator sources, see the **RCC2** register.

3:2	reserved	RO	0x0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1	IOSCDIS	R/W	0	Internal Oscillator Disable 0: Internal oscillator (IOSC) is enabled. 1: Internal oscillator is disabled.

Bit/Field	Name	Type	Reset	Description
0	MOSCDIS	R/W	1	Main Oscillator Disable 0: Main oscillator is enabled . 1: Main oscillator is disabled (default).

Register 9: XTAL to PLL Translation (PLLCFG), offset 0x064

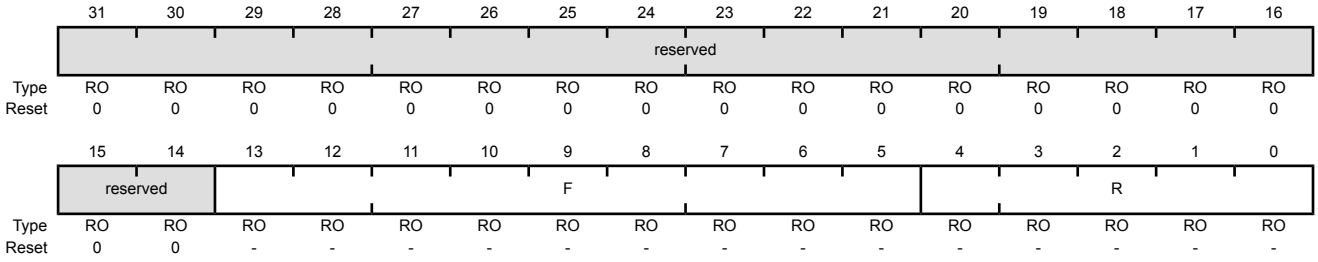
This register provides a means of translating external crystal frequencies into the appropriate PLL settings. This register is initialized during the reset sequence and updated anytime that the XTAL field changes in the **Run-Mode Clock Configuration (RCC)** register (see page 83).

The PLL frequency is calculated using the **PLLCFG** field values, as follows:

```
PLLFreq = OSCFreq * F / (R + 1)
```

XTAL to PLL Translation (PLLCFG)

Base 0x400F.E000
Offset 0x064
Type RO, reset -



Bit/Field	Name	Type	Reset	Description
31:14	reserved	RO	0x0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
13:5	F	RO	-	PLL F Value This field specifies the value supplied to the PLL's F input.
4:0	R	RO	-	PLL R Value This field specifies the value supplied to the PLL's R input.

Register 10: Run-Mode Clock Configuration 2 (RCC2), offset 0x070

This register overrides the **RCC** equivalent register fields when the USERCC2 bit is set, allowing the extended capabilities of the **RCC2** register to be used while also providing a means to be backward-compatible to previous parts. The fields within the **RCC2** register occupy the same bit positions as they do within the **RCC** register as LSB-justified.

The SYSDIV2 field is 2 bits wider than the SYSDIV field in the **RCC** register so that additional larger divisors are possible, allowing a lower system clock frequency for improved Deep Sleep power consumption. The PLL VCO frequency is 400 MHz.

Run-Mode Clock Configuration 2 (RCC2)

Base 0x400F.E000
Offset 0x070
Type R/W, reset 0x0780.2810

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
USERCC2	reserved		SYSDIV2						reserved						
Type R/W	RO	RO	R/W	R/W	R/W	R/W	R/W	R/W	RO	RO	RO	RO	RO	RO	RO
Reset 0	0	0	0	0	1	1	1	1	0	0	0	0	0	0	0

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
reserved		PWRDN2	reserved	BYPASS2	reserved				OSCSRC2			reserved			
Type RO	RO	R/W	RO	R/W	RO	RO	RO	RO	R/W	R/W	R/W	RO	RO	RO	RO
Reset 0	0	1	0	1	0	0	0	0	0	0	1	0	0	0	0

Bit/Field	Name	Type	Reset	Description
31	USERCC2	R/W	0	Use RCC2 When set, overrides the **RCC** register fields.
30:29	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
28:23	SYSDIV2	R/W	0x0F	System Clock Divisor Specifies which divisor is used to generate the system clock from the PLL output. Although the PLL VCO frequency is 400 MHz, it is predivided by 2 before the divisor is applied. This field is wider than the **RCC** register SYSDIV field in order to provide additional divisor values. This permits the system clock to be run at much lower frequencies during Deep Sleep mode. For example, where the **RCC** register SYSDIV encoding of 1111 provides /16, the **RCC2** register SYSDIV2 encoding of 111111 provides /64.
22:14	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
13	PWRDN2	R/W	1	Power-Down PLL When set, powers down the PLL.
12	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

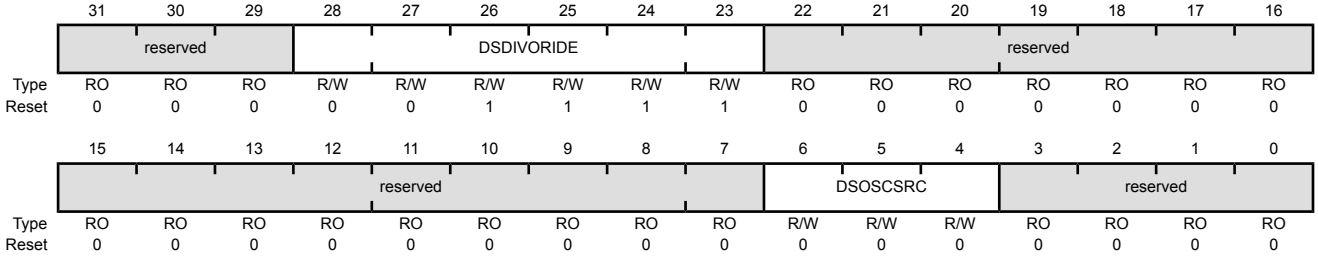
Bit/Field	Name	Type	Reset	Description
11	BYPASS2	R/W	1	Bypass PLL When set, bypasses the PLL for the clock source.
10:7	reserved	RO	0x0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
6:4	OSCSRC2	R/W	0x1	Oscillator Source Selects the input source for the OSC. The values are: Value Description 0x0 MOSC Main oscillator 0x1 IOSC Internal oscillator 0x2 IOSC/4 Internal oscillator / 4 0x3 30 kHz 30-kHz internal oscillator 0x4 Reserved 0x5 Reserved 0x6 Reserved 0x7 Reserved
3:0	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Register 11: Deep Sleep Clock Configuration (DSLPCLKCFG), offset 0x144

This register provides configuration information for the hardware control of Deep Sleep Mode.

Deep Sleep Clock Configuration (DSLPCLKCFG)

Base 0x400F.E000
Offset 0x144
Type R/W, reset 0x0780.0000



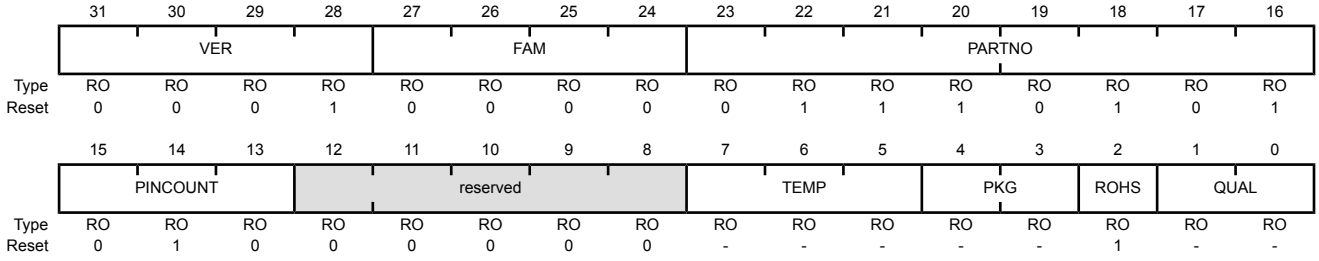
Bit/Field	Name	Type	Reset	Description
31:29	reserved	RO	0x0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
28:23	DSDIVORIDE	R/W	0x0F	Divider Field Override 6-bit system divider field to override when Deep-Sleep occurs with PLL running.
22:7	reserved	RO	0x0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
6:4	DSOSCSRC	R/W	0x0	Clock Source Specifies the clock source during Deep-Sleep mode. Value Description 0x0 MOSC Use main oscillator as source. 0x1 IOSC Use internal 12-MHz oscillator as source. 0x2 Reserved 0x3 30 kHz Use 30-kHz internal oscillator as source. 0x4 Reserved 0x5 Reserved 0x6 Reserved 0x7 Reserved
3:0	reserved	RO	0x0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Register 12: Device Identification 1 (DID1), offset 0x004

This register identifies the device family, part number, temperature range, pin count, and package type.

Device Identification 1 (DID1)

Base 0x400F.E000
Offset 0x004
Type RO, reset -



Bit/Field	Name	Type	Reset	Description
31:28	VER	RO	0x1	DID1 Version This field defines the **DID1** register format version. The version number is numeric. The value of the VER field is encoded as follows (all other encodings are reserved): Value Description 0x1 Second version of the **DID1** register format.
27:24	FAM	RO	0x0	Family This field provides the family identification of the device within the Luminary Micro product portfolio. The value is encoded as follows (all other encodings are reserved): Value Description 0x0 Stellaris family of microcontollers, that is, all devices with external part numbers starting with LM3S.
23:16	PARTNO	RO	0x75	Part Number This field provides the part number of the device within the family. The value is encoded as follows (all other encodings are reserved): Value Description 0x75 LM3S6432
15:13	PINCOUNT	RO	0x2	Package Pin Count This field specifies the number of pins on the device package. The value is encoded as follows (all other encodings are reserved): Value Description 0x2 100-pin or 108-ball package

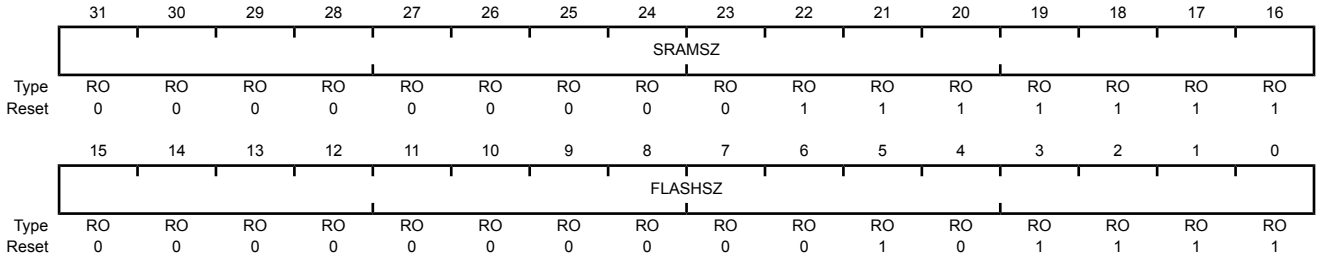
Bit/Field	Name	Type	Reset	Description
12:8	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:5	TEMP	RO	-	Temperature Range This field specifies the temperature rating of the device. The value is encoded as follows (all other encodings are reserved): Value Description 0x0 Commercial temperature range (0°C to 70°C) 0x1 Industrial temperature range (-40°C to 85°C) 0x2 Extended temperature range (-40°C to 105°C)
4:3	PKG	RO	-	Package Type This field specifies the package type. The value is encoded as follows (all other encodings are reserved): Value Description 0x0 SOIC package 0x1 LQFP package 0x2 BGA package
2	ROHS	RO	1	RoHS-Compliance This bit specifies whether the device is RoHS-compliant. A 1 indicates the part is RoHS-compliant.
1:0	QUAL	RO	-	Qualification Status This field specifies the qualification status of the device. The value is encoded as follows (all other encodings are reserved): Value Description 0x0 Engineering Sample (unqualified) 0x1 Pilot Production (unqualified) 0x2 Fully Qualified

Register 13: Device Capabilities 0 (DC0), offset 0x008

This register is predefined by the part and can be used to verify features.

Device Capabilities 0 (DC0)

Base 0x400F.E000
Offset 0x008
Type RO, reset 0x007F.002F



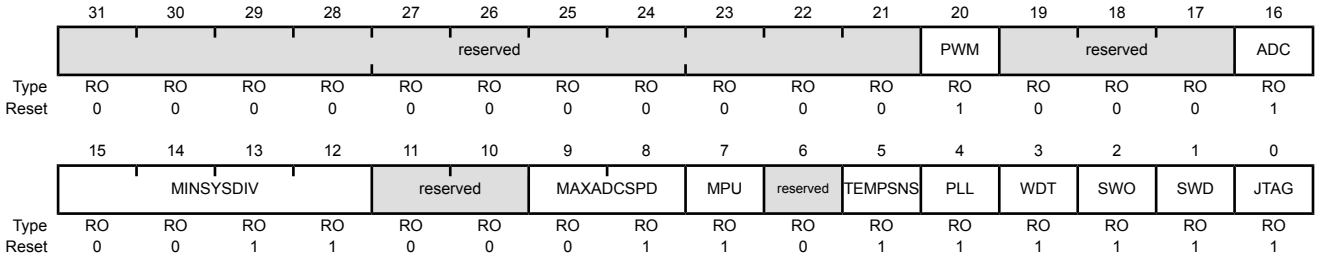
Bit/Field	Name	Type	Reset	Description
31:16	SRAMSZ	RO	0x007F	SRAM Size Indicates the size of the on-chip SRAM memory. Value Description 0x007F 32 KB of SRAM
15:0	FLASHSZ	RO	0x002F	Flash Size Indicates the size of the on-chip flash memory. Value Description 0x002F 96 KB of Flash

Register 14: Device Capabilities 1 (DC1), offset 0x010

This register provides a list of features available in the system. The Stellaris family uses this register format to indicate the availability of the following family features in the specific device: CANs, PWM, ADC, Watchdog timer, Hibernation module, and debug capabilities. This register also indicates the maximum clock frequency and maximum ADC sample rate. The format of this register is consistent with the **RCGC0**, **SCGC0**, and **DCGC0** clock control registers and the **SRCR0** software reset control register.

Device Capabilities 1 (DC1)

Base 0x400F.E000
Offset 0x010
Type RO, reset 0x0011.31BF



Bit/Field	Name	Type	Reset	Description
31:21	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
20	PWM	RO	1	PWM Module Present When set, indicates that the PWM module is present.
19:17	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
16	ADC	RO	1	ADC Module Present When set, indicates that the ADC module is present.
15:12	MINSYSDIV	RO	0x3	System Clock Divider Minimum 4-bit divider value for system clock. The reset value is hardware-dependent. See the **RCC** register for how to change the system clock divisor using the SYSDIV bit. Value Description 0x3 Specifies a 50-MHz CPU clock with a PLL divider of 4.
11:10	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Bit/Field	Name	Type	Reset	Description
9:8	MAXADCSPD	RO	0x1	Max ADC Speed Indicates the maximum rate at which the ADC samples data. Value Description 0x1 250K samples/second
7	MPU	RO	1	MPU Present When set, indicates that the Cortex-M3 Memory Protection Unit (MPU) module is present. See the ARM Cortex-M3 Technical Reference Manual for details on the MPU.
6	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
5	TEMPSNS	RO	1	Temp Sensor Present When set, indicates that the on-chip temperature sensor is present.
4	PLL	RO	1	PLL Present When set, indicates that the on-chip Phase Locked Loop (PLL) is present.
3	WDT	RO	1	Watchdog Timer Present When set, indicates that a watchdog timer is present.
2	SWO	RO	1	SWO Trace Port Present When set, indicates that the Serial Wire Output (SWO) trace port is present.
1	SWD	RO	1	SWD Present When set, indicates that the Serial Wire Debugger (SWD) is present.
0	JTAG	RO	1	JTAG Present When set, indicates that the JTAG debugger interface is present.

Register 15: Device Capabilities 2 (DC2), offset 0x014

This register provides a list of features available in the system. The Stellaris family uses this register format to indicate the availability of the following family features in the specific device: Analog Comparators, General-Purpose Timers, I2Cs, QEIs, SSIs, and UARTs. The format of this register is consistent with the **RCGC1**, **SCGC1**, and **DCGC1** clock control registers and the **SRCR1** software reset control register.

Device Capabilities 2 (DC2)

Base 0x400F.E000
Offset 0x014
Type RO, reset 0x0307.1013

	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	reserved						COMP1	COMP0	reserved					TIMER2	TIMER1	TIMER0
Type	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO
Reset	0	0	0	0	0	0	1	1	0	0	0	0	0	1	1	1

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	reserved			I2C0	reserved							SSI0	reserved		UART1	UART0
Type	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO
Reset	0	0	0	1	0	0	0	0	0	0	0	1	0	0	1	1

Bit/Field	Name	Type	Reset	Description
31:26	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
25	COMP1	RO	1	Analog Comparator 1 Present When set, indicates that analog comparator 1 is present.
24	COMP0	RO	1	Analog Comparator 0 Present When set, indicates that analog comparator 0 is present.
23:19	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
18	TIMER2	RO	1	Timer 2 Present When set, indicates that General-Purpose Timer module 2 is present.
17	TIMER1	RO	1	Timer 1 Present When set, indicates that General-Purpose Timer module 1 is present.
16	TIMER0	RO	1	Timer 0 Present When set, indicates that General-Purpose Timer module 0 is present.
15:13	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
12	I2C0	RO	1	I2C Module 0 Present When set, indicates that I2C module 0 is present.

Bit/Field	Name	Type	Reset	Description
11:5	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
4	SSI0	RO	1	SSI0 Present When set, indicates that SSI module 0 is present.
3:2	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1	UART1	RO	1	UART1 Present When set, indicates that UART module 1 is present.
0	UART0	RO	1	UART0 Present When set, indicates that UART module 0 is present.

Register 16: Device Capabilities 3 (DC3), offset 0x018

This register provides a list of features available in the system. The Stellaris family uses this register format to indicate the availability of the following family features in the specific device: Analog Comparator I/Os, CCP I/Os, ADC I/Os, and PWM I/Os.

Device Capabilities 3 (DC3)

Base 0x400F.E000
Offset 0x018
Type RO, reset 0x8F07.87C3

	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	32KHZ	reserved			CCP3	CCP2	CCP1	CCP0	reserved					ADC2	ADC1	ADC0
Type	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO
Reset	1	0	0	0	1	1	1	1	0	0	0	0	0	1	1	1

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PWMFAULT	reserved				C1PLUS	C1MINUS	C0O	C0PLUS	C0MINUS	reserved				PWM1	PWM0
Type	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO
Reset	1	0	0	0	0	1	1	1	1	1	0	0	0	0	1	1

Bit/Field	Name	Type	Reset	Description
31	32KHZ	RO	1	32KHz Input Clock Available When set, indicates an even CCP pin is present and can be used as a 32-KHz input clock.
30:28	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
27	CCP3	RO	1	CCP3 Pin Present When set, indicates that Capture/Compare/PWM pin 3 is present.
26	CCP2	RO	1	CCP2 Pin Present When set, indicates that Capture/Compare/PWM pin 2 is present.
25	CCP1	RO	1	CCP1 Pin Present When set, indicates that Capture/Compare/PWM pin 1 is present.
24	CCP0	RO	1	CCP0 Pin Present When set, indicates that Capture/Compare/PWM pin 0 is present.
23:19	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
18	ADC2	RO	1	ADC2 Pin Present When set, indicates that ADC pin 2 is present.
17	ADC1	RO	1	ADC1 Pin Present When set, indicates that ADC pin 1 is present.
16	ADC0	RO	1	ADC0 Pin Present When set, indicates that ADC pin 0 is present.

Bit/Field	Name	Type	Reset	Description
15	PWMFAULT	RO	1	PWM Fault Pin Present When set, indicates that the PWM Fault pin is present.
14:11	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
10	C1PLUS	RO	1	C1+ Pin Present When set, indicates that the analog comparator 1 (+) input pin is present.
9	C1MINUS	RO	1	C1- Pin Present When set, indicates that the analog comparator 1 (-) input pin is present.
8	C0O	RO	1	C0o Pin Present When set, indicates that the analog comparator 0 output pin is present.
7	C0PLUS	RO	1	C0+ Pin Present When set, indicates that the analog comparator 0 (+) input pin is present.
6	C0MINUS	RO	1	C0- Pin Present When set, indicates that the analog comparator 0 (-) input pin is present.
5:2	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1	PWM1	RO	1	PWM1 Pin Present When set, indicates that the PWM pin 1 is present.
0	PWM0	RO	1	PWM0 Pin Present When set, indicates that the PWM pin 0 is present.

Register 17: Device Capabilities 4 (DC4), offset 0x01C

This register provides a list of features available in the system. The Stellaris family uses this register format to indicate the availability of the following family features in the specific device: Ethernet MAC and PHY, GPIOs, and CCP I/Os. The format of this register is consistent with the **RCGC2**, **SCGC2**, and **DCGC2** clock control registers and the **SRCR2** software reset control register.

Device Capabilities 4 (DC4)

Base 0x400F.E000
Offset 0x01C
Type RO, reset 0x5000.007F

	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	reserved	EPHY0	reserved	EMAC0	reserved											
Type	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO
Reset	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	reserved									GPIOG	GPIOF	GPIOE	GPIOD	GPIOC	GPIOB	GPIOA
Type	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO
Reset	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1

Bit/Field	Name	Type	Reset	Description
31	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
30	EPHY0	RO	1	Ethernet PHY0 Present When set, indicates that Ethernet PHY module 0 is present.
29	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
28	EMAC0	RO	1	Ethernet MAC0 Present When set, indicates that Ethernet MAC module 0 is present.
27:7	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
6	GPIOG	RO	1	GPIO Port G Present When set, indicates that GPIO Port G is present.
5	GPIOF	RO	1	GPIO Port F Present When set, indicates that GPIO Port F is present.
4	GPIOE	RO	1	GPIO Port E Present When set, indicates that GPIO Port E is present.
3	GPIOD	RO	1	GPIO Port D Present When set, indicates that GPIO Port D is present.
2	GPIOC	RO	1	GPIO Port C Present When set, indicates that GPIO Port C is present.

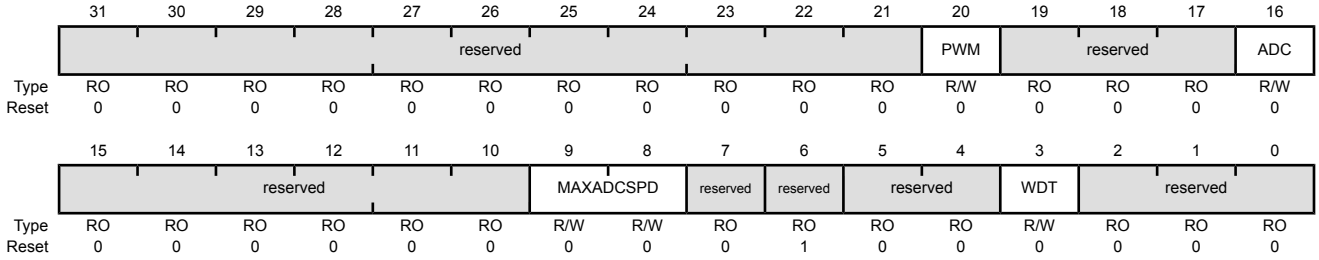
Bit/Field	Name	Type	Reset	Description
1	GPIOB	RO	1	GPIO Port B Present When set, indicates that GPIO Port B is present.
0	GPIOA	RO	1	GPIO Port A Present When set, indicates that GPIO Port A is present.

Register 18: Run Mode Clock Gating Control Register 0 (RCGC0), offset 0x100

This register controls the clock gating logic. Each bit controls a clock enable for a given interface, function, or unit. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled (saving power). If the unit is unclocked, reads or writes to the unit will generate a bus fault. The reset state of these bits is 0 (unclocked) unless otherwise noted, so that all functional units are disabled. It is the responsibility of software to enable the ports necessary for the application. Note that these registers may contain more bits than there are interfaces, functions, or units to control. This is to assure reasonable code compatibility with other family and future parts. **RCGC0** is the clock configuration register for running operation, **SCGC0** for Sleep operation, and **DCGC0** for Deep-Sleep operation. Setting the ACG bit in the **Run-Mode Clock Configuration (RCC)** register specifies that the system uses sleep modes.

Run Mode Clock Gating Control Register 0 (RCGC0)

Base 0x400F.E000
Offset 0x100
Type R/W, reset 0x00000040



Bit/Field	Name	Type	Reset	Description
31:21	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
20	PWM	R/W	0	PWM Clock Gating Control This bit controls the clock gating for the PWM module. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, a read or write to the unit generates a bus fault.
19:17	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
16	ADC	R/W	0	ADC0 Clock Gating Control This bit controls the clock gating for SAR ADC module 0. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, a read or write to the unit generates a bus fault.
15:10	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

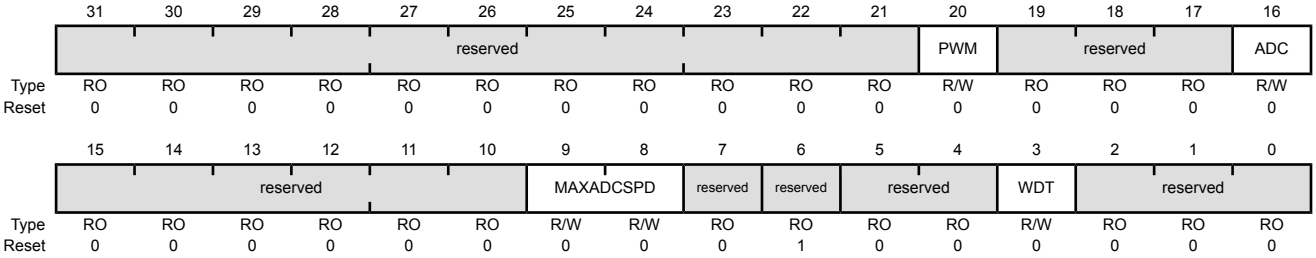
Bit/Field	Name	Type	Reset	Description
9:8	MAXADCSPD	R/W	0	ADC Sample Speed This field sets the rate at which the ADC samples data. You cannot set the rate higher than the maximum rate. You can set the sample rate by setting the MAXADCSPD bit as follows: Value Description 0x1 250K samples/second 0x0 125K samples/second
7	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
6	reserved	RO	1	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
5:4	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
3	WDT	R/W	0	WDT Clock Gating Control This bit controls the clock gating for the WDT module. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, a read or write to the unit generates a bus fault.
2:0	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Register 19: Sleep Mode Clock Gating Control Register 0 (SCGC0), offset 0x110

This register controls the clock gating logic. Each bit controls a clock enable for a given interface, function, or unit. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled (saving power). If the unit is unclocked, reads or writes to the unit will generate a bus fault. The reset state of these bits is 0 (unclocked) unless otherwise noted, so that all functional units are disabled. It is the responsibility of software to enable the ports necessary for the application. Note that these registers may contain more bits than there are interfaces, functions, or units to control. This is to assure reasonable code compatibility with other family and future parts. **RCGC0** is the clock configuration register for running operation, **SCGC0** for Sleep operation, and **DCGC0** for Deep-Sleep operation. Setting the ACG bit in the **Run-Mode Clock Configuration (RCC)** register specifies that the system uses sleep modes.

Sleep Mode Clock Gating Control Register 0 (SCGC0)

Base 0x400F.E000
Offset 0x110
Type R/W, reset 0x00000040



Bit/Field	Name	Type	Reset	Description
31:21	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
20	PWM	R/W	0	PWM Clock Gating Control This bit controls the clock gating for the PWM module. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, a read or write to the unit generates a bus fault.
19:17	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
16	ADC	R/W	0	ADC0 Clock Gating Control This bit controls the clock gating for SAR ADC module 0. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, a read or write to the unit generates a bus fault.
15:10	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Bit/Field	Name	Type	Reset	Description
9:8	MAXADCSPD	R/W	0	ADC Sample Speed This field sets the rate at which the ADC samples data. You cannot set the rate higher than the maximum rate. You can set the sample rate by setting the MAXADCSPD bit as follows: Value Description 0x1 250K samples/second 0x0 125K samples/second
7	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
6	reserved	RO	1	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
5:4	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
3	WDT	R/W	0	WDT Clock Gating Control This bit controls the clock gating for the WDT module. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, a read or write to the unit generates a bus fault.
2:0	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Register 20: Deep Sleep Mode Clock Gating Control Register 0 (DCGC0), offset 0x120

This register controls the clock gating logic. Each bit controls a clock enable for a given interface, function, or unit. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled (saving power). If the unit is unclocked, reads or writes to the unit will generate a bus fault. The reset state of these bits is 0 (unclocked) unless otherwise noted, so that all functional units are disabled. It is the responsibility of software to enable the ports necessary for the application. Note that these registers may contain more bits than there are interfaces, functions, or units to control. This is to assure reasonable code compatibility with other family and future parts. **RCGC0** is the clock configuration register for running operation, **SCGC0** for Sleep operation, and **DCGC0** for Deep-Sleep operation. Setting the ACG bit in the **Run-Mode Clock Configuration (RCC)** register specifies that the system uses sleep modes.

Deep Sleep Mode Clock Gating Control Register 0 (DCGC0)

Base 0x400F.E000
Offset 0x120
Type R/W, reset 0x00000040

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
reserved											PWM	reserved			ADC
RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	R/W	RO	RO	RO	R/W
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
reserved									reserved	reserved		WDT	reserved		
RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	R/W	RO	RO	RO
0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0

Bit/Field	Name	Type	Reset	Description
31:21	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
20	PWM	R/W	0	PWM Clock Gating Control This bit controls the clock gating for the PWM module. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, a read or write to the unit generates a bus fault.
19:17	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
16	ADC	R/W	0	ADC0 Clock Gating Control This bit controls the clock gating for SAR ADC module 0. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, a read or write to the unit generates a bus fault.
15:7	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
6	reserved	RO	1	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Bit/Field	Name	Type	Reset	Description
5:4	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
3	WDT	R/W	0	WDT Clock Gating Control This bit controls the clock gating for the WDT module. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, a read or write to the unit generates a bus fault.
2:0	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Register 21: Run Mode Clock Gating Control Register 1 (RCGC1), offset 0x104

This register controls the clock gating logic. Each bit controls a clock enable for a given interface, function, or unit. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled (saving power). If the unit is unclocked, reads or writes to the unit will generate a bus fault. The reset state of these bits is 0 (unclocked) unless otherwise noted, so that all functional units are disabled. It is the responsibility of software to enable the ports necessary for the application. Note that these registers may contain more bits than there are interfaces, functions, or units to control. This is to assure reasonable code compatibility with other family and future parts. **RCGC1** is the clock configuration register for running operation, **SCGC1** for Sleep operation, and **DCGC1** for Deep-Sleep operation. Setting the ACG bit in the **Run-Mode Clock Configuration (RCC)** register specifies that the system uses sleep modes.

Run Mode Clock Gating Control Register 1 (RCGC1)

Base 0x400F.E000
Offset 0x104
Type R/W, reset 0x00000000

	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	reserved						COMP1	COMP0	reserved					TIMER2	TIMER1	TIMER0
Type	RO	RO	RO	RO	RO	RO	R/W	R/W	RO	RO	RO	RO	RO	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	reserved			I2C0	reserved							SSI0	reserved		UART1	UART0
Type	RO	RO	RO	R/W	RO	RO	RO	RO	RO	RO	RO	R/W	RO	RO	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit/Field	Name	Type	Reset	Description
31:26	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
25	COMP1	R/W	0	Analog Comparator 1 Clock Gating This bit controls the clock gating for analog comparator 1. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
24	COMP0	R/W	0	Analog Comparator 0 Clock Gating This bit controls the clock gating for analog comparator 0. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
23:19	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
18	TIMER2	R/W	0	Timer 2 Clock Gating Control This bit controls the clock gating for General-Purpose Timer module 2. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.

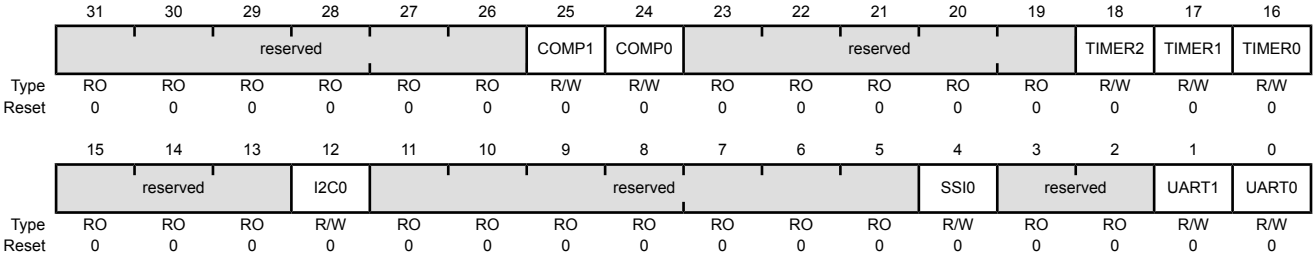
Bit/Field	Name	Type	Reset	Description
17	TIMER1	R/W	0	Timer 1 Clock Gating Control This bit controls the clock gating for General-Purpose Timer module 1. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
16	TIMER0	R/W	0	Timer 0 Clock Gating Control This bit controls the clock gating for General-Purpose Timer module 0. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
15:13	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
12	I2C0	R/W	0	I2C0 Clock Gating Control This bit controls the clock gating for I2C module 0. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
11:5	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
4	SSI0	R/W	0	SSI0 Clock Gating Control This bit controls the clock gating for SSI module 0. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
3:2	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1	UART1	R/W	0	UART1 Clock Gating Control This bit controls the clock gating for UART module 1. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
0	UART0	R/W	0	UART0 Clock Gating Control This bit controls the clock gating for UART module 0. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.

Register 22: Sleep Mode Clock Gating Control Register 1 (SCGC1), offset 0x114

This register controls the clock gating logic. Each bit controls a clock enable for a given interface, function, or unit. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled (saving power). If the unit is unclocked, reads or writes to the unit will generate a bus fault. The reset state of these bits is 0 (unclocked) unless otherwise noted, so that all functional units are disabled. It is the responsibility of software to enable the ports necessary for the application. Note that these registers may contain more bits than there are interfaces, functions, or units to control. This is to assure reasonable code compatibility with other family and future parts. **RCGC1** is the clock configuration register for running operation, **SCGC1** for Sleep operation, and **DCGC1** for Deep-Sleep operation. Setting the ACG bit in the **Run-Mode Clock Configuration (RCC)** register specifies that the system uses sleep modes.

Sleep Mode Clock Gating Control Register 1 (SCGC1)

Base 0x400F.E000
Offset 0x114
Type R/W, reset 0x00000000



Bit/Field	Name	Type	Reset	Description
31:26	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
25	COMP1	R/W	0	Analog Comparator 1 Clock Gating This bit controls the clock gating for analog comparator 1. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
24	COMP0	R/W	0	Analog Comparator 0 Clock Gating This bit controls the clock gating for analog comparator 0. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
23:19	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
18	TIMER2	R/W	0	Timer 2 Clock Gating Control This bit controls the clock gating for General-Purpose Timer module 2. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.

Bit/Field	Name	Type	Reset	Description
17	TIMER1	R/W	0	Timer 1 Clock Gating Control This bit controls the clock gating for General-Purpose Timer module 1. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
16	TIMER0	R/W	0	Timer 0 Clock Gating Control This bit controls the clock gating for General-Purpose Timer module 0. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
15:13	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
12	I2C0	R/W	0	I2C0 Clock Gating Control This bit controls the clock gating for I2C module 0. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
11:5	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
4	SSI0	R/W	0	SSI0 Clock Gating Control This bit controls the clock gating for SSI module 0. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
3:2	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1	UART1	R/W	0	UART1 Clock Gating Control This bit controls the clock gating for UART module 1. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
0	UART0	R/W	0	UART0 Clock Gating Control This bit controls the clock gating for UART module 0. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.

Register 23: Deep Sleep Mode Clock Gating Control Register 1 (DCGC1), offset 0x124

This register controls the clock gating logic. Each bit controls a clock enable for a given interface, function, or unit. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled (saving power). If the unit is unclocked, reads or writes to the unit will generate a bus fault. The reset state of these bits is 0 (unclocked) unless otherwise noted, so that all functional units are disabled. It is the responsibility of software to enable the ports necessary for the application. Note that these registers may contain more bits than there are interfaces, functions, or units to control. This is to assure reasonable code compatibility with other family and future parts. **RCGC1** is the clock configuration register for running operation, **SCGC1** for Sleep operation, and **DCGC1** for Deep-Sleep operation. Setting the ACG bit in the **Run-Mode Clock Configuration (RCC)** register specifies that the system uses sleep modes.

Deep Sleep Mode Clock Gating Control Register 1 (DCGC1)

Base 0x400F.E000
Offset 0x124
Type R/W, reset 0x00000000

	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	reserved						COMP1	COMP0	reserved					TIMER2	TIMER1	TIMER0
Type	RO	RO	RO	RO	RO	RO	R/W	R/W	RO	RO	RO	RO	RO	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	reserved			I2C0	reserved							SSI0	reserved		UART1	UART0
Type	RO	RO	RO	R/W	RO	RO	RO	RO	RO	RO	RO	R/W	RO	RO	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit/Field	Name	Type	Reset	Description
31:26	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
25	COMP1	R/W	0	Analog Comparator 1 Clock Gating This bit controls the clock gating for analog comparator 1. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
24	COMP0	R/W	0	Analog Comparator 0 Clock Gating This bit controls the clock gating for analog comparator 0. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
23:19	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
18	TIMER2	R/W	0	Timer 2 Clock Gating Control This bit controls the clock gating for General-Purpose Timer module 2. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.

Bit/Field	Name	Type	Reset	Description
17	TIMER1	R/W	0	Timer 1 Clock Gating Control This bit controls the clock gating for General-Purpose Timer module 1. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
16	TIMER0	R/W	0	Timer 0 Clock Gating Control This bit controls the clock gating for General-Purpose Timer module 0. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
15:13	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
12	I2C0	R/W	0	I2C0 Clock Gating Control This bit controls the clock gating for I2C module 0. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
11:5	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
4	SSI0	R/W	0	SSI0 Clock Gating Control This bit controls the clock gating for SSI module 0. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
3:2	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1	UART1	R/W	0	UART1 Clock Gating Control This bit controls the clock gating for UART module 1. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
0	UART0	R/W	0	UART0 Clock Gating Control This bit controls the clock gating for UART module 0. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.

Register 24: Run Mode Clock Gating Control Register 2 (RCGC2), offset 0x108

This register controls the clock gating logic. Each bit controls a clock enable for a given interface, function, or unit. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled (saving power). If the unit is unclocked, reads or writes to the unit will generate a bus fault. The reset state of these bits is 0 (unclocked) unless otherwise noted, so that all functional units are disabled. It is the responsibility of software to enable the ports necessary for the application. Note that these registers may contain more bits than there are interfaces, functions, or units to control. This is to assure reasonable code compatibility with other family and future parts. **RCGC2** is the clock configuration register for running operation, **SCGC2** for Sleep operation, and **DCGC2** for Deep-Sleep operation. Setting the ACG bit in the **Run-Mode Clock Configuration (RCC)** register specifies that the system uses sleep modes.

Run Mode Clock Gating Control Register 2 (RCGC2)

Base 0x400F.E000
Offset 0x108
Type R/W, reset 0x00000000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
reserved	EPHY0	reserved	EMAC0	reserved											
RO	R/W	RO	R/W	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
reserved									GPIOG	GPIOF	GPIOE	GPIOD	GPIOC	GPIOB	GPIOA
RO	RO	RO	RO	RO	RO	RO	RO	RO	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit/Field	Name	Type	Reset	Description
31	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
30	EPHY0	R/W	0	PHY0 Clock Gating Control This bit controls the clock gating for Ethernet PHY unit 0. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
29	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
28	EMAC0	R/W	0	MAC0 Clock Gating Control This bit controls the clock gating for Ethernet MAC unit 0. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
27:7	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
6	GPIOG	R/W	0	Port G Clock Gating Control This bit controls the clock gating for Port G. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.

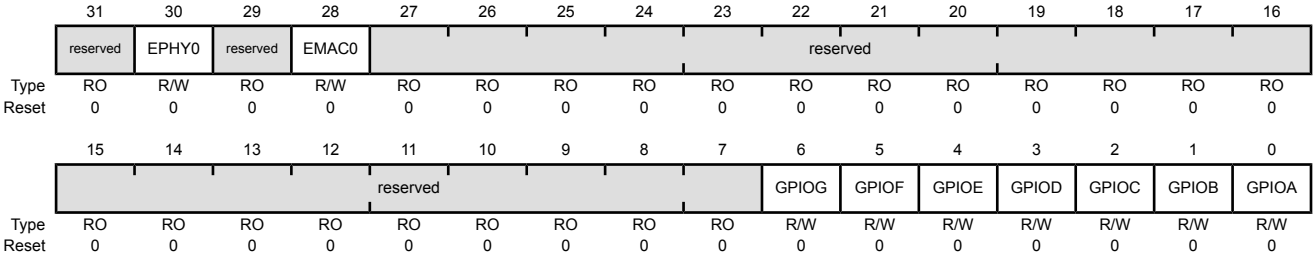
Bit/Field	Name	Type	Reset	Description
5	GPIOF	R/W	0	Port F Clock Gating Control This bit controls the clock gating for Port F. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
4	GPIOE	R/W	0	Port E Clock Gating Control This bit controls the clock gating for Port E. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
3	GPIOD	R/W	0	Port D Clock Gating Control This bit controls the clock gating for Port D. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
2	GPIOC	R/W	0	Port C Clock Gating Control This bit controls the clock gating for Port C. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
1	GPIOB	R/W	0	Port B Clock Gating Control This bit controls the clock gating for Port B. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
0	GPIOA	R/W	0	Port A Clock Gating Control This bit controls the clock gating for Port A. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.

Register 25: Sleep Mode Clock Gating Control Register 2 (SCGC2), offset 0x118

This register controls the clock gating logic. Each bit controls a clock enable for a given interface, function, or unit. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled (saving power). If the unit is unclocked, reads or writes to the unit will generate a bus fault. The reset state of these bits is 0 (unclocked) unless otherwise noted, so that all functional units are disabled. It is the responsibility of software to enable the ports necessary for the application. Note that these registers may contain more bits than there are interfaces, functions, or units to control. This is to assure reasonable code compatibility with other family and future parts. **RCGC2** is the clock configuration register for running operation, **SCGC2** for Sleep operation, and **DCGC2** for Deep-Sleep operation. Setting the ACG bit in the **Run-Mode Clock Configuration (RCC)** register specifies that the system uses sleep modes.

Sleep Mode Clock Gating Control Register 2 (SCGC2)

Base 0x400F.E000
Offset 0x118
Type R/W, reset 0x00000000



Bit/Field	Name	Type	Reset	Description
31	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
30	EPHY0	R/W	0	PHY0 Clock Gating Control This bit controls the clock gating for Ethernet PHY unit 0. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
29	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
28	EMAC0	R/W	0	MAC0 Clock Gating Control This bit controls the clock gating for Ethernet MAC unit 0. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
27:7	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Bit/Field	Name	Type	Reset	Description
6	GPIOG	R/W	0	Port G Clock Gating Control This bit controls the clock gating for Port G. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
5	GPIOF	R/W	0	Port F Clock Gating Control This bit controls the clock gating for Port F. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
4	GPIOE	R/W	0	Port E Clock Gating Control This bit controls the clock gating for Port E. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
3	GPIOD	R/W	0	Port D Clock Gating Control This bit controls the clock gating for Port D. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
2	GPIOC	R/W	0	Port C Clock Gating Control This bit controls the clock gating for Port C. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
1	GPIOB	R/W	0	Port B Clock Gating Control This bit controls the clock gating for Port B. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
0	GPIOA	R/W	0	Port A Clock Gating Control This bit controls the clock gating for Port A. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.

Register 26: Deep Sleep Mode Clock Gating Control Register 2 (DCGC2), offset 0x128

This register controls the clock gating logic. Each bit controls a clock enable for a given interface, function, or unit. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled (saving power). If the unit is unclocked, reads or writes to the unit will generate a bus fault. The reset state of these bits is 0 (unclocked) unless otherwise noted, so that all functional units are disabled. It is the responsibility of software to enable the ports necessary for the application. Note that these registers may contain more bits than there are interfaces, functions, or units to control. This is to assure reasonable code compatibility with other family and future parts. **RCGC2** is the clock configuration register for running operation, **SCGC2** for Sleep operation, and **DCGC2** for Deep-Sleep operation. Setting the ACG bit in the **Run-Mode Clock Configuration (RCC)** register specifies that the system uses sleep modes.

Deep Sleep Mode Clock Gating Control Register 2 (DCGC2)

Base 0x400F.E000
Offset 0x128
Type R/W, reset 0x00000000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
reserved	EPHY0	reserved	EMAC0	reserved											
Type RO	R/W	RO	R/W	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO
Reset 0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
reserved									GPIOG	GPIOF	GPIOE	GPIOD	GPIOC	GPIOB	GPIOA
Type RO	RO	RO	RO	RO	RO	RO	RO	RO	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset 0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit/Field	Name	Type	Reset	Description
31	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
30	EPHY0	R/W	0	PHY0 Clock Gating Control This bit controls the clock gating for Ethernet PHY unit 0. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
29	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
28	EMAC0	R/W	0	MAC0 Clock Gating Control This bit controls the clock gating for Ethernet MAC unit 0. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
27:7	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

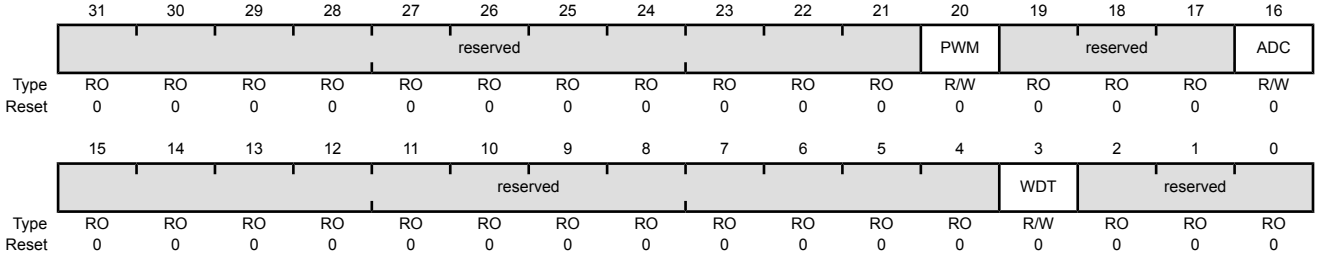
Bit/Field	Name	Type	Reset	Description
6	GPIOG	R/W	0	Port G Clock Gating Control This bit controls the clock gating for Port G. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
5	GPIOF	R/W	0	Port F Clock Gating Control This bit controls the clock gating for Port F. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
4	GPIOE	R/W	0	Port E Clock Gating Control This bit controls the clock gating for Port E. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
3	GPIOD	R/W	0	Port D Clock Gating Control This bit controls the clock gating for Port D. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
2	GPIOC	R/W	0	Port C Clock Gating Control This bit controls the clock gating for Port C. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
1	GPIOB	R/W	0	Port B Clock Gating Control This bit controls the clock gating for Port B. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.
0	GPIOA	R/W	0	Port A Clock Gating Control This bit controls the clock gating for Port A. If set, the unit receives a clock and functions. Otherwise, the unit is unclocked and disabled. If the unit is unclocked, reads or writes to the unit will generate a bus fault.

Register 27: Software Reset Control 0 (SRCR0), offset 0x040

Writes to this register are masked by the bits in the **Device Capabilities 1 (DC1)** register.

Software Reset Control 0 (SRCR0)

Base 0x400F.E000
Offset 0x040
Type R/W, reset 0x00000000



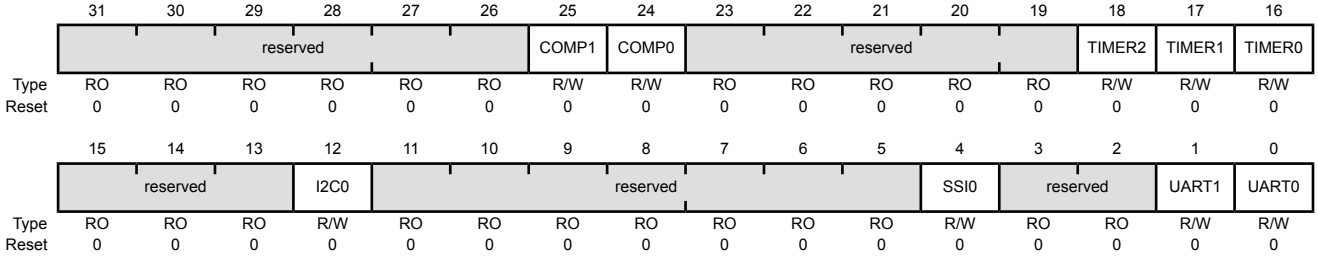
Bit/Field	Name	Type	Reset	Description
31:21	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
20	PWM	R/W	0	PWM Reset Control Reset control for PWM module.
19:17	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
16	ADC	R/W	0	ADC0 Reset Control Reset control for SAR ADC module 0.
15:4	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
3	WDT	R/W	0	WDT Reset Control Reset control for Watchdog unit.
2:0	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Register 28: Software Reset Control 1 (SRCR1), offset 0x044

Writes to this register are masked by the bits in the **Device Capabilities 2 (DC2)** register.

Software Reset Control 1 (SRCR1)

Base 0x400F.E000
Offset 0x044
Type R/W, reset 0x00000000



Bit/Field	Name	Type	Reset	Description
31:26	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
25	COMP1	R/W	0	Analog Comp 1 Reset Control Reset control for analog comparator 1.
24	COMP0	R/W	0	Analog Comp 0 Reset Control Reset control for analog comparator 0.
23:19	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
18	TIMER2	R/W	0	Timer 2 Reset Control Reset control for General-Purpose Timer module 2.
17	TIMER1	R/W	0	Timer 1 Reset Control Reset control for General-Purpose Timer module 1.
16	TIMER0	R/W	0	Timer 0 Reset Control Reset control for General-Purpose Timer module 0.
15:13	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
12	I2C0	R/W	0	I2C0 Reset Control Reset control for I2C unit 0.
11:5	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
4	SSI0	R/W	0	SSI0 Reset Control Reset control for SSI unit 0.

Bit/Field	Name	Type	Reset	Description
3:2	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1	UART1	R/W	0	UART1 Reset Control Reset control for UART unit 1.
0	UART0	R/W	0	UART0 Reset Control Reset control for UART unit 0.

Register 29: Software Reset Control 2 (SRCR2), offset 0x048

Writes to this register are masked by the bits in the **Device Capabilities 4 (DC4)** register.

Software Reset Control 2 (SRCR2)

Base 0x400F.E000
Offset 0x048
Type R/W, reset 0x00000000

	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	reserved	EPHY0	reserved	EMAC0	reserved											
Type	RO	R/W	RO	R/W	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	reserved									GPIOG	GPIOF	GPIOE	GPIOD	GPIOC	GPIOB	GPIOA
Type	RO	RO	RO	RO	RO	RO	RO	RO	RO	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit/Field	Name	Type	Reset	Description
31	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
30	EPHY0	R/W	0	PHY0 Reset Control Reset control for Ethernet PHY unit 0.
29	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
28	EMAC0	R/W	0	MAC0 Reset Control Reset control for Ethernet MAC unit 0.
27:7	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
6	GPIOG	R/W	0	Port G Reset Control Reset control for GPIO Port G.
5	GPIOF	R/W	0	Port F Reset Control Reset control for GPIO Port F.
4	GPIOE	R/W	0	Port E Reset Control Reset control for GPIO Port E.
3	GPIOD	R/W	0	Port D Reset Control Reset control for GPIO Port D.
2	GPIOC	R/W	0	Port C Reset Control Reset control for GPIO Port C.
1	GPIOB	R/W	0	Port B Reset Control Reset control for GPIO Port B.

Bit/Field	Name	Type	Reset	Description
0	GPIOA	R/W	0	Port A Reset Control Reset control for GPIO Port A.

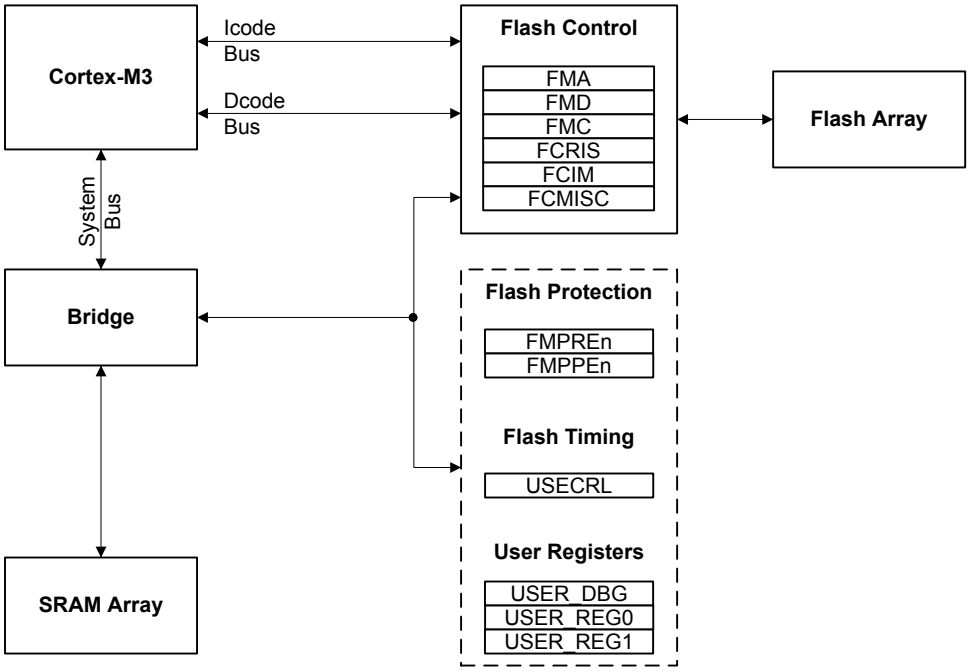
7 Internal Memory

The LM3S6432 microcontroller comes with 32 KB of bit-banded SRAM and 96 KB of flash memory. The flash controller provides a user-friendly interface, making flash programming a simple task. Flash protection can be applied to the flash memory on a 2-KB block basis.

7.1 Block Diagram

Figure 7-1 on page 126 illustrates the Flash functions. The dashed boxes in the figure indicate registers residing in the System Control module rather than the Flash Control module.

Figure 7-1. Flash Block Diagram



7.2 Functional Description

This section describes the functionality of the SRAM and Flash memories.

7.2.1 SRAM Memory

The internal SRAM of the Stellaris® devices is located at address 0x2000.0000 of the device memory map. To reduce the number of time consuming read-modify-write (RMW) operations, ARM has introduced *bit-banding* technology in the Cortex-M3 processor. With a bit-band-enabled processor, certain regions in the memory map (SRAM and peripheral space) can use address aliases to access individual bits in a single, atomic operation.

The bit-band alias is calculated by using the formula:

```
bit-band alias = bit-band base + (byte offset * 32) + (bit number * 4)
```

For example, if bit 3 at address 0x2000.1000 is to be modified, the bit-band alias is calculated as:

```
0x2200.0000 + (0x1000 * 32) + (3 * 4) = 0x2202.000C
```

With the alias address calculated, an instruction performing a read/write to address 0x2202.000C allows direct access to only bit 3 of the byte at address 0x2000.1000.

For details about bit-banding, please refer to Chapter 4, “Memory Map” in the *ARM® Cortex™-M3 Technical Reference Manual*.

7.2.2 Flash Memory

The flash is organized as a set of 1-KB blocks that can be individually erased. Erasing a block causes the entire contents of the block to be reset to all 1s. An individual 32-bit word can be programmed to change bits that are currently 1 to a 0. These blocks are paired into a set of 2-KB blocks that can be individually protected. The protection allows blocks to be marked as read-only or execute-only, providing different levels of code protection. Read-only blocks cannot be erased or programmed, protecting the contents of those blocks from being modified. Execute-only blocks cannot be erased or programmed, and can only be read by the controller instruction fetch mechanism, protecting the contents of those blocks from being read by either the controller or by a debugger.

See also “Serial Flash Loader” on page 543 for a preprogrammed flash-resident utility used to download code to the flash memory of a device without the use of a debug interface.

7.2.2.1 Flash Memory Timing

The timing for the flash is automatically handled by the flash controller. However, in order to do so, it must know the clock rate of the system in order to time its internal signals properly. The number of clock cycles per microsecond must be provided to the flash controller for it to accomplish this timing. It is software's responsibility to keep the flash controller updated with this information via the **USec Reload (USECRL)** register.

On reset, the **USECRL** register is loaded with a value that configures the flash timing so that it works with the maximum clock rate of the part. If software changes the system operating frequency, the new operating frequency minus 1 (in MHz) must be loaded into **USECRL** before any flash modifications are attempted. For example, if the device is operating at a speed of 20 MHz, a value of 0x13 (20-1) must be written to the **USECRL** register.

7.2.2.2 Flash Memory Protection

The user is provided two forms of flash protection per 2-KB flash blocks in two pairs of 32-bit wide registers. The protection policy for each form is controlled by individual bits (per policy per block) in the **FMPPEn** and **FMPREn** registers.

- **Flash Memory Protection Program Enable (FMPPEn)**: If set, the block may be programmed (written) or erased. If cleared, the block may not be changed.
- **Flash Memory Protection Read Enable (FMPREn)**: If a bit is set, the corresponding block may be executed or read by software or debuggers. If a bit is cleared, the corresponding block may only be executed, and contents of the memory block are prohibited from being read as data.

The policies may be combined as shown in Table 7-1 on page 127.

Table 7-1. Flash Protection Policy Combinations

FMPPEn	FMPREn	Protection
0	0	Execute-only protection. The block may only be executed and may not be written or erased. This mode is used to protect code.
1	0	The block may be written, erased or executed, but not read. This combination is unlikely to be used.

Table 7-1. Flash Protection Policy Combinations *(continued)*

FMPPEn	FMPREn	Protection
0	1	Read-only protection. The block may be read or executed but may not be written or erased. This mode is used to lock the block from further modification while allowing any read or execute access.
1	1	No protection. The block may be written, erased, executed or read.

A Flash memory access that attempts to read a read-protected block (**FMPREn** bit is set) is prohibited and generates a bus fault. A Flash memory access that attempts to program or erase a program-protected block (**FMPPEn** bit is set) is prohibited and can optionally generate an interrupt (by setting the AMASK bit in the **Flash Controller Interrupt Mask (FCIM)** register) to alert software developers of poorly behaving software during the development and debug phases.

The factory settings for the **FMPREn** and **FMPPEn** registers are a value of 1 for all implemented banks. These settings create a policy of open access and programmability. The register bits may be changed by clearing the specific register bit. The changes are not permanent until the register is committed (saved), at which point the bit change is permanent. If a bit is changed from a 1 to a 0 and not committed, it may be restored by executing a power-on reset sequence. The changes are committed using the **Flash Memory Control (FMC)** register. Details on programming these bits are discussed in “Nonvolatile Register Programming” on page 129.

7.2.2.3 Interrupts

The Flash memory controller can generate interrupts when the following conditions are observed:

- Programming Interrupt - signals when a program or erase action is complete.
- Access Interrupt - signals when a program or erase action has been attempted on a 2-kB block of memory that is protected by its corresponding **FMPPEn** bit.

The interrupt events that can trigger a controller-level interrupt are defined in the **Flash Controller Masked Interrupt Status (FCMIS)** register (see page 136) by setting the corresponding MASK bits. If interrupts are not used, the raw interrupt status is always visible via the **Flash Controller Raw Interrupt Status (FCRIS)** register (see page 135).

Interrupts are always cleared (for both the **FCMIS** and **FCRIS** registers) by writing a 1 to the corresponding bit in the **Flash Controller Masked Interrupt Status and Clear (FCMISC)** register (see page 137).

7.3 Flash Memory Initialization and Configuration

7.3.1 Flash Programming

The Stellaris® devices provide a user-friendly interface for flash programming. All erase/program operations are handled via three registers: **FMA**, **FMD**, and **FMC**.

7.3.1.1 To program a 32-bit word

1. Write source data to the **FMD** register.
2. Write the target address to the **FMA** register.
3. Write the flash write key and the WRITE bit (a value of 0xA442.0001) to the **FMC** register.
4. Poll the **FMC** register until the WRITE bit is cleared.

7.3.1.2 To perform an erase of a 1-KB page

1. Write the page address to the **FMA** register.
2. Write the flash write key and the ERASE bit (a value of 0xA442.0002) to the **FMC** register.
3. Poll the **FMC** register until the ERASE bit is cleared.

7.3.1.3 To perform a mass erase of the flash

1. Write the flash write key and the MERASE bit (a value of 0xA442.0004) to the **FMC** register.
2. Poll the **FMC** register until the MERASE bit is cleared.

7.3.2 Nonvolatile Register Programming

This section discusses how to update registers that are resident within the Flash memory itself. These registers exist in a separate space from the main Flash memory array and are not affected by an ERASE or MASS ERASE operation. The bits in these registers can be changed from 1 to 0 with a write operation. Prior to being committed, the register contents are unaffected by any reset condition except power-on reset, which returns the register contents to the original value. By committing the register values using the COMT bit in the **FMC** register, the register contents become nonvolatile and are therefore retained following power cycling. Once the register contents are committed, the contents are permanent, and they cannot be restored to their factory default values.

With the exception of the **USER_DBG** register, the settings in these registers can be tested before committing them to Flash memory. For the **USER_DBG** register, the data to be written is loaded into the **FMD** register before it is committed. The **FMD** register is read only and does not allow the **USER_DBG** operation to be tried before committing it to nonvolatile memory.

Important: These registers can only have bits changed from 1 to 0 by user programming. Once committed, these registers cannot be restored to their factory default values.

In addition, the **USER_REG0**, **USER_REG1**, **USER_REG2**, **USER_REG3**, and **USER_DBG** registers each use bit 31 (NW) to indicate that they have not been committed and bits in the register may be changed from 1 to 0. These five registers can only be committed once whereas the Flash memory protection registers may be committed multiple times. Table 7-2 on page 129 provides the **FMA** address required for commitment of each of the registers and the source of the data to be written when the **FMC** register is written with a value of 0xA442.0008. After writing the COMT bit, the user may poll the **FMC** register to wait for the commit operation to complete.

Table 7-2. User-Programmable Flash Memory Resident Registers

Register to be Committed	FMA Value	Data Source
FMPRE0	0x0000.0000	FMPRE0
FMPRE1	0x0000.0002	FMPRE1
FMPPE0	0x0000.0001	FMPPE0
FMPPE1	0x0000.0003	FMPPE1
USER_REG0	0x8000.0000	USER_REG0
USER_REG1	0x8000.0001	USER_REG1
USER_REG2	0x8000.0002	USER_REG2
USER_REG3	0x8000.0003	USER_REG3
USER_DBG	0x7510.0000	FMD

7.4 Register Map

Table 7-3 on page 130 lists the Flash memory and control registers. The offset listed is a hexadecimal increment to the register's address. The **FMA**, **FMD**, **FMC**, **FCRIS**, **FCIM**, and **FCMISC** register offsets are relative to the Flash memory control base address of 0x400F.D000. The Flash memory protection register offsets are relative to the System Control base address of 0x400F.E000.

Table 7-3. Flash Register Map

Offset	Name	Type	Reset	Description	See page
Flash Memory Control Registers (Flash Control Offset)					
0x000	FMA	R/W	0x0000.0000	Flash Memory Address	131
0x004	FMD	R/W	0x0000.0000	Flash Memory Data	132
0x008	FMC	R/W	0x0000.0000	Flash Memory Control	133
0x00C	FCRIS	RO	0x0000.0000	Flash Controller Raw Interrupt Status	135
0x010	FCIM	R/W	0x0000.0000	Flash Controller Interrupt Mask	136
0x014	FCMISC	R/W1C	0x0000.0000	Flash Controller Masked Interrupt Status and Clear	137
Flash Memory Protection Registers (System Control Offset)					
0x130	FMPRE0	R/W	0xFFFF.FFFF	Flash Memory Protection Read Enable 0	140
0x200	FMPRE0	R/W	0xFFFF.FFFF	Flash Memory Protection Read Enable 0	140
0x134	FMPPE0	R/W	0xFFFF.FFFF	Flash Memory Protection Program Enable 0	141
0x400	FMPPE0	R/W	0xFFFF.FFFF	Flash Memory Protection Program Enable 0	141
0x140	USECRL	R/W	0x31	USec Reload	139
0x1D0	USER_DBG	R/W	0xFFFF.FFFE	User Debug	142
0x1E0	USER_REG0	R/W	0xFFFF.FFFF	User Register 0	143
0x1E4	USER_REG1	R/W	0xFFFF.FFFF	User Register 1	144
0x204	FMPRE1	R/W	0x0000.FFFF	Flash Memory Protection Read Enable 1	145
0x208	FMPRE2	R/W	0x0000.0000	Flash Memory Protection Read Enable 2	146
0x20C	FMPRE3	R/W	0x0000.0000	Flash Memory Protection Read Enable 3	147
0x404	FMPPE1	R/W	0x0000.FFFF	Flash Memory Protection Program Enable 1	148
0x408	FMPPE2	R/W	0x0000.0000	Flash Memory Protection Program Enable 2	149
0x40C	FMPPE3	R/W	0x0000.0000	Flash Memory Protection Program Enable 3	150

7.5 Flash Register Descriptions (Flash Control Offset)

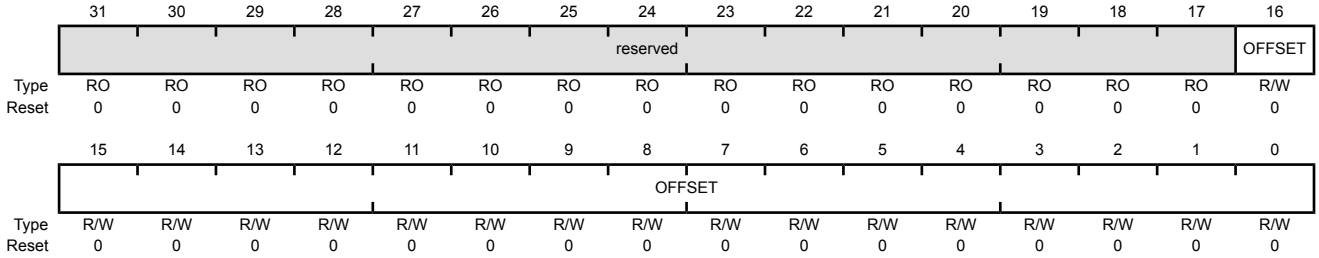
This section lists and describes the Flash Memory registers, in numerical order by address offset. Registers in this section are relative to the Flash control base address of 0x400F.D000.

Register 1: Flash Memory Address (FMA), offset 0x000

During a write operation, this register contains a 4-byte-aligned address and specifies where the data is written. During erase operations, this register contains a 1 KB-aligned address and specifies which page is erased. Note that the alignment requirements must be met by software or the results of the operation are unpredictable.

Flash Memory Address (FMA)

Base 0x400F.D000
Offset 0x000
Type R/W, reset 0x0000.0000



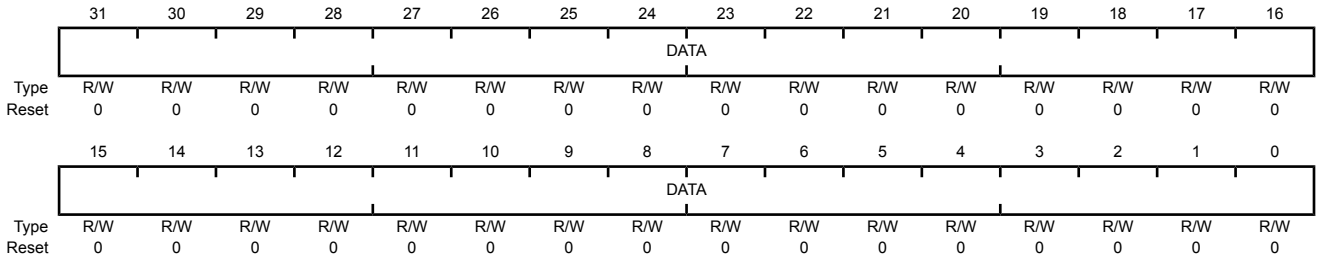
Bit/Field	Name	Type	Reset	Description
31:17	reserved	RO	0x0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
16:0	OFFSET	R/W	0x0	Address Offset Address offset in flash where operation is performed, except for nonvolatile registers (see “Nonvolatile Register Programming” on page 129 for details on values for this field).

Register 2: Flash Memory Data (FMD), offset 0x004

This register contains the data to be written during the programming cycle or read during the read cycle. Note that the contents of this register are undefined for a read access of an execute-only block. This register is not used during the erase cycles.

Flash Memory Data (FMD)

Base 0x400F.D000
Offset 0x004
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:0	DATA	R/W	0x0	Data Value Data value for write operation.

Register 3: Flash Memory Control (FMC), offset 0x008

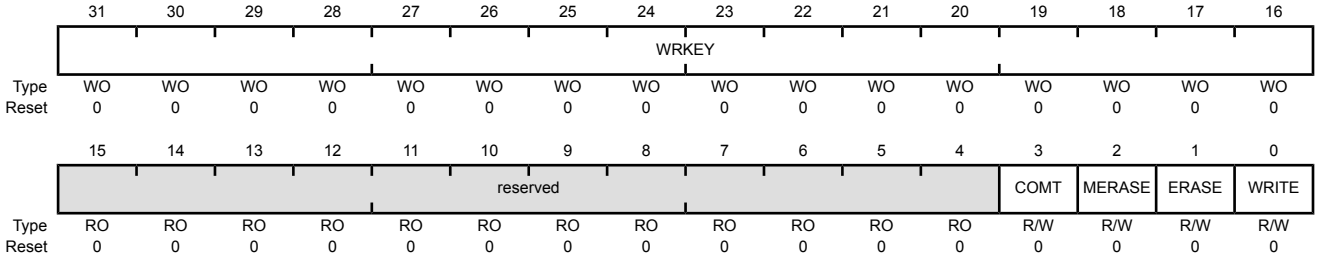
When this register is written, the flash controller initiates the appropriate access cycle for the location specified by the **Flash Memory Address (FMA)** register (see page 131). If the access is a write access, the data contained in the **Flash Memory Data (FMD)** register (see page 132) is written.

This is the final register written and initiates the memory operation. There are four control bits in the lower byte of this register that, when set, initiate the memory operation. The most used of these register bits are the ERASE and WRITE bits.

It is a programming error to write multiple control bits and the results of such an operation are unpredictable.

Flash Memory Control (FMC)

Base 0x400F.D000
Offset 0x008
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:16	WRKEY	WO	0x0	Flash Write Key This field contains a write key, which is used to minimize the incidence of accidental flash writes. The value 0xA442 must be written into this field for a write to occur. Writes to the **FMC** register without this WRKEY value are ignored. A read of this field returns the value 0.
15:4	reserved	RO	0x0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
3	COMT	R/W	0	Commit Register Value Commit (write) of register value to nonvolatile storage. A write of 0 has no effect on the state of this bit. If read, the state of the previous commit access is provided. If the previous commit access is complete, a 0 is returned; otherwise, if the commit access is not complete, a 1 is returned. This can take up to 50 µs.
2	MERASE	R/W	0	Mass Erase Flash Memory If this bit is set, the flash main memory of the device is all erased. A write of 0 has no effect on the state of this bit. If read, the state of the previous mass erase access is provided. If the previous mass erase access is complete, a 0 is returned; otherwise, if the previous mass erase access is not complete, a 1 is returned. This can take up to 250 ms.

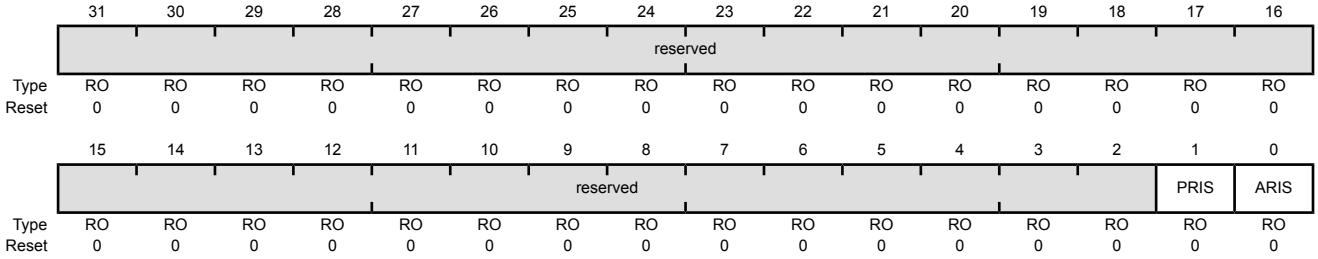
Bit/Field	Name	Type	Reset	Description
1	ERASE	R/W	0	Erase a Page of Flash Memory If this bit is set, the page of flash main memory as specified by the contents of **FMA** is erased. A write of 0 has no effect on the state of this bit. If read, the state of the previous erase access is provided. If the previous erase access is complete, a 0 is returned; otherwise, if the previous erase access is not complete, a 1 is returned. This can take up to 25 ms.
0	WRITE	R/W	0	Write a Word into Flash Memory If this bit is set, the data stored in **FMD** is written into the location as specified by the contents of **FMA**. A write of 0 has no effect on the state of this bit. If read, the state of the previous write update is provided. If the previous write access is complete, a 0 is returned; otherwise, if the write access is not complete, a 1 is returned. This can take up to 50 µs.

Register 4: Flash Controller Raw Interrupt Status (FCRIS), offset 0x00C

This register indicates that the flash controller has an interrupt condition. An interrupt is only signaled if the corresponding **FCIM** register bit is set.

Flash Controller Raw Interrupt Status (FCRIS)

Base 0x400F.D000
Offset 0x00C
Type RO, reset 0x0000.0000



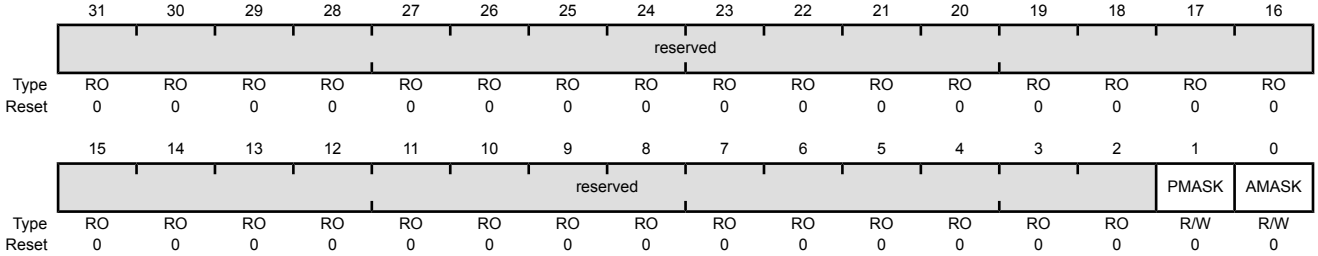
Bit/Field	Name	Type	Reset	Description
31:2	reserved	RO	0x0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1	PRIS	RO	0	Programming Raw Interrupt Status This bit provides status on programming cycles which are write or erase actions generated through the **FMC** register bits (see page 133). Value Description 1 The programming cycle has completed. 0 The programming cycle has not completed. This status is sent to the interrupt controller when the PMASK bit in the **FCIM** register is set. This bit is cleared by writing a 1 to the PMISC bit in the **FCMISC** register.
0	ARIS	RO	0	Access Raw Interrupt Status Value Description 1 A program or erase action was attempted on a block of Flash memory that contradicts the protection policy for that block as set in the **FMPPEn** registers. 0 No access has tried to improperly program or erase the Flash memory. This status is sent to the interrupt controller when the AMASK bit in the **FCIM** register is set. This bit is cleared by writing a 1 to the AMISC bit in the **FCMISC** register.

Register 5: Flash Controller Interrupt Mask (FCIM), offset 0x010

This register controls whether the flash controller generates interrupts to the controller.

Flash Controller Interrupt Mask (FCIM)

Base 0x400F.D000
Offset 0x010
Type R/W, reset 0x0000.0000



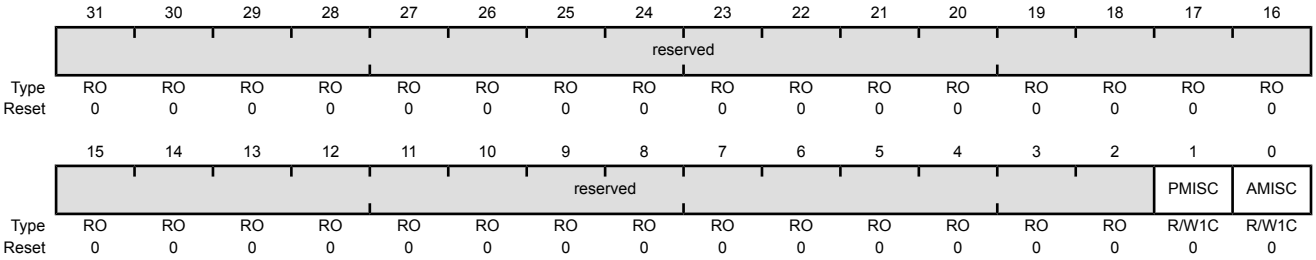
Bit/Field	Name	Type	Reset	Description
31:2	reserved	RO	0x0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1	PMASK	R/W	0	Programming Interrupt Mask This bit controls the reporting of the programming raw interrupt status to the interrupt controller. Value Description 1 An interrupt is sent to the interrupt controller when the PRIS bit is set. 0 The PRIS interrupt is suppressed and not sent to the interrupt controller.
0	AMASK	R/W	0	Access Interrupt Mask This bit controls the reporting of the access raw interrupt status to the interrupt controller. Value Description 1 An interrupt is sent to the interrupt controller when the ARIS bit is set. 0 The ARIS interrupt is suppressed and not sent to the interrupt controller.

Register 6: Flash Controller Masked Interrupt Status and Clear (FCMISC), offset 0x014

This register provides two functions. First, it reports the cause of an interrupt by indicating which interrupt source or sources are signalling the interrupt. Second, it serves as the method to clear the interrupt reporting.

Flash Controller Masked Interrupt Status and Clear (FCMISC)

Base 0x400F.D000
Offset 0x014
Type R/W1C, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:2	reserved	RO	0x0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1	PMISC	R/W1C	0	Programming Masked Interrupt Status and Clear Value Description 1 When read, a 1 indicates that an unmasked interrupt was signaled because a programming cycle completed. Writing a 1 to this bit clears `PMISC` and also the `PRIS` bit in the **FCRIS** register (see page 135). 0 When read, a 0 indicates that a programming cycle complete interrupt has not occurred. A write of 0 has no effect on the state of this bit.
0	AMISC	R/W1C	0	Access Masked Interrupt Status and Clear Value Description 1 When read, a 1 indicates that an unmasked interrupt was signaled because a program or erase action was attempted on a block of Flash memory that contradicts the protection policy for that block as set in the **FMPPEn** registers. Writing a 1 to this bit clears `AMISC` and also the `ARIS` bit in the **FCRIS** register (see page 135). 0 When read, a 0 indicates that no improper accesses have occurred. A write of 0 has no effect on the state of this bit.

7.6 Flash Register Descriptions (System Control Offset)

The remainder of this section lists and describes the Flash Memory registers, in numerical order by address offset. Registers in this section are relative to the System Control base address of 0x400F.E000.

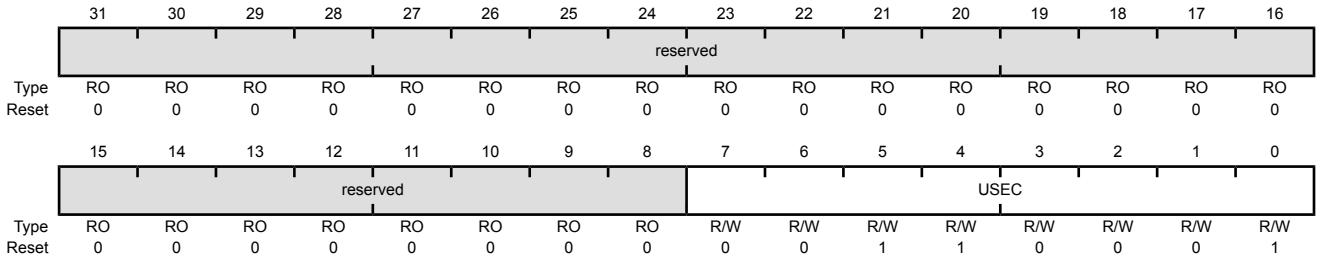
Register 7: USec Reload (USECRL), offset 0x140

Note: Offset is relative to System Control base address of 0x400F.E000

This register is provided as a means of creating a 1-µs tick divider reload value for the flash controller. The internal flash has specific minimum and maximum requirements on the length of time the high voltage write pulse can be applied. It is required that this register contain the operating frequency (in MHz -1) whenever the flash is being erased or programmed. The user is required to change this value if the clocking conditions are changed for a flash erase/program operation.

USec Reload (USECRL)

Base 0x400F.E000
Offset 0x140
Type R/W, reset 0x31



Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	USEC	R/W	0x31	Microsecond Reload Value MHz -1 of the controller clock when the flash is being erased or programmed. If the maximum system frequency is being used, USEC should be set to 0x31 (50 MHz) whenever the flash is being erased or programmed.

Register 8: Flash Memory Protection Read Enable 0 (FMPRE0), offset 0x130 and 0x200

Note: This register is aliased for backwards compatability.

Note: Offset is relative to System Control base address of 0x400FE000.

This register stores the read-only protection bits for each 2-KB flash block (**FMPPEn** stores the execute-only bits). Flash memory up to a total of 64 KB is controlled by this register. Other **FMPREn** registers (if any) provide protection for other 64K blocks. This register is loaded during the power-on reset sequence. The factory settings for the **FMPREn** and **FMPPEn** registers are a value of 1 for all implemented banks. This achieves a policy of open access and programmability. The register bits may be changed by writing the specific register bit. However, this register is R/W0; the user can only change the protection bit from a 1 to a 0 (and may NOT change a 0 to a 1). The changes are not permanent until the register is committed (saved), at which point the bit change is permanent. If a bit is changed from a 1 to a 0 and not committed, it may be restored by executing a power-on reset sequence. The reset value shown only applies to power-on reset; any other type of reset does not affect this register. For additional information, see the "Flash Memory Protection" section.

Flash Memory Protection Read Enable 0 (FMPRE0)

Base 0x400F.E000
Offset 0x130 and 0x200
Type R/W, reset 0xFFFF.FFFF

	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	READ_ENABLE															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	READ_ENABLE															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit/Field	Name	Type	Reset	Description
31:0	READ_ENABLE	R/W	0xFFFFFFFF	Flash Read Enable. Enables 2-KB Flash memory blocks to be executed or read. The policies may be combined as shown in the table “Flash Protection Policy Combinations”.

Value	Description
0xFFFFFFFF	Bits [31:0] each enable protection on a 2-KB block of Flash memory up to the total of 64 KB.

Register 9: Flash Memory Protection Program Enable 0 (FMPPE0), offset 0x134 and 0x400

Note: This register is aliased for backwards compatability.

Note: Offset is relative to System Control base address of 0x400FE000.

This register stores the execute-only protection bits for each 2-KB flash block (**FMPREn** stores the execute-only bits). Flash memory up to a total of 64 KB is controlled by this register. Other **FMPPEn** registers (if any) provide protection for other 64K blocks. This register is loaded during the power-on reset sequence. The factory settings for the **FMPREn** and **FMPPEn** registers are a value of 1 for all implemented banks. This achieves a policy of open access and programmability. The register bits may be changed by writing the specific register bit. However, this register is R/W0; the user can only change the protection bit from a 1 to a 0 (and may NOT change a 0 to a 1). The changes are not permanent until the register is committed (saved), at which point the bit change is permanent. If a bit is changed from a 1 to a 0 and not committed, it may be restored by executing a power-on reset sequence. The reset value shown only applies to power-on reset; any other type of reset does not affect this register. For additional information, see the "Flash Memory Protection" section.

Flash Memory Protection Program Enable 0 (FMPPE0)

Base 0x400F.E000
Offset 0x134 and 0x400
Type R/W, reset 0xFFFF.FFFF

	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	PROG_ENABLE															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PROG_ENABLE															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit/Field	Name	Type	Reset	Description
31:0	PROG_ENABLE	R/W	0xFFFFFFFF	Flash Programming Enable

Configures 2-KB flash blocks to be execute only. The policies may be combined as shown in the table “Flash Protection Policy Combinations”.

Value	Description
0xFFFFFFFF	Bits [31:0] each enable protection on a 2-KB block of Flash memory up to the total of 64 KB.

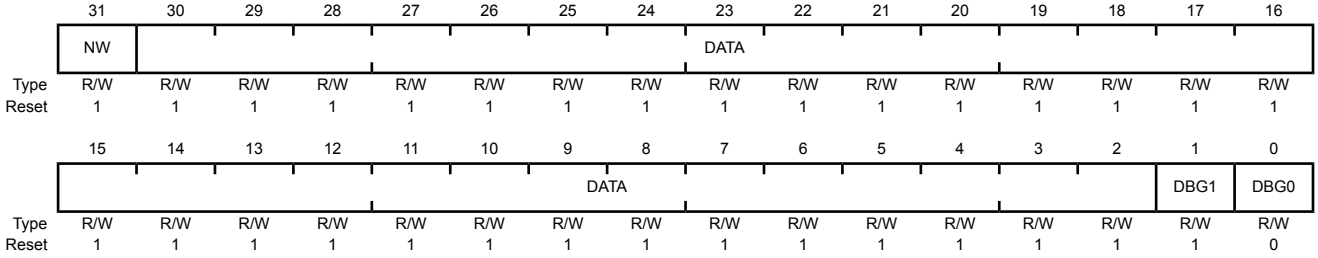
Register 10: User Debug (USER_DBG), offset 0x1D0

Note: Offset is relative to System Control base address of 0x400FE000.

This register provides a write-once mechanism to disable external debugger access to the device in addition to 27 additional bits of user-defined data. The DBG0 bit (bit 0) is set to 0 from the factory and the DBG1 bit (bit 1) is set to 1, which enables external debuggers. Changing the DBG1 bit to 0 disables any external debugger access to the device permanently, starting with the next power-up cycle of the device. The NW bit (bit 31) indicates that the register has not yet been committed and is controlled through hardware to ensure that the register is only committed once. Prior to being committed, bits can only be changed from 1 to 0. The reset value shown only applies to power-on reset; any other type of reset does not affect this register. Once committed, this register cannot be restored to the factory default value.

User Debug (USER_DBG)

Base 0x400F.E000
Offset 0x1D0
Type R/W, reset 0xFFFF.FFFE



Bit/Field	Name	Type	Reset	Description
31	NW	R/W	1	User Debug Not Written When set, this bit indicates that this 32-bit register has not been committed. When clear, this bit specifies that this register has been committed and may not be committed again.
30:2	DATA	R/W	0x1FFFFFFF	User Data Contains the user data value. This field is initialized to all 1s and can only be committed once.
1	DBG1	R/W	1	Debug Control 1 The DBG1 bit must be 1 and DBG0 must be 0 for debug to be available.
0	DBG0	R/W	0	Debug Control 0 The DBG1 bit must be 1 and DBG0 must be 0 for debug to be available.

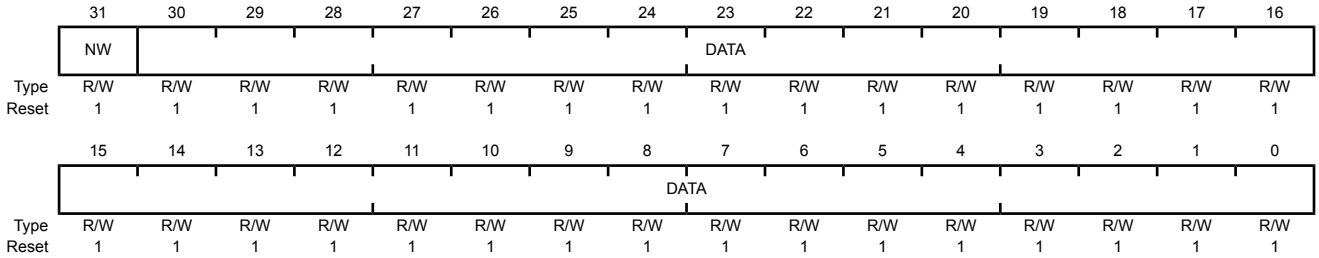
Register 11: User Register 0 (USER_REG0), offset 0x1E0

Note: Offset is relative to System Control base address of 0x400FE000.

This register provides 31 bits of user-defined data that is non-volatile and can only be committed once. Bit 31 indicates that the register is available to be committed and is controlled through hardware to ensure that the register is only committed once. Prior to being committed, bits can only be changed from 1 to 0. The reset value shown only applies to power-on reset; any other type of reset does not affect this register. The write-once characteristics of this register are useful for keeping static information like communication addresses that need to be unique per part and would otherwise require an external EEPROM or other non-volatile device. Once committed, this register cannot be restored to the factory default value.

User Register 0 (USER_REG0)

Base 0x400F.E000
Offset 0x1E0
Type R/W, reset 0xFFFF.FFFF



Bit/Field	Name	Type	Reset	Description
31	NW	R/W	1	Not Written When set, this bit indicates that this 32-bit register has not been committed. When clear, this bit specifies that this register has been committed and may not be committed again.
30:0	DATA	R/W	0x7FFFFFFF	User Data Contains the user data value. This field is initialized to all 1s and can only be committed once.

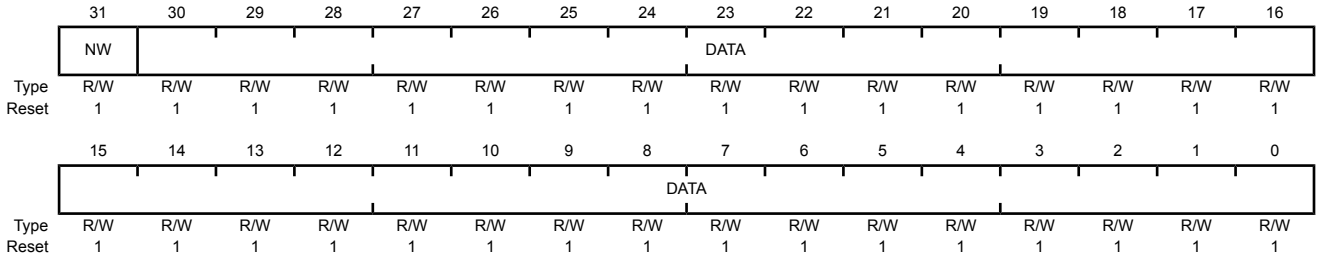
Register 12: User Register 1 (USER_REG1), offset 0x1E4

Note: Offset is relative to System Control base address of 0x400FE000.

This register provides 31 bits of user-defined data that is non-volatile and can only be committed once. Bit 31 indicates that the register is available to be committed and is controlled through hardware to ensure that the register is only committed once. Prior to being committed, bits can only be changed from 1 to 0. The reset value shown only applies to power-on reset; any other type of reset does not affect this register. The write-once characteristics of this register are useful for keeping static information like communication addresses that need to be unique per part and would otherwise require an external EEPROM or other non-volatile device. Once committed, this register cannot be restored to the factory default value.

User Register 1 (USER_REG1)

Base 0x400F.E000
Offset 0x1E4
Type R/W, reset 0xFFFF.FFFF



Bit/Field	Name	Type	Reset	Description
31	NW	R/W	1	Not Written When set, this bit indicates that this 32-bit register has not been committed. When clear, this bit specifies that this register has been committed and may not be committed again.
30:0	DATA	R/W	0x7FFFFFFF	User Data Contains the user data value. This field is initialized to all 1s and can only be committed once.

Register 13: Flash Memory Protection Read Enable 1 (FMPRE1), offset 0x204

Note: Offset is relative to System Control base address of 0x400FE000.

This register stores the read-only protection bits for each 2-KB flash block (**FMPPEn** stores the execute-only bits). Flash memory up to a total of 64 KB is controlled by this register. Other **FMPREn** registers (if any) provide protection for other 64K blocks. This register is loaded during the power-on reset sequence. The factory settings for the **FMPREn** and **FMPPEn** registers are a value of 1 for all implemented banks. This achieves a policy of open access and programmability. The register bits may be changed by writing the specific register bit. However, this register is R/W0; the user can only change the protection bit from a 1 to a 0 (and may NOT change a 0 to a 1). The changes are not permanent until the register is committed (saved), at which point the bit change is permanent. If a bit is changed from a 1 to a 0 and not committed, it may be restored by executing a power-on reset sequence. The reset value shown only applies to power-on reset; any other type of reset does not affect this register. If the Flash memory size on the device is less than 64 KB, this register usually reads as zeroes, but software should not rely on these bits to be zero. For additional information, see the "Flash Memory Protection" section.

Flash Memory Protection Read Enable 1 (FMPRE1)

Base 0x400F.E000
Offset 0x204
Type R/W, reset 0x0000.FFFF

	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	READ_ENABLE															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	READ_ENABLE															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit/Field	Name	Type	Reset	Description
31:0	READ_ENABLE	R/W	0x0000FFFF	Flash Read Enable. Enables 2-KB Flash memory blocks to be executed or read. The policies may be combined as shown in the table “Flash Protection Policy Combinations”.

Value	Description
0x0000FFFF	Bits [31:0] each enable protection on a 2-KB block of Flash memory in memory range from 65 to 128 KB.

Register 14: Flash Memory Protection Read Enable 2 (FMPRE2), offset 0x208

Note: Offset is relative to System Control base address of 0x400FE000.

This register stores the read-only protection bits for each 2-KB flash block (**FMPPEn** stores the execute-only bits). This register is loaded during the power-on reset sequence. The factory settings for the **FMPREn** and **FMPPEn** registers are a value of 1 for all implemented banks. This achieves a policy of open access and programmability. The register bits may be changed by writing the specific register bit. However, this register is R/W0; the user can only change the protection bit from a 1 to a 0 (and may NOT change a 0 to a 1). For additional information, see the "Flash Memory Protection" section.

Flash Memory Protection Read Enable 2 (FMPRE2)

Base 0x400F.E000
Offset 0x208
Type R/W, reset 0x0000.0000

	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	READ_ENABLE															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	READ_ENABLE															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit/Field	Name	Type	Reset	Description
31:0	READ_ENABLE	R/W	0x00000000	Flash Read Enable Enables 2-KB flash blocks to be executed or read. The policies may be combined as shown in the table “Flash Protection Policy Combinations”. Value Description 0x00000000 Enables 96 KB of flash.

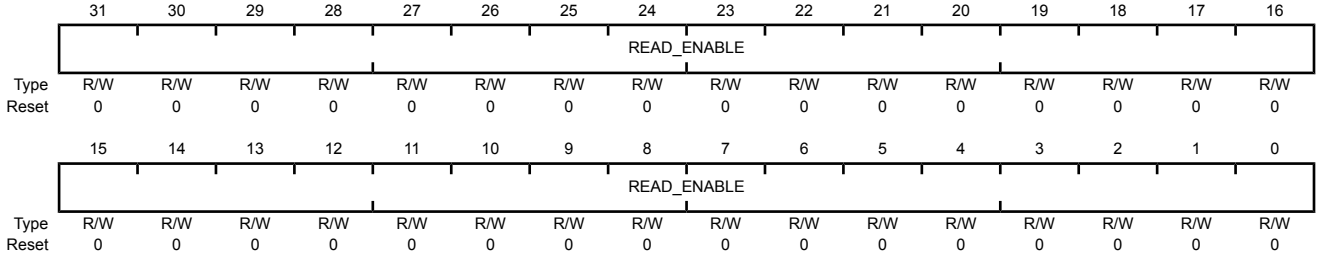
Register 15: Flash Memory Protection Read Enable 3 (FMPRE3), offset 0x20C

Note: Offset is relative to System Control base address of 0x400FE000.

This register stores the read-only protection bits for each 2-KB flash block (**FMPPEn** stores the execute-only bits). This register is loaded during the power-on reset sequence. The factory settings for the **FMPREn** and **FMPPEn** registers are a value of 1 for all implemented banks. This achieves a policy of open access and programmability. The register bits may be changed by writing the specific register bit. However, this register is R/W0; the user can only change the protection bit from a 1 to a 0 (and may NOT change a 0 to a 1). The changes are not permanent until the register is committed (saved), at which point the bit change is permanent. If a bit is changed from a 1 to a 0 and not committed, it may be restored by executing a power-on reset sequence. For additional information, see the "Flash Memory Protection" section.

Flash Memory Protection Read Enable 3 (FMPRE3)

Base 0x400F.E000
Offset 0x20C
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:0	READ_ENABLE	R/W	0x00000000	Flash Read Enable Enables 2-KB flash blocks to be executed or read. The policies may be combined as shown in the table “Flash Protection Policy Combinations”. Value Description 0x00000000 Enables 96 KB of flash.

Register 16: Flash Memory Protection Program Enable 1 (FMPPE1), offset 0x404

Note: Offset is relative to System Control base address of 0x400FE000.

This register stores the execute-only protection bits for each 2-KB flash block (**FMPREn** stores the execute-only bits). Flash memory up to a total of 64 KB is controlled by this register. Other **FMPPEn** registers (if any) provide protection for other 64K blocks. This register is loaded during the power-on reset sequence. The factory settings for the **FMPREn** and **FMPPEn** registers are a value of 1 for all implemented banks. This achieves a policy of open access and programmability. The register bits may be changed by writing the specific register bit. However, this register is R/W0; the user can only change the protection bit from a 1 to a 0 (and may NOT change a 0 to a 1). The changes are not permanent until the register is committed (saved), at which point the bit change is permanent. If a bit is changed from a 1 to a 0 and not committed, it may be restored by executing a power-on reset sequence. The reset value shown only applies to power-on reset; any other type of reset does not affect this register. If the Flash memory size on the device is less than 64 KB, this register usually reads as zeroes, but software should not rely on these bits to be zero. For additional information, see the "Flash Memory Protection" section.

Flash Memory Protection Program Enable 1 (FMPPE1)

Base 0x400F.E000
Offset 0x404
Type R/W, reset 0x0000.FFFF

	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	PROG_ENABLE															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PROG_ENABLE															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit/Field	Name	Type	Reset	Description
31:0	PROG_ENABLE	R/W	0x0000FFFF	Flash Programming Enable

Value	Description
0x0000FFFF	Bits [31:0] each enable protection on a 2-KB block of Flash memory in memory range from 65 to 128 KB.

Register 17: Flash Memory Protection Program Enable 2 (FMPPE2), offset 0x408

Note: Offset is relative to System Control base address of 0x400FE000.

This register stores the execute-only protection bits for each 2-KB flash block (**FMPREn** stores the execute-only bits). This register is loaded during the power-on reset sequence. The factory settings for the **FMPREn** and **FMPPEn** registers are a value of 1 for all implemented banks. This achieves a policy of open access and programmability. The register bits may be changed by writing the specific register bit. However, this register is R/W0; the user can only change the protection bit from a 1 to a 0 (and may NOT change a 0 to a 1). The changes are not permanent until the register is committed (saved), at which point the bit change is permanent. If a bit is changed from a 1 to a 0 and not committed, it may be restored by executing a power-on reset sequence. For additional information, see the "Flash Memory Protection" section.

Flash Memory Protection Program Enable 2 (FMPPE2)

Base 0x400F.E000
Offset 0x408
Type R/W, reset 0x0000.0000

	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	PROG_ENABLE															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PROG_ENABLE															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit/Field	Name	Type	Reset	Description
31:0	PROG_ENABLE	R/W	0x00000000	Flash Programming Enable Configures 2-KB flash blocks to be execute only. The policies may be combined as shown in the table “Flash Protection Policy Combinations”. Value Description 0x00000000 Enables 96 KB of flash.

Register 18: Flash Memory Protection Program Enable 3 (FMPPE3), offset 0x40C

Note: Offset is relative to System Control base address of 0x400FE000.

This register stores the execute-only protection bits for each 2-KB flash block (**FMPREn** stores the execute-only bits). This register is loaded during the power-on reset sequence. The factory settings for the **FMPREn** and **FMPPEn** registers are a value of 1 for all implemented banks. This achieves a policy of open access and programmability. The register bits may be changed by writing the specific register bit. However, this register is R/W0; the user can only change the protection bit from a 1 to a 0 (and may NOT change a 0 to a 1). The changes are not permanent until the register is committed (saved), at which point the bit change is permanent. If a bit is changed from a 1 to a 0 and not committed, it may be restored by executing a power-on reset sequence. For additional information, see the "Flash Memory Protection" section.

Flash Memory Protection Program Enable 3 (FMPPE3)

Base 0x400F.E000
Offset 0x40C
Type R/W, reset 0x0000.0000

	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	PROG_ENABLE															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PROG_ENABLE															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit/Field	Name	Type	Reset	Description
31:0	PROG_ENABLE	R/W	0x00000000	Flash Programming Enable Configures 2-KB flash blocks to be execute only. The policies may be combined as shown in the table “Flash Protection Policy Combinations”. Value Description 0x00000000 Enables 96 KB of flash.

8 General-Purpose Input/Outputs (GPIOs)

The GPIO module is composed of seven physical GPIO blocks, each corresponding to an individual GPIO port (Port A, Port B, Port C, Port D, Port E, Port F, Port G). The GPIO module supports 14-43 programmable input/output pins, depending on the peripherals being used.

The GPIO module has the following features:

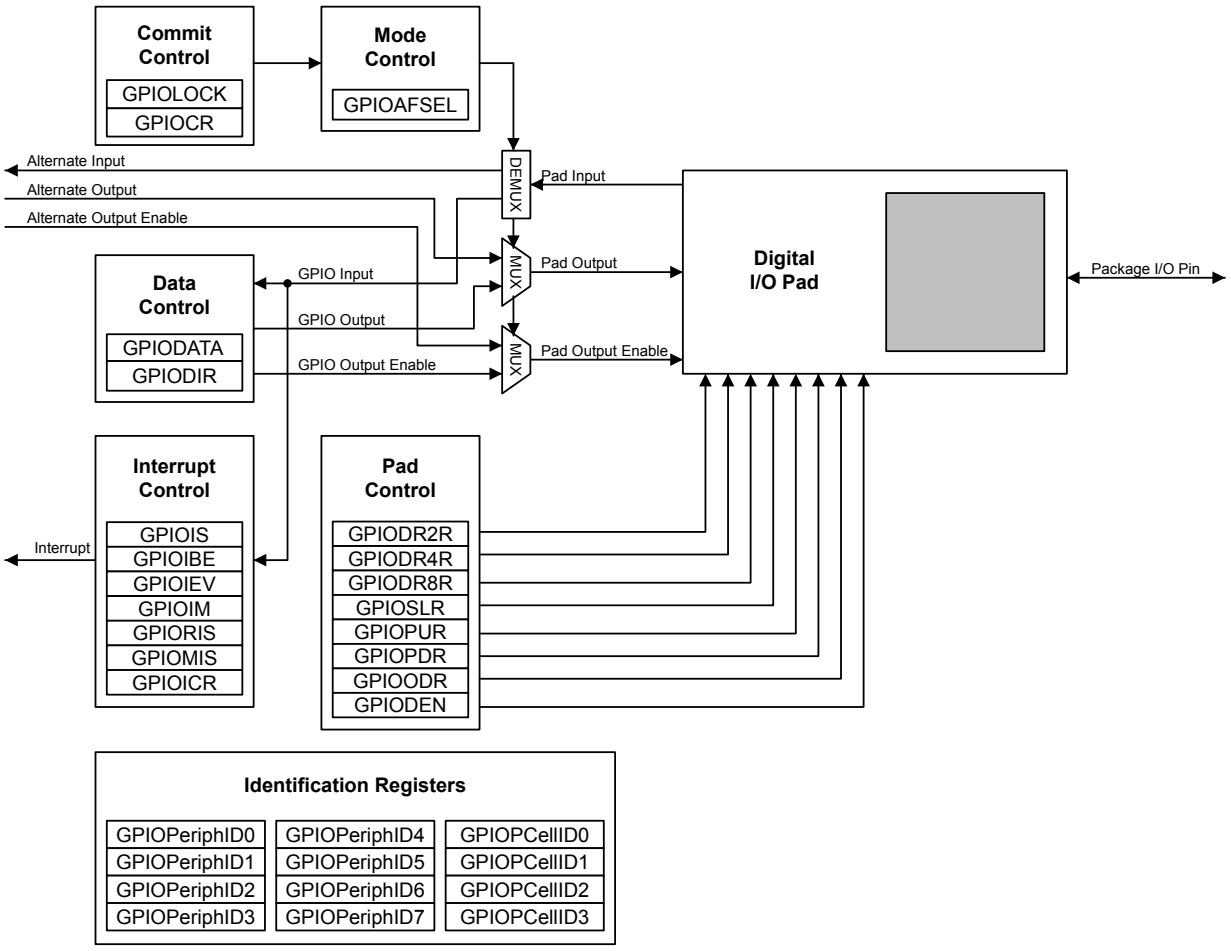
- 14-43 GPIOs, depending on configuration
- 5-V-tolerant input/outputs
- Programmable control for GPIO interrupts
  - Interrupt generation masking
  - Edge-triggered on rising, falling, or both
  - Level-sensitive on High or Low values
- Bit masking in both read and write operations through address lines
- Can initiate an ADC sample sequence
- Pins configured as digital inputs are Schmitt-triggered.
- Programmable control for GPIO pad configuration
  - Weak pull-up or pull-down resistors
  - 2-mA, 4-mA, and 8-mA pad drive for digital communication; up to four pads can be configured with an 18-mA pad drive for high-current applications
  - Slew rate control for the 8-mA drive
  - Open drain enables
  - Digital input enables

8.1 Functional Description

Important: All GPIO pins are tri-stated by default (**GPIOAFSEL**=0, **GPIODEN**=0, **GPIOPDR**=0, and **GPIOPUR**=0), with the exception of the five JTAG/SWD pins (PB7 and PC[3:0]). The JTAG/SWD pins default to their JTAG/SWD functionality (**GPIOAFSEL**=1, **GPIODEN=1** and **GPIOPUR**=1). A Power-On-Reset ($\overline{\text{POR}}$) or asserting $\overline{\text{RST}}$ puts both groups of pins back to their default state.

Each GPIO port is a separate hardware instantiation of the same physical block (see Figure 8-1 on page 152). The LM3S6432 microcontroller contains seven ports and thus seven of these physical GPIO blocks.

Figure 8-1. GPIO Port Block Diagram



8.1.1 Data Control

The data control registers allow software to configure the operational modes of the GPIOs. The data direction register configures the GPIO as an input or an output while the data register either captures incoming data or drives it out to the pads.

8.1.1.1 Data Direction Operation

The **GPIO Direction (GPIODIR)** register (see page 160) is used to configure each individual pin as an input or output. When the data direction bit is set to 0, the GPIO is configured as an input and the corresponding data register bit will capture and store the value on the GPIO port. When the data direction bit is set to 1, the GPIO is configured as an output and the corresponding data register bit will be driven out on the GPIO port.

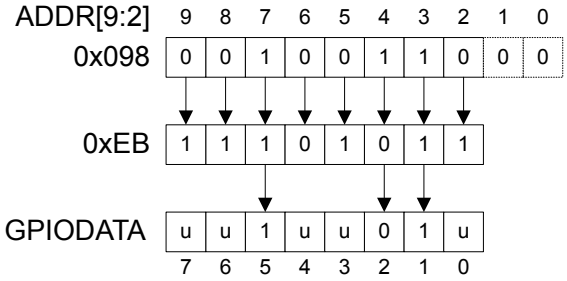
8.1.1.2 Data Register Operation

To aid in the efficiency of software, the GPIO ports allow for the modification of individual bits in the **GPIO Data (GPIODATA)** register (see page 159) by using bits [9:2] of the address bus as a mask. This allows software drivers to modify individual GPIO pins in a single instruction, without affecting the state of the other pins. This is in contrast to the "typical" method of doing a read-modify-write operation to set or clear an individual GPIO pin. To accommodate this feature, the **GPIODATA** register covers 256 locations in the memory map.

During a write, if the address bit associated with that data bit is set to 1, the value of the **GPIODATA** register is altered. If it is cleared to 0, it is left unchanged.

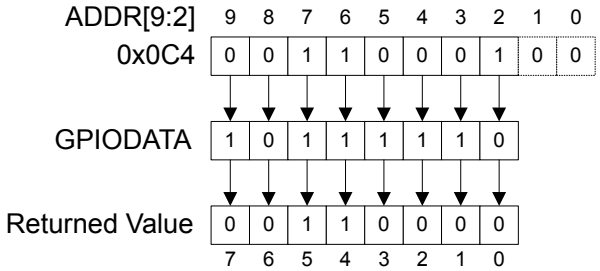
For example, writing a value of 0xEB to the address GPIODATA + 0x098 would yield as shown in Figure 8-2 on page 153, where u is data unchanged by the write.

Figure 8-2. GPIODATA Write Example



During a read, if the address bit associated with the data bit is set to 1, the value is read. If the address bit associated with the data bit is set to 0, it is read as a zero, regardless of its actual value. For example, reading address GPIODATA + 0x0C4 yields as shown in Figure 8-3 on page 153.

Figure 8-3. GPIODATA Read Example



8.1.2 Interrupt Control

The interrupt capabilities of each GPIO port are controlled by a set of seven registers. With these registers, it is possible to select the source of the interrupt, its polarity, and the edge properties. When one or more GPIO inputs cause an interrupt, a single interrupt output is sent to the interrupt controller for the entire GPIO port. For edge-triggered interrupts, software must clear the interrupt to enable any further interrupts. For a level-sensitive interrupt, it is assumed that the external source holds the level constant for the interrupt to be recognized by the controller.

Three registers are required to define the edge or sense that causes interrupts:

- **GPIO Interrupt Sense (GPIOIS)** register (see page 161)
- **GPIO Interrupt Both Edges (GPIOIBE)** register (see page 162)
- **GPIO Interrupt Event (GPIOIEV)** register (see page 163)

Interrupts are enabled/disabled via the **GPIO Interrupt Mask (GPIOIM)** register (see page 164).

When an interrupt condition occurs, the state of the interrupt signal can be viewed in two locations: the **GPIO Raw Interrupt Status (GPIORIS)** and **GPIO Masked Interrupt Status (GPIOMIS)** registers (see page 165 and page 166). As the name implies, the **GPIOMIS** register only shows interrupt

conditions that are allowed to be passed to the controller. The **GPIORIS** register indicates that a GPIO pin meets the conditions for an interrupt, but has not necessarily been sent to the controller.

In addition to providing GPIO functionality, PB4 can also be used as an external trigger for the ADC. If PB4 is configured as a non-masked interrupt pin (the appropriate bit of GPIOIM is set to 1), not only is an interrupt for PortB generated, but an external trigger signal is sent to the ADC. If the **ADC Event Multiplexer Select (ADCEMUX)** register is configured to use the external trigger, an ADC conversion is initiated.

If no other PortB pins are being used to generate interrupts, the ARM Integrated Nested Vectored Interrupt Controller (NVIC) Interrupt Set Enable (SETNA) register can disable the PortB interrupts and the ADC interrupt can be used to read back the converted data. Otherwise, the PortB interrupt handler needs to ignore and clear interrupts on B4, and wait for the ADC interrupt or the ADC interrupt needs to be disabled in the SETNA register and the PortB interrupt handler polls the ADC registers until the conversion is completed.

Interrupts are cleared by writing a 1 to the appropriate bit of the **GPIO Interrupt Clear (GPIOICR)** register (see page 167).

When programming the following interrupt control registers, the interrupts should be masked (**GPIOIM** set to 0). Writing any value to an interrupt control register (**GPIOIS**, **GPIOIBE**, or **GPIOIEV**) can generate a spurious interrupt if the corresponding bits are enabled.

8.1.3 Mode Control

The GPIO pins can be controlled by either hardware or software. When hardware control is enabled via the **GPIO Alternate Function Select (GPIOAFSEL)** register (see page 168), the pin state is controlled by its alternate function (that is, the peripheral). Software control corresponds to GPIO mode, where the **GPIODATA** register is used to read/write the corresponding pins.

8.1.4 Commit Control

The GPIO commit control registers provide a layer of protection against accidental programming of critical hardware peripherals. Protection is currently provided for the five JTAG/SWD pins (PB7 and PC[3:0]). Writes to protected bits of the **GPIO Alternate Function Select (GPIOAFSEL)** register (see page 168) are not committed to storage unless the **GPIO Lock (GPIOLOCK)** register (see page 178) has been unlocked and the appropriate bits of the **GPIO Commit (GPIOCR)** register (see page 179) have been set to 1.

8.1.5 Pad Control

The pad control registers allow for GPIO pad configuration by software based on the application requirements. The pad control registers include the **GPIODR2R**, **GPIODR4R**, **GPIODR8R**, **GPIOODR**, **GPIOPUR**, **GPIOPDR**, **GPIOSLR**, and **GPIODEN** registers. These registers control drive strength, open-drain configuration, pull-up and pull-down resistors, slew-rate control and digital input enable.

For special high-current applications, the GPIO output buffers may be used with the following restrictions. With the GPIO pins configured as 8-mA output drivers, a total of four GPIO outputs may be used to sink current loads up to 18 mA each. At 18-mA sink current loading, the V_{OL} value is specified as 1.2 V. The high-current GPIO package pins must be selected such that there are only a maximum of two per side of the physical package or BGA pin group with the total number of high-current GPIO outputs not exceeding four for the entire package.

8.1.6 Identification

The identification registers configured at reset allow software to detect and identify the module as a GPIO block. The identification registers include the **GPIOPeriphID0**-**GPIOPeriphID7** registers as well as the **GPIOPCellID0**-**GPIOPCellID3** registers.

8.2 Initialization and Configuration

To use the GPIO, the peripheral clock must be enabled by setting the appropriate GPIO Port bit field (`GPIOn`) in the **RCGC2** register.

On reset, all GPIO pins (except for the five JTAG pins) are configured out of reset to be undriven (tristate): **GPIOAFSEL**=0, **GPIODEN**=0, **GPIOPDR**=0, and **GPIOPUR**=0. Table 8-1 on page 155 shows all possible configurations of the GPIO pads and the control register settings required to achieve them. Table 8-2 on page 155 shows how a rising edge interrupt would be configured for pin 2 of a GPIO port.

Table 8-1. GPIO Pad Configuration Examples

Configuration	GPIO Register Bit Value[a]									
	AFSEL	DIR	ODR	DEN	PUR	PDR	DR2R	DR4R	DR8R	SLR
Digital Input (GPIO)	0	0	0	1	?	?	X	X	X	X
Digital Output (GPIO)	0	1	0	1	?	?	?	?	?	?
Open Drain Output (GPIO)	0	1	1	1	X	X	?	?	?	?
Open Drain Input/Output (I^2C)	1	X	1	1	X	X	?	?	?	?
Digital Input (Timer CCP)	1	X	0	1	?	?	X	X	X	X
Digital Output (PWM)	1	X	0	1	?	?	?	?	?	?
Digital Output (Timer PWM)	1	X	0	1	?	?	?	?	?	?
Digital Input/Output (SSI)	1	X	0	1	?	?	?	?	?	?
Digital Input/Output (UART)	1	X	0	1	?	?	?	?	?	?
Analog Input (Comparator)	0	0	0	0	0	0	X	X	X	X
Digital Output (Comparator)	1	X	0	1	?	?	?	?	?	?

a. X=Ignored (don't care bit)

?=Can be either 0 or 1, depending on the configuration

Table 8-2. GPIO Interrupt Configuration Example

Register	Desired Interrupt Event Trigger	Pin 2 Bit Value[a]							
		7	6	5	4	3	2	1	0
GPIOIS	0=edge 1=level	X	X	X	X	X	0	X	X

Table 8-2. GPIO Interrupt Configuration Example *(continued)*

Register	Desired Interrupt Event Trigger	Pin 2 Bit Value[a]							
		7	6	5	4	3	2	1	0
GPIOIBE	0=single edge 1=both edges	X	X	X	X	X	0	X	X
GPIOIEV	0=Low level, or negative edge 1=High level, or positive edge	X	X	X	X	X	1	X	X
GPIOIM	0=masked 1=not masked	0	0	0	0	0	1	0	0

a. X=Ignored (don't care bit)

8.3 Register Map

Table 8-3 on page 157 lists the GPIO registers. The offset listed is a hexadecimal increment to the register's address, relative to that GPIO port's base address:

- GPIO Port A: 0x4000.4000
- GPIO Port B: 0x4000.5000
- GPIO Port C: 0x4000.6000
- GPIO Port D: 0x4000.7000
- GPIO Port E: 0x4002.4000
- GPIO Port F: 0x4002.5000
- GPIO Port G: 0x4002.6000

Important: The GPIO registers in this chapter are duplicated in each GPIO block; however, depending on the block, all eight bits may not be connected to a GPIO pad. In those cases, writing to those unconnected bits has no effect, and reading those unconnected bits returns no meaningful data.

Note: The default reset value for the **GPIOAFSEL**, **GPIOPUR**, and **GPIODEN** registers are 0x0000.0000 for all GPIO pins, with the exception of the five JTAG/SWD pins (PB7 and PC[3:0]). These five pins default to JTAG/SWD functionality. Because of this, the default reset value of these registers for GPIO Port B is 0x0000.0080 while the default reset value for Port C is 0x0000.000F.

The default register type for the **GPIOCR** register is RO for all GPIO pins with the exception of the five JTAG/SWD pins (PB7 and PC[3:0]). These five pins are currently the only GPIOs that are protected by the **GPIOCR** register. Because of this, the register type for GPIO Port B7 and GPIO Port C[3:0] is R/W.

The default reset value for the **GPIOCR** register is 0x0000.00FF for all GPIO pins, with the exception of the five JTAG/SWD pins (PB7 and PC[3:0]). To ensure that the JTAG port is not accidentally programmed as a GPIO, these five pins default to non-committable.

Because of this, the default reset value of **GPIOCR** for GPIO Port B is 0x0000.007F while the default reset value of GPIOCR for Port C is 0x0000.00F0.

Table 8-3. GPIO Register Map

Offset	Name	Type	Reset	Description	See page
0x000	GPIODATA	R/W	0x0000.0000	GPIO Data	159
0x400	GPIODIR	R/W	0x0000.0000	GPIO Direction	160
0x404	GPIOIS	R/W	0x0000.0000	GPIO Interrupt Sense	161
0x408	GPIOIBE	R/W	0x0000.0000	GPIO Interrupt Both Edges	162
0x40C	GPIOIEV	R/W	0x0000.0000	GPIO Interrupt Event	163
0x410	GPIOIM	R/W	0x0000.0000	GPIO Interrupt Mask	164
0x414	GPIORIS	RO	0x0000.0000	GPIO Raw Interrupt Status	165
0x418	GPIOMIS	RO	0x0000.0000	GPIO Masked Interrupt Status	166
0x41C	GPIOICR	W1C	0x0000.0000	GPIO Interrupt Clear	167
0x420	GPIOAFSEL	R/W	-	GPIO Alternate Function Select	168
0x500	GPIODR2R	R/W	0x0000.00FF	GPIO 2-mA Drive Select	170
0x504	GPIODR4R	R/W	0x0000.0000	GPIO 4-mA Drive Select	171
0x508	GPIODR8R	R/W	0x0000.0000	GPIO 8-mA Drive Select	172
0x50C	GPIOODR	R/W	0x0000.0000	GPIO Open Drain Select	173
0x510	GPIOPUR	R/W	-	GPIO Pull-Up Select	174
0x514	GPIOPDR	R/W	0x0000.0000	GPIO Pull-Down Select	175
0x518	GPIOSLR	R/W	0x0000.0000	GPIO Slew Rate Control Select	176
0x51C	GPIODEN	R/W	-	GPIO Digital Enable	177
0x520	GPIOLOCK	R/W	0x0000.0001	GPIO Lock	178
0x524	GPIOCR	-	-	GPIO Commit	179
0xFD0	GPIOPeriphID4	RO	0x0000.0000	GPIO Peripheral Identification 4	181
0xFD4	GPIOPeriphID5	RO	0x0000.0000	GPIO Peripheral Identification 5	182
0xFD8	GPIOPeriphID6	RO	0x0000.0000	GPIO Peripheral Identification 6	183
0xFDC	GPIOPeriphID7	RO	0x0000.0000	GPIO Peripheral Identification 7	184
0xFE0	GPIOPeriphID0	RO	0x0000.0061	GPIO Peripheral Identification 0	185
0xFE4	GPIOPeriphID1	RO	0x0000.0000	GPIO Peripheral Identification 1	186
0xFE8	GPIOPeriphID2	RO	0x0000.0018	GPIO Peripheral Identification 2	187
0xFEC	GPIOPeriphID3	RO	0x0000.0001	GPIO Peripheral Identification 3	188
0xFF0	GPIOPCellID0	RO	0x0000.000D	GPIO PrimeCell Identification 0	189
0xFF4	GPIOPCellID1	RO	0x0000.00F0	GPIO PrimeCell Identification 1	190

Table 8-3. GPIO Register Map *(continued)*

Offset	Name	Type	Reset	Description	See page
0xFF8	GPIOPCellID2	RO	0x0000.0005	GPIO PrimeCell Identification 2	191
0xFFC	GPIOPCellID3	RO	0x0000.00B1	GPIO PrimeCell Identification 3	192

8.4 Register Descriptions

The remainder of this section lists and describes the GPIO registers, in numerical order by address offset.

Register 1: GPIO Data (GPIODATA), offset 0x000

The **GPIODATA** register is the data register. In software control mode, values written in the **GPIODATA** register are transferred onto the GPIO port pins if the respective pins have been configured as outputs through the **GPIO Direction (GPIODIR)** register (see page 160).

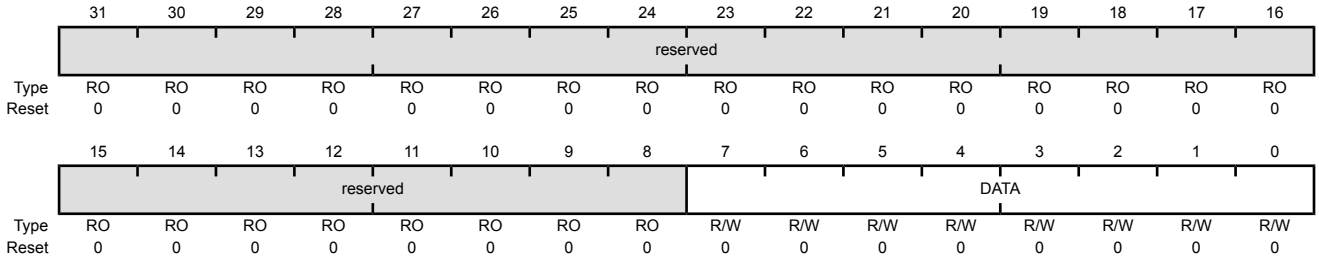
In order to write to **GPIODATA**, the corresponding bits in the mask, resulting from the address bus bits [9:2], must be High. Otherwise, the bit values remain unchanged by the write.

Similarly, the values read from this register are determined for each bit by the mask bit derived from the address used to access the data register, bits [9:2]. Bits that are 1 in the address mask cause the corresponding bits in **GPIODATA** to be read, and bits that are 0 in the address mask cause the corresponding bits in **GPIODATA** to be read as 0, regardless of their value.

A read from **GPIODATA** returns the last bit value written if the respective pins are configured as outputs, or it returns the value on the corresponding input pin when these are configured as inputs. All bits are cleared by a reset.

GPIO Data (GPIODATA)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0x000
Type R/W, reset 0x0000.0000



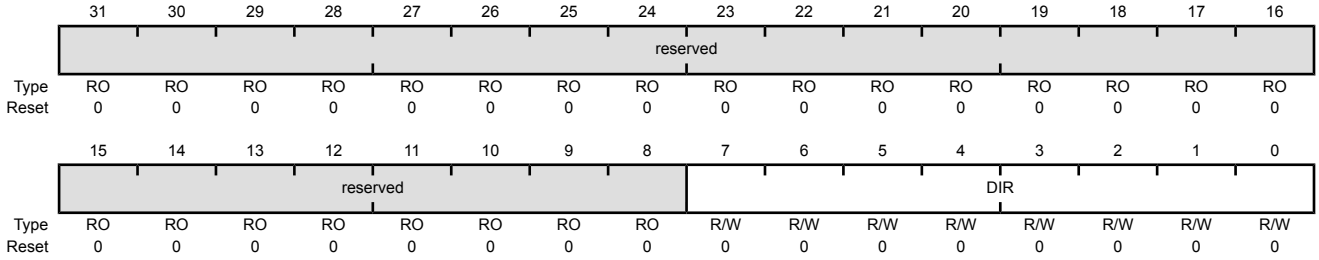
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	DATA	R/W	0x00	GPIO Data This register is virtually mapped to 256 locations in the address space. To facilitate the reading and writing of data to these registers by independent drivers, the data read from and the data written to the registers are masked by the eight address lines `ipaddr[9:2]`. Reads from this register return its current state. Writes to this register only affect bits that are not masked by `ipaddr[9:2]` and are configured as outputs. See “Data Register Operation” on page 152 for examples of reads and writes.

Register 2: GPIO Direction (GPIODIR), offset 0x400

The **GPIODIR** register is the data direction register. Bits set to 1 in the **GPIODIR** register configure the corresponding pin to be an output, while bits set to 0 configure the pins to be inputs. All bits are cleared by a reset, meaning all GPIO pins are inputs by default.

GPIO Direction (GPIODIR)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0x400
Type R/W, reset 0x0000.0000



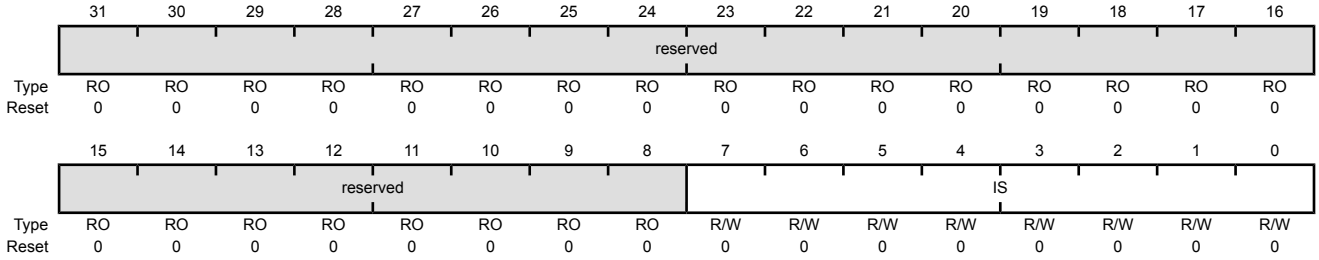
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	DIR	R/W	0x00	GPIO Data Direction The `DIR` values are defined as follows: Value Description 0 Pins are inputs. 1 Pins are outputs.

Register 3: GPIO Interrupt Sense (GPIOIS), offset 0x404

The **GPIOIS** register is the interrupt sense register. Bits set to 1 in **GPIOIS** configure the corresponding pins to detect levels, while bits set to 0 configure the pins to detect edges. All bits are cleared by a reset.

GPIO Interrupt Sense (GPIOIS)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0x404
Type R/W, reset 0x0000.0000



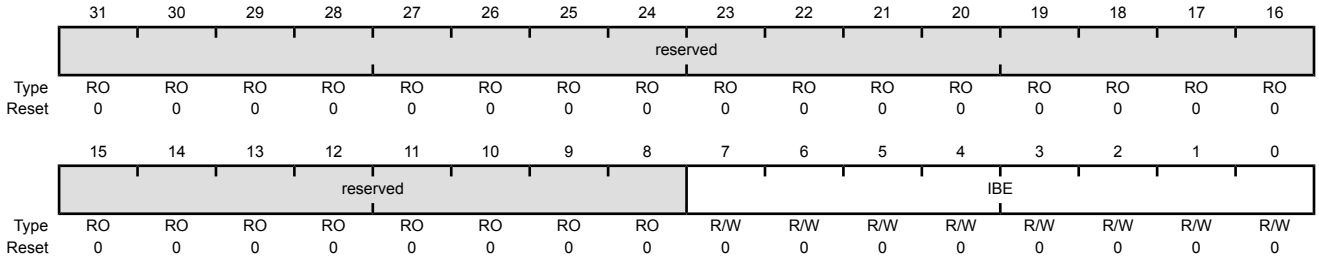
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	IS	R/W	0x00	GPIO Interrupt Sense The `IS` values are defined as follows: Value Description 0 Edge on corresponding pin is detected (edge-sensitive). 1 Level on corresponding pin is detected (level-sensitive).

Register 4: GPIO Interrupt Both Edges (GPIOIBE), offset 0x408

The **GPIOIBE** register is the interrupt both-edges register. When the corresponding bit in the **GPIO Interrupt Sense (GPIOIS)** register (see page 161) is set to detect edges, bits set to High in **GPIOIBE** configure the corresponding pin to detect both rising and falling edges, regardless of the corresponding bit in the **GPIO Interrupt Event (GPIOIEV)** register (see page 163). Clearing a bit configures the pin to be controlled by **GPIOIEV**. All bits are cleared by a reset.

GPIO Interrupt Both Edges (GPIOIBE)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0x408
Type R/W, reset 0x0000.0000



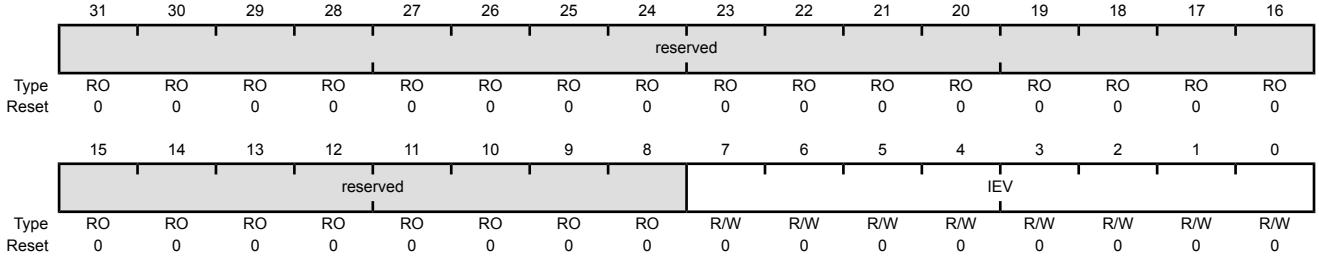
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	IBE	R/W	0x00	GPIO Interrupt Both Edges The `IBE` values are defined as follows: Value Description 0 Interrupt generation is controlled by the **GPIO Interrupt Event (GPIOIEV)** register (see page 163). 1 Both edges on the corresponding pin trigger an interrupt. **Note:** Single edge is determined by the corresponding bit in **GPIOIEV**.

Register 5: GPIO Interrupt Event (GPIOIEV), offset 0x40C

The **GPIOIEV** register is the interrupt event register. Bits set to High in **GPIOIEV** configure the corresponding pin to detect rising edges or high levels, depending on the corresponding bit value in the **GPIO Interrupt Sense (GPIOIS)** register (see page 161). Clearing a bit configures the pin to detect falling edges or low levels, depending on the corresponding bit value in **GPIOIS**. All bits are cleared by a reset.

GPIO Interrupt Event (GPIOIEV)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0x40C
Type R/W, reset 0x0000.0000



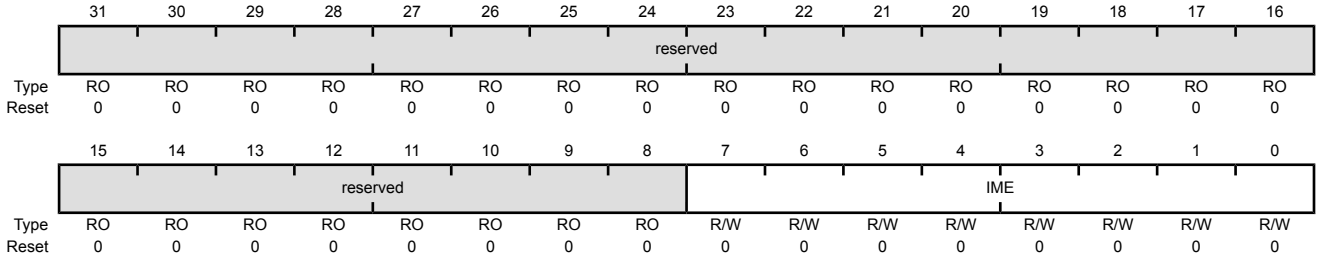
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	IEV	R/W	0x00	GPIO Interrupt Event The `IEV` values are defined as follows: Value Description 0 Falling edge or Low levels on corresponding pins trigger interrupts. 1 Rising edge or High levels on corresponding pins trigger interrupts.

Register 6: GPIO Interrupt Mask (GPIOIM), offset 0x410

The **GPIOIM** register is the interrupt mask register. Bits set to High in **GPIOIM** allow the corresponding pins to trigger their individual interrupts and the combined **GPIOINTR** line. Clearing a bit disables interrupt triggering on that pin. All bits are cleared by a reset.

GPIO Interrupt Mask (GPIOIM)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0x410
Type R/W, reset 0x0000.0000



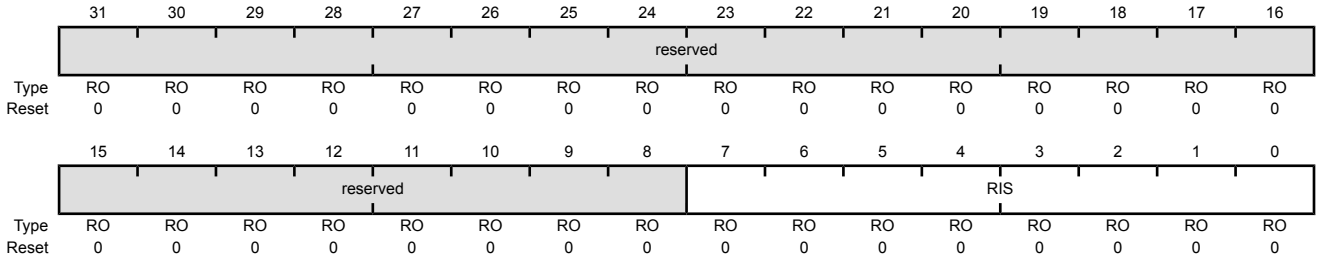
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	IME	R/W	0x00	GPIO Interrupt Mask Enable The IME values are defined as follows: Value Description 0 Corresponding pin interrupt is masked. 1 Corresponding pin interrupt is not masked.

Register 7: GPIO Raw Interrupt Status (GPIORIS), offset 0x414

The **GPIORIS** register is the raw interrupt status register. Bits read High in **GPIORIS** reflect the status of interrupt trigger conditions detected (raw, prior to masking), indicating that all the requirements have been met, before they are finally allowed to trigger by the **GPIO Interrupt Mask (GPIOIM)** register (see page 164). Bits read as zero indicate that corresponding input pins have not initiated an interrupt. All bits are cleared by a reset.

GPIO Raw Interrupt Status (GPIORIS)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0x414
Type RO, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	RIS	RO	0x00	GPIO Interrupt Raw Status Reflects the status of interrupt trigger condition detection on pins (raw, prior to masking). The `RIS` values are defined as follows: Value Description 0 Corresponding pin interrupt requirements not met. 1 Corresponding pin interrupt has met requirements.

Register 8: GPIO Masked Interrupt Status (GPIOMIS), offset 0x418

The **GPIOMIS** register is the masked interrupt status register. Bits read High in **GPIOMIS** reflect the status of input lines triggering an interrupt. Bits read as Low indicate that either no interrupt has been generated, or the interrupt is masked.

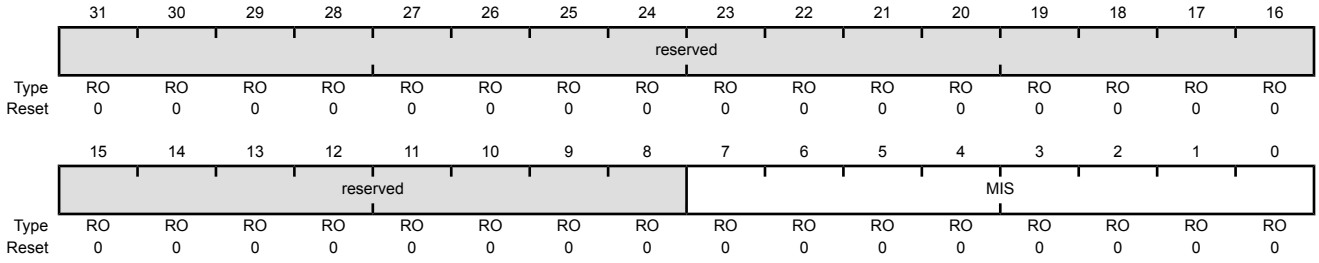
In addition to providing GPIO functionality, PB4 can also be used as an external trigger for the ADC. If PB4 is configured as a non-masked interrupt pin (the appropriate bit of GPIOIM is set to 1), not only is an interrupt for PortB generated, but an external trigger signal is sent to the ADC. If the **ADC Event Multiplexer Select (ADCEMUX)** register is configured to use the external trigger, an ADC conversion is initiated.

If no other PortB pins are being used to generate interrupts, the ARM Integrated Nested Vectored Interrupt Controller (NVIC) Interrupt Set Enable (SETNA) register can disable the PortB interrupts and the ADC interrupt can be used to read back the converted data. Otherwise, the PortB interrupt handler needs to ignore and clear interrupts on B4, and wait for the ADC interrupt or the ADC interrupt needs to be disabled in the SETNA register and the PortB interrupt handler polls the ADC registers until the conversion is completed.

GPIOMIS is the state of the interrupt after masking.

GPIO Masked Interrupt Status (GPIOMIS)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0x418
Type RO, reset 0x0000.0000



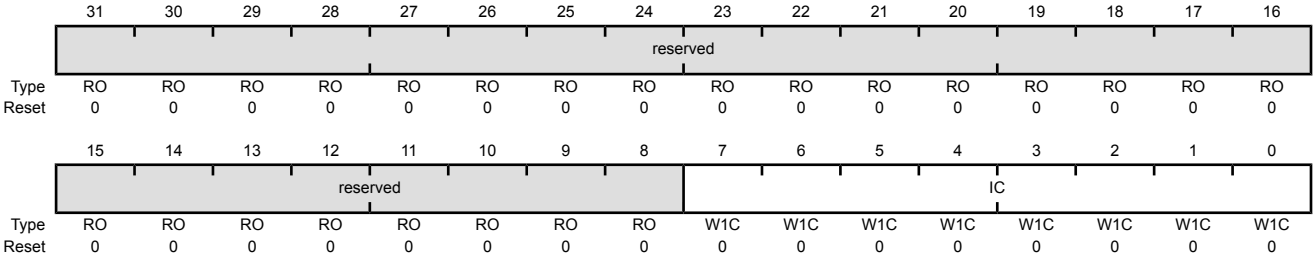
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	MIS	RO	0x00	GPIO Masked Interrupt Status Masked value of interrupt due to corresponding pin. The MIS values are defined as follows: Value Description 0 Corresponding GPIO line interrupt not active. 1 Corresponding GPIO line asserting interrupt.

Register 9: GPIO Interrupt Clear (GPIOICR), offset 0x41C

The **GPIOICR** register is the interrupt clear register. Writing a 1 to a bit in this register clears the corresponding interrupt edge detection logic register. Writing a 0 has no effect.

GPIO Interrupt Clear (GPIOICR)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0x41C
Type W1C, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	IC	W1C	0x00	GPIO Interrupt Clear The `IC` values are defined as follows: Value Description 0 Corresponding interrupt is unaffected. 1 Corresponding interrupt is cleared.

Register 10: GPIO Alternate Function Select (GPIOAFSEL), offset 0x420

The **GPIOAFSEL** register is the mode control select register. Writing a 1 to any bit in this register selects the hardware control for the corresponding GPIO line. All bits are cleared by a reset, therefore no GPIO line is set to hardware control by default.

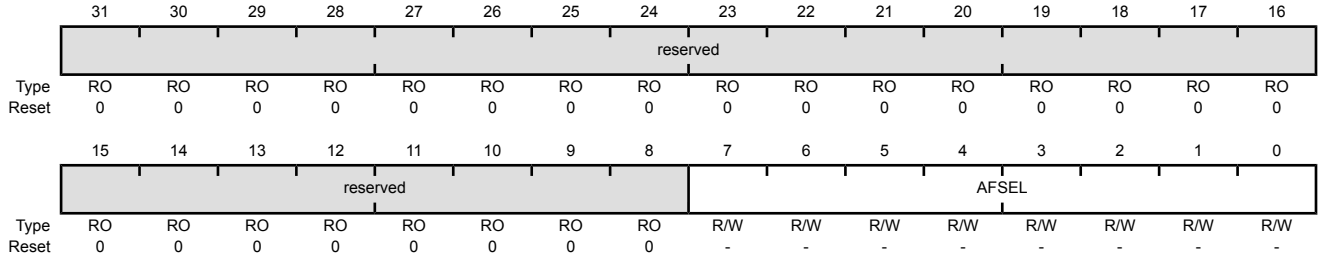
The GPIO commit control registers provide a layer of protection against accidental programming of critical hardware peripherals. Protection is currently provided for the five JTAG/SWD pins (PB7 and PC[3:0]). Writes to protected bits of the **GPIO Alternate Function Select (GPIOAFSEL)** register (see page 168) are not committed to storage unless the **GPIO Lock (GPIOLOCK)** register (see page 178) has been unlocked and the appropriate bits of the **GPIO Commit (GPIOCR)** register (see page 179) have been set to 1.

Important: All GPIO pins are tri-stated by default (**GPIOAFSEL**=0, **GPIODEN**=0, **GPIOPDR**=0, and **GPIOPUR**=0), with the exception of the five JTAG/SWD pins (PB7 and PC[3:0]). The JTAG/SWD pins default to their JTAG/SWD functionality (**GPIOAFSEL**=1, **GPIODEN=1** and **GPIOPUR**=1). A Power-On-Reset (POR) or asserting RST puts both groups of pins back to their default state.

Caution – It is possible to create a software sequence that prevents the debugger from connecting to the Stellaris® microcontroller. If the program code loaded into flash immediately changes the JTAG pins to their GPIO functionality, the debugger may not have enough time to connect and halt the controller before the JTAG pin functionality switches. This may lock the debugger out of the part. This can be avoided with a software routine that restores JTAG functionality based on an external or software trigger.

GPIO Alternate Function Select (GPIOAFSEL)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0x420
Type R/W, reset -



Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

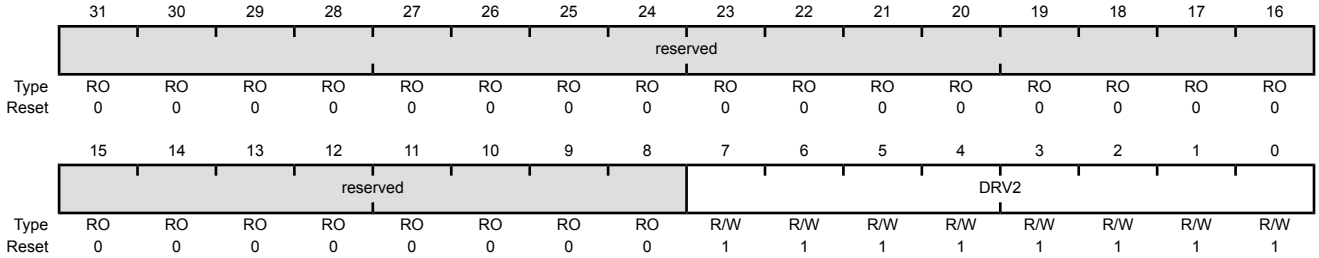
Bit/Field	Name	Type	Reset	Description
7:0	AFSEL	R/W	-	GPIO Alternate Function Select The AFSEL values are defined as follows: Value Description 0 Software control of corresponding GPIO line (GPIO mode). 1 Hardware control of corresponding GPIO line (alternate hardware function). **Note:** The default reset value for the **GPIOAFSEL**, **GPIOPUR**, and **GPIODEN** registers are 0x0000.0000 for all GPIO pins, with the exception of the five JTAG/SWD pins (PB7 and PC[3:0]). These five pins default to JTAG/SWD functionality. Because of this, the default reset value of these registers for GPIO Port B is 0x0000.0080 while the default reset value for Port C is 0x0000.000F.

Register 11: GPIO 2-mA Drive Select (GPIODR2R), offset 0x500

The **GPIODR2R** register is the 2-mA drive control register. It allows for each GPIO signal in the port to be individually configured without affecting the other pads. When writing a DRV2 bit for a GPIO signal, the corresponding DRV4 bit in the **GPIODR4R** register and the DRV8 bit in the **GPIODR8R** register are automatically cleared by hardware.

GPIO 2-mA Drive Select (GPIODR2R)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0x500
Type R/W, reset 0x0000.00FF



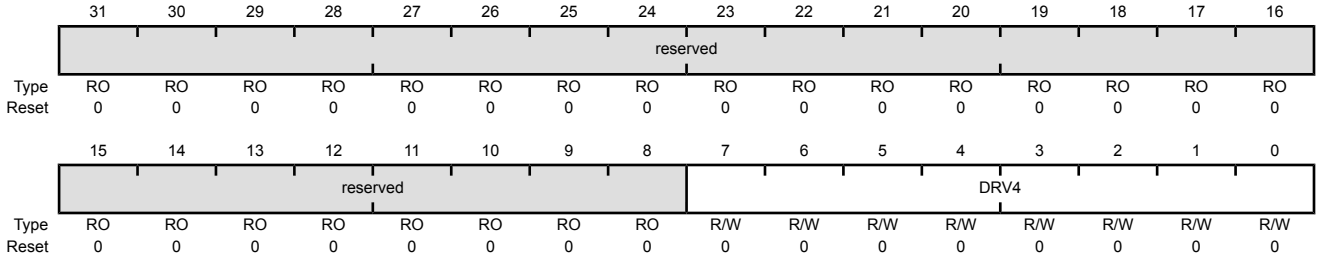
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	DRV2	R/W	0xFF	Output Pad 2-mA Drive Enable A write of 1 to either **GPIODR4[n]** or **GPIODR8[n]** clears the corresponding 2-mA enable bit. The change is effective on the second clock cycle after the write.

Register 12: GPIO 4-mA Drive Select (GPIODR4R), offset 0x504

The **GPIODR4R** register is the 4-mA drive control register. It allows for each GPIO signal in the port to be individually configured without affecting the other pads. When writing the DRV4 bit for a GPIO signal, the corresponding DRV2 bit in the **GPIODR2R** register and the DRV8 bit in the **GPIODR8R** register are automatically cleared by hardware.

GPIO 4-mA Drive Select (GPIODR4R)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0x504
Type R/W, reset 0x0000.0000



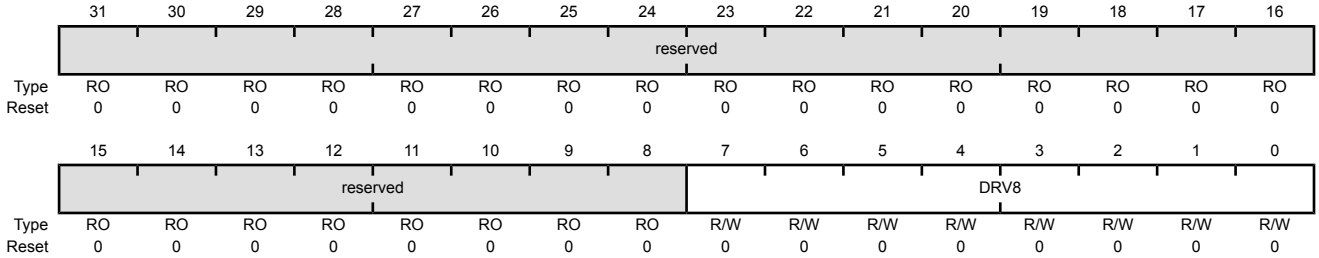
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	DRV4	R/W	0x00	Output Pad 4-mA Drive Enable A write of 1 to either **GPIODR2[n]** or **GPIODR8[n]** clears the corresponding 4-mA enable bit. The change is effective on the second clock cycle after the write.

Register 13: GPIO 8-mA Drive Select (GPIODR8R), offset 0x508

The **GPIODR8R** register is the 8-mA drive control register. It allows for each GPIO signal in the port to be individually configured without affecting the other pads. When writing the DRV8 bit for a GPIO signal, the corresponding DRV2 bit in the **GPIODR2R** register and the DRV4 bit in the **GPIODR4R** register are automatically cleared by hardware.

GPIO 8-mA Drive Select (GPIODR8R)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0x508
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	DRV8	R/W	0x00	Output Pad 8-mA Drive Enable A write of 1 to either **GPIODR2[n]** or **GPIODR4[n]** clears the corresponding 8-mA enable bit. The change is effective on the second clock cycle after the write.

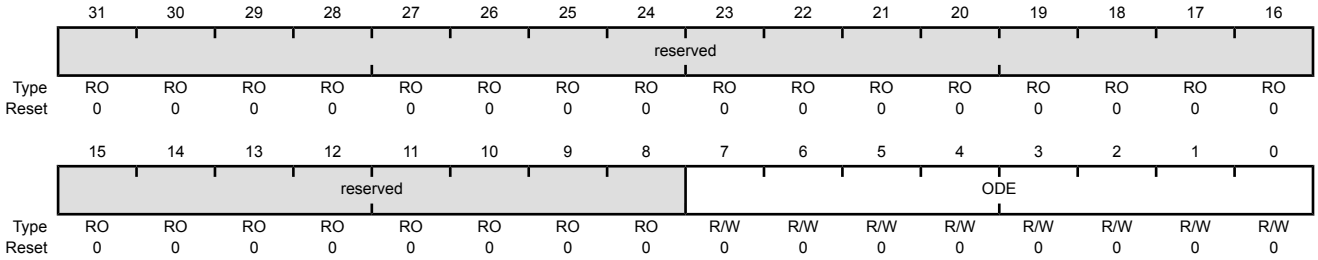
Register 14: GPIO Open Drain Select (GPIOODR), offset 0x50C

The **GPIOODR** register is the open drain control register. Setting a bit in this register enables the open drain configuration of the corresponding GPIO pad. When open drain mode is enabled, the corresponding bit should also be set in the **GPIO Digital Input Enable (GPIODEN)** register (see page 177). Corresponding bits in the drive strength registers (**GPIODR2R**, **GPIODR4R**, **GPIODR8R**, and **GPIOSLR**) can be set to achieve the desired rise and fall times. The GPIO acts as an open-drain input if the corresponding bit in the **GPIODIR** register is cleared. If open drain is selected while the GPIO is configured as an input, the GPIO will remain an input and the open-drain selection has no effect until the GPIO is changed to an output.

When using the I^2C module, in addition to configuring the pin to open drain, the **GPIO Alternate Function Select (GPIOAFSEL)** register bits for the I^2C clock and data pins should be set to 1 (see examples in “Initialization and Configuration” on page 155).

GPIO Open Drain Select (GPIOODR)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0x50C
Type R/W, reset 0x0000.0000



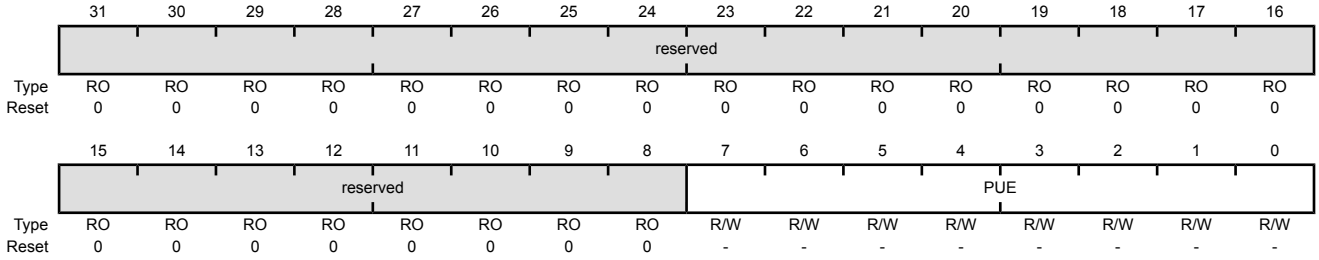
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	ODE	R/W	0x00	Output Pad Open Drain Enable The ODE values are defined as follows: Value Description 0 Open drain configuration is disabled. 1 Open drain configuration is enabled.

Register 15: GPIO Pull-Up Select (GPIOPUR), offset 0x510

The **GPIOPUR** register is the pull-up control register. When a bit is set to 1, it enables a weak pull-up resistor on the corresponding GPIO signal. Setting a bit in **GPIOPUR** automatically clears the corresponding bit in the **GPIO Pull-Down Select (GPIOPDR)** register (see page 175).

GPIO Pull-Up Select (GPIOPUR)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0x510
Type R/W, reset -



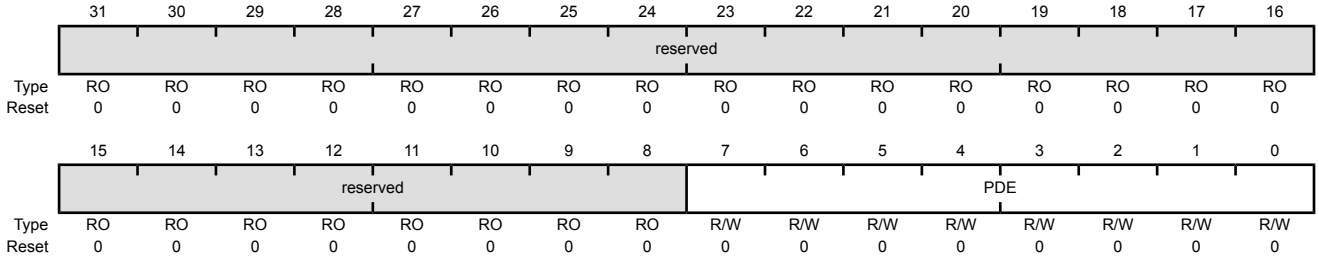
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PUE	R/W	-	Pad Weak Pull-Up Enable A write of 1 to **GPIOPDR[n]** clears the corresponding **GPIOPUR[n]** enables. The change is effective on the second clock cycle after the write. **Note:** The default reset value for the **GPIOAFSEL**, **GPIOPUR**, and **GPIODEN** registers are 0x0000.0000 for all GPIO pins, with the exception of the five JTAG/SWD pins (PB7 and PC[3:0]). These five pins default to JTAG/SWD functionality. Because of this, the default reset value of these registers for GPIO Port B is 0x0000.0080 while the default reset value for Port C is 0x0000.000F.

Register 16: GPIO Pull-Down Select (GPIOPDR), offset 0x514

The **GPIOPDR** register is the pull-down control register. When a bit is set to 1, it enables a weak pull-down resistor on the corresponding GPIO signal. Setting a bit in **GPIOPDR** automatically clears the corresponding bit in the **GPIO Pull-Up Select (GPIOPUR)** register (see page 174).

GPIO Pull-Down Select (GPIOPDR)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0x514
Type R/W, reset 0x0000.0000



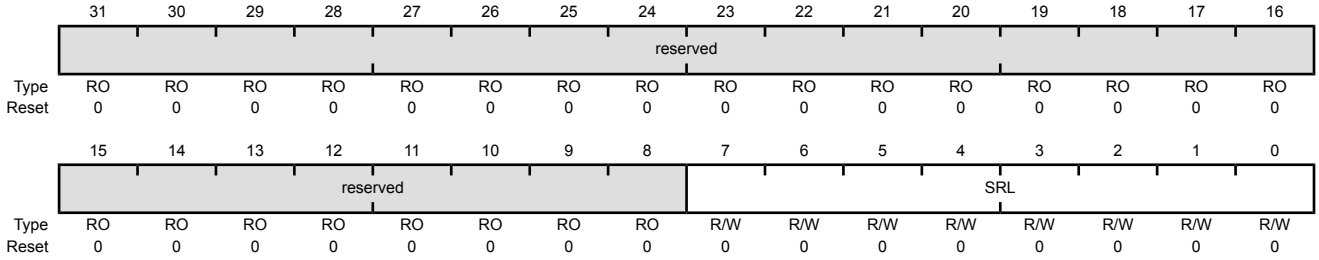
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PDE	R/W	0x00	Pad Weak Pull-Down Enable A write of 1 to **GPIOPUR[n]** clears the corresponding **GPIOPDR[n]** enables. The change is effective on the second clock cycle after the write.

Register 17: GPIO Slew Rate Control Select (GPIOSLR), offset 0x518

The **GPIOSLR** register is the slew rate control register. Slew rate control is only available when using the 8-mA drive strength option via the **GPIO 8-mA Drive Select (GPIODR8R)** register (see page 172).

GPIO Slew Rate Control Select (GPIOSLR)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0x518
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	SRL	R/W	0x00	Slew Rate Limit Enable (8-mA drive only) The SRL values are defined as follows: Value Description 0 Slew rate control disabled. 1 Slew rate control enabled.

Register 18: GPIO Digital Enable (GPIODEN), offset 0x51C

Note: Pins configured as digital inputs are Schmitt-triggered.

The **GPIODEN** register is the digital enable register. By default, with the exception of the GPIO signals used for JTAG/SWD function, all other GPIO signals are configured out of reset to be undriven (tristate). Their digital function is disabled; they do not drive a logic value on the pin and they do not allow the pin voltage into the GPIO receiver. To use the pin in a digital function (either GPIO or alternate function), the corresponding GPIODEN bit must be set.

GPIO Digital Enable (GPIODEN)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0x51C
Type R/W, reset -

	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	reserved															
Type	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	reserved								DEN							
Type	RO	RO	RO	RO	RO	RO	RO	RO	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	-	-	-	-	-	-	-	-

Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	DEN	R/W	-	Digital Enable The DEN values are defined as follows: Value Description 0 Digital functions disabled. 1 Digital functions enabled. **Note:** The default reset value for the **GPIOAFSEL**, **GPIOPUR**, and **GPIODEN** registers are 0x0000.0000 for all GPIO pins, with the exception of the five JTAG/SWD pins (PB7 and PC[3:0]). These five pins default to JTAG/SWD functionality. Because of this, the default reset value of these registers for GPIO Port B is 0x0000.0080 while the default reset value for Port C is 0x0000.000F.

Register 19: GPIO Lock (GPIOLOCK), offset 0x520

The **GPIOLOCK** register enables write access to the **GPIOCR** register (see page 179). Writing 0x1ACC.E551 to the **GPIOLOCK** register will unlock the **GPIOCR** register. Writing any other value to the **GPIOLOCK** register re-enables the locked state. Reading the **GPIOLOCK** register returns the lock status rather than the 32-bit value that was previously written. Therefore, when write accesses are disabled, or locked, reading the **GPIOLOCK** register returns 0x00000001. When write accesses are enabled, or unlocked, reading the **GPIOLOCK** register returns 0x00000000.

GPIO Lock (GPIOLOCK)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0x520
Type R/W, reset 0x0000.0001

	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	LOCK															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	LOCK															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bit/Field	Name	Type	Reset	Description
31:0	LOCK	R/W	0x0000.0001	GPIO Lock

A write of the value 0x1ACC.E551 unlocks the **GPIO Commit (GPIOCR)** register for write access.

A write of any other value or a write to the **GPIOCR** register reapplies the lock, preventing any register updates. A read of this register returns the following values:

Value	Description
0x0000.0001	locked
0x0000.0000	unlocked

Register 20: GPIO Commit (GPIOCR), offset 0x524

The **GPIOCR** register is the commit register. The value of the **GPIOCR** register determines which bits of the **GPIOAFSEL** register are committed when a write to the **GPIOAFSEL** register is performed. If a bit in the **GPIOCR** register is a zero, the data being written to the corresponding bit in the **GPIOAFSEL** register will not be committed and will retain its previous value. If a bit in the **GPIOCR** register is a one, the data being written to the corresponding bit of the **GPIOAFSEL** register will be committed to the register and will reflect the new value.

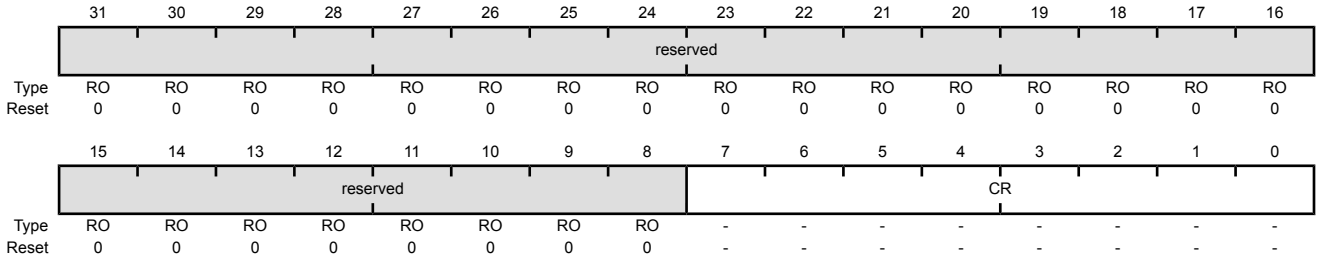
The contents of the **GPIOCR** register can only be modified if the **GPIOLOCK** register is unlocked. Writes to the **GPIOCR** register are ignored if the **GPIOLOCK** register is locked.

Important: This register is designed to prevent accidental programming of the registers that control connectivity to the JTAG/SWD debug hardware. By initializing the bits of the **GPIOCR** register to 0 for PB7 and PC[3:0], the JTAG/SWD debug port can only be converted to GPIOs through a deliberate set of writes to the **GPIOLOCK**, **GPIOCR**, and the corresponding registers.

Because this protection is currently only implemented on the JTAG/SWD pins on PB7 and PC[3:0], all of the other bits in the **GPIOCR** registers cannot be written with 0x0. These bits are hardwired to 0x1, ensuring that it is always possible to commit new values to the **GPIOAFSEL**register bits of these other pins.

GPIO Commit (GPIOCR)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0x524
Type -, reset -



Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

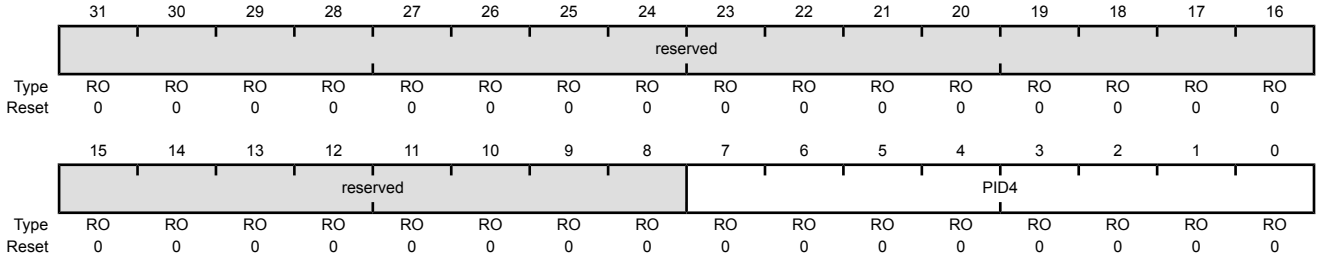
Bit/Field	Name	Type	Reset	Description
7:0	CR	-	-	GPIO Commit On a bit-wise basis, any bit set allows the corresponding **GPIOAFSEL** bit to be set to its alternate function. **Note:** The default register type for the **GPIOCR** register is RO for all GPIO pins with the exception of the five JTAG/SWD pins (PB7 and PC[3:0]). These five pins are currently the only GPIOs that are protected by the **GPIOCR** register. Because of this, the register type for GPIO Port B7 and GPIO Port C[3:0] is R/W. The default reset value for the **GPIOCR** register is 0x0000.00FF for all GPIO pins, with the exception of the five JTAG/SWD pins (PB7 and PC[3:0]). To ensure that the JTAG port is not accidentally programmed as a GPIO, these five pins default to non-committable. Because of this, the default reset value of **GPIOCR** for GPIO Port B is 0x0000.007F while the default reset value of GPIOCR for Port C is 0x0000.00F0.

Register 21: GPIO Peripheral Identification 4 (GPIOPeriphID4), offset 0xFD0

The **GPIOPeriphID4**, **GPIOPeriphID5**, **GPIOPeriphID6**, and **GPIOPeriphID7** registers can conceptually be treated as one 32-bit register; each register contains eight bits of the 32-bit register, used by software to identify the peripheral.

GPIO Peripheral Identification 4 (GPIOPeriphID4)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0xFD0
Type RO, reset 0x0000.0000



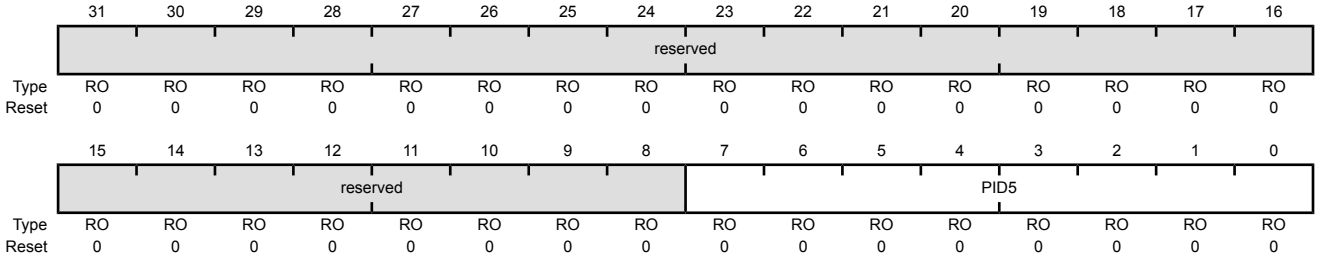
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID4	RO	0x00	GPIO Peripheral ID Register[7:0]

Register 22: GPIO Peripheral Identification 5 (GPIOPeriphID5), offset 0xFD4

The **GPIOPeriphID4**, **GPIOPeriphID5**, **GPIOPeriphID6**, and **GPIOPeriphID7** registers can conceptually be treated as one 32-bit register; each register contains eight bits of the 32-bit register, used by software to identify the peripheral.

GPIO Peripheral Identification 5 (GPIOPeriphID5)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0xFD4
Type RO, reset 0x0000.0000



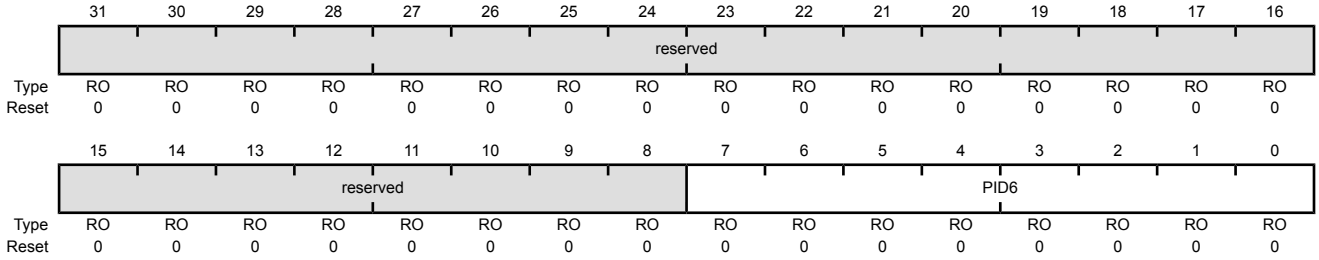
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID5	RO	0x00	GPIO Peripheral ID Register[15:8]

Register 23: GPIO Peripheral Identification 6 (GPIOPeriphID6), offset 0xFD8

The **GPIOPeriphID4**, **GPIOPeriphID5**, **GPIOPeriphID6**, and **GPIOPeriphID7** registers can conceptually be treated as one 32-bit register; each register contains eight bits of the 32-bit register, used by software to identify the peripheral.

GPIO Peripheral Identification 6 (GPIOPeriphID6)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0xFD8
Type RO, reset 0x0000.0000



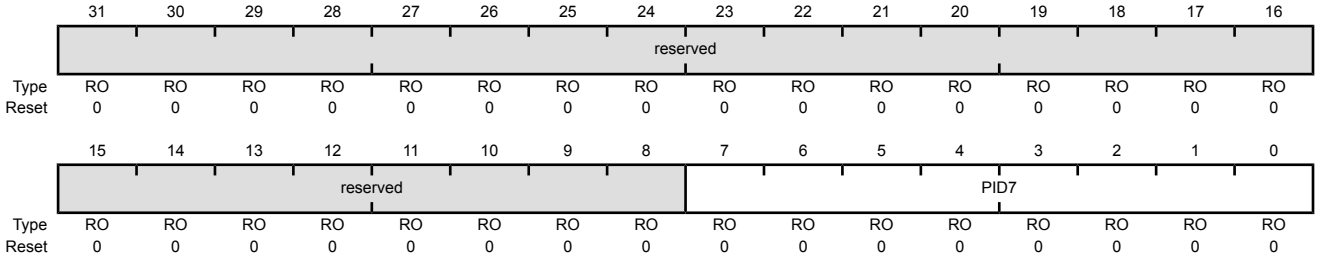
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID6	RO	0x00	GPIO Peripheral ID Register[23:16]

Register 24: GPIO Peripheral Identification 7 (GPIOPeriphID7), offset 0xFDC

The **GPIOPeriphID4**, **GPIOPeriphID5**, **GPIOPeriphID6**, and **GPIOPeriphID7** registers can conceptually be treated as one 32-bit register; each register contains eight bits of the 32-bit register, used by software to identify the peripheral.

GPIO Peripheral Identification 7 (GPIOPeriphID7)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0xFDC
Type RO, reset 0x0000.0000



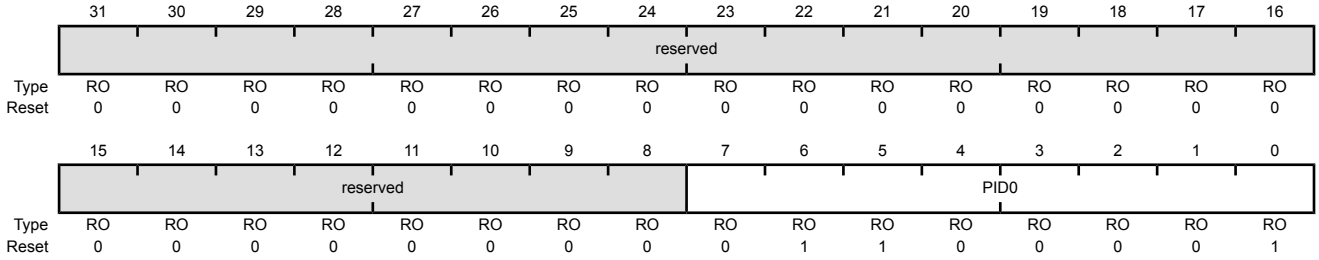
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID7	RO	0x00	GPIO Peripheral ID Register[31:24]

Register 25: GPIO Peripheral Identification 0 (GPIOPeriphID0), offset 0xFE0

The **GPIOPeriphID0**, **GPIOPeriphID1**, **GPIOPeriphID2**, and **GPIOPeriphID3** registers can conceptually be treated as one 32-bit register; each register contains eight bits of the 32-bit register, used by software to identify the peripheral.

GPIO Peripheral Identification 0 (GPIOPeriphID0)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0xFE0
Type RO, reset 0x0000.0061



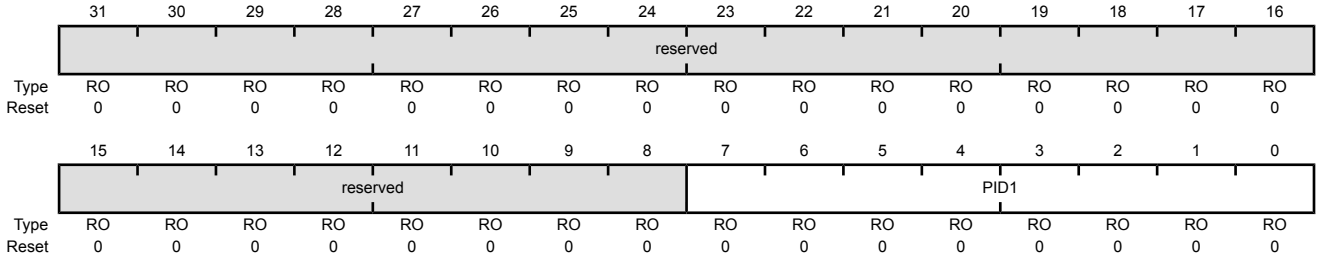
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID0	RO	0x61	GPIO Peripheral ID Register[7:0] Can be used by software to identify the presence of this peripheral.

Register 26: GPIO Peripheral Identification 1 (GPIOPeriphID1), offset 0xFE4

The **GPIOPeriphID0**, **GPIOPeriphID1**, **GPIOPeriphID2**, and **GPIOPeriphID3** registers can conceptually be treated as one 32-bit register; each register contains eight bits of the 32-bit register, used by software to identify the peripheral.

GPIO Peripheral Identification 1 (GPIOPeriphID1)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0xFE4
Type RO, reset 0x0000.0000



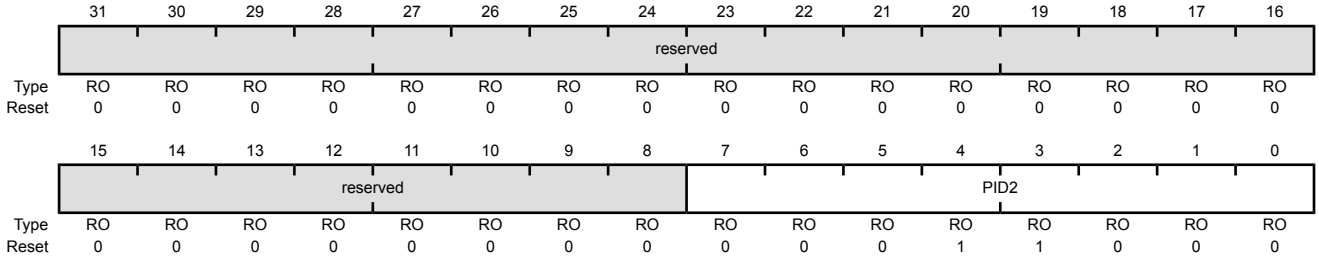
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID1	RO	0x00	GPIO Peripheral ID Register[15:8] Can be used by software to identify the presence of this peripheral.

Register 27: GPIO Peripheral Identification 2 (GPIOPeriphID2), offset 0xFE8

The **GPIOPeriphID0**, **GPIOPeriphID1**, **GPIOPeriphID2**, and **GPIOPeriphID3** registers can conceptually be treated as one 32-bit register; each register contains eight bits of the 32-bit register, used by software to identify the peripheral.

GPIO Peripheral Identification 2 (GPIOPeriphID2)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0xFE8
Type RO, reset 0x0000.0018



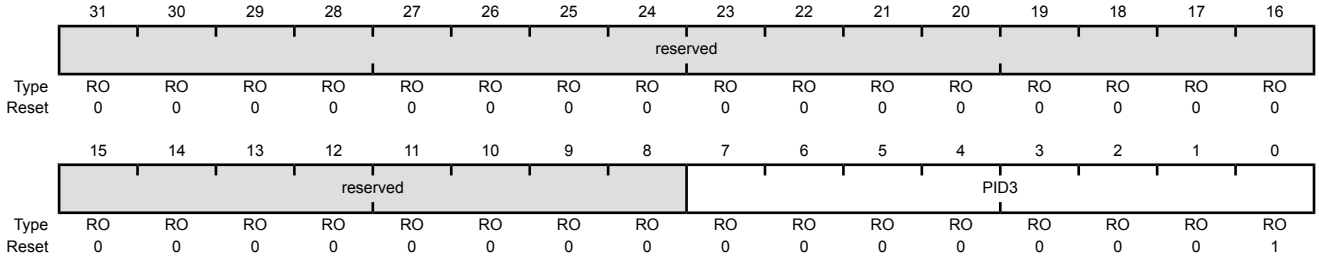
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID2	RO	0x18	GPIO Peripheral ID Register[23:16] Can be used by software to identify the presence of this peripheral.

Register 28: GPIO Peripheral Identification 3 (GPIOPeriphID3), offset 0xFEC

The **GPIOPeriphID0**, **GPIOPeriphID1**, **GPIOPeriphID2**, and **GPIOPeriphID3** registers can conceptually be treated as one 32-bit register; each register contains eight bits of the 32-bit register, used by software to identify the peripheral.

GPIO Peripheral Identification 3 (GPIOPeriphID3)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0xFEC
Type RO, reset 0x0000.0001



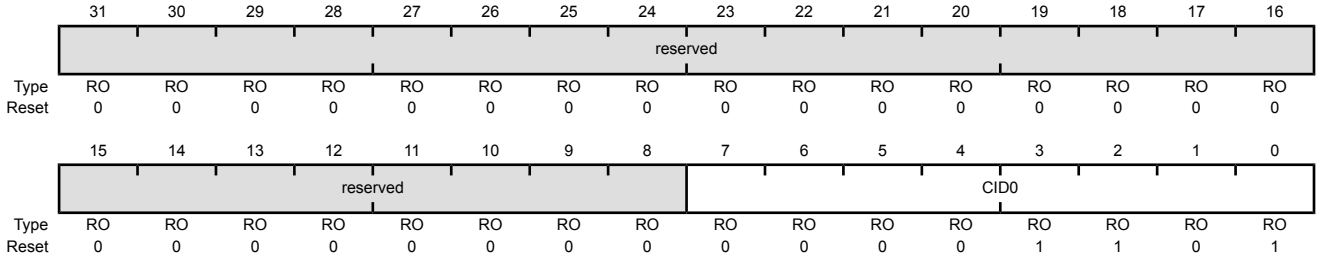
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID3	RO	0x01	GPIO Peripheral ID Register[31:24] Can be used by software to identify the presence of this peripheral.

Register 29: GPIO PrimeCell Identification 0 (GPIOPCellID0), offset 0xFF0

The **GPIOPCellID0**, **GPIOPCellID1**, **GPIOPCellID2**, and **GPIOPCellID3** registers are four 8-bit wide registers, that can conceptually be treated as one 32-bit register. The register is used as a standard cross-peripheral identification system.

GPIO PrimeCell Identification 0 (GPIOPCellID0)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0xFF0
Type RO, reset 0x0000.000D



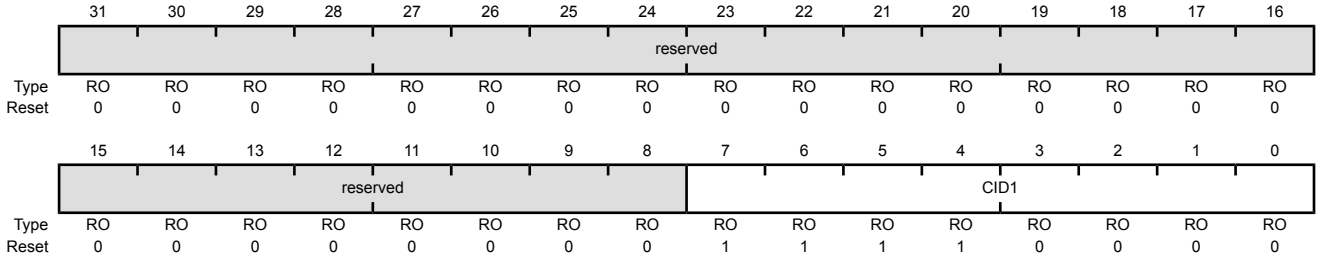
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	CID0	RO	0x0D	GPIO PrimeCell ID Register[7:0] Provides software a standard cross-peripheral identification system.

Register 30: GPIO PrimeCell Identification 1 (GPIOPCellID1), offset 0xFF4

The **GPIOPCellID0**, **GPIOPCellID1**, **GPIOPCellID2**, and **GPIOPCellID3** registers are four 8-bit wide registers, that can conceptually be treated as one 32-bit register. The register is used as a standard cross-peripheral identification system.

GPIO PrimeCell Identification 1 (GPIOPCellID1)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0xFF4
Type RO, reset 0x0000.00F0



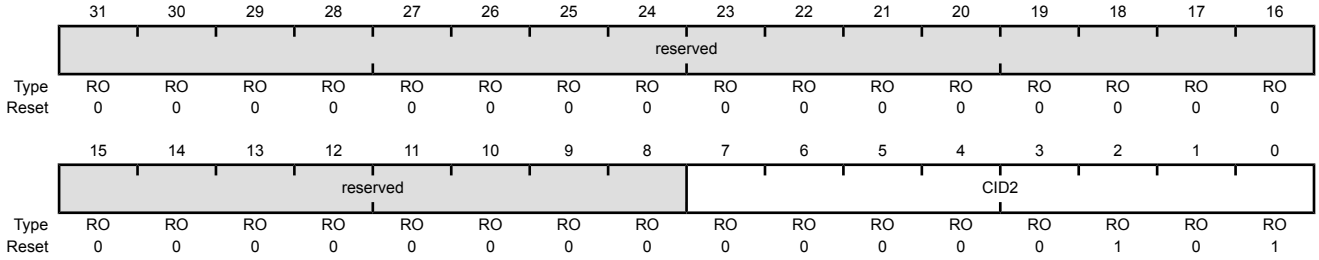
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	CID1	RO	0xF0	GPIO PrimeCell ID Register[15:8] Provides software a standard cross-peripheral identification system.

Register 31: GPIO PrimeCell Identification 2 (GPIOPCellID2), offset 0xFF8

The **GPIOPCellID0**, **GPIOPCellID1**, **GPIOPCellID2**, and **GPIOPCellID3** registers are four 8-bit wide registers, that can conceptually be treated as one 32-bit register. The register is used as a standard cross-peripheral identification system.

GPIO PrimeCell Identification 2 (GPIOPCellID2)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0xFF8
Type RO, reset 0x0000.0005



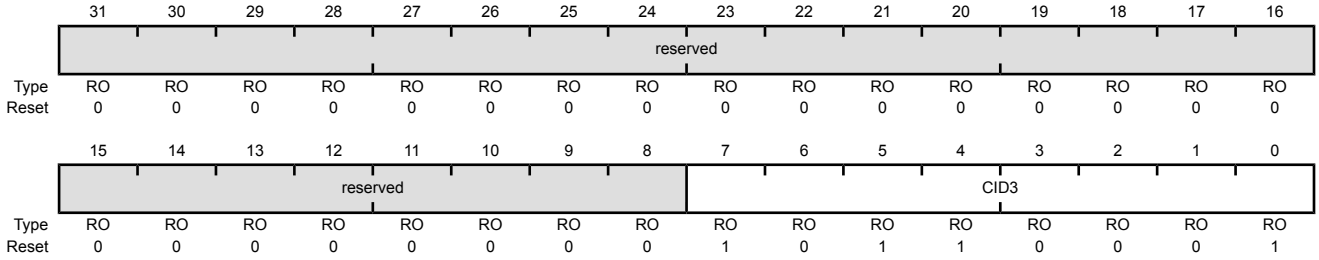
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	CID2	RO	0x05	GPIO PrimeCell ID Register[23:16] Provides software a standard cross-peripheral identification system.

Register 32: GPIO PrimeCell Identification 3 (GPIOPCellID3), offset 0xFFC

The **GPIOPCellID0**, **GPIOPCellID1**, **GPIOPCellID2**, and **GPIOPCellID3** registers are four 8-bit wide registers, that can conceptually be treated as one 32-bit register. The register is used as a standard cross-peripheral identification system.

GPIO PrimeCell Identification 3 (GPIOPCellID3)

GPIO Port A base: 0x4000.4000
GPIO Port B base: 0x4000.5000
GPIO Port C base: 0x4000.6000
GPIO Port D base: 0x4000.7000
GPIO Port E base: 0x4002.4000
GPIO Port F base: 0x4002.5000
GPIO Port G base: 0x4002.6000
Offset 0xFFC
Type RO, reset 0x0000.00B1



Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	CID3	RO	0xB1	GPIO PrimeCell ID Register[31:24] Provides software a standard cross-peripheral identification system.

9 General-Purpose Timers

Programmable timers can be used to count or time external events that drive the Timer input pins. The Stellaris® General-Purpose Timer Module (GPTM) contains three GPTM blocks (Timer0, Timer1, and Timer 2). Each GPTM block provides two 16-bit timers/counters (referred to as TimerA and TimerB) that can be configured to operate independently as timers or event counters, or configured to operate as one 32-bit timer or one 32-bit Real-Time Clock (RTC).

In addition, timers can be used to trigger analog-to-digital conversions (ADC). The ADC trigger signals from all of the general-purpose timers are ORed together before reaching the ADC module, so only one timer should be used to trigger ADC events.

The GPT Module is one timing resource available on the Stellaris® microcontrollers. Other timer resources include the System Timer (SysTick) (see “System Timer (SysTick)” on page 44) and the PWM timer in the PWM module (see “PWM Timer” on page 465).

The General-Purpose Timers provide the following features:

- Three General-Purpose Timer Modules (GPTM), each of which provides two 16-bit timers/counters. Each GPTM can be configured to operate independently:
  - As a single 32-bit timer
  - As one 32-bit Real-Time Clock (RTC) to event capture
  - For Pulse Width Modulation (PWM)
  - To trigger analog-to-digital conversions
- 32-bit Timer modes
  - Programmable one-shot timer
  - Programmable periodic timer
  - Real-Time Clock when using an external 32.768-KHz clock as the input
  - User-enabled stalling when the controller asserts CPU Halt flag during debug
  - ADC event trigger
- 16-bit Timer modes
  - General-purpose timer function with an 8-bit prescaler (for one-shot and periodic modes only)
  - Programmable one-shot timer
  - Programmable periodic timer
  - User-enabled stalling when the controller asserts CPU Halt flag during debug
  - ADC event trigger
- 16-bit Input Capture modes
  - Input edge count capture

– Input edge time capture

■ 16-bit PWM mode

– Simple PWM mode with software-programmable output inversion of the PWM signal

9.1 Block Diagram

Note: In Figure 9-1 on page 194, the specific CCP pins available depend on the Stellaris® device. See Table 9-1 on page 194 for the available CCPs.

Figure 9-1. GPTM Module Block Diagram

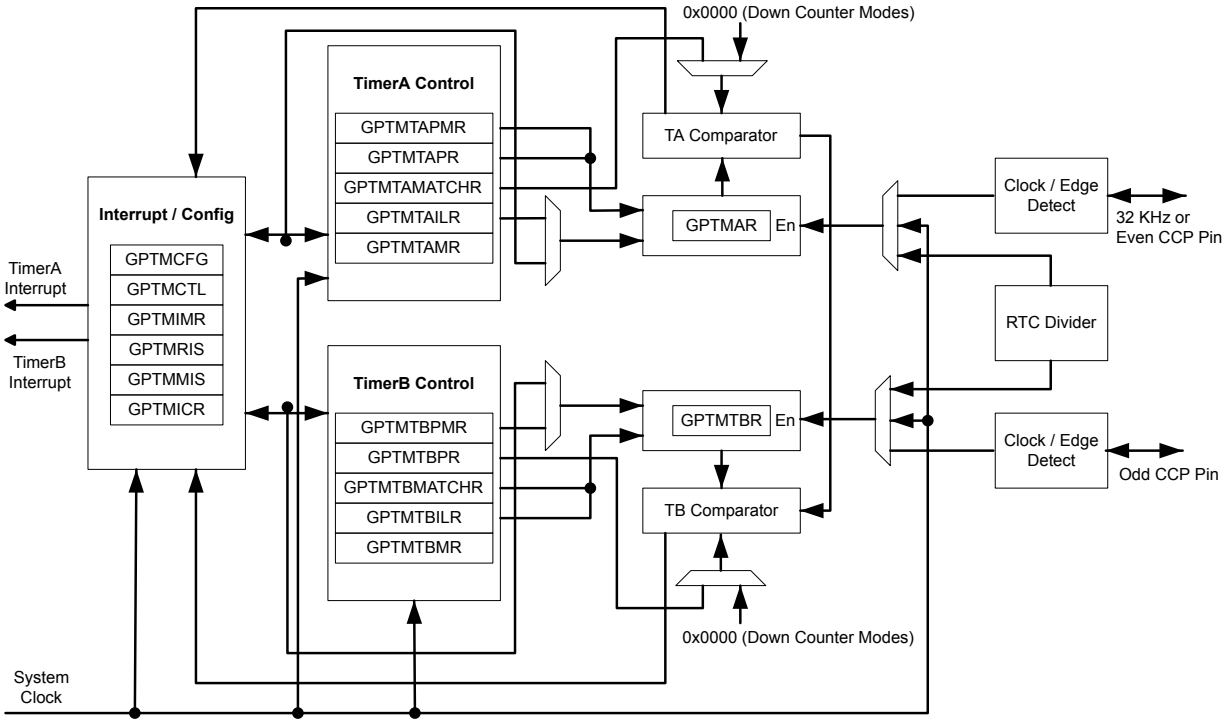


Table 9-1. Available CCP Pins

Timer	16-Bit Up/Down Counter	Even CCP Pin	Odd CCP Pin
Timer 0	TimerA	CCP0	-
	TimerB	-	CCP1
Timer 1	TimerA	CCP2	-
	TimerB	-	CCP3
Timer 2	TimerA	-	-
	TimerB	-	-

9.2 Functional Description

The main components of each GPTM block are two free-running 16-bit up/down counters (referred to as TimerA and TimerB), two 16-bit match registers, two prescaler match registers, and two 16-bit

load/initialization registers and their associated control functions. The exact functionality of each GPTM is controlled by software and configured through the register interface.

Software configures the GPTM using the **GPTM Configuration (GPTMCFG)** register (see page 205), the **GPTM TimerA Mode (GPTMTAMR)** register (see page 206), and the **GPTM TimerB Mode (GPTMTBMR)** register (see page 208). When in one of the 32-bit modes, the timer can only act as a 32-bit timer. However, when configured in 16-bit mode, the GPTM can have its two 16-bit timers configured in any combination of the 16-bit modes.

9.2.1 GPTM Reset Conditions

After reset has been applied to the GPTM module, the module is in an inactive state, and all control registers are cleared and in their default states. Counters TimerA and TimerB are initialized to 0xFFFF, along with their corresponding load registers: the **GPTM TimerA Interval Load (GPTMTAILR)** register (see page 219) and the **GPTM TimerB Interval Load (GPTMTBILR)** register (see page 220). The prescale counters are initialized to 0x00: the **GPTM TimerA Prescale (GPTMTAPR)** register (see page 223) and the **GPTM TimerB Prescale (GPTMTBPR)** register (see page 224).

9.2.2 32-Bit Timer Operating Modes

This section describes the three GPTM 32-bit timer modes (One-Shot, Periodic, and RTC) and their configuration.

The GPTM is placed into 32-bit mode by writing a 0 (One-Shot/Periodic 32-bit timer mode) or a 1 (RTC mode) to the **GPTM Configuration (GPTMCFG)** register. In both configurations, certain GPTM registers are concatenated to form pseudo 32-bit registers. These registers include:

- **GPTM TimerA Interval Load (GPTMTAILR)** register [15:0], see page 219
- **GPTM TimerB Interval Load (GPTMTBILR)** register [15:0], see page 220
- **GPTM TimerA (GPTMTAR)** register [15:0], see page 227
- **GPTM TimerB (GPTMTBR)** register [15:0], see page 228

In the 32-bit modes, the GPTM translates a 32-bit write access to **GPTMTAILR** into a write access to both **GPTMTAILR** and **GPTMTBILR**. The resulting word ordering for such a write operation is:

```
GPTMTBILR[15:0]:GPTMTAILR[15:0]
```

Likewise, a read access to **GPTMTAR** returns the value:

```
GPTMTBR[15:0]:GPTMTAR[15:0]
```

9.2.2.1 32-Bit One-Shot/Periodic Timer Mode

In 32-bit one-shot and periodic timer modes, the concatenated versions of the TimerA and TimerB registers are configured as a 32-bit down-counter. The selection of one-shot or periodic mode is determined by the value written to the TAMR field of the **GPTM TimerA Mode (GPTMTAMR)** register (see page 206), and there is no need to write to the **GPTM TimerB Mode (GPTMTBMR)** register.

When software writes the TAEN bit in the **GPTM Control (GPTMCTL)** register (see page 210), the timer begins counting down from its preloaded value. Once the 0x0000.0000 state is reached, the timer reloads its start value from the concatenated **GPTMTAILR** on the next cycle. If configured to be a one-shot timer, the timer stops counting and clears the TAEN bit in the **GPTMCTL** register. If configured as a periodic timer, it continues counting.

In addition to reloading the count value, the GPTM generates interrupts and triggers when it reaches the 0x000.0000 state. The GPTM sets the TATORIS bit in the **GPTM Raw Interrupt Status (GPTMRIS)** register (see page 215), and holds it until it is cleared by writing the **GPTM Interrupt Clear (GPTMICR)** register (see page 217). If the time-out interrupt is enabled in the **GPTM Interrupt Mask (GPTIMR)** register (see page 213), the GPTM also sets the TATOMIS bit in the **GPTM Masked Interrupt Status (GPTMMIS)** register (see page 216). The ADC trigger is enabled by setting the TAOTE bit in **GPTMCTL**.

If software reloads the **GPTMTAILR** register while the counter is running, the counter loads the new value on the next clock cycle and continues counting from the new value.

If the TASTALL bit in the **GPTMCTL** register is set, the timer freezes counting while the processor is halted by the debugger. The timer resumes counting when the processor resumes execution.

9.2.2.2 32-Bit Real-Time Clock Timer Mode

In Real-Time Clock (RTC) mode, the concatenated versions of the TimerA and TimerB registers are configured as a 32-bit up-counter. When RTC mode is selected for the first time, the counter is loaded with a value of 0x0000.0001. All subsequent load values must be written to the **GPTM TimerA Match (GPTMTAMATCHR)** register (see page 221) by the controller.

The input clock on an even CCP input is required to be 32.768 KHz in RTC mode. The clock signal is then divided down to a 1 Hz rate and is passed along to the input of the 32-bit counter.

When software writes the TAEN bit inthe **GPTMCTL** register, the counter starts counting up from its preloaded value of 0x0000.0001. When the current count value matches the preloaded value in the **GPTMTAMATCHR** register, it rolls over to a value of 0x0000.0000 and continues counting until either a hardware reset, or it is disabled by software (clearing the TAEN bit). When a match occurs, the GPTM asserts the RTCRIS bit in **GPTMRIS**. If the RTC interrupt is enabled in **GPTIMR**, the GPTM also sets the RTCMIS bit in **GPTMISR** and generates a controller interrupt. The status flags are cleared by writing the RTCCINT bit in **GPTMICR**.

If the TASTALL and/or TBSTALL bits in the **GPTMCTL** register are set, the timer does not freeze if the RTCEN bit is set in **GPTMCTL**.

9.2.3 16-Bit Timer Operating Modes

The GPTM is placed into global 16-bit mode by writing a value of 0x4 to the **GPTM Configuration (GPTMCFG)** register (see page 205). This section describes each of the GPTM 16-bit modes of operation. TimerA and TimerB have identical modes, so a single description is given using an **n** to reference both.

9.2.3.1 16-Bit One-Shot/Periodic Timer Mode

In 16-bit one-shot and periodic timer modes, the timer is configured as a 16-bit down-counter with an optional 8-bit prescaler that effectively extends the counting range of the timer to 24 bits. The selection of one-shot or periodic mode is determined by the value written to the TnMR field of the **GPTMTnMR** register. The optional prescaler is loaded into the **GPTM Timern Prescale (GPTMTnPR)** register.

When software writes the TnEN bit in the **GPTMCTL** register, the timer begins counting down from its preloaded value. Once the 0x0000 state is reached, the timer reloads its start value from **GPTMTnILR** and **GPTMTnPR** on the next cycle. If configured to be a one-shot timer, the timer stops counting and clears the TnEN bit in the **GPTMCTL** register. If configured as a periodic timer, it continues counting.

In addition to reloading the count value, the timer generates interrupts and triggers when it reaches the 0x0000 state. The GPTM sets the TnTORIS bit in the **GPTMRIS** register, and holds it until it is cleared by writing the **GPTMICR** register. If the time-out interrupt is enabled in **GPTIMR**, the GPTM also sets the TnTOMIS bit in **GPTMISR** and generates a controller interrupt. The ADC trigger is enabled by setting the TnOTE bit in the **GPTMCTL** register.

If software reloads the **GPTMTAILR** register while the counter is running, the counter loads the new value on the next clock cycle and continues counting from the new value.

If the TnSTALL bit in the **GPTMCTL** register is set, the timer freezes counting while the processor is halted by the debugger. The timer resumes counting when the processor resumes execution.

The following example shows a variety of configurations for a 16-bit free running timer while using the prescaler. All values assume a 50-MHz clock with Tc=20 ns (clock period).

Table 9-2. 16-Bit Timer With Prescaler Configurations

Prescale	#Clock (T c)[a]	Max Time	Units
00000000	1	1.3107	mS
00000001	2	2.6214	mS
00000010	3	3.9322	mS
----------	--	--	--
11111101	254	332.9229	mS
11111110	255	334.2336	mS
11111111	256	335.5443	mS

a. Tc is the clock period.

9.2.3.2 16-Bit Input Edge Count Mode

Note: For rising-edge detection, the input signal must be High for at least two system clock periods following the rising edge. Similarly, for falling-edge detection, the input signal must be Low for at least two system clock periods following the falling edge. Based on this criteria, the maximum input frequency for edge detection is 1/4 of the system frequency.

Note: The prescaler is not available in 16-Bit Input Edge Count mode.

In Edge Count mode, the timer is configured as a down-counter capable of capturing three types of events: rising edge, falling edge, or both. To place the timer in Edge Count mode, the TnCMR bit of the **GPTMTnMR** register must be set to 0. The type of edge that the timer counts is determined by the TnEVENT fields of the **GPTMCTL** register. During initialization, the **GPTM Timern Match (GPTMTnMATCHR)** register is configured so that the difference between the value in the **GPTMTnILR** register and the **GPTMTnMATCHR** register equals the number of edge events that must be counted.

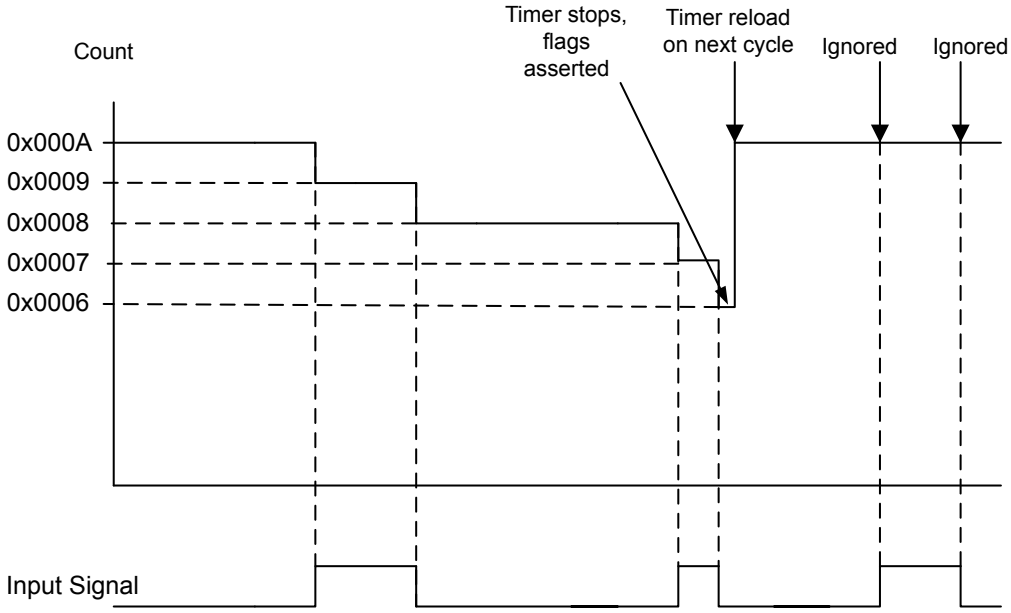
When software writes the TnEN bit in the **GPTM Control (GPTMCTL)** register, the timer is enabled for event capture. Each input event on the CCP pin decrements the counter by 1 until the event count matches **GPTMTnMATCHR**. When the counts match, the GPTM asserts the CnMRIS bit in the **GPTMRIS** register (and the CnMMIS bit, if the interrupt is not masked).

The counter is then reloaded using the value in **GPTMTnILR**, and stopped since the GPTM automatically clears the TnEN bit in the **GPTMCTL** register. Once the event count has been reached, all further events are ignored until TnEN is re-enabled by software.

Figure 9-2 on page 198 shows how input edge count mode works. In this case, the timer start value is set to **GPTMnILR** =0x000A and the match value is set to **GPTMnMATCHR** =0x0006 so that four edge events are counted. The counter is configured to detect both edges of the input signal.

Note that the last two edges are not counted since the timer automatically clears the TnEN bit after the current count matches the value in the **GPTMnMR** register.

Figure 9-2. 16-Bit Input Edge Count Mode Example



9.2.3.3 16-Bit Input Edge Time Mode

Note: For rising-edge detection, the input signal must be High for at least two system clock periods following the rising edge. Similarly, for falling edge detection, the input signal must be Low for at least two system clock periods following the falling edge. Based on this criteria, the maximum input frequency for edge detection is 1/4 of the system frequency.

Note: The prescaler is not available in 16-Bit Input Edge Time mode.

In Edge Time mode, the timer is configured as a free-running down-counter initialized to the value loaded in the **GPTMTnILR** register (or 0xFFFF at reset). This mode allows for event capture of either rising or falling edges, but not both. The timer is placed into Edge Time mode by setting the TnCMR bit in the **GPTMTnMR** register, and the type of event that the timer captures is determined by the TnEVENT fields of the **GPTMCnTL** register.

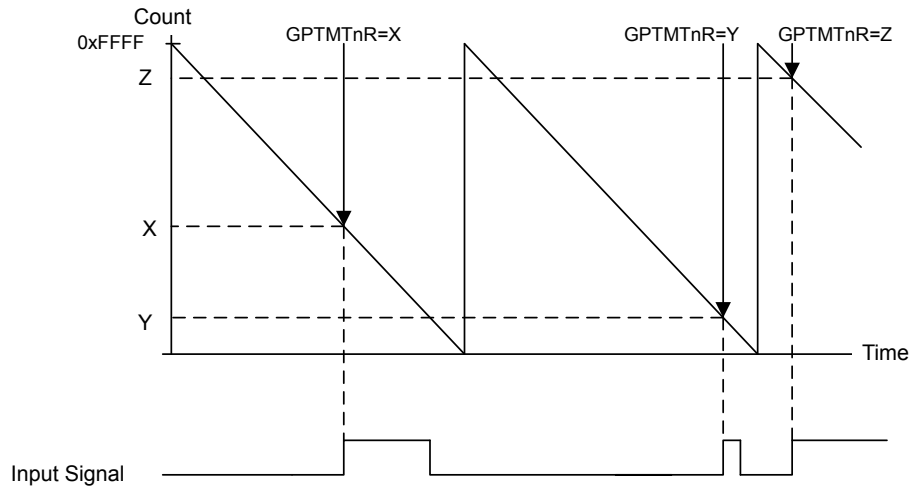
When software writes the TnEN bit in the **GPTMCTL** register, the timer is enabled for event capture. When the selected input event is detected, the current Tn counter value is captured in the **GPTMTnR** register and is available to be read by the controller. The GPTM then asserts the CnERIS bit (and the CnEMIS bit, if the interrupt is not masked).

After an event has been captured, the timer does not stop counting. It continues to count until the TnEN bit is cleared. When the timer reaches the 0x0000 state, it is reloaded with the value from the **GPTMnILR** register.

Figure 9-3 on page 199 shows how input edge timing mode works. In the diagram, it is assumed that the start value of the timer is the default value of 0xFFFF, and the timer is configured to capture rising edge events.

Each time a rising edge event is detected, the current count value is loaded into the **GPTMTnR** register, and is held there until another rising edge is detected (at which point the new count value is loaded into **GPTMTnR**).

Figure 9-3. 16-Bit Input Edge Time Mode Example



9.2.3.4 16-Bit PWM Mode

Note: The prescaler is not available in 16-Bit PWM mode.

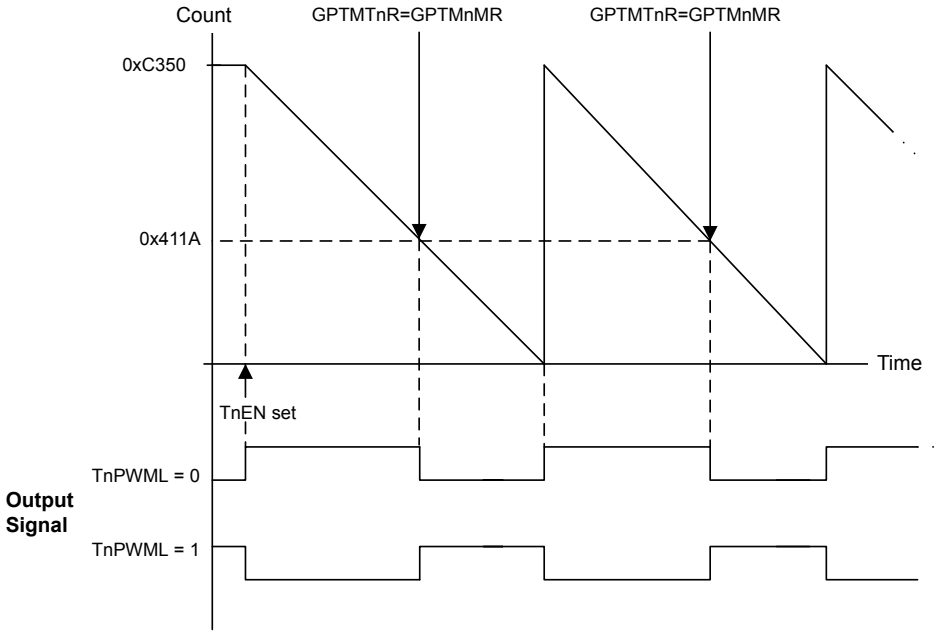
The GPTM supports a simple PWM generation mode. In PWM mode, the timer is configured as a down-counter with a start value (and thus period) defined by **GPTMTnILR**. PWM mode is enabled with the **GPTMTnMR** register by setting the TnAMS bit to 0x1, the TnCMR bit to 0x0, and the TnMR field to 0x2.

When software writes the TnEN bit in the **GPTMCTL** register, the counter begins counting down until it reaches the 0x0000 state. On the next counter cycle, the counter reloads its start value from **GPTMTnILR** and continues counting until disabled by software clearing the TnEN bit in the **GPTMCTL** register. No interrupts or status bits are asserted in PWM mode.

The output PWM signal asserts when the counter is at the value of the **GPTMTnILR** register (its start state), and is deasserted when the counter value equals the value in the **GPTM Timern Match Register (GPTMnMATCHR)**. Software has the capability of inverting the output PWM signal by setting the TnPWML bit in the **GPTMCTL** register.

Figure 9-4 on page 200 shows how to generate an output PWM with a 1-ms period and a 66% duty cycle assuming a 50-MHz input clock and **TnPWML** =0 (duty cycle would be 33% for the **TnPWML** =1 configuration). For this example, the start value is **GPTMnIRL**=0xC350 and the match value is **GPTMnMR**=0x411A.

Figure 9-4. 16-Bit PWM Mode Example



9.3 Initialization and Configuration

To use the general-purpose timers, the peripheral clock must be enabled by setting the TIMER0, TIMER1, and TIMER2 bits in the **RCGC1** register.

This section shows module initialization and configuration examples for each of the supported timer modes.

9.3.1 32-Bit One-Shot/Periodic Timer Mode

The GPTM is configured for 32-bit One-Shot and Periodic modes by the following sequence:

1. Ensure the timer is disabled (the TAEN bit in the **GPTMCTL** register is cleared) before making any changes.
2. Write the **GPTM Configuration Register (GPTMCFG)** with a value of 0x0.
3. Set the TAMR field in the **GPTM TimerA Mode Register (GPTMTAMR)**:
   a. Write a value of 0x1 for One-Shot mode.
   b. Write a value of 0x2 for Periodic mode.
4. Load the start value into the **GPTM TimerA Interval Load Register (GPTMTAILR)**.
5. If interrupts are required, set the TATOIM bit in the **GPTM Interrupt Mask Register (GPTMIMR)**.
6. Set the TAEN bit in the **GPTMCTL** register to enable the timer and start counting.

7. Poll the TATORIS bit in the **GPTMRIS** register or wait for the interrupt to be generated (if enabled). In both cases, the status flags are cleared by writing a 1 to the TATOCINT bit of the **GPTM Interrupt Clear Register (GPTMICR)**.

In One-Shot mode, the timer stops counting after step 7 on page 201. To re-enable the timer, repeat the sequence. A timer configured in Periodic mode does not stop counting after it times out.

9.3.2 32-Bit Real-Time Clock (RTC) Mode

To use the RTC mode, the timer must have a 32.768-KHz input signal on an even CCP input. To enable the RTC feature, follow these steps:

1. Ensure the timer is disabled (the TAEN bit is cleared) before making any changes.
2. Write the **GPTM Configuration Register (GPTMCFG)** with a value of 0x1.
3. Write the desired match value to the **GPTM TimerA Match Register (GPTMTAMATCHR)**.
4. Set/clear the RTCEN bit in the **GPTM Control Register (GPTMCTL)** as desired.
5. If interrupts are required, set the RTCIM bit in the **GPTM Interrupt Mask Register (GPTMIMR)**.
6. Set the TAEN bit in the **GPTMCTL** register to enable the timer and start counting.

When the timer count equals the value in the **GPTMTAMATCHR** register, the counter is re-loaded with 0x0000.0000 and begins counting. If an interrupt is enabled, it does not have to be cleared.

9.3.3 16-Bit One-Shot/Periodic Timer Mode

A timer is configured for 16-bit One-Shot and Periodic modes by the following sequence:

1. Ensure the timer is disabled (the TnEN bit is cleared) before making any changes.
2. Write the **GPTM Configuration Register (GPTMCFG)** with a value of 0x4.
3. Set the TnMR field in the **GPTM Timer Mode (GPTMTnMR)** register:
   a. Write a value of 0x1 for One-Shot mode.
   b. Write a value of 0x2 for Periodic mode.
4. If a prescaler is to be used, write the prescale value to the **GPTM Timern Prescale Register (GPTMTnPR)**.
5. Load the start value into the **GPTM Timer Interval Load Register (GPTMTnILR)**.
6. If interrupts are required, set the TnTOIM bit in the **GPTM Interrupt Mask Register (GPTMIMR)**.
7. Set the TnEN bit in the **GPTM Control Register (GPTMCTL)** to enable the timer and start counting.
8. Poll the TnTORIS bit in the **GPTMRIS** register or wait for the interrupt to be generated (if enabled). In both cases, the status flags are cleared by writing a 1 to the TnTOCINT bit of the **GPTM Interrupt Clear Register (GPTMICR)**.

In One-Shot mode, the timer stops counting after step 8 on page 201. To re-enable the timer, repeat the sequence. A timer configured in Periodic mode does not stop counting after it times out.

9.3.4 16-Bit Input Edge Count Mode

A timer is configured to Input Edge Count mode by the following sequence:

1. Ensure the timer is disabled (the TnEN bit is cleared) before making any changes.
2. Write the **GPTM Configuration (GPTMCFG)** register with a value of 0x4.
3. In the **GPTM Timer Mode (GPTMTnMR)** register, write the TnCMR field to 0x0 and the TnMR field to 0x3.
4. Configure the type of event(s) that the timer captures by writing the TnEVENT field of the **GPTM Control (GPTMCTL)** register.
5. Load the timer start value into the **GPTM Timern Interval Load (GPTMTnILR)** register.
6. Load the desired event count into the **GPTM Timern Match (GPTMTnMATCHR)** register.
7. If interrupts are required, set the CnMIM bit in the **GPTM Interrupt Mask (GPTMIMR)** register.
8. Set the TnEN bit in the **GPTMCTL** register to enable the timer and begin waiting for edge events.
9. Poll the CnMRIS bit in the **GPTMRIS** register or wait for the interrupt to be generated (if enabled). In both cases, the status flags are cleared by writing a 1 to the CnMCINT bit of the **GPTM Interrupt Clear (GPTMICR)** register.

In Input Edge Count Mode, the timer stops after the desired number of edge events has been detected. To re-enable the timer, ensure that the TnEN bit is cleared and repeat step 4 on page 202 through step 9 on page 202.

9.3.5 16-Bit Input Edge Timing Mode

A timer is configured to Input Edge Timing mode by the following sequence:

1. Ensure the timer is disabled (the TnEN bit is cleared) before making any changes.
2. Write the **GPTM Configuration (GPTMCFG)** register with a value of 0x4.
3. In the **GPTM Timer Mode (GPTMTnMR)** register, write the TnCMR field to 0x1 and the TnMR field to 0x3.
4. Configure the type of event that the timer captures by writing the TnEVENT field of the **GPTM Control (GPTMCTL)** register.
5. Load the timer start value into the **GPTM Timern Interval Load (GPTMTnILR)** register.
6. If interrupts are required, set the CnEIM bit in the **GPTM Interrupt Mask (GPTMIMR)** register.
7. Set the TnEN bit in the **GPTM Control (GPTMCTL)** register to enable the timer and start counting.
8. Poll the CnERIS bit in the **GPTMRIS** register or wait for the interrupt to be generated (if enabled). In both cases, the status flags are cleared by writing a 1 to the CnECINT bit of the **GPTM**

Interrupt Clear (GPTMICR) register. The time at which the event happened can be obtained by reading the **GPTM Timern (GPTMTnR)** register.

In Input Edge Timing mode, the timer continues running after an edge event has been detected, but the timer interval can be changed at any time by writing the **GPTMTnILR** register. The change takes effect at the next cycle after the write.

9.3.6 16-Bit PWM Mode

A timer is configured to PWM mode using the following sequence:

1. Ensure the timer is disabled (the TnEN bit is cleared) before making any changes.
2. Write the **GPTM Configuration (GPTMCFG)** register with a value of 0x4.
3. In the **GPTM Timer Mode (GPTMTnMR)** register, set the TnAMS bit to 0x1, the TnCMR bit to 0x0, and the TnMR field to 0x2.
4. Configure the output state of the PWM signal (whether or not it is inverted) in the TnEVENT field of the **GPTM Control (GPTMCTL)** register.
5. Load the timer start value into the **GPTM Timern Interval Load (GPTMTnILR)** register.
6. Load the **GPTM Timern Match (GPTMTnMATCHR)** register with the desired value.
7. Set the TnEN bit in the **GPTM Control (GPTMCTL)** register to enable the timer and begin generation of the output PWM signal.

In PWM Timing mode, the timer continues running after the PWM signal has been generated. The PWM period can be adjusted at any time by writing the **GPTMTnILR** register, and the change takes effect at the next cycle after the write.

9.4 Register Map

Table 9-3 on page 203 lists the GPTM registers. The offset listed is a hexadecimal increment to the register's address, relative to that timer's base address:

- Timer0: 0x4003.0000
- Timer1: 0x4003.1000
- Timer2: 0x4003.2000

Table 9-3. Timers Register Map

Offset	Name	Type	Reset	Description	See page
0x000	GPTMCFG	R/W	0x0000.0000	GPTM Configuration	205
0x004	GPTMTAMR	R/W	0x0000.0000	GPTM TimerA Mode	206
0x008	GPTMTBMR	R/W	0x0000.0000	GPTM TimerB Mode	208
0x00C	GPTMCTL	R/W	0x0000.0000	GPTM Control	210
0x018	GPTMIMR	R/W	0x0000.0000	GPTM Interrupt Mask	213
0x01C	GPTMRIS	RO	0x0000.0000	GPTM Raw Interrupt Status	215

Table 9-3. Timers Register Map *(continued)*

Offset	Name	Type	Reset	Description	See page
0x020	GPTMMIS	RO	0x0000.0000	GPTM Masked Interrupt Status	216
0x024	GPTMICR	W1C	0x0000.0000	GPTM Interrupt Clear	217
0x028	GPTMTAILR	R/W	0xFFFF.FFFF	GPTM TimerA Interval Load	219
0x02C	GPTMTBILR	R/W	0x0000.FFFF	GPTM TimerB Interval Load	220
0x030	GPTMTAMATCHR	R/W	0xFFFF.FFFF	GPTM TimerA Match	221
0x034	GPTMTBMATCHR	R/W	0x0000.FFFF	GPTM TimerB Match	222
0x038	GPTMTAPR	R/W	0x0000.0000	GPTM TimerA Prescale	223
0x03C	GPTMTBPR	R/W	0x0000.0000	GPTM TimerB Prescale	224
0x040	GPTMTAPMR	R/W	0x0000.0000	GPTM TimerA Prescale Match	225
0x044	GPTMTBPMR	R/W	0x0000.0000	GPTM TimerB Prescale Match	226
0x048	GPTMTAR	RO	0xFFFF.FFFF	GPTM TimerA	227
0x04C	GPTMTBR	RO	0x0000.FFFF	GPTM TimerB	228

9.5 Register Descriptions

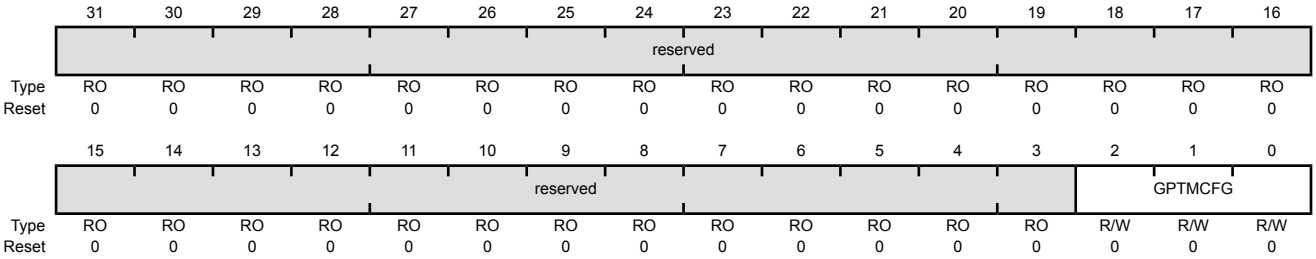
The remainder of this section lists and describes the GPTM registers, in numerical order by address offset.

Register 1: GPTM Configuration (GPTMCFG), offset 0x000

This register configures the global operation of the GPTM module. The value written to this register determines whether the GPTM is in 32- or 16-bit mode.

GPTM Configuration (GPTMCFG)

Timer0 base: 0x4003.0000
Timer1 base: 0x4003.1000
Timer2 base: 0x4003.2000
Offset 0x000
Type R/W, reset 0x0000.0000



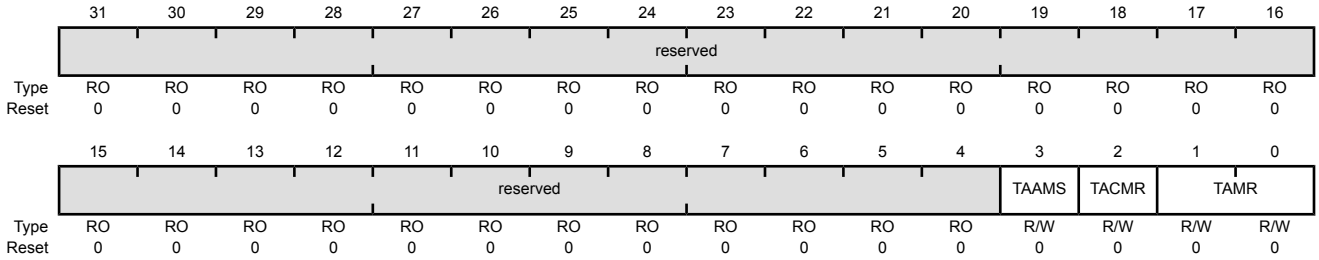
Bit/Field	Name	Type	Reset	Description
31:3	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
2:0	GPTMCFG	R/W	0x0	GPTM Configuration The GPTMCFG values are defined as follows: Value Description 0x0 32-bit timer configuration. 0x1 32-bit real-time clock (RTC) counter configuration. 0x2 Reserved 0x3 Reserved 0x4-0x7 16-bit timer configuration, function is controlled by bits 1:0 of **GPTMTAMR** and **GPTMTBMR**.

Register 2: GPTM TimerA Mode (GPTMTAMR), offset 0x004

This register configures the GPTM based on the configuration selected in the **GPTMCFG** register. When in 16-bit PWM mode, set the TAAMS bit to 0x1, the TACMR bit to 0x0, and the TAMR field to 0x2.

GPTM TimerA Mode (GPTMTAMR)

Timer0 base: 0x4003.0000
Timer1 base: 0x4003.1000
Timer2 base: 0x4003.2000
Offset 0x004
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:4	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
3	TAAMS	R/W	0	GPTM TimerA Alternate Mode Select The TAAMS values are defined as follows: Value Description 0 Capture mode is enabled. 1 PWM mode is enabled. **Note:** To enable PWM mode, you must also clear the TACMR bit and set the TAMR field to 0x2.
2	TACMR	R/W	0	GPTM TimerA Capture Mode The TACMR values are defined as follows: Value Description 0 Edge-Count mode 1 Edge-Time mode

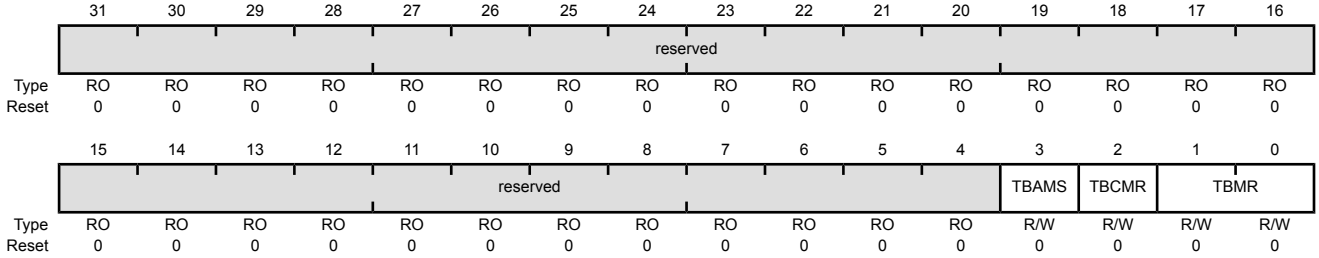
Bit/Field	Name	Type	Reset	Description
1:0	TAMR	R/W	0x0	GPTM TimerA Mode The TAMR values are defined as follows: Value Description 0x0 Reserved 0x1 One-Shot Timer mode 0x2 Periodic Timer mode 0x3 Capture mode The Timer mode is based on the timer configuration defined by bits 2:0 in the **GPTMCFG** register (16-or 32-bit). In 16-bit timer configuration, TAMR controls the 16-bit timer modes for TimerA. In 32-bit timer configuration, this register controls the mode and the contents of **GPTMTBMR** are ignored.

Register 3: GPTM TimerB Mode (GPTMTBMR), offset 0x008

This register configures the GPTM based on the configuration selected in the **GPTMCFG** register. When in 16-bit PWM mode, set the TBAMS bit to 0x1, the TBCMR bit to 0x0, and the TBMR field to 0x2.

GPTM TimerB Mode (GPTMTBMR)

Timer0 base: 0x4003.0000
Timer1 base: 0x4003.1000
Timer2 base: 0x4003.2000
Offset 0x008
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:4	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
3	TBAMS	R/W	0	GPTM TimerB Alternate Mode Select The TBAMS values are defined as follows: Value Description 0 Capture mode is enabled. 1 PWM mode is enabled. **Note:** To enable PWM mode, you must also clear the TBCMR bit and set the TBMR field to 0x2.
2	TBCMR	R/W	0	GPTM TimerB Capture Mode The TBCMR values are defined as follows: Value Description 0 Edge-Count mode 1 Edge-Time mode

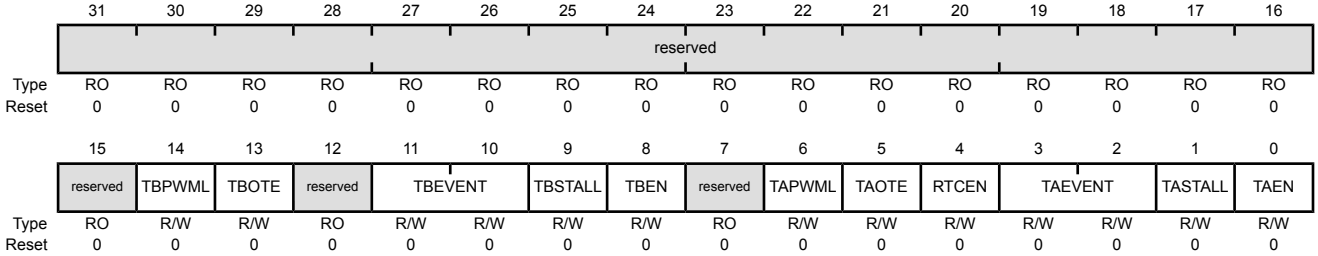
Bit/Field	Name	Type	Reset	Description
1:0	TBMR	R/W	0x0	GPTM TimerB Mode The TBMR values are defined as follows: Value Description 0x0 Reserved 0x1 One-Shot Timer mode 0x2 Periodic Timer mode 0x3 Capture mode The timer mode is based on the timer configuration defined by bits 2:0 in the **GPTMCFG** register. In 16-bit timer configuration, these bits control the 16-bit timer modes for TimerB. In 32-bit timer configuration, this register's contents are ignored and **GPTMTAMR** is used.

Register 4: GPTM Control (GPTMCTL), offset 0x00C

This register is used alongside the **GPTMCFG** and **GMTMTnMR** registers to fine-tune the timer configuration, and to enable other features such as timer stall and the output trigger. The output trigger can be used to initiate transfers on the ADC module.

GPTM Control (GPTMCTL)

Timer0 base: 0x4003.0000
Timer1 base: 0x4003.1000
Timer2 base: 0x4003.2000
Offset 0x00C
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:15	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
14	TBPWML	R/W	0	GPTM TimerB PWM Output Level The TBPWML values are defined as follows: Value Description 0 Output is unaffected. 1 Output is inverted.
13	TBOTE	R/W	0	GPTM TimerB Output Trigger Enable The TBOTE values are defined as follows: Value Description 0 The output TimerB ADC trigger is disabled. 1 The output TimerB ADC trigger is enabled. In addition, the ADC must be enabled and the timer selected as a trigger source with the EMn bit in the **ADCEMUX** register (see page 268).
12	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Bit/Field	Name	Type	Reset	Description
11:10	TBEVENT	R/W	0x0	GPTM TimerB Event Mode The TBEVENT values are defined as follows: Value Description 0x0 Positive edge 0x1 Negative edge 0x2 Reserved 0x3 Both edges
9	TBSTALL	R/W	0	GPTM Timer B Stall Enable The TBSTALL values are defined as follows: Value Description 0 Timer B continues counting while the processor is halted by the debugger. 1 Timer B freezes counting while the processor is halted by the debugger. If the processor is executing normally, the TBSTALL bit is ignored.
8	TBEN	R/W	0	GPTM TimerB Enable The TBEN values are defined as follows: Value Description 0 TimerB is disabled. 1 TimerB is enabled and begins counting or the capture logic is enabled based on the **GPTMCFG** register.
7	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
6	TAPWML	R/W	0	GPTM TimerA PWM Output Level The TAPWML values are defined as follows: Value Description 0 Output is unaffected. 1 Output is inverted.
5	TAOTE	R/W	0	GPTM TimerA Output Trigger Enable The TAOTE values are defined as follows: Value Description 0 The output TimerA ADC trigger is disabled. 1 The output TimerA ADC trigger is enabled. In addition, the ADC must be enabled and the timer selected as a trigger source with the EMn bit in the **ADCEMUX** register (see page 268).

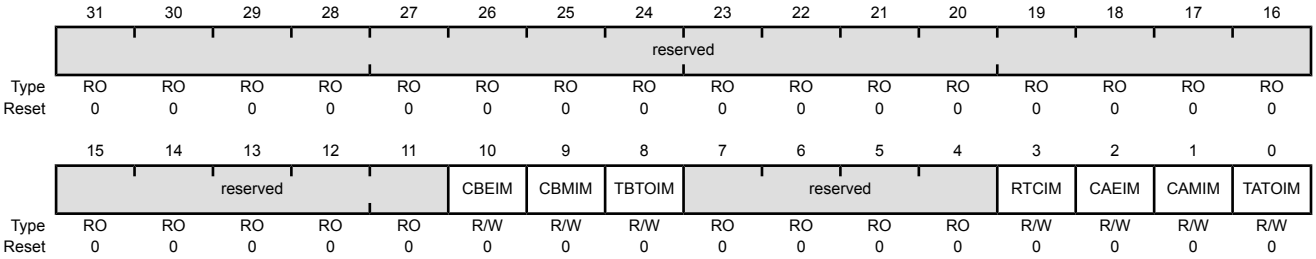
Bit/Field	Name	Type	Reset	Description
4	RTCEN	R/W	0	GPTM RTC Enable The RTCEN values are defined as follows: Value Description 0 RTC counting is disabled. 1 RTC counting is enabled.
3:2	TAEVENT	R/W	0x0	GPTM TimerA Event Mode The TAEVENT values are defined as follows: Value Description 0x0 Positive edge 0x1 Negative edge 0x2 Reserved 0x3 Both edges
1	TASTALL	R/W	0	GPTM Timer A Stall Enable The TASTALL values are defined as follows: Value Description 0 Timer A continues counting while the processor is halted by the debugger. 1 Timer A freezes counting while the processor is halted by the debugger. If the processor is executing normally, the TASTALL bit is ignored.
0	TAEN	R/W	0	GPTM TimerA Enable The TAEN values are defined as follows: Value Description 0 TimerA is disabled. 1 TimerA is enabled and begins counting or the capture logic is enabled based on the **GPTMCFG** register.

Register 5: GPTM Interrupt Mask (GPTMIMR), offset 0x018

This register allows software to enable/disable GPTM controller-level interrupts. Writing a 1 enables the interrupt, while writing a 0 disables it.

GPTM Interrupt Mask (GPTMIMR)

Timer0 base: 0x4003.0000
Timer1 base: 0x4003.1000
Timer2 base: 0x4003.2000
Offset 0x018
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:11	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
10	CBEIM	R/W	0	GPTM CaptureB Event Interrupt Mask The CBEIM values are defined as follows: Value Description 0 Interrupt is disabled. 1 Interrupt is enabled.
9	CBMIM	R/W	0	GPTM CaptureB Match Interrupt Mask The CBMIM values are defined as follows: Value Description 0 Interrupt is disabled. 1 Interrupt is enabled.
8	TBTOIM	R/W	0	GPTM TimerB Time-Out Interrupt Mask The TBTOIM values are defined as follows: Value Description 0 Interrupt is disabled. 1 Interrupt is enabled.
7:4	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

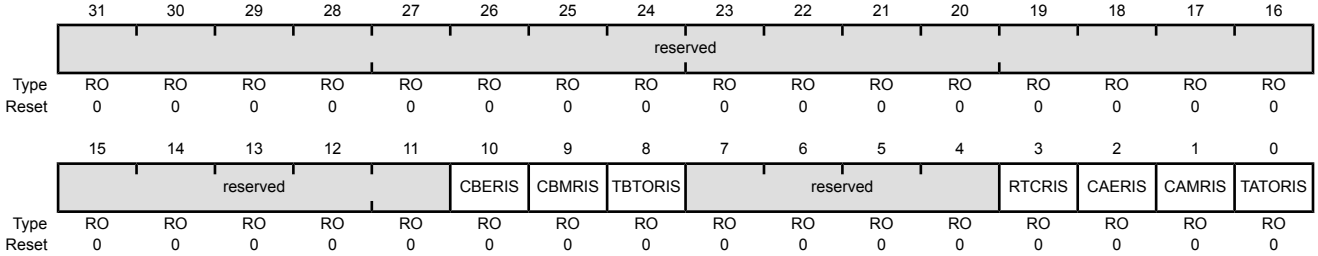
Bit/Field	Name	Type	Reset	Description
3	RTCIM	R/W	0	GPTM RTC Interrupt Mask The RTCIM values are defined as follows: Value Description 0 Interrupt is disabled. 1 Interrupt is enabled.
2	CAEIM	R/W	0	GPTM CaptureA Event Interrupt Mask The CAEIM values are defined as follows: Value Description 0 Interrupt is disabled. 1 Interrupt is enabled.
1	CAMIM	R/W	0	GPTM CaptureA Match Interrupt Mask The CAMIM values are defined as follows: Value Description 0 Interrupt is disabled. 1 Interrupt is enabled.
0	TATOIM	R/W	0	GPTM TimerA Time-Out Interrupt Mask The TATOIM values are defined as follows: Value Description 0 Interrupt is disabled. 1 Interrupt is enabled.

Register 6: GPTM Raw Interrupt Status (GPTMRIS), offset 0x01C

This register shows the state of the GPTM's internal interrupt signal. These bits are set whether or not the interrupt is masked in the **GPTMIMR** register. Each bit can be cleared by writing a 1 to its corresponding bit in **GPTMICR**.

GPTM Raw Interrupt Status (GPTMRIS)

Timer0 base: 0x4003.0000
Timer1 base: 0x4003.1000
Timer2 base: 0x4003.2000
Offset 0x01C
Type RO, reset 0x0000.0000



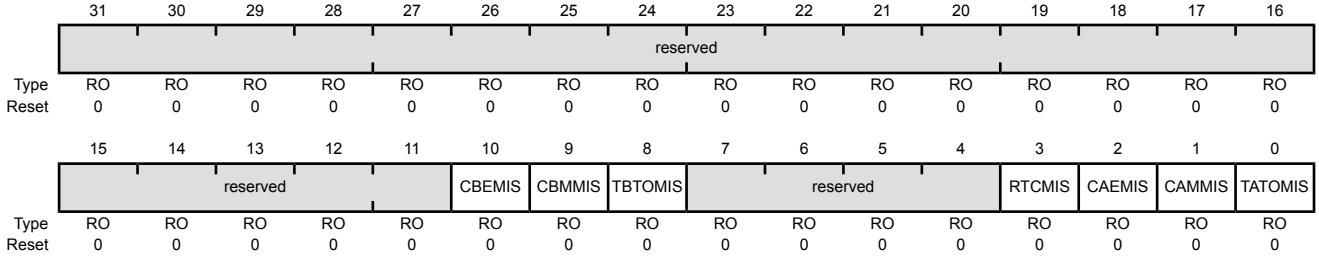
Bit/Field	Name	Type	Reset	Description
31:11	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
10	CBERIS	RO	0	GPTM CaptureB Event Raw Interrupt This is the CaptureB Event interrupt status prior to masking.
9	CBMRIS	RO	0	GPTM CaptureB Match Raw Interrupt This is the CaptureB Match interrupt status prior to masking.
8	TBTORIS	RO	0	GPTM TimerB Time-Out Raw Interrupt This is the TimerB time-out interrupt status prior to masking.
7:4	reserved	RO	0x0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
3	RTCRIS	RO	0	GPTM RTC Raw Interrupt This is the RTC Event interrupt status prior to masking.
2	CAERIS	RO	0	GPTM CaptureA Event Raw Interrupt This is the CaptureA Event interrupt status prior to masking.
1	CAMRIS	RO	0	GPTM CaptureA Match Raw Interrupt This is the CaptureA Match interrupt status prior to masking.
0	TATORIS	RO	0	GPTM TimerA Time-Out Raw Interrupt This the TimerA time-out interrupt status prior to masking.

Register 7: GPTM Masked Interrupt Status (GPTMMIS), offset 0x020

This register show the state of the GPTM's controller-level interrupt. If an interrupt is unmasked in **GPTMIMR**, and there is an event that causes the interrupt to be asserted, the corresponding bit is set in this register. All bits are cleared by writing a 1 to the corresponding bit in **GPTMICR**.

GPTM Masked Interrupt Status (GPTMMIS)

Timer0 base: 0x4003.0000
Timer1 base: 0x4003.1000
Timer2 base: 0x4003.2000
Offset 0x020
Type RO, reset 0x0000.0000



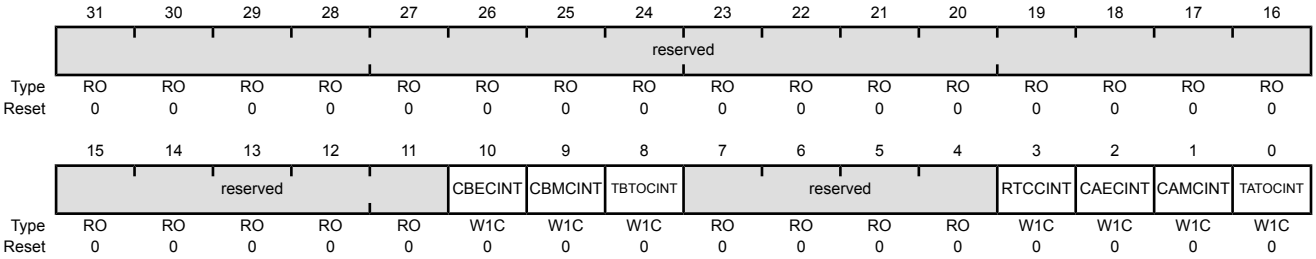
Bit/Field	Name	Type	Reset	Description
31:11	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
10	CBEMIS	RO	0	GPTM CaptureB Event Masked Interrupt This is the CaptureB event interrupt status after masking.
9	CBMMIS	RO	0	GPTM CaptureB Match Masked Interrupt This is the CaptureB match interrupt status after masking.
8	TBTOMIS	RO	0	GPTM TimerB Time-Out Masked Interrupt This is the TimerB time-out interrupt status after masking.
7:4	reserved	RO	0x0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
3	RTCMIS	RO	0	GPTM RTC Masked Interrupt This is the RTC event interrupt status after masking.
2	CAEMIS	RO	0	GPTM CaptureA Event Masked Interrupt This is the CaptureA event interrupt status after masking.
1	CAMMIS	RO	0	GPTM CaptureA Match Masked Interrupt This is the CaptureA match interrupt status after masking.
0	TATOMIS	RO	0	GPTM TimerA Time-Out Masked Interrupt This is the TimerA time-out interrupt status after masking.

Register 8: GPTM Interrupt Clear (GPTMICR), offset 0x024

This register is used to clear the status bits in the **GPTMRIS** and **GPTMMIS** registers. Writing a 1 to a bit clears the corresponding bit in the **GPTMRIS** and **GPTMMIS** registers.

GPTM Interrupt Clear (GPTMICR)

Timer0 base: 0x4003.0000
Timer1 base: 0x4003.1000
Timer2 base: 0x4003.2000
Offset 0x024
Type W1C, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:11	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
10	CBECINT	W1C	0	GPTM CaptureB Event Interrupt Clear The CBECINT values are defined as follows: Value Description 0 The interrupt is unaffected. 1 The interrupt is cleared.
9	CBMCINT	W1C	0	GPTM CaptureB Match Interrupt Clear The CBMCINT values are defined as follows: Value Description 0 The interrupt is unaffected. 1 The interrupt is cleared.
8	TBTOCINT	W1C	0	GPTM TimerB Time-Out Interrupt Clear The TBTOCINT values are defined as follows: Value Description 0 The interrupt is unaffected. 1 The interrupt is cleared.
7:4	reserved	RO	0x0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Bit/Field	Name	Type	Reset	Description
3	RTCCINT	W1C	0	GPTM RTC Interrupt Clear The RTCCINT values are defined as follows: Value Description 0 The interrupt is unaffected. 1 The interrupt is cleared.
2	CAECINT	W1C	0	GPTM CaptureA Event Interrupt Clear The CAECINT values are defined as follows: Value Description 0 The interrupt is unaffected. 1 The interrupt is cleared.
1	CAMCINT	W1C	0	GPTM CaptureA Match Raw Interrupt This is the CaptureA match interrupt status after masking.
0	TATOCINT	W1C	0	GPTM TimerA Time-Out Raw Interrupt The TATOCINT values are defined as follows: Value Description 0 The interrupt is unaffected. 1 The interrupt is cleared.

Register 9: GPTM TimerA Interval Load (GPTMTAILR), offset 0x028

This register is used to load the starting count value into the timer. When GPTM is configured to one of the 32-bit modes, **GPTMTAILR** appears as a 32-bit register (the upper 16-bits correspond to the contents of the **GPTM TimerB Interval Load (GPTMTBILR)** register). In 16-bit mode, the upper 16 bits of this register read as 0s and have no effect on the state of **GPTMTBILR**.

GPTM TimerA Interval Load (GPTMTAILR)

Timer0 base: 0x4003.0000
Timer1 base: 0x4003.1000
Timer2 base: 0x4003.2000
Offset 0x028
Type R/W, reset 0xFFFF.FFFF

	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	TAILRH															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	TAILRL															
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

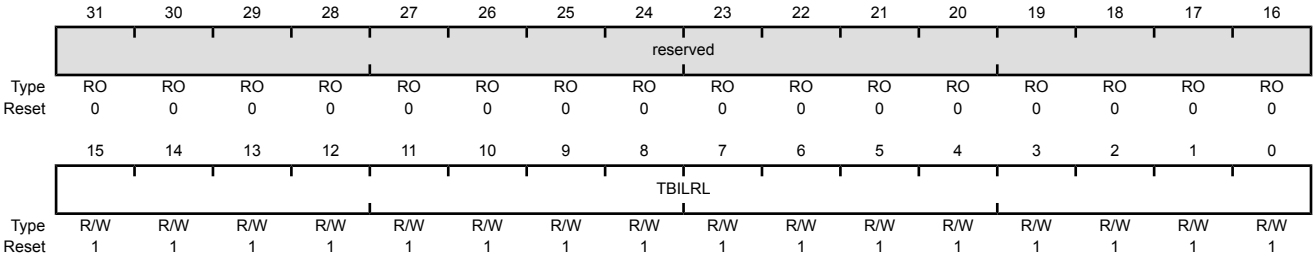
Bit/Field	Name	Type	Reset	Description
31:16	TAILRH	R/W	0xFFFF	GPTM TimerA Interval Load Register High When configured for 32-bit mode via the **GPTMCFG** register, the **GPTM TimerB Interval Load (GPTMTBILR)** register loads this value on a write. A read returns the current value of **GPTMTBILR**. In 16-bit mode, this field reads as 0 and does not have an effect on the state of **GPTMTBILR**.
15:0	TAILRL	R/W	0xFFFF	GPTM TimerA Interval Load Register Low For both 16- and 32-bit modes, writing this field loads the counter for TimerA. A read returns the current value of **GPTMTAILR**.

Register 10: GPTM TimerB Interval Load (GPTMTBILR), offset 0x02C

This register is used to load the starting count value into TimerB. When the GPTM is configured to a 32-bit mode, **GPTMTBILR** returns the current value of TimerB and ignores writes.

GPTM TimerB Interval Load (GPTMTBILR)

Timer0 base: 0x4003.0000
Timer1 base: 0x4003.1000
Timer2 base: 0x4003.2000
Offset 0x02C
Type R/W, reset 0x0000.FFFF



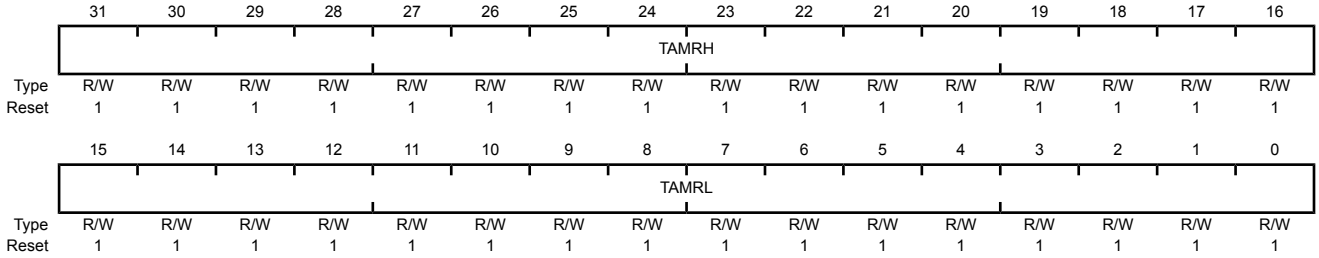
Bit/Field	Name	Type	Reset	Description
31:16	reserved	RO	0x0000	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
15:0	TBILRL	R/W	0xFFFF	GPTM TimerB Interval Load Register When the GPTM is not configured as a 32-bit timer, a write to this field updates **GPTMTBILR**. In 32-bit mode, writes are ignored, and reads return the current value of **GPTMTBILR**.

Register 11: GPTM TimerA Match (GPTMTAMATCHR), offset 0x030

This register is used in 32-bit Real-Time Clock mode and 16-bit PWM and Input Edge Count modes.

GPTM TimerA Match (GPTMTAMATCHR)

Timer0 base: 0x4003.0000
Timer1 base: 0x4003.1000
Timer2 base: 0x4003.2000
Offset 0x030
Type R/W, reset 0xFFFF.FFFF



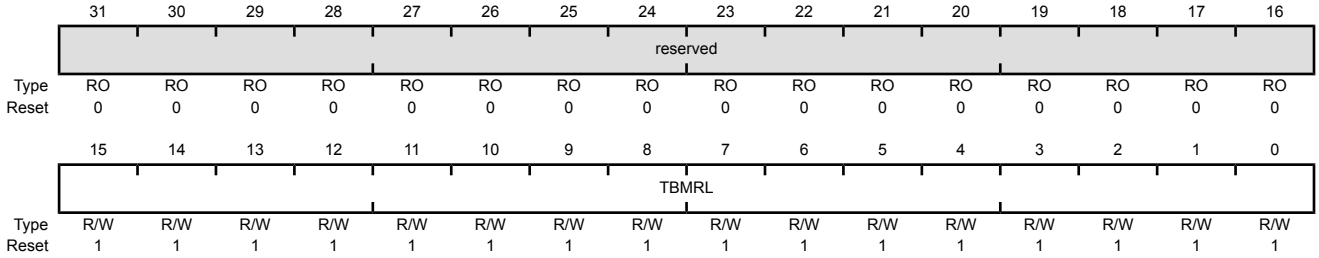
Bit/Field	Name	Type	Reset	Description
31:16	TAMRH	R/W	0xFFFF	GPTM TimerA Match Register High When configured for 32-bit Real-Time Clock (RTC) mode via the **GPTMCFG** register, this value is compared to the upper half of **GPTMTAR**, to determine match events. In 16-bit mode, this field reads as 0 and does not have an effect on the state of **GPTMTBMATCHR**.
15:0	TAMRL	R/W	0xFFFF	GPTM TimerA Match Register Low When configured for 32-bit Real-Time Clock (RTC) mode via the **GPTMCFG** register, this value is compared to the lower half of **GPTMTAR**, to determine match events. When configured for PWM mode, this value along with **GPTMTAILR**, determines the duty cycle of the output PWM signal. When configured for Edge Count mode, this value along with **GPTMTAILR**, determines how many edge events are counted. The total number of edge events counted is equal to the value in **GPTMTAILR** minus this value.

Register 12: GPTM TimerB Match (GPTMTBMATCHR), offset 0x034

This register is used in 16-bit PWM and Input Edge Count modes.

GPTM TimerB Match (GPTMTBMATCHR)

Timer0 base: 0x4003.0000
Timer1 base: 0x4003.1000
Timer2 base: 0x4003.2000
Offset 0x034
Type R/W, reset 0x0000.FFFF



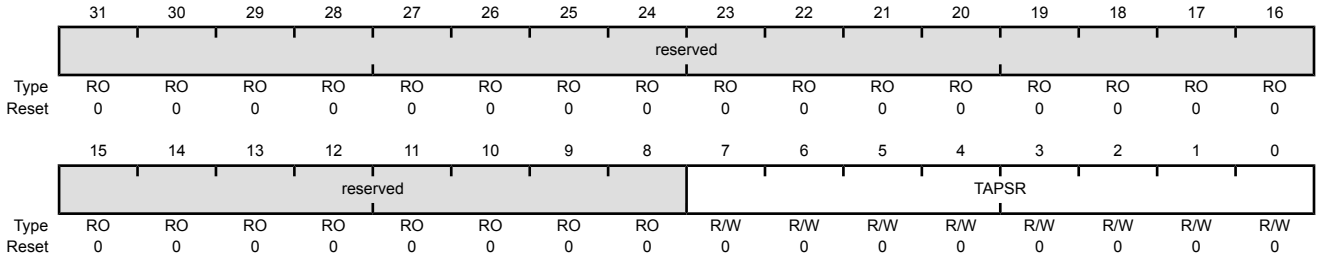
Bit/Field	Name	Type	Reset	Description
31:16	reserved	RO	0x0000	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
15:0	TBMRL	R/W	0xFFFF	GPTM TimerB Match Register Low When configured for PWM mode, this value along with **GPTMTBILR**, determines the duty cycle of the output PWM signal. When configured for Edge Count mode, this value along with **GPTMTBILR**, determines how many edge events are counted. The total number of edge events counted is equal to the value in **GPTMTBILR** minus this value.

Register 13: GPTM TimerA Prescale (GPTMTAPR), offset 0x038

This register allows software to extend the range of the 16-bit timers when operating in one-shot or periodic mode.

GPTM TimerA Prescale (GPTMTAPR)

Timer0 base: 0x4003.0000
Timer1 base: 0x4003.1000
Timer2 base: 0x4003.2000
Offset 0x038
Type R/W, reset 0x0000.0000



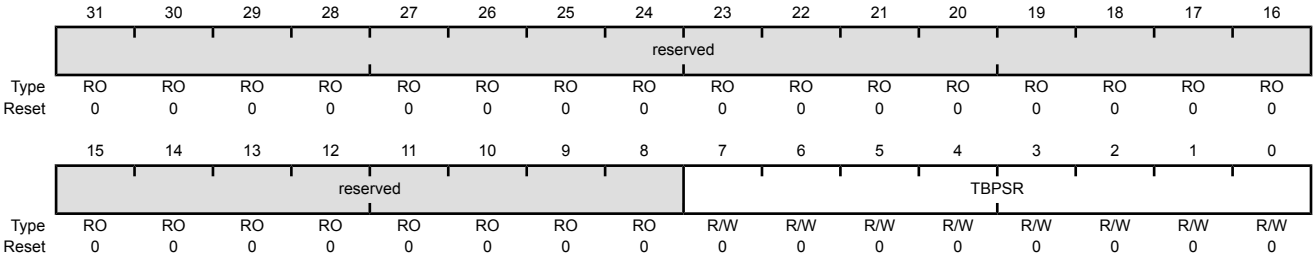
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	TAPSR	R/W	0x00	GPTM TimerA Prescale The register loads this value on a write. A read returns the current value of the register. Refer to Table 9-2 on page 197 for more details and an example.

Register 14: GPTM TimerB Prescale (GPTMTBPR), offset 0x03C

This register allows software to extend the range of the 16-bit timers when operating in one-shot or periodic mode.

GPTM TimerB Prescale (GPTMTBPR)

Timer0 base: 0x4003.0000
Timer1 base: 0x4003.1000
Timer2 base: 0x4003.2000
Offset 0x03C
Type R/W, reset 0x0000.0000



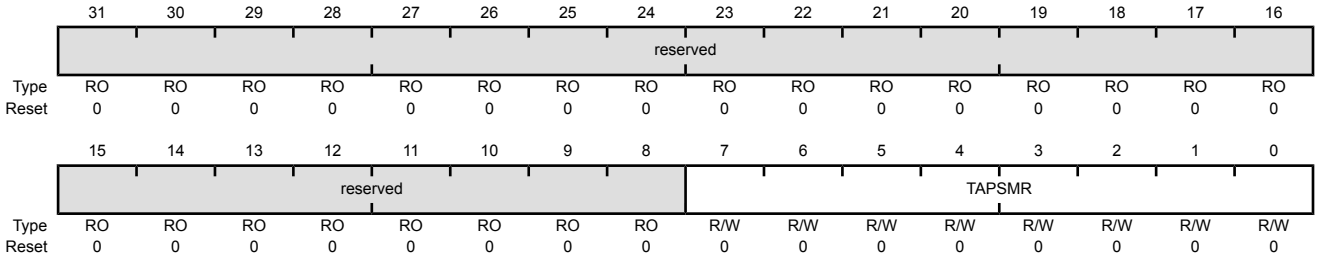
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	TBPSR	R/W	0x00	GPTM TimerB Prescale The register loads this value on a write. A read returns the current value of this register. Refer to Table 9-2 on page 197 for more details and an example.

Register 15: GPTM TimerA Prescale Match (GPTMTAPMR), offset 0x040

This register effectively extends the range of **GPTMTAMATCHR** to 24 bits when operating in 16-bit one-shot or periodic mode.

GPTM TimerA Prescale Match (GPTMTAPMR)

Timer0 base: 0x4003.0000
Timer1 base: 0x4003.1000
Timer2 base: 0x4003.2000
Offset 0x040
Type R/W, reset 0x0000.0000



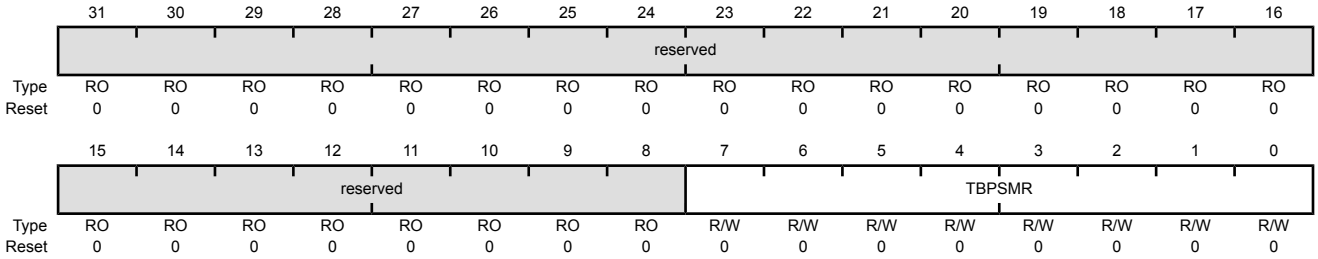
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	TAPSMR	R/W	0x00	GPTM TimerA Prescale Match This value is used alongside **GPTMTAMATCHR** to detect timer match events while using a prescaler.

Register 16: GPTM TimerB Prescale Match (GPTMTBPMR), offset 0x044

This register effectively extends the range of **GPTMTBMATCHR** to 24 bits when operating in 16-bit one-shot or periodic mode.

GPTM TimerB Prescale Match (GPTMTBPMR)

Timer0 base: 0x4003.0000
Timer1 base: 0x4003.1000
Timer2 base: 0x4003.2000
Offset 0x044
Type R/W, reset 0x0000.0000



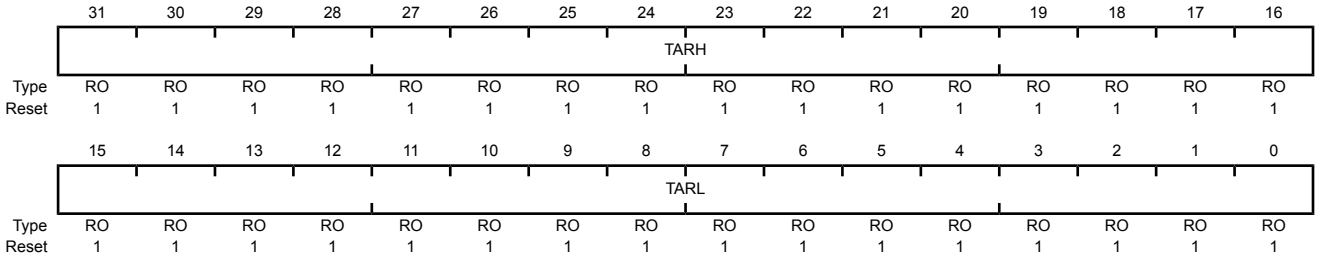
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	TBPSMR	R/W	0x00	GPTM TimerB Prescale Match This value is used alongside **GPTMTBMATCHR** to detect timer match events while using a prescaler.

Register 17: GPTM TimerA (GPTMTAR), offset 0x048

This register shows the current value of the TimerA counter in all cases except for Input Edge Count mode. When in this mode, this register contains the time at which the last edge event took place.

GPTM TimerA (GPTMTAR)

Timer0 base: 0x4003.0000
Timer1 base: 0x4003.1000
Timer2 base: 0x4003.2000
Offset 0x048
Type RO, reset 0xFFFF.FFFF



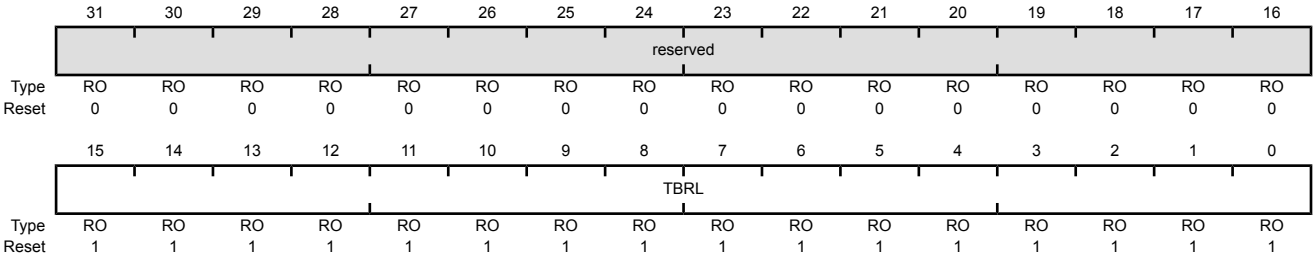
Bit/Field	Name	Type	Reset	Description
31:16	TARH	RO	0xFFFF	GPTM TimerA Register High If the **GPTMCFG** is in a 32-bit mode, TimerB value is read. If the **GPTMCFG** is in a 16-bit mode, this is read as zero.
15:0	TARL	RO	0xFFFF	GPTM TimerA Register Low A read returns the current value of the **GPTM TimerA Count Register**, except in Input Edge Count mode, when it returns the timestamp from the last edge event.

Register 18: GPTM TimerB (GPTMTBR), offset 0x04C

This register shows the current value of the TimerB counter in all cases except for Input Edge Count mode. When in this mode, this register contains the time at which the last edge event took place.

GPTM TimerB (GPTMTBR)

Timer0 base: 0x4003.0000
Timer1 base: 0x4003.1000
Timer2 base: 0x4003.2000
Offset 0x04C
Type RO, reset 0x0000.FFFF



Bit/Field	Name	Type	Reset	Description
31:16	reserved	RO	0x0000	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
15:0	TBRL	RO	0xFFFF	GPTM TimerB A read returns the current value of the **GPTM TimerB Count Register**, except in Input Edge Count mode, when it returns the timestamp from the last edge event.

10 Watchdog Timer

A watchdog timer can generate nonmaskable interrupts (NMIs) or a reset when a time-out value is reached. The watchdog timer is used to regain control when a system has failed due to a software error or due to the failure of an external device to respond in the expected way.

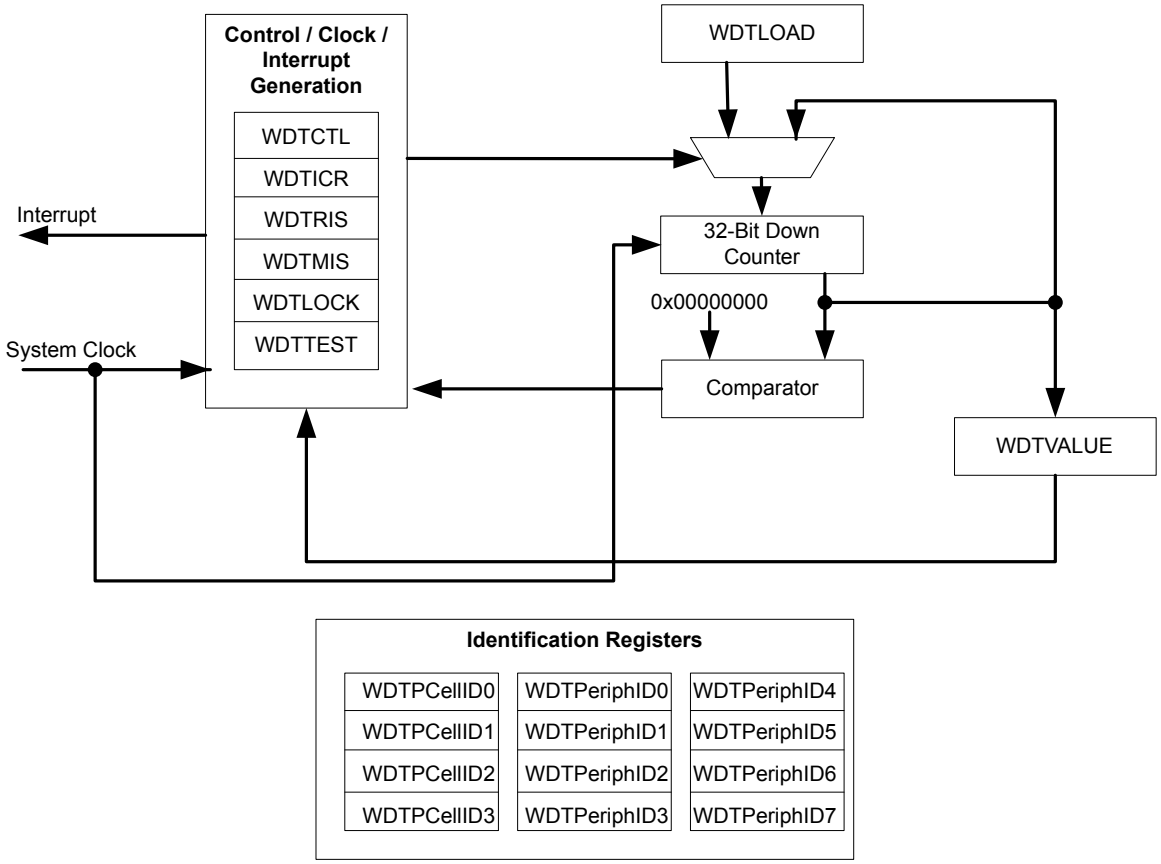
The Stellaris® Watchdog Timer module has the following features:

- 32-bit down counter with a programmable load register
- Separate watchdog clock with an enable
- Programmable interrupt generation logic with interrupt masking
- Lock register protection from runaway software
- Reset generation logic with an enable/disable
- User-enabled stalling when the controller asserts the CPU Halt flag during debug

The Watchdog Timer can be configured to generate an interrupt to the controller on its first time-out, and to generate a reset signal on its second time-out. Once the Watchdog Timer has been configured, the lock register can be written to prevent the timer configuration from being inadvertently altered.

10.1 Block Diagram

Figure 10-1. WDT Module Block Diagram



10.2 Functional Description

The Watchdog Timer module generates the first time-out signal when the 32-bit counter reaches the zero state after being enabled; enabling the counter also enables the watchdog timer interrupt. After the first time-out event, the 32-bit counter is re-loaded with the value of the **Watchdog Timer Load (WDTLOAD)** register, and the timer resumes counting down from that value. Once the Watchdog Timer has been configured, the **Watchdog Timer Lock (WDTLOCK)** register is written, which prevents the timer configuration from being inadvertently altered by software.

If the timer counts down to its zero state again before the first time-out interrupt is cleared, and the reset signal has been enabled (via the `WatchdogResetEnable` function), the Watchdog timer asserts its reset signal to the system. If the interrupt is cleared before the 32-bit counter reaches its second time-out, the 32-bit counter is loaded with the value in the **WDTLOAD** register, and counting resumes from that value.

If **WDTLOAD** is written with a new value while the Watchdog Timer counter is counting, then the counter is loaded with the new value and continues counting.

Writing to **WDTLOAD** does not clear an active interrupt. An interrupt must be specifically cleared by writing to the **Watchdog Interrupt Clear (WDTICR)** register.

The Watchdog module interrupt and reset generation can be enabled or disabled as required. When the interrupt is re-enabled, the 32-bit counter is preloaded with the load register value and not its last state.

10.3 Initialization and Configuration

To use the WDT, its peripheral clock must be enabled by setting the WDT bit in the **RCGC0** register. The Watchdog Timer is configured using the following sequence:

1. Load the **WDTLOAD** register with the desired timer load value.
2. If the Watchdog is configured to trigger system resets, set the RESEN bit in the **WDTCTL** register.
3. Set the INTEN bit in the **WDTCTL** register to enable the Watchdog and lock the control register.

If software requires that all of the watchdog registers are locked, the Watchdog Timer module can be fully locked by writing any value to the **WDTLOCK** register. To unlock the Watchdog Timer, write a value of 0x1ACC.E551.

10.4 Register Map

Table 10-1 on page 231 lists the Watchdog registers. The offset listed is a hexadecimal increment to the register's address, relative to the Watchdog Timer base address of 0x4000.0000.

Table 10-1. Watchdog Timer Register Map

Offset	Name	Type	Reset	Description	See page
0x000	WDTLOAD	R/W	0xFFFF.FFFF	Watchdog Load	233
0x004	WDTVALUE	RO	0xFFFF.FFFF	Watchdog Value	234
0x008	WDTCTL	R/W	0x0000.0000	Watchdog Control	235
0x00C	WDTICR	WO	-	Watchdog Interrupt Clear	236
0x010	WDTRIS	RO	0x0000.0000	Watchdog Raw Interrupt Status	237
0x014	WDTMIS	RO	0x0000.0000	Watchdog Masked Interrupt Status	238
0x418	WDTTEST	R/W	0x0000.0000	Watchdog Test	239
0xC00	WDTLOCK	R/W	0x0000.0000	Watchdog Lock	240
0xFD0	WDTPeriphID4	RO	0x0000.0000	Watchdog Peripheral Identification 4	241
0xFD4	WDTPeriphID5	RO	0x0000.0000	Watchdog Peripheral Identification 5	242
0xFD8	WDTPeriphID6	RO	0x0000.0000	Watchdog Peripheral Identification 6	243
0xFDC	WDTPeriphID7	RO	0x0000.0000	Watchdog Peripheral Identification 7	244
0xFE0	WDTPeriphID0	RO	0x0000.0005	Watchdog Peripheral Identification 0	245
0xFE4	WDTPeriphID1	RO	0x0000.0018	Watchdog Peripheral Identification 1	246
0xFE8	WDTPeriphID2	RO	0x0000.0018	Watchdog Peripheral Identification 2	247

Table 10-1. Watchdog Timer Register Map *(continued)*

Offset	Name	Type	Reset	Description	See page
0xFEC	WDTPeriphID3	RO	0x0000.0001	Watchdog Peripheral Identification 3	248
0xFF0	WDTPCellID0	RO	0x0000.000D	Watchdog PrimeCell Identification 0	249
0xFF4	WDTPCellID1	RO	0x0000.00F0	Watchdog PrimeCell Identification 1	250
0xFF8	WDTPCellID2	RO	0x0000.0005	Watchdog PrimeCell Identification 2	251
0xFFC	WDTPCellID3	RO	0x0000.00B1	Watchdog PrimeCell Identification 3	252

10.5 Register Descriptions

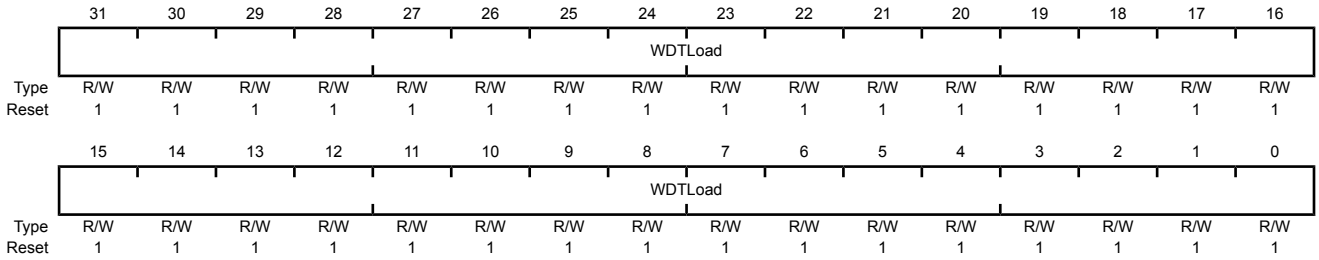
The remainder of this section lists and describes the WDT registers, in numerical order by address offset.

Register 1: Watchdog Load (WDTLOAD), offset 0x000

This register is the 32-bit interval value used by the 32-bit counter. When this register is written, the value is immediately loaded and the counter restarts counting down from the new value. If the **WDTLOAD** register is loaded with 0x0000.0000, an interrupt is immediately generated.

Watchdog Load (WDTLOAD)

Base 0x4000.0000
Offset 0x000
Type R/W, reset 0xFFFF.FFFF



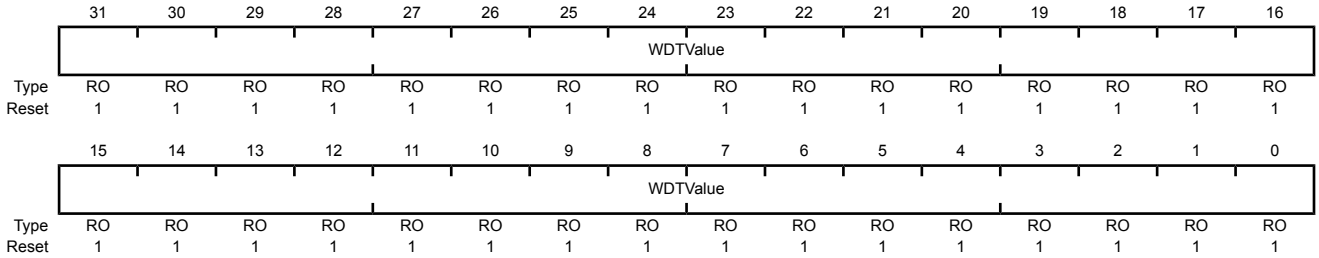
Bit/Field	Name	Type	Reset	Description
31:0	WDTLoad	R/W	0xFFFF.FFFF	Watchdog Load Value

Register 2: Watchdog Value (WDTVALUE), offset 0x004

This register contains the current count value of the timer.

Watchdog Value (WDTVALUE)

Base 0x4000.0000
Offset 0x004
Type RO, reset 0xFFFF.FFFF



Bit/Field	Name	Type	Reset	Description
31:0	WDTValue	RO	0xFFFF.FFFF	Watchdog Value Current value of the 32-bit down counter.

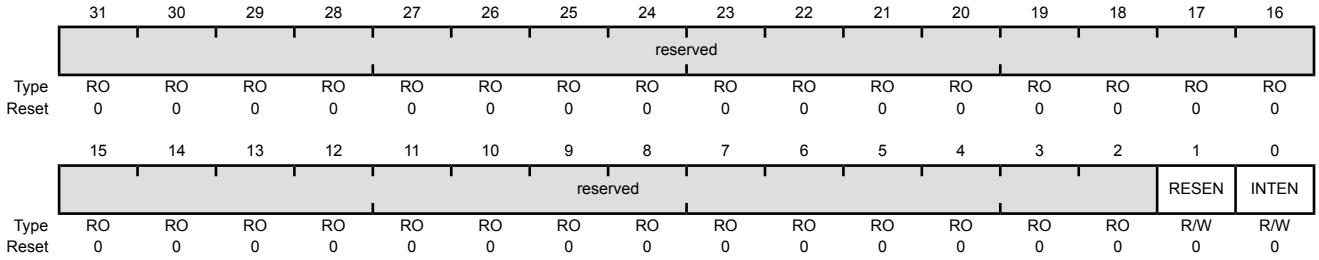
Register 3: Watchdog Control (WDTCTL), offset 0x008

This register is the watchdog control register. The watchdog timer can be configured to generate a reset signal (on second time-out) or an interrupt on time-out.

When the watchdog interrupt has been enabled, all subsequent writes to the control register are ignored. The only mechanism that can re-enable writes is a hardware reset.

Watchdog Control (WDTCTL)

Base 0x4000.0000
Offset 0x008
Type R/W, reset 0x0000.0000



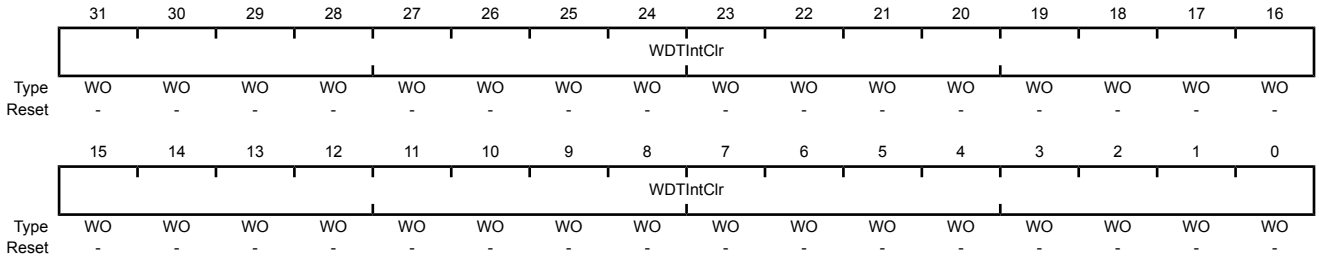
Bit/Field	Name	Type	Reset	Description
31:2	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1	RESEN	R/W	0	Watchdog Reset Enable The RESEN values are defined as follows: Value Description 0 Disabled. 1 Enable the Watchdog module reset output.
0	INTEN	R/W	0	Watchdog Interrupt Enable The INTEN values are defined as follows: Value Description 0 Interrupt event disabled (once this bit is set, it can only be cleared by a hardware reset). 1 Interrupt event enabled. Once enabled, all writes are ignored.

Register 4: Watchdog Interrupt Clear (WDTICR), offset 0x00C

This register is the interrupt clear register. A write of any value to this register clears the Watchdog interrupt and reloads the 32-bit counter from the **WDTLOAD** register. Value for a read or reset is indeterminate.

Watchdog Interrupt Clear (WDTICR)

Base 0x4000.0000
Offset 0x00C
Type WO, reset -



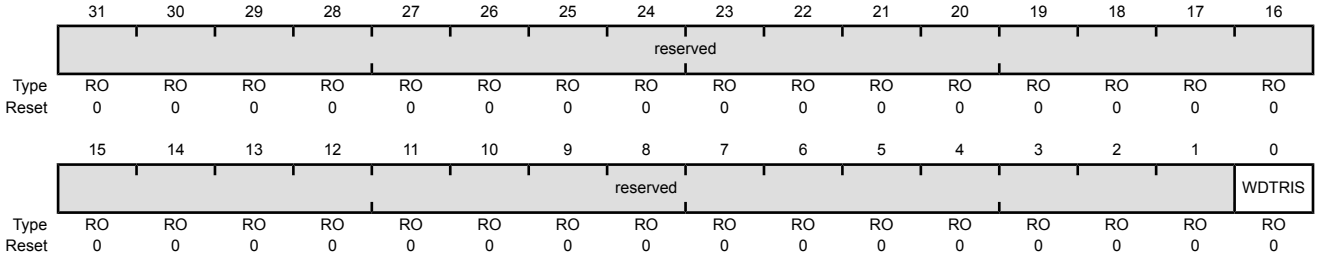
Bit/Field	Name	Type	Reset	Description
31:0	WDTIntClr	WO	-	Watchdog Interrupt Clear

Register 5: Watchdog Raw Interrupt Status (WDTRIS), offset 0x010

This register is the raw interrupt status register. Watchdog interrupt events can be monitored via this register if the controller interrupt is masked.

Watchdog Raw Interrupt Status (WDTRIS)

Base 0x4000.0000
Offset 0x010
Type RO, reset 0x0000.0000



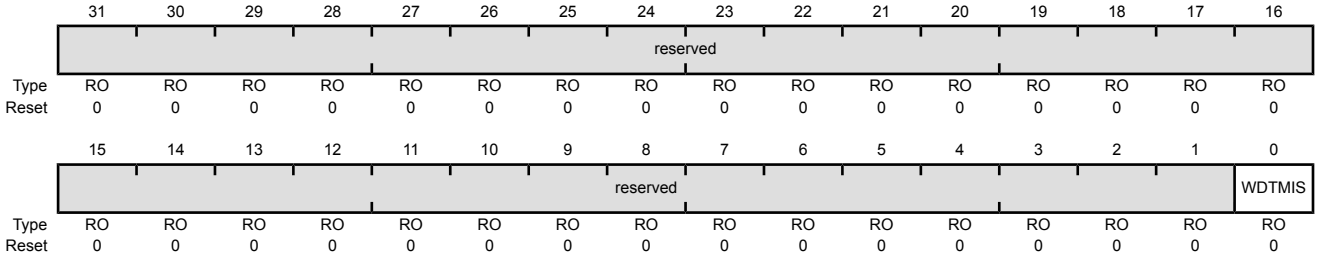
Bit/Field	Name	Type	Reset	Description
31:1	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
0	WDTRIS	RO	0	Watchdog Raw Interrupt Status Gives the raw interrupt state (prior to masking) of **WDTINTR**.

Register 6: Watchdog Masked Interrupt Status (WDTMIS), offset 0x014

This register is the masked interrupt status register. The value of this register is the logical AND of the raw interrupt bit and the Watchdog interrupt enable bit.

Watchdog Masked Interrupt Status (WDTMIS)

Base 0x4000.0000
Offset 0x014
Type RO, reset 0x0000.0000



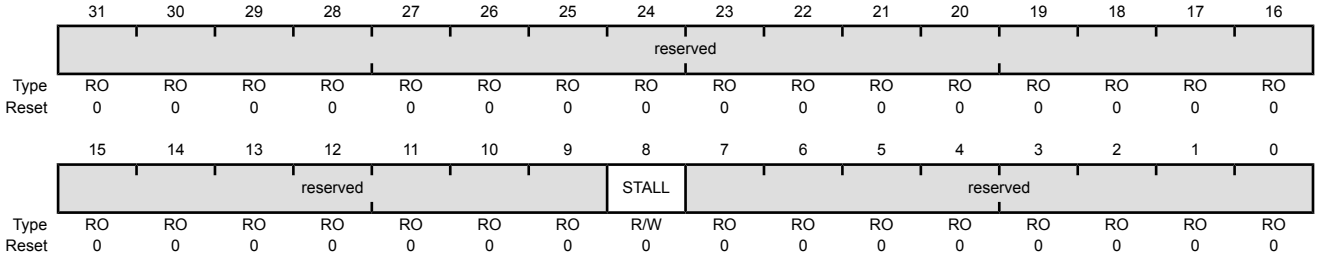
Bit/Field	Name	Type	Reset	Description
31:1	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
0	WDTMIS	RO	0	Watchdog Masked Interrupt Status Gives the masked interrupt state (after masking) of the **WDTINTR** interrupt.

Register 7: Watchdog Test (WDTTEST), offset 0x418

This register provides user-enabled stalling when the microcontroller asserts the CPU halt flag during debug.

Watchdog Test (WDTTEST)

Base 0x4000.0000
Offset 0x418
Type R/W, reset 0x0000.0000



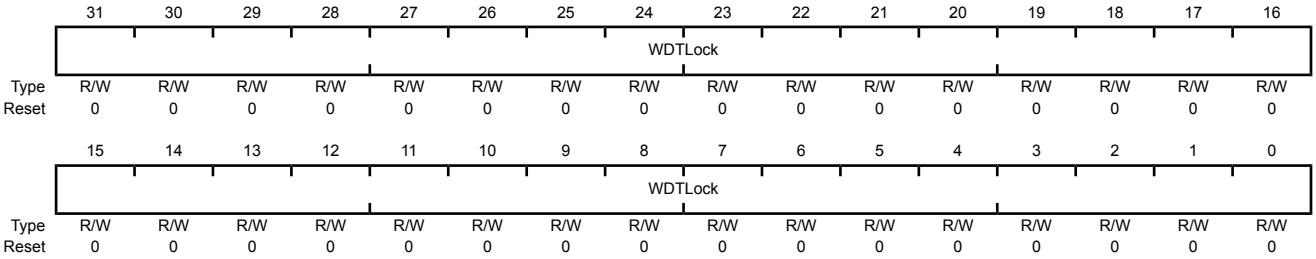
Bit/Field	Name	Type	Reset	Description
31:9	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
8	STALL	R/W	0	Watchdog Stall Enable When set to 1, if the Stellaris® microcontroller is stopped with a debugger, the watchdog timer stops counting. Once the microcontroller is restarted, the watchdog timer resumes counting.
7:0	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Register 8: Watchdog Lock (WDTLOCK), offset 0xC00

Writing 0x1ACC.E551 to the **WDTLOCK** register enables write access to all other registers. Writing any other value to the **WDTLOCK** register re-enables the locked state for register writes to all the other registers. Reading the **WDTLOCK** register returns the lock status rather than the 32-bit value written. Therefore, when write accesses are disabled, reading the **WDTLOCK** register returns 0x0000.0001 (when locked; otherwise, the returned value is 0x0000.0000 (unlocked)).

Watchdog Lock (WDTLOCK)

Base 0x4000.0000
Offset 0xC00
Type R/W, reset 0x0000.0000



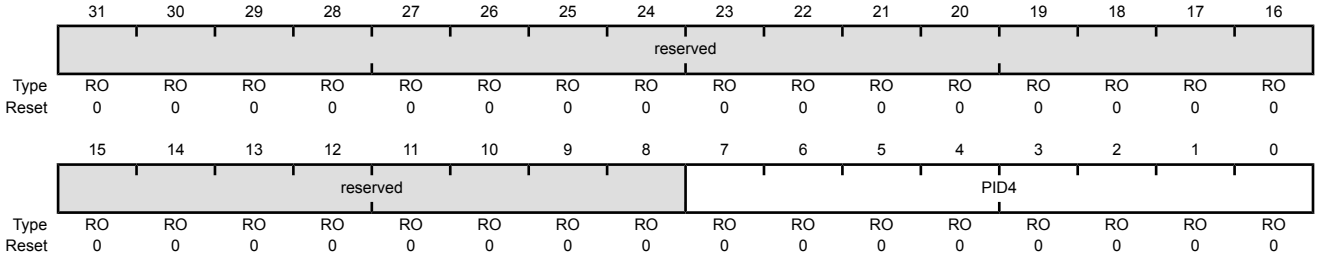
Bit/Field	Name	Type	Reset	Description
31:0	WDTLock	R/W	0x0000	Watchdog Lock A write of the value 0x1ACC.E551 unlocks the watchdog registers for write access. A write of any other value reapplies the lock, preventing any register updates. A read of this register returns the following values: Value Description 0x0000.0001 Locked 0x0000.0000 Unlocked

Register 9: Watchdog Peripheral Identification 4 (WDTPeriphID4), offset 0xFD0

The **WDTPeriphIDn** registers are hard-coded and the fields within the register determine the reset value.

Watchdog Peripheral Identification 4 (WDTPeriphID4)

Base 0x4000.0000
Offset 0xFD0
Type RO, reset 0x0000.0000



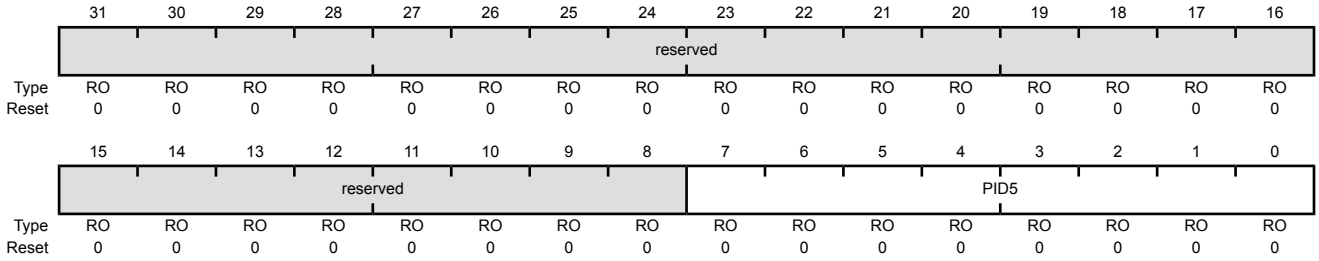
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID4	RO	0x00	WDT Peripheral ID Register[7:0]

Register 10: Watchdog Peripheral Identification 5 (WDTPeriphID5), offset 0xFD4

The **WDTPeriphIDn** registers are hard-coded and the fields within the register determine the reset value.

Watchdog Peripheral Identification 5 (WDTPeriphID5)

Base 0x4000.0000
Offset 0xFD4
Type RO, reset 0x0000.0000



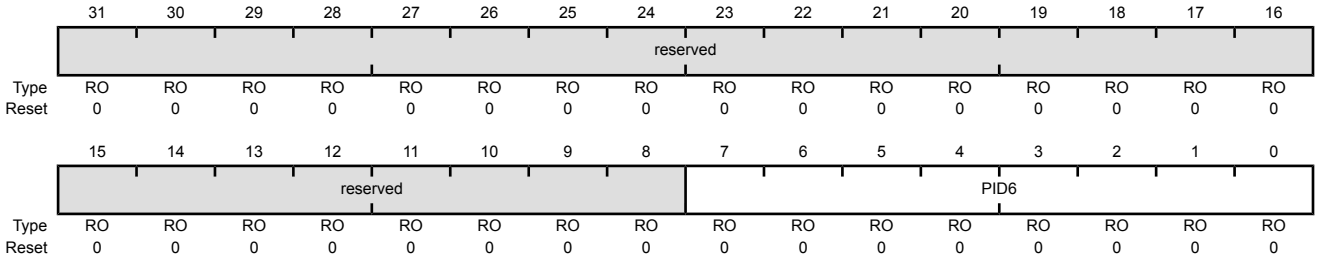
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID5	RO	0x00	WDT Peripheral ID Register[15:8]

Register 11: Watchdog Peripheral Identification 6 (WDTPeriphID6), offset 0xFD8

The **WDTPeriphIDn** registers are hard-coded and the fields within the register determine the reset value.

Watchdog Peripheral Identification 6 (WDTPeriphID6)

Base 0x4000.0000
Offset 0xFD8
Type RO, reset 0x0000.0000



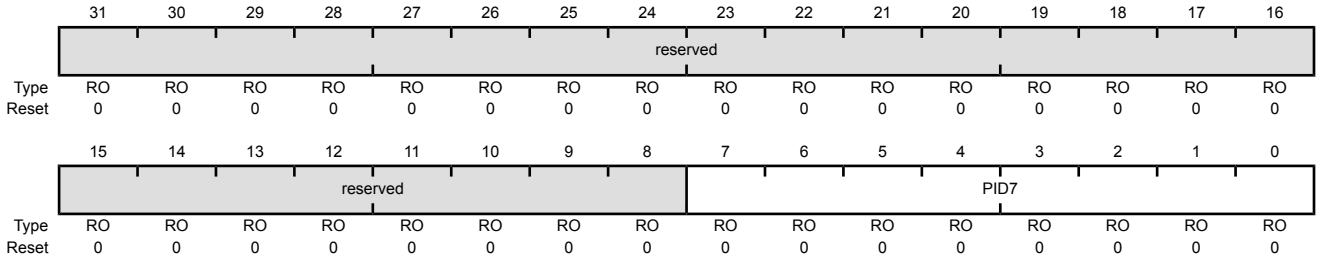
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID6	RO	0x00	WDT Peripheral ID Register[23:16]

Register 12: Watchdog Peripheral Identification 7 (WDTPeriphID7), offset 0xFDC

The **WDTPeriphIDn** registers are hard-coded and the fields within the register determine the reset value.

Watchdog Peripheral Identification 7 (WDTPeriphID7)

Base 0x4000.0000
Offset 0xFDC
Type RO, reset 0x0000.0000



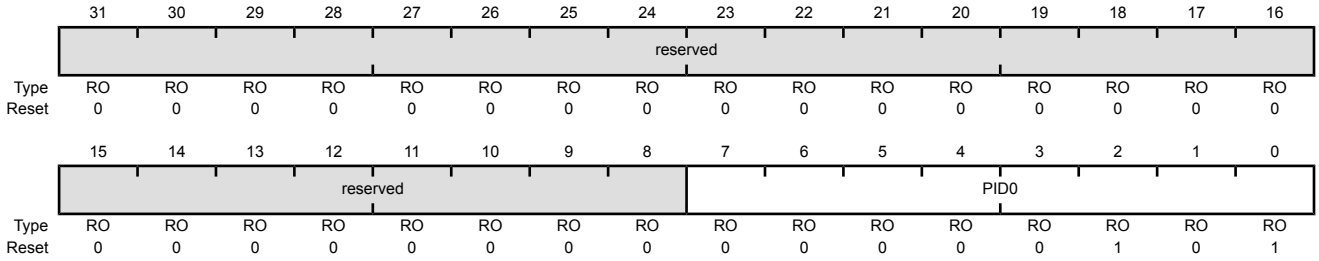
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID7	RO	0x00	WDT Peripheral ID Register[31:24]

Register 13: Watchdog Peripheral Identification 0 (WDTPeriphID0), offset 0xFE0

The **WDTPeriphIDn** registers are hard-coded and the fields within the register determine the reset value.

Watchdog Peripheral Identification 0 (WDTPeriphID0)

Base 0x4000.0000
Offset 0xFE0
Type RO, reset 0x0000.0005



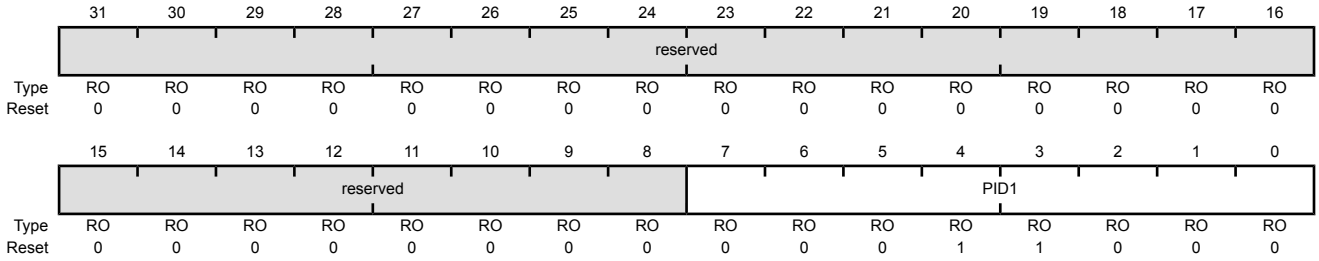
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID0	RO	0x05	Watchdog Peripheral ID Register[7:0]

Register 14: Watchdog Peripheral Identification 1 (WDTPeriphID1), offset 0xFE4

The **WDTPeriphIDn** registers are hard-coded and the fields within the register determine the reset value.

Watchdog Peripheral Identification 1 (WDTPeriphID1)

Base 0x4000.0000
Offset 0xFE4
Type RO, reset 0x0000.0018



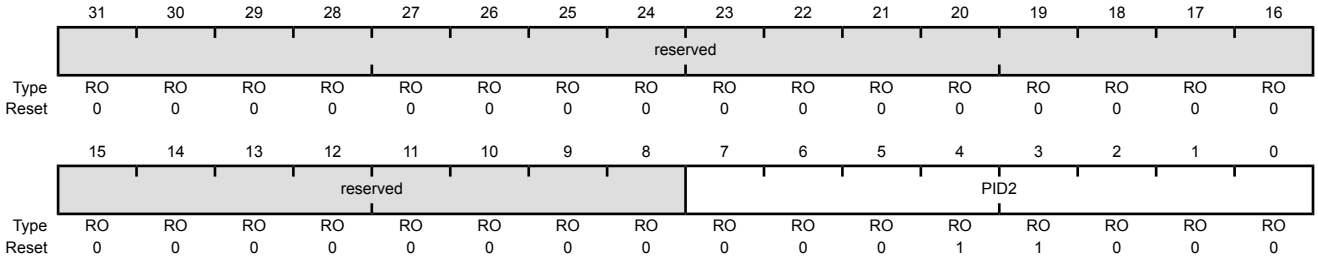
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID1	RO	0x18	Watchdog Peripheral ID Register[15:8]

Register 15: Watchdog Peripheral Identification 2 (WDTPeriphID2), offset 0xFE8

The **WDTPeriphIDn** registers are hard-coded and the fields within the register determine the reset value.

Watchdog Peripheral Identification 2 (WDTPeriphID2)

Base 0x4000.0000
Offset 0xFE8
Type RO, reset 0x0000.0018



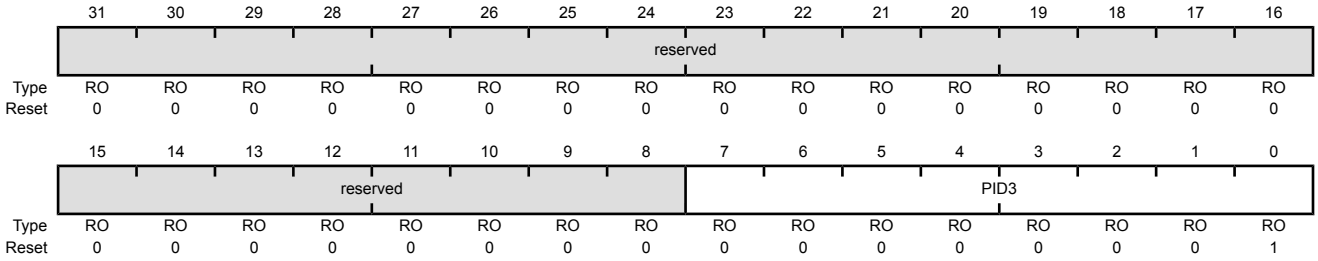
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID2	RO	0x18	Watchdog Peripheral ID Register[23:16]

Register 16: Watchdog Peripheral Identification 3 (WDTPeriphID3), offset 0xFEC

The **WDTPeriphIDn** registers are hard-coded and the fields within the register determine the reset value.

Watchdog Peripheral Identification 3 (WDTPeriphID3)

Base 0x4000.0000
Offset 0xFEC
Type RO, reset 0x0000.0001



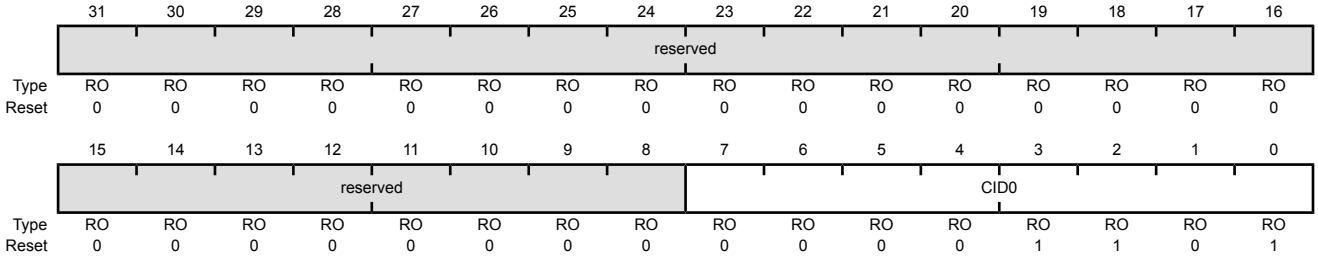
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID3	RO	0x01	Watchdog Peripheral ID Register[31:24]

Register 17: Watchdog PrimeCell Identification 0 (WDTPCellID0), offset 0xFF0

The **WDTPCellIDn** registers are hard-coded and the fields within the register determine the reset value.

Watchdog PrimeCell Identification 0 (WDTPCellID0)

Base 0x4000.0000
Offset 0xFF0
Type RO, reset 0x0000.000D



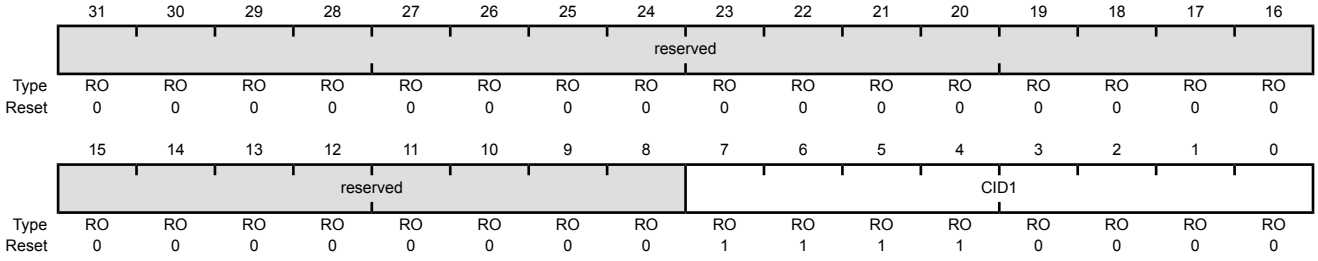
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	CID0	RO	0x0D	Watchdog PrimeCell ID Register[7:0]

Register 18: Watchdog PrimeCell Identification 1 (WDTPCellID1), offset 0xFF4

The **WDTPCellIDn** registers are hard-coded and the fields within the register determine the reset value.

Watchdog PrimeCell Identification 1 (WDTPCellID1)

Base 0x4000.0000
Offset 0xFF4
Type RO, reset 0x0000.00F0



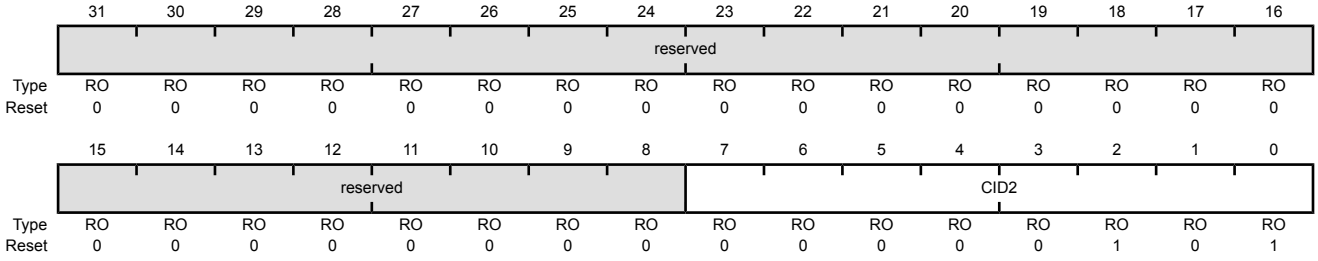
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	CID1	RO	0xF0	Watchdog PrimeCell ID Register[15:8]

Register 19: Watchdog PrimeCell Identification 2 (WDTPCellID2), offset 0xFF8

The **WDTPCellIDn** registers are hard-coded and the fields within the register determine the reset value.

Watchdog PrimeCell Identification 2 (WDTPCellID2)

Base 0x4000.0000
Offset 0xFF8
Type RO, reset 0x0000.0005



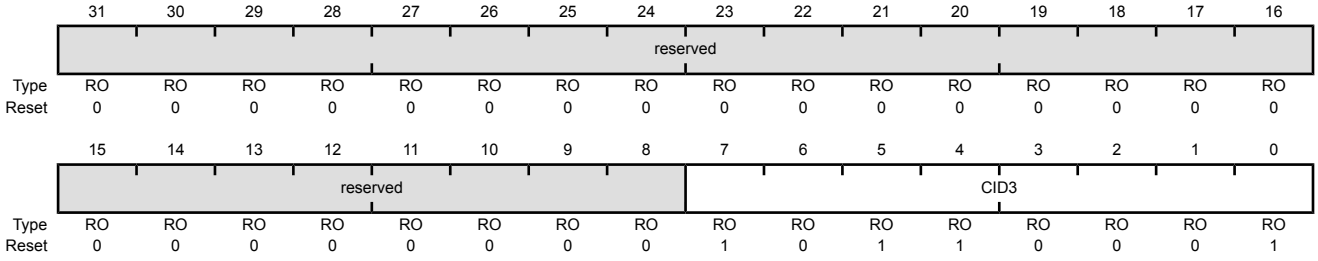
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	CID2	RO	0x05	Watchdog PrimeCell ID Register[23:16]

Register 20: Watchdog PrimeCell Identification 3 (WDTPCellID3), offset 0xFFC

The **WDTPCellIDn** registers are hard-coded and the fields within the register determine the reset value.

Watchdog PrimeCell Identification 3 (WDTPCellID3)

Base 0x4000.0000
Offset 0xFFC
Type RO, reset 0x0000.00B1



Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	CID3	RO	0xB1	Watchdog PrimeCell ID Register[31:24]

11 Analog-to-Digital Converter (ADC)

An analog-to-digital converter (ADC) is a peripheral that converts a continuous analog voltage to a discrete digital number.

The Stellaris® ADC module features 10-bit conversion resolution and supports three input channels, plus an internal temperature sensor. The ADC module contains four programmable sequencer which allows for the sampling of multiple analog input sources without controller intervention. Each sample sequence provides flexible programming with fully configurable input source, trigger events, interrupt generation, and sequence priority.

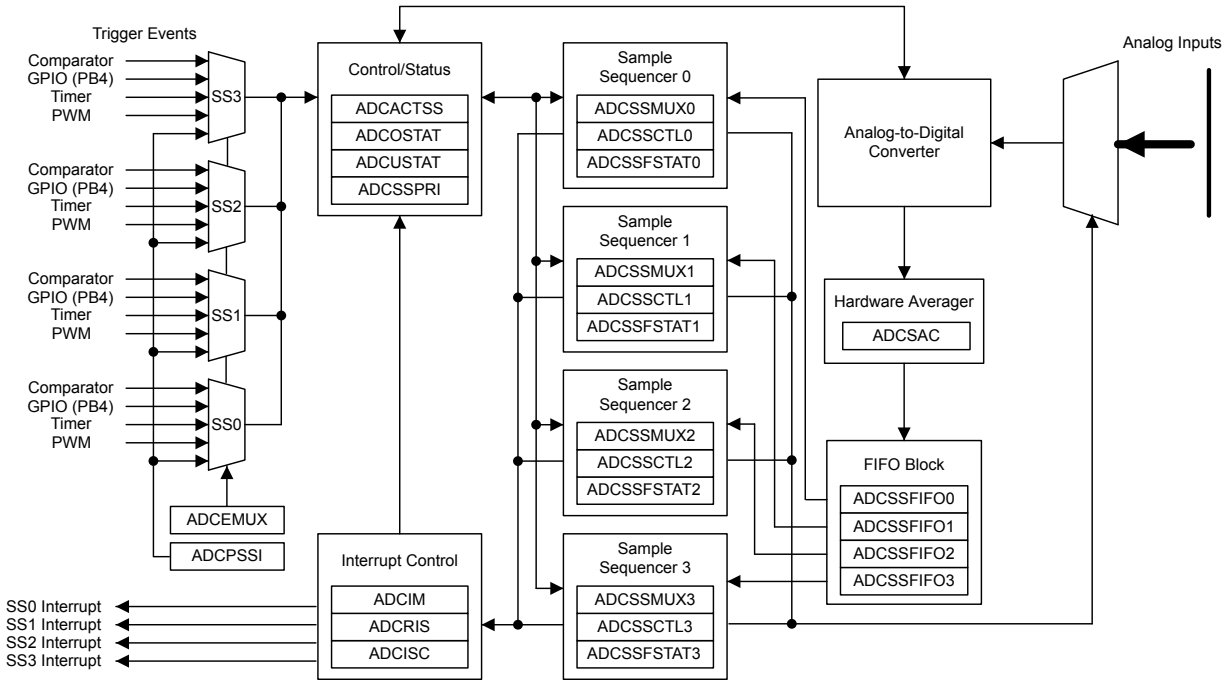
The Stellaris® ADC module provides the following features:

- Three analog input channels
- Single-ended and differential-input configurations
- On-chip internal temperature sensor
- Sample rate of 250 thousand samples/second
- Flexible, configurable analog-to-digital conversion
- Four programmable sample conversion sequences from one to eight entries long, with corresponding conversion result FIFOs
- Flexible trigger control
  - Controller (software)
  - Timers
  - Analog Comparators
  - PWM
  - GPIO
- Hardware averaging of up to 64 samples for improved accuracy
- Converter uses an internal 3-V reference
- Power and ground for the analog circuitry is separate from the digital power and ground

11.1 Block Diagram

Figure 11-1 on page 254 provides details on the internal configuration of the ADC controls and data registers.

Figure 11-1. ADC Module Block Diagram



11.2 Functional Description

The Stellaris® ADC collects sample data by using a programmable sequence-based approach instead of the traditional single or double-sampling approaches found on many ADC modules. Each *sample sequence* is a fully programmed series of consecutive (back-to-back) samples, allowing the ADC to collect data from multiple input sources without having to be re-configured or serviced by the controller. The programming of each sample in the sample sequence includes parameters such as the input source and mode (differential versus single-ended input), interrupt generation on sample completion, and the indicator for the last sample in the sequence.

11.2.1 Sample Sequencers

The sampling control and data capture is handled by the sample sequencers. All of the sequencers are identical in implementation except for the number of samples that can be captured and the depth of the FIFO. Table 11-1 on page 254 shows the maximum number of samples that each sequencer can capture and its corresponding FIFO depth. In this implementation, each FIFO entry is a 32-bit word, with the lower 10 bits containing the conversion result.

Table 11-1. Samples and FIFO Depth of Sequencers

Sequencer	Number of Samples	Depth of FIFO
SS3	1	1
SS2	4	4
SS1	4	4
SS0	8	8

For a given sample sequence, each sample is defined by two 4-bit nibbles in the **ADC Sample Sequence Input Multiplexer Select (ADCSSMUXn)** and **ADC Sample Sequence Control**

(ADCSSCTLn) registers, where "n" corresponds to the sequence number. The **ADCSSMUXn** nibbles select the input pin, while the **ADCSSCTLn** nibbles contain the sample control bits corresponding to parameters such as temperature sensor selection, interrupt enable, end of sequence, and differential input mode. Sample sequencers are enabled by setting the respective ASENn bit in the **ADC Active Sample Sequencer (ADCACTSS)** register, and should be configured before being enabled.

When configuring a sample sequence, multiple uses of the same input pin within the same sequence is allowed. In the **ADCSSCTLn** register, the IEn bits can be set for any combination of samples, allowing interrupts to be generated after every sample in the sequence if necessary. Also, the END bit can be set at any point within a sample sequence. For example, if Sequencer 0 is used, the END bit can be set in the nibble associated with the fifth sample, allowing Sequencer 0 to complete execution of the sample sequence after the fifth sample.

After a sample sequence completes execution, the result data can be retrieved from the **ADC Sample Sequence Result FIFO (ADCSSFIFOn)** registers. The FIFOs are simple circular buffers that read a single address to "pop" result data. For software debug purposes, the positions of the FIFO head and tail pointers are visible in the **ADC Sample Sequence FIFO Status (ADCSSFSTATn)** registers along with FULL and EMPTY status flags. Overflow and underflow conditions are monitored using the **ADCOSTAT** and **ADCUSTAT** registers.

11.2.2 Module Control

Outside of the sample sequencers, the remainder of the control logic is responsible for tasks such as:

- Interrupt generation
- Sequence prioritization
- Trigger configuration

Most of the ADC control logic runs at the ADC clock rate of 14-18 MHz. The internal ADC divider is configured automatically by hardware when the system XTAL is selected. The automatic clock divider configuration targets 16.667 MHz operation for all Stellaris® devices.

11.2.2.1 Interrupts

The register configurations of the sample sequencers dictate which events generate raw interrupts, but do not have control over whether the interrupt is actually sent to the interrupt controller. The ADC module's interrupt signals are controlled by the state of the MASK bits in the **ADC Interrupt Mask (ADCIM)** register. Interrupt status can be viewed at two locations: the **ADC Raw Interrupt Status (ADCRIS)** register, which shows the raw status of the various interrupt signals, and the **ADC Interrupt Status and Clear (ADCISC)** register, which shows active interrupts that are enabled by the **ADCIM** register. Sequencer interrupts are cleared by writing a 1 to the corresponding IN bit in **ADCISC**.

11.2.2.2 Prioritization

When sampling events (triggers) happen concurrently, they are prioritized for processing by the values in the **ADC Sample Sequencer Priority (ADCSSPRI)** register. Valid priority values are in the range of 0-3, with 0 being the highest priority and 3 being the lowest. Multiple active sample sequencer units with the same priority do not provide consistent results, so software must ensure that all active sample sequencer units have a unique priority value.

11.2.2.3 Sampling Events

Sample triggering for each sample sequencer is defined in the **ADC Event Multiplexer Select (ADCEMUX)** register. The external peripheral triggering sources vary by Stellaris® family member, but all devices share the "Controller" and "Always" triggers. Software can initiate sampling by setting the SSx bits in the **ADC Processor Sample Sequence Initiate (ADCPSSI)** register.

Care must be taken when using the "Always" trigger. If a sequence's priority is too high, it is possible to starve other lower priority sequences.

11.2.3 Hardware Sample Averaging Circuit

Higher precision results can be generated using the hardware averaging circuit, however, the improved results are at the cost of throughput. Up to 64 samples can be accumulated and averaged to form a single data entry in the sequencer FIFO. Throughput is decreased proportionally to the number of samples in the averaging calculation. For example, if the averaging circuit is configured to average 16 samples, the throughput is decreased by a factor of 16.

By default the averaging circuit is off and all data from the converter passes through to the sequencer FIFO. The averaging hardware is controlled by the **ADC Sample Averaging Control (ADCSAC)** register (see page 276). There is a single averaging circuit and all input channels receive the same amount of averaging whether they are single-ended or differential.

11.2.4 Analog-to-Digital Converter

The converter itself generates a 10-bit output value for selected analog input. Special analog pads are used to minimize the distortion on the input. An internal 3 V reference is used by the converter resulting in sample values ranging from 0x000 at 0 V input to 0x3FF at 3 V input when in single-ended input mode.

11.2.5 Differential Sampling

In addition to traditional single-ended sampling, the ADC module supports differential sampling of two analog input channels. To enable differential sampling, software must set the Dn bit in the **ADCSSCTL0n** register in a step's configuration nibble.

When a sequence step is configured for differential sampling, its corresponding value in the **ADCSSMUXn** register must be set to one of the four differential pairs, numbered 0-3. Differential pair 0 samples analog inputs 0 and 1; differential pair 1 samples analog inputs 2 and 3; and so on (see Table 11-2 on page 256). The ADC does not support other differential pairings such as analog input 0 with analog input 3. The number of differential pairs supported is dependent on the number of analog inputs (see Table 11-2 on page 256).

Table 11-2. Differential Sampling Pairs

Differential Pair	Analog Inputs
0	0 and 1
1	2 and 3

The voltage sampled in differential mode is the difference between the odd and even channels:

ΔV (differential voltage) = V_{IN_EVEN} (even channels) – V_{IN_ODD} (odd channels), therefore:

- If $\Delta V = 0$, then the conversion result = 0x1FF
- If $\Delta V > 0$, then the conversion result > 0x1FF (range is 0x1FF–0x3FF)

- If ΔV < 0, then the conversion result < 0x1FF (range is 0–0x1FF)

The differential pairs assign polarities to the analog inputs: the even-numbered input is always positive, and the odd-numbered input is always negative. In order for a valid conversion result to appear, the negative input must be in the range of ± 1.5 V of the positive input. If an analog input is greater than 3 V or less than 0 V (the valid range for analog inputs), the input voltage is clipped, meaning it appears as either 3 V or 0 V, respectively, to the ADC.

Figure 11-2 on page 257 shows an example of the negative input centered at 1.5 V. In this configuration, the differential range spans from -1.5 V to 1.5 V. Figure 11-3 on page 258 shows an example where the negative input is centered at -0.75 V, meaning inputs on the positive input saturate past a differential voltage of -0.75 V since the input voltage is less than 0 V. Figure 11-4 on page 258 shows an example of the negative input centered at 2.25 V, where inputs on the positive channel saturate past a differential voltage of 0.75 V since the input voltage would be greater than 3 V.

Figure 11-2. Differential Sampling Range, V_{IN_ODD} = 1.5 V

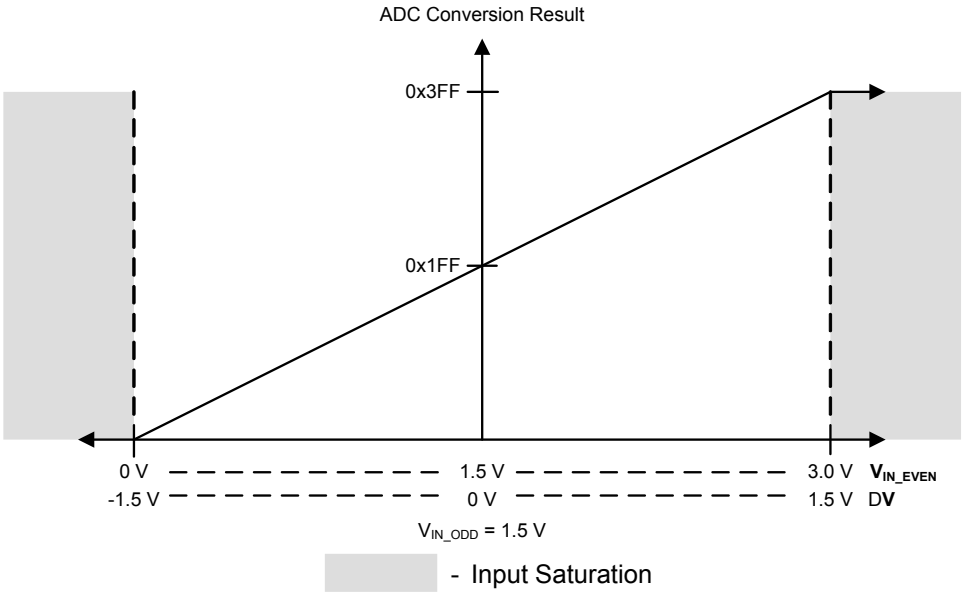


Figure 11-3. Differential Sampling Range, V_{IN_ODD} = 0.75 V

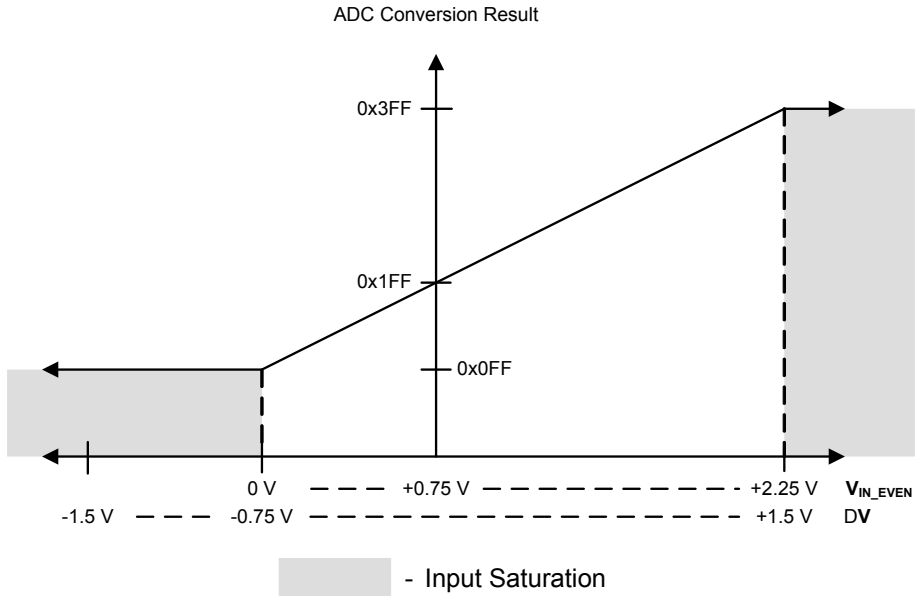
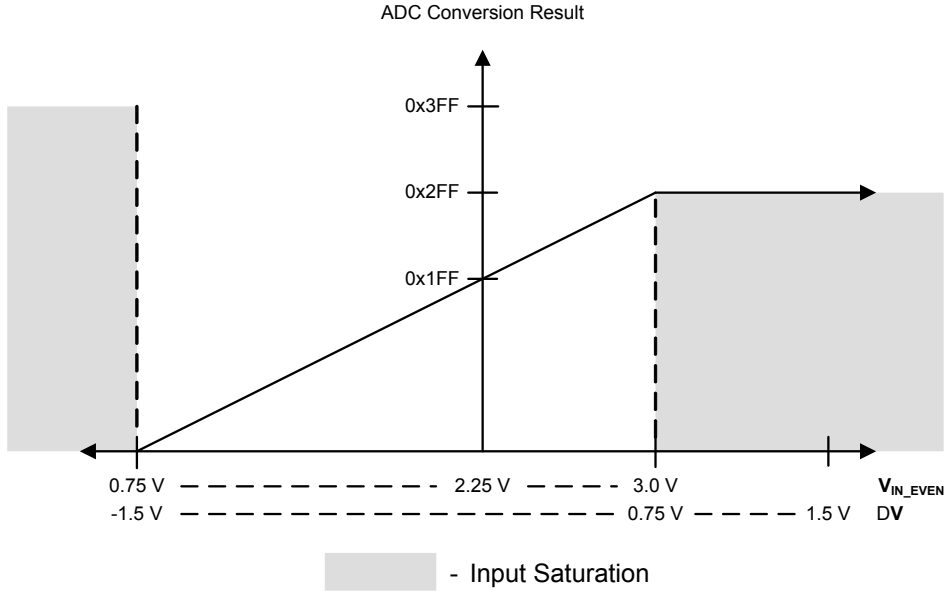


Figure 11-4. Differential Sampling Range, V_{IN_ODD} = 2.25 V



11.2.6 Test Modes

There is a user-available test mode that allows for loopback operation within the digital portion of the ADC module. This can be useful for debugging software without having to provide actual analog stimulus. This mode is available through the **ADC Test Mode Loopback (ADCTMLB)** register (see page 289).

11.2.7 Internal Temperature Sensor

The temperature sensor's primary purpose is to notify the system that the internal temperature is too high or low for reliable operation.

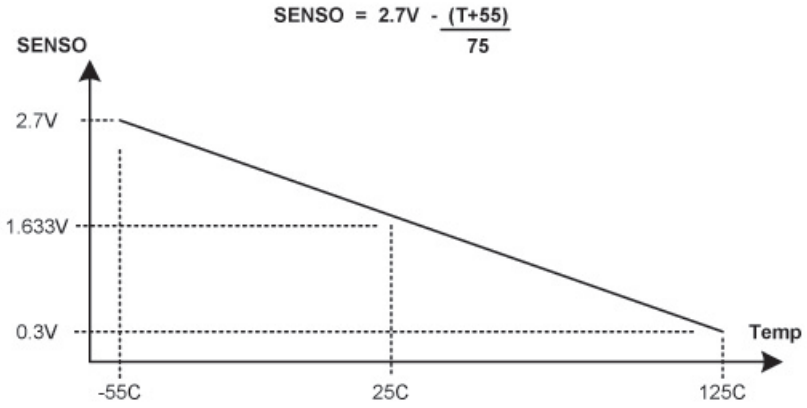
The temperature sensor does not have a separate enable, since it also contains the bandgap reference and must always be enabled. The reference is supplied to other analog modules; not just the ADC.

The internal temperature sensor provides an analog temperature reading as well as a reference voltage. The voltage at the output terminal SENSO is given by the following equation:

$$SENSO = 2.7 - ((T + 55) / 75)$$

This relation is shown in Figure 11-5 on page 259.

Figure 11-5. Internal Temperature Sensor Characteristic



11.3 Initialization and Configuration

In order for the ADC module to be used, the PLL must be enabled and using a supported crystal frequency (see the **RCC** register). Using unsupported frequencies can cause faulty operation in the ADC module.

11.3.1 Module Initialization

Initialization of the ADC module is a simple process with very few steps. The main steps include enabling the clock to the ADC and reconfiguring the sample sequencer priorities (if needed).

The initialization sequence for the ADC is as follows:

1. Enable the ADC clock by writing a value of 0x0001.0000 to the **RCGC0** register (see page 103).
2. If required by the application, reconfigure the sample sequencer priorities in the **ADCSSPRI** register. The default configuration has Sample Sequencer 0 with the highest priority, and Sample Sequencer 3 as the lowest priority.

11.3.2 Sample Sequencer Configuration

Configuration of the sample sequencers is slightly more complex than the module initialization since each sample sequence is completely programmable.

The configuration for each sample sequencer should be as follows:

1. Ensure that the sample sequencer is disabled by writing a 0 to the corresponding ASENn bit in the **ADCACTSS** register. Programming of the sample sequencers is allowed without having them enabled. Disabling the sequencer during programming prevents erroneous execution if a trigger event were to occur during the configuration process.
2. Configure the trigger event for the sample sequencer in the **ADCEMUX** register.
3. For each sample in the sample sequence, configure the corresponding input source in the **ADCSSMUXn** register.
4. For each sample in the sample sequence, configure the sample control bits in the corresponding nibble in the **ADCSSCTLn** register. When programming the last nibble, ensure that the END bit is set. Failure to set the END bit causes unpredictable behavior.
5. If interrupts are to be used, write a 1 to the corresponding MASK bit in the **ADCIM** register.
6. Enable the sample sequencer logic by writing a 1 to the corresponding ASENn bit in the **ADCACTSS** register.

11.4 Register Map

Table 11-3 on page 260 lists the ADC registers. The offset listed is a hexadecimal increment to the register's address, relative to the ADC base address of 0x4003.8000.

Table 11-3. ADC Register Map

Offset	Name	Type	Reset	Description	See page
0x000	ADCACTSS	R/W	0x0000.0000	ADC Active Sample Sequencer	262
0x004	ADCRIS	RO	0x0000.0000	ADC Raw Interrupt Status	263
0x008	ADCIM	R/W	0x0000.0000	ADC Interrupt Mask	264
0x00C	ADCISC	R/W1C	0x0000.0000	ADC Interrupt Status and Clear	265
0x010	ADCOSTAT	R/W1C	0x0000.0000	ADC Overflow Status	267
0x014	ADCEMUX	R/W	0x0000.0000	ADC Event Multiplexer Select	268
0x018	ADCUSTAT	R/W1C	0x0000.0000	ADC Underflow Status	272
0x020	ADCSSPRI	R/W	0x0000.3210	ADC Sample Sequencer Priority	273
0x028	ADCPSSI	WO	-	ADC Processor Sample Sequence Initiate	275
0x030	ADCSAC	R/W	0x0000.0000	ADC Sample Averaging Control	276
0x040	ADCSSMUX0	R/W	0x0000.0000	ADC Sample Sequence Input Multiplexer Select 0	277
0x044	ADCSSCTL0	R/W	0x0000.0000	ADC Sample Sequence Control 0	279
0x048	ADCSSFIFO0	RO	-	ADC Sample Sequence Result FIFO 0	282

Table 11-3. ADC Register Map *(continued)*

Offset	Name	Type	Reset	Description	See page
0x04C	ADCSSFSTAT0	RO	0x0000.0100	ADC Sample Sequence FIFO 0 Status	283
0x060	ADCSSMUX1	R/W	0x0000.0000	ADC Sample Sequence Input Multiplexer Select 1	284
0x064	ADCSSCTL1	R/W	0x0000.0000	ADC Sample Sequence Control 1	285
0x068	ADCSSFIFO1	RO	-	ADC Sample Sequence Result FIFO 1	282
0x06C	ADCSSFSTAT1	RO	0x0000.0100	ADC Sample Sequence FIFO 1 Status	283
0x080	ADCSSMUX2	R/W	0x0000.0000	ADC Sample Sequence Input Multiplexer Select 2	284
0x084	ADCSSCTL2	R/W	0x0000.0000	ADC Sample Sequence Control 2	285
0x088	ADCSSFIFO2	RO	-	ADC Sample Sequence Result FIFO 2	282
0x08C	ADCSSFSTAT2	RO	0x0000.0100	ADC Sample Sequence FIFO 2 Status	283
0x0A0	ADCSSMUX3	R/W	0x0000.0000	ADC Sample Sequence Input Multiplexer Select 3	287
0x0A4	ADCSSCTL3	R/W	0x0000.0002	ADC Sample Sequence Control 3	288
0x0A8	ADCSSFIFO3	RO	-	ADC Sample Sequence Result FIFO 3	282
0x0AC	ADCSSFSTAT3	RO	0x0000.0100	ADC Sample Sequence FIFO 3 Status	283
0x100	ADCTMLB	R/W	0x0000.0000	ADC Test Mode Loopback	289

11.5 Register Descriptions

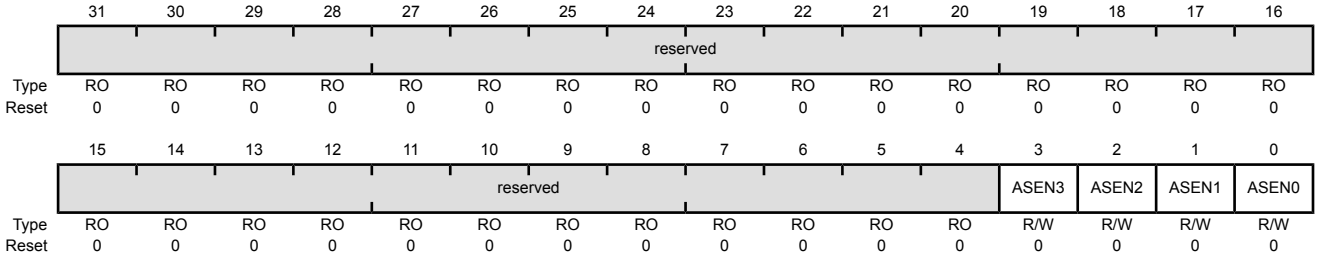
The remainder of this section lists and describes the ADC registers, in numerical order by address offset.

Register 1: ADC Active Sample Sequencer (ADCACTSS), offset 0x000

This register controls the activation of the sample sequencers. Each sample sequencer can be enabled or disabled independently.

ADC Active Sample Sequencer (ADCACTSS)

Base 0x4003.8000
Offset 0x000
Type R/W, reset 0x0000.0000



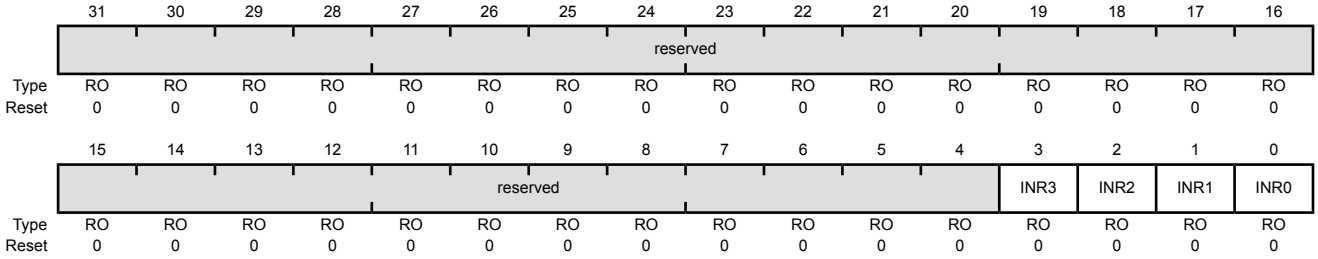
Bit/Field	Name	Type	Reset	Description
31:4	reserved	RO	0x0000.000	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
3	ASEN3	R/W	0	ADC SS3 Enable Specifies whether Sample Sequencer 3 is enabled. If set, the sample sequence logic for Sequencer 3 is active. Otherwise, the sequencer is inactive.
2	ASEN2	R/W	0	ADC SS2 Enable Specifies whether Sample Sequencer 2 is enabled. If set, the sample sequence logic for Sequencer 2 is active. Otherwise, the sequencer is inactive.
1	ASEN1	R/W	0	ADC SS1 Enable Specifies whether Sample Sequencer 1 is enabled. If set, the sample sequence logic for Sequencer 1 is active. Otherwise, the sequencer is inactive.
0	ASEN0	R/W	0	ADC SS0 Enable Specifies whether Sample Sequencer 0 is enabled. If set, the sample sequence logic for Sequencer 0 is active. Otherwise, the sequencer is inactive.

Register 2: ADC Raw Interrupt Status (ADCRIS), offset 0x004

This register shows the status of the raw interrupt signal of each sample sequencer. These bits may be polled by software to look for interrupt conditions without having to generate controller interrupts.

ADC Raw Interrupt Status (ADCRIS)

Base 0x4003.8000
Offset 0x004
Type RO, reset 0x0000.0000



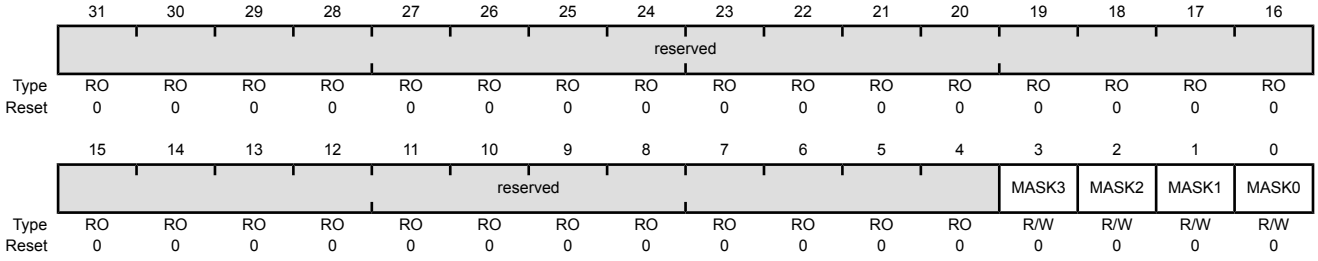
Bit/Field	Name	Type	Reset	Description
31:4	reserved	RO	0x000	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
3	INR3	RO	0	SS3 Raw Interrupt Status This bit is set by hardware when a sample with its respective **ADCSSCTL3** `IE` bit has completed conversion. This bit is cleared by setting the `IN3` bit in the **ADCISC** register.
2	INR2	RO	0	SS2 Raw Interrupt Status This bit is set by hardware when a sample with its respective **ADCSSCTL2** `IE` bit has completed conversion. This bit is cleared by setting the `IN2` bit in the **ADCISC** register.
1	INR1	RO	0	SS1 Raw Interrupt Status This bit is set by hardware when a sample with its respective **ADCSSCTL1** `IE` bit has completed conversion. This bit is cleared by setting the `IN1` bit in the **ADCISC** register.
0	INR0	RO	0	SS0 Raw Interrupt Status This bit is set by hardware when a sample with its respective **ADCSSCTL0** `IE` bit has completed conversion. This bit is cleared by setting the `IN30` bit in the **ADCISC** register.

Register 3: ADC Interrupt Mask (ADCIM), offset 0x008

This register controls whether the sample sequencer raw interrupt signals are promoted to controller interrupts. Each raw interrupt signal can be masked independently.

ADC Interrupt Mask (ADCIM)

Base 0x4003.8000
Offset 0x008
Type R/W, reset 0x0000.0000



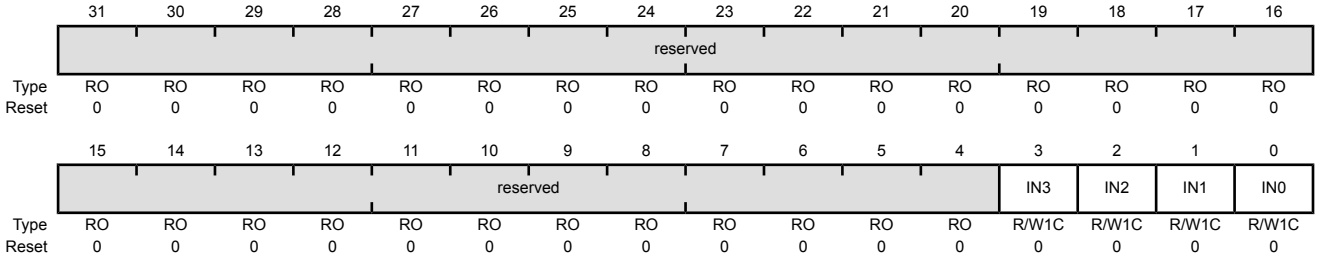
Bit/Field	Name	Type	Reset	Description
31:4	reserved	RO	0x000	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
3	MASK3	R/W	0	SS3 Interrupt Mask When set, this bit allows the raw interrupt signal from Sample Sequencer 3 (**ADCRIS** register INR3 bit) to be promoted to a controller interrupt. When clear, the status of Sample Sequencer 3 does not affect the SS3 interrupt status.
2	MASK2	R/W	0	SS2 Interrupt Mask When set, this bit allows the raw interrupt signal from Sample Sequencer 2 (**ADCRIS** register INR2 bit) to be promoted to a controller interrupt. When clear, the status of Sample Sequencer 2 does not affect the SS2 interrupt status.
1	MASK1	R/W	0	SS1 Interrupt Mask When set, this bit allows the raw interrupt signal from Sample Sequencer 1 (**ADCRIS** register INR1 bit) to be promoted to a controller interrupt. When clear, the status of Sample Sequencer 1 does not affect the SS1 interrupt status.
0	MASK0	R/W	0	SS0 Interrupt Mask When set, this bit allows the raw interrupt signal from Sample Sequencer 0 (**ADCRIS** register INR0 bit) to be promoted to a controller interrupt. When clear, the status of Sample Sequencer 0 does not affect the SS0 interrupt status.

Register 4: ADC Interrupt Status and Clear (ADCISC), offset 0x00C

This register provides the mechanism for clearing sample sequence interrupt conditions and shows the status of controller interrupts generated by the sample sequencers. When read, each bit field is the logical AND of the respective INR and MASK bits. Sample sequence nterrupts are cleared by setting the corresponding bit position. If software is polling the **ADCRIS** instead of generating interrupts, the sample sequence INR bits are still cleared via the **ADCISC** register, even if the IN bit is not set.

ADC Interrupt Status and Clear (ADCISC)

Base 0x4003.8000
Offset 0x00C
Type R/W1C, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:4	reserved	RO	0x000	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
3	IN3	R/W1C	0	SS3 Interrupt Status and Clear This bit is set when both the INR3 bit in the **ADCRIS** register and the MASK3 bit in the **ADCIM** register are set, providing a level-based interrupt to the controller. This bit is cleared by writing a 1. Clearing this bit also clears the INR3 bit.
2	IN2	R/W1C	0	SS2 Interrupt Status and Clear This bit is set when both the INR2 bit in the **ADCRIS** register and the MASK2 bit in the **ADCIM** register are set, providing a level-based interrupt to the controller. This bit is cleared by writing a 1. Clearing this bit also clears the INR2 bit.
1	IN1	R/W1C	0	SS1 Interrupt Status and Clear This bit is set when both the INR1 bit in the **ADCRIS** register and the MASK1 bit in the **ADCIM** register are set, providing a level-based interrupt to the controller. This bit is cleared by writing a 1. Clearing this bit also clears the INR1 bit.

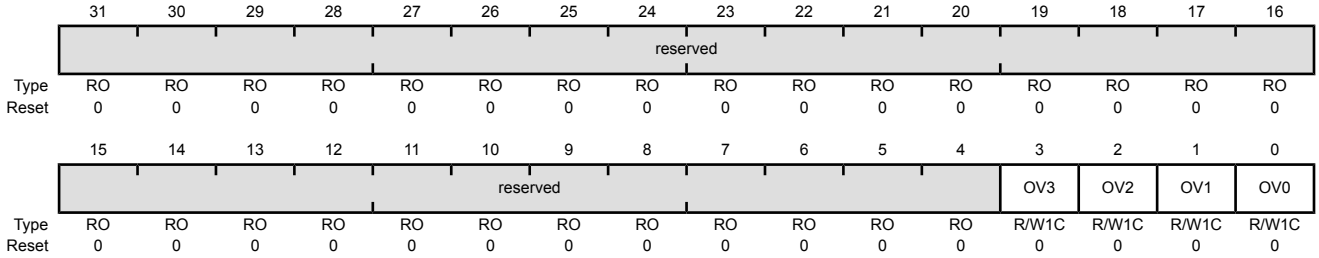
Bit/Field	Name	Type	Reset	Description
0	IN0	R/W1C	0	SS0 Interrupt Status and Clear This bit is set when both the INR0 bit in the **ADCRIS** register and the MASK0 bit in the **ADCIM** register are set, providing a level-based interrupt to the controller. This bit is cleared by writing a 1. Clearing this bit also clears the INR0 bit.

Register 5: ADC Overflow Status (ADCOSTAT), offset 0x010

This register indicates overflow conditions in the sample sequencer FIFOs. Once the overflow condition has been handled by software, the condition can be cleared by writing a 1 to the corresponding bit position.

ADC Overflow Status (ADCOSTAT)

Base 0x4003.8000
Offset 0x010
Type R/W1C, reset 0x0000.0000



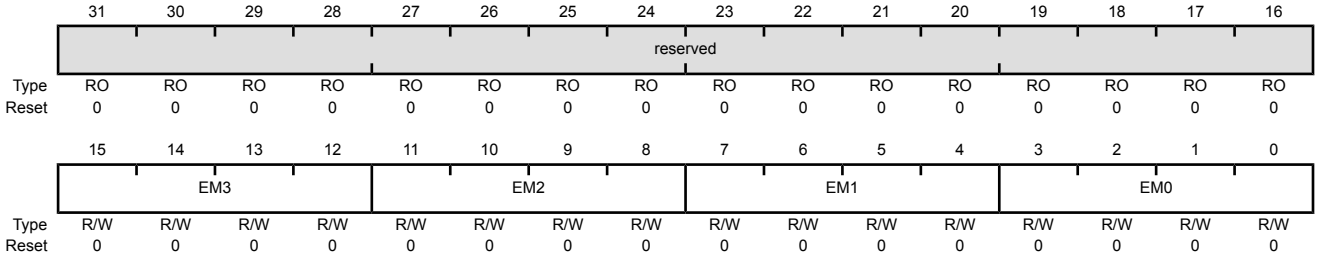
Bit/Field	Name	Type	Reset	Description
31:4	reserved	RO	0x0000.000	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
3	OV3	R/W1C	0	SS3 FIFO Overflow When set, this bit specifies that the FIFO for Sample Sequencer 3 has hit an overflow condition where the FIFO is full and a write was requested. When an overflow is detected, the most recent write is dropped. This bit is cleared by writing a 1.
2	OV2	R/W1C	0	SS2 FIFO Overflow When set, this bit specifies that the FIFO for Sample Sequencer 2 has hit an overflow condition where the FIFO is full and a write was requested. When an overflow is detected, the most recent write is dropped. This bit is cleared by writing a 1.
1	OV1	R/W1C	0	SS1 FIFO Overflow When set, this bit specifies that the FIFO for Sample Sequencer 1 has hit an overflow condition where the FIFO is full and a write was requested. When an overflow is detected, the most recent write is dropped. This bit is cleared by writing a 1.
0	OV0	R/W1C	0	SS0 FIFO Overflow When set, this bit specifies that the FIFO for Sample Sequencer 0 has hit an overflow condition where the FIFO is full and a write was requested. When an overflow is detected, the most recent write is dropped. This bit is cleared by writing a 1.

Register 6: ADC Event Multiplexer Select (ADCEMUX), offset 0x014

The **ADCEMUX** selects the event (trigger) that initiates sampling for each sample sequencer. Each sample sequencer can be configured with a unique trigger source.

ADC Event Multiplexer Select (ADCEMUX)

Base 0x4003.8000
Offset 0x014
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:16	reserved	RO	0x0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
15:12	EM3	R/W	0x0	SS3 Trigger Select This field selects the trigger source for Sample Sequencer 3. The valid configurations for this field are:

Value	Event
0x0	Controller (default)
0x1	Analog Comparator 0
0x2	Analog Comparator 1
0x3	Reserved
0x4	External (GPIO PB4)
0x5	Timer In addition, the trigger must be enabled with the TnOTE bit in the **GPTMCTL** register (see page 210).
0x6	PWM0 The PWM module 0 trigger can be configured with the **PWM0 Interrupt and Trigger Enable (PWM0INTEN)** register, see page 482.
0x7	PWM1 The PWM module 1 trigger can be configured with the **PWM1INTEN** register, see page 482.
0x8	reserved
0x9-0xE	reserved
0xF	Always (continuously sample)

Bit/Field	Name	Type	Reset	Description
11:8	EM2	R/W	0x0	SS2 Trigger Select This field selects the trigger source for Sample Sequencer 2. The valid configurations for this field are:

Value	Event
0x0	Controller (default)
0x1	Analog Comparator 0
0x2	Analog Comparator 1
0x3	Reserved
0x4	External (GPIO PB4)
0x5	Timer In addition, the trigger must be enabled with the TnOTE bit in the **GPTMCTL** register (see page 210).
0x6	PWM0 The PWM module 0 trigger can be configured with the **PWM0 Interrupt and Trigger Enable (PWM0INTEN)** register, see page 482.
0x7	PWM1 The PWM module 1 trigger can be configured with the **PWM1INTEN** register, see page 482.
0x8	reserved
0x9-0xE	reserved
0xF	Always (continuously sample)

Bit/Field	Name	Type	Reset	Description
7:4	EM1	R/W	0x0	SS1 Trigger Select This field selects the trigger source for Sample Sequencer 1. The valid configurations for this field are:

Value	Event
0x0	Controller (default)
0x1	Analog Comparator 0
0x2	Analog Comparator 1
0x3	Reserved
0x4	External (GPIO PB4)
0x5	Timer In addition, the trigger must be enabled with the TnOTE bit in the **GPTMCTL** register (see page 210).
0x6	PWM0 The PWM module 0 trigger can be configured with the **PWM0 Interrupt and Trigger Enable (PWM0INTEN)** register, see page 482.
0x7	PWM1 The PWM module 1 trigger can be configured with the **PWM1INTEN** register, see page 482.
0x8	reserved
0x9-0xE	reserved
0xF	Always (continuously sample)

Bit/Field	Name	Type	Reset	Description
3:0	EM0	R/W	0x0	SS0 Trigger Select This field selects the trigger source for Sample Sequencer 0. The valid configurations for this field are:

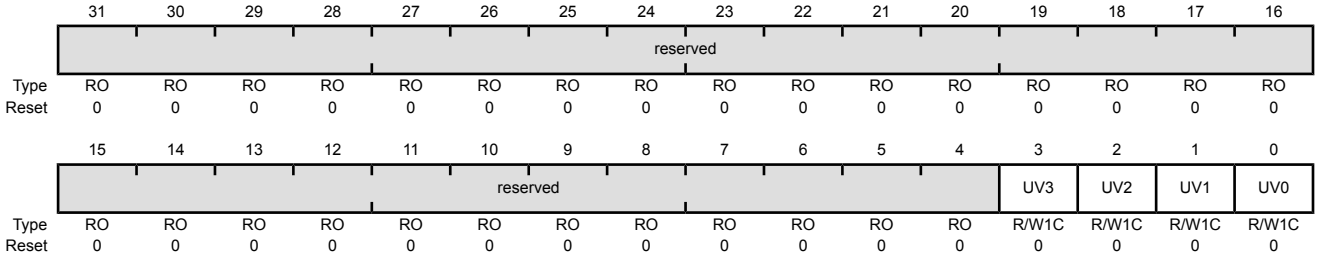
Value	Event
0x0	Controller (default)
0x1	Analog Comparator 0
0x2	Analog Comparator 1
0x3	Reserved
0x4	External (GPIO PB4)
0x5	Timer In addition, the trigger must be enabled with the TnOTE bit in the **GPTMCTL** register (see page 210).
0x6	PWM0 The PWM module 0 trigger can be configured with the **PWM0 Interrupt and Trigger Enable (PWM0INTEN)** register, see page 482.
0x7	PWM1 The PWM module 1 trigger can be configured with the **PWM1INTEN** register, see page 482.
0x8	reserved
0x9-0xE	reserved
0xF	Always (continuously sample)

Register 7: ADC Underflow Status (ADCUSTAT), offset 0x018

This register indicates underflow conditions in the sample sequencer FIFOs. The corresponding underflow condition is cleared by writing a 1 to the relevant bit position.

ADC Underflow Status (ADCUSTAT)

Base 0x4003.8000
Offset 0x018
Type R/W1C, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:4	reserved	RO	0x0000.000	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
3	UV3	R/W1C	0	SS3 FIFO Underflow When set, this bit specifies that the FIFO for Sample Sequencer 3 has hit an underflow condition where the FIFO is empty and a read was requested. The problematic read does not move the FIFO pointers, and 0s are returned. This bit is cleared by writing a 1.
2	UV2	R/W1C	0	SS2 FIFO Underflow When set, this bit specifies that the FIFO for Sample Sequencer 2 has hit an underflow condition where the FIFO is empty and a read was requested. The problematic read does not move the FIFO pointers, and 0s are returned. This bit is cleared by writing a 1.
1	UV1	R/W1C	0	SS1 FIFO Underflow When set, this bit specifies that the FIFO for Sample Sequencer 1 has hit an underflow condition where the FIFO is empty and a read was requested. The problematic read does not move the FIFO pointers, and 0s are returned. This bit is cleared by writing a 1.
0	UV0	R/W1C	0	SS0 FIFO Underflow When set, this bit specifies that the FIFO for Sample Sequencer 0 has hit an underflow condition where the FIFO is empty and a read was requested. The problematic read does not move the FIFO pointers, and 0s are returned. This bit is cleared by writing a 1.

Register 8: ADC Sample Sequencer Priority (ADCSSPRI), offset 0x020

This register sets the priority for each of the sample sequencers. Out of reset, Sequencer 0 has the highest priority, and Sequencer 3 has the lowest priority. When reconfiguring sequence priorities, each sequence must have a unique priority for the ADC to operate properly.

ADC Sample Sequencer Priority (ADCSSPRI)

Base 0x4003.8000
Offset 0x020
Type R/W, reset 0x0000.3210



Bit/Field	Name	Type	Reset	Description
31:14	reserved	RO	0x0000.0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
13:12	SS3	R/W	0x3	SS3 Priority This field contains a binary-encoded value that specifies the priority encoding of Sample Sequencer 3. A priority encoding of 0 is highest and 3 is lowest. The priorities assigned to the sequencers must be uniquely mapped. The ADC may not operate properly if two or more fields are equal.
11:10	reserved	RO	0x0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
9:8	SS2	R/W	0x2	SS2 Priority This field contains a binary-encoded value that specifies the priority encoding of Sample Sequencer 2. A priority encoding of 0 is highest and 3 is lowest. The priorities assigned to the sequencers must be uniquely mapped. The ADC may not operate properly if two or more fields are equal.
7:6	reserved	RO	0x0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
5:4	SS1	R/W	0x1	SS1 Priority This field contains a binary-encoded value that specifies the priority encoding of Sample Sequencer 1. A priority encoding of 0 is highest and 3 is lowest. The priorities assigned to the sequencers must be uniquely mapped. The ADC may not operate properly if two or more fields are equal.
3:2	reserved	RO	0x0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Bit/Field	Name	Type	Reset	Description
1:0	SS0	R/W	0x0	SS0 Priority This field contains a binary-encoded value that specifies the priority encoding of Sample Sequencer 0. A priority encoding of 0 is highest and 3 is lowest. The priorities assigned to the sequencers must be uniquely mapped. The ADC may not operate properly if two or more fields are equal.

Register 9: ADC Processor Sample Sequence Initiate (ADCPSSI), offset 0x028

This register provides a mechanism for application software to initiate sampling in the sample sequencers. Sample sequences can be initiated individually or in any combination. When multiple sequences are triggered simultaneously, the priority encodings in **ADCSSPRI** dictate execution order.

ADC Processor Sample Sequence Initiate (ADCPSSI)

Base 0x4003.8000
Offset 0x028
Type WO, reset -

	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	reserved															
Type	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	reserved												SS3	SS2	SS1	SS0
Type	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	WO	WO	WO	WO
Reset	0	0	0	0	0	0	0	0	0	0	0	0	-	-	-	-

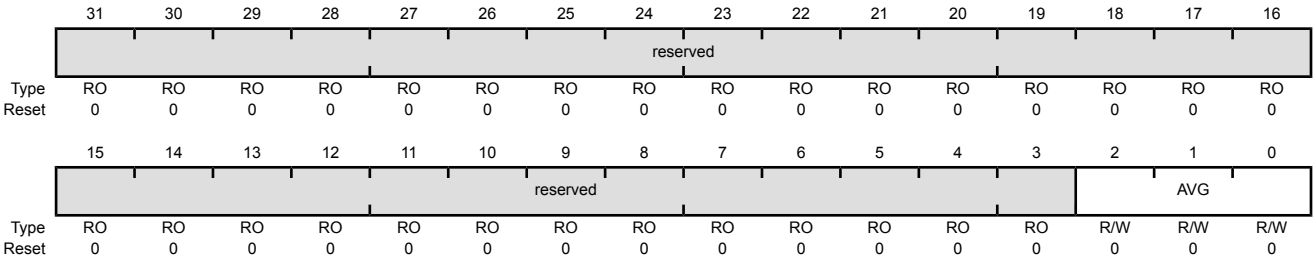
Bit/Field	Name	Type	Reset	Description
31:4	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
3	SS3	WO	-	SS3 Initiate When set, this bit triggers sampling on Sample Sequencer 3 if the sequencer is enabled in the **ADCACTSS** register. Only a write by software is valid; a read of this register returns no meaningful data.
2	SS2	WO	-	SS2 Initiate When set, this bit triggers sampling on Sample Sequencer 2 if the sequencer is enabled in the **ADCACTSS** register. Only a write by software is valid; a read of this register returns no meaningful data.
1	SS1	WO	-	SS1 Initiate When set, this bit triggers sampling on Sample Sequencer 1 if the sequencer is enabled in the **ADCACTSS** register. Only a write by software is valid; a read of this register returns no meaningful data.
0	SS0	WO	-	SS0 Initiate When set, this bit triggers sampling on Sample Sequencer 0 if the sequencer is enabled in the **ADCACTSS** register. Only a write by software is valid; a read of this register returns no meaningful data.

Register 10: ADC Sample Averaging Control (ADCSAC), offset 0x030

This register controls the amount of hardware averaging applied to conversion results. The final conversion result stored in the FIFO is averaged from 2^{AVG} consecutive ADC samples at the specified ADC speed. If AVG is 0, the sample is passed directly through without any averaging. If AVG=6, then 64 consecutive ADC samples are averaged to generate one result in the sequencer FIFO. An AVG = 7 provides unpredictable results.

ADC Sample Averaging Control (ADCSAC)

Base 0x4003.8000
Offset 0x030
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:3	reserved	RO	0x0000.000	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
2:0	AVG	R/W	0x0	Hardware Averaging Control Specifies the amount of hardware averaging that will be applied to ADC samples. The AVG field can be any value between 0 and 6. Entering a value of 7 creates unpredictable results. Value Description 0x0 No hardware oversampling 0x1 2x hardware oversampling 0x2 4x hardware oversampling 0x3 8x hardware oversampling 0x4 16x hardware oversampling 0x5 32x hardware oversampling 0x6 64x hardware oversampling 0x7 Reserved

Register 11: ADC Sample Sequence Input Multiplexer Select 0 (ADCSSMUX0), offset 0x040

This register defines the analog input configuration for each sample in a sequence executed with Sample Sequencer 0. This register is 32 bits wide and contains information for eight possible samples.

ADC Sample Sequence Input Multiplexer Select 0 (ADCSSMUX0)

Base 0x4003.8000
Offset 0x040
Type R/W, reset 0x0000.0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
reserved		MUX7		reserved		MUX6		reserved		MUX5		reserved		MUX4	
Type	RO	RO	R/W	R/W	RO	RO	R/W	R/W	RO	RO	R/W	R/W	RO	RO	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
reserved		MUX3		reserved		MUX2		reserved		MUX1		reserved		MUX0	
Type	RO	RO	R/W	R/W	RO	RO	R/W	R/W	RO	RO	R/W	R/W	RO	RO	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit/Field	Name	Type	Reset	Description
31:30	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
29:28	MUX7	R/W	0x0	8th Sample Input Select The MUX7 field is used during the eighth sample of a sequence executed with the sample sequencer. It specifies which of the analog inputs is sampled for the analog-to-digital conversion. The value set here indicates the corresponding pin, for example, a value of 1 indicates the input is ADC1.
27:26	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
25:24	MUX6	R/W	0x0	7th Sample Input Select The MUX6 field is used during the seventh sample of a sequence executed with the sample sequencer. It specifies which of the analog inputs is sampled for the analog-to-digital conversion.
23:22	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
21:20	MUX5	R/W	0x0	6th Sample Input Select The MUX5 field is used during the sixth sample of a sequence executed with the sample sequencer. It specifies which of the analog inputs is sampled for the analog-to-digital conversion.
19:18	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Bit/Field	Name	Type	Reset	Description
17:16	MUX4	R/W	0x0	5th Sample Input Select The MUX4 field is used during the fifth sample of a sequence executed with the sample sequencer. It specifies which of the analog inputs is sampled for the analog-to-digital conversion.
15:14	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
13:12	MUX3	R/W	0x0	4th Sample Input Select The MUX3 field is used during the fourth sample of a sequence executed with the sample sequencer. It specifies which of the analog inputs is sampled for the analog-to-digital conversion.
11:10	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
9:8	MUX2	R/W	0x0	3rd Sample Input Select The MUX72 field is used during the third sample of a sequence executed with the sample sequencer. It specifies which of the analog inputs is sampled for the analog-to-digital conversion.
7:6	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
5:4	MUX1	R/W	0x0	2nd Sample Input Select The MUX1 field is used during the second sample of a sequence executed with the sample sequencer. It specifies which of the analog inputs is sampled for the analog-to-digital conversion.
3:2	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1:0	MUX0	R/W	0x0	1st Sample Input Select The MUX0 field is used during the first sample of a sequence executed with the sample sequencer. It specifies which of the analog inputs is sampled for the analog-to-digital conversion.

Register 12: ADC Sample Sequence Control 0 (ADCSSCTL0), offset 0x044

This register contains the configuration information for each sample for a sequence executed with a sample sequencer. When configuring a sample sequence, the END bit must be set at some point, whether it be after the first sample, last sample, or any sample in between. This register is 32-bits wide and contains information for eight possible samples.

ADC Sample Sequence Control 0 (ADCSSCTL0)

Base 0x4003.8000
Offset 0x044
Type R/W, reset 0x0000.0000

	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	TS7	IE7	END7	D7	TS6	IE6	END6	D6	TS5	IE5	END5	D5	TS4	IE4	END4	D4
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	TS3	IE3	END3	D3	TS2	IE2	END2	D2	TS1	IE1	END1	D1	TS0	IE0	END0	D0
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit/Field	Name	Type	Reset	Description
31	TS7	R/W	0	8th Sample Temp Sensor Select This bit is used during the eighth sample of the sample sequence and and specifies the input source of the sample. When set, the temperature sensor is read. When clear, the input pin specified by the **ADCSSMUX** register is read.
30	IE7	R/W	0	8th Sample Interrupt Enable This bit is used during the eighth sample of the sample sequence and specifies whether the raw interrupt signal (INR0 bit) is asserted at the end of the sample's conversion. If the MASK0 bit in the **ADCIM** register is set, the interrupt is promoted to a controller-level interrupt. When this bit is set, the raw interrupt is asserted. When this bit is clear, the raw interrupt is not asserted. It is legal to have multiple samples within a sequence generate interrupts.
29	END7	R/W	0	8th Sample is End of Sequence The END7 bit indicates that this is the last sample of the sequence. It is possible to end the sequence on any sample position. Samples defined after the sample containing a set END are not requested for conversion even though the fields may be non-zero. It is required that software write the END bit somewhere within the sequence. (Sample Sequencer 3, which only has a single sample in the sequence, is hardwired to have the END0 bit set.) Setting this bit indicates that this sample is the last in the sequence.
28	D7	R/W	0	8th Sample Diff Input Select The D7 bit indicates that the analog input is to be differentially sampled. The corresponding **ADCSSMUXx** nibble must be set to the pair number "i", where the paired inputs are "2i and 2i+1". The temperature sensor does not have a differential option. When set, the analog inputs are differentially sampled.

Bit/Field	Name	Type	Reset	Description
27	TS6	R/W	0	7th Sample Temp Sensor Select Same definition as TS7 but used during the seventh sample.
26	IE6	R/W	0	7th Sample Interrupt Enable Same definition as IE7 but used during the seventh sample.
25	END6	R/W	0	7th Sample is End of Sequence Same definition as END7 but used during the seventh sample.
24	D6	R/W	0	7th Sample Diff Input Select Same definition as D7 but used during the seventh sample.
23	TS5	R/W	0	6th Sample Temp Sensor Select Same definition as TS7 but used during the sixth sample.
22	IE5	R/W	0	6th Sample Interrupt Enable Same definition as IE7 but used during the sixth sample.
21	END5	R/W	0	6th Sample is End of Sequence Same definition as END7 but used during the sixth sample.
20	D5	R/W	0	6th Sample Diff Input Select Same definition as D7 but used during the sixth sample.
19	TS4	R/W	0	5th Sample Temp Sensor Select Same definition as TS7 but used during the fifth sample.
18	IE4	R/W	0	5th Sample Interrupt Enable Same definition as IE7 but used during the fifth sample.
17	END4	R/W	0	5th Sample is End of Sequence Same definition as END7 but used during the fifth sample.
16	D4	R/W	0	5th Sample Diff Input Select Same definition as D7 but used during the fifth sample.
15	TS3	R/W	0	4th Sample Temp Sensor Select Same definition as TS7 but used during the fourth sample.
14	IE3	R/W	0	4th Sample Interrupt Enable Same definition as IE7 but used during the fourth sample.
13	END3	R/W	0	4th Sample is End of Sequence Same definition as END7 but used during the fourth sample.
12	D3	R/W	0	4th Sample Diff Input Select Same definition as D7 but used during the fourth sample.

Bit/Field	Name	Type	Reset	Description
11	TS2	R/W	0	3rd Sample Temp Sensor Select Same definition as TS7 but used during the third sample.
10	IE2	R/W	0	3rd Sample Interrupt Enable Same definition as IE7 but used during the third sample.
9	END2	R/W	0	3rd Sample is End of Sequence Same definition as END7 but used during the third sample.
8	D2	R/W	0	3rd Sample Diff Input Select Same definition as D7 but used during the third sample.
7	TS1	R/W	0	2nd Sample Temp Sensor Select Same definition as TS7 but used during the second sample.
6	IE1	R/W	0	2nd Sample Interrupt Enable Same definition as IE7 but used during the second sample.
5	END1	R/W	0	2nd Sample is End of Sequence Same definition as END7 but used during the second sample.
4	D1	R/W	0	2nd Sample Diff Input Select Same definition as D7 but used during the second sample.
3	TS0	R/W	0	1st Sample Temp Sensor Select Same definition as TS7 but used during the first sample.
2	IE0	R/W	0	1st Sample Interrupt Enable Same definition as IE7 but used during the first sample.
1	END0	R/W	0	1st Sample is End of Sequence Same definition as END7 but used during the first sample.
0	D0	R/W	0	1st Sample Diff Input Select Same definition as D7 but used during the first sample.

Register 13: ADC Sample Sequence Result FIFO 0 (ADCSSFIFO0), offset 0x048

Register 14: ADC Sample Sequence Result FIFO 1 (ADCSSFIFO1), offset 0x068

Register 15: ADC Sample Sequence Result FIFO 2 (ADCSSFIFO2), offset 0x088

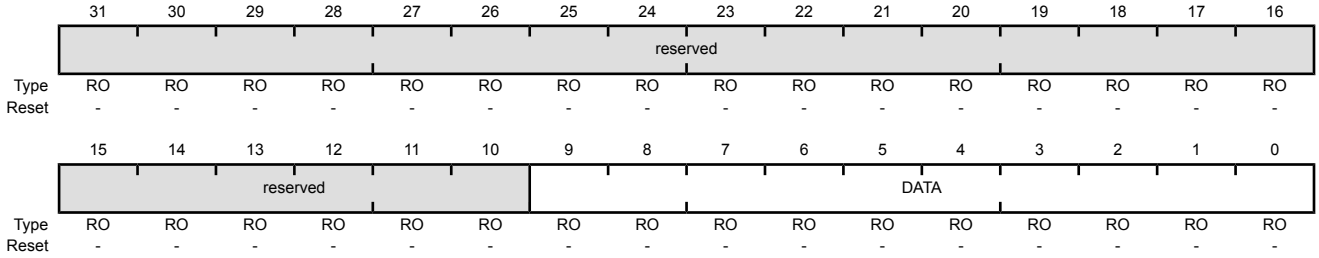
Register 16: ADC Sample Sequence Result FIFO 3 (ADCSSFIFO3), offset 0x0A8

Important: Use caution when reading this register. Performing a read may change bit status.

This register contains the conversion results for samples collected with the sample sequencer (the **ADCSSFIFO0** register is used for Sample Sequencer 0, **ADCSSFIFO1** for Sequencer 1, **ADCSSFIFO2** for Sequencer 2, and **ADCSSFIFO3** for Sequencer 3). Reads of this register return conversion result data in the order sample 0, sample 1, and so on, until the FIFO is empty. If the FIFO is not properly handled by software, overflow and underflow conditions are registered in the **ADCOSTAT** and **ADCUSTAT** registers.

ADC Sample Sequence Result FIFO 0 (ADCSSFIFO0)

Base 0x4003.8000
Offset 0x048
Type RO, reset -



Bit/Field	Name	Type	Reset	Description
31:10	reserved	RO	-	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
9:0	DATA	RO	-	Conversion Result Data

Register 17: ADC Sample Sequence FIFO 0 Status (ADCSSFSTAT0), offset 0x04C

Register 18: ADC Sample Sequence FIFO 1 Status (ADCSSFSTAT1), offset 0x06C

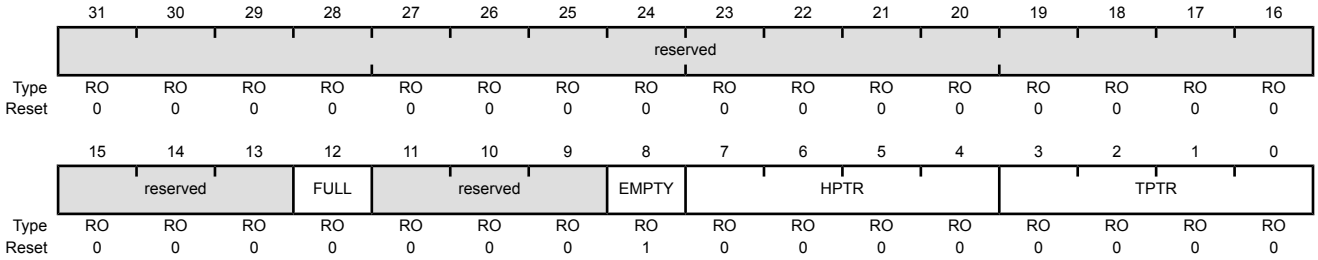
Register 19: ADC Sample Sequence FIFO 2 Status (ADCSSFSTAT2), offset 0x08C

Register 20: ADC Sample Sequence FIFO 3 Status (ADCSSFSTAT3), offset 0x0AC

This register provides a window into the sample sequencer, providing full/empty status information as well as the positions of the head and tail pointers. The reset value of 0x100 indicates an empty FIFO. The **ADCSSFSTAT0** register provides status on FIFO0, **ADCSSFSTAT1** on FIFO1, **ADCSSFSTAT2** on FIFO2, and **ADCSSFSTAT3** on FIFO3.

ADC Sample Sequence FIFO 0 Status (ADCSSFSTAT0)

Base 0x4003.8000
Offset 0x04C
Type RO, reset 0x0000.0100



Bit/Field	Name	Type	Reset	Description
31:13	reserved	RO	0x0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
12	FULL	RO	0	FIFO Full When set, this bit indicates that the FIFO is currently full.
11:9	reserved	RO	0x0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
8	EMPTY	RO	1	FIFO Empty When set, this bit indicates that the FIFO is currently empty.
7:4	HPTR	RO	0x0	FIFO Head Pointer This field contains the current "head" pointer index for the FIFO, that is, the next entry to be written.
3:0	TPTR	RO	0x0	FIFO Tail Pointer This field contains the current "tail" pointer index for the FIFO, that is, the next entry to be read.

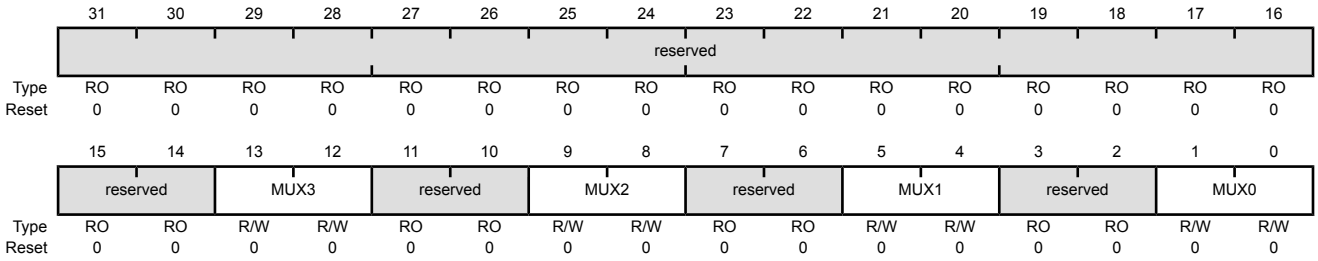
Register 21: ADC Sample Sequence Input Multiplexer Select 1 (ADCSSMUX1), offset 0x060

Register 22: ADC Sample Sequence Input Multiplexer Select 2 (ADCSSMUX2), offset 0x080

This register defines the analog input configuration for each sample in a sequence executed with Sample Sequencer 1 or 2. These registers are 16-bits wide and contain information for four possible samples. See the **ADCSSMUX0** register on page 277 for detailed bit descriptions. The **ADCSSMUX1** register affects Sample Sequencer 1 and the **ADCSSMUX2** register affects Sample Sequencer 2.

ADC Sample Sequence Input Multiplexer Select 1 (ADCSSMUX1)

Base 0x4003.8000
Offset 0x060
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:14	reserved	RO	0x0000	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
13:12	MUX3	R/W	0x0	4th Sample Input Select
11:10	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
9:8	MUX2	R/W	0x0	3rd Sample Input Select
7:6	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
5:4	MUX1	R/W	0x0	2nd Sample Input Select
3:2	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1:0	MUX0	R/W	0x0	1st Sample Input Select

Register 23: ADC Sample Sequence Control 1 (ADCSSCTL1), offset 0x064

Register 24: ADC Sample Sequence Control 2 (ADCSSCTL2), offset 0x084

These registers contain the configuration information for each sample for a sequence executed with Sample Sequencer 1 or 2. When configuring a sample sequence, the END bit must be set at some point, whether it be after the first sample, last sample, or any sample in between. These registers are 16-bits wide and contain information for four possible samples. See the **ADCSSCTL0** register on page 279 for detailed bit descriptions. The **ADCSSCTL1** register configures Sample Sequencer 1 and the **ADCSSCTL2** register configures Sample Sequencer 2.

ADC Sample Sequence Control 1 (ADCSSCTL1)

Base 0x4003.8000
Offset 0x064
Type R/W, reset 0x0000.0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
reserved															
RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TS3	IE3	END3	D3	TS2	IE2	END2	D2	TS1	IE1	END1	D1	TS0	IE0	END0	D0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit/Field	Name	Type	Reset	Description
31:16	reserved	RO	0x0000	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
15	TS3	R/W	0	4th Sample Temp Sensor Select Same definition as TS7 but used during the fourth sample.
14	IE3	R/W	0	4th Sample Interrupt Enable Same definition as IE7 but used during the fourth sample.
13	END3	R/W	0	4th Sample is End of Sequence Same definition as END7 but used during the fourth sample.
12	D3	R/W	0	4th Sample Diff Input Select Same definition as D7 but used during the fourth sample.
11	TS2	R/W	0	3rd Sample Temp Sensor Select Same definition as TS7 but used during the third sample.
10	IE2	R/W	0	3rd Sample Interrupt Enable Same definition as IE7 but used during the third sample.
9	END2	R/W	0	3rd Sample is End of Sequence Same definition as END7 but used during the third sample.
8	D2	R/W	0	3rd Sample Diff Input Select Same definition as D7 but used during the third sample.

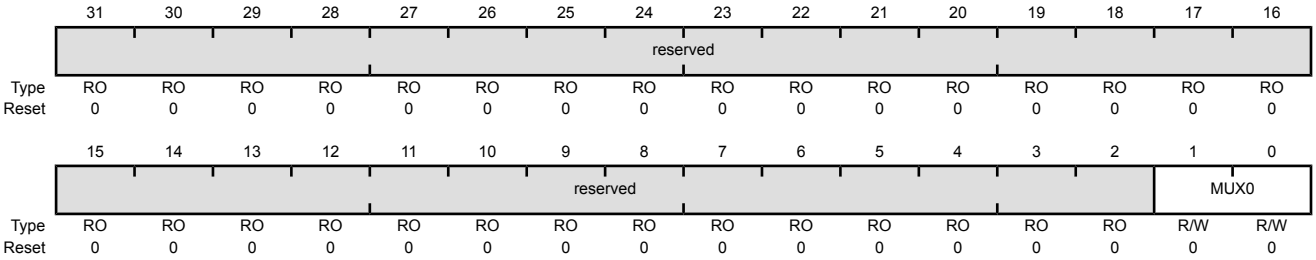
Bit/Field	Name	Type	Reset	Description
7	TS1	R/W	0	2nd Sample Temp Sensor Select Same definition as TS7 but used during the second sample.
6	IE1	R/W	0	2nd Sample Interrupt Enable Same definition as IE7 but used during the second sample.
5	END1	R/W	0	2nd Sample is End of Sequence Same definition as END7 but used during the second sample.
4	D1	R/W	0	2nd Sample Diff Input Select Same definition as D7 but used during the second sample.
3	TS0	R/W	0	1st Sample Temp Sensor Select Same definition as TS7 but used during the first sample.
2	IE0	R/W	0	1st Sample Interrupt Enable Same definition as IE7 but used during the first sample.
1	END0	R/W	0	1st Sample is End of Sequence Same definition as END7 but used during the first sample.
0	D0	R/W	0	1st Sample Diff Input Select Same definition as D7 but used during the first sample.

Register 25: ADC Sample Sequence Input Multiplexer Select 3 (ADCSSMUX3), offset 0x0A0

This register defines the analog input configuration for a sample executed with Sample Sequencer 3. This register is 4-bits wide and contains information for one possible sample. See the **ADCSSMUX0** register on page 277 for detailed bit descriptions.

ADC Sample Sequence Input Multiplexer Select 3 (ADCSSMUX3)

Base 0x4003.8000
Offset 0x0A0
Type R/W, reset 0x0000.0000



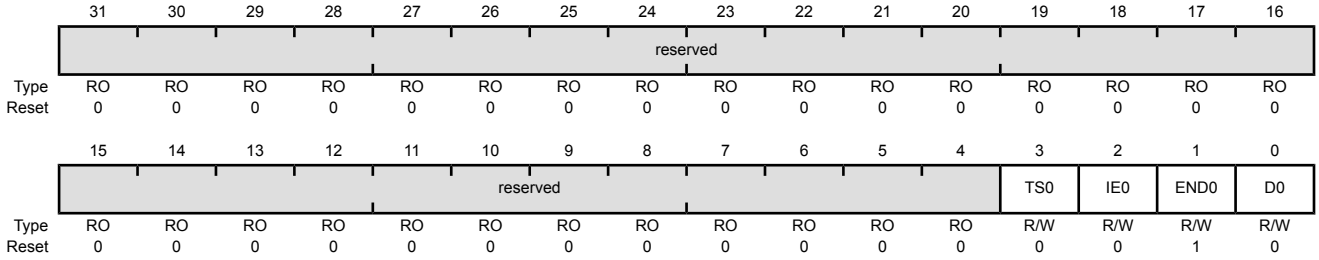
Bit/Field	Name	Type	Reset	Description
31:2	reserved	RO	0x0000.000	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1:0	MUX0	R/W	0	1st Sample Input Select

Register 26: ADC Sample Sequence Control 3 (ADCSSCTL3), offset 0x0A4

This register contains the configuration information for a sample executed with Sample Sequencer 3. The END bit is always set since there is only one sample in this sequencer. This register is 4-bits wide and contains information for one possible sample. See the **ADCSSCTL0** register on page 279 for detailed bit descriptions.

ADC Sample Sequence Control 3 (ADCSSCTL3)

Base 0x4003.8000
Offset 0x0A4
Type R/W, reset 0x0000.0002



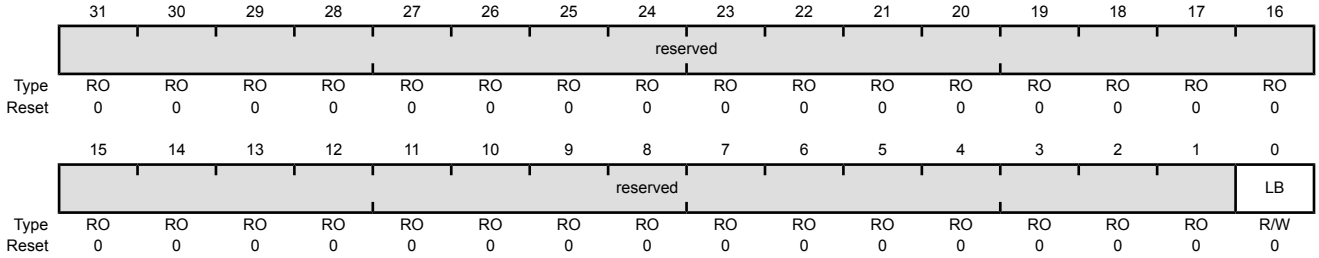
Bit/Field	Name	Type	Reset	Description
31:4	reserved	RO	0x0000.000	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
3	TS0	R/W	0	1st Sample Temp Sensor Select Same definition as TS7 but used during the first sample.
2	IE0	R/W	0	1st Sample Interrupt Enable Same definition as IE7 but used during the first sample.
1	END0	R/W	1	1st Sample is End of Sequence Same definition as END7 but used during the first sample. Since this sequencer has only one entry, this bit must be set.
0	D0	R/W	0	1st Sample Diff Input Select Same definition as D7 but used during the first sample.

Register 27: ADC Test Mode Loopback (ADCTMLB), offset 0x100

This register provides loopback operation within the digital logic of the ADC, which can be useful in debugging software without having to provide actual analog stimulus. This test mode is entered by writing a value of 0x0000.0001 to this register. When data is read from the FIFO in loopback mode, the read-only portion of this register is returned.

ADC Test Mode Loopback (ADCTMLB)

Base 0x4003.8000
Offset 0x100
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:1	reserved	RO	0x0000.000	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
0	LB	R/W	0	Loopback Mode Enable When set, forces a loopback within the digital block to provide information on input and unique numbering. The **ADCSSFIFOn** registers do not provide sample data, but instead provide the 10-bit loopback data as shown below.

Bit/Field	Name	Description
9:6	CNT	Continuous Sample Counter Continuous sample counter that is initialized to 0 and counts each sample as it processed. This helps provide a unique value for the data received.
5	CONT	Continuation Sample Indicator When set, indicates that this is a continuation sample. For example, if two sequencers were to run back-to-back, this indicates that the controller kept continuously sampling at full rate.
4	DIFF	Differential Sample Indicator When set, indicates that this is a differential sample.
3	TS	Temp Sensor Sample Indicator When set, indicates that this is a temperature sensor sample.
2:0	MUX	Analog Input Indicator Indicates which analog input is to be sampled.

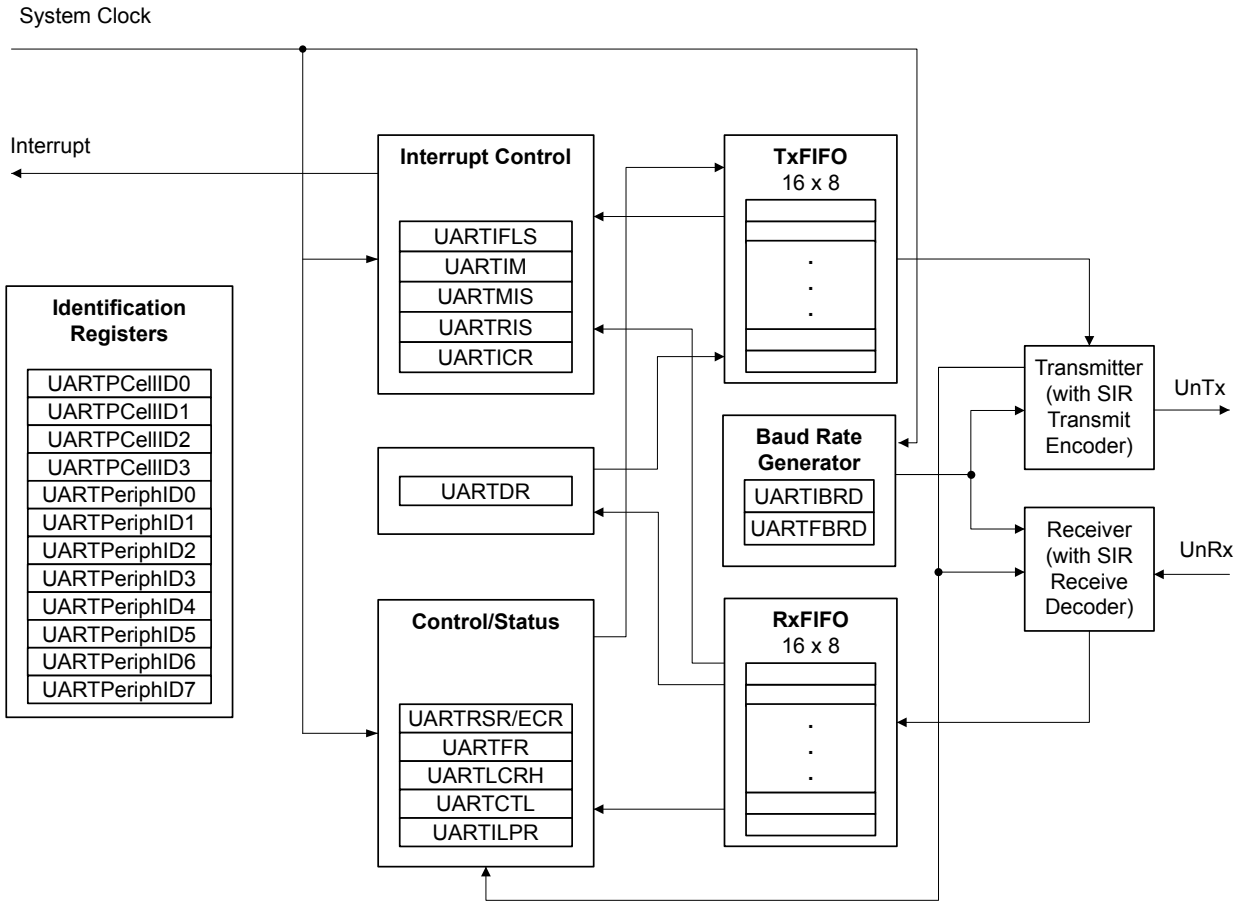
12 Universal Asynchronous Receivers/Transmitters (UARTs)

Each Stellaris® Universal Asynchronous Receiver/Transmitter (UART) has the following features:

- Two fully programmable 16C550-type UARTs with IrDA support
- Separate 16x8 transmit (TX) and receive (RX) FIFOs to reduce CPU interrupt service loading
- Programmable baud-rate generator allowing speeds up to 3.125 Mbps
- Programmable FIFO length, including 1-byte deep operation providing conventional double-buffered interface
- FIFO trigger levels of 1/8, 1/4, 1/2, 3/4, and 7/8
- Standard asynchronous communication bits for start, stop, and parity
- False-start bit detection
- Line-break generation and detection
- Fully programmable serial interface characteristics
  - 5, 6, 7, or 8 data bits
  - Even, odd, stick, or no-parity bit generation/detection
  - 1 or 2 stop bit generation
- IrDA serial-IR (SIR) encoder/decoder providing
  - Programmable use of IrDA Serial Infrared (SIR) or UART input/output
  - Support of IrDA SIR encoder/decoder functions for data rates up to 115.2 Kbps half-duplex
  - Support of normal 3/16 and low-power (1.41-2.23 µs) bit durations
  - Programmable internal clock generator enabling division of reference clock by 1 to 256 for low-power mode bit duration

12.1 Block Diagram

Figure 12-1. UART Module Block Diagram



12.2 Functional Description

Each Stellaris® UART performs the functions of parallel-to-serial and serial-to-parallel conversions. It is similar in functionality to a 16C550 UART, but is not register compatible.

The UART is configured for transmit and/or receive via the TXE and RXE bits of the **UART Control (UARTCTL)** register (see page 309). Transmit and receive are both enabled out of reset. Before any control registers are programmed, the UART must be disabled by clearing the UARTEN bit in **UARTCTL**. If the UART is disabled during a TX or RX operation, the current transaction is completed prior to the UART stopping.

The UART peripheral also includes a serial IR (SIR) encoder/decoder block that can be connected to an infrared transceiver to implement an IrDA SIR physical layer. The SIR function is programmed using the UARTCTL register.

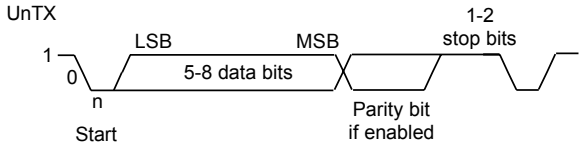
12.2.1 Transmit/Receive Logic

The transmit logic performs parallel-to-serial conversion on the data read from the transmit FIFO. The control logic outputs the serial bit stream beginning with a start bit, and followed by the data

bits (LSB first), parity bit, and the stop bits according to the programmed configuration in the control registers. See Figure 12-2 on page 292 for details.

The receive logic performs serial-to-parallel conversion on the received bit stream after a valid start pulse has been detected. Overrun, parity, frame error checking, and line-break detection are also performed, and their status accompanies the data that is written to the receive FIFO.

Figure 12-2. UART Character Frame



12.2.2 Baud-Rate Generation

The baud-rate divisor is a 22-bit number consisting of a 16-bit integer and a 6-bit fractional part. The number formed by these two values is used by the baud-rate generator to determine the bit period. Having a fractional baud-rate divider allows the UART to generate all the standard baud rates.

The 16-bit integer is loaded through the **UART Integer Baud-Rate Divisor (UARTIBRD)** register (see page 305) and the 6-bit fractional part is loaded with the **UART Fractional Baud-Rate Divisor (UARTFBRD)** register (see page 306). The baud-rate divisor (BRD) has the following relationship to the system clock (where *BRDI* is the integer part of the BRD and *BRDF* is the fractional part, separated by a decimal place.)

```
BRD = BRDI + BRDF = UARTSysClk / (16 * Baud Rate)
```

where UARTSysClk is the system clock connected to the UART.

The 6-bit fractional number (that is to be loaded into the DIVFRAC bit field in the **UARTFBRD** register) can be calculated by taking the fractional part of the baud-rate divisor, multiplying it by 64, and adding 0.5 to account for rounding errors:

```
UARTFBRD[DIVFRAC] = integer(BRDF * 64 + 0.5)
```

The UART generates an internal baud-rate reference clock at 16x the baud-rate (referred to as Baud16). This reference clock is divided by 16 to generate the transmit clock, and is used for error detection during receive operations.

Along with the **UART Line Control, High Byte (UARTLCRH)** register (see page 307), the **UARTIBRD** and **UARTFBRD** registers form an internal 30-bit register. This internal register is only updated when a write operation to **UARTLCRH** is performed, so any changes to the baud-rate divisor must be followed by a write to the **UARTLCRH** register for the changes to take effect.

To update the baud-rate registers, there are four possible sequences:

- **UARTIBRD** write, **UARTFBRD** write, and **UARTLCRH** write
- **UARTFBRD** write, **UARTIBRD** write, and **UARTLCRH** write
- **UARTIBRD** write and **UARTLCRH** write
- **UARTFBRD** write and **UARTLCRH** write

12.2.3 Data Transmission

Data received or transmitted is stored in two 16-byte FIFOs, though the receive FIFO has an extra four bits per character for status information. For transmission, data is written into the transmit FIFO. If the UART is enabled, it causes a data frame to start transmitting with the parameters indicated in the **UARTLCRH** register. Data continues to be transmitted until there is no data left in the transmit FIFO. The BUSY bit in the **UART Flag (UARTFR)** register (see page 302) is asserted as soon as data is written to the transmit FIFO (that is, if the FIFO is non-empty) and remains asserted while data is being transmitted. The BUSY bit is negated only when the transmit FIFO is empty, and the last character has been transmitted from the shift register, including the stop bits. The UART can indicate that it is busy even though the UART may no longer be enabled.

When the receiver is idle (the UnRx is continuously 1) and the data input goes Low (a start bit has been received), the receive counter begins running and data is sampled on the eighth cycle of Baud16 (described in “Transmit/Receive Logic” on page 291).

The start bit is valid if UnRx is still low on the eighth cycle of Baud16, otherwise a false start bit is detected and it is ignored. Start bit errors can be viewed in the **UART Receive Status (UARTRSR)** register (see page 300). If the start bit was valid, successive data bits are sampled on every 16th cycle of Baud16 (that is, one bit period later) according to the programmed length of the data characters. The parity bit is then checked if parity mode was enabled. Data length and parity are defined in the **UARTLCRH** register.

Lastly, a valid stop bit is confirmed if UnRx is High, otherwise a framing error has occurred. When a full word is received, the data is stored in the receive FIFO, with any error bits associated with that word.

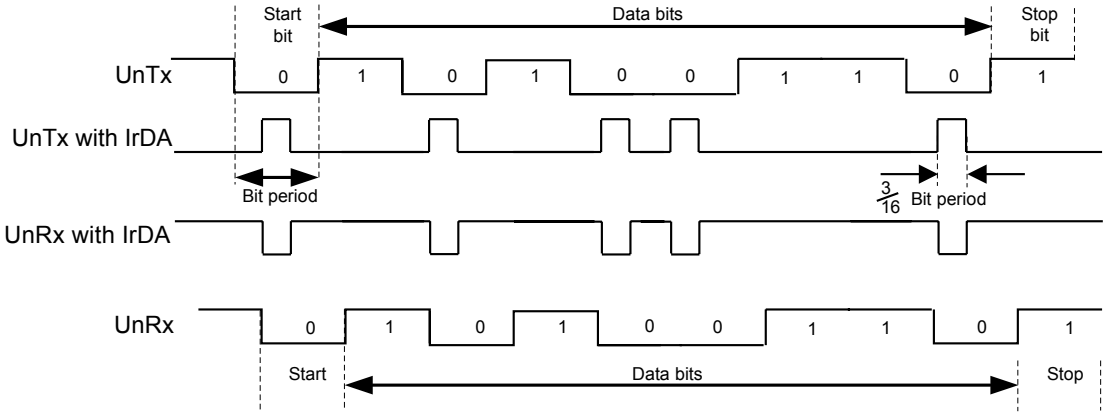
12.2.4 Serial IR (SIR)

The UART peripheral includes an IrDA serial-IR (SIR) encoder/decoder block. The IrDA SIR block provides functionality that converts between an asynchronous UART data stream, and half-duplex serial SIR interface. No analog processing is performed on-chip. The role of the SIR block is to provide a digital encoded output and decoded input to the UART. The UART signal pins can be connected to an infrared transceiver to implement an IrDA SIR physical layer link. The SIR block has two modes of operation:

- In normal IrDA mode, a zero logic level is transmitted as high pulse of 3/16th duration of the selected baud rate bit period on the output pin, while logic one levels are transmitted as a static LOW signal. These levels control the driver of an infrared transmitter, sending a pulse of light for each zero. On the reception side, the incoming light pulses energize the photo transistor base of the receiver, pulling its output LOW. This drives the UART input pin LOW.
- In low-power IrDA mode, the width of the transmitted infrared pulse is set to three times the period of the internally generated IrLPBaud16 signal (1.63 µs, assuming a nominal 1.8432 MHz frequency) by changing the appropriate bit in the **UARTCR** register. See page 304 for more information on IrDA low-power pulse-duration configuration.

Figure 12-3 on page 294 shows the UART transmit and receive signals, with and without IrDA modulation.

Figure 12-3. IrDA Data Modulation



In both normal and low-power IrDA modes:

- During transmission, the UART data bit is used as the base for encoding
- During reception, the decoded bits are transferred to the UART receive logic

The IrDA SIR physical layer specifies a half-duplex communication link, with a minimum 10 ms delay between transmission and reception. This delay must be generated by software because it is not automatically supported by the UART. The delay is required because the infrared receiver electronics might become biased, or even saturated from the optical power coupled from the adjacent transmitter LED. This delay is known as latency, or receiver setup time.

If the application does not require the use of the UnRx signal, the GPIO pin that has the UnRx signal as an alternate function must be configured as the UnRx signal and pulled High.

12.2.5 FIFO Operation

The UART has two 16-entry FIFOs; one for transmit and one for receive. Both FIFOs are accessed via the **UART Data (UARTDR)** register (see page 298). Read operations of the **UARTDR** register return a 12-bit value consisting of 8 data bits and 4 error flags while write operations place 8-bit data in the transmit FIFO.

Out of reset, both FIFOs are disabled and act as 1-byte-deep holding registers. The FIFOs are enabled by setting the FEN bit in **UARTLCRH** (page 307).

FIFO status can be monitored via the **UART Flag (UARTFR)** register (see page 302) and the **UART Receive Status (UARTRSR)** register. Hardware monitors empty, full and overrun conditions. The **UARTFR** register contains empty and full flags (TXFE, TXFF, RXFE, and RXFF bits) and the **UARTRSR** register shows overrun status via the OE bit.

The trigger points at which the FIFOs generate interrupts is controlled via the **UART Interrupt FIFO Level Select (UARTIFLS)** register (see page 311). Both FIFOs can be individually configured to trigger interrupts at different levels. Available configurations include 1/8, ¼, ½, ¾, and 7/8. For example, if the ¼ option is selected for the receive FIFO, the UART generates a receive interrupt after 4 data bytes are received. Out of reset, both FIFOs are configured to trigger an interrupt at the ½ mark.

12.2.6 Interrupts

The UART can generate interrupts when the following conditions are observed:

- Overrun Error
- Break Error
- Parity Error
- Framing Error
- Receive Timeout
- Transmit (when condition defined in the TXIFLSEL bit in the **UARTIFLS** register is met)
- Receive (when condition defined in the RXIFLSEL bit in the **UARTIFLS** register is met)

All of the interrupt events are ORed together before being sent to the interrupt controller, so the UART can only generate a single interrupt request to the controller at any given time. Software can service multiple interrupt events in a single interrupt service routine by reading the **UART Masked Interrupt Status (UARTMIS)** register (see page 316).

The interrupt events that can trigger a controller-level interrupt are defined in the **UART Interrupt Mask (UARTIM**) register (see page 313) by setting the corresponding IM bit to 1. If interrupts are not used, the raw interrupt status is always visible via the **UART Raw Interrupt Status (UARTRIS)** register (see page 315).

Interrupts are always cleared (for both the **UARTMIS** and **UARTRIS** registers) by setting the corresponding bit in the **UART Interrupt Clear (UARTICR)** register (see page 317).

The receive timeout interrupt is asserted when the receive FIFO is not empty, and no further data is received over a 32-bit period. The receive timeout interrupt is cleared either when the FIFO becomes empty through reading all the data (or by reading the holding register), or when a 1 is written to the corresponding bit in the **UARTICR** register.

12.2.7 Loopback Operation

The UART can be placed into an internal loopback mode for diagnostic or debug work. This is accomplished by setting the LBE bit in the **UARTCTL** register (see page 309). In loopback mode, data transmitted on UnTx is received on the UnRx input.

12.2.8 IrDA SIR block

The IrDA SIR block contains an IrDA serial IR (SIR) protocol encoder/decoder. When enabled, the SIR block uses the UnTx and UnRx pins for the SIR protocol, which should be connected to an IR transceiver.

The SIR block can receive and transmit, but it is only half-duplex so it cannot do both at the same time. Transmission must be stopped before data can be received. The IrDA SIR physical layer specifies a minimum 10-ms delay between transmission and reception.

12.3 Initialization and Configuration

To use the UARTs, the peripheral clock must be enabled by setting the UART0 or UART1 bits in the **RCGC1** register.

This section discusses the steps that are required to use a UART module. For this example, the UART clock is assumed to be 20 MHz and the desired UART configuration is:

- 115200 baud rate

- Data length of 8 bits
- One stop bit
- No parity
- FIFOs disabled
- No interrupts

The first thing to consider when programming the UART is the baud-rate divisor (BRD), since the **UARTIBRD** and **UARTFBRD** registers must be written before the **UARTLCRH** register. Using the equation described in "Baud-Rate Generation" on page 292, the BRD can be calculated:

```
BRD = 20,000,000 / (16 * 115,200) = 10.8507
```

which means that the DIVINT field of the **UARTIBRD** register (see page 305) should be set to 10. The value to be loaded into the **UARTFBRD** register (see page 306) is calculated by the equation:

```
UARTFBRD[DIVFRAC] = integer(0.8507 * 64 + 0.5) = 54
```

With the BRD values in hand, the UART configuration is written to the module in the following order:

1. Disable the UART by clearing the UARTEN bit in the **UARTCTL** register.
2. Write the integer portion of the BRD to the **UARTIBRD** register.
3. Write the fractional portion of the BRD to the **UARTFBRD** register.
4. Write the desired serial parameters to the **UARTLCRH** register (in this case, a value of 0x0000.0060).
5. Enable the UART by setting the UARTEN bit in the **UARTCTL** register.

12.4 Register Map

Table 12-1 on page 296 lists the UART registers. The offset listed is a hexadecimal increment to the register's address, relative to that UART's base address:

- UART0: 0x4000.C000
- UART1: 0x4000.D000

Note: The UART must be disabled (see the UARTEN bit in the **UARTCTL** register on page 309) before any of the control registers are reprogrammed. When the UART is disabled during a TX or RX operation, the current transaction is completed prior to the UART stopping.

Table 12-1. UART Register Map

Offset	Name	Type	Reset	Description	See page
0x000	UARTDR	R/W	0x0000.0000	UART Data	298
0x004	UARTRSR/UARTECR	R/W	0x0000.0000	UART Receive Status/Error Clear	300
0x018	UARTFR	RO	0x0000.0090	UART Flag	302

Table 12-1. UART Register Map *(continued)*

Offset	Name	Type	Reset	Description	See page
0x020	UARTILPR	R/W	0x0000.0000	UART IrDA Low-Power Register	304
0x024	UARTIBRD	R/W	0x0000.0000	UART Integer Baud-Rate Divisor	305
0x028	UARTFBRD	R/W	0x0000.0000	UART Fractional Baud-Rate Divisor	306
0x02C	UARTLCRH	R/W	0x0000.0000	UART Line Control	307
0x030	UARTCTL	R/W	0x0000.0300	UART Control	309
0x034	UARTIFLS	R/W	0x0000.0012	UART Interrupt FIFO Level Select	311
0x038	UARTIM	R/W	0x0000.0000	UART Interrupt Mask	313
0x03C	UARTRIS	RO	0x0000.000F	UART Raw Interrupt Status	315
0x040	UARTMIS	RO	0x0000.0000	UART Masked Interrupt Status	316
0x044	UARTICR	W1C	0x0000.0000	UART Interrupt Clear	317
0xFD0	UARTPeriphID4	RO	0x0000.0000	UART Peripheral Identification 4	319
0xFD4	UARTPeriphID5	RO	0x0000.0000	UART Peripheral Identification 5	320
0xFD8	UARTPeriphID6	RO	0x0000.0000	UART Peripheral Identification 6	321
0xFDC	UARTPeriphID7	RO	0x0000.0000	UART Peripheral Identification 7	322
0xFE0	UARTPeriphID0	RO	0x0000.0011	UART Peripheral Identification 0	323
0xFE4	UARTPeriphID1	RO	0x0000.0000	UART Peripheral Identification 1	324
0xFE8	UARTPeriphID2	RO	0x0000.0018	UART Peripheral Identification 2	325
0xFEC	UARTPeriphID3	RO	0x0000.0001	UART Peripheral Identification 3	326
0xFF0	UARTPCellID0	RO	0x0000.000D	UART PrimeCell Identification 0	327
0xFF4	UARTPCellID1	RO	0x0000.00F0	UART PrimeCell Identification 1	328
0xFF8	UARTPCellID2	RO	0x0000.0005	UART PrimeCell Identification 2	329
0xFFC	UARTPCellID3	RO	0x0000.00B1	UART PrimeCell Identification 3	330

12.5 Register Descriptions

The remainder of this section lists and describes the UART registers, in numerical order by address offset.

Register 1: UART Data (UARTDR), offset 0x000

Important: Use caution when reading this register. Performing a read may change bit status.

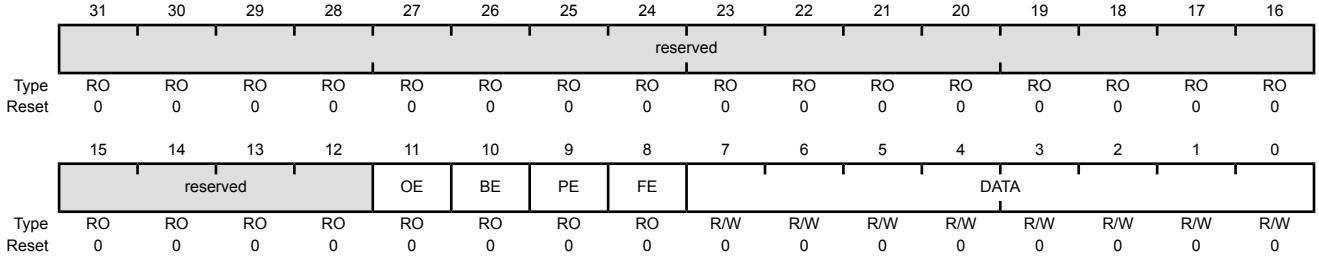
This register is the data register (the interface to the FIFOs).

When FIFOs are enabled, data written to this location is pushed onto the transmit FIFO. If FIFOs are disabled, data is stored in the transmitter holding register (the bottom word of the transmit FIFO). A write to this register initiates a transmission from the UART.

For received data, if the FIFO is enabled, the data byte and the 4-bit status (break, frame, parity, and overrun) is pushed onto the 12-bit wide receive FIFO. If FIFOs are disabled, the data byte and status are stored in the receiving holding register (the bottom word of the receive FIFO). The received data can be retrieved by reading this register.

UART Data (UARTDR)

UART0 base: 0x4000.C000
UART1 base: 0x4000.D000
Offset 0x000
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:12	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
11	OE	RO	0	UART Overrun Error The OE values are defined as follows: Value Description 0 There has been no data loss due to a FIFO overrun. 1 New data was received when the FIFO was full, resulting in data loss.
10	BE	RO	0	UART Break Error This bit is set to 1 when a break condition is detected, indicating that the receive data input was held Low for longer than a full-word transmission time (defined as start, data, parity, and stop bits). In FIFO mode, this error is associated with the character at the top of the FIFO. When a break occurs, only one 0 character is loaded into the FIFO. The next character is only enabled after the received data input goes to a 1 (marking state) and the next valid start bit is received.

Bit/Field	Name	Type	Reset	Description
9	PE	RO	0	UART Parity Error This bit is set to 1 when the parity of the received data character does not match the parity defined by bits 2 and 7 of the **UARTLCRH** register. In FIFO mode, this error is associated with the character at the top of the FIFO.
8	FE	RO	0	UART Framing Error This bit is set to 1 when the received character does not have a valid stop bit (a valid stop bit is 1).
7:0	DATA	R/W	0	Data Transmitted or Received When written, the data that is to be transmitted via the UART. When read, the data that was received by the UART.

Register 2: UART Receive Status/Error Clear (UARTRSR/UARTECR), offset 0x004

The **UARTRSR/UARTECR** register is the receive status register/error clear register.

In addition to the **UARTDR** register, receive status can also be read from the **UARTRSR** register. If the status is read from this register, then the status information corresponds to the entry read from **UARTDR** prior to reading **UARTRSR**. The status information for overrun is set immediately when an overrun condition occurs.

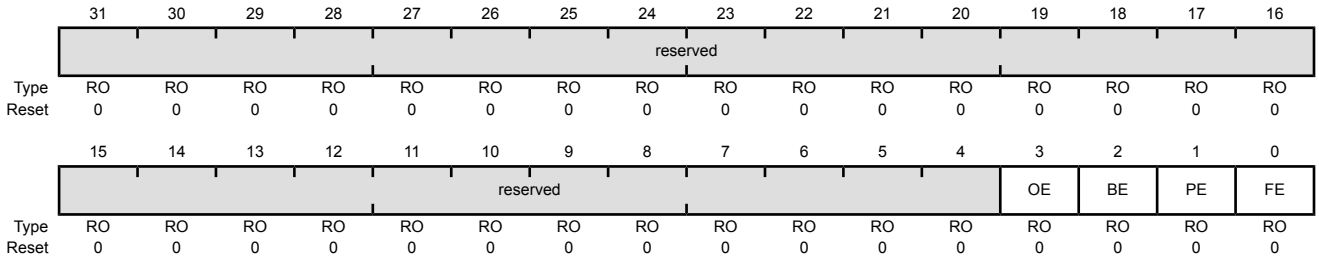
The **UARTRSR** register cannot be written.

A write of any value to the **UARTECR** register clears the framing, parity, break, and overrun errors. All the bits are cleared to 0 on reset.

Reads

UART Receive Status/Error Clear (UARTRSR/UARTECR)

UART0 base: 0x4000.C000
UART1 base: 0x4000.D000
Offset 0x004
Type RO, reset 0x0000.0000



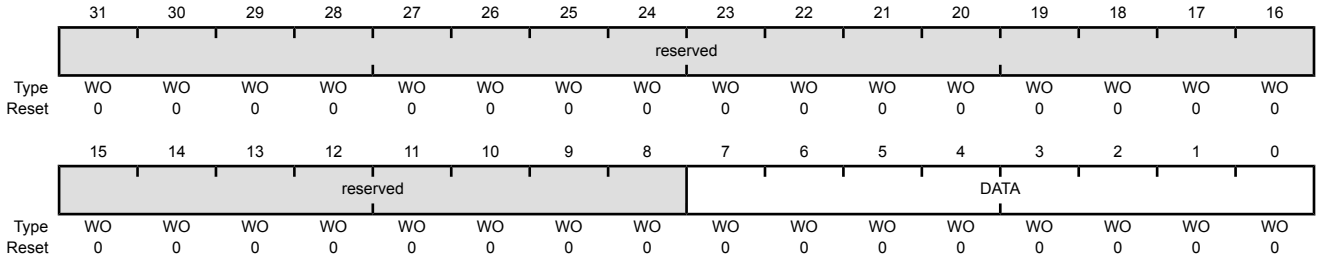
Bit/Field	Name	Type	Reset	Description
31:4	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
3	OE	RO	0	UART Overrun Error When this bit is set to 1, data is received and the FIFO is already full. This bit is cleared to 0 by a write to **UARTECR**. The FIFO contents remain valid since no further data is written when the FIFO is full, only the contents of the shift register are overwritten. The CPU must now read the data in order to empty the FIFO.
2	BE	RO	0	UART Break Error This bit is set to 1 when a break condition is detected, indicating that the received data input was held Low for longer than a full-word transmission time (defined as start, data, parity, and stop bits). This bit is cleared to 0 by a write to **UARTECR**. In FIFO mode, this error is associated with the character at the top of the FIFO. When a break occurs, only one 0 character is loaded into the FIFO. The next character is only enabled after the receive data input goes to a 1 (marking state) and the next valid start bit is received.

Bit/Field	Name	Type	Reset	Description
1	PE	RO	0	UART Parity Error This bit is set to 1 when the parity of the received data character does not match the parity defined by bits 2 and 7 of the **UARTLCRH** register. This bit is cleared to 0 by a write to **UARTECR**.
0	FE	RO	0	UART Framing Error This bit is set to 1 when the received character does not have a valid stop bit (a valid stop bit is 1). This bit is cleared to 0 by a write to **UARTECR**. In FIFO mode, this error is associated with the character at the top of the FIFO.

Writes

UART Receive Status/Error Clear (UARTRSR/UARTECR)

UART0 base: 0x4000.C000
UART1 base: 0x4000.D000
Offset 0x004
Type WO, reset 0x0000.0000

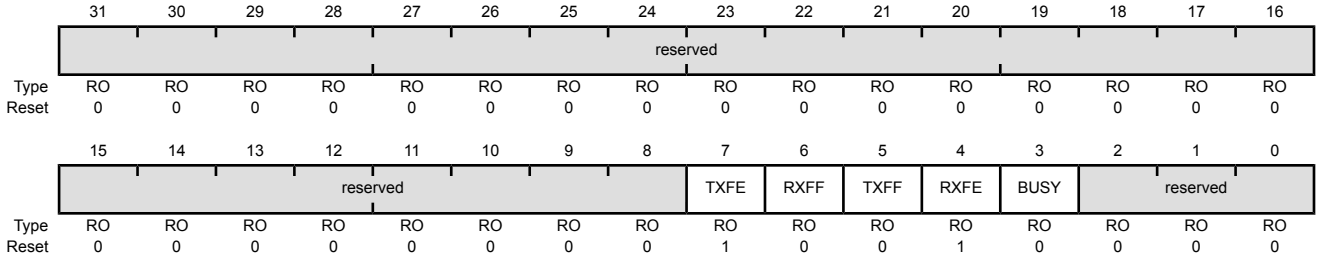


Bit/Field	Name	Type	Reset	Description
31:8	reserved	WO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	DATA	WO	0	Error Clear A write to this register of any data clears the framing, parity, break, and overrun flags.

Register 3: UART Flag (UARTFR), offset 0x018

The **UARTFR** register is the flag register. After reset, the TXFF, RXFF, and BUSY bits are 0, and TXFE and RXFE bits are 1.

UART Flag (UARTFR)
UART0 base: 0x4000.C000
UART1 base: 0x4000.D000
Offset 0x018
Type RO, reset 0x0000.0090



Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7	TXFE	RO	1	UART Transmit FIFO Empty The meaning of this bit depends on the state of the FEN bit in the **UARTLCRH** register. If the FIFO is disabled (FEN is 0), this bit is set when the transmit holding register is empty. If the FIFO is enabled (FEN is 1), this bit is set when the transmit FIFO is empty.
6	RXFF	RO	0	UART Receive FIFO Full The meaning of this bit depends on the state of the FEN bit in the **UARTLCRH** register. If the FIFO is disabled, this bit is set when the receive holding register is full. If the FIFO is enabled, this bit is set when the receive FIFO is full.
5	TXFF	RO	0	UART Transmit FIFO Full The meaning of this bit depends on the state of the FEN bit in the **UARTLCRH** register. If the FIFO is disabled, this bit is set when the transmit holding register is full. If the FIFO is enabled, this bit is set when the transmit FIFO is full.

Bit/Field	Name	Type	Reset	Description
4	RXFE	RO	1	UART Receive FIFO Empty The meaning of this bit depends on the state of the FEN bit in the **UARTLCRH** register. If the FIFO is disabled, this bit is set when the receive holding register is empty. If the FIFO is enabled, this bit is set when the receive FIFO is empty.
3	BUSY	RO	0	UART Busy When this bit is 1, the UART is busy transmitting data. This bit remains set until the complete byte, including all stop bits, has been sent from the shift register. This bit is set as soon as the transmit FIFO becomes non-empty (regardless of whether UART is enabled).
2:0	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Register 4: UART IrDA Low-Power Register (UARTILPR), offset 0x020

The **UARTILPR** register is an 8-bit read/write register that stores the low-power counter divisor value used to derive the low-power SIR pulse width clock by dividing down the system clock (SysClk). All the bits are cleared to 0 when reset.

The internal `IrLPBaud16` clock is generated by dividing down SysClk according to the low-power divisor value written to **UARTILPR**. The duration of SIR pulses generated when low-power mode is enabled is three times the period of the `IrLPBaud16` clock. The low-power divisor value is calculated as follows:

$$\texttt{ILPDVSR} = \texttt{SysClk} / F_{\texttt{IrLPBaud16}}$$

where $F_{\texttt{IrLPBaud16}}$ is nominally 1.8432 MHz.

You must choose the divisor so that 1.42 MHz < $F_{\texttt{IrLPBaud16}}$ < 2.12 MHz, which results in a low-power pulse duration of 1.41–2.11 µs (three times the period of `IrLPBaud16`). The minimum frequency of `IrLPBaud16` ensures that pulses less than one period of `IrLPBaud16` are rejected, but that pulses greater than 1.4 µs are accepted as valid pulses.

Note: Zero is an illegal value. Programming a zero value results in no `IrLPBaud16` pulses being generated.

UART IrDA Low-Power Register (UARTILPR)

UART0 base: 0x4000.C000
UART1 base: 0x4000.D000
Offset 0x020
Type R/W, reset 0x0000.0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	reserved															
Type	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	reserved								ILPDVSR							
Type	RO	RO	RO	RO	RO	RO	RO	RO	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

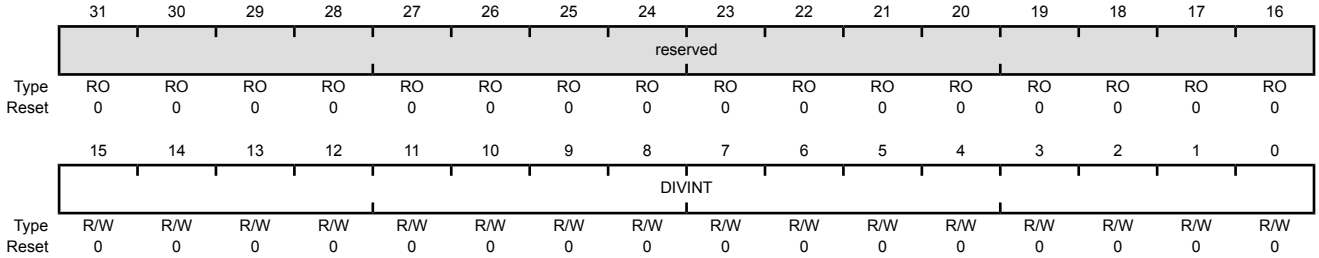
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	ILPDVSR	R/W	0x00	IrDA Low-Power Divisor This is an 8-bit low-power divisor value.

Register 5: UART Integer Baud-Rate Divisor (UARTIBRD), offset 0x024

The **UARTIBRD** register is the integer part of the baud-rate divisor value. All the bits are cleared on reset. The minimum possible divide ratio is 1 (when **UARTIBRD**=0), in which case the **UARTFBRD** register is ignored. When changing the **UARTIBRD** register, the new value does not take effect until transmission/reception of the current character is complete. Any changes to the baud-rate divisor must be followed by a write to the **UARTLCRH** register. See “Baud-Rate Generation” on page 292 for configuration details.

UART Integer Baud-Rate Divisor (UARTIBRD)

UART0 base: 0x4000.C000
UART1 base: 0x4000.D000
Offset 0x024
Type R/W, reset 0x0000.0000



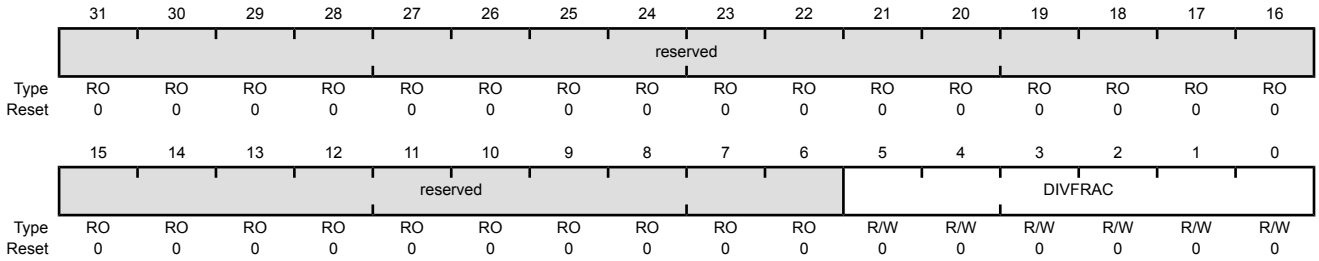
Bit/Field	Name	Type	Reset	Description
31:16	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
15:0	DIVINT	R/W	0x0000	Integer Baud-Rate Divisor

Register 6: UART Fractional Baud-Rate Divisor (UARTFBRD), offset 0x028

The **UARTFBRD** register is the fractional part of the baud-rate divisor value. All the bits are cleared on reset. When changing the **UARTFBRD** register, the new value does not take effect until transmission/reception of the current character is complete. Any changes to the baud-rate divisor must be followed by a write to the **UARTLCRH** register. See “Baud-Rate Generation” on page 292 for configuration details.

UART Fractional Baud-Rate Divisor (UARTFBRD)

UART0 base: 0x4000.C000
UART1 base: 0x4000.D000
Offset 0x028
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:6	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
5:0	DIVFRAC	R/W	0x000	Fractional Baud-Rate Divisor

Register 7: UART Line Control (UARTLCRH), offset 0x02C

The **UARTLCRH** register is the line control register. Serial parameters such as data length, parity, and stop bit selection are implemented in this register.

When updating the baud-rate divisor (**UARTIBRD** and/or **UARTIFRD**), the **UARTLCRH** register must also be written. The write strobe for the baud-rate divisor registers is tied to the **UARTLCRH** register.

UART Line Control (UARTLCRH)

UART0 base: 0x4000.C000
UART1 base: 0x4000.D000
Offset 0x02C
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7	SPS	R/W	0	UART Stick Parity Select When bits 1, 2, and 7 of **UARTLCRH** are set, the parity bit is transmitted and checked as a 0. When bits 1 and 7 are set and 2 is cleared, the parity bit is transmitted and checked as a 1. When this bit is cleared, stick parity is disabled.
6:5	WLEN	R/W	0	UART Word Length The bits indicate the number of data bits transmitted or received in a frame as follows: Value Description 0x3 8 bits 0x2 7 bits 0x1 6 bits 0x0 5 bits (default)
4	FEN	R/W	0	UART Enable FIFOs If this bit is set to 1, transmit and receive FIFO buffers are enabled (FIFO mode). When cleared to 0, FIFOs are disabled (Character mode). The FIFOs become 1-byte-deep holding registers.

Bit/Field	Name	Type	Reset	Description
3	STP2	R/W	0	UART Two Stop Bits Select If this bit is set to 1, two stop bits are transmitted at the end of a frame. The receive logic does not check for two stop bits being received.
2	EPS	R/W	0	UART Even Parity Select If this bit is set to 1, even parity generation and checking is performed during transmission and reception, which checks for an even number of 1s in data and parity bits. When cleared to 0, then odd parity is performed, which checks for an odd number of 1s. This bit has no effect when parity is disabled by the PEN bit.
1	PEN	R/W	0	UART Parity Enable If this bit is set to 1, parity checking and generation is enabled; otherwise, parity is disabled and no parity bit is added to the data frame.
0	BRK	R/W	0	UART Send Break If this bit is set to 1, a Low level is continually output on the UnTX output, after completing transmission of the current character. For the proper execution of the break command, the software must set this bit for at least two frames (character periods). For normal use, this bit must be cleared to 0.

Register 8: UART Control (UARTCTL), offset 0x030

The **UARTCTL** register is the control register. All the bits are cleared on reset except for the `Transmit Enable (TXE)` and `Receive Enable (RXE)` bits, which are set to 1.

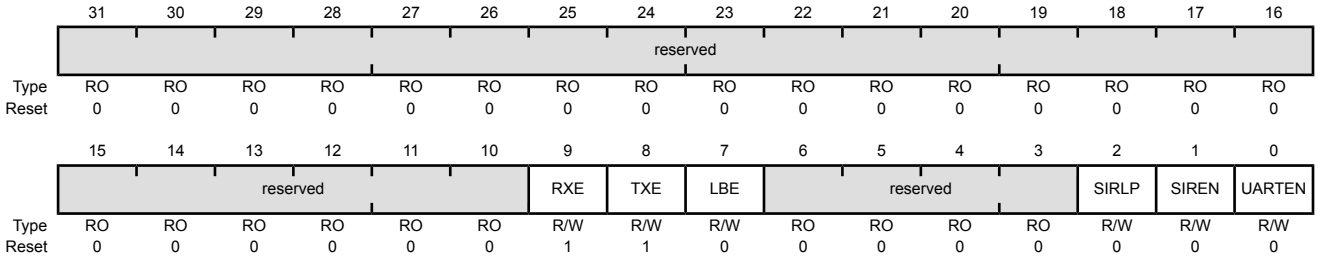
To enable the UART module, the `UARTEN` bit must be set to 1. If software requires a configuration change in the module, the `UARTEN` bit must be cleared before the configuration changes are written. If the UART is disabled during a transmit or receive operation, the current transaction is completed prior to the UART stopping.

Note: The **UARTCTL** register should not be changed while the UART is enabled or else the results are unpredictable. The following sequence is recommended for making changes to the **UARTCTL** register.

1. Disable the UART.
2. Wait for the end of transmission or reception of the current character.
3. Flush the transmit FIFO by disabling bit 4 (`FEN`) in the line control register (**UARTLCRH**).
4. Reprogram the control register.
5. Enable the UART.

UART Control (UARTCTL)

UART0 base: 0x4000.C000
UART1 base: 0x4000.D000
Offset 0x030
Type R/W, reset 0x0000.0300



Bit/Field	Name	Type	Reset	Description
31:10	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
9	RXE	R/W	1	UART Receive Enable If this bit is set to 1, the receive section of the UART is enabled. When the UART is disabled in the middle of a receive, it completes the current character before stopping. **Note:** To enable reception, the `UARTEN` bit must also be set.
8	TXE	R/W	1	UART Transmit Enable If this bit is set to 1, the transmit section of the UART is enabled. When the UART is disabled in the middle of a transmission, it completes the current character before stopping. **Note:** To enable transmission, the `UARTEN` bit must also be set.

Bit/Field	Name	Type	Reset	Description
7	LBE	R/W	0	UART Loop Back Enable If this bit is set to 1, the UnTX path is fed through the UnRX path.
6:3	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
2	SIRLP	R/W	0	UART SIR Low Power Mode This bit selects the IrDA encoding mode. If this bit is cleared to 0, low-level bits are transmitted as an active High pulse with a width of 3/16th of the bit period. If this bit is set to 1, low-level bits are transmitted with a pulse width which is 3 times the period of the IrLPBaud16 input signal, regardless of the selected bit rate. Setting this bit uses less power, but might reduce transmission distances. See page 304 for more information.
1	SIREN	R/W	0	UART SIR Enable If this bit is set to 1, the IrDA SIR block is enabled, and the UART will transmit and receive data using SIR protocol.
0	UARTEN	R/W	0	UART Enable If this bit is set to 1, the UART is enabled. When the UART is disabled in the middle of transmission or reception, it completes the current character before stopping.

Register 9: UART Interrupt FIFO Level Select (UARTIFLS), offset 0x034

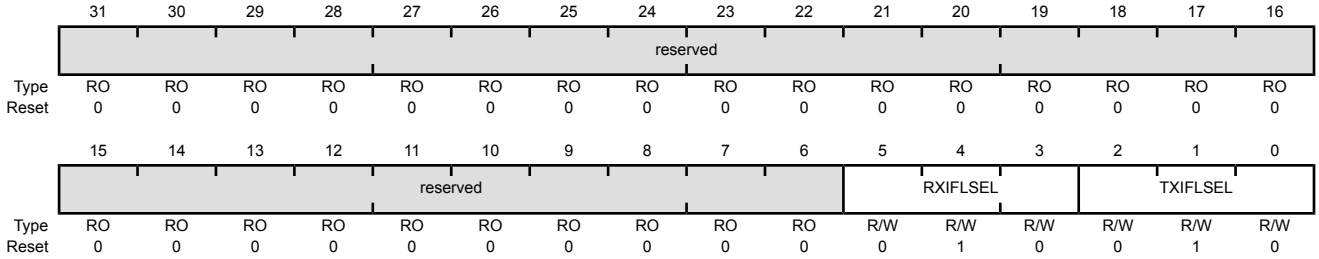
The **UARTIFLS** register is the interrupt FIFO level select register. You can use this register to define the FIFO level at which the TXRIS and RXRIS bits in the **UARTRIS** register are triggered.

The interrupts are generated based on a transition through a level rather than being based on the level. That is, the interrupts are generated when the fill level progresses through the trigger level. For example, if the receive trigger level is set to the half-way mark, the interrupt is triggered as the module is receiving the 9th character.

Out of reset, the TXIFLSEL and RXIFLSEL bits are configured so that the FIFOs trigger an interrupt at the half-way mark.

UART Interrupt FIFO Level Select (UARTIFLS)

UART0 base: 0x4000.C000
UART1 base: 0x4000.D000
Offset 0x034
Type R/W, reset 0x0000.0012



Bit/Field	Name	Type	Reset	Description
31:6	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
5:3	RXIFLSEL	R/W	0x2	UART Receive Interrupt FIFO Level Select The trigger points for the receive interrupt are as follows: Value Description 0x0 RX FIFO ≥ 1/8 full 0x1 RX FIFO ≥ ¼ full 0x2 RX FIFO ≥ ½ full (default) 0x3 RX FIFO ≥ ¾ full 0x4 RX FIFO ≥ 7/8 full 0x5-0x7 Reserved

Bit/Field	Name	Type	Reset	Description
2:0	TXIFLSEL	R/W	0x2	UART Transmit Interrupt FIFO Level Select The trigger points for the transmit interrupt are as follows: Value Description 0x0 TX FIFO ≤ 1/8 full 0x1 TX FIFO ≤ ¼ full 0x2 TX FIFO ≤ ½ full (default) 0x3 TX FIFO ≤ ¾ full 0x4 TX FIFO ≤ 7/8 full 0x5-0x7 Reserved

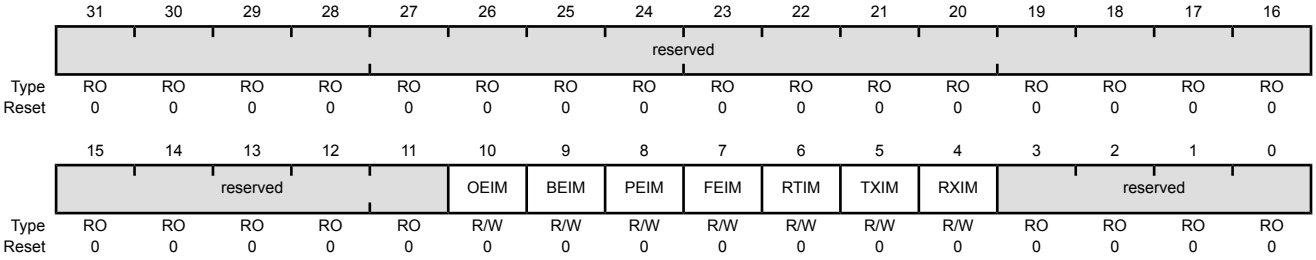
Register 10: UART Interrupt Mask (UARTIM), offset 0x038

The **UARTIM** register is the interrupt mask set/clear register.

On a read, this register gives the current value of the mask on the relevant interrupt. Writing a 1 to a bit allows the corresponding raw interrupt signal to be routed to the interrupt controller. Writing a 0 prevents the raw interrupt signal from being sent to the interrupt controller.

UART Interrupt Mask (UARTIM)

UART0 base: 0x4000.C000
UART1 base: 0x4000.D000
Offset 0x038
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:11	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
10	OEIM	R/W	0	UART Overrun Error Interrupt Mask On a read, the current mask for the `OEIM` interrupt is returned. Setting this bit to 1 promotes the `OEIM` interrupt to the interrupt controller.
9	BEIM	R/W	0	UART Break Error Interrupt Mask On a read, the current mask for the `BEIM` interrupt is returned. Setting this bit to 1 promotes the `BEIM` interrupt to the interrupt controller.
8	PEIM	R/W	0	UART Parity Error Interrupt Mask On a read, the current mask for the `PEIM` interrupt is returned. Setting this bit to 1 promotes the `PEIM` interrupt to the interrupt controller.
7	FEIM	R/W	0	UART Framing Error Interrupt Mask On a read, the current mask for the `FEIM` interrupt is returned. Setting this bit to 1 promotes the `FEIM` interrupt to the interrupt controller.
6	RTIM	R/W	0	UART Receive Time-Out Interrupt Mask On a read, the current mask for the `RTIM` interrupt is returned. Setting this bit to 1 promotes the `RTIM` interrupt to the interrupt controller.
5	TXIM	R/W	0	UART Transmit Interrupt Mask On a read, the current mask for the `TXIM` interrupt is returned. Setting this bit to 1 promotes the `TXIM` interrupt to the interrupt controller.

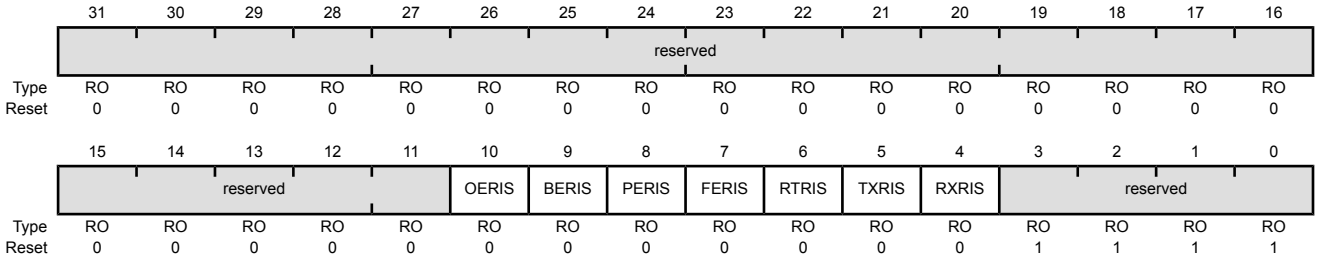
Bit/Field	Name	Type	Reset	Description
4	RXIM	R/W	0	UART Receive Interrupt Mask On a read, the current mask for the RXIM interrupt is returned. Setting this bit to 1 promotes the RXIM interrupt to the interrupt controller.
3:0	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Register 11: UART Raw Interrupt Status (UARTRIS), offset 0x03C

The **UARTRIS** register is the raw interrupt status register. On a read, this register gives the current raw status value of the corresponding interrupt. A write has no effect.

UART Raw Interrupt Status (UARTRIS)

UART0 base: 0x4000.C000
UART1 base: 0x4000.D000
Offset 0x03C
Type RO, reset 0x0000.000F



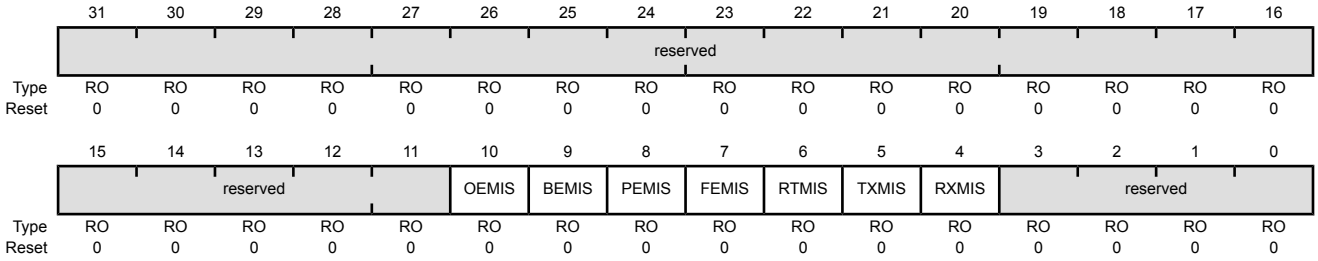
Bit/Field	Name	Type	Reset	Description
31:11	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
10	OERIS	RO	0	UART Overrun Error Raw Interrupt Status Gives the raw interrupt state (prior to masking) of this interrupt.
9	BERIS	RO	0	UART Break Error Raw Interrupt Status Gives the raw interrupt state (prior to masking) of this interrupt.
8	PERIS	RO	0	UART Parity Error Raw Interrupt Status Gives the raw interrupt state (prior to masking) of this interrupt.
7	FERIS	RO	0	UART Framing Error Raw Interrupt Status Gives the raw interrupt state (prior to masking) of this interrupt.
6	RTRIS	RO	0	UART Receive Time-Out Raw Interrupt Status Gives the raw interrupt state (prior to masking) of this interrupt.
5	TXRIS	RO	0	UART Transmit Raw Interrupt Status Gives the raw interrupt state (prior to masking) of this interrupt.
4	RXRIS	RO	0	UART Receive Raw Interrupt Status Gives the raw interrupt state (prior to masking) of this interrupt.
3:0	reserved	RO	0xF	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Register 12: UART Masked Interrupt Status (UARTMIS), offset 0x040

The **UARTMIS** register is the masked interrupt status register. On a read, this register gives the current masked status value of the corresponding interrupt. A write has no effect.

UART Masked Interrupt Status (UARTMIS)

UART0 base: 0x4000.C000
UART1 base: 0x4000.D000
Offset 0x040
Type RO, reset 0x0000.0000



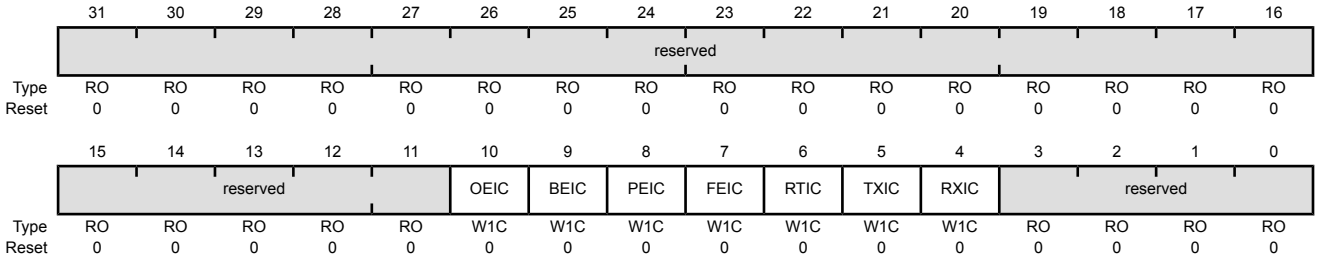
Bit/Field	Name	Type	Reset	Description
31:11	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
10	OEMIS	RO	0	UART Overrun Error Masked Interrupt Status Gives the masked interrupt state of this interrupt.
9	BEMIS	RO	0	UART Break Error Masked Interrupt Status Gives the masked interrupt state of this interrupt.
8	PEMIS	RO	0	UART Parity Error Masked Interrupt Status Gives the masked interrupt state of this interrupt.
7	FEMIS	RO	0	UART Framing Error Masked Interrupt Status Gives the masked interrupt state of this interrupt.
6	RTMIS	RO	0	UART Receive Time-Out Masked Interrupt Status Gives the masked interrupt state of this interrupt.
5	TXMIS	RO	0	UART Transmit Masked Interrupt Status Gives the masked interrupt state of this interrupt.
4	RXMIS	RO	0	UART Receive Masked Interrupt Status Gives the masked interrupt state of this interrupt.
3:0	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Register 13: UART Interrupt Clear (UARTICR), offset 0x044

The **UARTICR** register is the interrupt clear register. On a write of 1, the corresponding interrupt (both raw interrupt and masked interrupt, if enabled) is cleared. A write of 0 has no effect.

UART Interrupt Clear (UARTICR)

UART0 base: 0x4000.C000
UART1 base: 0x4000.D000
Offset 0x044
Type W1C, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:11	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
10	OEIC	W1C	0	Overrun Error Interrupt Clear The `OEIC` values are defined as follows: Value Description 0 No effect on the interrupt. 1 Clears interrupt.
9	BEIC	W1C	0	Break Error Interrupt Clear The `BEIC` values are defined as follows: Value Description 0 No effect on the interrupt. 1 Clears interrupt.
8	PEIC	W1C	0	Parity Error Interrupt Clear The `PEIC` values are defined as follows: Value Description 0 No effect on the interrupt. 1 Clears interrupt.

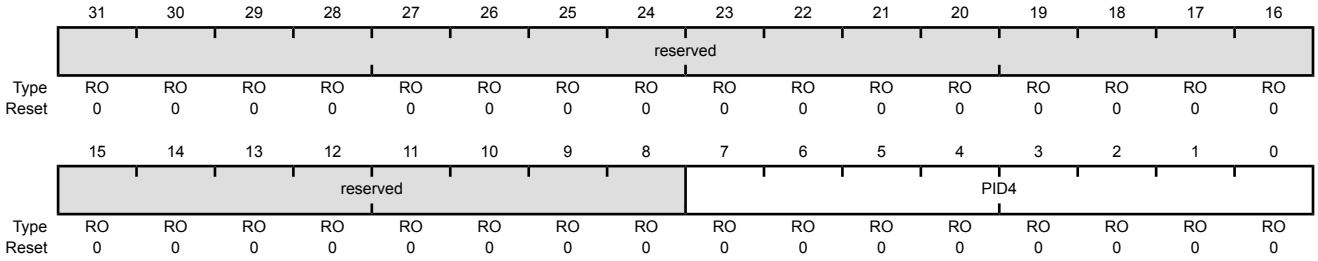
Bit/Field	Name	Type	Reset	Description
7	FEIC	W1C	0	Framing Error Interrupt Clear The FEIC values are defined as follows: Value Description 0 No effect on the interrupt. 1 Clears interrupt.
6	RTIC	W1C	0	Receive Time-Out Interrupt Clear The RTIC values are defined as follows: Value Description 0 No effect on the interrupt. 1 Clears interrupt.
5	TXIC	W1C	0	Transmit Interrupt Clear The TXIC values are defined as follows: Value Description 0 No effect on the interrupt. 1 Clears interrupt.
4	RXIC	W1C	0	Receive Interrupt Clear The RXIC values are defined as follows: Value Description 0 No effect on the interrupt. 1 Clears interrupt.
3:0	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Register 14: UART Peripheral Identification 4 (UARTPeriphID4), offset 0xFD0

The **UARTPeriphIDn** registers are hard-coded and the fields within the registers determine the reset values.

UART Peripheral Identification 4 (UARTPeriphID4)

UART0 base: 0x4000.C000
UART1 base: 0x4000.D000
Offset 0xFD0
Type RO, reset 0x0000.0000



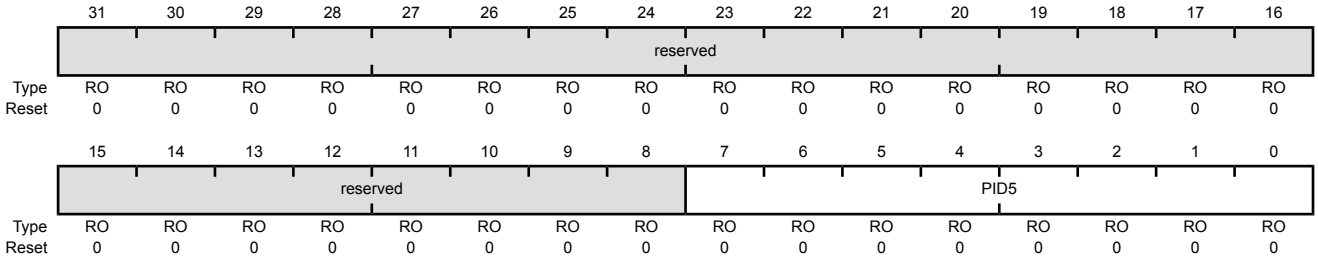
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID4	RO	0x0000	UART Peripheral ID Register[7:0] Can be used by software to identify the presence of this peripheral.

Register 15: UART Peripheral Identification 5 (UARTPeriphID5), offset 0xFD4

The **UARTPeriphIDn** registers are hard-coded and the fields within the registers determine the reset values.

UART Peripheral Identification 5 (UARTPeriphID5)

UART0 base: 0x4000.C000
UART1 base: 0x4000.D000
Offset 0xFD4
Type RO, reset 0x0000.0000



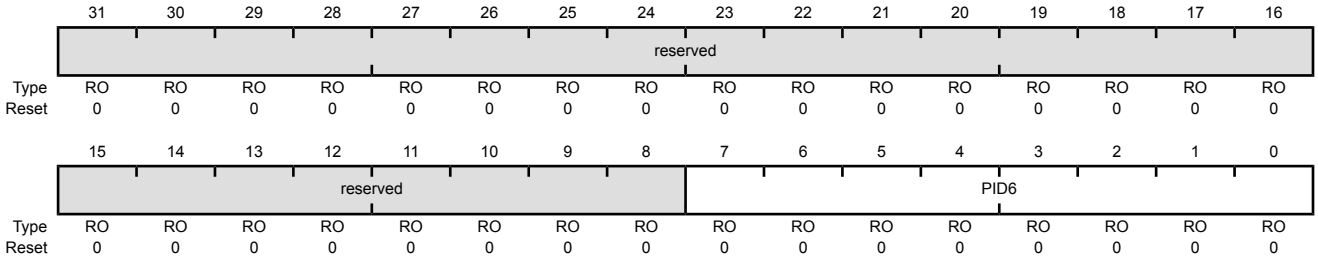
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID5	RO	0x0000	UART Peripheral ID Register[15:8] Can be used by software to identify the presence of this peripheral.

Register 16: UART Peripheral Identification 6 (UARTPeriphID6), offset 0xFD8

The **UARTPeriphIDn** registers are hard-coded and the fields within the registers determine the reset values.

UART Peripheral Identification 6 (UARTPeriphID6)

UART0 base: 0x4000.C000
UART1 base: 0x4000.D000
Offset 0xFD8
Type RO, reset 0x0000.0000



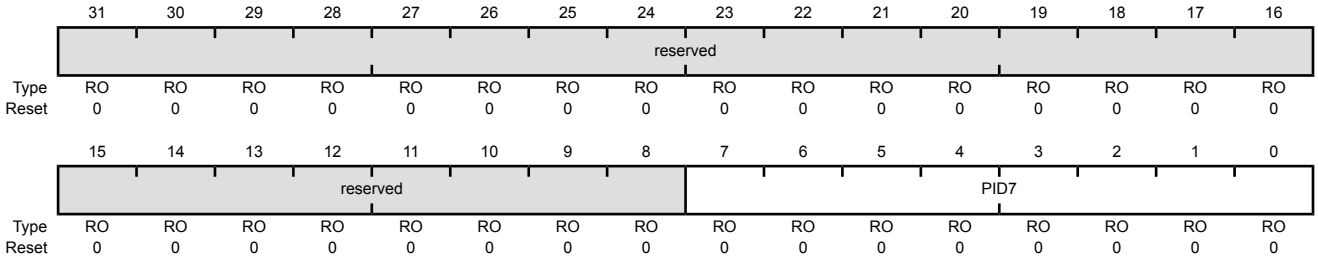
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID6	RO	0x0000	UART Peripheral ID Register[23:16] Can be used by software to identify the presence of this peripheral.

Register 17: UART Peripheral Identification 7 (UARTPeriphID7), offset 0xFDC

The **UARTPeriphIDn** registers are hard-coded and the fields within the registers determine the reset values.

UART Peripheral Identification 7 (UARTPeriphID7)

UART0 base: 0x4000.C000
UART1 base: 0x4000.D000
Offset 0xFDC
Type RO, reset 0x0000.0000



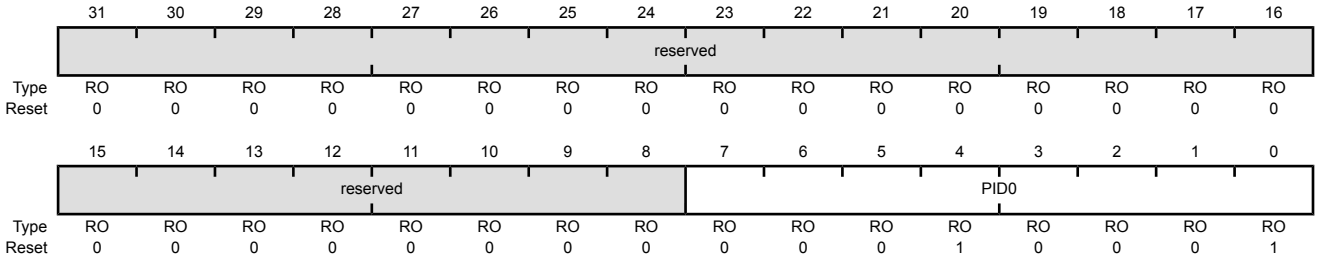
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID7	RO	0x0000	UART Peripheral ID Register[31:24] Can be used by software to identify the presence of this peripheral.

Register 18: UART Peripheral Identification 0 (UARTPeriphID0), offset 0xFE0

The **UARTPeriphIDn** registers are hard-coded and the fields within the registers determine the reset values.

UART Peripheral Identification 0 (UARTPeriphID0)

UART0 base: 0x4000.C000
UART1 base: 0x4000.D000
Offset 0xFE0
Type RO, reset 0x0000.0011



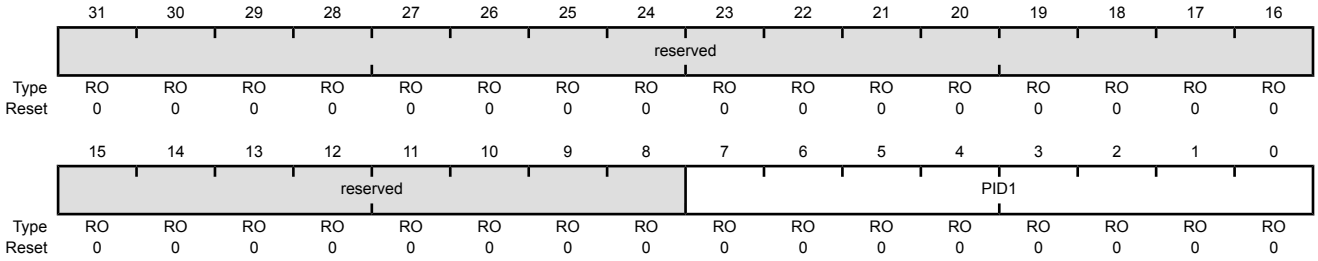
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID0	RO	0x11	UART Peripheral ID Register[7:0] Can be used by software to identify the presence of this peripheral.

Register 19: UART Peripheral Identification 1 (UARTPeriphID1), offset 0xFE4

The **UARTPeriphIDn** registers are hard-coded and the fields within the registers determine the reset values.

UART Peripheral Identification 1 (UARTPeriphID1)

UART0 base: 0x4000.C000
UART1 base: 0x4000.D000
Offset 0xFE4
Type RO, reset 0x0000.0000



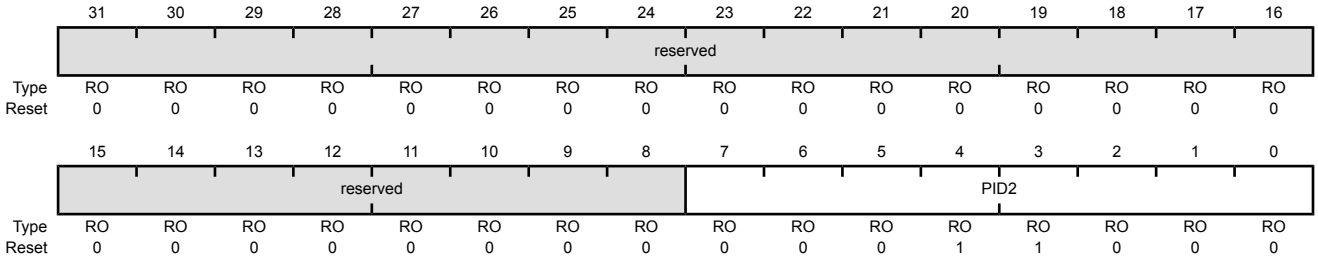
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID1	RO	0x00	UART Peripheral ID Register[15:8] Can be used by software to identify the presence of this peripheral.

Register 20: UART Peripheral Identification 2 (UARTPeriphID2), offset 0xFE8

The **UARTPeriphIDn** registers are hard-coded and the fields within the registers determine the reset values.

UART Peripheral Identification 2 (UARTPeriphID2)

UART0 base: 0x4000.C000
UART1 base: 0x4000.D000
Offset 0xFE8
Type RO, reset 0x0000.0018



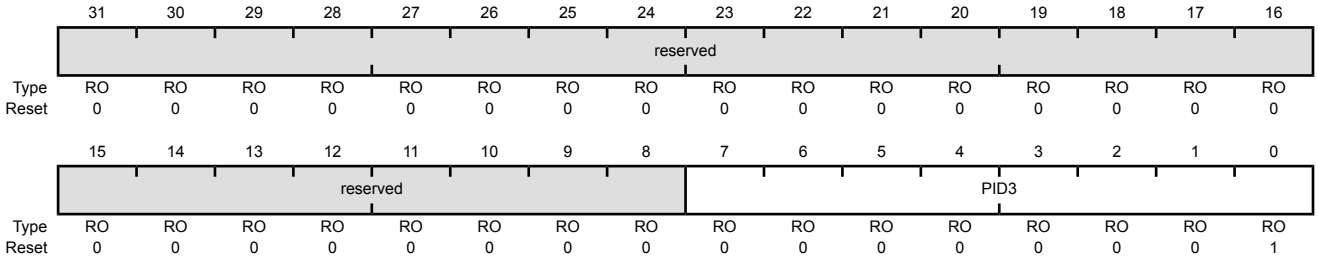
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID2	RO	0x18	UART Peripheral ID Register[23:16] Can be used by software to identify the presence of this peripheral.

Register 21: UART Peripheral Identification 3 (UARTPeriphID3), offset 0xFEC

The **UARTPeriphIDn** registers are hard-coded and the fields within the registers determine the reset values.

UART Peripheral Identification 3 (UARTPeriphID3)

UART0 base: 0x4000.C000
UART1 base: 0x4000.D000
Offset 0xFEC
Type RO, reset 0x0000.0001



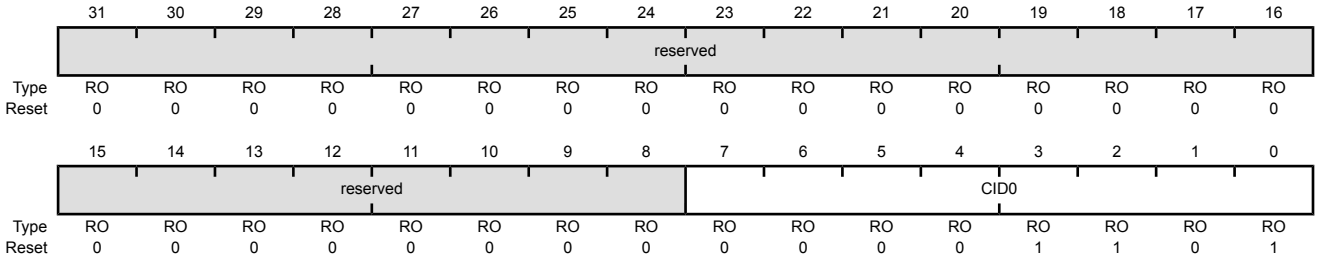
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID3	RO	0x01	UART Peripheral ID Register[31:24] Can be used by software to identify the presence of this peripheral.

Register 22: UART PrimeCell Identification 0 (UARTPCellID0), offset 0xFF0

The **UARTPCellIDn** registers are hard-coded and the fields within the registers determine the reset values.

UART PrimeCell Identification 0 (UARTPCellID0)

UART0 base: 0x4000.C000
UART1 base: 0x4000.D000
Offset 0xFF0
Type RO, reset 0x0000.000D



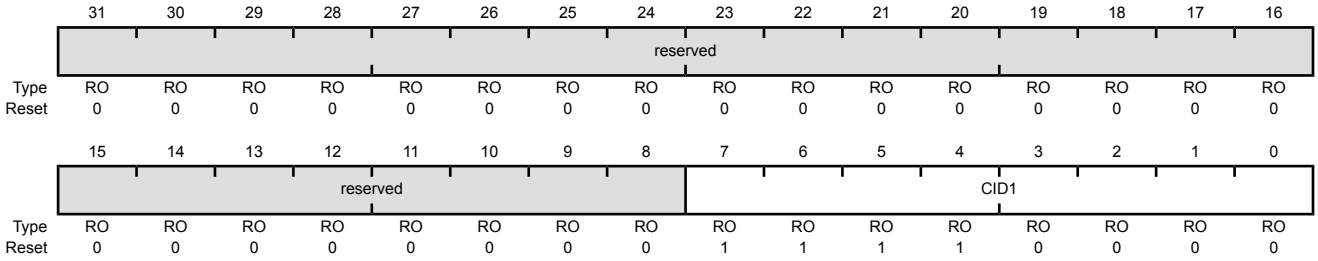
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	CID0	RO	0x0D	UART PrimeCell ID Register[7:0] Provides software a standard cross-peripheral identification system.

Register 23: UART PrimeCell Identification 1 (UARTPCellID1), offset 0xFF4

The **UARTPCellIDn** registers are hard-coded and the fields within the registers determine the reset values.

UART PrimeCell Identification 1 (UARTPCellID1)

UART0 base: 0x4000.C000
UART1 base: 0x4000.D000
Offset 0xFF4
Type RO, reset 0x0000.00F0



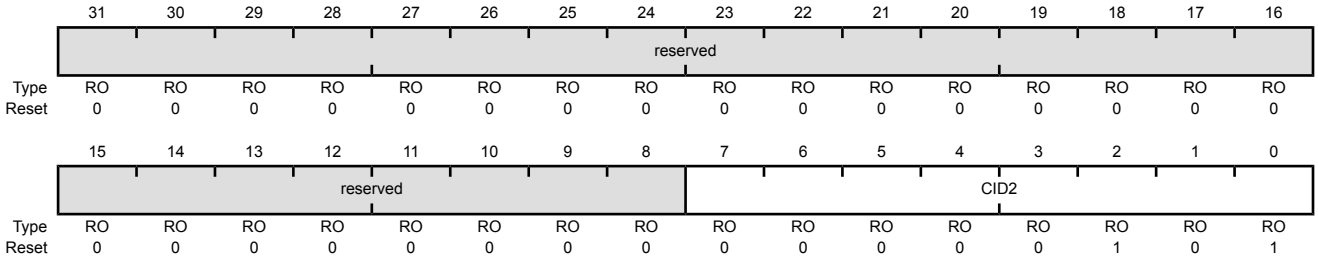
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	CID1	RO	0xF0	UART PrimeCell ID Register[15:8] Provides software a standard cross-peripheral identification system.

Register 24: UART PrimeCell Identification 2 (UARTPCellID2), offset 0xFF8

The **UARTPCellIDn** registers are hard-coded and the fields within the registers determine the reset values.

UART PrimeCell Identification 2 (UARTPCellID2)

UART0 base: 0x4000.C000
UART1 base: 0x4000.D000
Offset 0xFF8
Type RO, reset 0x0000.0005



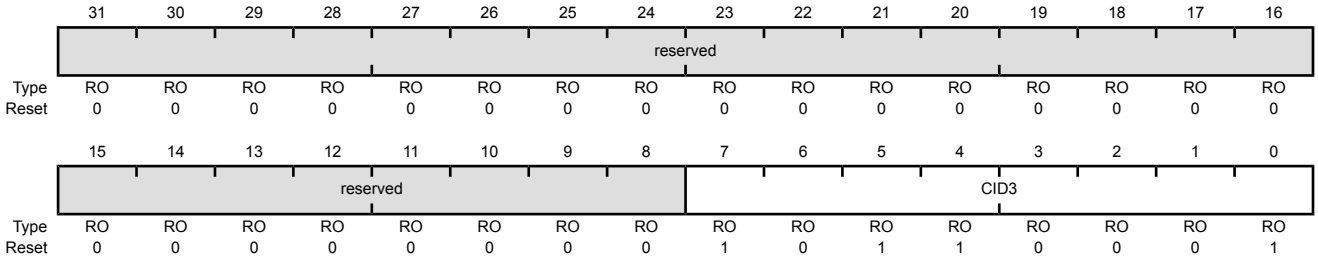
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	CID2	RO	0x05	UART PrimeCell ID Register[23:16] Provides software a standard cross-peripheral identification system.

Register 25: UART PrimeCell Identification 3 (UARTPCellID3), offset 0xFFC

The **UARTPCellIDn** registers are hard-coded and the fields within the registers determine the reset values.

UART PrimeCell Identification 3 (UARTPCellID3)

UART0 base: 0x4000.C000
UART1 base: 0x4000.D000
Offset 0xFFC
Type RO, reset 0x0000.00B1



Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	CID3	RO	0xB1	UART PrimeCell ID Register[31:24] Provides software a standard cross-peripheral identification system.

13 Synchronous Serial Interface (SSI)

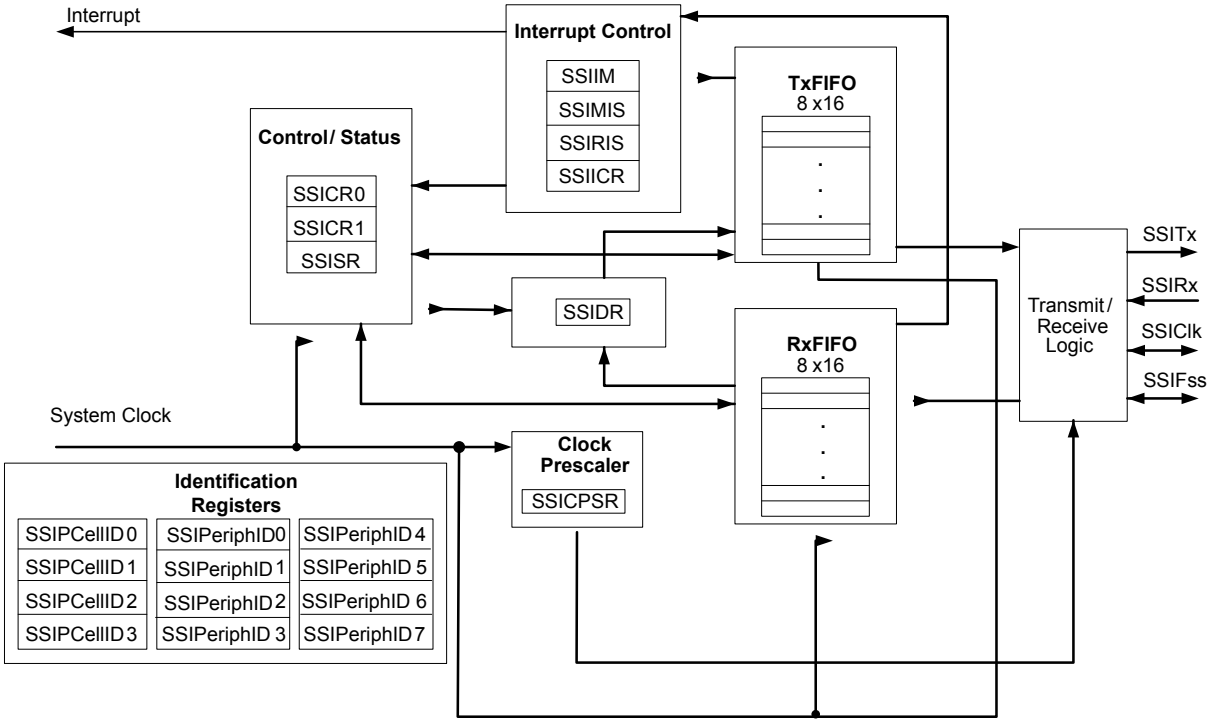
The Stellaris® Synchronous Serial Interface (SSI) is a master or slave interface for synchronous serial communication with peripheral devices that have either Freescale SPI, MICROWIRE, or Texas Instruments synchronous serial interfaces.

The Stellaris® SSI module has the following features:

- Master or slave operation
- Programmable clock bit rate and prescale
- Separate transmit and receive FIFOs, 16 bits wide, 8 locations deep
- Programmable interface operation for Freescale SPI, MICROWIRE, or Texas Instruments synchronous serial interfaces
- Programmable data frame size from 4 to 16 bits
- Internal loopback test mode for diagnostic/debug testing

13.1 Block Diagram

Figure 13-1. SSI Module Block Diagram



13.2 Functional Description

The SSI performs serial-to-parallel conversion on data received from a peripheral device. The CPU accesses data, control, and status information. The transmit and receive paths are buffered with

internal FIFO memories allowing up to eight 16-bit values to be stored independently in both transmit and receive modes.

13.2.1 Bit Rate Generation

The SSI includes a programmable bit rate clock divider and prescaler to generate the serial output clock. Bit rates are supported to 2 MHz and higher, although maximum bit rate is determined by peripheral devices.

The serial bit rate is derived by dividing down the input clock (FSysClk). The clock is first divided by an even prescale value CPSDVSR from 2 to 254, which is programmed in the **SSI Clock Prescale (SSICPSR)** register (see page 350). The clock is further divided by a value from 1 to 256, which is 1 + *SCR*, where *SCR* is the value programmed in the **SSI Control0 (SSICR0)** register (see page 343).

The frequency of the output clock SSIClk is defined by:

```
SSIClk = FSysClk / (CPSDVSR * (1 + SCR))
```

Note: For master mode, the system clock must be at least two times faster than the SSIClk. For slave mode, the system clock must be at least 12 times faster than the SSIClk.

See “Synchronous Serial Interface (SSI)” on page 536 to view SSI timing parameters.

13.2.2 FIFO Operation

13.2.2.1 Transmit FIFO

The common transmit FIFO is a 16-bit wide, 8-locations deep, first-in, first-out memory buffer. The CPU writes data to the FIFO by writing the **SSI Data (SSIDR)** register (see page 347), and data is stored in the FIFO until it is read out by the transmission logic.

When configured as a master or a slave, parallel data is written into the transmit FIFO prior to serial conversion and transmission to the attached slave or master, respectively, through the SSITx pin.

In slave mode, the SSI transmits data each time the master initiates a transaction. If the transmit FIFO is empty and the master initiates, the slave transmits the 8th most recent value in the transmit FIFO. If less than 8 values have been written to the transmit FIFO since the SSI module clock was enabled using the SSI bit in the **RGCG1** register, then 0 is transmitted. Care should be taken to ensure that valid data is in the FIFO as needed. The SSI can be configured to generate an interrupt or a µDMA request when the FIFO is empty.

13.2.2.2 Receive FIFO

The common receive FIFO is a 16-bit wide, 8-locations deep, first-in, first-out memory buffer. Received data from the serial interface is stored in the buffer until read out by the CPU, which accesses the read FIFO by reading the **SSIDR** register.

When configured as a master or slave, serial data received through the SSIRx pin is registered prior to parallel loading into the attached slave or master receive FIFO, respectively.

13.2.3 Interrupts

The SSI can generate interrupts when the following conditions are observed:

- Transmit FIFO service
- Receive FIFO service

- Receive FIFO time-out
- Receive FIFO overrun

All of the interrupt events are ORed together before being sent to the interrupt controller, so the SSI can only generate a single interrupt request to the controller at any given time. You can mask each of the four individual maskable interrupts by setting the appropriate bits in the **SSI Interrupt Mask (SSIIM)** register (see page 351). Setting the appropriate mask bit to 1 enables the interrupt.

Provision of the individual outputs, as well as a combined interrupt output, allows use of either a global interrupt service routine, or modular device drivers to handle interrupts. The transmit and receive dynamic dataflow interrupts have been separated from the status interrupts so that data can be read or written in response to the FIFO trigger levels. The status of the individual interrupt sources can be read from the **SSI Raw Interrupt Status (SSIRIS)** and **SSI Masked Interrupt Status (SSIMIS)** registers (see page 353 and page 354, respectively).

13.2.4 Frame Formats

Each data frame is between 4 and 16 bits long, depending on the size of data programmed, and is transmitted starting with the MSB. There are three basic frame types that can be selected:

- Texas Instruments synchronous serial
- Freescale SPI
- MICROWIRE

For all three formats, the serial clock (SSIClk) is held inactive while the SSI is idle, and SSIClk transitions at the programmed frequency only during active transmission or reception of data. The idle state of SSIClk is utilized to provide a receive timeout indication that occurs when the receive FIFO still contains data after a timeout period.

For Freescale SPI and MICROWIRE frame formats, the serial frame (SSIFss) pin is active Low, and is asserted (pulled down) during the entire transmission of the frame.

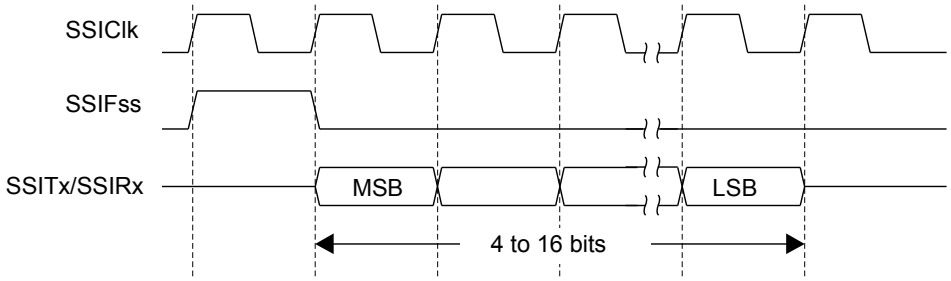
For Texas Instruments synchronous serial frame format, the SSIFss pin is pulsed for one serial clock period starting at its rising edge, prior to the transmission of each frame. For this frame format, both the SSI and the off-chip slave device drive their output data on the rising edge of SSIClk, and latch data from the other device on the falling edge.

Unlike the full-duplex transmission of the other two frame formats, the MICROWIRE format uses a special master-slave messaging technique, which operates at half-duplex. In this mode, when a frame begins, an 8-bit control message is transmitted to the off-chip slave. During this transmit, no incoming data is received by the SSI. After the message has been sent, the off-chip slave decodes it and, after waiting one serial clock after the last bit of the 8-bit control message has been sent, responds with the requested data. The returned data can be 4 to 16 bits in length, making the total frame length anywhere from 13 to 25 bits.

13.2.4.1 Texas Instruments Synchronous Serial Frame Format

Figure 13-2 on page 334 shows the Texas Instruments synchronous serial frame format for a single transmitted frame.

Figure 13-2. TI Synchronous Serial Frame Format (Single Transfer)

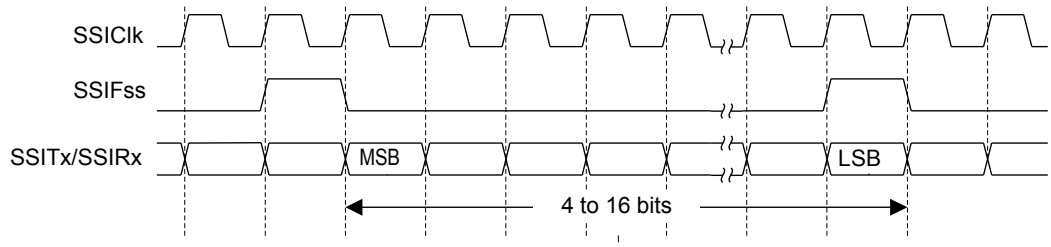


In this mode, SSIClk and SSIFss are forced Low, and the transmit data line SSITx is tristated whenever the SSI is idle. Once the bottom entry of the transmit FIFO contains data, SSIFss is pulsed High for one SSIClk period. The value to be transmitted is also transferred from the transmit FIFO to the serial shift register of the transmit logic. On the next rising edge of SSIClk, the MSB of the 4 to 16-bit data frame is shifted out on the SSITx pin. Likewise, the MSB of the received data is shifted onto the SSIRx pin by the off-chip serial slave device.

Both the SSI and the off-chip serial slave device then clock each data bit into their serial shifter on the falling edge of each SSIClk. The received data is transferred from the serial shifter to the receive FIFO on the first rising edge of SSIClk after the LSB has been latched.

Figure 13-3 on page 334 shows the Texas Instruments synchronous serial frame format when back-to-back frames are transmitted.

Figure 13-3. TI Synchronous Serial Frame Format (Continuous Transfer)



13.2.4.2 Freescale SPI Frame Format

The Freescale SPI interface is a four-wire interface where the SSIFss signal behaves as a slave select. The main feature of the Freescale SPI format is that the inactive state and phase of the SSIClk signal are programmable through the SPO and SPH bits within the **SSISCR0** control register.

SPO Clock Polarity Bit

When the SPO clock polarity control bit is Low, it produces a steady state Low value on the SSIClk pin. If the SPO bit is High, a steady state High value is placed on the SSIClk pin when data is not being transferred.

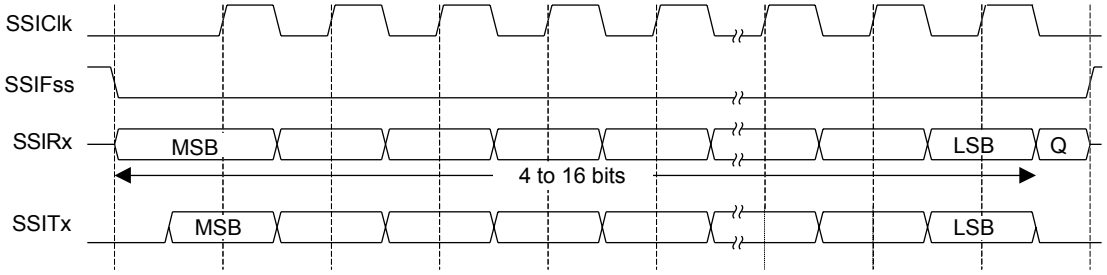
SPH Phase Control Bit

The SPH phase control bit selects the clock edge that captures data and allows it to change state. It has the most impact on the first bit transmitted by either allowing or not allowing a clock transition before the first data capture edge. When the SPH phase control bit is Low, data is captured on the first clock edge transition. If the SPH bit is High, data is captured on the second clock edge transition.

13.2.4.3 Freescale SPI Frame Format with SPO=0 and SPH=0

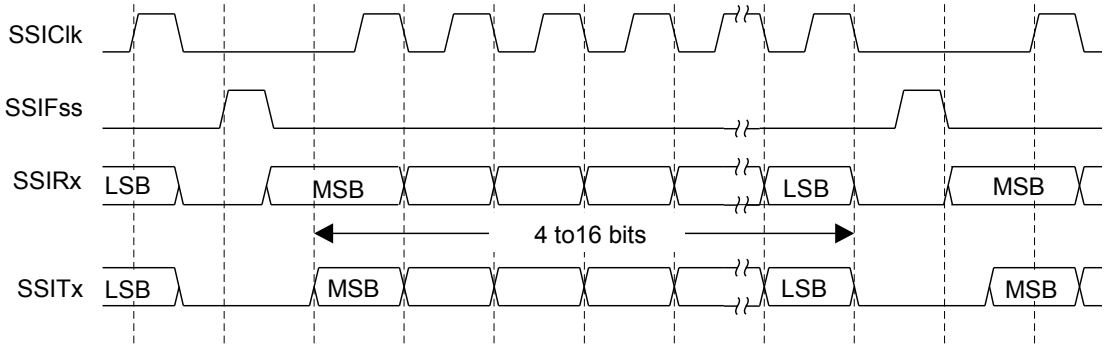
Single and continuous transmission signal sequences for Freescale SPI format with SPO=0 and SPH=0 are shown in Figure 13-4 on page 335 and Figure 13-5 on page 335.

Figure 13-4. Freescale SPI Format (Single Transfer) with SPO=0 and SPH=0



Note: Q is undefined.

Figure 13-5. Freescale SPI Format (Continuous Transfer) with SPO=0 and SPH=0



In this configuration, during idle periods:

- SSIClk is forced Low
- SSIFss is forced High
- The transmit data line SSITx is arbitrarily forced Low
- When the SSI is configured as a master, it enables the SSIClk pad
- When the SSI is configured as a slave, it disables the SSIClk pad

If the SSI is enabled and there is valid data within the transmit FIFO, the start of transmission is signified by the SSIFss master signal being driven Low. This causes slave data to be enabled onto the SSIRx input line of the master. The master SSITx output pad is enabled.

One half SSIClk period later, valid master data is transferred to the SSITx pin. Now that both the master and slave data have been set, the SSIClk master clock pin goes High after one further half SSIClk period.

The data is now captured on the rising and propagated on the falling edges of the SSIClk signal.

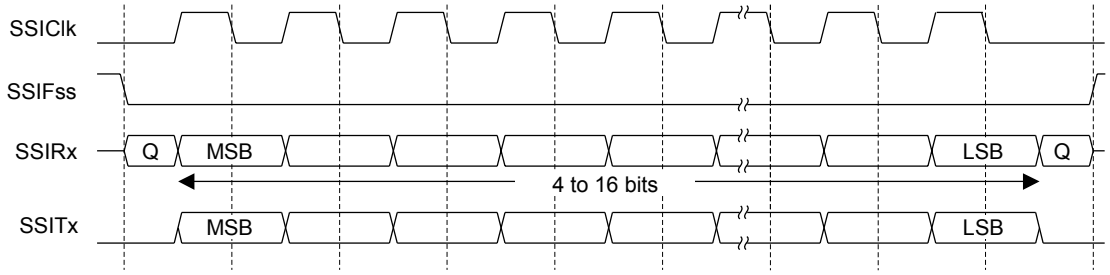
In the case of a single word transmission, after all bits of the data word have been transferred, the SSIFss line is returned to its idle High state one SSIClk period after the last bit has been captured.

However, in the case of continuous back-to-back transmissions, the SSIFss signal must be pulsed High between each data word transfer. This is because the slave select pin freezes the data in its serial peripheral register and does not allow it to be altered if the SPH bit is logic zero. Therefore, the master device must raise the SSIFss pin of the slave device between each data transfer to enable the serial peripheral data write. On completion of the continuous transfer, the SSIFss pin is returned to its idle state one SSIClk period after the last bit has been captured.

13.2.4.4 Freescale SPI Frame Format with SPO=0 and SPH=1

The transfer signal sequence for Freescale SPI format with SPO=0 and SPH=1 is shown in Figure 13-6 on page 336, which covers both single and continuous transfers.

Figure 13-6. Freescale SPI Frame Format with SPO=0 and SPH=1



Note: Q is undefined.

In this configuration, during idle periods:

- SSIClk is forced Low
- SSIFss is forced High
- The transmit data line SSITx is arbitrarily forced Low
- When the SSI is configured as a master, it enables the SSIClk pad
- When the SSI is configured as a slave, it disables the SSIClk pad

If the SSI is enabled and there is valid data within the transmit FIFO, the start of transmission is signified by the SSIFss master signal being driven Low. The master SSITx output is enabled. After a further one half SSIClk period, both master and slave valid data is enabled onto their respective transmission lines. At the same time, the SSIClk is enabled with a rising edge transition.

Data is then captured on the falling edges and propagated on the rising edges of the SSIClk signal.

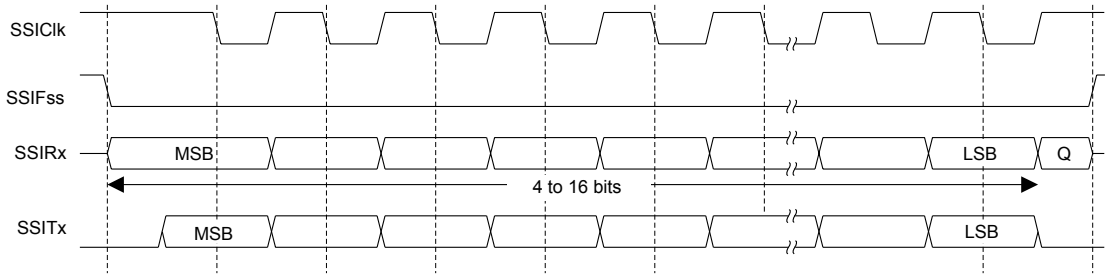
In the case of a single word transfer, after all bits have been transferred, the SSIFss line is returned to its idle High state one SSIClk period after the last bit has been captured.

For continuous back-to-back transfers, the SSIFss pin is held Low between successive data words and termination is the same as that of the single word transfer.

13.2.4.5 Freescale SPI Frame Format with SPO=1 and SPH=0

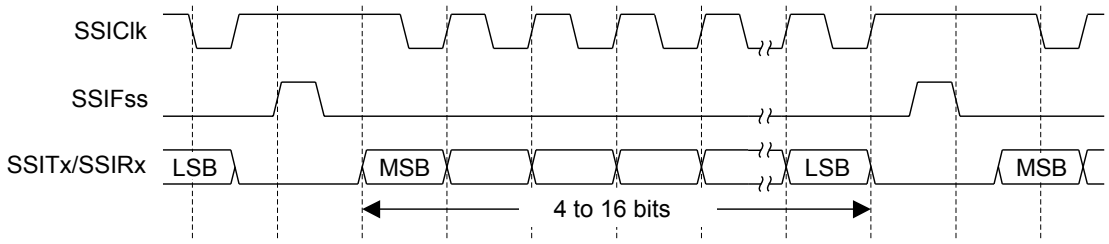
Single and continuous transmission signal sequences for Freescale SPI format with SPO=1 and SPH=0 are shown in Figure 13-7 on page 337 and Figure 13-8 on page 337.

Figure 13-7. Freescale SPI Frame Format (Single Transfer) with SPO=1 and SPH=0



Note: Q is undefined.

Figure 13-8. Freescale SPI Frame Format (Continuous Transfer) with SPO=1 and SPH=0



In this configuration, during idle periods:

- SSIClk is forced High
- SSIFss is forced High
- The transmit data line SSITx is arbitrarily forced Low
- When the SSI is configured as a master, it enables the SSIClk pad
- When the SSI is configured as a slave, it disables the SSIClk pad

If the SSI is enabled and there is valid data within the transmit FIFO, the start of transmission is signified by the SSIFss master signal being driven Low, which causes slave data to be immediately transferred onto the SSIRx line of the master. The master SSITx output pad is enabled.

One half period later, valid master data is transferred to the SSITx line. Now that both the master and slave data have been set, the SSIClk master clock pin becomes Low after one further half SSIClk period. This means that data is captured on the falling edges and propagated on the rising edges of the SSIClk signal.

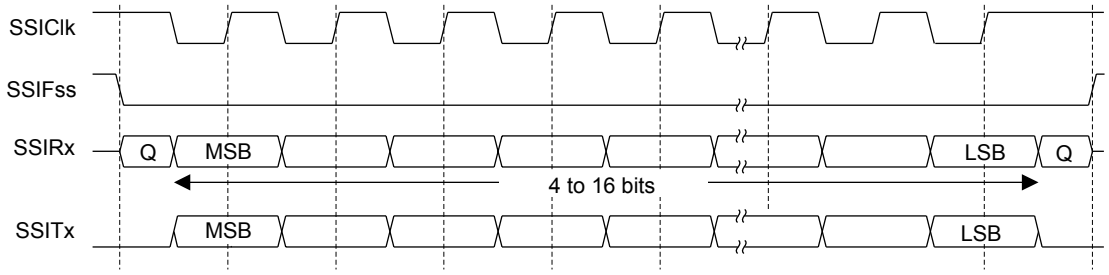
In the case of a single word transmission, after all bits of the data word are transferred, the SSIFss line is returned to its idle High state one SSIClk period after the last bit has been captured.

However, in the case of continuous back-to-back transmissions, the SSIFss signal must be pulsed High between each data word transfer. This is because the slave select pin freezes the data in its serial peripheral register and does not allow it to be altered if the SPH bit is logic zero. Therefore, the master device must raise the SSIFss pin of the slave device between each data transfer to enable the serial peripheral data write. On completion of the continuous transfer, the SSIFss pin is returned to its idle state one SSIClk period after the last bit has been captured.

13.2.4.6 Freescale SPI Frame Format with SPO=1 and SPH=1

The transfer signal sequence for Freescale SPI format with SPO=1 and SPH=1 is shown in Figure 13-9 on page 338, which covers both single and continuous transfers.

Figure 13-9. Freescale SPI Frame Format with SPO=1 and SPH=1



Note: Q is undefined.

In this configuration, during idle periods:

- SSIClk is forced High
- SSIFss is forced High
- The transmit data line SSITx is arbitrarily forced Low
- When the SSI is configured as a master, it enables the SSIClk pad
- When the SSI is configured as a slave, it disables the SSIClk pad

If the SSI is enabled and there is valid data within the transmit FIFO, the start of transmission is signified by the SSIFss master signal being driven Low. The master SSITx output pad is enabled. After a further one-half SSIClk period, both master and slave data are enabled onto their respective transmission lines. At the same time, SSIClk is enabled with a falling edge transition. Data is then captured on the rising edges and propagated on the falling edges of the SSIClk signal.

After all bits have been transferred, in the case of a single word transmission, the SSIFss line is returned to its idle high state one SSIClk period after the last bit has been captured.

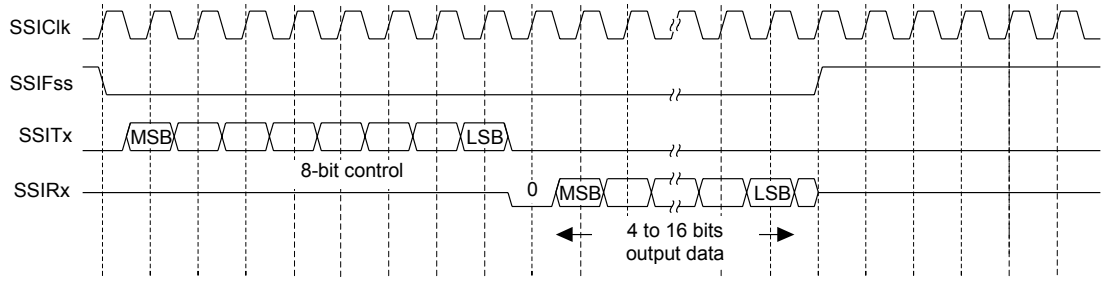
For continuous back-to-back transmissions, the SSIFss pin remains in its active Low state, until the final bit of the last word has been captured, and then returns to its idle state as described above.

For continuous back-to-back transfers, the SSIFss pin is held Low between successive data words and termination is the same as that of the single word transfer.

13.2.4.7 MICROWIRE Frame Format

Figure 13-10 on page 339 shows the MICROWIRE frame format, again for a single frame. Figure 13-11 on page 340 shows the same format when back-to-back frames are transmitted.

Figure 13-10. MICROWIRE Frame Format (Single Frame)



MICROWIRE format is very similar to SPI format, except that transmission is half-duplex instead of full-duplex, using a master-slave message passing technique. Each serial transmission begins with an 8-bit control word that is transmitted from the SSI to the off-chip slave device. During this transmission, no incoming data is received by the SSI. After the message has been sent, the off-chip slave decodes it and, after waiting one serial clock after the last bit of the 8-bit control message has been sent, responds with the required data. The returned data is 4 to 16 bits in length, making the total frame length anywhere from 13 to 25 bits.

In this configuration, during idle periods:

- SSIClk is forced Low
- SSIFss is forced High
- The transmit data line SSITx is arbitrarily forced Low

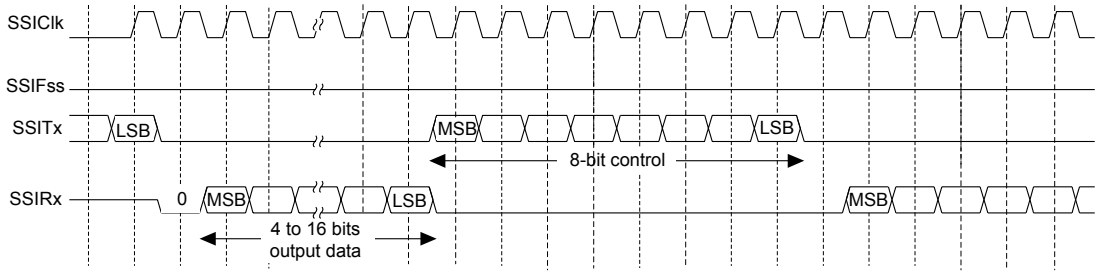
A transmission is triggered by writing a control byte to the transmit FIFO. The falling edge of SSIFss causes the value contained in the bottom entry of the transmit FIFO to be transferred to the serial shift register of the transmit logic, and the MSB of the 8-bit control frame to be shifted out onto the SSITx pin. SSIFss remains Low for the duration of the frame transmission. The SSIRx pin remains tristated during this transmission.

The off-chip serial slave device latches each control bit into its serial shifter on the rising edge of each SSIClk. After the last bit is latched by the slave device, the control byte is decoded during a one clock wait-state, and the slave responds by transmitting data back to the SSI. Each bit is driven onto the SSIRx line on the falling edge of SSIClk. The SSI in turn latches each bit on the rising edge of SSIClk. At the end of the frame, for single transfers, the SSIFss signal is pulled High one clock period after the last bit has been latched in the receive serial shifter, which causes the data to be transferred to the receive FIFO.

Note: The off-chip slave device can tristate the receive line either on the falling edge of SSIClk after the LSB has been latched by the receive shifter, or when the SSIFss pin goes High.

For continuous transfers, data transmission begins and ends in the same manner as a single transfer. However, the SSIFss line is continuously asserted (held Low) and transmission of data occurs back-to-back. The control byte of the next frame follows directly after the LSB of the received data from the current frame. Each of the received values is transferred from the receive shifter on the falling edge of SSIClk, after the LSB of the frame has been latched into the SSI.

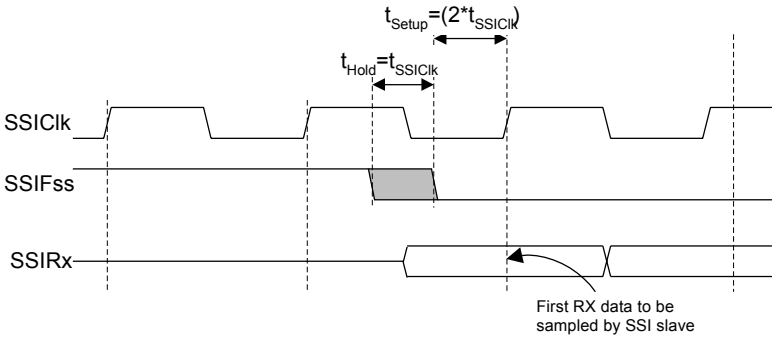
Figure 13-11. MICROWIRE Frame Format (Continuous Transfer)



In the MICROWIRE mode, the SSI slave samples the first bit of receive data on the rising edge of SSIClk after SSIFss has gone Low. Masters that drive a free-running SSIClk must ensure that the SSIFss signal has sufficient setup and hold margins with respect to the rising edge of SSIClk.

Figure 13-12 on page 340 illustrates these setup and hold time requirements. With respect to the SSIClk rising edge on which the first bit of receive data is to be sampled by the SSI slave, SSIFss must have a setup of at least two times the period of SSIClk on which the SSI operates. With respect to the SSIClk rising edge previous to this edge, SSIFss must have a hold of at least one SSIClk period.

Figure 13-12. MICROWIRE Frame Format, SSIFss Input Setup and Hold Requirements



13.3 Initialization and Configuration

To use the SSI, its peripheral clock must be enabled by setting the SSI bit in the **RCGC1** register.

For each of the frame formats, the SSI is configured using the following steps:

1. Ensure that the SSE bit in the **SSICR1** register is disabled before making any configuration changes.
2. Select whether the SSI is a master or slave:
   a. For master operations, set the **SSICR1** register to 0x0000.0000.
   b. For slave mode (output enabled), set the **SSICR1** register to 0x0000.0004.
   c. For slave mode (output disabled), set the **SSICR1** register to 0x0000.000C.
3. Configure the clock prescale divisor by writing the **SSICPSR** register.
4. Write the **SSICR0** register with the following configuration:

- Serial clock rate (SCR)
- Desired clock phase/polarity, if using Freescale SPI mode (SPH and SPO)
- The protocol mode: Freescale SPI, TI SSF, MICROWIRE (FRF)
- The data size (DSS)

5. Enable the SSI by setting the SSE bit in the **SSICR1** register.

As an example, assume the SSI must be configured to operate with the following parameters:

- Master operation
- Freescale SPI mode (SPO=1, SPH=1)
- 1 Mbps bit rate
- 8 data bits

Assuming the system clock is 20 MHz, the bit rate calculation would be:

```
FSSIClk = FSysClk / (CPSDVSR * (1 + SCR))
1x106 = 20x106 / (CPSDVSR * (1 + SCR))
```

In this case, if CPSDVSR=2, SCR must be 9.

The configuration sequence would be as follows:

1. Ensure that the SSE bit in the **SSICR1** register is disabled.
2. Write the **SSICR1** register with a value of 0x0000.0000.
3. Write the **SSICPSR** register with a value of 0x0000.0002.
4. Write the **SSICR0** register with a value of 0x0000.09C7.
5. The SSI is then enabled by setting the SSE bit in the **SSICR1** register to 1.

13.4 Register Map

Table 13-1 on page 341 lists the SSI registers. The offset listed is a hexadecimal increment to the register's address, relative to that SSI module's base address:

- SSI0: 0x4000.8000

Note: The SSI must be disabled (see the SSE bit in the **SSICR1** register) before any of the control registers are reprogrammed.

Table 13-1. SSI Register Map

Offset	Name	Type	Reset	Description	See page
0x000	SSICR0	R/W	0x0000.0000	SSI Control 0	343
0x004	SSICR1	R/W	0x0000.0000	SSI Control 1	345

Table 13-1. SSI Register Map *(continued)*

Offset	Name	Type	Reset	Description	See page
0x008	SSIDR	R/W	0x0000.0000	SSI Data	347
0x00C	SSISR	RO	0x0000.0003	SSI Status	348
0x010	SSICPSR	R/W	0x0000.0000	SSI Clock Prescale	350
0x014	SSIIM	R/W	0x0000.0000	SSI Interrupt Mask	351
0x018	SSIRIS	RO	0x0000.0008	SSI Raw Interrupt Status	353
0x01C	SSIMIS	RO	0x0000.0000	SSI Masked Interrupt Status	354
0x020	SSIICR	W1C	0x0000.0000	SSI Interrupt Clear	355
0xFD0	SSIPeriphID4	RO	0x0000.0000	SSI Peripheral Identification 4	356
0xFD4	SSIPeriphID5	RO	0x0000.0000	SSI Peripheral Identification 5	357
0xFD8	SSIPeriphID6	RO	0x0000.0000	SSI Peripheral Identification 6	358
0xFDC	SSIPeriphID7	RO	0x0000.0000	SSI Peripheral Identification 7	359
0xFE0	SSIPeriphID0	RO	0x0000.0022	SSI Peripheral Identification 0	360
0xFE4	SSIPeriphID1	RO	0x0000.0000	SSI Peripheral Identification 1	361
0xFE8	SSIPeriphID2	RO	0x0000.0018	SSI Peripheral Identification 2	362
0xFEC	SSIPeriphID3	RO	0x0000.0001	SSI Peripheral Identification 3	363
0xFF0	SSIPCellID0	RO	0x0000.000D	SSI PrimeCell Identification 0	364
0xFF4	SSIPCellID1	RO	0x0000.00F0	SSI PrimeCell Identification 1	365
0xFF8	SSIPCellID2	RO	0x0000.0005	SSI PrimeCell Identification 2	366
0xFFC	SSIPCellID3	RO	0x0000.00B1	SSI PrimeCell Identification 3	367

13.5 Register Descriptions

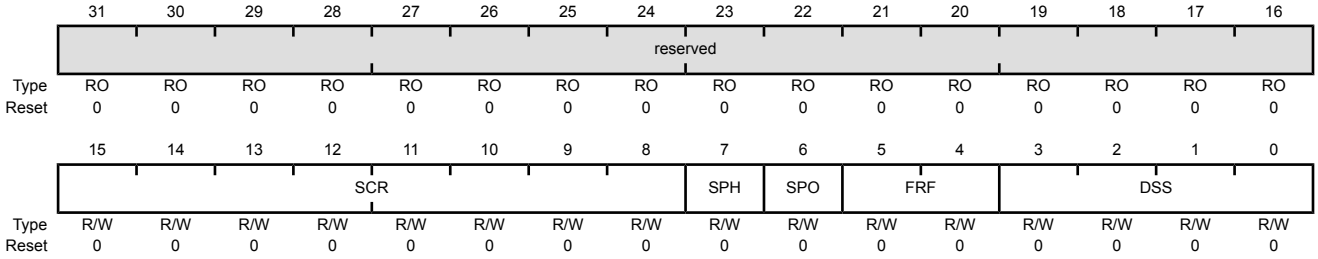
The remainder of this section lists and describes the SSI registers, in numerical order by address offset.

Register 1: SSI Control 0 (SSICR0), offset 0x000

SSICR0 is control register 0 and contains bit fields that control various functions within the SSI module. Functionality such as protocol mode, clock rate, and data size are configured in this register.

SSI Control 0 (SSICR0)

SSI0 base: 0x4000.8000
Offset 0x000
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:16	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
15:8	SCR	R/W	0x0000	SSI Serial Clock Rate The value SCR is used to generate the transmit and receive bit rate of the SSI. The bit rate is: `BR=FSSIClk/(CPSDVSR * (1 + SCR))` where CPSDVSR is an even value from 2-254 programmed in the **SSICPSR** register, and SCR is a value from 0-255.
7	SPH	R/W	0	SSI Serial Clock Phase This bit is only applicable to the Freescale SPI Format. The SPH control bit selects the clock edge that captures data and allows it to change state. It has the most impact on the first bit transmitted by either allowing or not allowing a clock transition before the first data capture edge. When the SPH bit is 0, data is captured on the first clock edge transition. If SPH is 1, data is captured on the second clock edge transition.
6	SPO	R/W	0	SSI Serial Clock Polarity This bit is only applicable to the Freescale SPI Format. When the SPO bit is 0, it produces a steady state Low value on the SSIClk pin. If SPO is 1, a steady state High value is placed on the SSIClk pin when data is not being transferred.

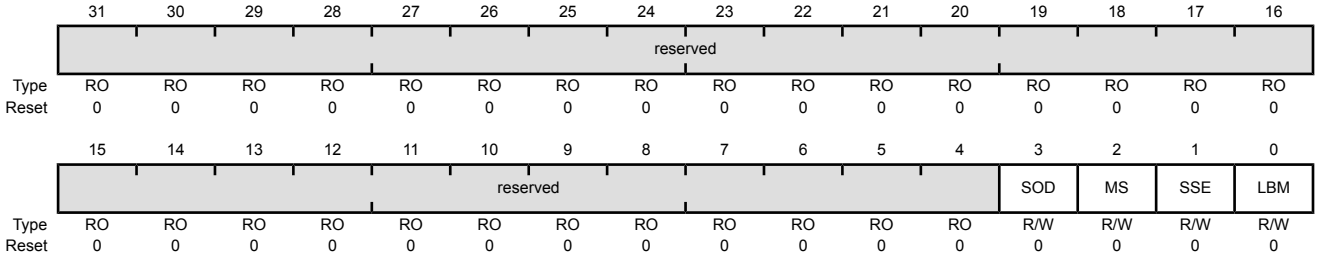
Bit/Field	Name	Type	Reset	Description
5:4	FRF	R/W	0x0	SSI Frame Format Select The FRF values are defined as follows: Value Frame Format 0x0 Freescale SPI Frame Format 0x1 Texas Instruments Synchronous Serial Frame Format 0x2 MICROWIRE Frame Format 0x3 Reserved
3:0	DSS	R/W	0x00	SSI Data Size Select The DSS values are defined as follows: Value Data Size 0x0-0x2 Reserved 0x3 4-bit data 0x4 5-bit data 0x5 6-bit data 0x6 7-bit data 0x7 8-bit data 0x8 9-bit data 0x9 10-bit data 0xA 11-bit data 0xB 12-bit data 0xC 13-bit data 0xD 14-bit data 0xE 15-bit data 0xF 16-bit data

Register 2: SSI Control 1 (SSICR1), offset 0x004

SSICR1 is control register 1 and contains bit fields that control various functions within the SSI module. Master and slave mode functionality is controlled by this register.

SSI Control 1 (SSICR1)

SSI0 base: 0x4000.8000
Offset 0x004
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:4	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
3	SOD	R/W	0	SSI Slave Mode Output Disable This bit is relevant only in the Slave mode (MS=1). In multiple-slave systems, it is possible for the SSI master to broadcast a message to all slaves in the system while ensuring that only one slave drives data onto the serial output line. In such systems, the TXD lines from multiple slaves could be tied together. To operate in such a system, the SOD bit can be configured so that the SSI slave does not drive the SSITx pin. The SOD values are defined as follows: Value Description 0 SSI can drive SSITx output in Slave Output mode. 1 SSI must not drive the SSITx output in Slave mode.
2	MS	R/W	0	SSI Master/Slave Select This bit selects Master or Slave mode and can be modified only when SSI is disabled (SSE=0). The MS values are defined as follows: Value Description 0 Device configured as a master. 1 Device configured as a slave.

Bit/Field	Name	Type	Reset	Description
1	SSE	R/W	0	SSI Synchronous Serial Port Enable Setting this bit enables SSI operation. The SSE values are defined as follows: Value Description 0 SSI operation disabled. 1 SSI operation enabled. **Note:** This bit must be set to 0 before any control registers are reprogrammed.
0	LBM	R/W	0	SSI Loopback Mode Setting this bit enables Loopback Test mode. The LBM values are defined as follows: Value Description 0 Normal serial port operation enabled. 1 Output of the transmit serial shift register is connected internally to the input of the receive serial shift register.

Register 3: SSI Data (SSIDR), offset 0x008

Important: Use caution when reading this register. Performing a read may change bit status.

SSIDR is the data register and is 16-bits wide. When **SSIDR** is read, the entry in the receive FIFO (pointed to by the current FIFO read pointer) is accessed. As data values are removed by the SSI receive logic from the incoming data frame, they are placed into the entry in the receive FIFO (pointed to by the current FIFO write pointer).

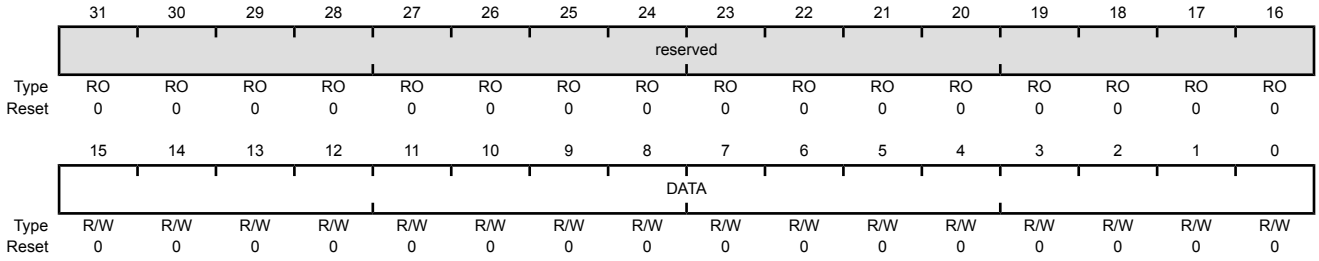
When **SSIDR** is written to, the entry in the transmit FIFO (pointed to by the write pointer) is written to. Data values are removed from the transmit FIFO one value at a time by the transmit logic. It is loaded into the transmit serial shifter, then serially shifted out onto the SSITx pin at the programmed bit rate.

When a data size of less than 16 bits is selected, the user must right-justify data written to the transmit FIFO. The transmit logic ignores the unused bits. Received data less than 16 bits is automatically right-justified in the receive buffer.

When the SSI is programmed for MICROWIRE frame format, the default size for transmit data is eight bits (the most significant byte is ignored). The receive data size is controlled by the programmer. The transmit FIFO and the receive FIFO are not cleared even when the SSE bit in the **SSICR1** register is set to zero. This allows the software to fill the transmit FIFO before enabling the SSI.

SSI Data (SSIDR)

SSI0 base: 0x4000.8000
Offset 0x008
Type R/W, reset 0x0000.0000



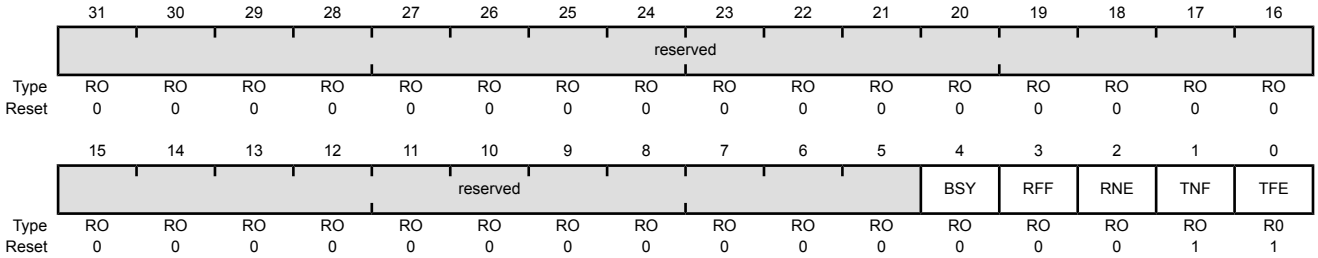
Bit/Field	Name	Type	Reset	Description
31:16	reserved	RO	0x0000	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
15:0	DATA	R/W	0x0000	SSI Receive/Transmit Data A read operation reads the receive FIFO. A write operation writes the transmit FIFO. Software must right-justify data when the SSI is programmed for a data size that is less than 16 bits. Unused bits at the top are ignored by the transmit logic. The receive logic automatically right-justifies the data.

Register 4: SSI Status (SSISR), offset 0x00C

SSISR is a status register that contains bits that indicate the FIFO fill status and the SSI busy status.

SSI Status (SSISR)

SSI0 base: 0x4000.8000
Offset 0x00C
Type RO, reset 0x0000.0003



Bit/Field	Name	Type	Reset	Description
31:5	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
4	BSY	RO	0	SSI Busy Bit The BSY values are defined as follows: Value Description 0 SSI is idle. 1 SSI is currently transmitting and/or receiving a frame, or the transmit FIFO is not empty.
3	RFF	RO	0	SSI Receive FIFO Full The RFF values are defined as follows: Value Description 0 Receive FIFO is not full. 1 Receive FIFO is full.
2	RNE	RO	0	SSI Receive FIFO Not Empty The RNE values are defined as follows: Value Description 0 Receive FIFO is empty. 1 Receive FIFO is not empty.
1	TNF	RO	1	SSI Transmit FIFO Not Full The TNF values are defined as follows: Value Description 0 Transmit FIFO is full. 1 Transmit FIFO is not full.

Bit/Field	Name	Type	Reset	Description
0	TFE	R0	1	SSI Transmit FIFO Empty The TFE values are defined as follows: Value Description 0 Transmit FIFO is not empty. 1 Transmit FIFO is empty.

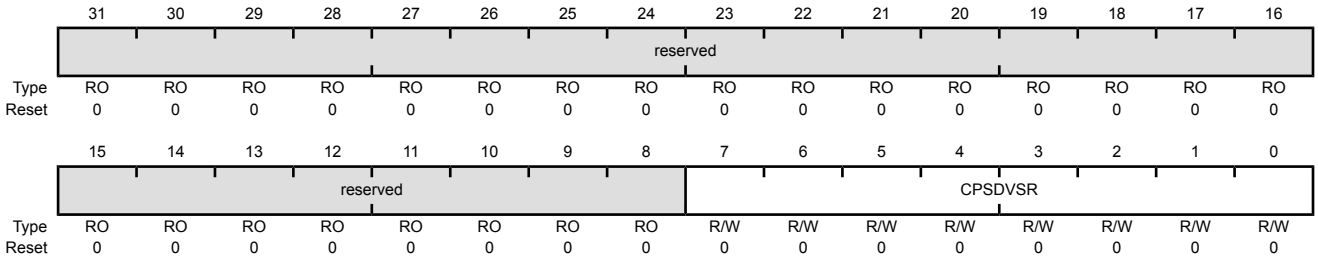
Register 5: SSI Clock Prescale (SSICPSR), offset 0x010

SSICPSR is the clock prescale register and specifies the division factor by which the system clock must be internally divided before further use.

The value programmed into this register must be an even number between 2 and 254. The least-significant bit of the programmed number is hard-coded to zero. If an odd number is written to this register, data read back from this register has the least-significant bit as zero.

SSI Clock Prescale (SSICPSR)

SSI0 base: 0x4000.8000
Offset 0x010
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	CPSDVSR	R/W	0x00	SSI Clock Prescale Divisor This value must be an even number from 2 to 254, depending on the frequency of SSIClk. The LSB always returns 0 on reads.

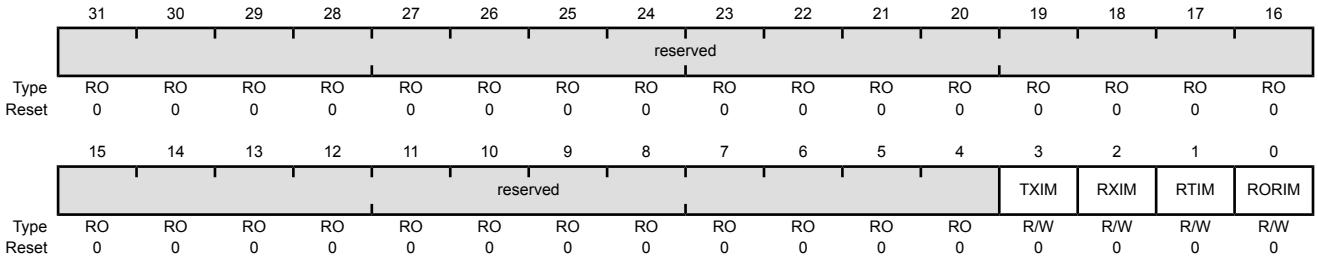
Register 6: SSI Interrupt Mask (SSIIM), offset 0x014

The **SSIIM** register is the interrupt mask set or clear register. It is a read/write register and all bits are cleared to 0 on reset.

On a read, this register gives the current value of the mask on the relevant interrupt. A write of 1 to the particular bit sets the mask, enabling the interrupt to be read. A write of 0 clears the corresponding mask.

SSI Interrupt Mask (SSIIM)

SSI0 base: 0x4000.8000
Offset 0x014
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:4	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
3	TXIM	R/W	0	SSI Transmit FIFO Interrupt Mask The `TXIM` values are defined as follows: Value Description 0 TX FIFO half-full or less condition interrupt is masked. 1 TX FIFO half-full or less condition interrupt is not masked.
2	RXIM	R/W	0	SSI Receive FIFO Interrupt Mask The `RXIM` values are defined as follows: Value Description 0 RX FIFO half-full or more condition interrupt is masked. 1 RX FIFO half-full or more condition interrupt is not masked.
1	RTIM	R/W	0	SSI Receive Time-Out Interrupt Mask The `RTIM` values are defined as follows: Value Description 0 RX FIFO time-out interrupt is masked. 1 RX FIFO time-out interrupt is not masked.

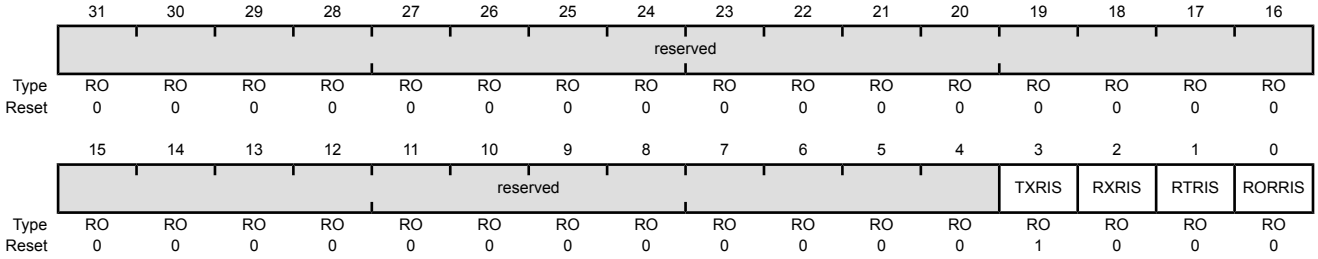
Bit/Field	Name	Type	Reset	Description
0	RORIM	R/W	0	SSI Receive Overrun Interrupt Mask The RORIM values are defined as follows: Value Description 0 RX FIFO overrun interrupt is masked. 1 RX FIFO overrun interrupt is not masked.

Register 7: SSI Raw Interrupt Status (SSIRIS), offset 0x018

The **SSIRIS** register is the raw interrupt status register. On a read, this register gives the current raw status value of the corresponding interrupt prior to masking. A write has no effect.

SSI Raw Interrupt Status (SSIRIS)

SSI0 base: 0x4000.8000
Offset 0x018
Type RO, reset 0x0000.0008



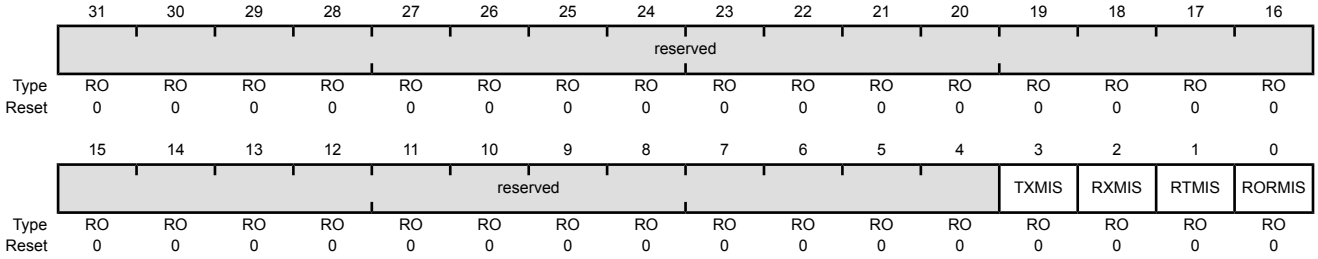
Bit/Field	Name	Type	Reset	Description
31:4	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
3	TXRIS	RO	1	SSI Transmit FIFO Raw Interrupt Status Indicates that the transmit FIFO is half full or less, when set.
2	RXRIS	RO	0	SSI Receive FIFO Raw Interrupt Status Indicates that the receive FIFO is half full or more, when set.
1	RTRIS	RO	0	SSI Receive Time-Out Raw Interrupt Status Indicates that the receive time-out has occurred, when set.
0	RORRIS	RO	0	SSI Receive Overrun Raw Interrupt Status Indicates that the receive FIFO has overflowed, when set.

Register 8: SSI Masked Interrupt Status (SSIMIS), offset 0x01C

The **SSIMIS** register is the masked interrupt status register. On a read, this register gives the current masked status value of the corresponding interrupt. A write has no effect.

SSI Masked Interrupt Status (SSIMIS)

SSI0 base: 0x4000.8000
Offset 0x01C
Type RO, reset 0x0000.0000



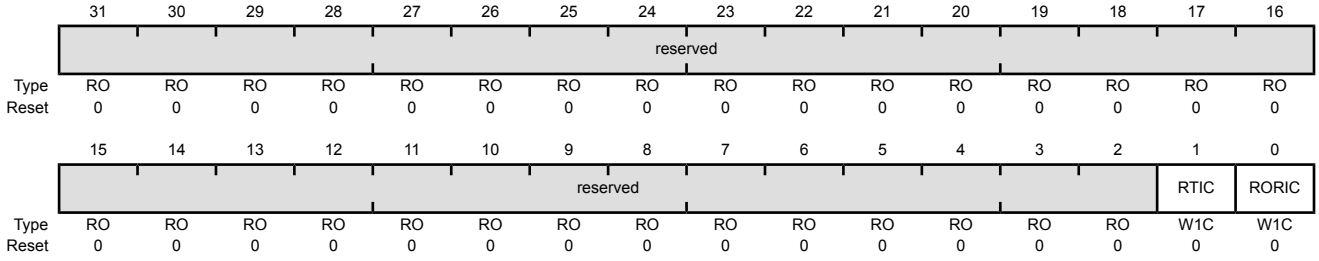
Bit/Field	Name	Type	Reset	Description
31:4	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
3	TXMIS	RO	0	SSI Transmit FIFO Masked Interrupt Status Indicates that the transmit FIFO is half full or less, when set.
2	RXMIS	RO	0	SSI Receive FIFO Masked Interrupt Status Indicates that the receive FIFO is half full or more, when set.
1	RTMIS	RO	0	SSI Receive Time-Out Masked Interrupt Status Indicates that the receive time-out has occurred, when set.
0	RORMIS	RO	0	SSI Receive Overrun Masked Interrupt Status Indicates that the receive FIFO has overflowed, when set.

Register 9: SSI Interrupt Clear (SSIICR), offset 0x020

The **SSIICR** register is the interrupt clear register. On a write of 1, the corresponding interrupt is cleared. A write of 0 has no effect.

SSI Interrupt Clear (SSIICR)

SSI0 base: 0x4000.8000
Offset 0x020
Type W1C, reset 0x0000.0000



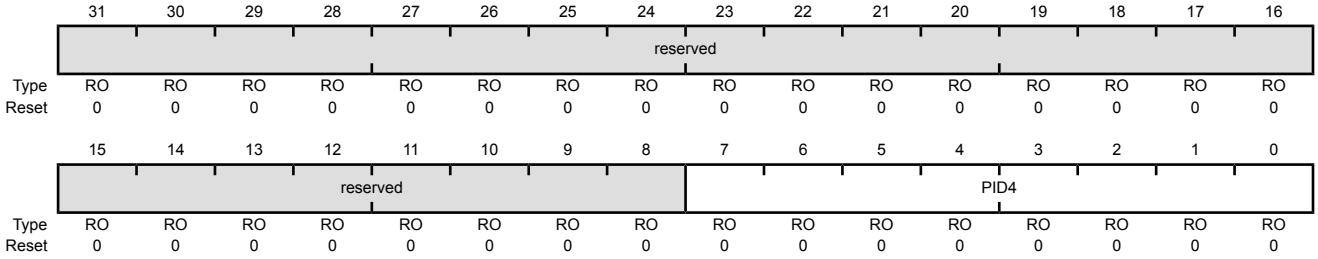
Bit/Field	Name	Type	Reset	Description
31:2	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1	RTIC	W1C	0	SSI Receive Time-Out Interrupt Clear The `RTIC` values are defined as follows: Value Description 0 No effect on interrupt. 1 Clears interrupt.
0	RORIC	W1C	0	SSI Receive Overrun Interrupt Clear The `RORIC` values are defined as follows: Value Description 0 No effect on interrupt. 1 Clears interrupt.

Register 10: SSI Peripheral Identification 4 (SSIPeriphID4), offset 0xFD0

The **SSIPeriphIDn** registers are hard-coded and the fields within the register determine the reset value.

SSI Peripheral Identification 4 (SSIPeriphID4)

SSI0 base: 0x4000.8000
Offset 0xFD0
Type RO, reset 0x0000.0000



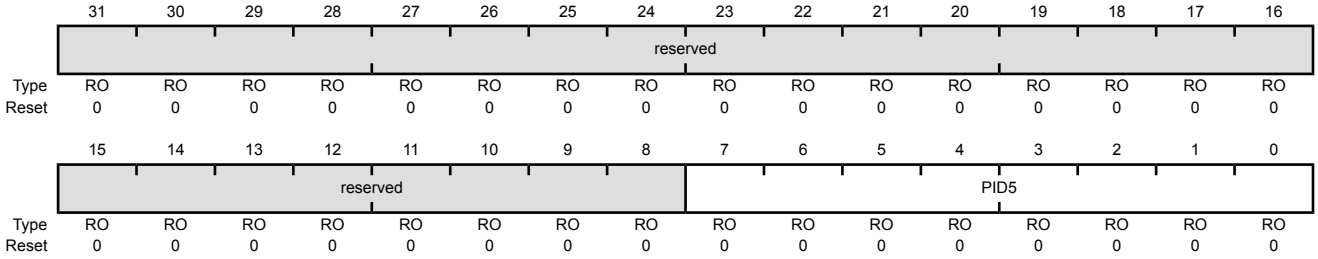
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID4	RO	0x00	SSI Peripheral ID Register[7:0] Can be used by software to identify the presence of this peripheral.

Register 11: SSI Peripheral Identification 5 (SSIPeriphID5), offset 0xFD4

The **SSIPeriphIDn** registers are hard-coded and the fields within the register determine the reset value.

SSI Peripheral Identification 5 (SSIPeriphID5)

SSI0 base: 0x4000.8000
Offset 0xFD4
Type RO, reset 0x0000.0000



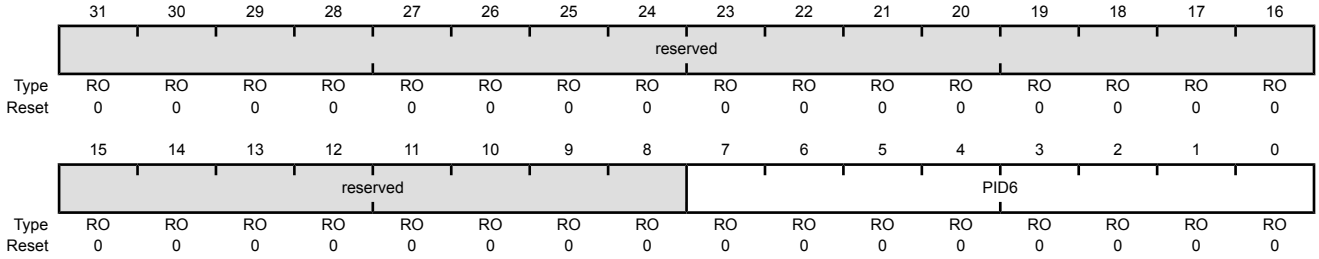
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID5	RO	0x00	SSI Peripheral ID Register[15:8] Can be used by software to identify the presence of this peripheral.

Register 12: SSI Peripheral Identification 6 (SSIPeriphID6), offset 0xFD8

The **SSIPeriphIDn** registers are hard-coded and the fields within the register determine the reset value.

SSI Peripheral Identification 6 (SSIPeriphID6)

SSI0 base: 0x4000.8000
Offset 0xFD8
Type RO, reset 0x0000.0000



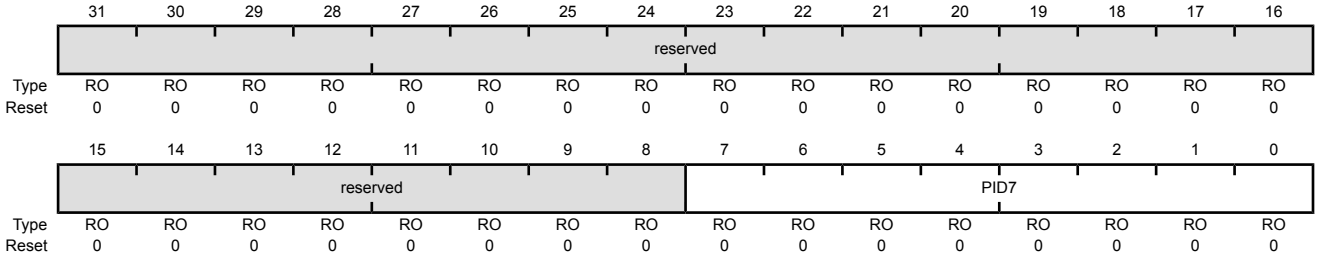
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID6	RO	0x00	SSI Peripheral ID Register[23:16] Can be used by software to identify the presence of this peripheral.

Register 13: SSI Peripheral Identification 7 (SSIPeriphID7), offset 0xFDC

The **SSIPeriphIDn** registers are hard-coded and the fields within the register determine the reset value.

SSI Peripheral Identification 7 (SSIPeriphID7)

SSI0 base: 0x4000.8000
Offset 0xFDC
Type RO, reset 0x0000.0000



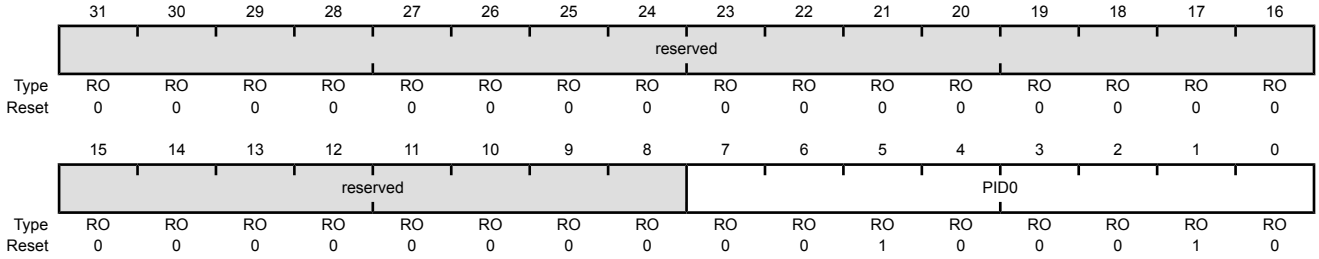
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID7	RO	0x00	SSI Peripheral ID Register[31:24] Can be used by software to identify the presence of this peripheral.

Register 14: SSI Peripheral Identification 0 (SSIPeriphID0), offset 0xFE0

The **SSIPeriphIDn** registers are hard-coded and the fields within the register determine the reset value.

SSI Peripheral Identification 0 (SSIPeriphID0)

SSI0 base: 0x4000.8000
Offset 0xFE0
Type RO, reset 0x0000.0022



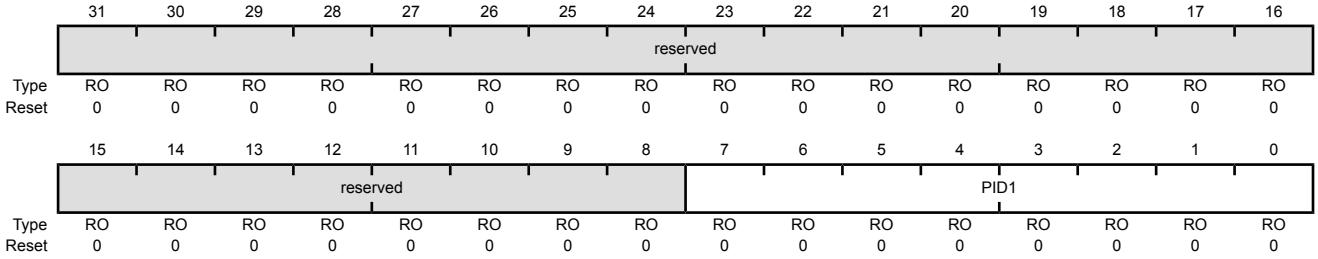
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID0	RO	0x22	SSI Peripheral ID Register[7:0] Can be used by software to identify the presence of this peripheral.

Register 15: SSI Peripheral Identification 1 (SSIPeriphID1), offset 0xFE4

The **SSIPeriphIDn** registers are hard-coded and the fields within the register determine the reset value.

SSI Peripheral Identification 1 (SSIPeriphID1)

SSI0 base: 0x4000.8000
Offset 0xFE4
Type RO, reset 0x0000.0000



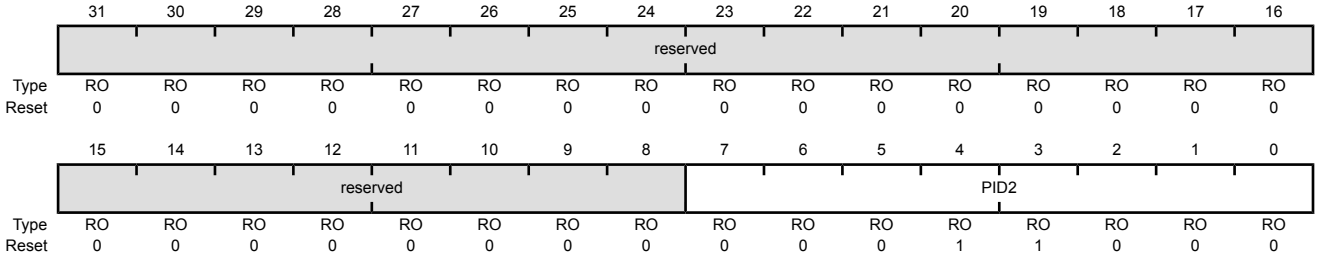
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID1	RO	0x00	SSI Peripheral ID Register [15:8] Can be used by software to identify the presence of this peripheral.

Register 16: SSI Peripheral Identification 2 (SSIPeriphID2), offset 0xFE8

The **SSIPeriphIDn** registers are hard-coded and the fields within the register determine the reset value.

SSI Peripheral Identification 2 (SSIPeriphID2)

SSI0 base: 0x4000.8000
Offset 0xFE8
Type RO, reset 0x0000.0018



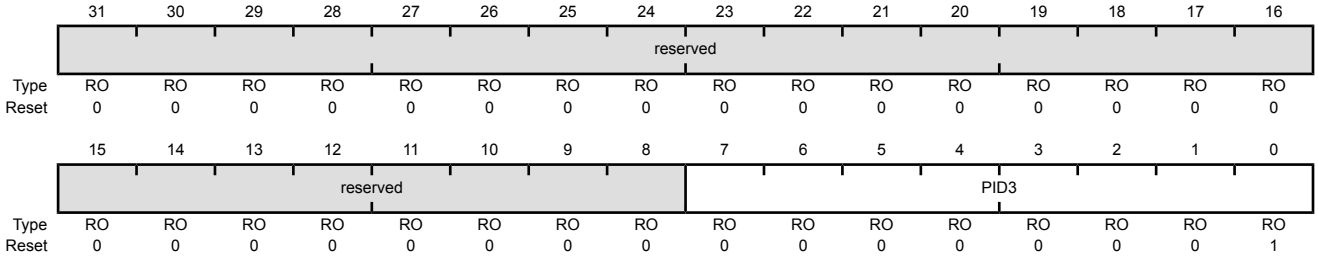
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID2	RO	0x18	SSI Peripheral ID Register [23:16] Can be used by software to identify the presence of this peripheral.

Register 17: SSI Peripheral Identification 3 (SSIPeriphID3), offset 0xFEC

The **SSIPeriphIDn** registers are hard-coded and the fields within the register determine the reset value.

SSI Peripheral Identification 3 (SSIPeriphID3)

SSI0 base: 0x4000.8000
Offset 0xFEC
Type RO, reset 0x0000.0001



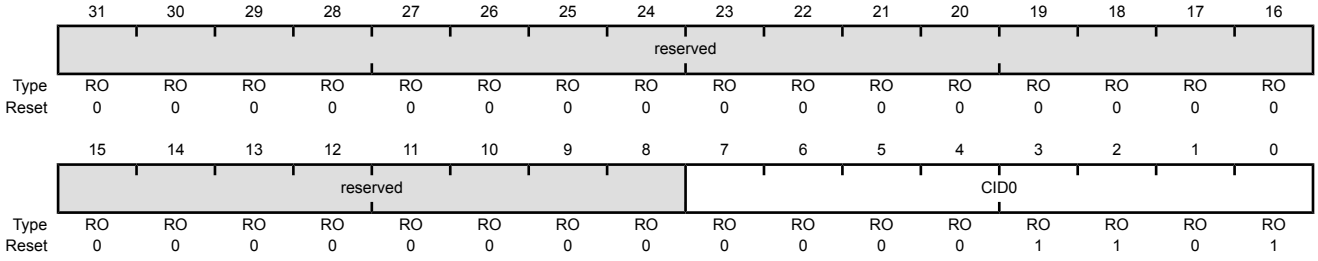
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	PID3	RO	0x01	SSI Peripheral ID Register [31:24] Can be used by software to identify the presence of this peripheral.

Register 18: SSI PrimeCell Identification 0 (SSIPCellID0), offset 0xFF0

The **SSIPCellIDn** registers are hard-coded, and the fields within the register determine the reset value.

SSI PrimeCell Identification 0 (SSIPCellID0)

SSI0 base: 0x4000.8000
Offset 0xFF0
Type RO, reset 0x0000.000D



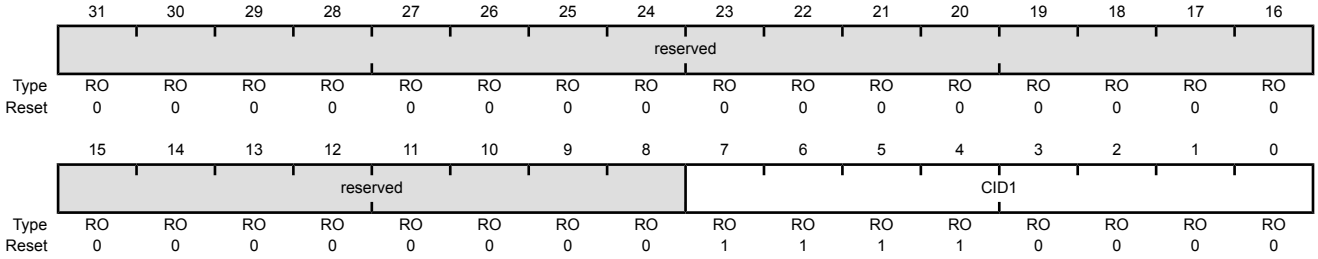
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	CID0	RO	0x0D	SSI PrimeCell ID Register [7:0] Provides software a standard cross-peripheral identification system.

Register 19: SSI PrimeCell Identification 1 (SSIPCellID1), offset 0xFF4

The **SSIPCellIDn** registers are hard-coded, and the fields within the register determine the reset value.

SSI PrimeCell Identification 1 (SSIPCellID1)

SSI0 base: 0x4000.8000
Offset 0xFF4
Type RO, reset 0x0000.00F0



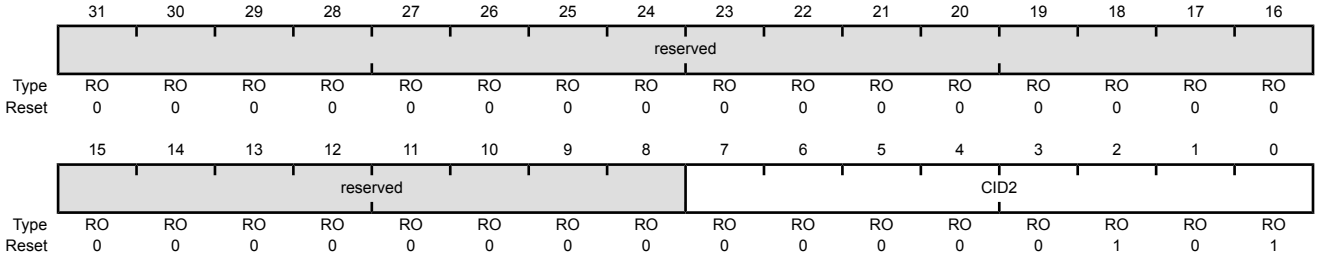
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	CID1	RO	0xF0	SSI PrimeCell ID Register [15:8] Provides software a standard cross-peripheral identification system.

Register 20: SSI PrimeCell Identification 2 (SSIPCellID2), offset 0xFF8

The **SSIPCellIDn** registers are hard-coded, and the fields within the register determine the reset value.

SSI PrimeCell Identification 2 (SSIPCellID2)

SSI0 base: 0x4000.8000
Offset 0xFF8
Type RO, reset 0x0000.0005



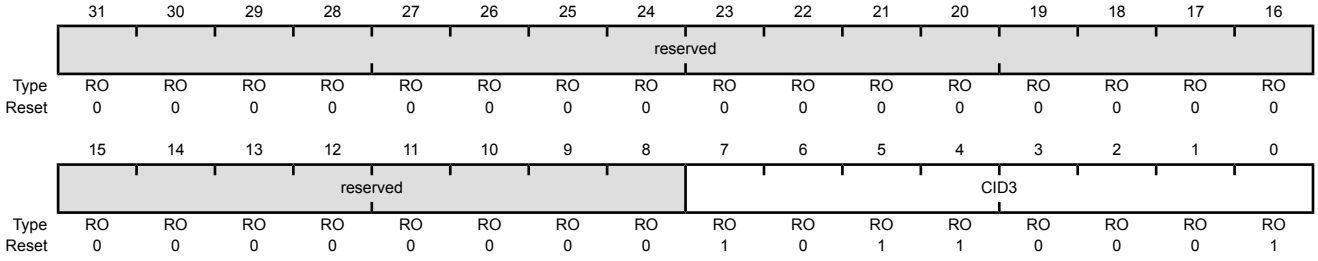
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	CID2	RO	0x05	SSI PrimeCell ID Register [23:16] Provides software a standard cross-peripheral identification system.

Register 21: SSI PrimeCell Identification 3 (SSIPCellID3), offset 0xFFC

The **SSIPCellIDn** registers are hard-coded, and the fields within the register determine the reset value.

SSI PrimeCell Identification 3 (SSIPCellID3)

SSI0 base: 0x4000.8000
Offset 0xFFC
Type RO, reset 0x0000.00B1



Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	CID3	RO	0xB1	SSI PrimeCell ID Register [31:24] Provides software a standard cross-peripheral identification system.

14 Inter-Integrated Circuit (I^2C) Interface

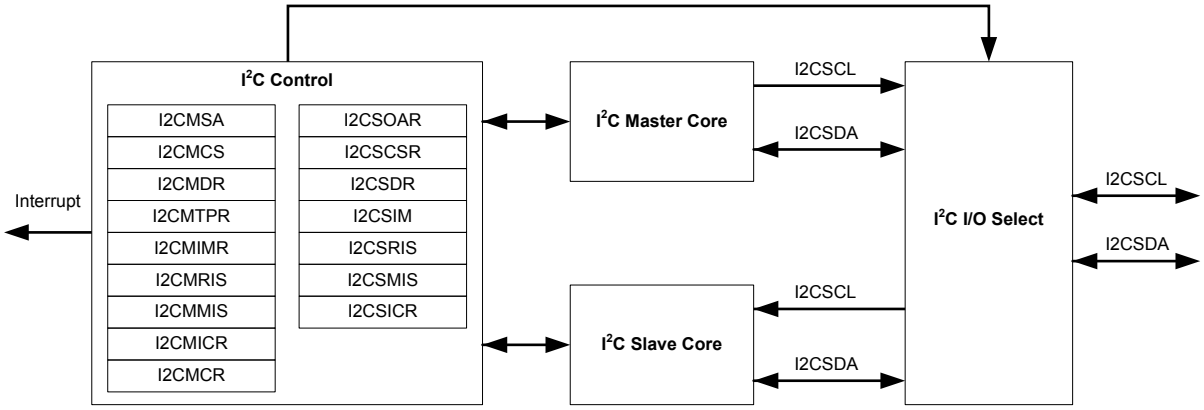
The Inter-Integrated Circuit (I^2C) bus provides bi-directional data transfer through a two-wire design (a serial data line SDA and a serial clock line SCL), and interfaces to external I^2C devices such as serial memory (RAMs and ROMs), networking devices, LCDs, tone generators, and so on. The I^2C bus may also be used for system testing and diagnostic purposes in product development and manufacture. The LM3S6432 microcontroller includes one I^2C module, providing the ability to interact (both send and receive) with other I^2C devices on the bus.

The Stellaris® I^2C interface has the following features:

- Devices on the I^2C bus can be designated as either a master or a slave
  - Supports both sending and receiving data as either a master or a slave
  - Supports simultaneous master and slave operation
- Four I^2C modes
  - Master transmit
  - Master receive
  - Slave transmit
  - Slave receive
- Two transmission speeds: Standard (100 Kbps) and Fast (400 Kbps)
- Master and slave interrupt generation
  - Master generates interrupts when a transmit or receive operation completes (or aborts due to an error)
  - Slave generates interrupts when data has been sent or requested by a master
- Master with arbitration and clock synchronization, multimaster support, and 7-bit addressing mode

14.1 Block Diagram

Figure 14-1. I²C Block Diagram

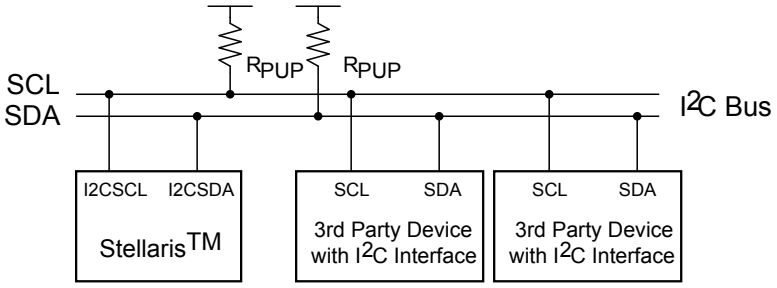


14.2 Functional Description

The I²C module is comprised of both master and slave functions which are implemented as separate peripherals. For proper operation, the SDA and SCL pins must be connected to bi-directional open-drain pads. A typical I²C bus configuration is shown in Figure 14-2 on page 369.

See "Inter-Integrated Circuit (I²C) Interface" on page 538 for I²C timing diagrams.

Figure 14-2. I²C Bus Configuration



14.2.1 I²C Bus Functional Overview

The I²C bus uses only two signals: SDA and SCL, named `I2CSDA` and `I2CSCL` on Stellaris® microcontrollers. SDA is the bi-directional serial data line and SCL is the bi-directional serial clock line. The bus is considered idle when both lines are High.

Every transaction on the I²C bus is nine bits long, consisting of eight data bits and a single acknowledge bit. The number of bytes per transfer (defined as the time between a valid START and STOP condition, described in "START and STOP Conditions" on page 370) is unrestricted, but each byte has to be followed by an acknowledge bit, and data must be transferred MSB first. When a receiver cannot receive another complete byte, it can hold the clock line SCL Low and force the transmitter into a wait state. The data transfer continues when the receiver releases the clock SCL.

14.2.1.1 START and STOP Conditions

The protocol of the I^2C bus defines two states to begin and end a transaction: START and STOP. A High-to-Low transition on the SDA line while the SCL is High is defined as a START condition, and a Low-to-High transition on the SDA line while SCL is High is defined as a STOP condition. The bus is considered busy after a START condition and free after a STOP condition. See Figure 14-3 on page 370.

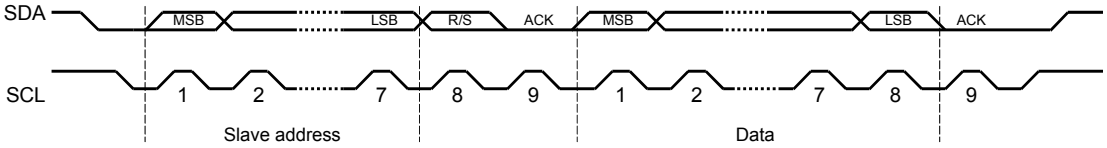
Figure 14-3. START and STOP Conditions



14.2.1.2 Data Format with 7-Bit Address

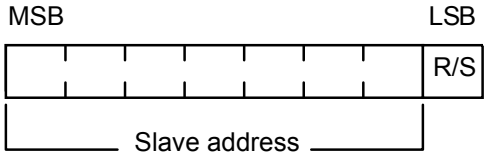
Data transfers follow the format shown in Figure 14-4 on page 370. After the START condition, a slave address is sent. This address is 7-bits long followed by an eighth bit, which is a data direction bit (R/S bit in the **I2CMSA** register). A zero indicates a transmit operation (send), and a one indicates a request for data (receive). A data transfer is always terminated by a STOP condition generated by the master, however, a master can initiate communications with another device on the bus by generating a repeated START condition and addressing another slave without first generating a STOP condition. Various combinations of receive/send formats are then possible within a single transfer.

Figure 14-4. Complete Data Transfer with a 7-Bit Address



The first seven bits of the first byte make up the slave address (see Figure 14-5 on page 370). The eighth bit determines the direction of the message. A zero in the R/S position of the first byte means that the master will write (send) data to the selected slave, and a one in this position means that the master will receive data from the slave.

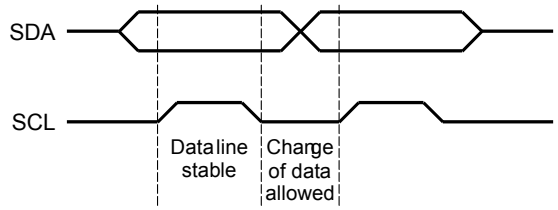
Figure 14-5. R/S Bit in First Byte



14.2.1.3 Data Validity

The data on the SDA line must be stable during the high period of the clock, and the data line can only change when SCL is Low (see Figure 14-6 on page 371).

Figure 14-6. Data Validity During Bit Transfer on the I²C Bus



14.2.1.4 Acknowledge

All bus transactions have a required acknowledge clock cycle that is generated by the master. During the acknowledge cycle, the transmitter (which can be the master or slave) releases the SDA line. To acknowledge the transaction, the receiver must pull down SDA during the acknowledge clock cycle. The data sent out by the receiver during the acknowledge cycle must comply with the data validity requirements described in “Data Validity” on page 370.

When a slave receiver does not acknowledge the slave address, SDA must be left High by the slave so that the master can generate a STOP condition and abort the current transfer. If the master device is acting as a receiver during a transfer, it is responsible for acknowledging each transfer made by the slave. Since the master controls the number of bytes in the transfer, it signals the end of data to the slave transmitter by not generating an acknowledge on the last data byte. The slave transmitter must then release SDA to allow the master to generate the STOP or a repeated START condition.

14.2.1.5 Arbitration

A master may start a transfer only if the bus is idle. It's possible for two or more masters to generate a START condition within minimum hold time of the START condition. In these situations, an arbitration scheme takes place on the SDA line, while SCL is High. During arbitration, the first of the competing master devices to place a '1' (High) on SDA while another master transmits a '0' (Low) will switch off its data output stage and retire until the bus is idle again.

Arbitration can take place over several bits. Its first stage is a comparison of address bits, and if both masters are trying to address the same device, arbitration continues on to the comparison of data bits.

14.2.2 Available Speed Modes

The I²C clock rate is determined by the parameters: `CLK_PRD`, `TIMER_PRD`, `SCL_LP`, and `SCL_HP`.

where:

`CLK_PRD` is the system clock period

`SCL_LP` is the low phase of SCL (fixed at 6)

`SCL_HP` is the high phase of SCL (fixed at 4)

`TIMER_PRD` is the programmed value in the **I²C Master Timer Period (I2CMTPR)** register (see page 389).

The I²C clock period is calculated as follows:

```
SCL_PERIOD = 2*(1 + TIMER_PRD)*(SCL_LP + SCL_HP)*CLK_PRD
```

For example:

```
CLK_PRD = 50 ns
TIMER_PRD = 2
SCL_LP=6
SCL_HP=4
```

yields a SCL frequency of:

```
1/T = 333 Khz
```

Table 14-1 on page 372 gives examples of timer period, system clock, and speed mode (Standard or Fast).

Table 14-1. Examples of I2C Master Timer Period versus Speed Mode

System Clock	Timer Period	Standard Mode	Timer Period	Fast Mode
4 MHz	0x01	100 Kbps	-	-
6 MHz	0x02	100 Kbps	-	-
12.5 MHz	0x06	89 Kbps	0x01	312 Kbps
16.7 MHz	0x08	93 Kbps	0x02	278 Kbps
20 MHz	0x09	100 Kbps	0x02	333 Kbps
25 MHz	0x0C	96.2 Kbps	0x03	312 Kbps
33 MHz	0x10	97.1 Kbps	0x04	330 Kbps
40 MHz	0x13	100 Kbps	0x04	400 Kbps
50 MHz	0x18	100 Kbps	0x06	357 Kbps

14.2.3 Interrupts

The I2C can generate interrupts when the following conditions are observed:

- Master transaction completed
- Master transaction error
- Slave transaction received
- Slave transaction requested

There is a separate interrupt signal for the I2C master and I2C slave modules. While both modules can generate interrupts for multiple conditions, only a single interrupt signal is sent to the interrupt controller.

14.2.3.1 I2C Master Interrupts

The I2C master module generates an interrupt when a transaction completes (either transmit or receive), or when an error occurs during a transaction. To enable the I2C master interrupt, software must write a '1' to the **I2C Master Interrupt Mask (I2CMIMR)** register. When an interrupt condition is met, software must check the ERROR bit in the **I2C Master Control/Status (I2CMCS)** register to verify that an error didn't occur during the last transaction. An error condition is asserted if the last transaction wasn't acknowledge by the slave or if the master was forced to give up ownership of the bus due to a lost arbitration round with another master. If an error is not detected, the application can proceed with the transfer. The interrupt is cleared by writing a '1' to the **I2C Master Interrupt Clear (I2CMICR)** register.

If the application doesn't require the use of interrupts, the raw interrupt status is always visible via the **I²C Master Raw Interrupt Status (I2CMRIS)** register.

14.2.3.2 I²C Slave Interrupts

The slave module can generate an interrupt when data has been received or requested. This interrupt is enabled by writing a 1 to the DATAIM bit in the **I²C Slave Interrupt Mask (I2CSIMR)** register. Software determines whether the module should write (transmit) or read (receive) data from the **I²C Slave Data (I2CSDR)** register, by checking the RREQ and TREQ bits of the **I²C Slave Control/Status (I2CSCSR)** register. If the slave module is in receive mode and the first byte of a transfer is received, the FBR bit is set along with the RREQ bit. The interrupt is cleared by writing a 1 to the DATAIC bit in the **I²C Slave Interrupt Clear (I2CSICR)** register.

If the application doesn't require the use of interrupts, the raw interrupt status is always visible via the **I²C Slave Raw Interrupt Status (I2CSRIS)** register.

14.2.4 Loopback Operation

The I²C modules can be placed into an internal loopback mode for diagnostic or debug work. This is accomplished by setting the LPBK bit in the **I²C Master Configuration (I2CMCR)** register. In loopback mode, the SDA and SCL signals from the master and slave modules are tied together.

14.2.5 Command Sequence Flow Charts

This section details the steps required to perform the various I²C transfer types in both master and slave mode.

14.2.5.1 I²C Master Command Sequences

The figures that follow show the command sequences available for the I²C master.

Figure 14-7. Master Single SEND

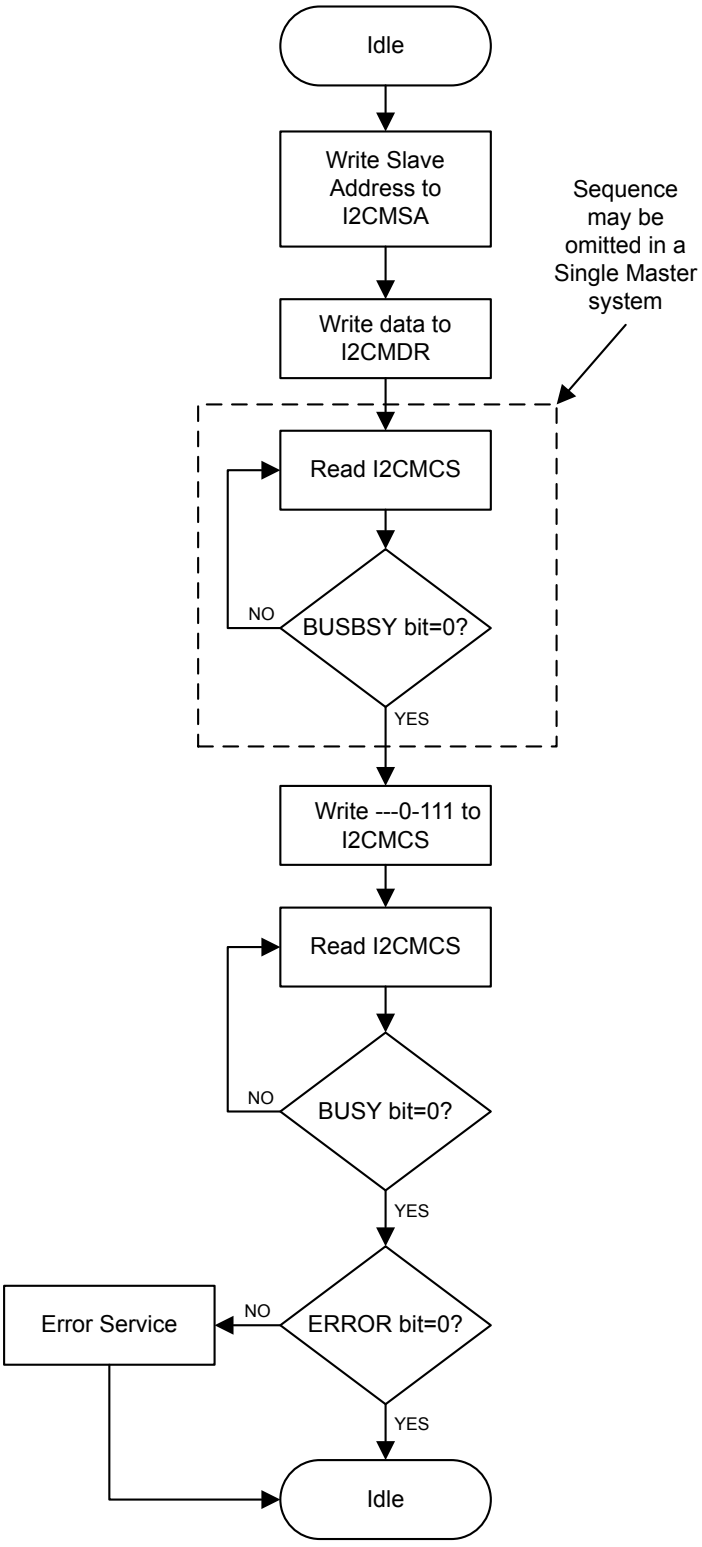


Figure 14-8. Master Single RECEIVE

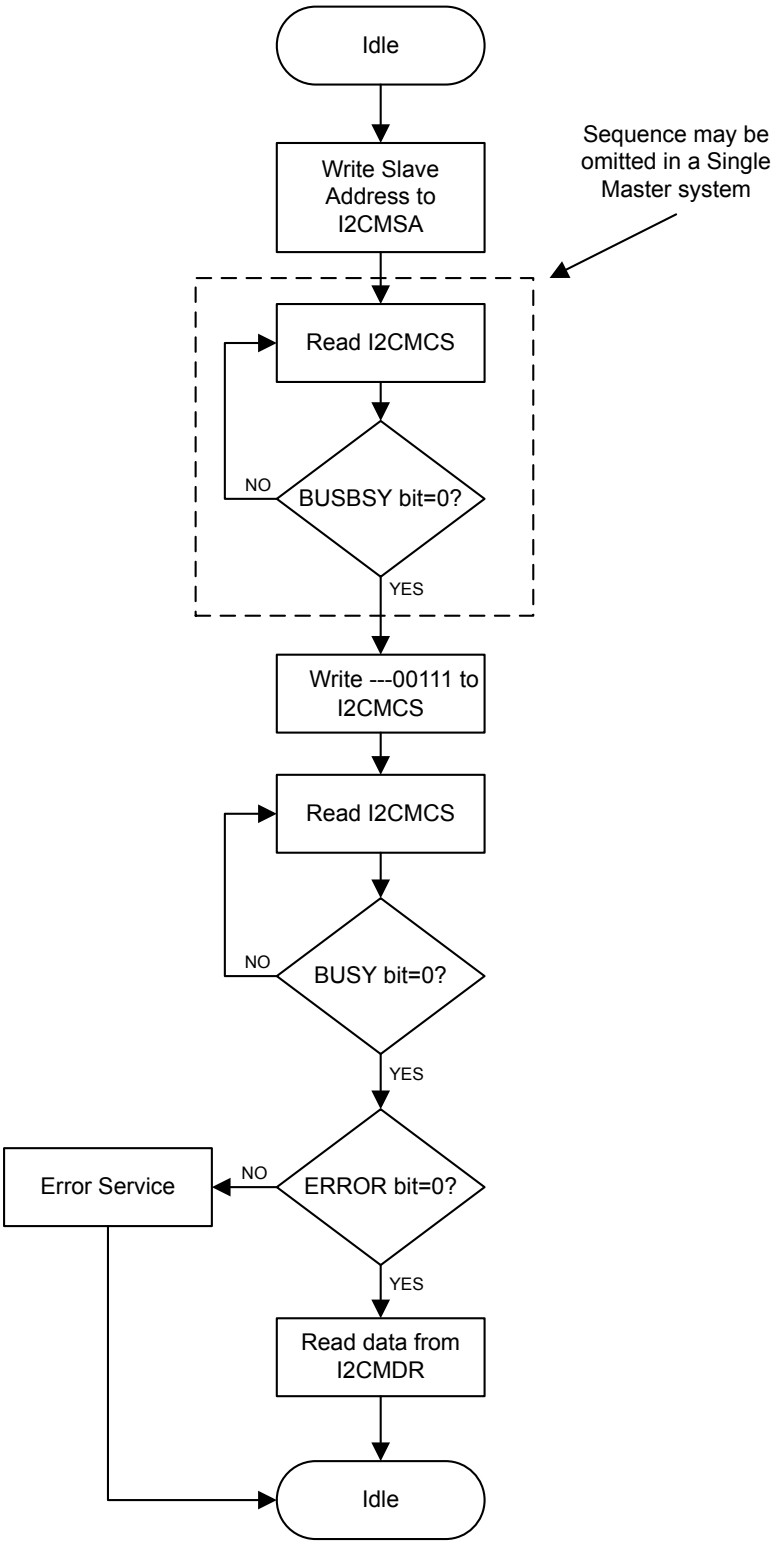


Figure 14-9. Master Burst SEND

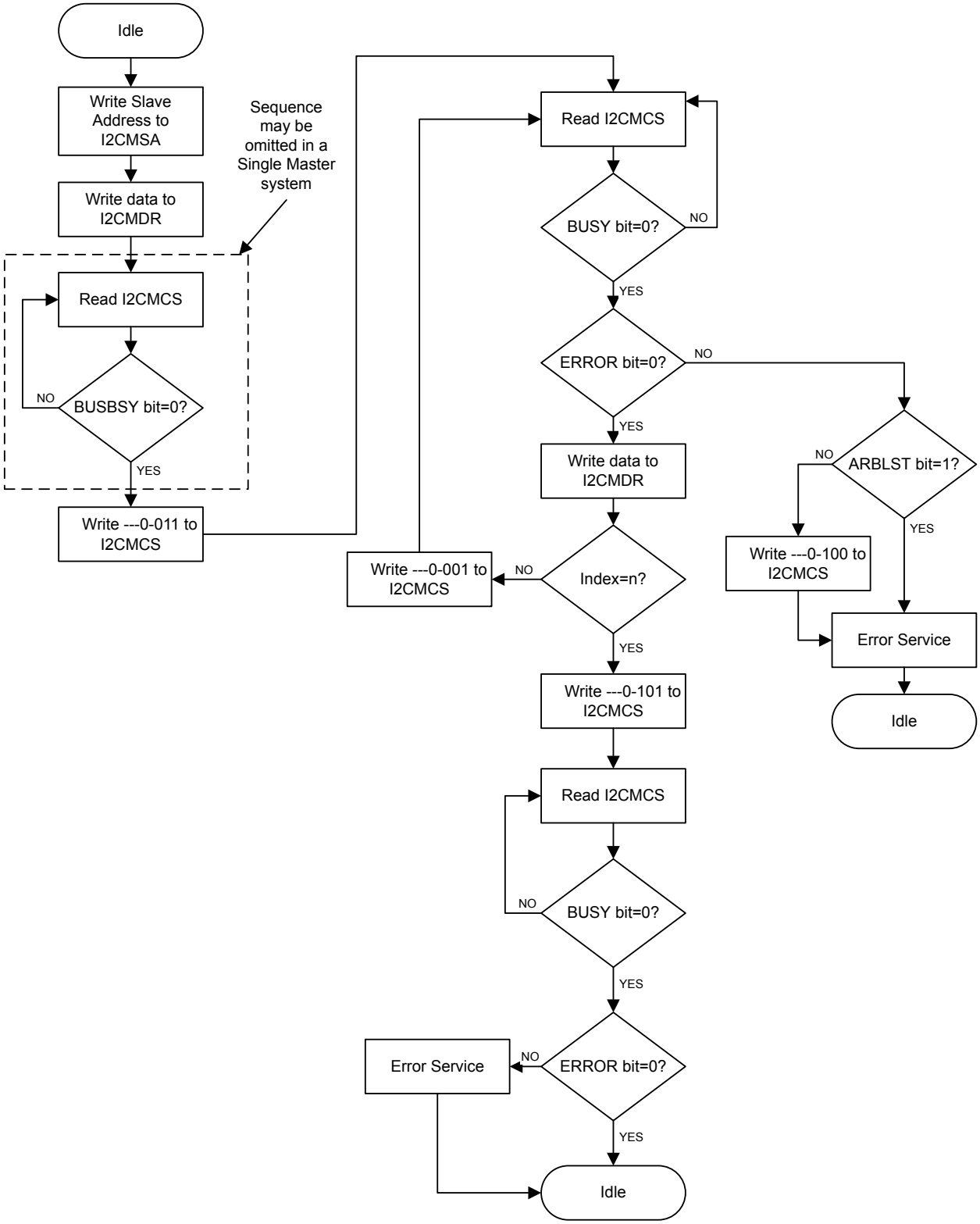


Figure 14-10. Master Burst RECEIVE

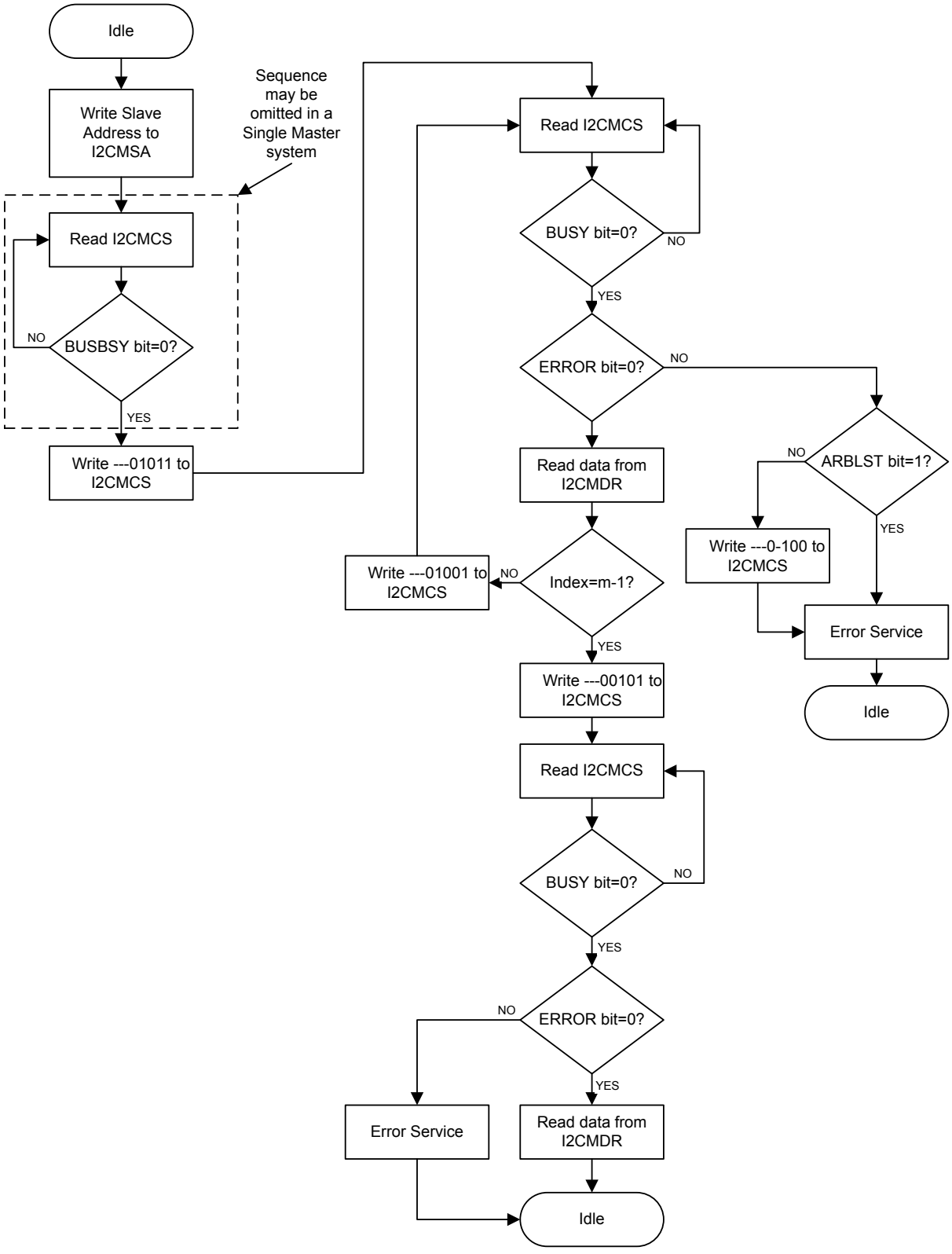


Figure 14-11. Master Burst RECEIVE after Burst SEND

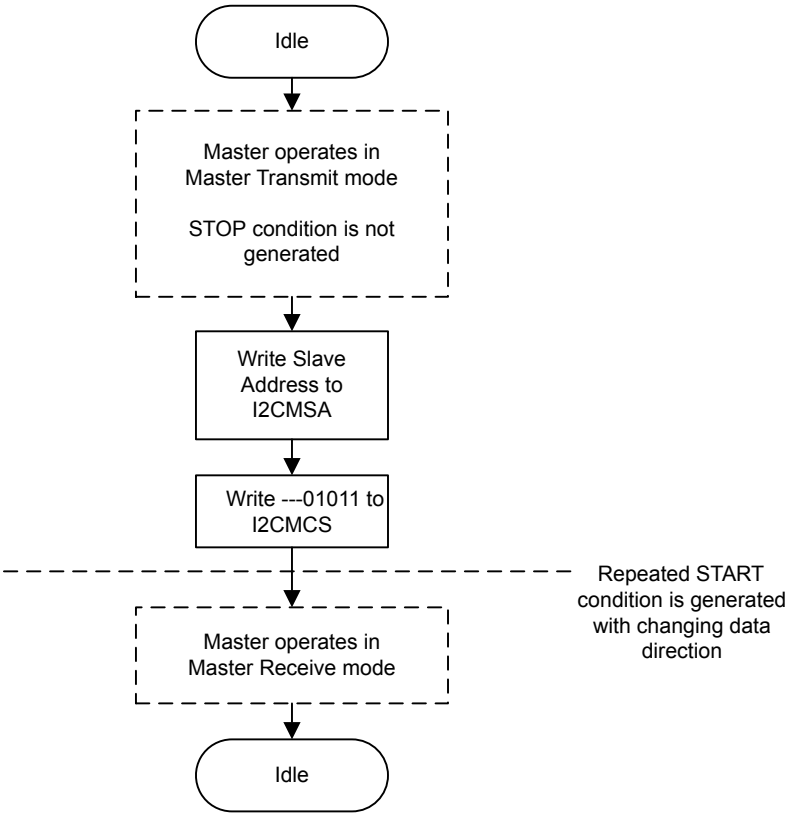
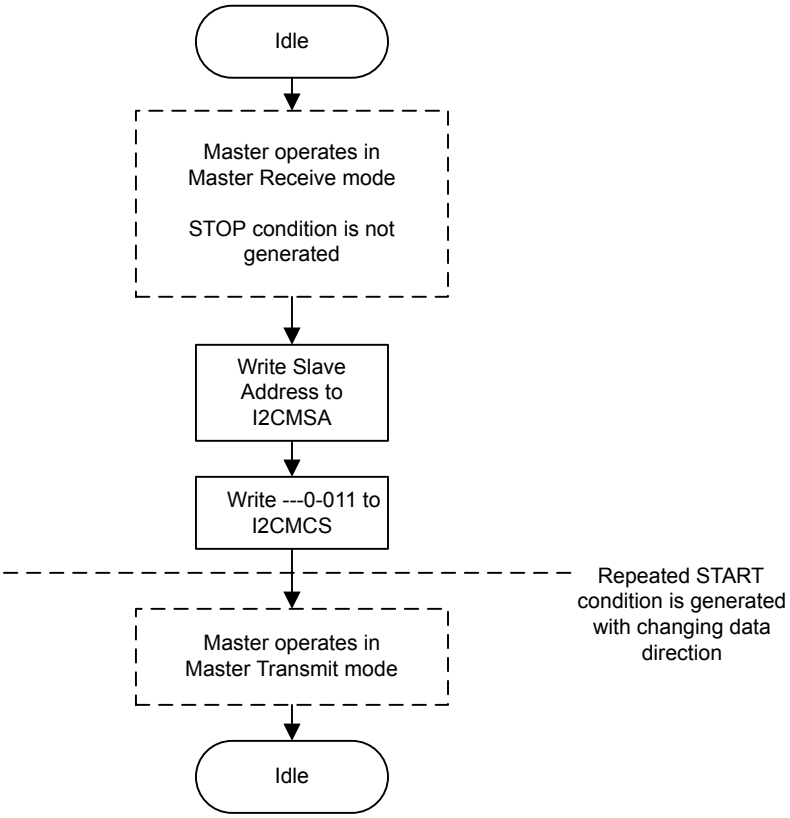


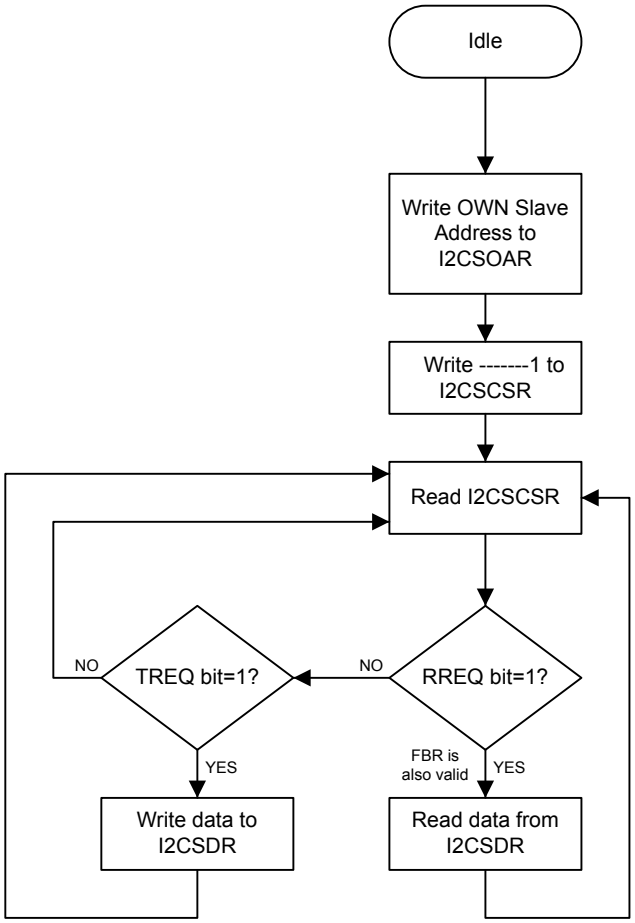
Figure 14-12. Master Burst SEND after Burst RECEIVE



14.2.5.2 I²C Slave Command Sequences

Figure 14-13 on page 380 presents the command sequence available for the I²C slave.

Figure 14-13. Slave Command Sequence



14.3 Initialization and Configuration

The following example shows how to configure the I2C module to send a single byte as a master. This assumes the system clock is 20 MHz.

1. Enable the I2C clock by writing a value of 0x0000.1000 to the **RCGC1** register in the System Control module.
2. Enable the clock to the appropriate GPIO module via the **RCGC2** register in the System Control module.
3. In the GPIO module, enable the appropriate pins for their alternate function using the **GPIOAFSEL** register. Also, be sure to enable the same pins for Open Drain operation.
4. Initialize the I2C Master by writing the **I2CMCR** register with a value of 0x0000.0020.
5. Set the desired SCL clock speed of 100 Kbps by writing the **I2CMTPR** register with the correct value. The value written to the **I2CMTPR** register represents the number of system clock periods in one SCL clock period. The TPR value is determined by the following equation:

```
TPR = (System Clock / (2 * (SCL_LP + SCL_HP) * SCL_CLK)) - 1;
TPR = (20MHz / (2 * (6 + 4) * 100000)) - 1;
TPR = 9
```

Write the **I2CMTPR** register with the value of 0x0000.0009.

6. Specify the slave address of the master and that the next operation will be a Send by writing the **I2CMSA** register with a value of 0x0000.0076. This sets the slave address to 0x3B.
7. Place data (byte) to be sent in the data register by writing the **I2CMDR** register with the desired data.
8. Initiate a single byte send of the data from Master to Slave by writing the **I2CMCS** register with a value of 0x0000.0007 (STOP, START, RUN).
9. Wait until the transmission completes by polling the **I2CMCS** register's BUSBSY bit until it has been cleared.

14.4 Register Map

Table 14-2 on page 381 lists the I^2C registers. All addresses given are relative to the I^2C base addresses for the master and slave:

- I^2C Master 0: 0x4002.0000
- I^2C Slave 0: 0x4002.0800

Table 14-2. Inter-Integrated Circuit (I^2C) Interface Register Map

Offset	Name	Type	Reset	Description	See page
I^2C Master					
0x000	I2CMSA	R/W	0x0000.0000	I2C Master Slave Address	383
0x004	I2CMCS	R/W	0x0000.0000	I2C Master Control/Status	384
0x008	I2CMDR	R/W	0x0000.0000	I2C Master Data	388
0x00C	I2CMTPR	R/W	0x0000.0001	I2C Master Timer Period	389
0x010	I2CMIMR	R/W	0x0000.0000	I2C Master Interrupt Mask	390
0x014	I2CMRIS	RO	0x0000.0000	I2C Master Raw Interrupt Status	391
0x018	I2CMMIS	RO	0x0000.0000	I2C Master Masked Interrupt Status	392
0x01C	I2CMICR	WO	0x0000.0000	I2C Master Interrupt Clear	393
0x020	I2CMCR	R/W	0x0000.0000	I2C Master Configuration	394
I^2C Slave					
0x000	I2CSOAR	R/W	0x0000.0000	I2C Slave Own Address	396
0x004	I2CSCSR	RO	0x0000.0000	I2C Slave Control/Status	397
0x008	I2CSDR	R/W	0x0000.0000	I2C Slave Data	399
0x00C	I2CSIMR	R/W	0x0000.0000	I2C Slave Interrupt Mask	400

Table 14-2. Inter-Integrated Circuit (I2C) Interface Register Map *(continued)*

Offset	Name	Type	Reset	Description	See page
0x010	I2CSRIS	RO	0x0000.0000	I2C Slave Raw Interrupt Status	401
0x014	I2CSMIS	RO	0x0000.0000	I2C Slave Masked Interrupt Status	402
0x018	I2CSICR	WO	0x0000.0000	I2C Slave Interrupt Clear	403

14.5 Register Descriptions (I2C Master)

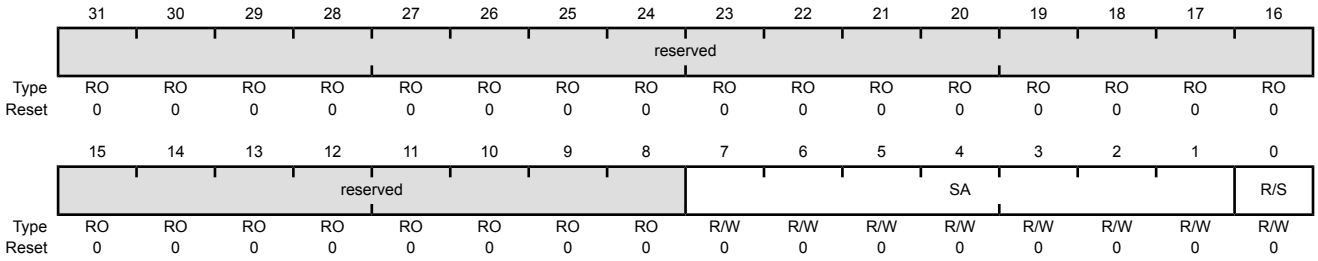
The remainder of this section lists and describes the I2C master registers, in numerical order by address offset. See also “Register Descriptions (I2C Slave)” on page 395.

Register 1: I²C Master Slave Address (I2CMSA), offset 0x000

This register consists of eight bits: seven address bits (A6-A0), and a Receive/Send bit, which determines if the next operation is a Receive (High), or Send (Low).

I2C Master Slave Address (I2CMSA)

I2C Master 0 base: 0x4002.0000
Offset 0x000
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:1	SA	R/W	0	I²C Slave Address This field specifies bits A6 through A0 of the slave address.
0	R/S	R/W	0	Receive/Send The R/S bit specifies if the next operation is a Receive (High) or Send (Low). Value Description 0 Send. 1 Receive.

Register 2: I2C Master Control/Status (I2CMCS), offset 0x004

This register accesses four control bits when written, and accesses seven status bits when read.

The status register consists of seven bits, which when read determine the state of the I2C bus controller.

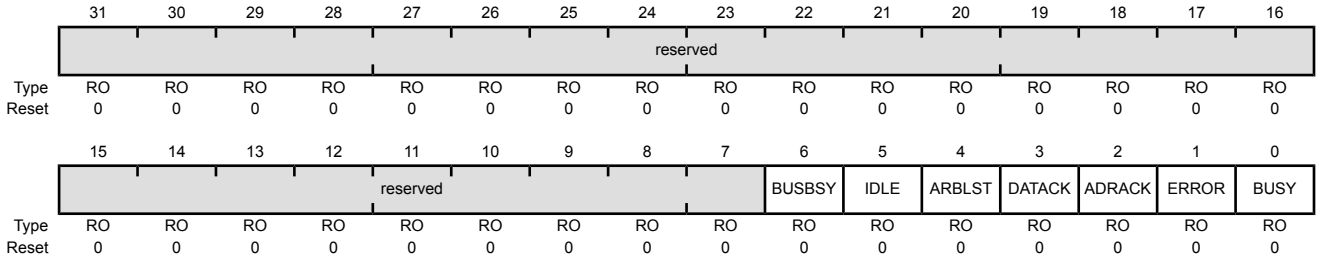
The control register consists of four bits: the RUN, START, STOP, and ACK bits. The START bit causes the generation of the START, or REPEATED START condition.

The STOP bit determines if the cycle stops at the end of the data cycle, or continues on to a burst. To generate a single send cycle, the **I2C Master Slave Address (I2CMSA)** register is written with the desired address, the R/S bit is set to 0, and the Control register is written with ACK=X (0 or 1), STOP=1, START=1, and RUN=1 to perform the operation and stop. When the operation is completed (or aborted due an error), the interrupt pin becomes active and the data may be read from the **I2CMDR** register. When the I2C module operates in Master receiver mode, the ACK bit must be set normally to logic 1. This causes the I2C bus controller to send an acknowledge automatically after each byte. This bit must be reset when the I2C bus controller requires no further data to be sent from the slave transmitter.

Reads

I2C Master Control/Status (I2CMCS)

I2C Master 0 base: 0x4002.0000
Offset 0x004
Type RO, reset 0x0000.0000



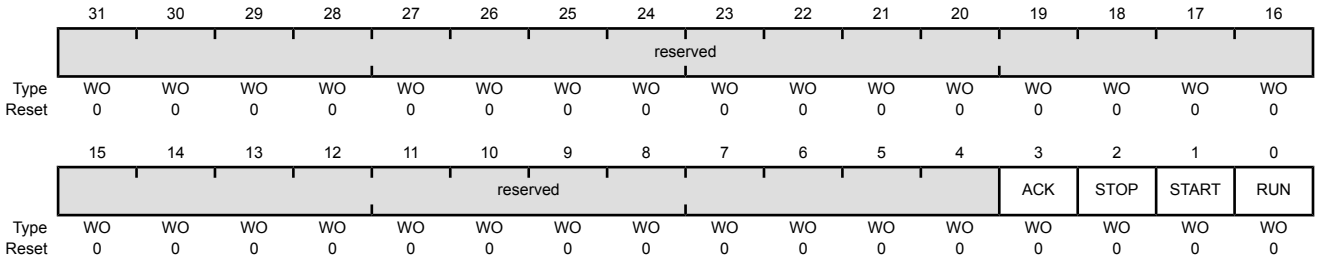
Bit/Field	Name	Type	Reset	Description
31:7	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
6	BUSBSY	RO	0	Bus Busy This bit specifies the state of the I2C bus. If set, the bus is busy; otherwise, the bus is idle. The bit changes based on the START and STOP conditions.
5	IDLE	RO	0	I2C Idle This bit specifies the I2C controller state. If set, the controller is idle; otherwise the controller is not idle.
4	ARBLST	RO	0	Arbitration Lost This bit specifies the result of bus arbitration. If set, the controller lost arbitration; otherwise, the controller won arbitration.

Bit/Field	Name	Type	Reset	Description
3	DATACK	RO	0	Acknowledge Data This bit specifies the result of the last data operation. If set, the transmitted data was not acknowledged; otherwise, the data was acknowledged.
2	ADRACK	RO	0	Acknowledge Address This bit specifies the result of the last address operation. If set, the transmitted address was not acknowledged; otherwise, the address was acknowledged.
1	ERROR	RO	0	Error This bit specifies the result of the last bus operation. If set, an error occurred on the last operation; otherwise, no error was detected. The error can be from the slave address not being acknowledged, the transmit data not being acknowledged, or because the controller lost arbitration.
0	BUSY	RO	0	I^2C Busy This bit specifies the state of the controller. If set, the controller is busy; otherwise, the controller is idle. When the BUSY bit is set, the other status bits are not valid.

Writes

I2C Master Control/Status (I2CMCS)

I2C Master 0 base: 0x4002.0000
Offset 0x004
Type WO, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:4	reserved	WO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
3	ACK	WO	0	Data Acknowledge Enable When set, causes received data byte to be acknowledged automatically by the master. See field decoding in Table 14-3 on page 386.
2	STOP	WO	0	Generate STOP When set, causes the generation of the STOP condition. See field decoding in Table 14-3 on page 386.

Bit/Field	Name	Type	Reset	Description
1	START	WO	0	Generate START When set, causes the generation of a START or repeated START condition. See field decoding in Table 14-3 on page 386.
0	RUN	WO	0	I2C Master Enable When set, allows the master to send or receive data. See field decoding in Table 14-3 on page 386.

Table 14-3. Write Field Decoding for I2CMCS[3:0] Field (Sheet 1 of 3)

Current State	I2CMSA[0]	I2CMCS[3:0]				Description
	R/S	ACK	STOP	START	RUN	
Idle	0	X[a]	0	1	1	START condition followed by SEND (master goes to the Master Transmit state).
	0	X	1	1	1	START condition followed by a SEND and STOP condition (master remains in Idle state).
	1	0	0	1	1	START condition followed by RECEIVE operation with negative ACK (master goes to the Master Receive state).
	1	0	1	1	1	START condition followed by RECEIVE and STOP condition (master remains in Idle state).
	1	1	0	1	1	START condition followed by RECEIVE (master goes to the Master Receive state).
	1	1	1	1	1	Illegal.
	All other combinations not listed are non-operations.					NOP.
Master Transmit	X	X	0	0	1	SEND operation (master remains in Master Transmit state).
	X	X	1	0	0	STOP condition (master goes to Idle state).
	X	X	1	0	1	SEND followed by STOP condition (master goes to Idle state).
	0	X	0	1	1	Repeated START condition followed by a SEND (master remains in Master Transmit state).
	0	X	1	1	1	Repeated START condition followed by SEND and STOP condition (master goes to Idle state).
	1	0	0	1	1	Repeated START condition followed by a RECEIVE operation with a negative ACK (master goes to Master Receive state).
	1	0	1	1	1	Repeated START condition followed by a SEND and STOP condition (master goes to Idle state).
	1	1	0	1	1	Repeated START condition followed by RECEIVE (master goes to Master Receive state).
	1	1	1	1	1	Illegal.
	All other combinations not listed are non-operations.					NOP.

Table 14-3. Write Field Decoding for I2CMCS[3:0] Field (Sheet 1 of 3) *(continued)*

Current State	I2CMSA[0]	I2CMCS[3:0]				Description
	R/S	ACK	STOP	START	RUN	
Master Receive	X	0	0	0	1	RECEIVE operation with negative ACK (master remains in Master Receive state).
	X	X	1	0	0	STOP condition (master goes to Idle state).[b]
	X	0	1	0	1	RECEIVE followed by STOP condition (master goes to Idle state).
	X	1	0	0	1	RECEIVE operation (master remains in Master Receive state).
	X	1	1	0	1	Illegal.
	1	0	0	1	1	Repeated START condition followed by RECEIVE operation with a negative ACK (master remains in Master Receive state).
	1	0	1	1	1	Repeated START condition followed by RECEIVE and STOP condition (master goes to Idle state).
	1	1	0	1	1	Repeated START condition followed by RECEIVE (master remains in Master Receive state).
	0	X	0	1	1	Repeated START condition followed by SEND (master goes to Master Transmit state).
	0	X	1	1	1	Repeated START condition followed by SEND and STOP condition (master goes to Idle state).
	All other combinations not listed are non-operations.					NOP.

a. An X in a table cell indicates the bit can be 0 or 1.

b. In Master Receive mode, a STOP condition should be generated only after a Data Negative Acknowledge executed by the master or an Address Negative Acknowledge executed by the slave.

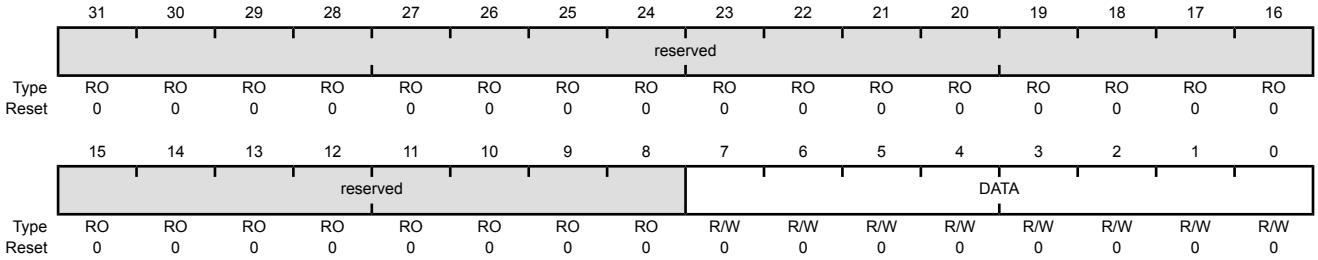
Register 3: I2C Master Data (I2CMDR), offset 0x008

Important: Use caution when reading this register. Performing a read may change bit status.

This register contains the data to be transmitted when in the Master Transmit state, and the data received when in the Master Receive state.

I2C Master Data (I2CMDR)

I2C Master 0 base: 0x4002.0000
Offset 0x008
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	DATA	R/W	0x00	Data Transferred Data transferred during transaction.

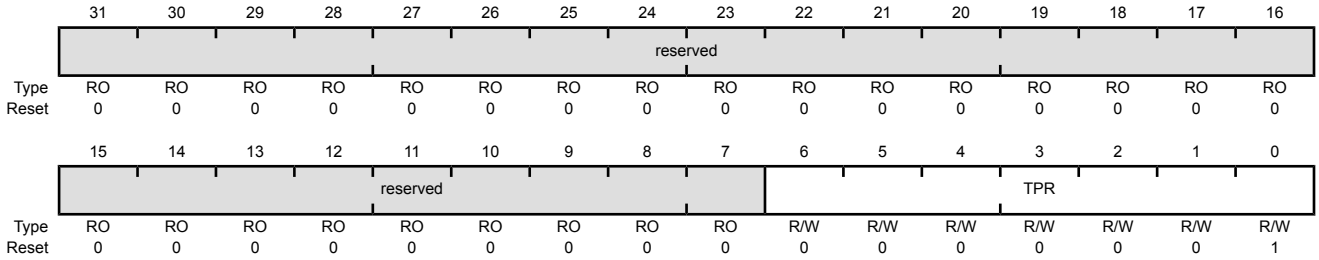
Register 4: I²C Master Timer Period (I2CMTPR), offset 0x00C

This register specifies the period of the SCL clock.

Caution – Take care not to set bit 7 when accessing this register as unpredictable behavior can occur.

I2C Master Timer Period (I2CMTPR)

I2C Master 0 base: 0x4002.0000
Offset 0x00C
Type R/W, reset 0x0000.0001



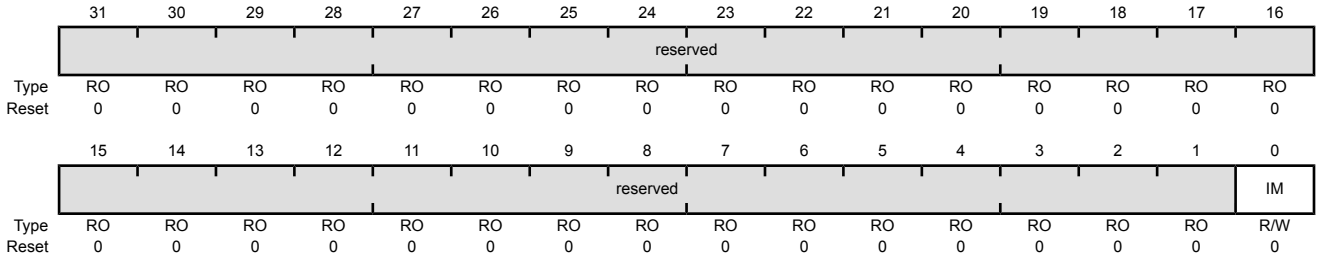
Bit/Field	Name	Type	Reset	Description
31:7	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
6:0	TPR	R/W	0x1	SCL Clock Period This field specifies the period of the SCL clock. `SCL_PRD = 2*(1 + TPR)*(SCL_LP + SCL_HP)*CLK_PRD` where: SCL_PRD is the SCL line period (I²C clock). TPR is the Timer Period register value (range of 1 to 127). SCL_LP is the SCL Low period (fixed at 6). SCL_HP is the SCL High period (fixed at 4).

Register 5: I²C Master Interrupt Mask (I2CMIMR), offset 0x010

This register controls whether a raw interrupt is promoted to a controller interrupt.

I2C Master Interrupt Mask (I2CMIMR)

I2C Master 0 base: 0x4002.0000
Offset 0x010
Type R/W, reset 0x0000.0000



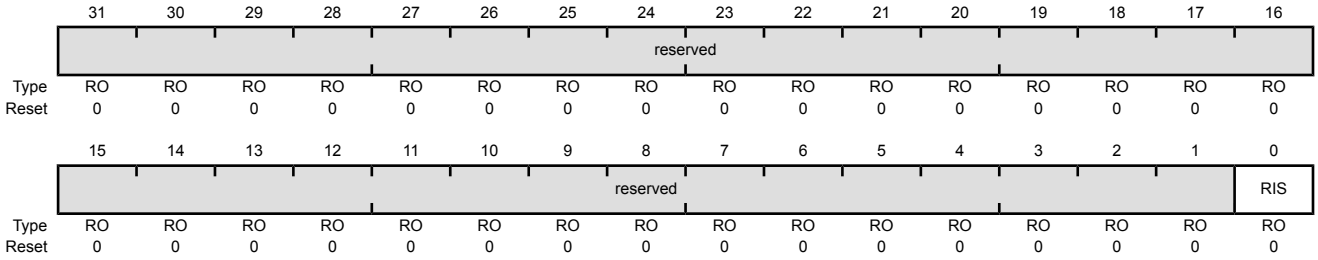
Bit/Field	Name	Type	Reset	Description
31:1	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
0	IM	R/W	0	Interrupt Mask This bit controls whether a raw interrupt is promoted to a controller interrupt. If set, the interrupt is not masked and the interrupt is promoted; otherwise, the interrupt is masked.

Register 6: I²C Master Raw Interrupt Status (I2CMRIS), offset 0x014

This register specifies whether an interrupt is pending.

I2C Master Raw Interrupt Status (I2CMRIS)

I2C Master 0 base: 0x4002.0000
Offset 0x014
Type RO, reset 0x0000.0000



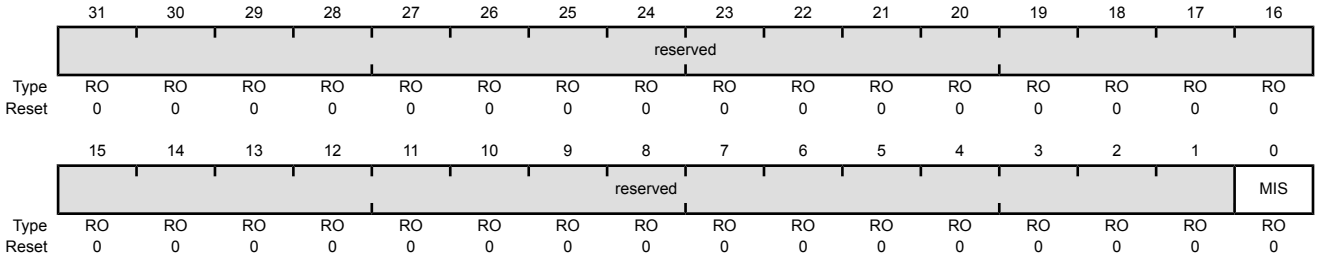
Bit/Field	Name	Type	Reset	Description
31:1	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
0	RIS	RO	0	Raw Interrupt Status This bit specifies the raw interrupt state (prior to masking) of the I²C master block. If set, an interrupt is pending; otherwise, an interrupt is not pending.

Register 7: I²C Master Masked Interrupt Status (I2CMMIS), offset 0x018

This register specifies whether an interrupt was signaled.

I2C Master Masked Interrupt Status (I2CMMIS)

I2C Master 0 base: 0x4002.0000
Offset 0x018
Type RO, reset 0x0000.0000



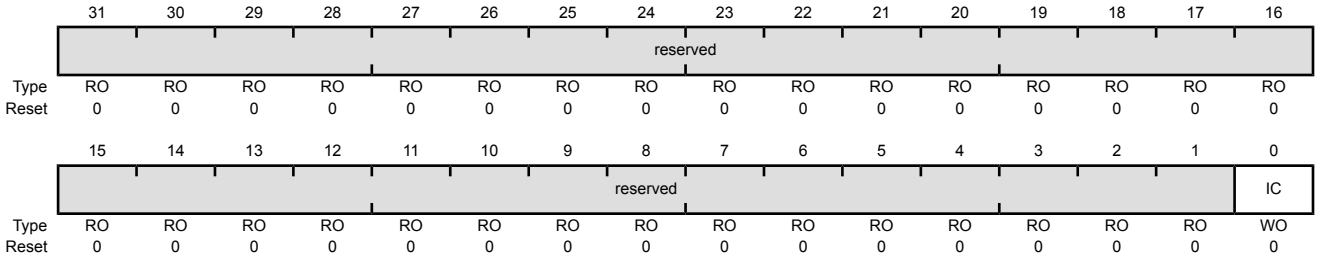
Bit/Field	Name	Type	Reset	Description
31:1	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
0	MIS	RO	0	Masked Interrupt Status This bit specifies the raw interrupt state (after masking) of the I²C master block. If set, an interrupt was signaled; otherwise, an interrupt has not been generated since the bit was last cleared.

Register 8: I²C Master Interrupt Clear (I2CMICR), offset 0x01C

This register clears the raw interrupt.

I2C Master Interrupt Clear (I2CMICR)

I2C Master 0 base: 0x4002.0000
Offset 0x01C
Type WO, reset 0x0000.0000



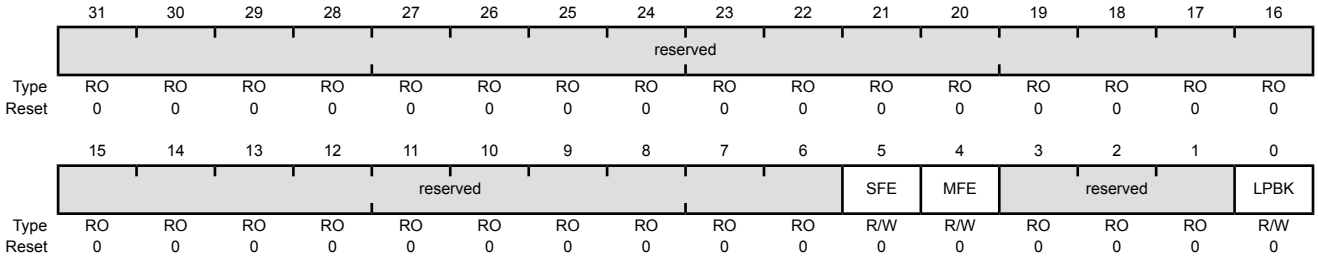
Bit/Field	Name	Type	Reset	Description
31:1	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
0	IC	WO	0	Interrupt Clear This bit controls the clearing of the raw interrupt. A write of 1 clears the interrupt; otherwise, a write of 0 has no affect on the interrupt state. A read of this register returns no meaningful data.

Register 9: I²C Master Configuration (I2CMCR), offset 0x020

This register configures the mode (Master or Slave) and sets the interface for test mode loopback.

I2C Master Configuration (I2CMCR)

I2C Master 0 base: 0x4002.0000
Offset 0x020
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:6	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
5	SFE	R/W	0	I²C Slave Function Enable This bit specifies whether the interface may operate in Slave mode. If set, Slave mode is enabled; otherwise, Slave mode is disabled.
4	MFE	R/W	0	I²C Master Function Enable This bit specifies whether the interface may operate in Master mode. If set, Master mode is enabled; otherwise, Master mode is disabled and the interface clock is disabled.
3:1	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
0	LPBK	R/W	0	I²C Loopback This bit specifies whether the interface is operating normally or in Loopback mode. If set, the device is put in a test mode loopback configuration; otherwise, the device operates normally.

14.6 Register Descriptions (I^2C Slave)

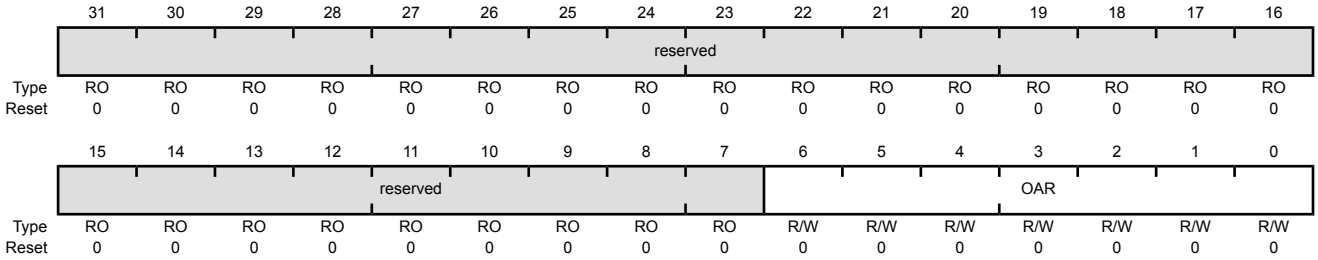
The remainder of this section lists and describes the I^2C slave registers, in numerical order by address offset. See also “Register Descriptions (I^2C Master)” on page 382.

Register 10: I2C Slave Own Address (I2CSOAR), offset 0x000

This register consists of seven address bits that identify the Stellaris® I2C device on the I2C bus.

I2C Slave Own Address (I2CSOAR)

I2C Slave 0 base: 0x4002.0800
Offset 0x000
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:7	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
6:0	OAR	R/W	0x00	I2C Slave Own Address This field specifies bits A6 through A0 of the slave address.

Register 11: I²C Slave Control/Status (I2CSCSR), offset 0x004

This register accesses one control bit when written, and three status bits when read.

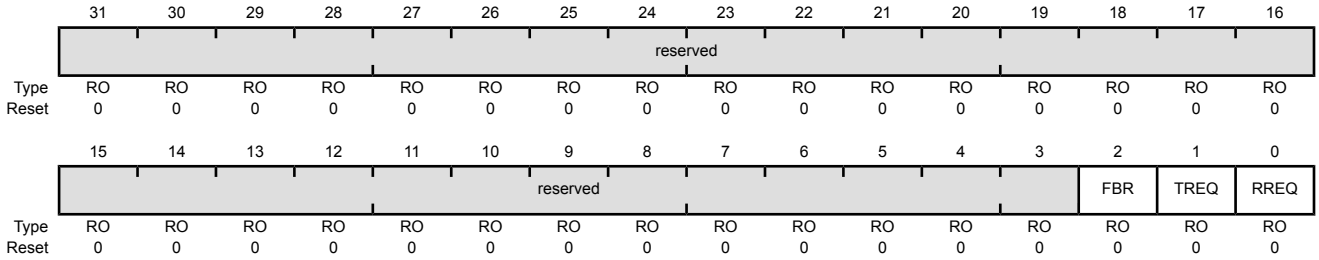
The read-only Status register consists of three bits: the FBR, RREQ, and TREQ bits. The First Byte Received (FBR) bit is set only after the Stellaris® device detects its own slave address and receives the first data byte from the I²C master. The Receive Request (RREQ) bit indicates that the Stellaris® I²C device has received a data byte from an I²C master. Read one data byte from the **I²C Slave Data (I2CSDR)** register to clear the RREQ bit. The Transmit Request (TREQ) bit indicates that the Stellaris® I²C device is addressed as a Slave Transmitter. Write one data byte into the **I²C Slave Data (I2CSDR)** register to clear the TREQ bit.

The write-only Control register consists of one bit: the DA bit. The DA bit enables and disables the Stellaris® I²C slave operation.

Reads

I2C Slave Control/Status (I2CSCSR)

I2C Slave 0 base: 0x4002.0800
Offset 0x004
Type RO, reset 0x0000.0000



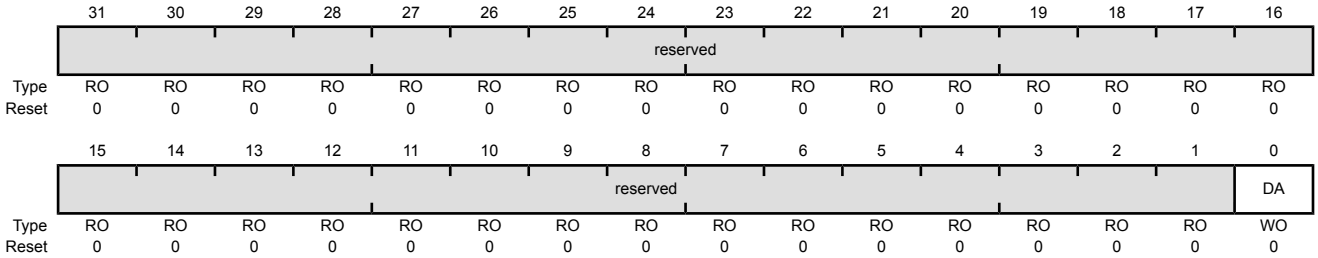
Bit/Field	Name	Type	Reset	Description
31:3	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
2	FBR	RO	0	First Byte Received Indicates that the first byte following the slave's own address is received. This bit is only valid when the RREQ bit is set, and is automatically cleared when data has been read from the **I2CSDR** register. **Note:** This bit is not used for slave transmit operations.
1	TREQ	RO	0	Transmit Request This bit specifies the state of the I²C slave with regards to outstanding transmit requests. If set, the I²C unit has been addressed as a slave transmitter and uses clock stretching to delay the master until data has been written to the **I2CSDR** register. Otherwise, there is no outstanding transmit request.

Bit/Field	Name	Type	Reset	Description
0	RREQ	RO	0	Receive Request This bit specifies the status of the I²C slave with regards to outstanding receive requests. If set, the I²C unit has outstanding receive data from the I²C master and uses clock stretching to delay the master until the data has been read from the **I2CSDR** register. Otherwise, no receive data is outstanding.

Writes

I2C Slave Control/Status (I2CSCSR)

I2C Slave 0 base: 0x4002.0800
Offset 0x004
Type WO, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:1	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
0	DA	WO	0	Device Active Value Description 0 Disables the I²C slave operation. 1 Enables the I²C slave operation.

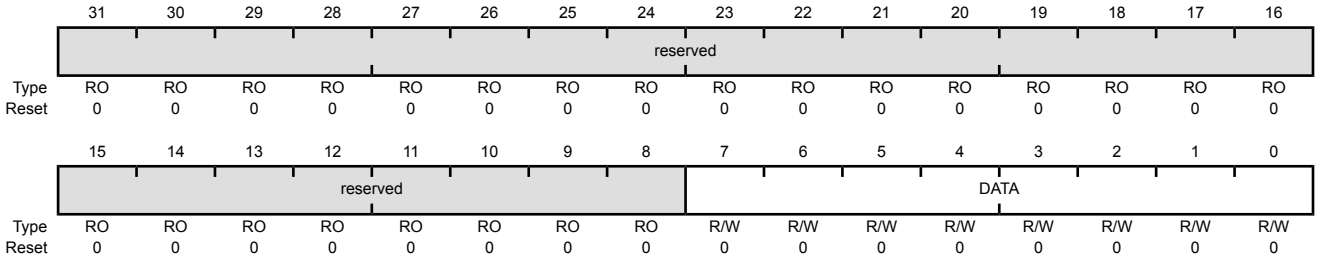
Register 12: I²C Slave Data (I2CSDR), offset 0x008

Important: Use caution when reading this register. Performing a read may change bit status.

This register contains the data to be transmitted when in the Slave Transmit state, and the data received when in the Slave Receive state.

I2C Slave Data (I2CSDR)

I2C Slave 0 base: 0x4002.0800
Offset 0x008
Type R/W, reset 0x0000.0000



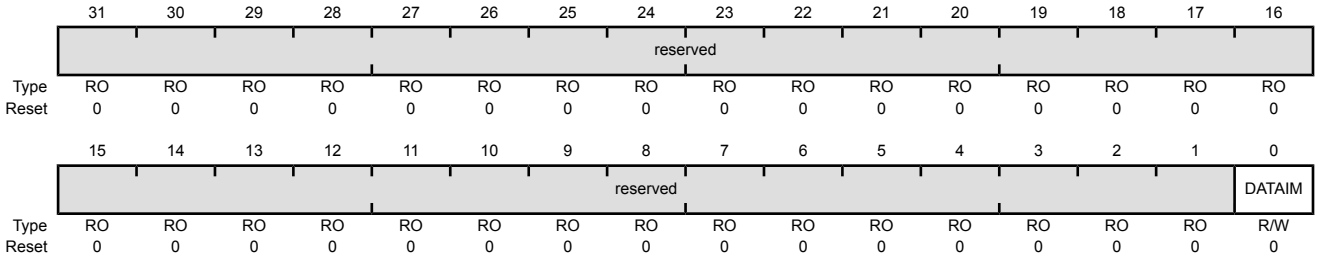
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	DATA	R/W	0x0	Data for Transfer This field contains the data for transfer during a slave receive or transmit operation.

Register 13: I²C Slave Interrupt Mask (I2CSIMR), offset 0x00C

This register controls whether a raw interrupt is promoted to a controller interrupt.

I2C Slave Interrupt Mask (I2CSIMR)

I2C Slave 0 base: 0x4002.0800
Offset 0x00C
Type R/W, reset 0x0000.0000



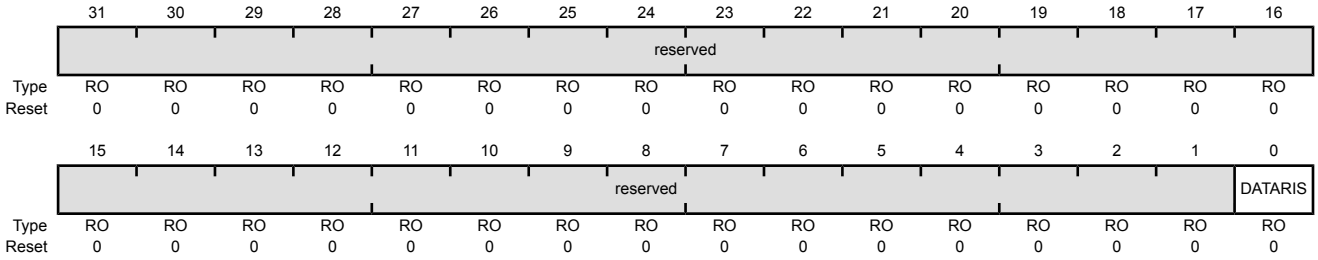
Bit/Field	Name	Type	Reset	Description
31:1	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
0	DATAIM	R/W	0	Data Interrupt Mask This bit controls whether the raw interrupt for data received and data requested is promoted to a controller interrupt. If set, the interrupt is not masked and the interrupt is promoted; otherwise, the interrupt is masked.

Register 14: I²C Slave Raw Interrupt Status (I2CSRIS), offset 0x010

This register specifies whether an interrupt is pending.

I2C Slave Raw Interrupt Status (I2CSRIS)

I2C Slave 0 base: 0x4002.0800
Offset 0x010
Type RO, reset 0x0000.0000



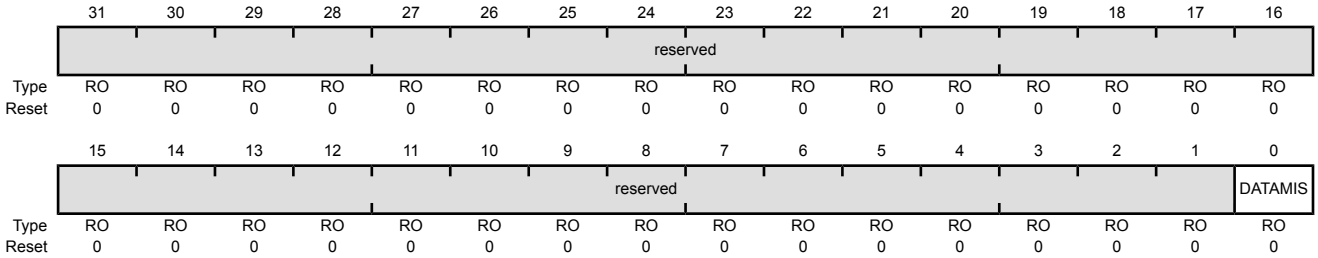
Bit/Field	Name	Type	Reset	Description
31:1	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
0	DATARIS	RO	0	Data Raw Interrupt Status This bit specifies the raw interrupt state for data received and data requested (prior to masking) of the I²C slave block. If set, an interrupt is pending; otherwise, an interrupt is not pending.

Register 15: I²C Slave Masked Interrupt Status (I2CSMIS), offset 0x014

This register specifies whether an interrupt was signaled.

I2C Slave Masked Interrupt Status (I2CSMIS)

I2C Slave 0 base: 0x4002.0800
Offset 0x014
Type RO, reset 0x0000.0000



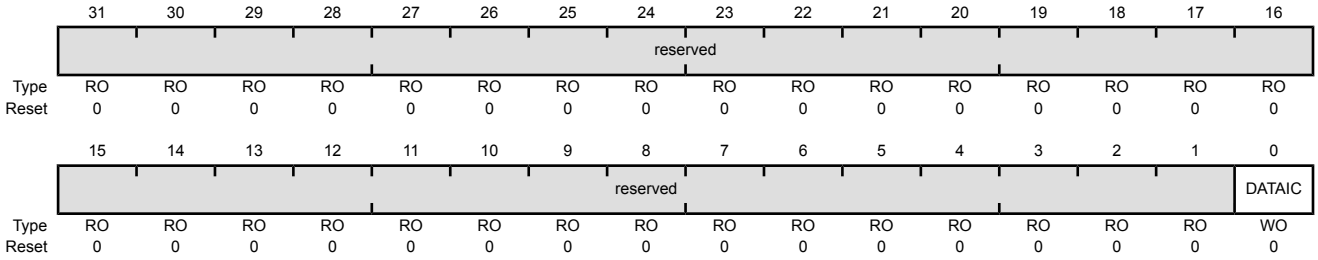
Bit/Field	Name	Type	Reset	Description
31:1	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
0	DATAMIS	RO	0	Data Masked Interrupt Status This bit specifies the interrupt state for data received and data requested (after masking) of the I²C slave block. If set, an interrupt was signaled; otherwise, an interrupt has not been generated since the bit was last cleared.

Register 16: I²C Slave Interrupt Clear (I2CSICR), offset 0x018

This register clears the raw interrupt. A read of this register returns no meaningful data.

I2C Slave Interrupt Clear (I2CSICR)

I2C Slave 0 base: 0x4002.0800
Offset 0x018
Type WO, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:1	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
0	DATAIC	WO	0	Data Interrupt Clear This bit controls the clearing of the raw interrupt for data received and data requested. When set, it clears the DATARIS interrupt bit; otherwise, it has no effect on the DATARIS bit value.

15 Ethernet Controller

The Stellaris® Ethernet Controller consists of a fully integrated media access controller (MAC) and network physical (PHY) interface. The Ethernet Controller conforms to *IEEE 802.3* specifications and fully supports 10BASE-T and 100BASE-TX standards.

The Stellaris® Ethernet Controller module has the following features:

- Conforms to the *IEEE 802.3-2002 specification*
  - 10BASE-T/100BASE-TX IEEE-802.3 compliant. Requires only a dual 1:1 isolation transformer interface to the line
  - 10BASE-T/100BASE-TX ENDEC, 100BASE-TX scrambler/descrambler
  - Full-featured auto-negotiation
- Multiple operational modes
  - Full- and half-duplex 100 Mbps
  - Full- and half-duplex 10 Mbps
  - Power-saving and power-down modes
- Highly configurable
  - Programmable MAC address
  - LED activity selection
  - Promiscuous mode support
  - CRC error-rejection control
  - User-configurable interrupts
- Physical media manipulation
  - Automatic MDI/MDI-X cross-over correction
  - Register-programmable transmit amplitude
  - Automatic polarity correction and 10BASE-T signal reception

15.1 Block Diagram

As shown in Figure 15-1 on page 405, the Ethernet Controller is functionally divided into two layers: the Media Access Controller (MAC) layer and the Network Physical (PHY) layer. These layers correspond to the OSI model layers 2 and 1. The CPU accesses the Ethernet Controller via the MAC layer. The MAC layer provides transmit and receive processing for Ethernet frames. The MAC layer also provides the interface to the PHY layer via an internal Media Independent Interface (MII). The PHY layer communicates with the Ethernet bus.

Figure 15-1. Ethernet Controller

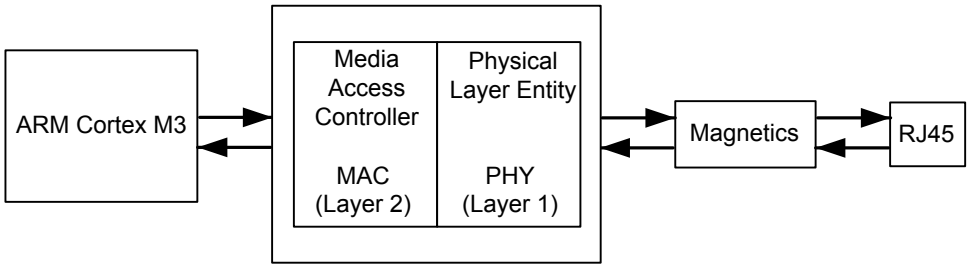
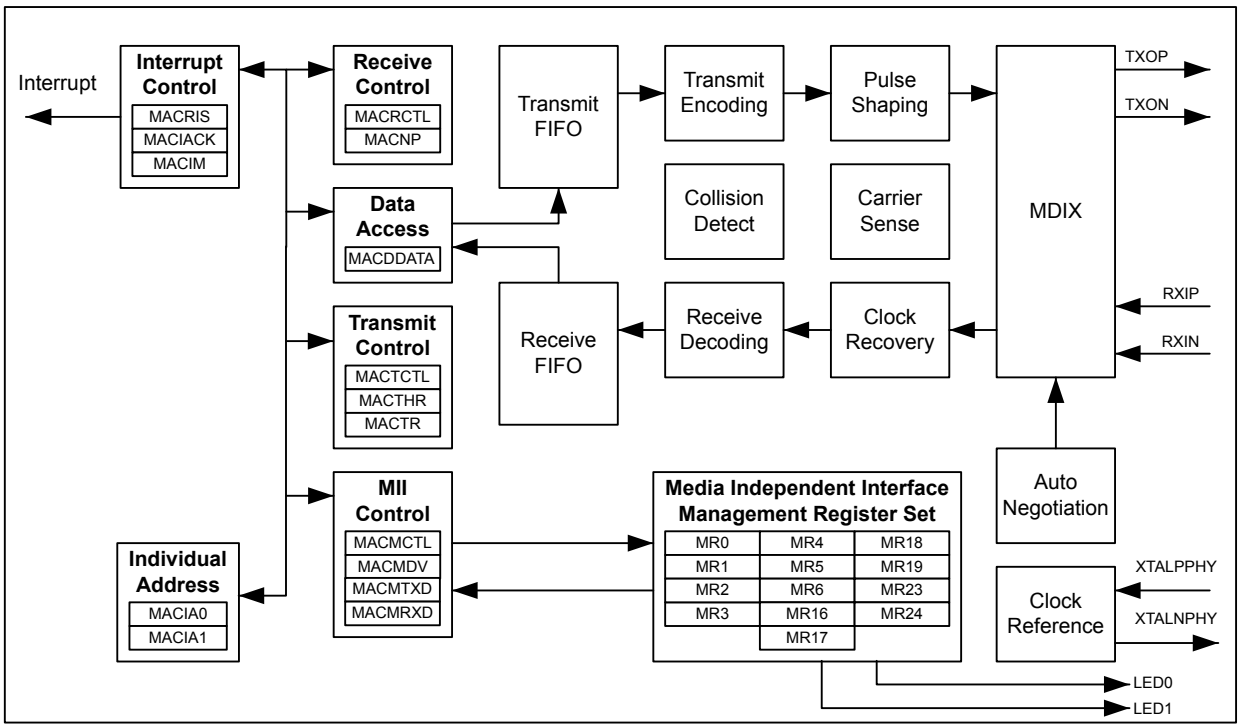


Figure 15-2 on page 405 shows more detail of the internal structure of the Ethernet Controller and how the register set relates to various functions.

Figure 15-2. Ethernet Controller Block Diagram



15.2 Functional Description

Note: A 12.4-kΩ resistor should be connected between the ERBIAS and ground. The 12.4-kΩ resistor should have a 1% tolerance and should be located in close proximity to the ERBIAS pin. Power dissipation in the resistor is low, so a chip resistor of any geometry may be used.

The functional description of the Ethernet Controller is discussed in the following sections.

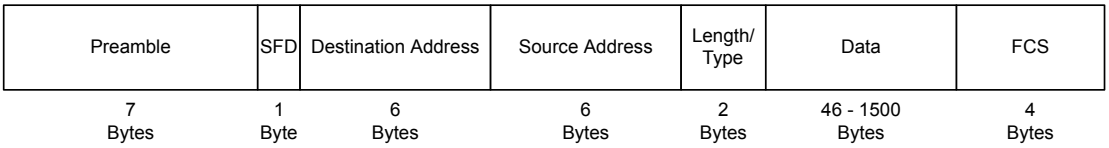
15.2.1 MAC Operation

The following sections decribe the operation of the MAC unit, including an overview of the Ethernet frame format, the MAC layer FIFOs, Ethernet transmission and reception options, and LED indicators.

15.2.1.1 Ethernet Frame Format

Ethernet data is carried by Ethernet frames. The basic frame format is shown in Figure 15-3 on page 406.

Figure 15-3. Ethernet Frame



The seven fields of the frame are transmitted from left to right. The bits within the frame are transmitted from least to most significant bit.

- Preamble

  The Preamble field is used to synchronize with the received frame's timing. The preamble is 7 octets long.

- Start Frame Delimiter (SFD)

  The SFD field follows the preamble pattern and indicates the start of the frame. Its value is 1010.1011.

- Destination Address (DA)

  This field specifies destination addresses for which the frame is intended. The LSB (bit 16 of DA oct 1 in the frame, see Table 15-1 on page 407) of the DA determines whether the address is an individual (0), or group/multicast (1) address.

- Source Address (SA)

  The source address field identifies the station from which the frame was initiated.

- Length/Type Field

  The meaning of this field depends on its numeric value. This field can be interpreted as length or type code. The maximum length of the data field is 1500 octets. If the value of the Length/Type field is less than or equal to 1500 decimal, it indicates the number of MAC client data octets. If the value of this field is greater than or equal to 1536 decimal, then it is type interpretation. The meaning of the Length/Type field when the value is between 1500 and 1536 decimal is unspecified by the IEEE 802.3 standard. However, the Ethernet Controller assumes type interpretation if the value of the Length/Type field is greater than 1500 decimal. The definition of the Type field is specified in the IEEE 802.3 standard. The first of the two octets in this field is most significant.

- Data

  The data field is a sequence of octets that is at least 46 in length, up to 1500 in length. Full data transparency is provided so any values can appear in this field. A minimum frame size of 46 octets is required to meet the IEEE standard. If the frame size is too small, the Ethernet Controller automatically appends extra bits (a pad), thus the pad can have a size of 0 to 46 octets. Data padding can be disabled by clearing the PADEN bit in the **Ethernet MAC Transmit Control (MACTCTL)** register.

  For the Ethernet Controller, data sent/received can be larger than 1500 bytes without causing a Frame Too Long error. Instead, a FIFO overrun error is reported using the F0V bit in the

Ethernet MAC Raw Interrupt Status(MACRIS) register when the frame received is too large to fit into the Ethernet Controller's 2K RAM.

- Frame Check Sequence (FCS)

  The frame check sequence carries the cyclic redundancy check (CRC) value. The CRC is computed over the destination address, source address, length/type, and data (including pad) fields using the CRC-32 algorithm. The Ethernet Controller computes the FCS value one nibble at a time. For transmitted frames, this field is automatically inserted by the MAC layer, unless disabled by clearing the CRC bit in the **MACTCTL** register. For received frames, this field is automatically checked. If the FCS does not pass, the frame is not placed in the RX FIFO, unless the FCS check is disabled by clearing the BADCRC bit in the **MACRCTL** register.

15.2.1.2 MAC Layer FIFOs

The Ethernet Controller is capable of simultaneous transmission and reception. This feature is enabled by setting the DUPLEX bit in the **MACTCTL** register.

For Ethernet frame transmission, a 2 KB transmit FIFO is provided that can be used to store a single frame. While the *IEEE 802.3 specification* limits the size of an Ethernet frame's payload section to 1500 Bytes, the Ethernet Controller places no such limit. The full buffer can be used, for a payload of up to 2032 bytes (as the first 16 bytes in the FIFO are reserved for destination address, source address and length/type information).

For Ethernet frame reception, a 2-KB receive FIFO is provided that can be used to store multiple frames, up to a maximum of 31 frames. If a frame is received, and there is insufficient space in the RX FIFO, an overflow error is indicated using the FOV bit in the **MACRIS** register.

For details regarding the TX and RX FIFO layout, refer to Table 15-1 on page 407. Please note the following difference between TX and RX FIFO layout. For the TX FIFO, the Data Length field in the first FIFO word refers to the Ethernet frame data payload, as shown in the 5th to nth FIFO positions. For the RX FIFO, the Frame Length field is the total length of the received Ethernet frame, including the Length/Type bytes and the FCS bits.

If FCS generation is disabled by clearing the CRC bit in the **MACTCTL** register, the last word in the TX FIFO must contain the FCS bytes for the frame that has been written to the FIFO.

Also note that if the length of the data payload section is not a multiple of 4, the FCS field is not be aligned on a word boundary in the FIFO. However, for the RX FIFO the beginning of the next frame is always on a word boundary.

Table 15-1. TX & RX FIFO Organization

FIFO Word Read/Write Sequence	Word Bit Fields	TX FIFO (Write)	RX FIFO (Read)
1st	7:0	Data Length Least Significant Byte	Frame Length Least Significant Byte
	15:8	Data Length Most Significant Byte	Frame Length Most Significant Byte
	23:16	DA oct 1	
	31:24	DA oct 2	
2nd	7:0	DA oct 3	
	15:8	DA oct 4	
	23:16	DA oct 5	
	31:24	DA oct 6	

Table 15-1. TX & RX FIFO Organization *(continued)*

FIFO Word Read/Write Sequence	Word Bit Fields	TX FIFO (Write)	RX FIFO (Read)
3rd	7:0	SA oct 1	
	15:8	SA oct 2	
	23:16	SA oct 3	
	31:24	SA oct 4	
4th	7:0	SA oct 5	
	15:8	SA oct 6	
	23:16	Len/Type Most Significant Byte	
	31:24	Len/Type Least Significant Byte	
5th to nth	7:0	data oct n	
	15:8	data oct n+1	
	23:16	data oct n+2	
	31:24	data oct n+3	
last	7:0	FCS 1	
	15:8	FCS 2	
	23:16	FCS 3	
	31:24	FCS 4	

Note: If the CRC bit in the **MACTCTL** register is clear, the FCS bytes must be written with the correct CRC. If the CRC bit is set, the Ethernet Controller automatically writes the FCS bytes.

15.2.1.3 Ethernet Transmission Options

At the MAC layer, the transmitter can be configured for both full-duplex and half-duplex operation by using the DUPLEX bit in the **MACTCTL** register.

The Ethernet Controller automatically generates and inserts the Frame Check Sequence (FCS) at the end of the transmit frame when the CRC bit in the **MACTCTL** register is set. However, for test purposes, this feature can be disabled in order to generate a frame with an invalid CRC by clearing the CRC bit.

The *IEEE 802.3 specification* requires that the Ethernet frame payload section be a minimum of 46 bytes. The Ethernet Controller automatically pads the data section if the payload data section loaded into the FIFO is less than the minimum 46 bytes when the PADEN bit in the **MACTCTL** register is set. This feature can be disabled by clearing the PADEN bit.

The transmitter must be enabled by setting the TXEN bit in the **TCTL** register.

15.2.1.4 Ethernet Reception Options

The Ethernet Controller RX FIFO should be cleared during software initialization. The receiver should first be disabled by clearing the RXEN bit in the **Ethernet MAC Receive Control (MACRCTL)** register, then the FIFO can be cleared by setting the RSTFIFO bit in the **MACRCTL** register.

The receiver automatically rejects frames that contain bad CRC values in the FCS field. In this case, a Receive Error interrupt is generated and the receive data is lost. To accept all frames, clear the BADCRC bit in the **MACRCTL** register.

In normal operating mode, the receiver accepts only those frames that have a destination address that matches the address programmed into the **Ethernet MAC Individual Address 0 (MACIA0)**

and **Ethernet MAC Individual Address 1 (MACIA1)** registers. However, the Ethernet receiver can also be configured for Promiscuous and Multicast modes by setting the PRMS and AMUL bits in the **MACRCTL** register.

15.2.2 Internal MII Operation

For the MII management interface to function properly, the MDIO signal must be connected through a 10k Ω pull-up resistor to the +3.3 V supply. Failure to connect this pull-up resistor prevents management transactions on this internal MII to function. Note that it is possible for data transmission across the MII to still function since the PHY layer auto-negotiates the link parameters by default.

For the MII management interface to function properly, the internal clock must be divided down from the system clock to a frequency no greater than 2.5 MHz. The **Ethernet MAC Management Divider (MACMDV)** register contains the divider used for scaling down the system clock. See page 428 for more details about the use of this register.

15.2.3 PHY Operation

The Physical Layer (PHY) in the Ethernet Controller includes integrated ENDECs, scrambler/descrambler, dual-speed clock recovery, and full-featured auto-negotiation functions. The transmitter includes an on-chip pulse shaper and a low-power line driver. The receiver has an adaptive equalizer and a baseline restoration circuit required for accurate clock and data recovery. The transceiver interfaces to Category-5 unshielded twisted pair (Cat-5 UTP) cabling for 100BASE-TX applications, and Category-3 unshielded twisted pair (Cat-3 UTP) for 10BASE-T applications. The Ethernet Controller is connected to the line media via dual 1:1 isolation transformers. No external filter is required.

15.2.3.1 Clock Selection

The Ethernet Controller has an on-chip crystal oscillator which can also be driven by an external oscillator. In this mode of operation, a 25-MHz crystal should be connected between the XTALPPHY and XTALNPHY pins. Alternatively, an external 25-MHz clock input can be connected to the XTALPPHY pin. In this mode of operation, a crystal is not required and the XTALNPHY pin must be tied to ground.

15.2.3.2 Auto-Negotiation

The Ethernet Controller supports the auto-negotiation functions of Clause 28 of the *IEEE 802.3* standard for 10/100 Mbps operation over copper wiring. This function is controlled via register settings. The auto-negotiation function is turned on by default, and the ANEGEN bit in the **Ethernet PHY Management Register 0 - Control (MR0)** is set after reset. Software can disable the auto-negotiation function by clearing the ANEGEN bit. The contents of the **Ethernet PHY Management Register - Auto-Negotiation Advertisement (MR4)** are reflected to the Ethernet Controller's link partner during auto-negotiation via fast-link pulse coding.

Once auto-negotiation is complete, the DPLX and RATE bits in the **Ethernet PHY Management Register 18 - Diagnostic (MR18)** register reflect the actual speed and duplex condition. If auto-negotiation fails to establish a link for any reason, the ANEGF bit in the **MR18** register reflects this and auto-negotiation restarts from the beginning. Setting the RANEG bit in the **MR0** register also causes auto-negotiation to restart.

15.2.3.3 Polarity Correction

The Ethernet Controller is capable of either automatic or manual polarity reversal for 10BASE-T and auto-negotiation functions. Bits 4 and 5 (RVSPOL and APOL) in the **Ethernet PHY Management Register 16 - Vendor-Specific (MR16)** control this feature. The default is automatic mode, where

APOL is clear and RVSPOL indicates if the detection circuitry has inverted the input signal. To enter manual mode, APOL should be set. In manual mode RVSPOL controls the signal polarity.

15.2.3.4 MDI/MDI-X Configuration

The Ethernet Controller supports the MDI/MDI-X configuration as defined in *IEEE 802.3-2002 specification*. The MDI/MDI-X configuration eliminates the need for cross-over cables when connecting to another device, such as a hub. The algorithm is controlled via settings in the **Ethernet PHY Management Register 24 - MDI/MIDIX Control (MR24)**. Refer to page 450 for additional details about these settings.

15.2.3.5 Power Management

The PHY has two power-saving modes:

- Power-Down
- Receive Power Management

Power-down mode is activated by setting the PWRDN bit in the **MR0** register. When the PHY is in power-down mode, it consumes minimum power. While in the power-down state, the Ethernet Controller still responds to management transactions.

Receive power management (RXCC mode) is activated by setting the RXCC bit in the **MR16** register. In this mode of operation, the adaptive equalizer, the clock recovery phase lock loop (PLL), and all other receive circuitry are powered down. As soon as a valid signal is detected, all circuits are automatically powered up to resume normal operation. Note that the RXCC mode is not supported during 10BASE-T operation.

15.2.3.6 LED Indicators

The Ethernet Controller supports two LED signals that can be used to indicate various states of operation. These signals are mapped to the LED0 and LED1 pins. By default, these pins are configured as GPIO signals (PF3 and PF2). For the PHY layer to drive these signals, they must be reconfigured to their alternate function. See “General-Purpose Input/Outputs (GPIOs)” on page 151 for additional details. The function of these pins is programmable via the PHY layer **Ethernet PHY Management Register 23 - LED Configuration (MR23)**. Refer to page 449 for additional details on how to program these LED functions.

15.2.4 Interrupts

The Ethernet Controller can generate an interrupt for one or more of the following conditions:

- A frame has been received into an empty RX FIFO
- A frame transmission error has occurred
- A frame has been transmitted successfully
- A frame has been received with inadequate room in the RX FIFO (overrun)
- A frame has been received with one or more error conditions (for example, FCS failed)
- An MII management transaction between the MAC and PHY layers has completed
- One or more of the following PHY layer conditions occurs:

– Auto-Negotiate Complete

– Remote Fault

– Link Status Change

– Link Partner Acknowledge

– Parallel Detect Fault

– Page Received

– Receive Error

– Jabber Event Detected

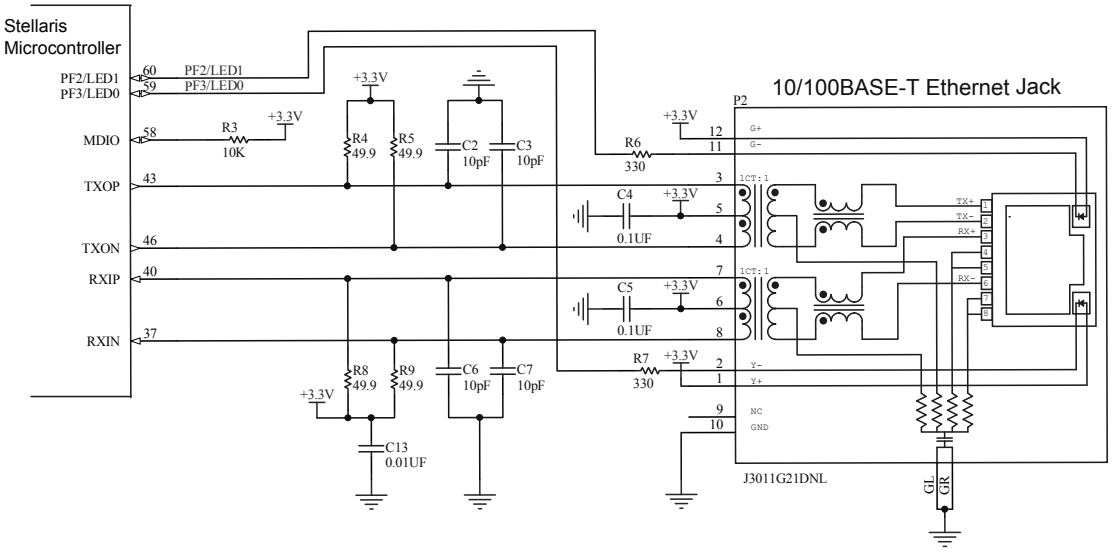
15.3 Initialization and Configuration

The following sections describe the hardware and software configuration required to set up the Ethernet Controller.

15.3.1 Hardware Configuration

Figure 15-4 on page 411 shows the proper method for interfacing the Ethernet Controller to a 10/100BASE-T Ethernet jack.

Figure 15-4. Interface to an Ethernet Jack



The following isolation transformers have been tested and are known to successfully interface to the Ethernet PHY layer.

■ Isolation Transformers
  – TDK TLA-6T103
  – Bel-Fuse S558-5999-46
  – Halo TG22-3506ND
  – Pulse PE-68515
  – Valor ST6118

  - YCL 20PMT04
- Isolation transformers in low profile packages (0.100 in/2.5 mm or less)
  - TDK TLA-6T118
  - Halo TG110-S050
  - PCA EPF8023G
- Isolation transformers with integrated RJ45 connector
  - TDK TLA-6T704
  - Delta RJS-1A08T089A
- Isolation transformers with integrated RJ45 connector, LEDs and termination resistors
  - Pulse J0011D21B/E
  - Pulse J3011G21DNL

15.3.2 Software Configuration

To use the Ethernet Controller, it must be enabled by setting the EPHY0 and EMAC0 bits in the **RCGC2** register (see page 115). The following steps can then be used to configure the Ethernet Controller for basic operation.

1. Program the **MACDIV** register to obtain a 2.5 MHz clock (or less) on the internal MII. Assuming a 20-MHz system clock, the **MACDIV** value should be 0x03 or greater.

2. Program the **MACIA0** and **MACIA1** register for address filtering.

3. Program the **MACTCTL** register for Auto CRC generation, padding, and full-duplex operation using a value of 0x16.

4. Program the **MACRCTL** register to flush the receive FIFO and reject frames with bad FCS using a value of 0x18.

5. Enable both the Transmitter and Receive by setting the LSB in both the **MACTCTL** and **MACRCTL** registers.

6. To transmit a frame, write the frame into the TX FIFO using the **Ethernet MAC Data (MACDATA)** register. Then set the NEWTX bit in the **Ethernet Mac Transmission Request (MACTR)** register to initiate the transmit process. When the NEWTX bit has been cleared, the TX FIFO is available for the next transmit frame.

7. To receive a frame, wait for the NPR field in the **Ethernet MAC Number of Packets (MACNP)** register to be non-zero. Then begin reading the frame from the RX FIFO by using the **MACDATA** register. To ensure that the entire packet is received, either use the DriverLib EthernetPacketGet() API or compare the number of bytes received to the Length field from the frame to determine when the packet has been completely read.

15.4 Ethernet Register Map

Table 15-2 on page 413 lists the Ethernet MAC registers. All addresses given are relative to the Ethernet MAC base address of 0x4004.8000.

The *IEEE 802.3* standard specifies a register set for controlling and gathering status from the PHY layer. The registers are collectively known as the MII Management registers and are detailed in Section 22.2.4 of the *IEEE 802.3 specification*. Table 15-2 on page 413 also lists these MII Management registers. *All addresses given are absolute and are written directly to the REGADR field of the **Ethernet MAC Management Control (MACMCTL)** register.* The format of registers 0 to 15 are defined by the IEEE specification and are common to all PHY layer implementations. The only variance allowed is for features that may or may not be supported by a specific PHY implementation.

Registers 16 to 31 are vendor-specific registers, used to support features that are specific to a vendor's PHY implementation. Vendor-specific registers not listed are reserved.

Table 15-2. Ethernet Register Map

Offset	Name	Type	Reset	Description	See page
Ethernet MAC					
0x000	MACRIS/MACIACK	R/W1C	0x0000.0000	Ethernet MAC Raw Interrupt Status/Acknowledge	415
0x004	MACIM	R/W	0x0000.007F	Ethernet MAC Interrupt Mask	418
0x008	MACRCTL	R/W	0x0000.0008	Ethernet MAC Receive Control	419
0x00C	MACTCTL	R/W	0x0000.0000	Ethernet MAC Transmit Control	420
0x010	MACDATA	R/W	0x0000.0000	Ethernet MAC Data	421
0x014	MACIA0	R/W	0x0000.0000	Ethernet MAC Individual Address 0	423
0x018	MACIA1	R/W	0x0000.0000	Ethernet MAC Individual Address 1	424
0x01C	MACTHR	R/W	0x0000.003F	Ethernet MAC Threshold	425
0x020	MACMCTL	R/W	0x0000.0000	Ethernet MAC Management Control	427
0x024	MACMDV	R/W	0x0000.0080	Ethernet MAC Management Divider	428
0x02C	MACMTXD	R/W	0x0000.0000	Ethernet MAC Management Transmit Data	429
0x030	MACMRXD	R/W	0x0000.0000	Ethernet MAC Management Receive Data	430
0x034	MACNP	RO	0x0000.0000	Ethernet MAC Number of Packets	431
0x038	MACTR	R/W	0x0000.0000	Ethernet MAC Transmission Request	432
MII Management					
-	MR0	R/W	0x3100	Ethernet PHY Management Register 0 – Control	433
-	MR1	RO	0x7849	Ethernet PHY Management Register 1 – Status	435
-	MR2	RO	0x000E	Ethernet PHY Management Register 2 – PHY Identifier 1	437
-	MR3	RO	0x7237	Ethernet PHY Management Register 3 – PHY Identifier 2	438
-	MR4	R/W	0x01E1	Ethernet PHY Management Register 4 – Auto-Negotiation Advertisement	439
-	MR5	RO	0x0000	Ethernet PHY Management Register 5 – Auto-Negotiation Link Partner Base Page Ability	441
-	MR6	RO	0x0000	Ethernet PHY Management Register 6 – Auto-Negotiation Expansion	442
-	MR16	R/W	0x0140	Ethernet PHY Management Register 16 – Vendor-Specific	443
-	MR17	R/W	0x0000	Ethernet PHY Management Register 17 – Interrupt Control/Status	445
-	MR18	RO	0x0000	Ethernet PHY Management Register 18 – Diagnostic	447

Table 15-2. Ethernet Register Map *(continued)*

Offset	Name	Type	Reset	Description	See page
-	MR19	R/W	0x4000	Ethernet PHY Management Register 19 – Transceiver Control	448
-	MR23	R/W	0x0010	Ethernet PHY Management Register 23 – LED Configuration	449
-	MR24	R/W	0x00C0	Ethernet PHY Management Register 24 –MDI/MDIX Control	450

15.5 Ethernet MAC Register Descriptions

The remainder of this section lists and describes the Ethernet MAC registers, in numerical order by address offset. Also see “MII Management Register Descriptions” on page 432.

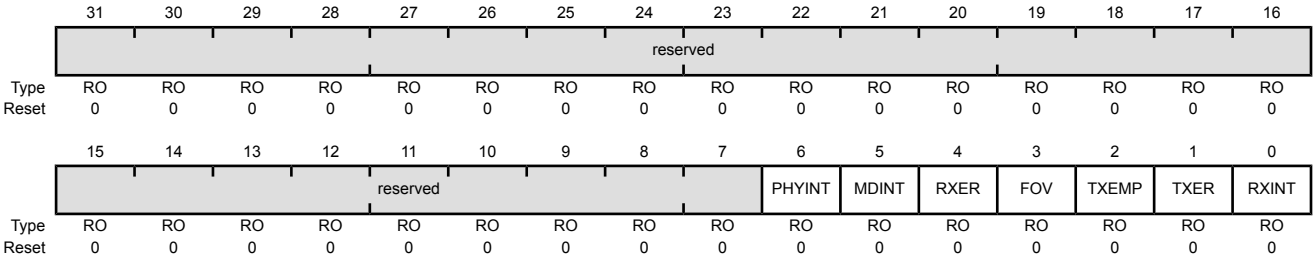
Register 1: Ethernet MAC Raw Interrupt Status/Acknowledge (MACRIS/MACIACK), offset 0x000

The **MACRIS/MACIACK** register is the interrupt status and acknowledge register. On a read, this register gives the current status value of the corresponding interrupt prior to masking. On a write, setting any bit clears the corresponding interrupt status bit.

Reads

Ethernet MAC Raw Interrupt Status/Acknowledge (MACRIS/MACIACK)

Base 0x4004.8000
Offset 0x000
Type RO, reset 0x0000.0000



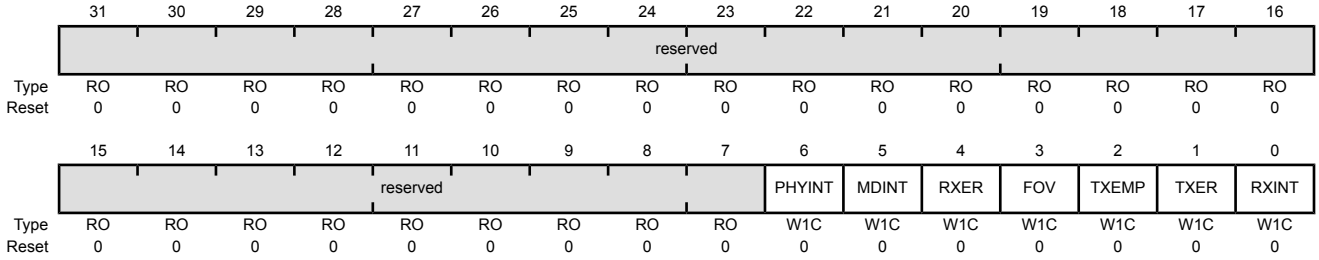
Bit/Field	Name	Type	Reset	Description
31:7	reserved	RO	0x0000.00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
6	PHYINT	RO	0	PHY Interrupt When set, indicates that an enabled interrupt in the PHY layer has occurred. **MR17** in the PHY must be read to determine the specific PHY event that triggered this interrupt.
5	MDINT	RO	0	MII Transaction Complete When set, indicates that a transaction (read or write) on the MII interface has completed successfully.
4	RXER	RO	0	Receive Error This bit indicates that an error was encountered on the receiver. The possible errors that can cause this interrupt bit to be set are: ■ A receive error occurs during the reception of a frame (100 Mb/s only). ■ The frame is not an integer number of bytes (dribble bits) due to an alignment error. ■ The CRC of the frame does not pass the FCS check. ■ The length/type field is inconsistent with the frame data size when interpreted as a length field.
3	FOV	RO	0	FIFO Overrun When set, indicates that an overrun was encountered on the receive FIFO.

Bit/Field	Name	Type	Reset	Description
2	TXEMP	RO	0	Transmit FIFO Empty When set, indicates that the packet was transmitted and that the TX FIFO is empty.
1	TXER	RO	0	Transmit Error When set, indicates that an error was encountered on the transmitter. The possible errors that can cause this interrupt bit to be set are: ■ The data length field stored in the TX FIFO exceeds 2032 decimal (buffer length - 16 bytes of header data). The frame is not sent when this error occurs. ■ The retransmission attempts during the backoff process have exceeded the maximum limit of 16 decimal.
0	RXINT	RO	0	Packet Received When set, indicates that at least one packet has been received and is stored in the receiver FIFO.

Writes

Ethernet MAC Raw Interrupt Status/Acknowledge (MACRIS/MACIACK)

Base 0x4004.8000
Offset 0x000
Type WO, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:7	reserved	RO	0x0000.00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
6	PHYINT	W1C	0	Clear PHY Interrupt Setting this bit clears the PHYINT interrupt in the **MACRIS** register.
5	MDINT	W1C	0	Clear MII Transaction Complete Setting this bit clears the MDINT interrupt in the **MACRIS** register.
4	RXER	W1C	0	Clear Receive Error Setting this bit clears the RXER interrupt in the **MACRIS** register.
3	FOV	W1C	0	Clear FIFO Overrun Setting this bit clears the FOV interrupt in the **MACRIS** register.

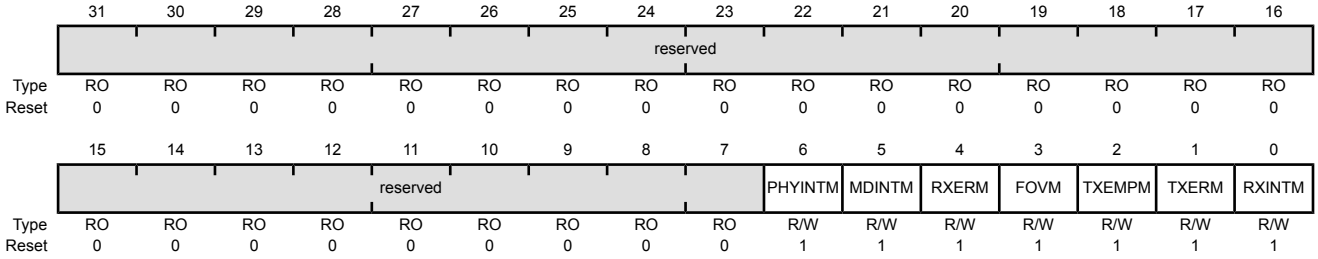
Bit/Field	Name	Type	Reset	Description
2	TXEMP	W1C	0	Clear Transmit FIFO Empty Setting this bit clears the TXEMP interrupt in the **MACRIS** register.
1	TXER	W1C	0	Clear Transmit Error Setting this bit clears the TXER interrupt in the **MACRIS** register and resets the TX FIFO write pointer.
0	RXINT	W1C	0	Clear Packet Received Setting this bit clears the RXINT interrupt in the **MACRIS** register.

Register 2: Ethernet MAC Interrupt Mask (MACIM), offset 0x004

This register allows software to enable/disable Ethernet MAC interrupts. Clearing a bit disables the interrupt, while setting the bit enables it.

Ethernet MAC Interrupt Mask (MACIM)

Base 0x4004.8000
Offset 0x004
Type R/W, reset 0x0000.007F



Bit/Field	Name	Type	Reset	Description
31:7	reserved	RO	0x0000.00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
6	PHYINTM	R/W	1	Mask PHY Interrupt Clearing this bit masks the PHYINT bit in the **MACRIS** register from being set.
5	MDINTM	R/W	1	Mask MII Transaction Complete Clearing this bit masks the MDINT bit in the **MACRIS** register from being set.
4	RXERM	R/W	1	Mask Receive Error Clearing this bit masks the RXER bit in the **MACRIS** register from being set.
3	FOVM	R/W	1	Mask FIFO Overrun Clearing this bit masks the FOV bit in the **MACRIS** register from being set.
2	TXEMPM	R/W	1	Mask Transmit FIFO Empty Clearing this bit masks the TXEMP bit in the **MACRIS** register from being set.
1	TXERM	R/W	1	Mask Transmit Error Clearing this bit masks the TXER bit in the **MACRIS** register from being set.
0	RXINTM	R/W	1	Mask Packet Received Clearing this bit masks the RXINT bit in the **MACRIS** register from being set.

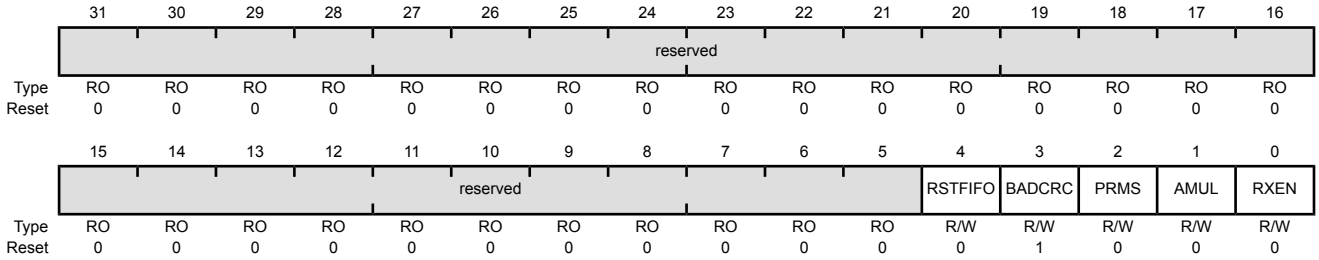
Register 3: Ethernet MAC Receive Control (MACRCTL), offset 0x008

This register configures the receiver and controls the types of frames that are received.

It is important to note that when the receiver is enabled, all valid frames with a broadcast address of FF-FF-FF-FF-FF-FF in the Destination Address field are received and stored in the RX FIFO, even if the AMUL bit is not set.

Ethernet MAC Receive Control (MACRCTL)

Base 0x4004.8000
Offset 0x008
Type R/W, reset 0x0000.0008



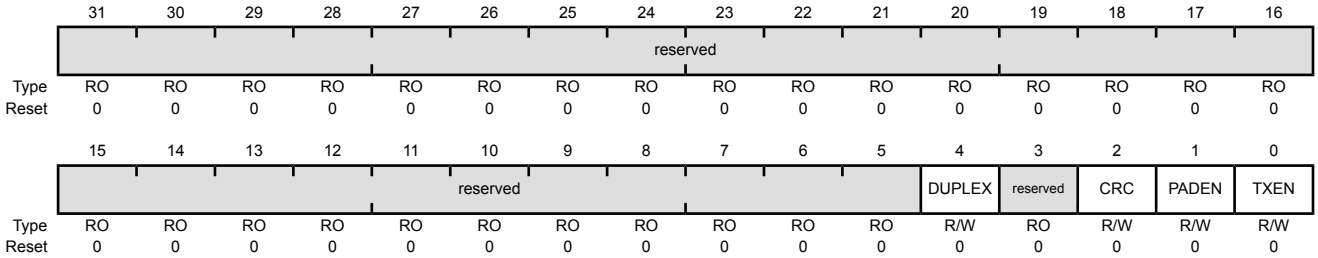
Bit/Field	Name	Type	Reset	Description
31:5	reserved	RO	0x0000.000	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
4	RSTFIFO	R/W	0	Clear Receive FIFO When set, this bit clears the receive FIFO. This should be done when software initialization is performed. It is recommended that the receiver be disabled (RXEN = 0), before a reset is initiated (RSTFIFO = 1). This sequence flushes and resets the RX FIFO. This bit is automatically cleared when read.
3	BADCRC	R/W	1	Enable Reject Bad CRC When set, the BADCRC bit enables the rejection of frames with an incorrectly calculated CRC. If a bad CRC is encountered, the RXER bit in the **MACRIS** register is set and the receiver FIFO is reset.
2	PRMS	R/W	0	Enable Promiscuous Mode When set, the PRMS bit enables Promiscuous mode, which accepts all valid frames, regardless of the specified Destination Address.
1	AMUL	R/W	0	Enable Multicast Frames When set, the AMUL bit enables the reception of multicast frames.
0	RXEN	R/W	0	Enable Receiver When set the RXEN bit enables the Ethernet receiver. When this bit is clear, the receiver is disabled and all frames are ignored.

Register 4: Ethernet MAC Transmit Control (MACTCTL), offset 0x00C

This register configures the transmitter and controls the frames that are transmitted.

Ethernet MAC Transmit Control (MACTCTL)

Base 0x4004.8000
Offset 0x00C
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:5	reserved	RO	0x0000.000	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
4	DUPLEX	R/W	0	Enable Duplex Mode When set, this bit enables Duplex mode, allowing simultaneous transmission and reception.
3	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
2	CRC	R/W	0	Enable CRC Generation When set this bit enables the automatic generation of the CRC and its placement at the end of the packet. If this bit is clear, the frames placed in the TX FIFO are sent exactly as they are written into the FIFO. Note that this bit should generally be set.
1	PADEN	R/W	0	Enable Packet Padding When set, this bit enables the automatic padding of packets that do not meet the minimum frame size. Note that this bit should generally be set.
0	TXEN	R/W	0	Enable Transmitter When set, this bit enables the transmitter. When this bit is clear, the transmitter is disabled.

Register 5: Ethernet MAC Data (MACDATA), offset 0x010

Important: Use caution when reading this register. Performing a read may change bit status.

This register enables software to access the TX and RX FIFOs.

Reads from this register return the data stored in the RX FIFO from the location indicated by the read pointer. The read pointer is then auto incremented to the next RX FIFO location. Reading from the RX FIFO when a frame has not been received or is in the process of being received will return indeterminate data and not increment the read pointer.

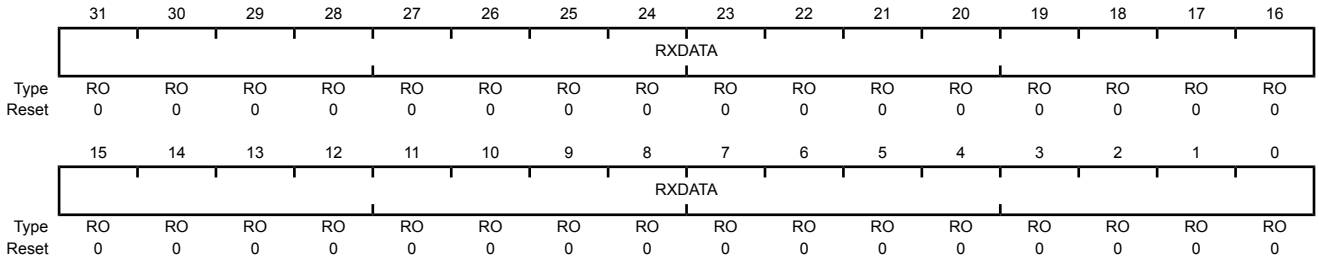
Writes to this register store the data in the TX FIFO at the location indicated by the write pointer. The write pointer is the auto incremented to the next TX FIFO location. Writing more data into the TX FIFO than indicated in the length field will result in the data being lost. Writing less data into the TX FIFO than indicated in the length field will result in indeterminate data being appended to the end of the frame to achieve the indicated length. Attempting to write the next frame into the TX FIFO before transmission of the first has completed will result in the data being lost.

There is no mechanism for randomly accessing bytes in either the RX or TX FIFOs. Data must be read from the RX FIFO sequentially and stored in a buffer for further processing. Once a read has been performed, the data in the FIFO cannot be re-read. Data must be written to the TX FIFO sequentially. If an error is made in placing the frame into the TX FIFO, the write pointer can be reset to the start of the TX FIFO by writing the TXER bit of the **MACIACK** register and then the data re-written.

Reads

Ethernet MAC Data (MACDATA)

Base 0x4004.8000
Offset 0x010
Type RO, reset 0x0000.0000

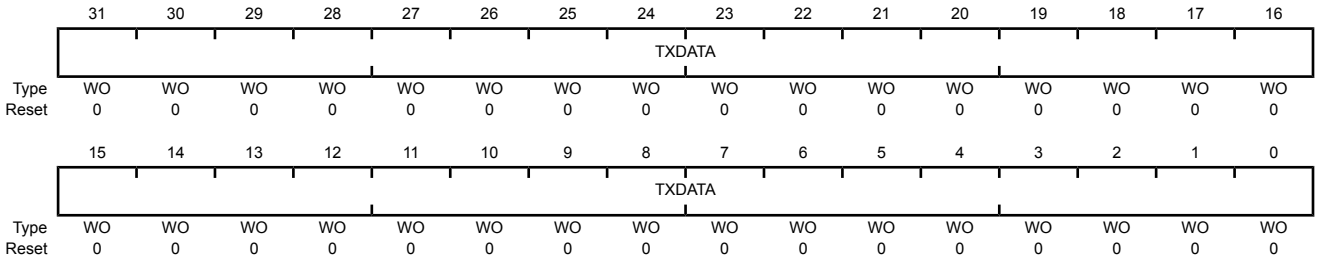


Bit/Field	Name	Type	Reset	Description
31:0	RXDATA	RO	0x0000.0000	Receive FIFO Data The RXDATA bits represent the next word of data stored in the RX FIFO.

Writes

Ethernet MAC Data (MACDATA)

Base 0x4004.8000
Offset 0x010
Type WO, reset 0x0000.0000



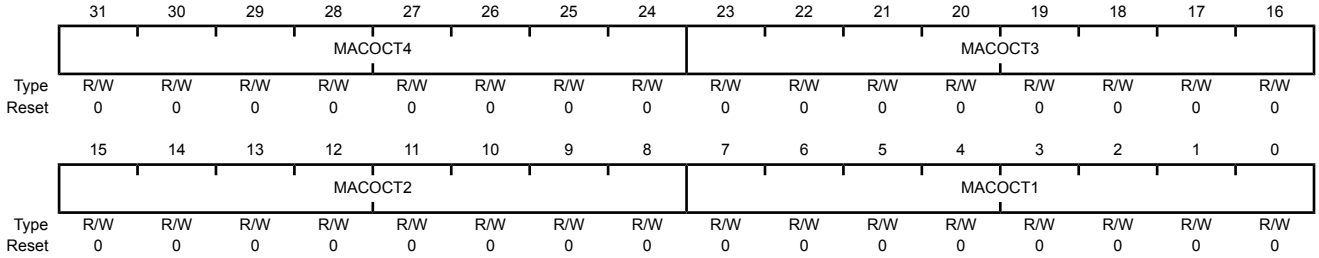
Bit/Field	Name	Type	Reset	Description
31:0	TXDATA	WO	0x0000.0000	Transmit FIFO Data The TXDATA bits represent the next word of data to place in the TX FIFO for transmission.

Register 6: Ethernet MAC Individual Address 0 (MACIA0), offset 0x014

This register enables software to program the first four bytes of the hardware MAC address of the Network Interface Card (NIC). (The last two bytes are in **MACIA1**). The 6-byte Individual Address is compared against the incoming Destination Address fields to determine whether the frame should be received.

Ethernet MAC Individual Address 0 (MACIA0)

Base 0x4004.8000
Offset 0x014
Type R/W, reset 0x0000.0000



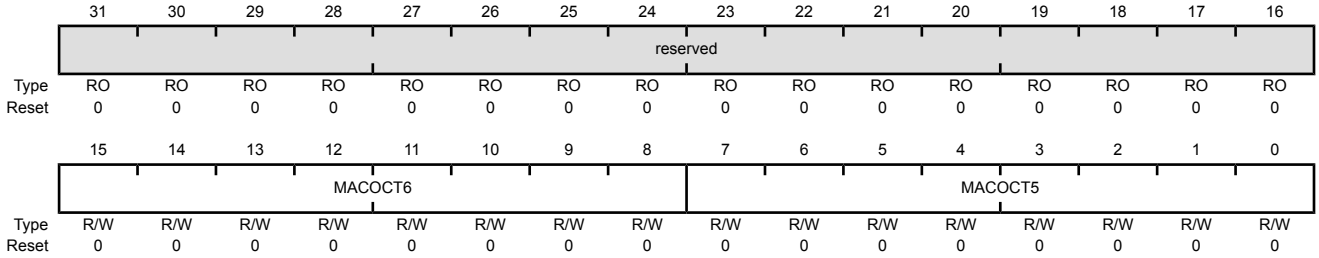
Bit/Field	Name	Type	Reset	Description
31:24	MACOCT4	R/W	0x00	MAC Address Octet 4 The MACOCT4 bits represent the fourth octet of the MAC address used to uniquely identify the Ethernet Controller.
23:16	MACOCT3	R/W	0x00	MAC Address Octet 3 The MACOCT3 bits represent the third octet of the MAC address used to uniquely identify the Ethernet Controller.
15:8	MACOCT2	R/W	0x00	MAC Address Octet 2 The MACOCT2 bits represent the second octet of the MAC address used to uniquely identify the Ethernet Controller.
7:0	MACOCT1	R/W	0x00	MAC Address Octet 1 The MACOCT1 bits represent the first octet of the MAC address used to uniquely identify the Ethernet Controller.

Register 7: Ethernet MAC Individual Address 1 (MACIA1), offset 0x018

This register enables software to program the last two bytes of the hardware MAC address of the Network Interface Card (NIC). (The first four bytes are in **MACIA0**). The 6-byte IAR is compared against the incoming Destination Address fields to determine whether the frame should be received.

Ethernet MAC Individual Address 1 (MACIA1)

Base 0x4004.8000
Offset 0x018
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:16	reserved	RO	0x0000	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
15:8	MACOCT6	R/W	0x00	MAC Address Octet 6 The MACOCT6 bits represent the sixth octet of the MAC address used to uniquely identify each Ethernet Controller.
7:0	MACOCT5	R/W	0x00	MAC Address Octet 5 The MACOCT5 bits represent the fifth octet of the MAC address used to uniquely identify the Ethernet Controller.

Register 8: Ethernet MAC Threshold (MACTHR), offset 0x01C

In order to increase the transmission rate, it is possible to program the Ethernet Controller to begin transmission of the next frame prior to the completion of the transmission of the current frame. Note: Extreme care must be used when implementing this function. Software must be able to guarantee that the complete frame is able to be stored in the transmission FIFO prior to the completion of the transmission frame.

This register enables software to set the threshold level at which the transmission of the frame begins. If the THRESH bits are set to 0x3F, which is the reset value, the early transmission feature is disabled, and transmission does not start until the NEWTX bit is set in the **MACTR** register.

Writing the THRESH bits to any value besides 0x3F enables the early transmission feature. Once the byte count of data in the TX FIFO reaches the value derived from the THRESH bits as shown below, transmission of the frame begins. When THRESH is set to all 0s, transmission of the frame begins after 4 bytes (a single write) are stored in the TX FIFO. Each increment of the THRESH bit field waits for an additional 32 bytes of data (eight writes) to be stored in the TX FIFO. Therefore, a value of 0x01 causes the transmitter to wait for 36 bytes of data to be written while a value of 0x02 makes the wait equal to 68 bytes of written data. In general, early transmission starts when:

```
Number of Bytes >= 4 (THRESH x 8 + 1)
```

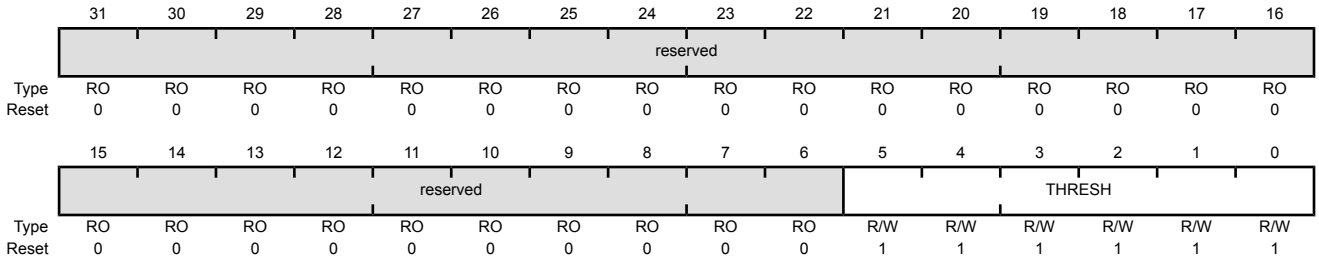
Reaching the threshold level has the same effect as setting the NEWTX bit in the **MACTR** register. Transmission of the frame begins and then the number of bytes indicated by the Data Length field is transmitted. Because under-run checking is not performed, if any event, such as an interrupt, delays the filling of the FIFO, the tail pointer may reach and pass the write pointer in the TX FIFO. In this event, indeterminate values are transmitted rather than the end of the frame. Therefore, sufficient bus bandwidth for writing to the TX FIFO must be guaranteed by the software.

If a frame smaller than the threshold level must be sent, the NEWTX bit in the **MACTR** register must be set with an explicit write. This initiates the transmission of the frame even though the threshold limit has not been reached.

If the threshold level is set too small, it is possible for the transmitter to underrun. If this occurs, the transmit frame is aborted, and a transmit error occurs. Note that in this case, the TXER bit in the **MACRIS** is not set meaning that the CPU receives no indication that a transmit error happened.

Ethernet MAC Threshold (MACTHR)

Base 0x4004.8000
Offset 0x01C
Type R/W, reset 0x0000.003F



Bit/Field	Name	Type	Reset	Description
31:6	reserved	RO	0x0000.00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Bit/Field	Name	Type	Reset	Description
5:0	THRESH	R/W	0x3F	Threshold Value The THRESH bits represent the early transmit threshold. Once the amount of data in the TX FIFO exceeds the value represented by the above equation, transmission of the packet begins.

Register 9: Ethernet MAC Management Control (MACMCTL), offset 0x020

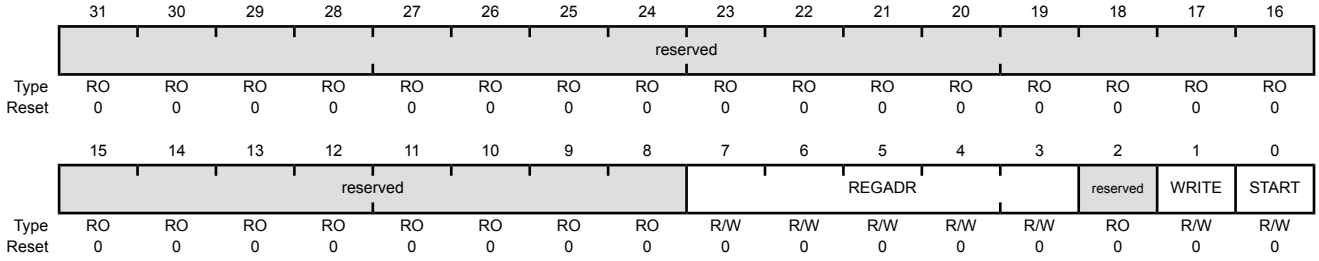
This register enables software to control the transfer of data to and from the MII Management registers in the Ethernet PHY layer. The address, name, type, reset configuration, and functional description of each of these registers can be found in Table 15-2 on page 413 and in "MII Management Register Descriptions" on page 432.

In order to initiate a *read* transaction from the MII Management registers, the WRITE bit must be cleared during the same cycle that the START bit is set.

In order to initiate a *write* transaction to the MII Management registers, the WRITE bit must be set during the same cycle that the START bit is set.

Ethernet MAC Management Control (MACMCTL)

Base 0x4004.8000
Offset 0x020
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x0000.00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:3	REGADR	R/W	0x0	MII Register Address The REGADR bit field represents the MII Management register address for the next MII management interface transaction. Refer to Table 15-2 on page 413 for the PHY register offsets. Note that any address that is not valid in the register map should not be written to and any data read should be ignored.
2	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1	WRITE	R/W	0	MII Register Transaction Type The WRITE bit represents the operation of the next MII management interface transaction. If WRITE is set, the next operation is a write; if WRITE is clear, the next transaction is a read.
0	START	R/W	0	MII Register Transaction Enable The START bit represents the initiation of the next MII management interface transaction. When this bit is set, the MII register located at REGADR is read (WRITE=0) or written (WRITE=1).

Register 10: Ethernet MAC Management Divider (MACMDV), offset 0x024

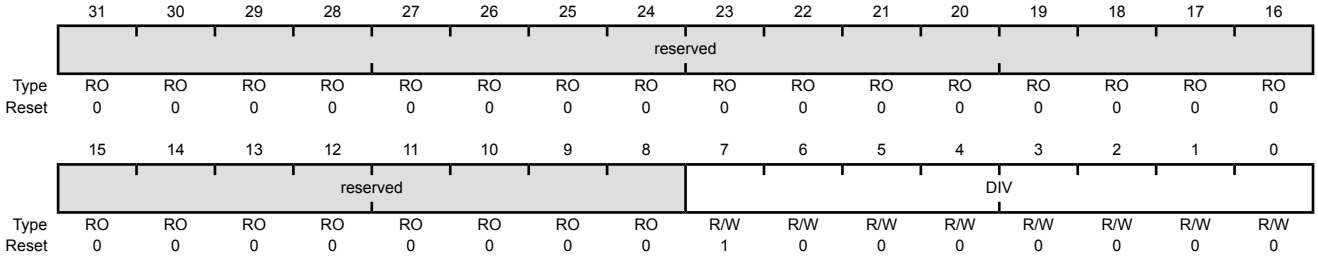
This register enables software to set the clock divider for the Management Data Clock (MDC). This clock is used to synchronize read and write transactions between the system and the MII Management registers. The frequency of the MDC clock can be calculated from the following formula:

$$F_{mdc} = \frac{F_{ipclk}}{2 \times (MACDVR + 1)}$$

The clock divider must be written with a value that ensures that the MDC clock does not exceed a frequency of 2.5 MHz.

Ethernet MAC Management Divider (MACMDV)

Base 0x4004.8000
Offset 0x024
Type R/W, reset 0x0000.0080



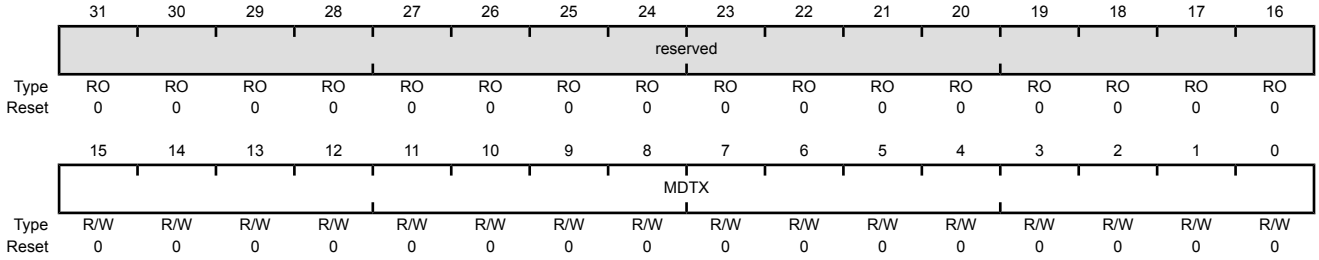
Bit/Field	Name	Type	Reset	Description
31:8	reserved	RO	0x0000.00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:0	DIV	R/W	0x80	Clock Divider The DIV bits are used to set the clock divider for the MDC clock used to transmit data between the MAC and PHY layers over the serial MII interface.

Register 11: Ethernet MAC Management Transmit Data (MACMTXD), offset 0x02C

This register holds the next value to be written to the MII Management registers.

Ethernet MAC Management Transmit Data (MACMTXD)

Base 0x4004.8000
Offset 0x02C
Type R/W, reset 0x0000.0000



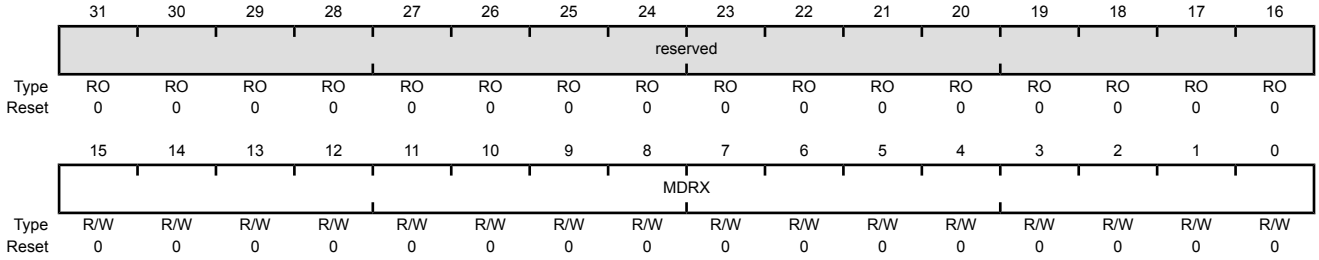
Bit/Field	Name	Type	Reset	Description
31:16	reserved	RO	0x0000	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
15:0	MDTX	R/W	0x0000	MII Register Transmit Data The MDTX bits represent the data that will be written in the next MII management transaction.

Register 12: Ethernet MAC Management Receive Data (MACMRXD), offset 0x030

This register holds the last value read from the MII Management registers.

Ethernet MAC Management Receive Data (MACMRXD)

Base 0x4004.8000
Offset 0x030
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:16	reserved	RO	0x0000	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
15:0	MDRX	R/W	0x0000	MII Register Receive Data The MDRX bits represent the data that was read in the previous MII management transaction.

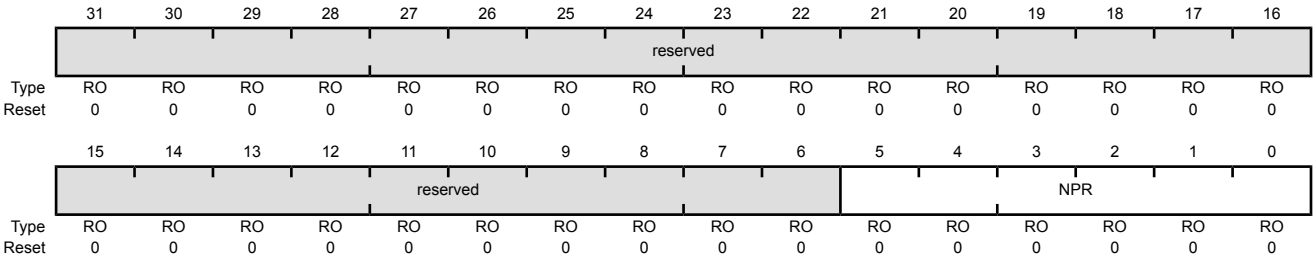
Register 13: Ethernet MAC Number of Packets (MACNP), offset 0x034

This register holds the number of frames that are currently in the RX FIFO. When NPR is 0, there are no frames in the RX FIFO, and the RXINT bit is clear. When NPR is any other value, at least one frame is in the RX FIFO, and the RXINT bit in the **MACRIS** register is set.

Note: The FCS bytes are not included in the NPR value. As a result, the NPR value could be zero before the FCS bytes are read from the FIFO. In addition, a new packet could be received before the NPR value reaches zero. To ensure that the entire packet is received, either use the DriverLib EthernetPacketGet() API or compare the number of bytes received to the Length field from the frame to determine when the packet has been completely read.

Ethernet MAC Number of Packets (MACNP)

Base 0x4004.8000
Offset 0x034
Type RO, reset 0x0000.0000



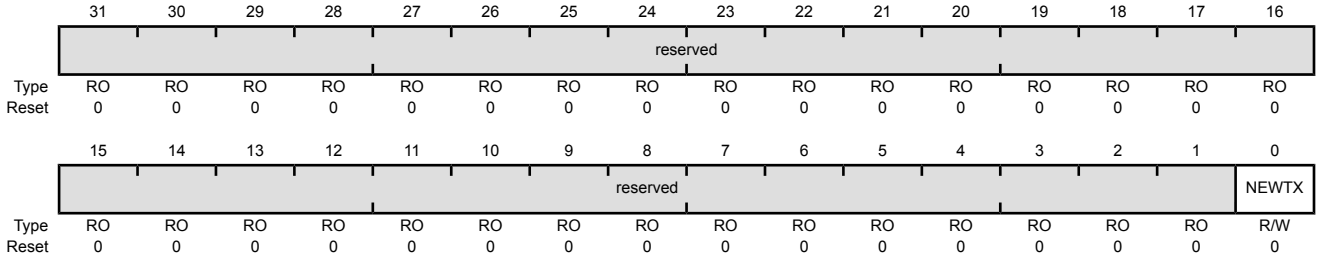
Bit/Field	Name	Type	Reset	Description
31:6	reserved	RO	0x0000.00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
5:0	NPR	RO	0x00	Number of Packets in Receive FIFO The NPR bits represent the number of packets stored in the RX FIFO. While the NPR field is greater than 0, the RXINT interrupt in the **MACRIS** register is set.

Register 14: Ethernet MAC Transmission Request (MACTR), offset 0x038

This register enables software to initiate the transmission of the frame currently located in the TX FIFO. Once the frame has been transmitted from the TX FIFO or a transmission error has been encountered, the NEWTX bit is automatically cleared.

Ethernet MAC Transmission Request (MACTR)

Base 0x4004.8000
Offset 0x038
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:1	reserved	RO	0x0000.000	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
0	NEWTX	R/W	0	New Transmission When set, the NEWTX bit initiates an Ethernet transmission once the packet has been placed in the TX FIFO. This bit is cleared once the transmission has been completed. If early transmission is being used (see the **MACTHR** register), this bit does not need to be set.

15.6 MII Management Register Descriptions

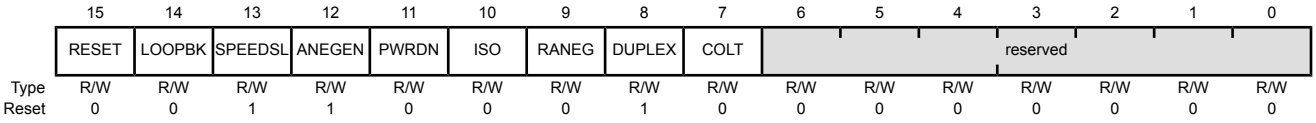
The *IEEE 802.3 standard* specifies a register set for controlling and gathering status from the PHY layer. The registers are collectively known as the MII Management registers. All addresses given are absolute. Addresses not listed are reserved; these addresses should not be written to and any data read should be ignored. Also see “Ethernet MAC Register Descriptions” on page 414.

Register 15: Ethernet PHY Management Register 0 – Control (MR0), address 0x00

This register enables software to configure the operation of the PHY layer. The default settings of these registers are designed to initialize the Ethernet Controller to a normal operational mode without configuration.

Ethernet PHY Management Register 0 – Control (MR0)

Base 0x4004.8000
Address 0x00
Type R/W, reset 0x3100



Bit/Field	Name	Type	Reset	Description
15	RESET	R/W	0	Reset Registers When set, this bit resets the PHY layer registers to their default state and reinitializes internal state machines. Once the reset operation has completed, this bit is cleared by hardware.
14	LOOPBK	R/W	0	Loopback Mode When set, this bit enables the Loopback mode of operation. The receiver ignores external inputs and receives the data that is transmitted by the transmitter.
13	SPEEDSL	R/W	1	Speed Select Value Description 1 Enables the 100 Mb/s mode of operation (100BASE-TX). 0 Enables the 10 Mb/s mode of operation (10BASE-T).
12	ANEGEN	R/W	1	Auto-Negotiation Enable When set, this bit enables the auto-negotiation process.
11	PWRDN	R/W	0	Power Down When set, this bit places the PHY layer into a low-power consuming state. All data on the data inputs is ignored.
10	ISO	R/W	0	Isolate When set, this bit isolates the transmit and receive data paths and ignores all data being transmitted and received.
9	RANEG	R/W	0	Restart Auto-Negotiation When set, this bit restarts the auto-negotiation process. Once the restart has initiated, this bit is cleared by hardware.

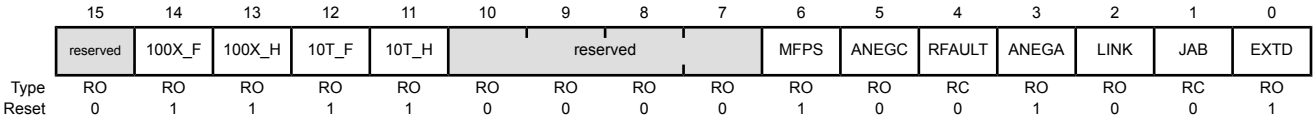
Bit/Field	Name	Type	Reset	Description
8	DUPLEX	R/W	1	Set Duplex Mode Value Description 1 Enables the Full-Duplex mode of operation. This bit can be set by software in a manual configuration process or by the auto-negotiation process. 0 Enables the Half-Duplex mode of operation.
7	COLT	R/W	0	Collision Test When set, this bit enables the Collision Test mode of operation. The COLT bit is set after the initiation of a transmission and is cleared once the transmission is halted.
6:0	reserved	R/W	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation. These bits should always be written as zero.

Register 16: Ethernet PHY Management Register 1 – Status (MR1), address 0x01

This register enables software to determine the capabilities of the PHY layer and perform its initialization and operation appropriately.

Ethernet PHY Management Register 1 – Status (MR1)

Base 0x4004.8000
Address 0x01
Type RO, reset 0x7849



Bit/Field	Name	Type	Reset	Description
15	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
14	100X_F	RO	1	100BASE-TX Full-Duplex Mode When set, this bit indicates that the Ethernet Controller is capable of supporting 100BASE-TX Full-Duplex mode.
13	100X_H	RO	1	100BASE-TX Half-Duplex Mode When set, this bit indicates that the Ethernet Controller is capable of supporting 100BASE-TX Half-Duplex mode.
12	10T_F	RO	1	10BASE-T Full-Duplex Mode When set, this bit indicates that the Ethernet Controller is capable of 10BASE-T Full-Duplex mode.
11	10T_H	RO	1	10BASE-T Half-Duplex Mode When set, this bit indicates that the Ethernet Controller is capable of supporting 10BASE-T Half-Duplex mode.
10:7	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
6	MFPS	RO	1	Management Frames with Preamble Suppressed When set, this bit indicates that the Management Interface is capable of receiving management frames with the preamble suppressed.
5	ANEGC	RO	0	Auto-Negotiation Complete When set, this bit indicates that the auto-negotiation process has been completed and that the extended registers defined by the auto-negotiation protocol are valid.
4	RFAULT	RC	0	Remote Fault When set, this bit indicates that a remote fault condition has been detected. This bit remains set until it is read, even if the condition no longer exists.

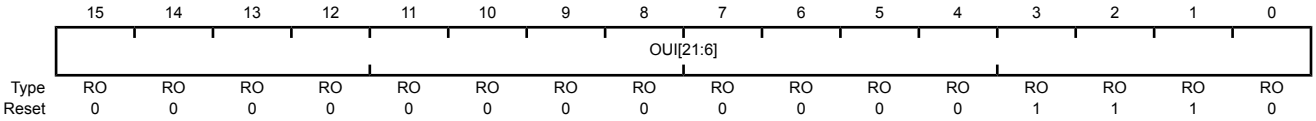
Bit/Field	Name	Type	Reset	Description
3	ANEGA	RO	1	Auto-Negotiation When set, this bit indicates that the Ethernet Controller has the ability to perform auto-negotiation.
2	LINK	RO	0	Link Made When set, this bit indicates that a valid link has been established by the Ethernet Controller.
1	JAB	RC	0	Jabber Condition When set, this bit indicates that a jabber condition has been detected by the Ethernet Controller. This bit remains set until it is read, even if the jabber condition no longer exists.
0	EXTD	RO	1	Extended Capabilities When set, this bit indicates that the Ethernet Controller provides an extended set of capabilities that can be accessed through the extended register set.

Register 17: Ethernet PHY Management Register 2 – PHY Identifier 1 (MR2), address 0x02

This register, along with **MR3**, provides a 32-bit value indicating the manufacturer, model, and revision information.

Ethernet PHY Management Register 2 – PHY Identifier 1 (MR2)

Base 0x4004.8000
Address 0x02
Type RO, reset 0x000E



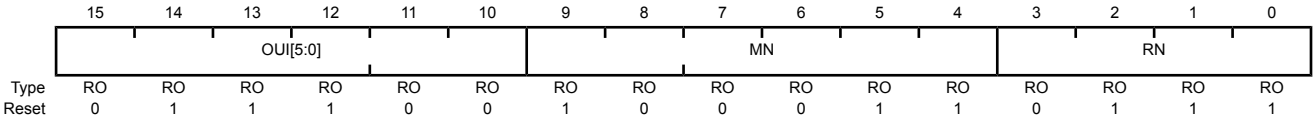
Bit/Field	Name	Type	Reset	Description
15:0	OUI[21:6]	RO	0x000E	Organizationally Unique Identifier[21:6] This field, along with the `OUI[5:0]` field in **MR3**, makes up the Organizationally Unique Identifier indicating the PHY manufacturer.

Register 18: Ethernet PHY Management Register 3 – PHY Identifier 2 (MR3), address 0x03

This register, along with **MR2**, provides a 32-bit value indicating the manufacturer, model, and revision information.

Ethernet PHY Management Register 3 – PHY Identifier 2 (MR3)

Base 0x4004.8000
Address 0x03
Type RO, reset 0x7237



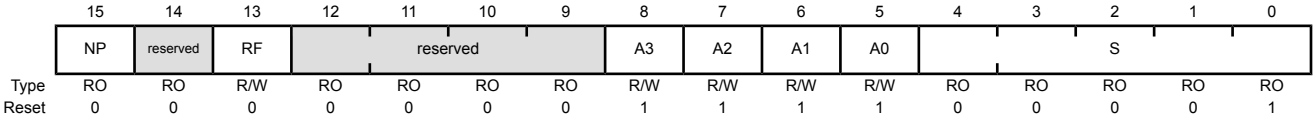
Bit/Field	Name	Type	Reset	Description
15:10	OUI[5:0]	RO	0x1C	Organizationally Unique Identifier[5:0] This field, along with the `OUI[21:6]` field in **MR2**, makes up the Organizationally Unique Identifier indicating the PHY manufacturer.
9:4	MN	RO	0x23	Model Number The `MN` field represents the Model Number of the PHY.
3:0	RN	RO	0x7	Revision Number The `RN` field represents the Revision Number of the PHY implementation.

Register 19: Ethernet PHY Management Register 4 – Auto-Negotiation Advertisement (MR4), address 0x04

This register provides the advertised abilities of the Ethernet Controller used during auto-negotiation. Bits 8:5 represent the Technology Ability Field bits. This field can be overwritten by software to auto-negotiate to an alternate common technology. Writing to this register has no effect until auto-negotiation is re-initiated by setting the RANEG bit in the **MR0** register.

Ethernet PHY Management Register 4 – Auto-Negotiation Advertisement (MR4)

Base 0x4004.8000
Address 0x04
Type R/W, reset 0x01E1



Bit/Field	Name	Type	Reset	Description
15	NP	RO	0	Next Page When set, this bit indicates the Ethernet Controller is capable of Next Page exchanges to provide more detailed information on the PHY layer's capabilities.
14	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
13	RF	R/W	0	Remote Fault When set, this bit indicates to the link partner that a Remote Fault condition has been encountered.
12:9	reserved	RO	0x0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
8	A3	R/W	1	Technology Ability Field[3] When set, this bit indicates that the Ethernet Controller supports the 100Base-TX full-duplex signaling protocol. If software wants to ensure that this mode is not used, this bit can be cleared and auto-negotiation re-initiated with the RANEG bit in the **MR0** register.
7	A2	R/W	1	Technology Ability Field[2] When set, this bit indicates that the Ethernet Controller supports the 100Base-TX half-duplex signaling protocol. If software wants to ensure that this mode is not used, this bit can be cleared and auto-negotiation re-initiated with the RANEG bit in the **MR0** register.
6	A1	R/W	1	Technology Ability Field[1] When set, this bit indicates that the Ethernet Controller supports the 10BASE-T full-duplex signaling protocol. If software wants to ensure that this mode is not used, this bit can be cleared and auto-negotiation re-initiated with the RANEG bit in the **MR0** register..

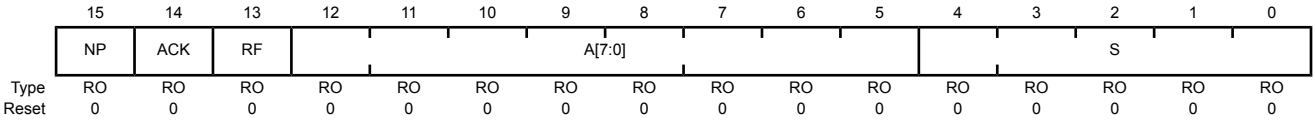
Bit/Field	Name	Type	Reset	Description
5	A0	R/W	1	Technology Ability Field[0] When set, this bit indicates that the Ethernet Controller supports the 10BASE-T half-duplex signaling protocol. If software wants to ensure that this mode is not used, this bit can be cleared and auto-negotiation re-initiated with the RANEG bit in the **MR0** register..
4:0	S	RO	0x1	Selector Field The S field encodes 32 possible messages for communicating between Ethernet Controllers. This field is hard-coded to 0x01, indicating that the Stellaris® Ethernet Controller is *IEEE 802.3* compliant.

Register 20: Ethernet PHY Management Register 5 – Auto-Negotiation Link Partner Base Page Ability (MR5), address 0x05

This register provides the advertised abilities of the link partner's Ethernet Controller that are received and stored during auto-negotiation.

Ethernet PHY Management Register 5 – Auto-Negotiation Link Partner Base Page Ability (MR5)

Base 0x4004.8000
Address 0x05
Type RO, reset 0x0000



Bit/Field	Name	Type	Reset	Description
15	NP	RO	0	Next Page When set, this bit indicates that the link partner's Ethernet Controller is capable of Next page exchanges to provide more detailed information on the Ethernet Controller's capabilities.
14	ACK	RO	0	Acknowledge When set, this bit indicates that the Ethernet Controller has successfully received the link partner's advertised abilities during auto-negotiation.
13	RF	RO	0	Remote Fault Used as a standard transport mechanism for transmitting simple fault information from the link partner.
12:5	A[7:0]	RO	0x00	Technology Ability Field The A[7:0] field encodes individual technologies that are supported by the Ethernet Controller. See the **MR4** register for definitions. Note that bits 12:9 describe functions that are not implemented on the Stellaris® Ethernet Controller. Refer to the IEEE 802.3 standard for definitions.
4:0	S	RO	0x00	Selector Field The S field encodes possible messages for communicating between Ethernet Controllers.

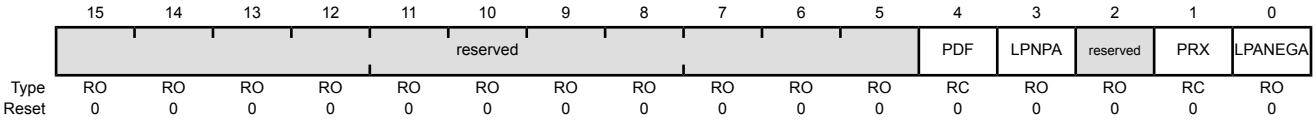
Value	Description
0x00	Reserved
0x01	IEEE Std 802.3
0x02	IEEE Std 802.9 ISLAN-16T
0x03	IEEE Std 802.5
0x04	IEEE Std 1394
0x05–0x1F	Reserved

Register 21: Ethernet PHY Management Register 6 – Auto-Negotiation Expansion (MR6), address 0x06

This register enables software to determine the auto-negotiation and next page capabilities of the Ethernet Controller and the link partner after auto-negotiation.

Ethernet PHY Management Register 6 – Auto-Negotiation Expansion (MR6)

Base 0x4004.8000
Address 0x06
Type RO, reset 0x0000



Bit/Field	Name	Type	Reset	Description
15:5	reserved	RO	0x000	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
4	PDF	RC	0	Parallel Detection Fault When set, this bit indicates that more than one technology has been detected at link up. This bit is cleared when read.
3	LPNPA	RO	0	Link Partner is Next Page Able When set, this bit indicates that the link partner is enabled to support next page.
2	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1	PRX	RC	0	New Page Received When set, this bit indicates that a new page has been received from the link partner and stored. This bit remains set until the register is read.
0	LPANEGA	RO	0	Link Partner is Auto-Negotiation Able When set, this bit indicates that the link partner is enabled to support auto-negotiation.

Register 22: Ethernet PHY Management Register 16 – Vendor-Specific (MR16), address 0x10

This register enables software to configure the operation of vendor-specific modes of the Ethernet Controller.

Ethernet PHY Management Register 16 – Vendor-Specific (MR16)

Base 0x4004.8000
Address 0x10
Type R/W, reset 0x0140

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	RPTR	INPOL	reserved	TXHIM	SQEI	NL10	reserved				APOL	RVSPOL	reserved		PCSBP	RXCC
Type	R/W	R/W0	RO	R/W	R/W	R/W	RO	RO	RO	RO	R/W	R/W	RO	RO	R/W	R/W
Reset	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0	0

Bit/Field	Name	Type	Reset	Description
15	RPTR	R/W	0	Repeater Mode When set, this bit enables the repeater mode of operation. In this mode, full-duplex is not allowed and the Carrier Sense signal only responds to receive activity.
14	INPOL	R/W0	0	Interrupt Polarity Value Description 1 Sets the polarity of the PHY interrupt to be active High. 0 Sets the polarity of the PHY interrupt to active Low. **Important:** Because the Media Access Controller expects active Low interrupts from the PHY, this bit must always be written with a 0 to ensure proper operation.
13	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
12	TXHIM	R/W	0	Transmit High Impedance Mode When set, this bit enables the transmitter High Impedance mode. In this mode, the TXOP and TXON transmitter pins are put into a high impedance state. The RXIP and RXIN pins remain fully functional.
11	SQEI	R/W	0	SQE Inhibit Testing When set, this bit prohibits 10BASE-T SQE testing. When clear, the SQE testing is performed by generating a collision pulse following the completion of the transmission of a frame.
10	NL10	R/W	0	Natural Loopback Mode When set, this bit enables the 10BASE-T Natural Loopback mode. In this mode, the transmission data received by the Ethernet Controller is looped back onto the receive data path when 10BASE-T mode is enabled.
9:6	reserved	RO	0x5	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Bit/Field	Name	Type	Reset	Description
5	APOL	R/W	0	Auto-Polarity Disable When set, this bit disables the Ethernet Controller's auto-polarity function. If this bit is clear, the Ethernet Controller automatically inverts the received signal due to a wrong polarity connection during auto-negotiation when in 10BASE-T mode.
4	RVSPOL	R/W	0	Receive Data Polarity This bit indicates whether the receive data pulses are being inverted. If the APOL bit is 0, then the RVSPOL bit is read-only and indicates whether the auto-polarity circuitry is reversing the polarity. In this case, if RVSPOL is set, it indicates that the receive data is inverted; if RVSPOL is clear, it indicates that the receive data is not inverted. If the APOL bit is 1, then the RVSPOL bit is writable and software can force the receive data to be inverted. Setting RVSPOL to 1 forces the receive data to be inverted; clearing RVSPOL does not invert the receive data.
3:2	reserved	RO	0x0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1	PCSBP	R/W	0	PCS Bypass When set, this bit enables the bypass of the PCS and scrambling/descrambling functions in 100BASE-TX mode. This mode is only valid when auto-negotiation is disabled and 100BASE-TX mode is enabled.
0	RXCC	R/W	0	Receive Clock Control When set, this bit enables the Receive Clock Control power saving mode if the Ethernet Controller is configured in 100BASE-TX mode. This mode shuts down the receive clock when no data is being received to save power. This mode should not be used when PCSBP is enabled and is automatically disabled when the LOOPBK bit in the **MR0** register is set.

Register 23: Ethernet PHY Management Register 17 – Interrupt Control/Status (MR17), address 0x11

This register provides the means for controlling and observing the events which trigger a PHY layer interrupt in the **MACRIS** register. This register can also be used in a polling mode via the Media Independent Interface as a means to observe key events within the PHY layer via one register address. Bits 0 through 7 are status bits which are each set based on an event. These bits are cleared after the register is read. Bits 8 through 15 of this register, when set, enable the corresponding bit in the lower byte to signal a PHY layer interrupt in the **MACRIS** register.

Ethernet PHY Management Register 17 – Interrupt Control/Status (MR17)

Base 0x4004.8000
Address 0x11
Type R/W, reset 0x0000

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	JABBER_IE	RXER_IE	PRX_IE	PDF_IE	LPACK_IE	LSCHG_IE	RFAULT_IE	ANEGCOMP_IE	JABBER_INT	RXER_INT	PRX_INT	PDF_INT	LPACK_INT	LSCHG_INT	RFAULT_INT	ANEGCOMP_INT
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	RC	RC	RC	RC	RC	RC	RC	RC
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit/Field	Name	Type	Reset	Description
15	JABBER_IE	R/W	0	Jabber Interrupt Enable When set, this bit enables system interrupts when a Jabber condition is detected by the Ethernet Controller.
14	RXER_IE	R/W	0	Receive Error Interrupt Enable When set, this bit enables system interrupts when a receive error is detected by the Ethernet Controller.
13	PRX_IE	R/W	0	Page Received Interrupt Enable When set, this bit enables system interrupts when a new page is received by the Ethernet Controller.
12	PDF_IE	R/W	0	Parallel Detection Fault Interrupt Enable When set, this bit enables system interrupts when a Parallel Detection Fault is detected by the Ethernet Controller.
11	LPACK_IE	R/W	0	LP Acknowledge Interrupt Enable When set, this bit enables system interrupts when FLP bursts are received with the ACK bit in the **MR5** register during auto-negotiation.
10	LSCHG_IE	R/W	0	Link Status Change Interrupt Enable When set, this bit enables system interrupts when the link status changes from OK to FAIL.
9	RFAULT_IE	R/W	0	Remote Fault Interrupt Enable When set, this bit enables system interrupts when a remote fault condition is signaled by the link partner.
8	ANEGCOMP_IE	R/W	0	Auto-Negotiation Complete Interrupt Enable When set, this bit enables system interrupts when the auto-negotiation sequence has completed successfully.

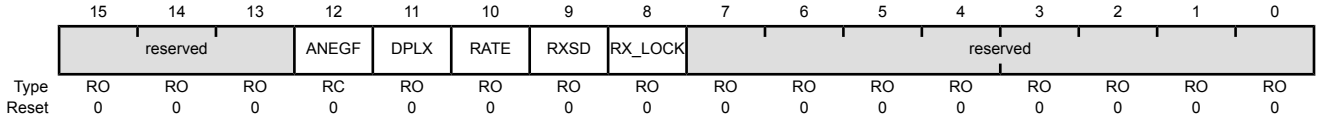
Bit/Field	Name	Type	Reset	Description
7	JABBER_INT	RC	0	Jabber Event Interrupt When set, this bit indicates that a Jabber event has been detected by the 10BASE-T circuitry.
6	RXER_INT	RC	0	Receive Error Interrupt When set, this bit indicates that a receive error has been detected by the Ethernet Controller.
5	PRX_INT	RC	0	Page Receive Interrupt When set, this bit indicates that a new page has been received from the link partner during auto-negotiation.
4	PDF_INT	RC	0	Parallel Detection Fault Interrupt When set, this bit indicates that a parallel detection fault has been detected by the Ethernet Controller during the auto-negotiation process.
3	LPACK_INT	RC	0	LP Acknowledge Interrupt When set, this bit indicates that an FLP burst has been received with the ACK bit set in the **MR5** register during auto-negotiation.
2	LSCHG_INT	RC	0	Link Status Change Interrupt When set, this bit indicates that the link status has changed from OK to FAIL.
1	RFAULT_INT	RC	0	Remote Fault Interrupt When set, this bit indicates that a remote fault condition has been signaled by the link partner.
0	ANEGCOMP_INT	RC	0	Auto-Negotiation Complete Interrupt When set, this bit indicates that the auto-negotiation sequence has completed successfully.

Register 24: Ethernet PHY Management Register 18 – Diagnostic (MR18), address 0x12

This register enables software to diagnose the results of the previous auto-negotiation.

Ethernet PHY Management Register 18 – Diagnostic (MR18)

Base 0x4004.8000
Address 0x12
Type RO, reset 0x0000



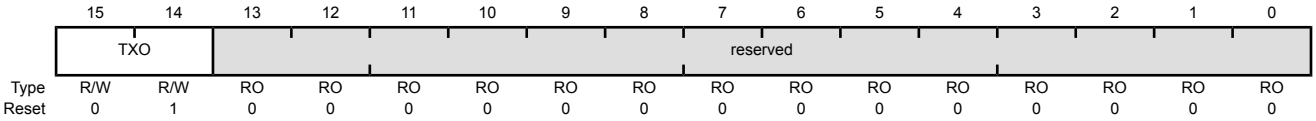
Bit/Field	Name	Type	Reset	Description
15:13	reserved	RO	0x0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
12	ANEGF	RC	0	Auto-Negotiation Failure When set, this bit indicates that no common technology was found during auto-negotiation and auto-negotiation has failed. This bit remains set until read.
11	DPLX	RO	0	Duplex Mode When set, this bit indicates that Full-Duplex was the highest common denominator found during the auto-negotiation process. Otherwise, Half-Duplex was the highest common denominator found.
10	RATE	RO	0	Rate When set, this bit indicates that 100BASE-TX was the highest common denominator found during the auto-negotiation process. Otherwise, 10BASE-T was the highest common denominator found.
9	RXSD	RO	0	Receive Detection When set, this bit indicates that receive signal detection has occurred (in 100BASE-TX mode) or that Manchester-encoded data has been detected (in 10BASE-T mode).
8	RX_LOCK	RO	0	Receive PLL Lock When set, this bit indicates that the Receive PLL has locked onto the receive signal for the selected speed of operation (10BASE-T or 100BASE-TX).
7:0	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Register 25: Ethernet PHY Management Register 19 – Transceiver Control (MR19), address 0x13

This register enables software to set the gain of the transmit output to compensate for transformer loss.

Ethernet PHY Management Register 19 – Transceiver Control (MR19)

Base 0x4004.8000
Address 0x13
Type R/W, reset 0x4000



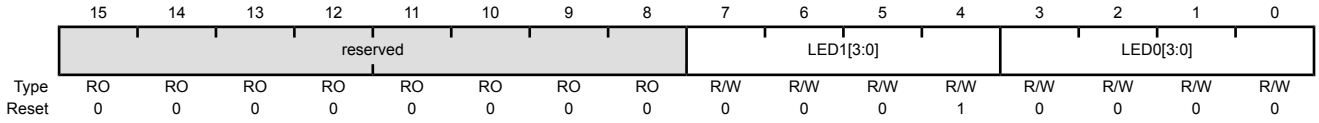
Bit/Field	Name	Type	Reset	Description
15:14	TXO	R/W	0x1	Transmit Amplitude Selection The TX0 field sets the transmit output amplitude to account for transmit transformer insertion loss. Value Description 0x0 Gain set for 0.0dB of insertion loss 0x1 Gain set for 0.4dB of insertion loss 0x2 Gain set for 0.8dB of insertion loss 0x3 Gain set for 1.2dB of insertion loss
13:0	reserved	RO	0x000	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

Register 26: Ethernet PHY Management Register 23 – LED Configuration (MR23), address 0x17

This register enables software to select the source that causes the LED1 and LED0 signals to toggle.

Ethernet PHY Management Register 23 – LED Configuration (MR23)

Base 0x4004.8000
Address 0x17
Type R/W, reset 0x0010



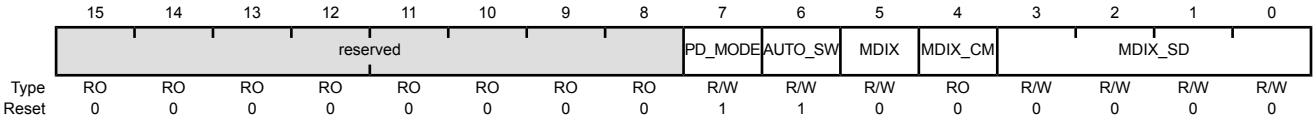
Bit/Field	Name	Type	Reset	Description
15:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7:4	LED1[3:0]	R/W	0x1	LED1 Source The LED1 field selects the source that toggles the LED1 signal. Value Description 0x0 Link OK 0x1 RX or TX Activity (Default LED1) 0x2 Reserved 0x3 Reserved 0x4 Reserved 0x5 100BASE-TX mode 0x6 10BASE-T mode 0x7 Full-Duplex 0x8 Link OK & Blink=RX or TX Activity
3:0	LED0[3:0]	R/W	0x0	LED0 Source The LED0 field selects the source that toggles the LED0 signal. Value Description 0x0 Link OK (Default LED0) 0x1 RX or TX Activity 0x2 Reserved 0x3 Reserved 0x4 Reserved 0x5 100BASE-TX mode 0x6 10BASE-T mode 0x7 Full-Duplex 0x8 Link OK & Blink=RX or TX Activity

Register 27: Ethernet PHY Management Register 24 –MDI/MDIX Control (MR24), address 0x18

This register enables software to control the behavior of the MDI/MDIX mux and its switching capabilities.

Ethernet PHY Management Register 24 –MDI/MDIX Control (MR24)

Base 0x4004.8000
Address 0x18
Type R/W, reset 0x00C0



Bit/Field	Name	Type	Reset	Description
15:8	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7	PD_MODE	R/W	1	Parallel Detection Mode When set, enables the Parallel Detection mode and allows auto-switching to work when auto-negotiation is not enabled.
6	AUTO_SW	R/W	1	Auto-Switching Enable When set, enables Auto-Switching of the MDI/MDIX mux.
5	MDIX	R/W	0	Auto-Switching Configuration When set, indicates that the MDI/MDIX mux is in the crossover (MDIX) configuration. When 0, it indicates that the mux is in the pass-through (MDI) configuration. When the AUTO_SW bit is 1, the MDIX bit is read-only. When the AUTO_SW bit is 0, the MDIX bit is read/write and can be configured manually.
4	MDIX_CM	RO	0	Auto-Switching Complete When set, indicates that the auto-switching sequence has completed. If 0, it indicates that the sequence has not completed or that auto-switching is disabled.
3:0	MDIX_SD	R/W	0x0	Auto-Switching Seed This field provides the initial seed for the switching algorithm. This seed directly affects the number of attempts [5,4] respectively to write bits [3:0]. A 0 sets the seed to 0x5.

16 Analog Comparators

An analog comparator is a peripheral that compares two analog voltages, and provides a logical output that signals the comparison result.

Note: Not all comparators have the option to drive an output pin.

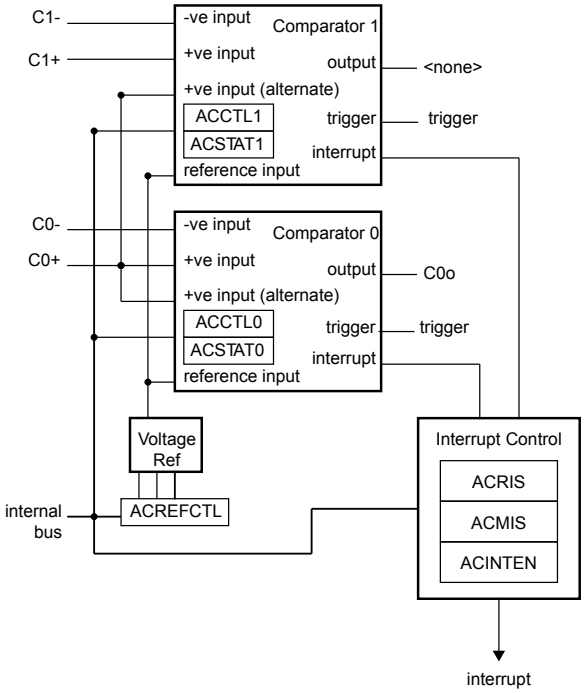
The comparator can provide its output to a device pin, acting as a replacement for an analog comparator on the board, or it can be used to signal the application via interrupts or triggers to the ADC to cause it to start capturing a sample sequence. The interrupt generation and ADC triggering logic is separate. This means, for example, that an interrupt can be generated on a rising edge and the ADC triggered on a falling edge.

The Stellaris® Analog Comparators module has the following features:

- Two independent integrated analog comparators
- Configurable for output to drive an output pin, generate an interrupt, or initiate an ADC sample sequence
- Compare external pin input to external pin input or to internal programmable voltage reference
- Compare a test voltage against any one of these voltages
  - An individual external reference voltage
  - A shared single external reference voltage
  - A shared internal reference voltage

16.1 Block Diagram

Figure 16-1. Analog Comparator Module Block Diagram



16.2 Functional Description

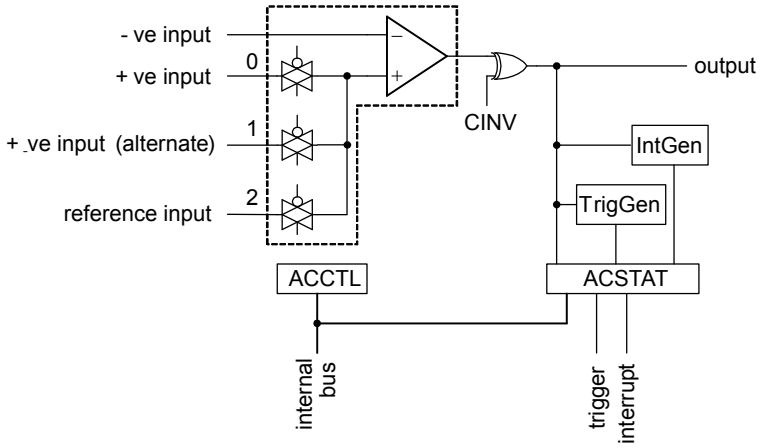
Important: It is recommended that the Digital-Input enable (the GPIODEN bit in the GPIO module) for the analog input pin be disabled to prevent excessive current draw from the I/O pads.

The comparator compares the VIN- and VIN+ inputs to produce an output, VOUT.

```
VIN- < VIN+, VOUT = 1
VIN- > VIN+, VOUT = 0
```

As shown in Figure 16-2 on page 453, the input source for VIN- is an external input. In addition to an external input, input sources for VIN+ can be the +ve input of comparator 0 or an internal reference.

Figure 16-2. Structure of Comparator Unit



A comparator is configured through two status/control registers (**ACCTL** and **ACSTAT**). The internal reference is configured through one control register (**ACREFCTL**). Interrupt status and control is configured through three registers (**ACMIS**, **ACRIS**, and **ACINTEN**).

Typically, the comparator output is used internally to generate controller interrupts. It may also be used to drive an external pin or generate an analog-to-digital converter (ADC) trigger.

Important: The ASRCP bits in the **ACCTLn** register must be set before using the analog comparators.

16.2.1 Internal Reference Programming

The structure of the internal reference is shown in Figure 16-3 on page 453. This is controlled by a single configuration register (**ACREFCTL**). Table 16-1 on page 453 shows the programming options to develop specific internal reference values, to compare an external voltage against a particular voltage generated internally.

Figure 16-3. Comparator Internal Reference Structure

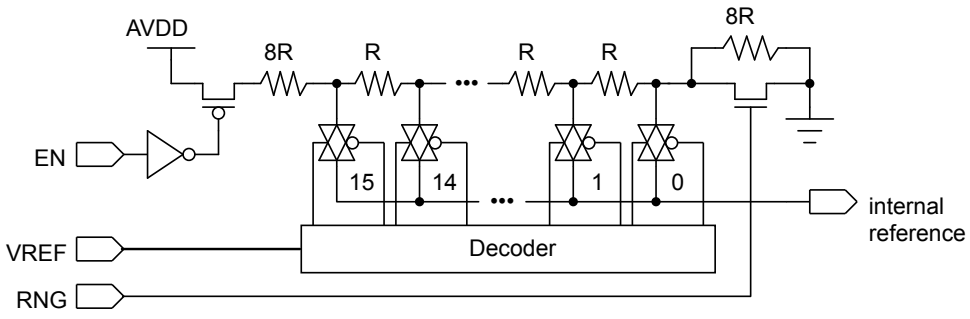


Table 16-1. Internal Reference Voltage and ACREFCTL Field Values

ACREFCTL Register		Output Reference Voltage Based on VREF Field Value
EN Bit Value	RNG Bit Value	
EN=0	RNG=X	0 V (GND) for any value of VREF; however, it is recommended that RNG=1 and VREF=0 for the least noisy ground reference.

Table 16-1. Internal Reference Voltage and ACREFCTL Field Values *(continued)*

ACREFCTL Register		Output Reference Voltage Based on VREF Field Value
EN Bit Value	**RNG Bit Value**	
EN=1	RNG=0	Total resistance in ladder is 31 R. $V_{REF} = AV_{DD} \times \frac{R_{VREF}}{R_T}$ $V_{REF} = AV_{DD} \times \frac{(VREF + 8)}{31}$ $V_{REF} = 0.85 + 0.106 \times VREF$ The range of internal reference in this mode is 0.85-2.448 V.
	RNG=1	Total resistance in ladder is 23 R. $V_{REF} = AV_{DD} \times \frac{R_{VREF}}{R_T}$ $V_{REF} = AV_{DD} \times \frac{VREF}{23}$ $V_{REF} = 0.143 \times VREF$ The range of internal reference for this mode is 0-2.152 V.

16.3 Initialization and Configuration

The following example shows how to configure an analog comparator to read back its output value from an internal register.

1. Enable the analog comparator 0 clock by writing a value of 0x0010.0000 to the **RCGC1** register in the System Control module.
2. In the GPIO module, enable the GPIO port/pin associated with C0- as a GPIO input.
3. Configure the internal voltage reference to 1.65 V by writing the **ACREFCTL** register with the value 0x0000.030C.
4. Configure comparator 0 to use the internal voltage reference and to *not* invert the output by writing the **ACCTL0** register with the value of 0x0000.040C.
5. Delay for some time.
6. Read the comparator output value by reading the **ACSTAT0** register's OVAL value.

Change the level of the signal input on C0- to see the OVAL value change.

16.4 Register Map

Table 16-2 on page 455 lists the comparator registers. The offset listed is a hexadecimal increment to the register's address, relative to the Analog Comparator base address of 0x4003.C000.

Table 16-2. Analog Comparators Register Map

Offset	Name	Type	Reset	Description	See page
0x000	ACMIS	R/W1C	0x0000.0000	Analog Comparator Masked Interrupt Status	456
0x004	ACRIS	RO	0x0000.0000	Analog Comparator Raw Interrupt Status	457
0x008	ACINTEN	R/W	0x0000.0000	Analog Comparator Interrupt Enable	458
0x010	ACREFCTL	R/W	0x0000.0000	Analog Comparator Reference Voltage Control	459
0x020	ACSTAT0	RO	0x0000.0000	Analog Comparator Status 0	460
0x024	ACCTL0	R/W	0x0000.0000	Analog Comparator Control 0	461
0x040	ACSTAT1	RO	0x0000.0000	Analog Comparator Status 1	460
0x044	ACCTL1	R/W	0x0000.0000	Analog Comparator Control 1	461

16.5 Register Descriptions

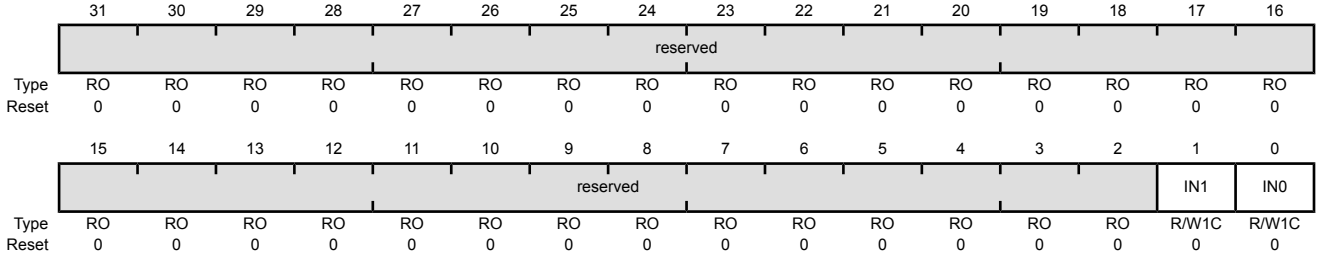
The remainder of this section lists and describes the Analog Comparator registers, in numerical order by address offset.

Register 1: Analog Comparator Masked Interrupt Status (ACMIS), offset 0x000

This register provides a summary of the interrupt status (masked) of the comparators.

Analog Comparator Masked Interrupt Status (ACMIS)

Base 0x4003.C000
Offset 0x000
Type R/W1C, reset 0x0000.0000



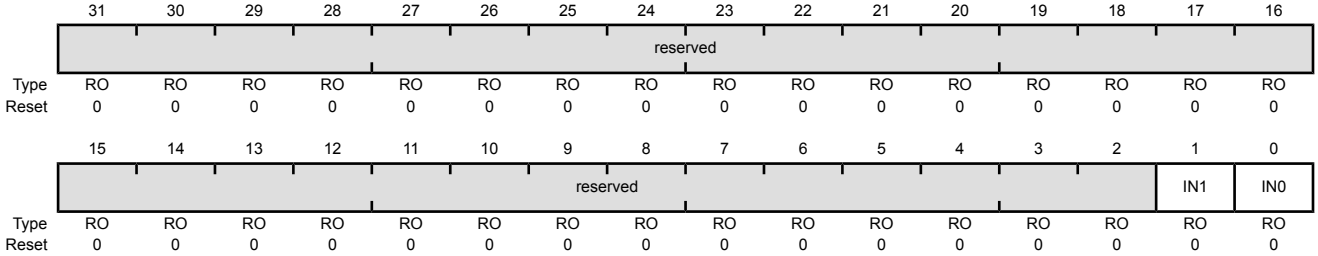
Bit/Field	Name	Type	Reset	Description
31:2	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1	IN1	R/W1C	0	Comparator 1 Masked Interrupt Status Gives the masked interrupt state of this interrupt. Write 1 to this bit to clear the pending interrupt.
0	IN0	R/W1C	0	Comparator 0 Masked Interrupt Status Gives the masked interrupt state of this interrupt. Write 1 to this bit to clear the pending interrupt.

Register 2: Analog Comparator Raw Interrupt Status (ACRIS), offset 0x004

This register provides a summary of the interrupt status (raw) of the comparators.

Analog Comparator Raw Interrupt Status (ACRIS)

Base 0x4003.C000
Offset 0x004
Type RO, reset 0x0000.0000



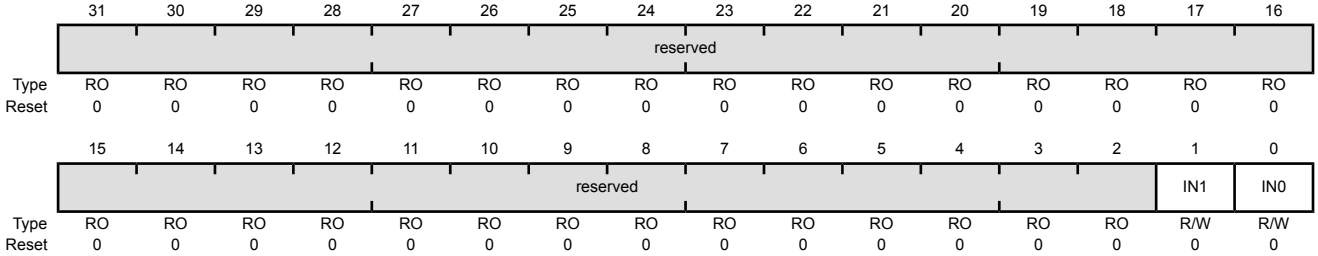
Bit/Field	Name	Type	Reset	Description
31:2	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1	IN1	RO	0	Comparator 1 Interrupt Status When set, indicates that an interrupt has been generated by comparator 1.
0	IN0	RO	0	Comparator 0 Interrupt Status When set, indicates that an interrupt has been generated by comparator 0.

Register 3: Analog Comparator Interrupt Enable (ACINTEN), offset 0x008

This register provides the interrupt enable for the comparators.

Analog Comparator Interrupt Enable (ACINTEN)

Base 0x4003.C000
Offset 0x008
Type R/W, reset 0x0000.0000



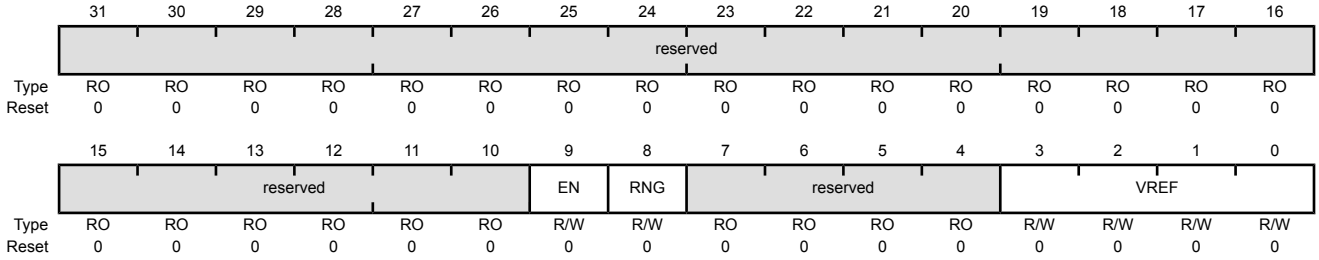
Bit/Field	Name	Type	Reset	Description
31:2	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1	IN1	R/W	0	Comparator 1 Interrupt Enable When set, enables the controller interrupt from the comparator 1 output.
0	IN0	R/W	0	Comparator 0 Interrupt Enable When set, enables the controller interrupt from the comparator 0 output.

Register 4: Analog Comparator Reference Voltage Control (ACREFCTL), offset 0x010

This register specifies whether the resistor ladder is powered on as well as the range and tap.

Analog Comparator Reference Voltage Control (ACREFCTL)

Base 0x4003.C000
Offset 0x010
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:10	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
9	EN	R/W	0	Resistor Ladder Enable The EN bit specifies whether the resistor ladder is powered on. If 0, the resistor ladder is unpowered. If 1, the resistor ladder is connected to the analog V_{DD}. This bit is reset to 0 so that the internal reference consumes the least amount of power if not used and programmed.
8	RNG	R/W	0	Resistor Ladder Range The RNG bit specifies the range of the resistor ladder. If 0, the resistor ladder has a total resistance of 31 R. If 1, the resistor ladder has a total resistance of 23 R.
7:4	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
3:0	VREF	R/W	0x00	Resistor Ladder Voltage Ref The VREF bit field specifies the resistor ladder tap that is passed through an analog multiplexer. The voltage corresponding to the tap position is the internal reference voltage available for comparison. See Table 16-1 on page 453 for some output reference voltage examples.

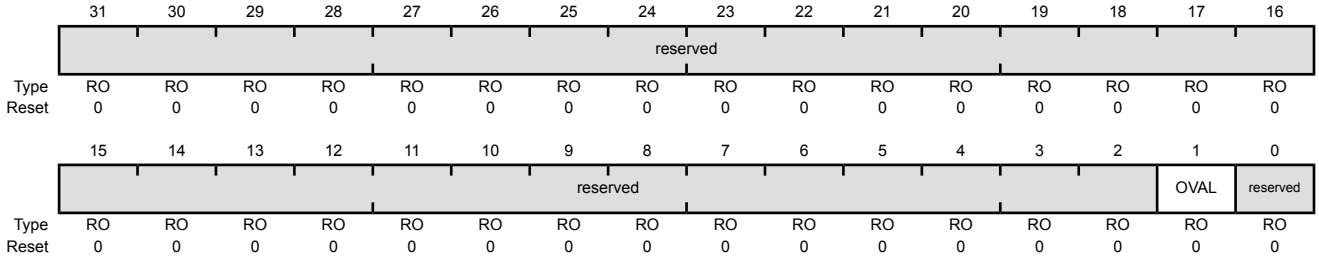
Register 5: Analog Comparator Status 0 (ACSTAT0), offset 0x020

Register 6: Analog Comparator Status 1 (ACSTAT1), offset 0x040

These registers specify the current output value of the comparator.

Analog Comparator Status 0 (ACSTAT0)

Base 0x4003.C000
Offset 0x020
Type RO, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:2	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1	OVAL	RO	0	Comparator Output Value The 0VAL bit specifies the current output value of the comparator.
0	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

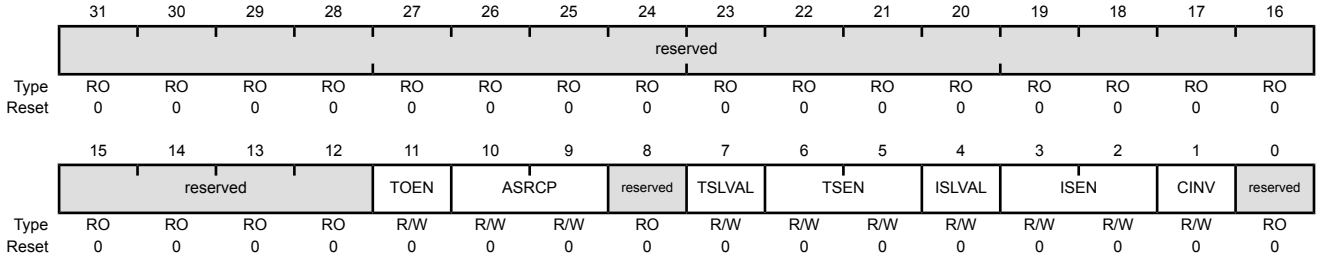
Register 7: Analog Comparator Control 0 (ACCTL0), offset 0x024

Register 8: Analog Comparator Control 1 (ACCTL1), offset 0x044

These registers configure the comparator's input and output.

Analog Comparator Control 0 (ACCTL0)

Base 0x4003.C000
Offset 0x024
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:12	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
11	TOEN	R/W	0	Trigger Output Enable The TOEN bit enables the ADC event transmission to the ADC. If 0, the event is suppressed and not sent to the ADC. If 1, the event is transmitted to the ADC.
10:9	ASRCP	R/W	0x00	Analog Source Positive The ASRCP field specifies the source of input voltage to the VIN+ terminal of the comparator. The encodings for this field are as follows: Value Function 0x0 Pin value 0x1 Pin value of C0+ 0x2 Internal voltage reference 0x3 Reserved
8	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
7	TSLVAL	R/W	0	Trigger Sense Level Value The TSLVAL bit specifies the sense value of the input that generates an ADC event if in Level Sense mode. If 0, an ADC event is generated if the comparator output is Low. Otherwise, an ADC event is generated if the comparator output is High.

Bit/Field	Name	Type	Reset	Description
6:5	TSEN	R/W	0x0	Trigger Sense The TSEN field specifies the sense of the comparator output that generates an ADC event. The sense conditioning is as follows: Value Function 0x0 Level sense, see TSLVAL 0x1 Falling edge 0x2 Rising edge 0x3 Either edge
4	ISLVAL	R/W	0	Interrupt Sense Level Value The ISLVAL bit specifies the sense value of the input that generates an interrupt if in Level Sense mode. If 0, an interrupt is generated if the comparator output is Low. Otherwise, an interrupt is generated if the comparator output is High.
3:2	ISEN	R/W	0x0	Interrupt Sense The ISEN field specifies the sense of the comparator output that generates an interrupt. The sense conditioning is as follows: Value Function 0x0 Level sense, see ISLVAL 0x1 Falling edge 0x2 Rising edge 0x3 Either edge
1	CINV	R/W	0	Comparator Output Invert The CINV bit conditionally inverts the output of the comparator. If 0, the output of the comparator is unchanged. If 1, the output of the comparator is inverted prior to being processed by hardware.
0	reserved	RO	0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.

17 Pulse Width Modulator (PWM)

Pulse width modulation (PWM) is a powerful technique for digitally encoding analog signal levels. High-resolution counters are used to generate a square wave, and the duty cycle of the square wave is modulated to encode an analog signal. Typical applications include switching power supplies and motor control.

The Stellaris® PWM module consists of one PWM generator block and a control block. The control block determines the polarity of the PWM signals, and which signals are passed through to the pins.

The PWM generator block produces two PWM signals that can either be independent signals (other than being based on the same timer and therefore having the same frequency) or a single pair of complementary signals with dead-band delays inserted. The output of the PWM generation block is managed by the output control block before being passed to the device pins.

The Stellaris® PWM module provides a great deal of flexibility. It can generate simple PWM signals, such as those required by a simple charge pump. It can also generate paired PWM signals with dead-band delays, such as those required by a half-H bridge driver.

The Stellaris® PWM module has the following features:

- One PWM generator blocks, each with one 16-bit counter, two PWM comparators, a PWM signal generator, a dead-band generator, and an interrupt/ADC-trigger selector
- One fault input in hardware to promote low-latency shutdown
- One 16-bit counter
  - Runs in Down or Up/Down mode
  - Output frequency controlled by a 16-bit load value
  - Load value updates can be synchronized
  - Produces output signals at zero and load value
- Two PWM comparators
  - Comparator value updates can be synchronized
  - Produces output signals on match
- PWM generator
  - Output PWM signal is constructed based on actions taken as a result of the counter and PWM comparator output signals
  - Produces two independent PWM signals
- Dead-band generator
  - Produces two PWM signals with programmable dead-band delays suitable for driving a half-H bridge
  - Can be bypassed, leaving input PWM signals unmodified

- Flexible output control block with PWM output enable of each PWM signal
  - PWM output enable of each PWM signal
  - Optional output inversion of each PWM signal (polarity control)
  - Optional fault handling for each PWM signal
  - Synchronization of timers in the PWM generator blocks
  - Synchronization of timer/comparator updates across the PWM generator blocks
  - Interrupt status summary of the PWM generator blocks
- Can initiate an ADC sample sequence

17.1 Block Diagram

Figure 17-1 on page 464 provides the Stellaris® PWM module unit diagram and Figure 17-2 on page 465 provides a more detailed diagram of a Stellaris® PWM generator. The LM3S6432 controller contains one generator block (PWM0) and generates two independent PWM signals or one paired PWM signal with dead-band delays inserted.

Figure 17-1. PWM Unit Diagram

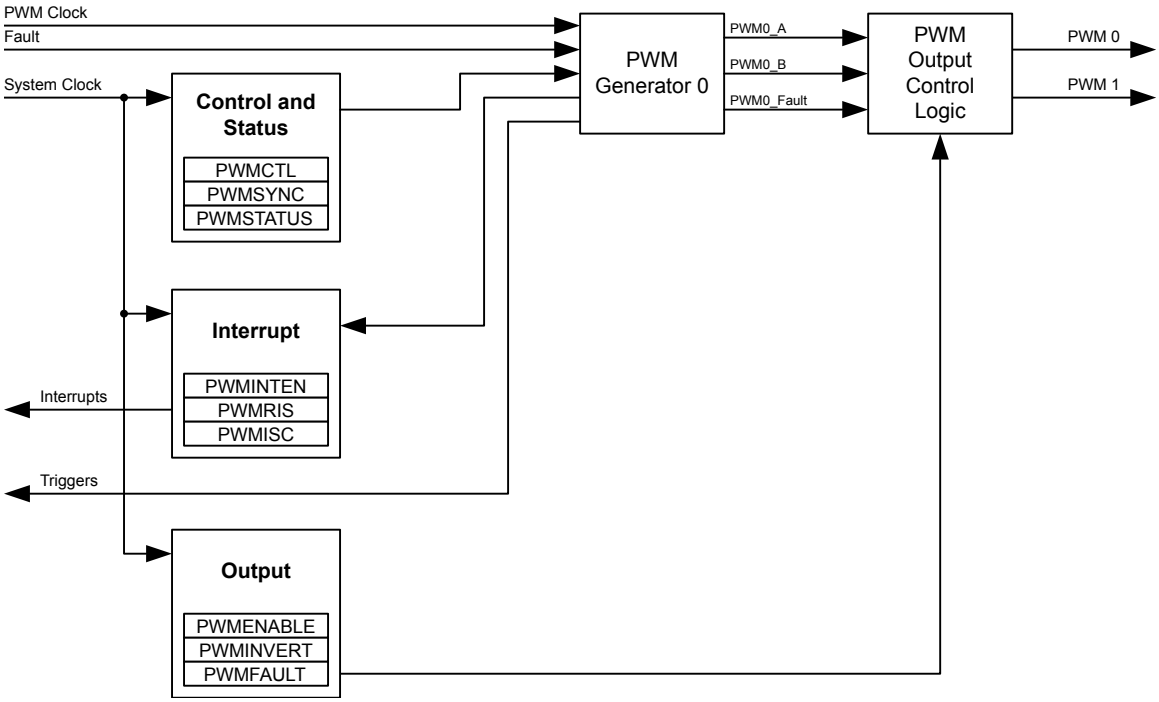
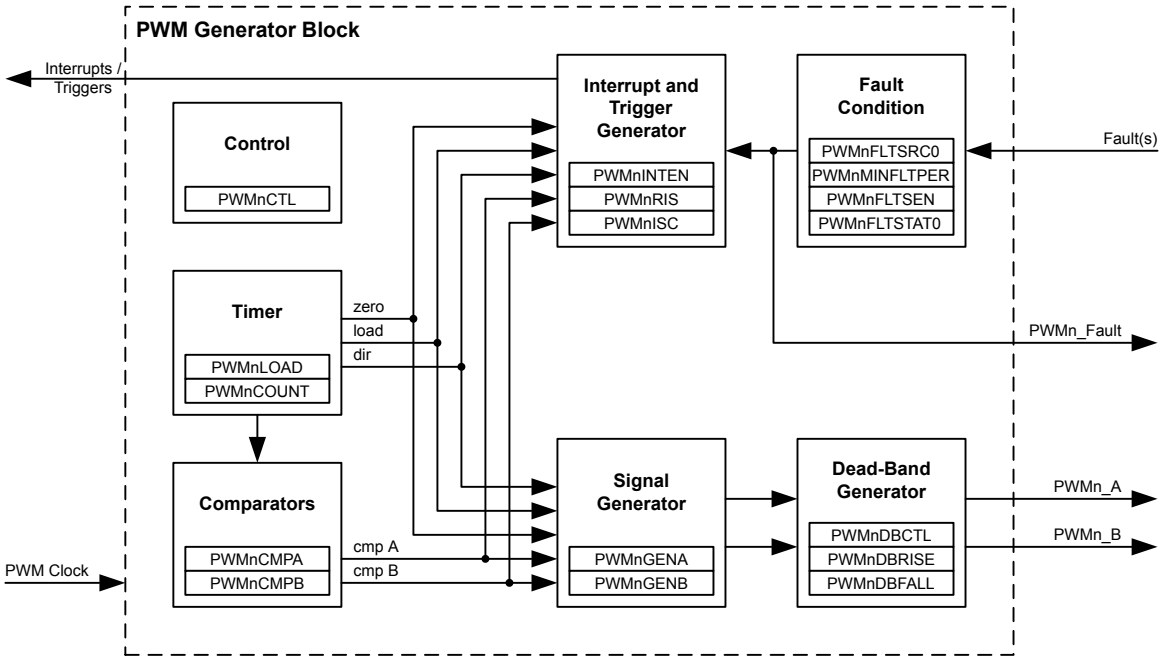


Figure 17-2. PWM Module Block Diagram



17.2 Functional Description

17.2.1 PWM Timer

The timer runs in one of two modes: Count-Down mode or Count-Up/Down mode. In Count-Down mode, the timer counts from the load value to zero, goes back to the load value, and continues counting down. In Count-Up/Down mode, the timer counts from zero up to the load value, back down to zero, back up to the load value, and so on. Generally, Count-Down mode is used for generating left- or right-aligned PWM signals, while the Count-Up/Down mode is used for generating center-aligned PWM signals.

The timers output three signals that are used in the PWM generation process: the direction signal (this is always Low in Count-Down mode, but alternates between Low and High in Count-Up/Down mode), a single-clock-cycle-width High pulse when the counter is zero, and a single-clock-cycle-width High pulse when the counter is equal to the load value. Note that in Count-Down mode, the zero pulse is immediately followed by the load pulse.

17.2.2 PWM Comparators

There are two comparators in the PWM generator that monitor the value of the counter; when either match the counter, they output a single-clock-cycle-width High pulse. When in Count-Up/Down mode, these comparators match both when counting up and when counting down; they are therefore qualified by the counter direction signal. These qualified pulses are used in the PWM generation process. If either comparator match value is greater than the counter load value, then that comparator never outputs a High pulse.

Figure 17-3 on page 466 shows the behavior of the counter and the relationship of these pulses when the counter is in Count-Down mode. Figure 17-4 on page 466 shows the behavior of the counter and the relationship of these pulses when the counter is in Count-Up/Down mode.

Figure 17-3. PWM Count-Down Mode

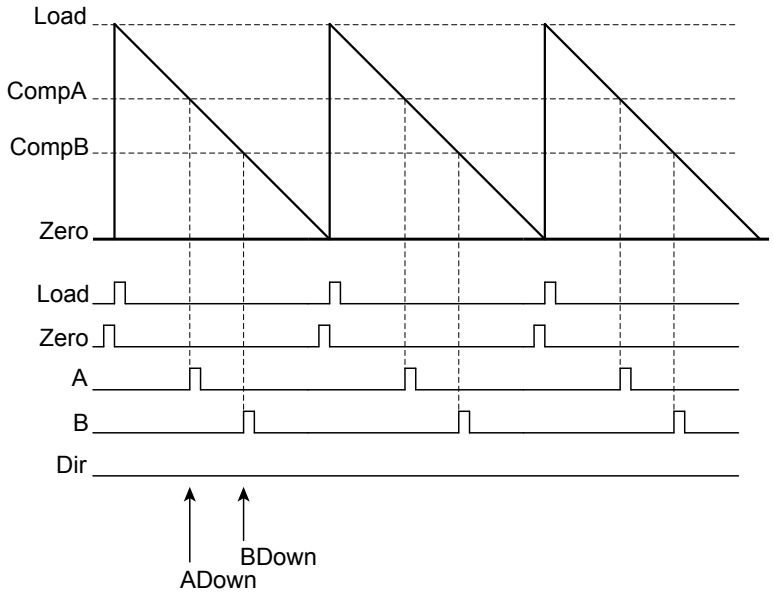
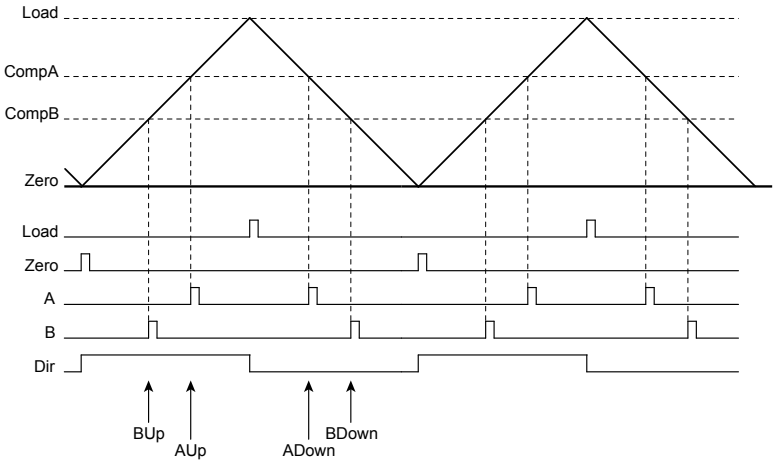


Figure 17-4. PWM Count-Up/Down Mode

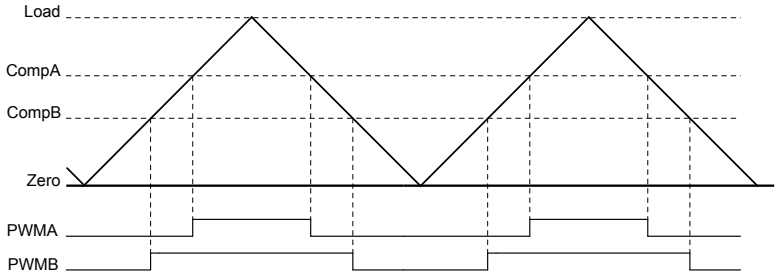


17.2.3 PWM Signal Generator

The PWM generator takes these pulses (qualified by the direction signal), and generates two PWM signals. In Count-Down mode, there are four events that can affect the PWM signal: zero, load, match A down, and match B down. In Count-Up/Down mode, there are six events that can affect the PWM signal: zero, load, match A down, match A up, match B down, and match B up. The match A or match B events are ignored when they coincide with the zero or load events. If the match A and match B events coincide, the first signal, PWMA, is generated based only on the match A event, and the second signal, PWMB, is generated based only on the match B event.

For each event, the effect on each output PWM signal is programmable: it can be left alone (ignoring the event), it can be toggled, it can be driven Low, or it can be driven High. These actions can be used to generate a pair of PWM signals of various positions and duty cycles, which do or do not overlap. Figure 17-5 on page 467 shows the use of Count-Up/Down mode to generate a pair of center-aligned, overlapped PWM signals that have different duty cycles.

Figure 17-5. PWM Generation Example In Count-Up/Down Mode



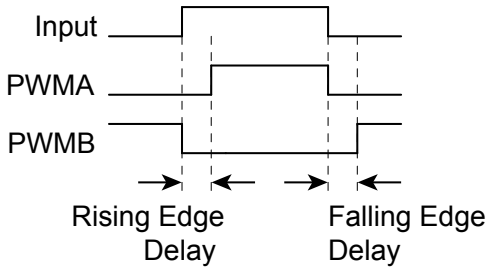
In this example, the first generator is set to drive High on match A up, drive Low on match A down, and ignore the other four events. The second generator is set to drive High on match B up, drive Low on match B down, and ignore the other four events. Changing the value of comparator A changes the duty cycle of the PWMA signal, and changing the value of comparator B changes the duty cycle of the PWMB signal.

17.2.4 Dead-Band Generator

The two PWM signals produced by the PWM generator are passed to the dead-band generator. If disabled, the PWM signals simply pass through unmodified. If enabled, the second PWM signal is lost and two PWM signals are generated based on the first PWM signal. The first output PWM signal is the input signal with the rising edge delayed by a programmable amount. The second output PWM signal is the inversion of the input signal with a programmable delay added between the falling edge of the input signal and the rising edge of this new signal.

This is therefore a pair of active High signals where one is always High, except for a programmable amount of time at transitions where both are Low. These signals are therefore suitable for driving a half-H bridge, with the dead-band delays preventing shoot-through current from damaging the power electronics. Figure 17-6 on page 467 shows the effect of the dead-band generator on an input PWM signal.

Figure 17-6. PWM Dead-Band Generator



17.2.5 Interrupt/ADC-Trigger Selector

The PWM generator also takes the same four (or six) counter events and uses them to generate an interrupt or an ADC trigger. Any of these events or a set of these events can be selected as a source for an interrupt; when any of the selected events occur, an interrupt is generated. Additionally, the same event, a different event, the same set of events, or a different set of events can be selected as a source for an ADC trigger; when any of these selected events occur, an ADC trigger pulse is generated. The selection of events allows the interrupt or ADC trigger to occur at a specific position within the PWM signal. Note that interrupts and ADC triggers are based on the raw events; delays in the PWM signal edges caused by the dead-band generator are not taken into account.

17.2.6 Synchronization Methods

There is a global reset capability that can reset the counter of the PWM generator.

The counter load values and comparator match values of the PWM generator can be updated in two ways. The first is immediate update mode, where a new value is used as soon as the counter reaches zero. By waiting for the counter to reach zero, a guaranteed behavior is defined, and overly short or overly long output PWM pulses are prevented.

The other update method is synchronous, where the new value is not used until a global synchronized update signal is asserted, at which point the new value is used as soon as the counter reaches zero. This second mode allows multiple items to be updated simultaneously without odd effects during the update; everything runs from the old values until a point at which they all run from the new values.

17.2.7 Fault Conditions

There are two external conditions that affect the PWM block; the signal input on the Fault pin and the stalling of the controller by a debugger. There are two mechanisms available to handle such conditions: the output signals can be forced into an inactive state and/or the PWM timers can be stopped.

Each output signal has a fault bit. If set, a fault input signal causes the corresponding output signal to go into the inactive state. If the inactive state is a safe condition for the signal to be in for an extended period of time, this keeps the output signal from driving the outside world in a dangerous manner during the fault condition. A fault condition can also generate a controller interrupt.

Each PWM generator can also be configured to stop counting during a stall condition. The user can select for the counters to run until they reach zero then stop, or to continue counting and reloading. A stall condition does not generate a controller interrupt.

17.2.8 Output Control Block

With the PWM generator block producing two raw PWM signals, the output control block takes care of the final conditioning of the PWM signals before they go to the pins. Via a single register, the set of PWM signals that are actually enabled to the pins can be modified; this can be used, for example, to perform commutation of a brushless DC motor with a single register write (and without modifying the individual PWM generators, which are modified by the feedback control loop). Similarly, fault control can disable any of the PWM signals as well. A final inversion can be applied to any of the PWM signals, making them active Low instead of the default active High.

17.3 Initialization and Configuration

The following example shows how to initialize the PWM Generator 0 with a 25-KHz frequency, and with a 25% duty cycle on the `PWM0` pin and a 75% duty cycle on the `PWM1` pin. This example assumes the system clock is 20 MHz.

1. Enable the PWM clock by writing a value of 0x0010.0000 to the **RCGC0** register in the System Control module.
2. Enable the clock to the appropriate GPIO module via the **RCGC2** register in the System Control module.
3. In the GPIO module, enable the appropriate pins for their alternate function using the **GPIOAFSEL** register.

4. Configure the **Run-Mode Clock Configuration (RCC)** register in the System Control module to use the PWM divide (USEPWMDIV) and set the divider (PWMDIV) to divide by 2 (000).

5. Configure the PWM generator for countdown mode with immediate updates to the parameters.
   - Write the **PWM0CTL** register with a value of 0x0000.0000.
   - Write the **PWM0GENA** register with a value of 0x0000.008C.
   - Write the **PWM0GENB** register with a value of 0x0000.080C.

6. Set the period. For a 25-KHz frequency, the period = 1/25,000, or 40 microseconds. The PWM clock source is 10 MHz; the system clock divided by 2. This translates to 400 clock ticks per period. Use this value to set the **PWM0LOAD** register. In Count-Down mode, set the Load field in the **PWM0LOAD** register to the requested period minus one.
   - Write the **PWM0LOAD** register with a value of 0x0000.018F.

7. Set the pulse width of the PWM0 pin for a 25% duty cycle.
   - Write the **PWM0CMPA** register with a value of 0x0000.012B.

8. Set the pulse width of the PWM1 pin for a 75% duty cycle.
   - Write the **PWM0CMPB** register with a value of 0x0000.0063.

9. Start the timers in PWM generator 0.
   - Write the **PWM0CTL** register with a value of 0x0000.0001.

10. Enable PWM outputs.
    - Write the **PWMENABLE** register with a value of 0x0000.0003.

17.4 Register Map

Table 17-1 on page 469 lists the PWM registers. The offset listed is a hexadecimal increment to the register's address, relative to the PWM base address of 0x4002.8000.

Table 17-1. PWM Register Map

Offset	Name	Type	Reset	Description	See page
0x000	PWMCTL	R/W	0x0000.0000	PWM Master Control	471
0x004	PWMSYNC	R/W	0x0000.0000	PWM Time Base Sync	472
0x008	PWMENABLE	R/W	0x0000.0000	PWM Output Enable	473
0x00C	PWMINVERT	R/W	0x0000.0000	PWM Output Inversion	474
0x010	PWMFAULT	R/W	0x0000.0000	PWM Output Fault	475
0x014	PWMINTEN	R/W	0x0000.0000	PWM Interrupt Enable	476
0x018	PWMRIS	RO	0x0000.0000	PWM Raw Interrupt Status	477
0x01C	PWMISC	R/W1C	0x0000.0000	PWM Interrupt Status and Clear	478

Table 17-1. PWM Register Map *(continued)*

Offset	Name	Type	Reset	Description	See page
0x020	PWMSTATUS	RO	0x0000.0000	PWM Status	479
0x040	PWM0CTL	R/W	0x0000.0000	PWM0 Control	480
0x044	PWM0INTEN	R/W	0x0000.0000	PWM0 Interrupt and Trigger Enable	482
0x048	PWM0RIS	RO	0x0000.0000	PWM0 Raw Interrupt Status	485
0x04C	PWM0ISC	R/W1C	0x0000.0000	PWM0 Interrupt Status and Clear	486
0x050	PWM0LOAD	R/W	0x0000.0000	PWM0 Load	487
0x054	PWM0COUNT	RO	0x0000.0000	PWM0 Counter	488
0x058	PWM0CMPA	R/W	0x0000.0000	PWM0 Compare A	489
0x05C	PWM0CMPB	R/W	0x0000.0000	PWM0 Compare B	490
0x060	PWM0GENA	R/W	0x0000.0000	PWM0 Generator A Control	491
0x064	PWM0GENB	R/W	0x0000.0000	PWM0 Generator B Control	494
0x068	PWM0DBCTL	R/W	0x0000.0000	PWM0 Dead-Band Control	497
0x06C	PWM0DBRISE	R/W	0x0000.0000	PWM0 Dead-Band Rising-Edge Delay	498
0x070	PWM0DBFALL	R/W	0x0000.0000	PWM0 Dead-Band Falling-Edge-Delay	499

17.5 Register Descriptions

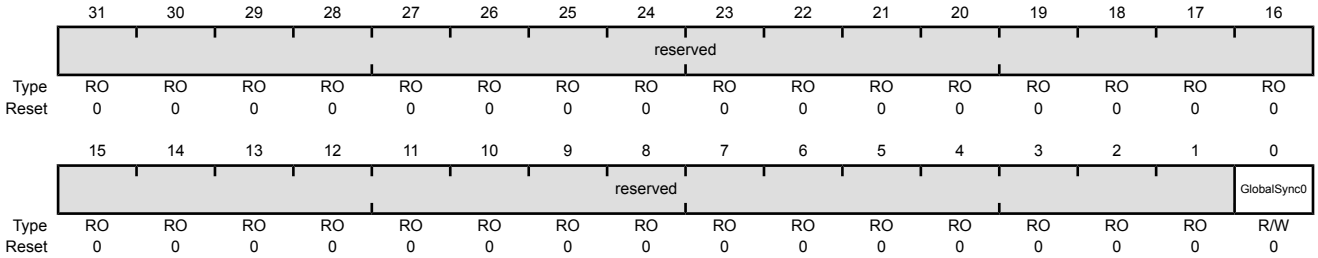
The remainder of this section lists and describes the PWM registers, in numerical order by address offset.

Register 1: PWM Master Control (PWMCTL), offset 0x000

This register provides master control over the PWM generation block.

PWM Master Control (PWMCTL)

Base 0x4002.8000
Offset 0x000
Type R/W, reset 0x0000.0000



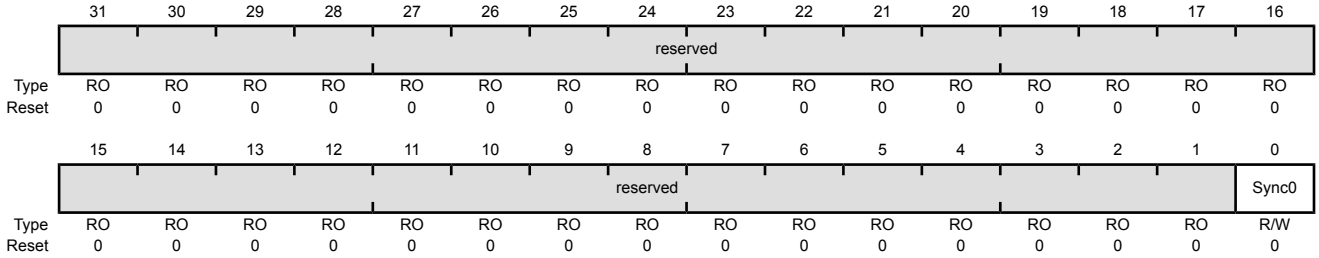
Bit/Field	Name	Type	Reset	Description
31:1	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
0	GlobalSync0	R/W	0	Update PWM Generator 0 Setting this bit causes any queued update to a load or comparator register in PWM generator 0 to be applied the next time the corresponding counter becomes zero. This bit automatically clears when the updates have completed; it cannot be cleared by software.

Register 2: PWM Time Base Sync (PWMSYNC), offset 0x004

This register provides a method to perform synchronization of the counters in the PWM generation blocks. Writing a bit in this register to 1 causes the specified counter to reset back to 0; writing multiple bits resets multiple counters simultaneously. The bits auto-clear after the reset has occurred; reading them back as zero indicates that the synchronization has completed.

PWM Time Base Sync (PWMSYNC)

Base 0x4002.8000
Offset 0x004
Type R/W, reset 0x0000.0000



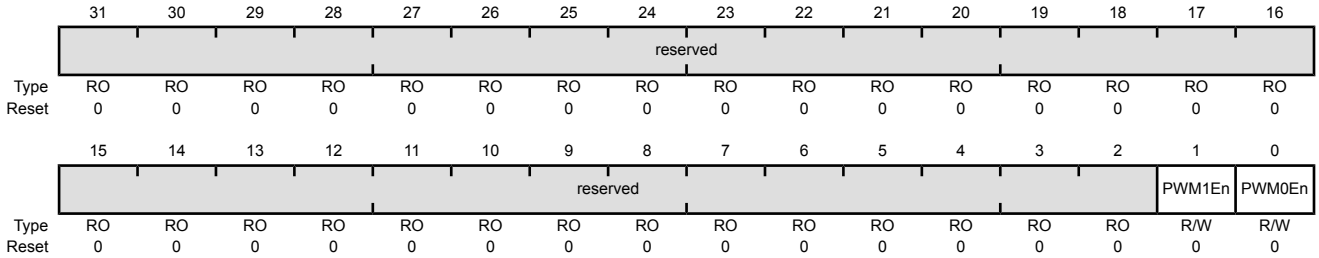
Bit/Field	Name	Type	Reset	Description
31:1	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
0	Sync0	R/W	0	Reset Generator 0 Counter Performs a reset of the PWM generator 0 counter.

Register 3: PWM Output Enable (PWMENABLE), offset 0x008

This register provides a master control of which generated PWM signals are output to device pins. By disabling a PWM output, the generation process can continue (for example, when the time bases are synchronized) without driving PWM signals to the pins. When bits in this register are set, the corresponding PWM signal is passed through to the output stage, which is controlled by the **PWMINVERT** register. When bits are not set, the PWM signal is replaced by a zero value which is also passed to the output stage.

PWM Output Enable (PWMENABLE)

Base 0x4002.8000
Offset 0x008
Type R/W, reset 0x0000.0000



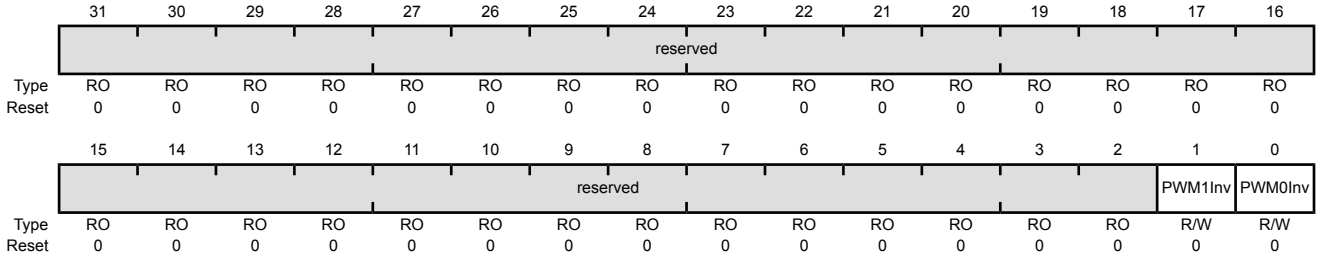
Bit/Field	Name	Type	Reset	Description
31:2	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1	PWM1En	R/W	0	PWM1 Output Enable When set, allows the generated `PWM1` signal to be passed to the device pin.
0	PWM0En	R/W	0	PWM0 Output Enable When set, allows the generated `PWM0` signal to be passed to the device pin.

Register 4: PWM Output Inversion (PWMINVERT), offset 0x00C

This register provides a master control of the polarity of the PWM signals on the device pins. The PWM signals generated by the PWM generator are active High; they can optionally be made active Low via this register. Disabled PWM channels are also passed through the output inverter (if so configured) so that inactive channels maintain the correct polarity.

PWM Output Inversion (PWMINVERT)

Base 0x4002.8000
Offset 0x00C
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:2	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1	PWM1Inv	R/W	0	Invert PWM1 Signal When set, the generated PWM1 signal is inverted.
0	PWM0Inv	R/W	0	Invert PWM0 Signal When set, the generated PWM0 signal is inverted.

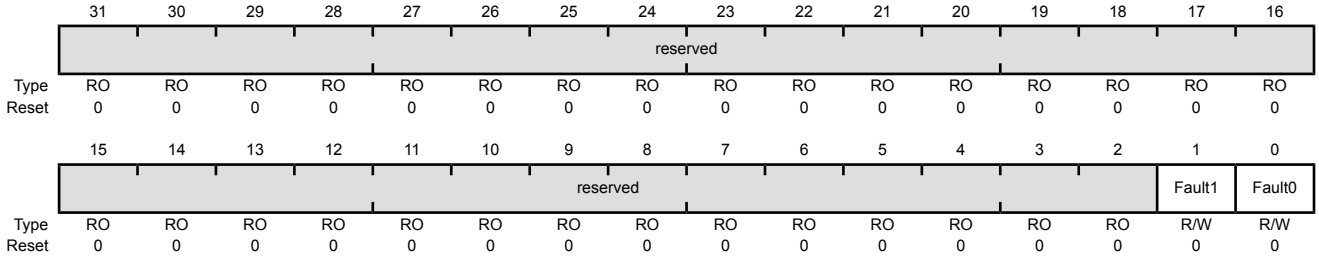
Register 5: PWM Output Fault (PWMFAULT), offset 0x010

This register controls the behavior of the PWM outputs in the presence of fault conditions. Both the fault inputs and debug events are considered fault conditions. On a fault condition, each PWM signal can be passed through unmodified or driven Low. For outputs that are configured for pass-through, the debug event handling on the corresponding PWM generator also determines if the PWM signal continues to be generated.

Fault condition control occurs before the output inverter, so PWM signals driven Low on fault are inverted if the channel is configured for inversion (therefore, the pin is driven High on a fault condition).

PWM Output Fault (PWMFAULT)

Base 0x4002.8000
Offset 0x010
Type R/W, reset 0x0000.0000



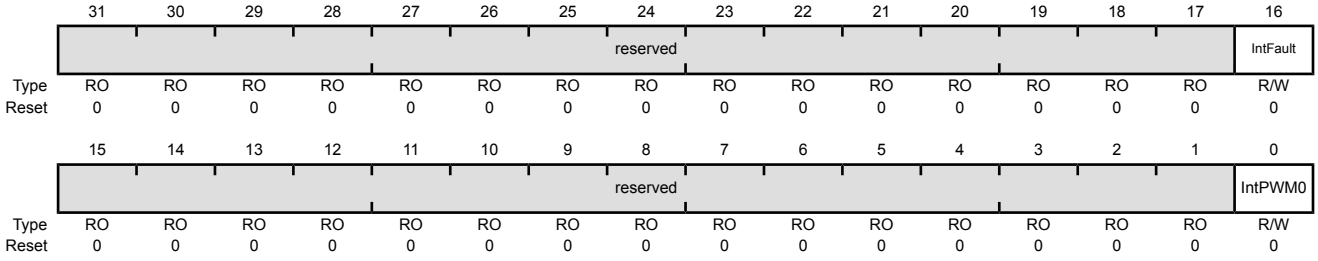
Bit/Field	Name	Type	Reset	Description
31:2	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
1	Fault1	R/W	0	PWM1 Fault When set, the PWM1 output signal is driven Low on a fault condition.
0	Fault0	R/W	0	PWM0 Fault When set, the PWM0 output signal is driven Low on a fault condition.

Register 6: PWM Interrupt Enable (PWMINTEN), offset 0x014

This register controls the global interrupt generation capabilities of the PWM module. The events that can cause an interrupt are the fault input and the individual interrupts from the PWM generator.

PWM Interrupt Enable (PWMINTEN)

Base 0x4002.8000
Offset 0x014
Type R/W, reset 0x0000.0000



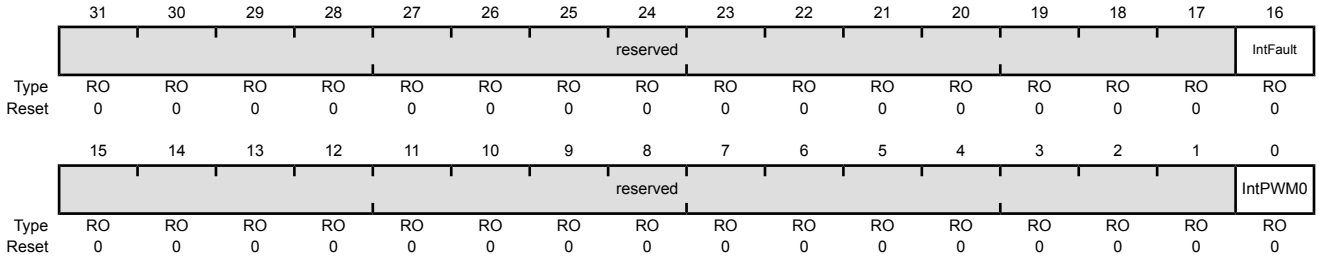
Bit/Field	Name	Type	Reset	Description
31:17	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
16	IntFault	R/W	0	Fault Interrupt Enable When set, an interrupt occurs when the fault input is asserted.
15:1	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
0	IntPWM0	R/W	0	PWM0 Interrupt Enable When set, an interrupt occurs when the PWM generator 0 block asserts an interrupt.

Register 7: PWM Raw Interrupt Status (PWMRIS), offset 0x018

This register provides the current set of interrupt sources that are asserted, regardless of whether they cause an interrupt to be asserted to the controller. The fault interrupt is latched on detection; it must be cleared through the **PWM Interrupt Status and Clear (PWMISC)** register (see page 478). The PWM generator interrupts simply reflect the status of the PWM generator; they are cleared via the interrupt status register in the PWM generator block. Bits set to 1 indicate the events that are active; zero bits indicate that the event in question is not active.

PWM Raw Interrupt Status (PWMRIS)

Base 0x4002.8000
Offset 0x018
Type RO, reset 0x0000.0000



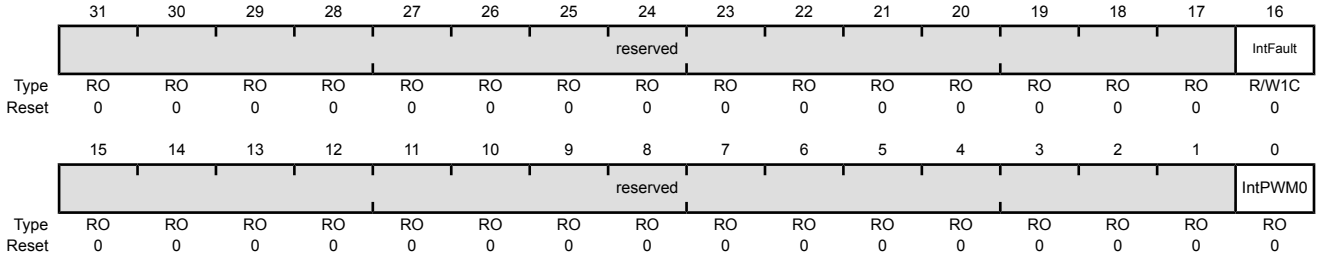
Bit/Field	Name	Type	Reset	Description
31:17	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
16	IntFault	RO	0	Fault Interrupt Asserted Indicates that the fault input is asserting.
15:1	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
0	IntPWM0	RO	0	PWM0 Interrupt Asserted Indicates that the PWM generator 0 block is asserting its interrupt.

Register 8: PWM Interrupt Status and Clear (PWMISC), offset 0x01C

This register provides a summary of the interrupt status of the PWM generator block. A bit set to 1 indicates that the generator block is asserting an interrupt. The individual interrupt status registers must be consulted to determine the reason for the interrupt, and used to clear the interrupt. For the fault interrupt, a write of 1 to that bit position clears the latched interrupt status.

PWM Interrupt Status and Clear (PWMISC)

Base 0x4002.8000
Offset 0x01C
Type R/W1C, reset 0x0000.0000



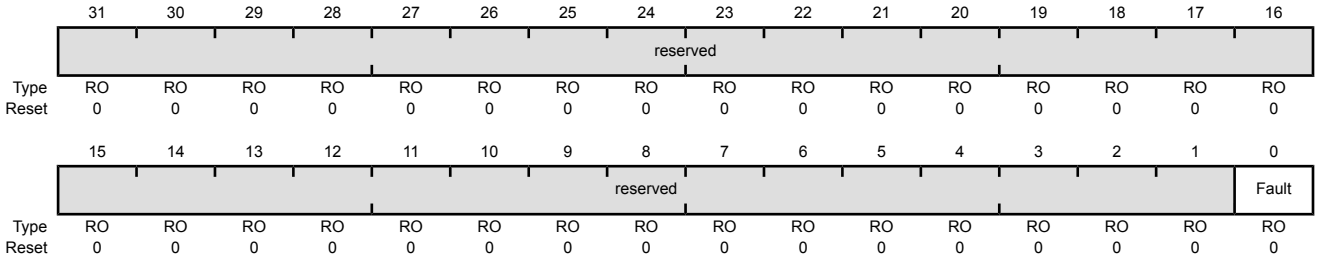
Bit/Field	Name	Type	Reset	Description
31:17	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
16	IntFault	R/W1C	0	Fault Interrupt Asserted Indicates that the fault input is asserting an interrupt.
15:1	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
0	IntPWM0	RO	0	PWM0 Interrupt Status Indicates if the PWM generator 0 block is asserting an interrupt.

Register 9: PWM Status (PWMSTATUS), offset 0x020

This register provides the status of the FAULT input signal.

PWM Status (PWMSTATUS)

Base 0x4002.8000
Offset 0x020
Type RO, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:1	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
0	Fault	RO	0	Fault Interrupt Status When set, indicates the fault input is asserted.

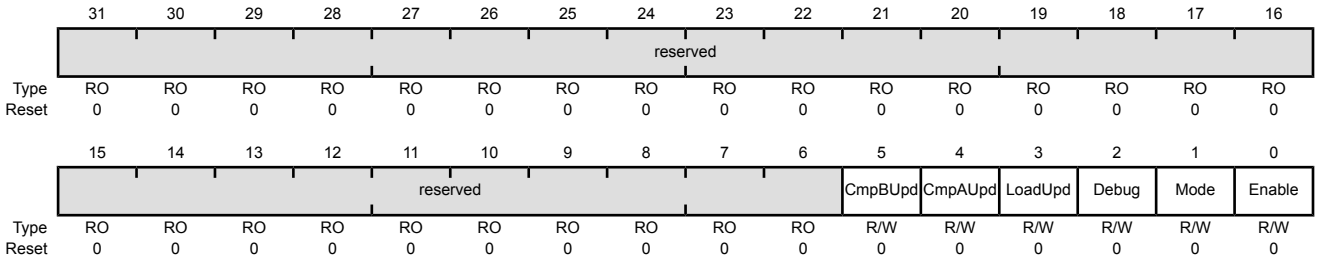
Register 10: PWM0 Control (PWM0CTL), offset 0x040

This register configures the PWM signal generation block. The Register Update mode, Debug mode, Counting mode, and Block Enable mode are all controlled via this register. The block produces the PWM signals, which can be either two independent PWM signals (from the same counter), or a paired set of PWM signals with dead-band delays added.

The PWM0 block produces the PWM0 and PWM1 outputs.

PWM0 Control (PWM0CTL)

Base 0x4002.8000
Offset 0x040
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:6	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
5	CmpBUpd	R/W	0	Comparator B Update Mode Same as CmpAUpd but for the comparator B register.
4	CmpAUpd	R/W	0	Comparator A Update Mode The Update mode for the comparator A register. When not set, updates to the register are reflected to the comparator the next time the counter is 0. When set, updates to the register are delayed until the next time the counter is 0 after a synchronous update has been requested through the **PWM Master Control (PWMCTL)** register (see page 471).
3	LoadUpd	R/W	0	Load Register Update Mode The Update mode for the load register. When not set, updates to the register are reflected to the counter the next time the counter is 0. When set, updates to the register are delayed until the next time the counter is 0 after a synchronous update has been requested through the **PWM Master Control (PWMCTL)** register.
2	Debug	R/W	0	Debug Mode The behavior of the counter in Debug mode. When not set, the counter stops running when it next reaches 0, and continues running again when no longer in Debug mode. When set, the counter always runs.
1	Mode	R/W	0	Counter Mode The mode for the counter. When not set, the counter counts down from the load value to 0 and then wraps back to the load value (Count-Down mode). When set, the counter counts up from 0 to the load value, back down to 0, and then repeats (Count-Up/Down mode).

Bit/Field	Name	Type	Reset	Description
0	Enable	R/W	0	PWM Block Enable Master enable for the PWM generation block. When not set, the entire block is disabled and not clocked. When set, the block is enabled and produces PWM signals.

Register 11: PWM0 Interrupt and Trigger Enable (PWM0INTEN), offset 0x044

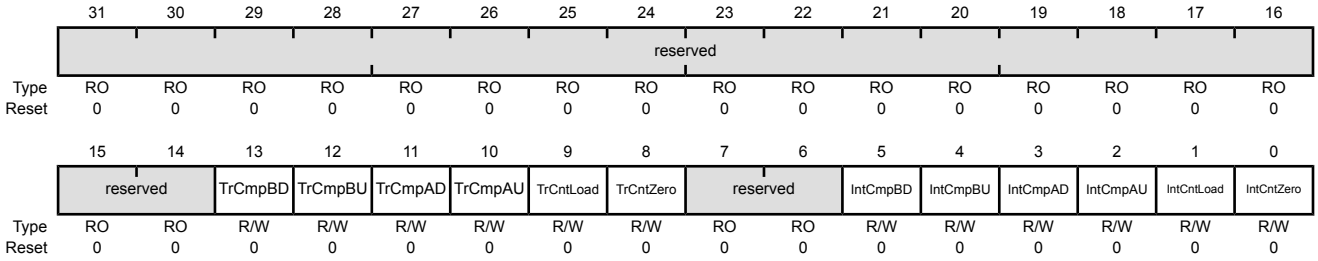
This register controls the interrupt and ADC trigger generation capabilities of the PWM generator. The events that can cause an interrupt or an ADC trigger are:

- The counter being equal to the load register
- The counter being equal to zero
- The counter being equal to the comparator A register while counting up
- The counter being equal to the comparator A register while counting down
- The counter being equal to the comparator B register while counting up
- The counter being equal to the comparator B register while counting down

Any combination of these events can generate either an interrupt, or an ADC trigger; though no determination can be made as to the actual event that caused an ADC trigger if more than one is specified.

PWM0 Interrupt and Trigger Enable (PWM0INTEN)

Base 0x4002.8000
Offset 0x044
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:14	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
13	TrCmpBD	R/W	0	Trigger for Counter=Comparator B Down Value Description 1 An ADC trigger pulse is output when the counter matches the value in the **PWMnCMPB** register value while counting down. 0 No ADC trigger is output.
12	TrCmpBU	R/W	0	Trigger for Counter=Comparator B Up Value Description 1 An ADC trigger pulse is output when the counter matches the value in the **PWMnCMPB** register value while counting up. 0 No ADC trigger is output.

Bit/Field	Name	Type	Reset	Description
11	TrCmpAD	R/W	0	Trigger for Counter=Comparator A Down Value Description 1 An ADC trigger pulse is output when the counter matches the value in the **PWMnCMPA** register value while counting down. 0 No ADC trigger is output.
10	TrCmpAU	R/W	0	Trigger for Counter=Comparator A Up Value Description 1 An ADC trigger pulse is output when the counter matches the value in the **PWMnCMPA** register value while counting up. 0 No ADC trigger is output.
9	TrCntLoad	R/W	0	Trigger for Counter=Load Value Description 1 An ADC trigger pulse is output when the counter matches the **PWMnLOAD** register. 0 No ADC trigger is output.
8	TrCntZero	R/W	0	Trigger for Counter=0 Value Description 1 An ADC trigger pulse is output when the counter is 0. 0 No ADC trigger is output.
7:6	reserved	RO	0x0	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
5	IntCmpBD	R/W	0	Interrupt for Counter=Comparator B Down Value Description 1 A raw interrupt occurs when the counter matches the value in the **PWMnCMPB** register value while counting down. 0 No interrupt.
4	IntCmpBU	R/W	0	Interrupt for Counter=Comparator B Up Value Description 1 A raw interrupt occurs when the counter matches the value in the **PWMnCMPB** register value while counting up. 0 No interrupt.

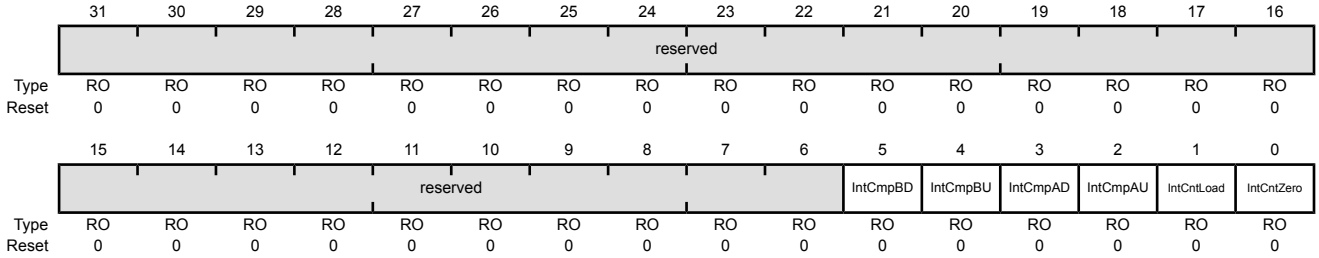
Bit/Field	Name	Type	Reset	Description
3	IntCmpAD	R/W	0	Interrupt for Counter=Comparator A Down Value Description 1 A raw interrupt occurs when the counter matches the value in the **PWMnCMPA** register value while counting down. 0 No interrupt.
2	IntCmpAU	R/W	0	Interrupt for Counter=Comparator A Up Value Description 1 A raw interrupt occurs when the counter matches the value in the **PWMnCMPA** register value while counting up. 0 No interrupt.
1	IntCntLoad	R/W	0	Interrupt for Counter=Load Value Description 1 A raw interrupt occurs when the counter matches the value in the **PWMnLOAD** register value. 0 No interrupt.
0	IntCntZero	R/W	0	Interrupt for Counter=0 Value Description 1 A raw interrupt occurs when the counter is zero. 0 No interrupt.

Register 12: PWM0 Raw Interrupt Status (PWM0RIS), offset 0x048

This register provides the current set of interrupt sources that are asserted, regardless of whether they cause an interrupt to be asserted to the controller. Bits set to 1 indicate the latched events that have occurred; bits set to 0 indicate that the event in question has not occurred.

PWM0 Raw Interrupt Status (PWM0RIS)

Base 0x4002.8000
Offset 0x048
Type RO, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:6	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
5	IntCmpBD	RO	0	Comparator B Down Interrupt Status Indicates that the counter has matched the comparator B value while counting down.
4	IntCmpBU	RO	0	Comparator B Up Interrupt Status Indicates that the counter has matched the comparator B value while counting up.
3	IntCmpAD	RO	0	Comparator A Down Interrupt Status Indicates that the counter has matched the comparator A value while counting down.
2	IntCmpAU	RO	0	Comparator A Up Interrupt Status Indicates that the counter has matched the comparator A value while counting up.
1	IntCntLoad	RO	0	Counter=Load Interrupt Status Indicates that the counter has matched the **PWMnLOAD** register.
0	IntCntZero	RO	0	Counter=0 Interrupt Status Indicates that the counter has matched 0.

Register 13: PWM0 Interrupt Status and Clear (PWM0ISC), offset 0x04C

This register provides the current set of interrupt sources that are asserted to the controller. Bits set to 1 indicate the latched events that have occurred; bits set to 0 indicate that the event in question has not occurred. These are R/W1C registers; writing a 1 to a bit position clears the corresponding interrupt reason.

PWM0 Interrupt Status and Clear (PWM0ISC)

Base 0x4002.8000
Offset 0x04C
Type R/W1C, reset 0x0000.0000

	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	reserved															
Type	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	reserved										IntCmpBD	IntCmpBU	IntCmpAD	IntCmpAU	IntCntLoad	IntCntZero
Type	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	R/W1C	R/W1C	R/W1C	R/W1C	R/W1C	R/W1C
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit/Field	Name	Type	Reset	Description
31:6	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
5	IntCmpBD	R/W1C	0	Comparator B Down Interrupt Indicates that the counter has matched the comparator B value while counting down.
4	IntCmpBU	R/W1C	0	Comparator B Up Interrupt Indicates that the counter has matched the comparator B value while counting up.
3	IntCmpAD	R/W1C	0	Comparator A Down Interrupt Indicates that the counter has matched the comparator A value while counting down.
2	IntCmpAU	R/W1C	0	Comparator A Up Interrupt Indicates that the counter has matched the comparator A value while counting up.
1	IntCntLoad	R/W1C	0	Counter=Load Interrupt Indicates that the counter has matched the **PWMnLOAD** register.
0	IntCntZero	R/W1C	0	Counter=0 Interrupt Indicates that the counter has matched 0.

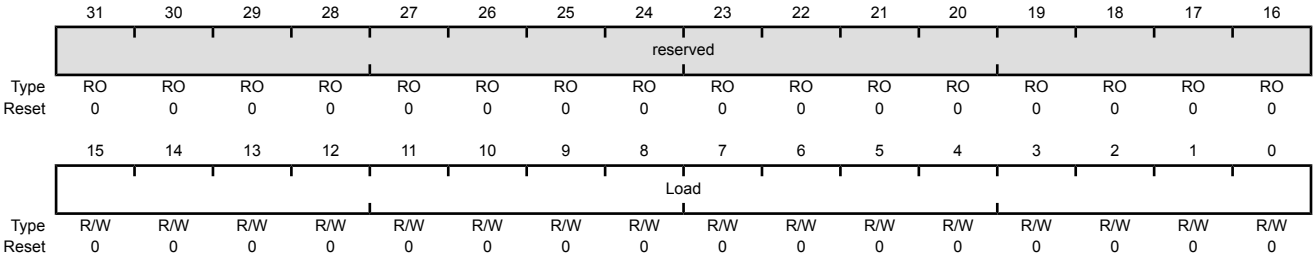
Register 14: PWM0 Load (PWM0LOAD), offset 0x050

This register contains the load value for the PWM counter. Based on the counter mode, either this value is loaded into the counter after it reaches zero, or it is the limit of up-counting after which the counter decrements back to zero.

If the Load Value Update mode is immediate, this value is used the next time the counter reaches zero; if the mode is synchronous, it is used the next time the counter reaches zero after a synchronous update has been requested through the **PWM Master Control (PWMCTL)** register (see page 471). If this register is re-written before the actual update occurs, the previous value is never used and is lost.

PWM0 Load (PWM0LOAD)

Base 0x4002.8000
Offset 0x050
Type R/W, reset 0x0000.0000



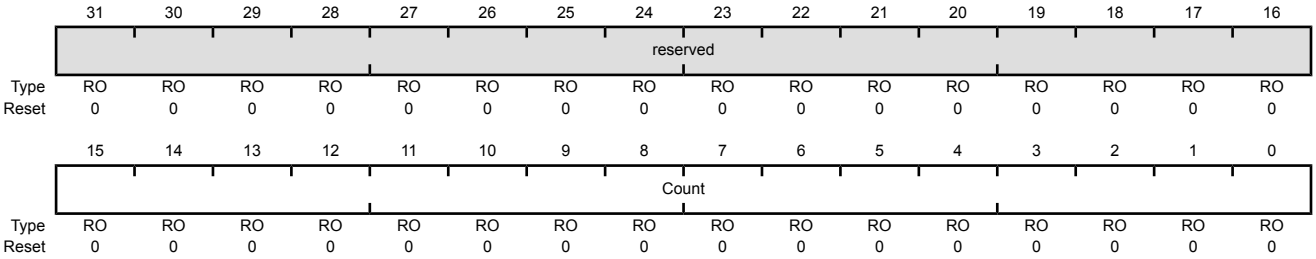
Bit/Field	Name	Type	Reset	Description
31:16	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
15:0	Load	R/W	0	Counter Load Value The counter load value.

Register 15: PWM0 Counter (PWM0COUNT), offset 0x054

This register contains the current value of the PWM counter. When this value matches the load register, a pulse is output; this can drive the generation of a PWM signal (via the **PWMnGENA/PWMnGENB** registers, see page 491 and page 494) or drive an interrupt or ADC trigger (via the **PWMnINTEN** register, see page 482). A pulse with the same capabilities is generated when this value is zero.

PWM0 Counter (PWM0COUNT)

Base 0x4002.8000
Offset 0x054
Type RO, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:16	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
15:0	Count	RO	0x00	Counter Value The current value of the counter.

Register 16: PWM0 Compare A (PWM0CMPA), offset 0x058

This register contains a value to be compared against the counter . When this value matches the counter, a pulse is output; this can drive the generation of a PWM signal (via the **PWMnGENA/PWMnGENB** registers) or drive an interrupt or ADC trigger (via the **PWMnINTEN** register). If the value of this register is greater than the **PWMnLOAD** register (see page 487), then no pulse is ever output.

If the comparator A update mode is immediate (based on the CmpAUpd bit in the **PWMnCTL** register), this 16-bit CompA value is used the next time the counter reaches zero. If the update mode is synchronous, it is used the next time the counter reaches zero after a synchronous update has been requested through the **PWM Master Control (PWMCTL)** register (see page 471). If this register is rewritten before the actual update occurs, the previous value is never used and is lost.

PWM0 Compare A (PWM0CMPA)

Base 0x4002.8000
Offset 0x058
Type R/W, reset 0x0000.0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
reserved															
Type RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO
Reset 0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CompA															
Type R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset 0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit/Field	Name	Type	Reset	Description
31:16	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
15:0	CompA	R/W	0x00	Comparator A Value The value to be compared against the counter.

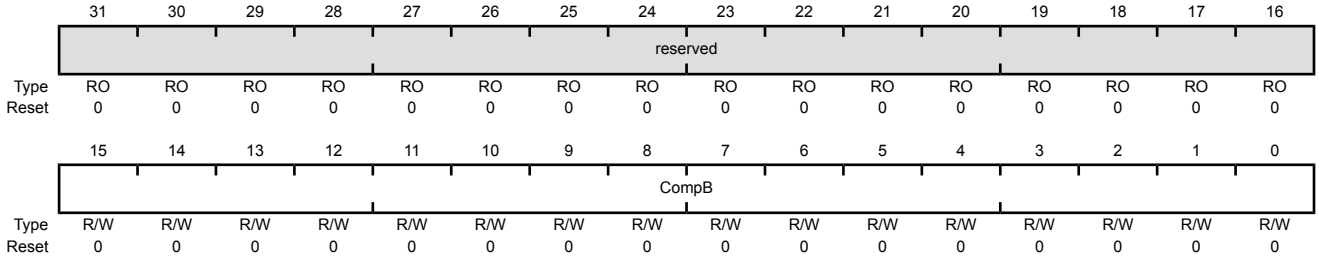
Register 17: PWM0 Compare B (PWM0CMPB), offset 0x05C

This register contains a value to be compared against the counter. When this value matches the counter, a pulse is output; this can drive the generation of a PWM signal (via the **PWMnGENA/PWMnGENB** registers) or drive an interrupt or ADC trigger (via the **PWMnINTEN** register). If the value of this register is greater than the **PWMnLOAD** register, no pulse is ever output.

If the comparator B update mode is immediate (based on the CmpBUpd bit in the **PWMnCTL** register), this 16-bit CompB value is used the next time the counter reaches zero. If the update mode is synchronous, it is used the next time the counter reaches zero after a synchronous update has been requested through the **PWM Master Control (PWMCTL)** register (see page 471). If this register is rewritten before the actual update occurs, the previous value is never used and is lost.

PWM0 Compare B (PWM0CMPB)

Base 0x4002.8000
Offset 0x05C
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:16	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
15:0	CompB	R/W	0x00	Comparator B Value The value to be compared against the counter.

Register 18: PWM0 Generator A Control (PWM0GENA), offset 0x060

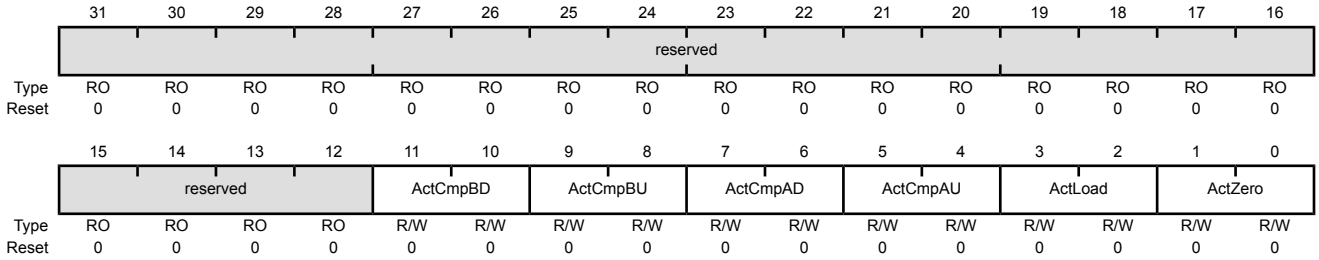
This register controls the generation of the PWMnA signal based on the load and zero output pulses from the counter, as well as the compare A and compare B pulses from the comparators. When the counter is running in Count-Down mode, only four of these events occur; when running in Count-Up/Down mode, all six occur. These events provide great flexibility in the positioning and duty cycle of the PWM signal that is produced.

The **PWM0GENA** register controls generation of the PWM0A signal.

If a zero or load event coincides with a compare A or compare B event, the zero or load action is taken and the compare A or compare B action is ignored. If a compare A event coincides with a compare B event, the compare A action is taken and the compare B action is ignored.

PWM0 Generator A Control (PWM0GENA)

Base 0x4002.8000
Offset 0x060
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:12	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
11:10	ActCmpBD	R/W	0x0	Action for Comparator B Down The action to be taken when the counter matches comparator B while counting down. The table below defines the effect of the event on the output signal. Value Description 0x0 Do nothing. 0x1 Invert the output signal. 0x2 Set the output signal to 0. 0x3 Set the output signal to 1.

Bit/Field	Name	Type	Reset	Description
9:8	ActCmpBU	R/W	0x0	Action for Comparator B Up The action to be taken when the counter matches comparator B while counting up. Occurs only when the Mode bit in the **PWMnCTL** register (see page 480) is set to 1. The table below defines the effect of the event on the output signal. Value Description 0x0 Do nothing. 0x1 Invert the output signal. 0x2 Set the output signal to 0. 0x3 Set the output signal to 1.
7:6	ActCmpAD	R/W	0x0	Action for Comparator A Down The action to be taken when the counter matches comparator A while counting down. The table below defines the effect of the event on the output signal. Value Description 0x0 Do nothing. 0x1 Invert the output signal. 0x2 Set the output signal to 0. 0x3 Set the output signal to 1.
5:4	ActCmpAU	R/W	0x0	Action for Comparator A Up The action to be taken when the counter matches comparator A while counting up. Occurs only when the Mode bit in the **PWMnCTL** register is set to 1. The table below defines the effect of the event on the output signal. Value Description 0x0 Do nothing. 0x1 Invert the output signal. 0x2 Set the output signal to 0. 0x3 Set the output signal to 1.
3:2	ActLoad	R/W	0x0	Action for Counter=Load The action to be taken when the counter matches the load value. The table below defines the effect of the event on the output signal. Value Description 0x0 Do nothing. 0x1 Invert the output signal. 0x2 Set the output signal to 0. 0x3 Set the output signal to 1.

Bit/Field	Name	Type	Reset	Description
1:0	ActZero	R/W	0x0	Action for Counter=0 The action to be taken when the counter is zero. The table below defines the effect of the event on the output signal. Value Description 0x0 Do nothing. 0x1 Invert the output signal. 0x2 Set the output signal to 0. 0x3 Set the output signal to 1.

Register 19: PWM0 Generator B Control (PWM0GENB), offset 0x064

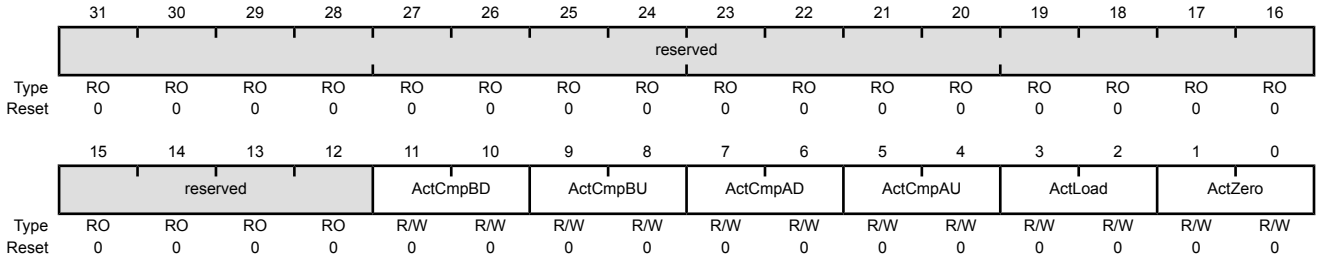
This register controls the generation of the PWMnB signal based on the load and zero output pulses from the counter, as well as the compare A and compare B pulses from the comparators. When the counter is running in Down mode, only four of these events occur; when running in Up/Down mode, all six occur. These events provide great flexibility in the positioning and duty cycle of the PWM signal that is produced.

The **PWM0GENB** register controls generation of the PWM0B signal.

If a zero or load event coincides with a compare A or compare B event, the zero or load action is taken and the compare A or compare B action is ignored. If a compare A event coincides with a compare B event, the compare B action is taken and the compare A action is ignored.

PWM0 Generator B Control (PWM0GENB)

Base 0x4002.8000
Offset 0x064
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:12	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
11:10	ActCmpBD	R/W	0x0	Action for Comparator B Down The action to be taken when the counter matches comparator B while counting down. The table below defines the effect of the event on the output signal. Value Description 0x0 Do nothing. 0x1 Invert the output signal. 0x2 Set the output signal to 0. 0x3 Set the output signal to 1.

Bit/Field	Name	Type	Reset	Description
9:8	ActCmpBU	R/W	0x0	Action for Comparator B Up The action to be taken when the counter matches comparator B while counting up. Occurs only when the Mode bit in the **PWMnCTL** register is set to 1. The table below defines the effect of the event on the output signal. Value Description 0x0 Do nothing. 0x1 Invert the output signal. 0x2 Set the output signal to 0. 0x3 Set the output signal to 1.
7:6	ActCmpAD	R/W	0x0	Action for Comparator A Down The action to be taken when the counter matches comparator A while counting down. The table below defines the effect of the event on the output signal. Value Description 0x0 Do nothing. 0x1 Invert the output signal. 0x2 Set the output signal to 0. 0x3 Set the output signal to 1.
5:4	ActCmpAU	R/W	0x0	Action for Comparator A Up The action to be taken when the counter matches comparator A while counting up. Occurs only when the Mode bit in the **PWMnCTL** register is set to 1. The table below defines the effect of the event on the output signal. Value Description 0x0 Do nothing. 0x1 Invert the output signal. 0x2 Set the output signal to 0. 0x3 Set the output signal to 1.
3:2	ActLoad	R/W	0x0	Action for Counter=Load The action to be taken when the counter matches the load value. The table below defines the effect of the event on the output signal. Value Description 0x0 Do nothing. 0x1 Invert the output signal. 0x2 Set the output signal to 0. 0x3 Set the output signal to 1.

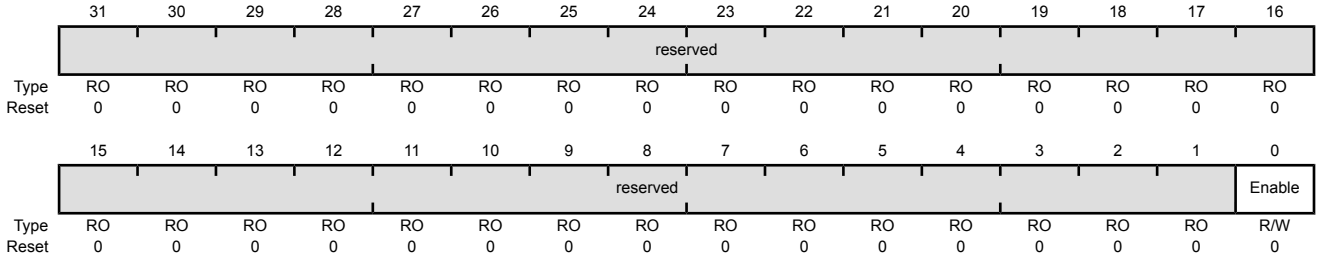
Bit/Field	Name	Type	Reset	Description
1:0	ActZero	R/W	0x0	Action for Counter=0 The action to be taken when the counter is 0. The table below defines the effect of the event on the output signal. Value Description 0x0 Do nothing. 0x1 Invert the output signal. 0x2 Set the output signal to 0. 0x3 Set the output signal to 1.

Register 20: PWM0 Dead-Band Control (PWM0DBCTL), offset 0x068

The **PWM0DBCTL** register controls the dead-band generator, which produces the PWM0 and PWM1 signals based on the PWM0A and PWM0B signals. When disabled, the PWM0A signal passes through to the PWM0 signal and the PWM0B signal passes through to the PWM1 signal. When enabled and inverting the resulting waveform, the PWM0B signal is ignored; the PWM0 signal is generated by delaying the rising edge(s) of the PWM0A signal by the value in the **PWM0DBRISE** register (see page 498), and the PWM1 signal is generated by delaying the falling edge(s) of the PWM0A signal by the value in the **PWM0DBFALL** register (see page 499).

PWM0 Dead-Band Control (PWM0DBCTL)

Base 0x4002.8000
Offset 0x068
Type R/W, reset 0x0000.0000



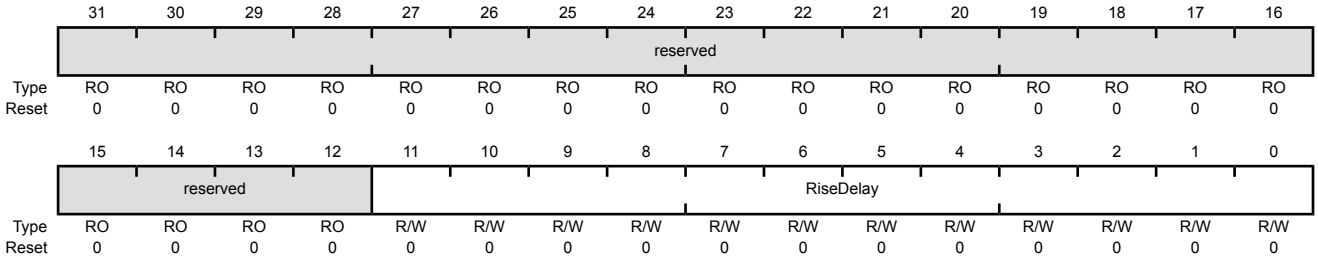
Bit/Field	Name	Type	Reset	Description
31:1	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
0	Enable	R/W	0	Dead-Band Generator Enable When set, the dead-band generator inserts dead bands into the output signals; when clear, it simply passes the PWM signals through.

Register 21: PWM0 Dead-Band Rising-Edge Delay (PWM0DBRISE), offset 0x06C

The **PWM0DBRISE** register contains the number of clock ticks to delay the rising edge of the PWM0A signal when generating the PWM0 signal. If the dead-band generator is disabled through the **PWMnDBCTL** register, the **PWM0DBRISE** register is ignored. If the value of this register is larger than the width of a High pulse on the input PWM signal, the rising-edge delay consumes the entire High time of the signal, resulting in no High time on the output. Care must be taken to ensure that the input High time always exceeds the rising-edge delay.

PWM0 Dead-Band Rising-Edge Delay (PWM0DBRISE)

Base 0x4002.8000
Offset 0x06C
Type R/W, reset 0x0000.0000



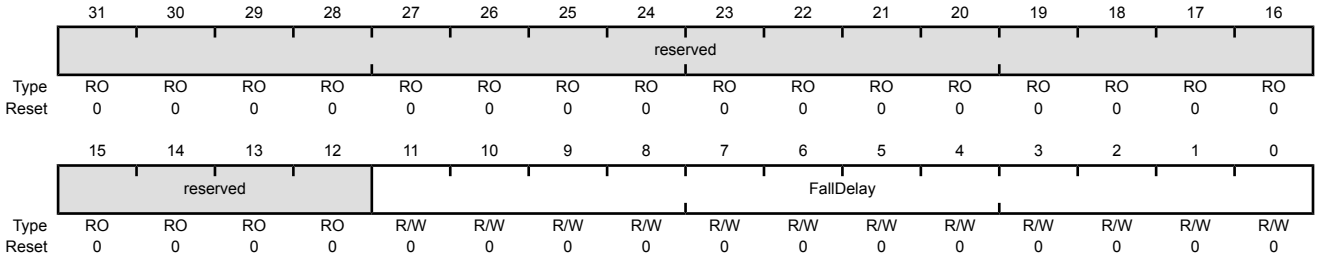
Bit/Field	Name	Type	Reset	Description
31:12	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
11:0	RiseDelay	R/W	0	Dead-Band Rise Delay The number of clock ticks to delay the rising edge.

Register 22: PWM0 Dead-Band Falling-Edge-Delay (PWM0DBFALL), offset 0x070

The **PWM0DBFALL** register contains the number of clock ticks to delay the falling edge of the PWM0A signal when generating the PWM1 signal. If the dead-band generator is disabled, this register is ignored. If the value of this register is larger than the width of a Low pulse on the input PWM signal, the falling-edge delay consumes the entire Low time of the signal, resulting in no Low time on the output. Care must be taken to ensure that the input Low time always exceeds the falling-edge delay.

PWM0 Dead-Band Falling-Edge-Delay (PWM0DBFALL)

Base 0x4002.8000
Offset 0x070
Type R/W, reset 0x0000.0000



Bit/Field	Name	Type	Reset	Description
31:12	reserved	RO	0x00	Software should not rely on the value of a reserved bit. To provide compatibility with future products, the value of a reserved bit should be preserved across a read-modify-write operation.
11:0	FallDelay	R/W	0x00	Dead-Band Fall Delay The number of clock ticks to delay the falling edge.

18 Pin Diagram

The LM3S6432 microcontroller pin diagrams are shown below.

Figure 18-1. 100-Pin LQFP Package Pin Diagram

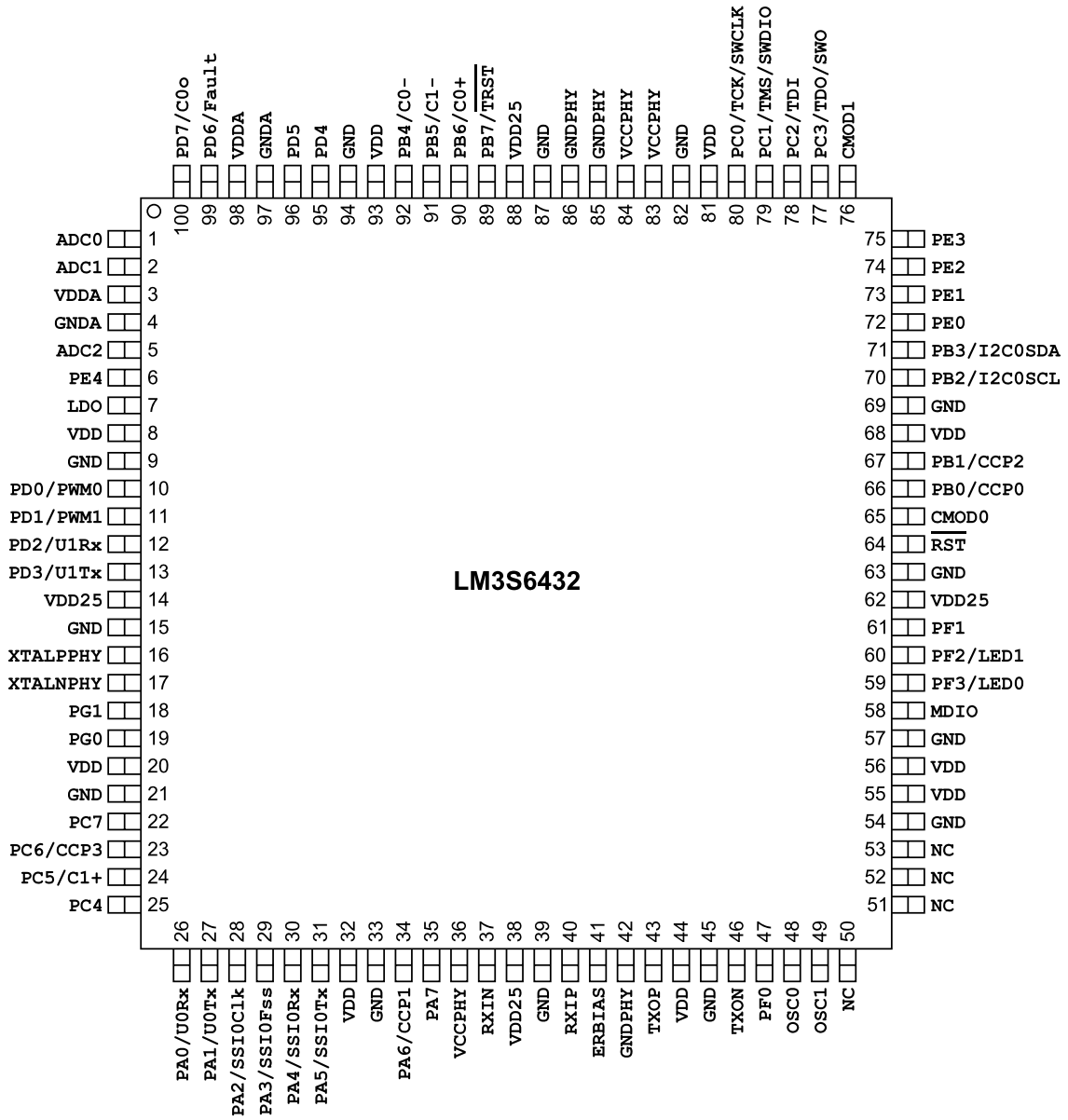
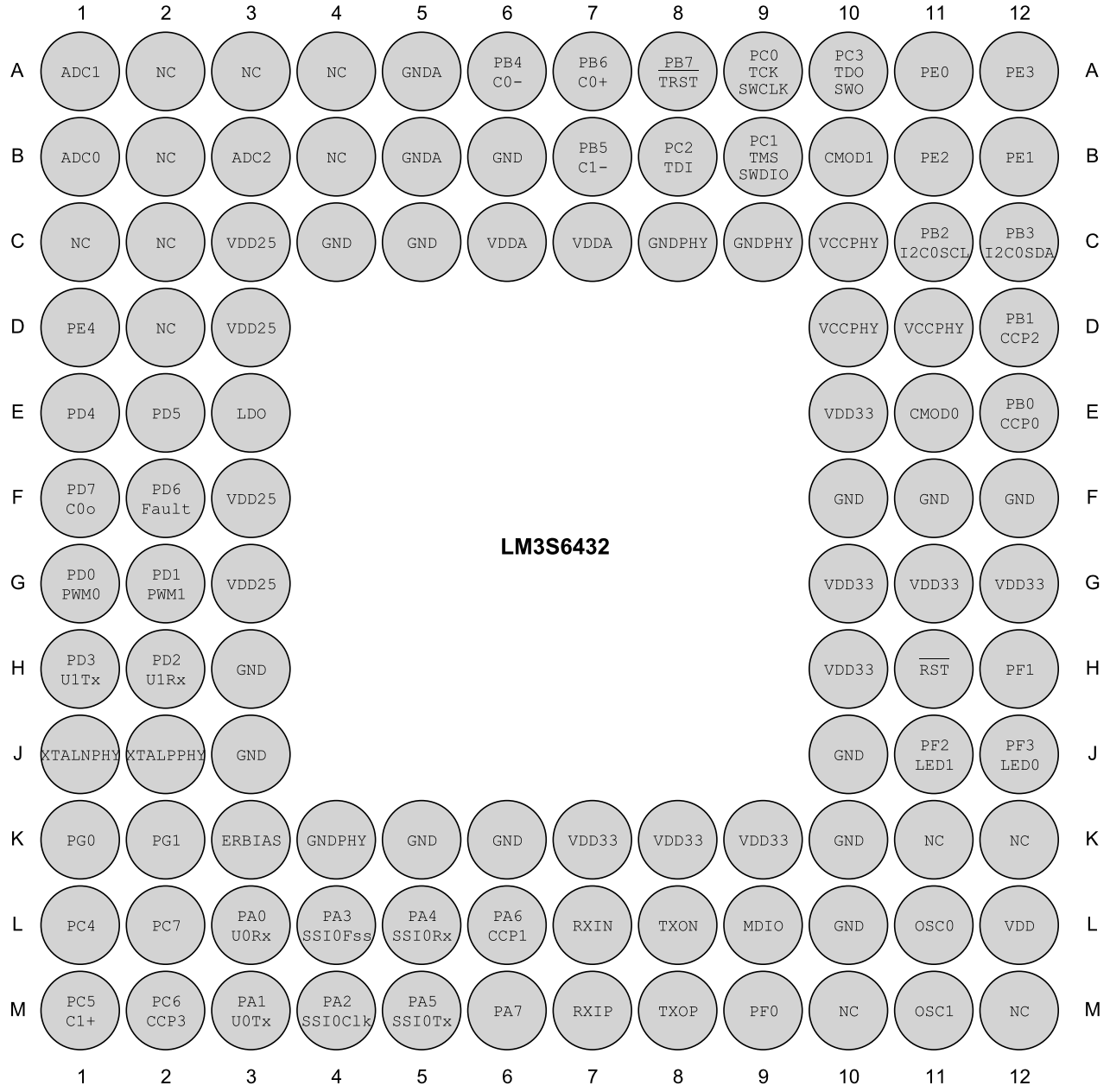


Figure 18-2. 108-Ball BGA Package Pin Diagram (Top View)



19 Signal Tables

The following tables list the signals available for each pin. Functionality is enabled by software with the **GPIOAFSEL** register.

Important: All multiplexed pins are GPIOs by default, with the exception of the five JTAG pins (PB7 and PC[3:0]) which default to the JTAG functionality.

Table 19-1 on page 502 shows the pin-to-signal-name mapping, including functional characteristics of the signals. Table 19-2 on page 506 lists the signals in alphabetical order by signal name.

Table 19-3 on page 509 groups the signals by functionality, except for GPIOs. Table 19-4 on page 512 lists the GPIO pins and their alternate functionality.

19.1 100-Pin LQFP Package Pin Tables

Table 19-1. Signals by Pin Number

Pin Number	Pin Name	Pin Type	Buffer Type[a]	Description
1	ADC0	I	Analog	Analog-to-digital converter input 0.
2	ADC1	I	Analog	Analog-to-digital converter input 1.
3	VDDA	-	Power	The positive supply (3.3 V) for the analog circuits (ADC, Analog Comparators, etc.). These are separated from VDD to minimize the electrical noise contained on VDD from affecting the analog functions. VDDA pins must be connected to 3.3 V, regardless of system implementation.
4	GNDA	-	Power	The ground reference for the analog circuits (ADC, Analog Comparators, etc.). These are separated from GND to minimize the electrical noise contained on VDD from affecting the analog functions.
5	ADC2	I	Analog	Analog-to-digital converter input 2.
6	PE4	I/O	TTL	GPIO port E bit 4.
7	LDO	-	Power	Low drop-out regulator output voltage. This pin requires an external capacitor between the pin and GND of 1 µF or greater. When the on-chip LDO is used to provide power to the logic, the LDO pin must also be connected to the VDD25 pins at the board level in addition to the decoupling capacitor(s).
8	VDD	-	Power	Positive supply for I/O and some logic.
9	GND	-	Power	Ground reference for logic and I/O pins.
10	PD0	I/O	TTL	GPIO port D bit 0.
	PWM0	O	TTL	PWM 0. This signal is controlled by PWM Generator 0.
11	PD1	I/O	TTL	GPIO port D bit 1.
	PWM1	O	TTL	PWM 1. This signal is controlled by PWM Generator 0.
12	PD2	I/O	TTL	GPIO port D bit 2.
	U1Rx	I	TTL	UART module 1 receive. When in IrDA mode, this signal has IrDA modulation.
13	PD3	I/O	TTL	GPIO port D bit 3.
	U1Tx	O	TTL	UART module 1 transmit. When in IrDA mode, this signal has IrDA modulation.
14	VDD25	-	Power	Positive supply for most of the logic function, including the processor core and most peripherals.

Table 19-1. Signals by Pin Number *(continued)*

Pin Number	Pin Name	Pin Type	Buffer Type[a]	Description
15	GND	-	Power	Ground reference for logic and I/O pins.
16	XTALPPHY	I	TTL	Ethernet PHY XTALP 25-MHz oscillator crystal input or external clock reference input.
17	XTALNPHY	O	TTL	Ethernet PHY XTALN 25-MHz oscillator crystal output. Leave unconnected when using a single-ended 25-MHz clock input connected to the XTALPPHY pin.
18	PG1	I/O	TTL	GPIO port G bit 1.
19	PG0	I/O	TTL	GPIO port G bit 0.
20	VDD	-	Power	Positive supply for I/O and some logic.
21	GND	-	Power	Ground reference for logic and I/O pins.
22	PC7	I/O	TTL	GPIO port C bit 7.
23	PC6	I/O	TTL	GPIO port C bit 6.
	CCP3	I/O	TTL	Capture/Compare/PWM 3.
24	PC5	I/O	TTL	GPIO port C bit 5.
	C1+	I	Analog	Analog comparator 1 positive input.
25	PC4	I/O	TTL	GPIO port C bit 4.
26	PA0	I/O	TTL	GPIO port A bit 0.
	U0Rx	I	TTL	UART module 0 receive. When in IrDA mode, this signal has IrDA modulation.
27	PA1	I/O	TTL	GPIO port A bit 1.
	U0Tx	O	TTL	UART module 0 transmit. When in IrDA mode, this signal has IrDA modulation.
28	PA2	I/O	TTL	GPIO port A bit 2.
	SSI0Clk	I/O	TTL	SSI module 0 clock.
29	PA3	I/O	TTL	GPIO port A bit 3.
	SSI0Fss	I/O	TTL	SSI module 0 frame.
30	PA4	I/O	TTL	GPIO port A bit 4.
	SSI0Rx	I	TTL	SSI module 0 receive.
31	PA5	I/O	TTL	GPIO port A bit 5.
	SSI0Tx	O	TTL	SSI module 0 transmit.
32	VDD	-	Power	Positive supply for I/O and some logic.
33	GND	-	Power	Ground reference for logic and I/O pins.
34	PA6	I/O	TTL	GPIO port A bit 6.
	CCP1	I/O	TTL	Capture/Compare/PWM 1.
35	PA7	I/O	TTL	GPIO port A bit 7.
36	VCCPHY	-	Power	VCC of the Ethernet PHY.
37	RXIN	I	Analog	RXIN of the Ethernet PHY.
38	VDD25	-	Power	Positive supply for most of the logic function, including the processor core and most peripherals.
39	GND	-	Power	Ground reference for logic and I/O pins.
40	RXIP	I	Analog	RXIP of the Ethernet PHY.
41	ERBIAS	I	Analog	12.4-kΩ resistor (1% precision) used internally for Ethernet PHY.
42	GNDPHY	-	Power	GND of the Ethernet PHY.

Table 19-1. Signals by Pin Number *(continued)*

Pin Number	Pin Name	Pin Type	Buffer Type[a]	Description
43	TXOP	O	Analog	TXOP of the Ethernet PHY.
44	VDD	-	Power	Positive supply for I/O and some logic.
45	GND	-	Power	Ground reference for logic and I/O pins.
46	TXON	O	Analog	TXON of the Ethernet PHY.
47	PF0	I/O	TTL	GPIO port F bit 0.
48	OSC0	I	Analog	Main oscillator crystal input or an external clock reference input.
49	OSC1	O	Analog	Main oscillator crystal output. Leave unconnected when using a single-ended clock source.
50	NC	-	-	No connect. Leave the pin electrically unconnected/isolated.
51	NC	-	-	No connect. Leave the pin electrically unconnected/isolated.
52	NC	-	-	No connect. Leave the pin electrically unconnected/isolated.
53	NC	-	-	No connect. Leave the pin electrically unconnected/isolated.
54	GND	-	Power	Ground reference for logic and I/O pins.
55	VDD	-	Power	Positive supply for I/O and some logic.
56	VDD	-	Power	Positive supply for I/O and some logic.
57	GND	-	Power	Ground reference for logic and I/O pins.
58	MDIO	I/O	TTL	MDIO of the Ethernet PHY.
59	PF3	I/O	TTL	GPIO port F bit 3.
	LED0	O	TTL	Ethernet LED 0.
60	PF2	I/O	TTL	GPIO port F bit 2.
	LED1	O	TTL	Ethernet LED 1.
61	PF1	I/O	TTL	GPIO port F bit 1.
62	VDD25	-	Power	Positive supply for most of the logic function, including the processor core and most peripherals.
63	GND	-	Power	Ground reference for logic and I/O pins.
64	$\overline{\text{RST}}$	I	TTL	System reset input.
65	CMOD0	I	TTL	CPU Mode bit 0. Input must be set to logic 0 (grounded); other encodings reserved.
66	PB0	I/O	TTL	GPIO port B bit 0.
	CCP0	I/O	TTL	Capture/Compare/PWM 0.
67	PB1	I/O	TTL	GPIO port B bit 1.
	CCP2	I/O	TTL	Capture/Compare/PWM 2.
68	VDD	-	Power	Positive supply for I/O and some logic.
69	GND	-	Power	Ground reference for logic and I/O pins.
70	PB2	I/O	TTL	GPIO port B bit 2.
	I2C0SCL	I/O	OD	I^2C module 0 clock.
71	PB3	I/O	TTL	GPIO port B bit 3.
	I2C0SDA	I/O	OD	I^2C module 0 data.
72	PE0	I/O	TTL	GPIO port E bit 0.
73	PE1	I/O	TTL	GPIO port E bit 1.
74	PE2	I/O	TTL	GPIO port E bit 2.
75	PE3	I/O	TTL	GPIO port E bit 3.

Table 19-1. Signals by Pin Number *(continued)*

Pin Number	Pin Name	Pin Type	Buffer Type[a]	Description
76	CMOD1	I	TTL	CPU Mode bit 1. Input must be set to logic 0 (grounded); other encodings reserved.
77	PC3	I/O	TTL	GPIO port C bit 3.
	SWO	O	TTL	JTAG TDO and SWO.
	TDO	O	TTL	JTAG TDO and SWO.
78	PC2	I/O	TTL	GPIO port C bit 2.
	TDI	I	TTL	JTAG TDI.
79	PC1	I/O	TTL	GPIO port C bit 1.
	SWDIO	I/O	TTL	JTAG TMS and SWDIO.
	TMS	I/O	TTL	JTAG TMS and SWDIO.
80	PC0	I/O	TTL	GPIO port C bit 0.
	SWCLK	I	TTL	JTAG/SWD CLK.
	TCK	I	TTL	JTAG/SWD CLK.
81	VDD	-	Power	Positive supply for I/O and some logic.
82	GND	-	Power	Ground reference for logic and I/O pins.
83	VCCPHY	-	Power	VCC of the Ethernet PHY.
84	VCCPHY	-	Power	VCC of the Ethernet PHY.
85	GNDPHY	-	Power	GND of the Ethernet PHY.
86	GNDPHY	-	Power	GND of the Ethernet PHY.
87	GND	-	Power	Ground reference for logic and I/O pins.
88	VDD25	-	Power	Positive supply for most of the logic function, including the processor core and most peripherals.
89	PB7	I/O	TTL	GPIO port B bit 7.
	TRST	I	TTL	JTAG TRST.
90	PB6	I/O	TTL	GPIO port B bit 6.
	C0+	I	Analog	Analog comparator 0 positive input.
91	PB5	I/O	TTL	GPIO port B bit 5.
	C1-	I	Analog	Analog comparator 1 negative input.
92	PB4	I/O	TTL	GPIO port B bit 4.
	C0-	I	Analog	Analog comparator 0 negative input.
93	VDD	-	Power	Positive supply for I/O and some logic.
94	GND	-	Power	Ground reference for logic and I/O pins.
95	PD4	I/O	TTL	GPIO port D bit 4.
96	PD5	I/O	TTL	GPIO port D bit 5.
97	GNDA	-	Power	The ground reference for the analog circuits (ADC, Analog Comparators, etc.). These are separated from GND to minimize the electrical noise contained on VDD from affecting the analog functions.
98	VDDA	-	Power	The positive supply (3.3 V) for the analog circuits (ADC, Analog Comparators, etc.). These are separated from VDD to minimize the electrical noise contained on VDD from affecting the analog functions. VDDA pins must be connected to 3.3 V, regardless of system implementation.

Table 19-1. Signals by Pin Number *(continued)*

Pin Number	Pin Name	Pin Type	Buffer Type[a]	Description
99	PD6	I/O	TTL	GPIO port D bit 6.
	Fault	I	TTL	PWM Fault.
100	PD7	I/O	TTL	GPIO port D bit 7.
	C0o	O	TTL	Analog comparator 0 output.

a. The TTL designation indicates the pin has TTL-compatible voltage levels.

Table 19-2. Signals by Signal Name

Pin Name	Pin Number	Pin Type	Buffer Type[a]	Description
ADC0	1	I	Analog	Analog-to-digital converter input 0.
ADC1	2	I	Analog	Analog-to-digital converter input 1.
ADC2	5	I	Analog	Analog-to-digital converter input 2.
C0+	90	I	Analog	Analog comparator 0 positive input.
C0-	92	I	Analog	Analog comparator 0 negative input.
C0o	100	O	TTL	Analog comparator 0 output.
C1+	24	I	Analog	Analog comparator 1 positive input.
C1-	91	I	Analog	Analog comparator 1 negative input.
CCP0	66	I/O	TTL	Capture/Compare/PWM 0.
CCP1	34	I/O	TTL	Capture/Compare/PWM 1.
CCP2	67	I/O	TTL	Capture/Compare/PWM 2.
CCP3	23	I/O	TTL	Capture/Compare/PWM 3.
CMOD0	65	I	TTL	CPU Mode bit 0. Input must be set to logic 0 (grounded); other encodings reserved.
CMOD1	76	I	TTL	CPU Mode bit 1. Input must be set to logic 0 (grounded); other encodings reserved.
ERBIAS	41	I	Analog	12.4-kΩ resistor (1% precision) used internally for Ethernet PHY.
Fault	99	I	TTL	PWM Fault.
GND	9 15 21 33 39 45 54 57 63 69 82 87 94	-	Power	Ground reference for logic and I/O pins.
GNDA	4 97	-	Power	The ground reference for the analog circuits (ADC, Analog Comparators, etc.). These are separated from GND to minimize the electrical noise contained on VDD from affecting the analog functions.
GNDPHY	42 85 86	-	Power	GND of the Ethernet PHY.

Table 19-2. Signals by Signal Name *(continued)*

Pin Name	Pin Number	Pin Type	Buffer Type[a]	Description
I2C0SCL	70	I/O	OD	I^2C module 0 clock.
I2C0SDA	71	I/O	OD	I^2C module 0 data.
LDO	7	-	Power	Low drop-out regulator output voltage. This pin requires an external capacitor between the pin and GND of 1 µF or greater. When the on-chip LDO is used to provide power to the logic, the LDO pin must also be connected to the VDD25 pins at the board level in addition to the decoupling capacitor(s).
LED0	59	O	TTL	Ethernet LED 0.
LED1	60	O	TTL	Ethernet LED 1.
MDIO	58	I/O	TTL	MDIO of the Ethernet PHY.
NC	50 51 52 53	-	-	No connect. Leave the pin electrically unconnected/isolated.
OSC0	48	I	Analog	Main oscillator crystal input or an external clock reference input.
OSC1	49	O	Analog	Main oscillator crystal output. Leave unconnected when using a single-ended clock source.
PA0	26	I/O	TTL	GPIO port A bit 0.
PA1	27	I/O	TTL	GPIO port A bit 1.
PA2	28	I/O	TTL	GPIO port A bit 2.
PA3	29	I/O	TTL	GPIO port A bit 3.
PA4	30	I/O	TTL	GPIO port A bit 4.
PA5	31	I/O	TTL	GPIO port A bit 5.
PA6	34	I/O	TTL	GPIO port A bit 6.
PA7	35	I/O	TTL	GPIO port A bit 7.
PB0	66	I/O	TTL	GPIO port B bit 0.
PB1	67	I/O	TTL	GPIO port B bit 1.
PB2	70	I/O	TTL	GPIO port B bit 2.
PB3	71	I/O	TTL	GPIO port B bit 3.
PB4	92	I/O	TTL	GPIO port B bit 4.
PB5	91	I/O	TTL	GPIO port B bit 5.
PB6	90	I/O	TTL	GPIO port B bit 6.
PB7	89	I/O	TTL	GPIO port B bit 7.
PC0	80	I/O	TTL	GPIO port C bit 0.
PC1	79	I/O	TTL	GPIO port C bit 1.
PC2	78	I/O	TTL	GPIO port C bit 2.
PC3	77	I/O	TTL	GPIO port C bit 3.
PC4	25	I/O	TTL	GPIO port C bit 4.
PC5	24	I/O	TTL	GPIO port C bit 5.
PC6	23	I/O	TTL	GPIO port C bit 6.
PC7	22	I/O	TTL	GPIO port C bit 7.
PD0	10	I/O	TTL	GPIO port D bit 0.

Table 19-2. Signals by Signal Name *(continued)*

Pin Name	Pin Number	Pin Type	Buffer Type[a]	Description
PD1	11	I/O	TTL	GPIO port D bit 1.
PD2	12	I/O	TTL	GPIO port D bit 2.
PD3	13	I/O	TTL	GPIO port D bit 3.
PD4	95	I/O	TTL	GPIO port D bit 4.
PD5	96	I/O	TTL	GPIO port D bit 5.
PD6	99	I/O	TTL	GPIO port D bit 6.
PD7	100	I/O	TTL	GPIO port D bit 7.
PE0	72	I/O	TTL	GPIO port E bit 0.
PE1	73	I/O	TTL	GPIO port E bit 1.
PE2	74	I/O	TTL	GPIO port E bit 2.
PE3	75	I/O	TTL	GPIO port E bit 3.
PE4	6	I/O	TTL	GPIO port E bit 4.
PF0	47	I/O	TTL	GPIO port F bit 0.
PF1	61	I/O	TTL	GPIO port F bit 1.
PF2	60	I/O	TTL	GPIO port F bit 2.
PF3	59	I/O	TTL	GPIO port F bit 3.
PG0	19	I/O	TTL	GPIO port G bit 0.
PG1	18	I/O	TTL	GPIO port G bit 1.
PWM0	10	O	TTL	PWM 0. This signal is controlled by PWM Generator 0.
PWM1	11	O	TTL	PWM 1. This signal is controlled by PWM Generator 0.
RST	64	I	TTL	System reset input.
RXIN	37	I	Analog	RXIN of the Ethernet PHY.
RXIP	40	I	Analog	RXIP of the Ethernet PHY.
SSI0Clk	28	I/O	TTL	SSI module 0 clock.
SSI0Fss	29	I/O	TTL	SSI module 0 frame.
SSI0Rx	30	I	TTL	SSI module 0 receive.
SSI0Tx	31	O	TTL	SSI module 0 transmit.
SWCLK	80	I	TTL	JTAG/SWD CLK.
SWDIO	79	I/O	TTL	JTAG TMS and SWDIO.
SWO	77	O	TTL	JTAG TDO and SWO.
TCK	80	I	TTL	JTAG/SWD CLK.
TDI	78	I	TTL	JTAG TDI.
TDO	77	O	TTL	JTAG TDO and SWO.
TMS	79	I/O	TTL	JTAG TMS and SWDIO.
TRST	89	I	TTL	JTAG TRST.
TXON	46	O	Analog	TXON of the Ethernet PHY.
TXOP	43	O	Analog	TXOP of the Ethernet PHY.
U0Rx	26	I	TTL	UART module 0 receive. When in IrDA mode, this signal has IrDA modulation.
U0Tx	27	O	TTL	UART module 0 transmit. When in IrDA mode, this signal has IrDA modulation.

Table 19-2. Signals by Signal Name *(continued)*

Pin Name	Pin Number	Pin Type	Buffer Type[a]	Description
U1Rx	12	I	TTL	UART module 1 receive. When in IrDA mode, this signal has IrDA modulation.
U1Tx	13	O	TTL	UART module 1 transmit. When in IrDA mode, this signal has IrDA modulation.
VCCPHY	36 83 84	-	Power	VCC of the Ethernet PHY.
VDD	8 20 32 44 55 56 68 81 93	-	Power	Positive supply for I/O and some logic.
VDD25	14 38 62 88	-	Power	Positive supply for most of the logic function, including the processor core and most peripherals.
VDDA	3 98	-	Power	The positive supply (3.3 V) for the analog circuits (ADC, Analog Comparators, etc.). These are separated from VDD to minimize the electrical noise contained on VDD from affecting the analog functions. VDDA pins must be connected to 3.3 V, regardless of system implementation.
XTALNPHY	17	O	TTL	Ethernet PHY XTALN 25-MHz oscillator crystal output. Leave unconnected when using a single-ended 25-MHz clock input connected to the XTALPPHY pin.
XTALPPHY	16	I	TTL	Ethernet PHY XTALP 25-MHz oscillator crystal input or external clock reference input.

a. The TTL designation indicates the pin has TTL-compatible voltage levels.

Table 19-3. Signals by Function, Except for GPIO

Function	Pin Name	Pin Number	Pin Type	Buffer Type[a]	Description
ADC	ADC0	1	I	Analog	Analog-to-digital converter input 0.
	ADC1	2	I	Analog	Analog-to-digital converter input 1.
	ADC2	5	I	Analog	Analog-to-digital converter input 2.
Analog Comparators	C0+	90	I	Analog	Analog comparator 0 positive input.
	C0-	92	I	Analog	Analog comparator 0 negative input.
	C0o	100	O	TTL	Analog comparator 0 output.
	C1+	24	I	Analog	Analog comparator 1 positive input.
	C1-	91	I	Analog	Analog comparator 1 negative input.

Table 19-3. Signals by Function, Except for GPIO *(continued)*

Function	Pin Name	Pin Number	Pin Type	Buffer Type[a]	Description
Ethernet	ERBIAS	41	I	Analog	12.4-kΩ resistor (1% precision) used internally for Ethernet PHY.
	GNDPHY	42 85 86	-	Power	GND of the Ethernet PHY.
	LED0	59	O	TTL	Ethernet LED 0.
	LED1	60	O	TTL	Ethernet LED 1.
	MDIO	58	I/O	TTL	MDIO of the Ethernet PHY.
	RXIN	37	I	Analog	RXIN of the Ethernet PHY.
	RXIP	40	I	Analog	RXIP of the Ethernet PHY.
	TXON	46	O	Analog	TXON of the Ethernet PHY.
	TXOP	43	O	Analog	TXOP of the Ethernet PHY.
	VCCPHY	36 83 84	-	Power	VCC of the Ethernet PHY.
	XTALNPHY	17	O	TTL	Ethernet PHY XTALN 25-MHz oscillator crystal output. Leave unconnected when using a single-ended 25-MHz clock input connected to the XTALPPHY pin.
	XTALPPHY	16	I	TTL	Ethernet PHY XTALP 25-MHz oscillator crystal input or external clock reference input.
General-Purpose Timers	CCP0	66	I/O	TTL	Capture/Compare/PWM 0.
	CCP1	34	I/O	TTL	Capture/Compare/PWM 1.
	CCP2	67	I/O	TTL	Capture/Compare/PWM 2.
	CCP3	23	I/O	TTL	Capture/Compare/PWM 3.
I2C	I2C0SCL	70	I/O	OD	I^2C module 0 clock.
	I2C0SDA	71	I/O	OD	I^2C module 0 data.
JTAG/SWD/SWO	SWCLK	80	I	TTL	JTAG/SWD CLK.
	SWDIO	79	I/O	TTL	JTAG TMS and SWDIO.
	SWO	77	O	TTL	JTAG TDO and SWO.
	TCK	80	I	TTL	JTAG/SWD CLK.
	TDI	78	I	TTL	JTAG TDI.
	TDO	77	O	TTL	JTAG TDO and SWO.
	TMS	79	I/O	TTL	JTAG TMS and SWDIO.
	$\overline{\text{TRST}}$	89	I	TTL	JTAG $\overline{\text{TRST}}$.
PWM	Fault	99	I	TTL	PWM Fault.
	PWM0	10	O	TTL	PWM 0. This signal is controlled by PWM Generator 0.
	PWM1	11	O	TTL	PWM 1. This signal is controlled by PWM Generator 0.

Table 19-3. Signals by Function, Except for GPIO *(continued)*

Function	Pin Name	Pin Number	Pin Type	Buffer Type[a]	Description
Power	GND	9 15 21 33 39 45 54 57 63 69 82 87 94	-	Power	Ground reference for logic and I/O pins.
	GNDA	4 97	-	Power	The ground reference for the analog circuits (ADC, Analog Comparators, etc.). These are separated from GND to minimize the electrical noise contained on VDD from affecting the analog functions.
	LDO	7	-	Power	Low drop-out regulator output voltage. This pin requires an external capacitor between the pin and GND of 1 µF or greater. When the on-chip LDO is used to provide power to the logic, the LDO pin must also be connected to the VDD25 pins at the board level in addition to the decoupling capacitor(s).
	VDD	8 20 32 44 55 56 68 81 93	-	Power	Positive supply for I/O and some logic.
	VDD25	14 38 62 88	-	Power	Positive supply for most of the logic function, including the processor core and most peripherals.
	VDDA	3 98	-	Power	The positive supply (3.3 V) for the analog circuits (ADC, Analog Comparators, etc.). These are separated from VDD to minimize the electrical noise contained on VDD from affecting the analog functions. VDDA pins must be connected to 3.3 V, regardless of system implementation.
SSI	SSI0Clk	28	I/O	TTL	SSI module 0 clock.
	SSI0Fss	29	I/O	TTL	SSI module 0 frame.
	SSI0Rx	30	I	TTL	SSI module 0 receive.
	SSI0Tx	31	O	TTL	SSI module 0 transmit.

Table 19-3. Signals by Function, Except for GPIO *(continued)*

Function	Pin Name	Pin Number	Pin Type	Buffer Type[a]	Description
System Control & Clocks	CMOD0	65	I	TTL	CPU Mode bit 0. Input must be set to logic 0 (grounded); other encodings reserved.
	CMOD1	76	I	TTL	CPU Mode bit 1. Input must be set to logic 0 (grounded); other encodings reserved.
	OSC0	48	I	Analog	Main oscillator crystal input or an external clock reference input.
	OSC1	49	O	Analog	Main oscillator crystal output. Leave unconnected when using a single-ended clock source.
	RST	64	I	TTL	System reset input.
UART	U0Rx	26	I	TTL	UART module 0 receive. When in IrDA mode, this signal has IrDA modulation.
	U0Tx	27	O	TTL	UART module 0 transmit. When in IrDA mode, this signal has IrDA modulation.
	U1Rx	12	I	TTL	UART module 1 receive. When in IrDA mode, this signal has IrDA modulation.
	U1Tx	13	O	TTL	UART module 1 transmit. When in IrDA mode, this signal has IrDA modulation.

a. The TTL designation indicates the pin has TTL-compatible voltage levels.

Table 19-4. GPIO Pins and Alternate Functions

IO	Pin Number	Multiplexed Function	Multiplexed Function
PA0	26	U0Rx	
PA1	27	U0Tx	
PA2	28	SSI0Clk	
PA3	29	SSI0Fss	
PA4	30	SSI0Rx	
PA5	31	SSI0Tx	
PA6	34	CCP1	
PA7	35		
PB0	66	CCP0	
PB1	67	CCP2	
PB2	70	I2C0SCL	
PB3	71	I2C0SDA	
PB4	92	C0-	
PB5	91	C1-	
PB6	90	C0+	
PB7	89	TRST	
PC0	80	TCK	SWCLK
PC1	79	TMS	SWDIO
PC2	78	TDI	
PC3	77	TDO	SWO
PC4	25		
PC5	24	C1+	
PC6	23	CCP3	

Table 19-4. GPIO Pins and Alternate Functions *(continued)*

IO	Pin Number	Multiplexed Function	Multiplexed Function
PC7	22		
PD0	10	PWM0	
PD1	11	PWM1	
PD2	12	U1Rx	
PD3	13	U1Tx	
PD4	95		
PD5	96		
PD6	99	Fault	
PD7	100	C0o	
PE0	72		
PE1	73		
PE2	74		
PE3	75		
PE4	6		
PF0	47		
PF1	61		
PF2	60	LED1	
PF3	59	LED0	
PG0	19		
PG1	18		

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Table 19-5. Signals by Pin Number

Pin Number	Pin Name	Pin Type	Buffer Type[a]	Description
A1	ADC1	I	Analog	Analog-to-digital converter input 1.
A2	NC	-	-	No connect. Leave the pin electrically unconnected/isolated.
A3	NC	-	-	No connect. Leave the pin electrically unconnected/isolated.
A4	NC	-	-	No connect. Leave the pin electrically unconnected/isolated.
A5	GNDA	-	Power	The ground reference for the analog circuits (ADC, Analog Comparators, etc.). These are separated from GND to minimize the electrical noise contained on VDD from affecting the analog functions.
A6	PB4	I/O	TTL	GPIO port B bit 4.
	C0-	I	Analog	Analog comparator 0 negative input.
A7	PB6	I/O	TTL	GPIO port B bit 6.
	C0+	I	Analog	Analog comparator 0 positive input.
A8	PB7	I/O	TTL	GPIO port B bit 7.
	TRST	I	TTL	JTAG TRST.
A9	PC0	I/O	TTL	GPIO port C bit 0.
	SWCLK	I	TTL	JTAG/SWD CLK.
	TCK	I	TTL	JTAG/SWD CLK.

Table 19-5. Signals by Pin Number *(continued)*

Pin Number	Pin Name	Pin Type	Buffer Type[a]	Description
A10	PC3	I/O	TTL	GPIO port C bit 3.
	SWO	O	TTL	JTAG TDO and SWO.
	TDO	O	TTL	JTAG TDO and SWO.
A11	PE0	I/O	TTL	GPIO port E bit 0.
A12	PE3	I/O	TTL	GPIO port E bit 3.
B1	ADC0	I	Analog	Analog-to-digital converter input 0.
B2	NC	-	-	No connect. Leave the pin electrically unconnected/isolated.
B3	ADC2	I	Analog	Analog-to-digital converter input 2.
B4	NC	-	-	No connect. Leave the pin electrically unconnected/isolated.
B5	GNDA	-	Power	The ground reference for the analog circuits (ADC, Analog Comparators, etc.). These are separated from GND to minimize the electrical noise contained on VDD from affecting the analog functions.
B6	GND	-	Power	Ground reference for logic and I/O pins.
B7	PB5	I/O	TTL	GPIO port B bit 5.
	C1-	I	Analog	Analog comparator 1 negative input.
B8	PC2	I/O	TTL	GPIO port C bit 2.
	TDI	I	TTL	JTAG TDI.
B9	PC1	I/O	TTL	GPIO port C bit 1.
	SWDIO	I/O	TTL	JTAG TMS and SWDIO.
	TMS	I/O	TTL	JTAG TMS and SWDIO.
B10	CMOD1	I	TTL	CPU Mode bit 1. Input must be set to logic 0 (grounded); other encodings reserved.
B11	PE2	I/O	TTL	GPIO port E bit 2.
B12	PE1	I/O	TTL	GPIO port E bit 1.
C1	NC	-	-	No connect. Leave the pin electrically unconnected/isolated.
C2	NC	-	-	No connect. Leave the pin electrically unconnected/isolated.
C3	VDD25	-	Power	Positive supply for most of the logic function, including the processor core and most peripherals.
C4	GND	-	Power	Ground reference for logic and I/O pins.
C5	GND	-	Power	Ground reference for logic and I/O pins.
C6	VDDA	-	Power	The positive supply (3.3 V) for the analog circuits (ADC, Analog Comparators, etc.). These are separated from VDD to minimize the electrical noise contained on VDD from affecting the analog functions. VDDA pins must be connected to 3.3 V, regardless of system implementation.
C7	VDDA	-	Power	The positive supply (3.3 V) for the analog circuits (ADC, Analog Comparators, etc.). These are separated from VDD to minimize the electrical noise contained on VDD from affecting the analog functions. VDDA pins must be connected to 3.3 V, regardless of system implementation.
C8	GNDPHY	-	Power	GND of the Ethernet PHY.
C9	GNDPHY	-	Power	GND of the Ethernet PHY.
C10	VCCPHY	-	Power	VCC of the Ethernet PHY.

Table 19-5. Signals by Pin Number *(continued)*

Pin Number	Pin Name	Pin Type	Buffer Type[a]	Description
C11	PB2	I/O	TTL	GPIO port B bit 2.
	I2C0SCL	I/O	OD	I^2C module 0 clock.
C12	PB3	I/O	TTL	GPIO port B bit 3.
	I2C0SDA	I/O	OD	I^2C module 0 data.
D1	PE4	I/O	TTL	GPIO port E bit 4.
D2	NC	-	-	No connect. Leave the pin electrically unconnected/isolated.
D3	VDD25	-	Power	Positive supply for most of the logic function, including the processor core and most peripherals.
D10	VCCPHY	-	Power	VCC of the Ethernet PHY.
D11	VCCPHY	-	Power	VCC of the Ethernet PHY.
D12	PB1	I/O	TTL	GPIO port B bit 1.
	CCP2	I/O	TTL	Capture/Compare/PWM 2.
E1	PD4	I/O	TTL	GPIO port D bit 4.
E2	PD5	I/O	TTL	GPIO port D bit 5.
E3	LDO	-	Power	Low drop-out regulator output voltage. This pin requires an external capacitor between the pin and GND of 1 µF or greater. When the on-chip LDO is used to provide power to the logic, the LDO pin must also be connected to the VDD25 pins at the board level in addition to the decoupling capacitor(s).
E10	VDD33	-	Power	Positive supply for I/O and some logic.
E11	CMOD0	I	TTL	CPU Mode bit 0. Input must be set to logic 0 (grounded); other encodings reserved.
E12	PB0	I/O	TTL	GPIO port B bit 0.
	CCP0	I/O	TTL	Capture/Compare/PWM 0.
F1	PD7	I/O	TTL	GPIO port D bit 7.
	C0o	O	TTL	Analog comparator 0 output.
F2	PD6	I/O	TTL	GPIO port D bit 6.
	Fault	I	TTL	PWM Fault.
F3	VDD25	-	Power	Positive supply for most of the logic function, including the processor core and most peripherals.
F10	GND	-	Power	Ground reference for logic and I/O pins.
F11	GND	-	Power	Ground reference for logic and I/O pins.
F12	GND	-	Power	Ground reference for logic and I/O pins.
G1	PD0	I/O	TTL	GPIO port D bit 0.
	PWM0	O	TTL	PWM 0. This signal is controlled by PWM Generator 0.
G2	PD1	I/O	TTL	GPIO port D bit 1.
	PWM1	O	TTL	PWM 1. This signal is controlled by PWM Generator 0.
G3	VDD25	-	Power	Positive supply for most of the logic function, including the processor core and most peripherals.
G10	VDD33	-	Power	Positive supply for I/O and some logic.
G11	VDD33	-	Power	Positive supply for I/O and some logic.
G12	VDD33	-	Power	Positive supply for I/O and some logic.

Table 19-5. Signals by Pin Number *(continued)*

Pin Number	Pin Name	Pin Type	Buffer Type[a]	Description
H1	PD3	I/O	TTL	GPIO port D bit 3.
	U1Tx	O	TTL	UART module 1 transmit. When in IrDA mode, this signal has IrDA modulation.
H2	PD2	I/O	TTL	GPIO port D bit 2.
	U1Rx	I	TTL	UART module 1 receive. When in IrDA mode, this signal has IrDA modulation.
H3	GND	-	Power	Ground reference for logic and I/O pins.
H10	VDD33	-	Power	Positive supply for I/O and some logic.
H11	$\overline{\text{RST}}$	I	TTL	System reset input.
H12	PF1	I/O	TTL	GPIO port F bit 1.
J1	XTALNPHY	O	TTL	Ethernet PHY XTALN 25-MHz oscillator crystal output. Leave unconnected when using a single-ended 25-MHz clock input connected to the XTALPPHY pin.
J2	XTALPPHY	I	TTL	Ethernet PHY XTALP 25-MHz oscillator crystal input or external clock reference input.
J3	GND	-	Power	Ground reference for logic and I/O pins.
J10	GND	-	Power	Ground reference for logic and I/O pins.
J11	PF2	I/O	TTL	GPIO port F bit 2.
	LED1	O	TTL	Ethernet LED 1.
J12	PF3	I/O	TTL	GPIO port F bit 3.
	LED0	O	TTL	Ethernet LED 0.
K1	PG0	I/O	TTL	GPIO port G bit 0.
K2	PG1	I/O	TTL	GPIO port G bit 1.
K3	ERBIAS	I	Analog	12.4-kΩ resistor (1% precision) used internally for Ethernet PHY.
K4	GNDPHY	-	Power	GND of the Ethernet PHY.
K5	GND	-	Power	Ground reference for logic and I/O pins.
K6	GND	-	Power	Ground reference for logic and I/O pins.
K7	VDD33	-	Power	Positive supply for I/O and some logic.
K8	VDD33	-	Power	Positive supply for I/O and some logic.
K9	VDD33	-	Power	Positive supply for I/O and some logic.
K10	GND	-	Power	Ground reference for logic and I/O pins.
K11	NC	-	-	No connect. Leave the pin electrically unconnected/isolated.
K12	NC	-	-	No connect. Leave the pin electrically unconnected/isolated.
L1	PC4	I/O	TTL	GPIO port C bit 4.
L2	PC7	I/O	TTL	GPIO port C bit 7.
L3	PA0	I/O	TTL	GPIO port A bit 0.
	U0Rx	I	TTL	UART module 0 receive. When in IrDA mode, this signal has IrDA modulation.
L4	PA3	I/O	TTL	GPIO port A bit 3.
	SSI0Fss	I/O	TTL	SSI module 0 frame.
L5	PA4	I/O	TTL	GPIO port A bit 4.
	SSI0Rx	I	TTL	SSI module 0 receive.

Table 19-5. Signals by Pin Number *(continued)*

Pin Number	Pin Name	Pin Type	Buffer Type[a]	Description
L6	PA6	I/O	TTL	GPIO port A bit 6.
	CCP1	I/O	TTL	Capture/Compare/PWM 1.
L7	RXIN	I	Analog	RXIN of the Ethernet PHY.
L8	TXON	O	Analog	TXON of the Ethernet PHY.
L9	MDIO	I/O	TTL	MDIO of the Ethernet PHY.
L10	GND	-	Power	Ground reference for logic and I/O pins.
L11	OSC0	I	Analog	Main oscillator crystal input or an external clock reference input.
L12	VDD	-	Power	Positive supply for I/O and some logic.
M1	PC5	I/O	TTL	GPIO port C bit 5.
	C1+	I	Analog	Analog comparator 1 positive input.
M2	PC6	I/O	TTL	GPIO port C bit 6.
	CCP3	I/O	TTL	Capture/Compare/PWM 3.
M3	PA1	I/O	TTL	GPIO port A bit 1.
	U0Tx	O	TTL	UART module 0 transmit. When in IrDA mode, this signal has IrDA modulation.
M4	PA2	I/O	TTL	GPIO port A bit 2.
	SSI0Clk	I/O	TTL	SSI module 0 clock.
M5	PA5	I/O	TTL	GPIO port A bit 5.
	SSI0Tx	O	TTL	SSI module 0 transmit.
M6	PA7	I/O	TTL	GPIO port A bit 7.
M7	RXIP	I	Analog	RXIP of the Ethernet PHY.
M8	TXOP	O	Analog	TXOP of the Ethernet PHY.
M9	PF0	I/O	TTL	GPIO port F bit 0.
M10	NC	-	-	No connect. Leave the pin electrically unconnected/isolated.
M11	OSC1	O	Analog	Main oscillator crystal output. Leave unconnected when using a single-ended clock source.
M12	NC	-	-	No connect. Leave the pin electrically unconnected/isolated.

a. The TTL designation indicates the pin has TTL-compatible voltage levels.

Table 19-6. Signals by Signal Name

Pin Name	Pin Number	Pin Type	Buffer Type[a]	Description
ADC0	B1	I	Analog	Analog-to-digital converter input 0.
ADC1	A1	I	Analog	Analog-to-digital converter input 1.
ADC2	B3	I	Analog	Analog-to-digital converter input 2.
C0+	A7	I	Analog	Analog comparator 0 positive input.
C0-	A6	I	Analog	Analog comparator 0 negative input.
C0o	F1	O	TTL	Analog comparator 0 output.
C1+	M1	I	Analog	Analog comparator 1 positive input.
C1-	B7	I	Analog	Analog comparator 1 negative input.
CCP0	E12	I/O	TTL	Capture/Compare/PWM 0.
CCP1	L6	I/O	TTL	Capture/Compare/PWM 1.
CCP2	D12	I/O	TTL	Capture/Compare/PWM 2.

Table 19-6. Signals by Signal Name *(continued)*

Pin Name	Pin Number	Pin Type	Buffer Type[a]	Description
CCP3	M2	I/O	TTL	Capture/Compare/PWM 3.
CMOD0	E11	I	TTL	CPU Mode bit 0. Input must be set to logic 0 (grounded); other encodings reserved.
CMOD1	B10	I	TTL	CPU Mode bit 1. Input must be set to logic 0 (grounded); other encodings reserved.
ERBIAS	K3	I	Analog	12.4-kΩ resistor (1% precision) used internally for Ethernet PHY.
Fault	F2	I	TTL	PWM Fault.
GND	B6 C4 C5 F10 F11 F12 H3 J3 J10 K5 K6 K10 L10	-	Power	Ground reference for logic and I/O pins.
GNDA	A5 B5	-	Power	The ground reference for the analog circuits (ADC, Analog Comparators, etc.). These are separated from GND to minimize the electrical noise contained on VDD from affecting the analog functions.
GNDPHY	C8 C9 K4	-	Power	GND of the Ethernet PHY.
I2C0SCL	C11	I/O	OD	I^2C module 0 clock.
I2C0SDA	C12	I/O	OD	I^2C module 0 data.
LDO	E3	-	Power	Low drop-out regulator output voltage. This pin requires an external capacitor between the pin and GND of 1 µF or greater. When the on-chip LDO is used to provide power to the logic, the LDO pin must also be connected to the VDD25 pins at the board level in addition to the decoupling capacitor(s).
LED0	J12	O	TTL	Ethernet LED 0.
LED1	J11	O	TTL	Ethernet LED 1.
MDIO	L9	I/O	TTL	MDIO of the Ethernet PHY.
NC	A2 A3 A4 B2 B4 C1 C2 D2 K11 K12 M10 M12	-	-	No connect. Leave the pin electrically unconnected/isolated.
OSC0	L11	I	Analog	Main oscillator crystal input or an external clock reference input.

Table 19-6. Signals by Signal Name *(continued)*

Pin Name	Pin Number	Pin Type	Buffer Type[a]	Description
OSC1	M11	O	Analog	Main oscillator crystal output. Leave unconnected when using a single-ended clock source.
PA0	L3	I/O	TTL	GPIO port A bit 0.
PA1	M3	I/O	TTL	GPIO port A bit 1.
PA2	M4	I/O	TTL	GPIO port A bit 2.
PA3	L4	I/O	TTL	GPIO port A bit 3.
PA4	L5	I/O	TTL	GPIO port A bit 4.
PA5	M5	I/O	TTL	GPIO port A bit 5.
PA6	L6	I/O	TTL	GPIO port A bit 6.
PA7	M6	I/O	TTL	GPIO port A bit 7.
PB0	E12	I/O	TTL	GPIO port B bit 0.
PB1	D12	I/O	TTL	GPIO port B bit 1.
PB2	C11	I/O	TTL	GPIO port B bit 2.
PB3	C12	I/O	TTL	GPIO port B bit 3.
PB4	A6	I/O	TTL	GPIO port B bit 4.
PB5	B7	I/O	TTL	GPIO port B bit 5.
PB6	A7	I/O	TTL	GPIO port B bit 6.
PB7	A8	I/O	TTL	GPIO port B bit 7.
PC0	A9	I/O	TTL	GPIO port C bit 0.
PC1	B9	I/O	TTL	GPIO port C bit 1.
PC2	B8	I/O	TTL	GPIO port C bit 2.
PC3	A10	I/O	TTL	GPIO port C bit 3.
PC4	L1	I/O	TTL	GPIO port C bit 4.
PC5	M1	I/O	TTL	GPIO port C bit 5.
PC6	M2	I/O	TTL	GPIO port C bit 6.
PC7	L2	I/O	TTL	GPIO port C bit 7.
PD0	G1	I/O	TTL	GPIO port D bit 0.
PD1	G2	I/O	TTL	GPIO port D bit 1.
PD2	H2	I/O	TTL	GPIO port D bit 2.
PD3	H1	I/O	TTL	GPIO port D bit 3.
PD4	E1	I/O	TTL	GPIO port D bit 4.
PD5	E2	I/O	TTL	GPIO port D bit 5.
PD6	F2	I/O	TTL	GPIO port D bit 6.
PD7	F1	I/O	TTL	GPIO port D bit 7.
PE0	A11	I/O	TTL	GPIO port E bit 0.
PE1	B12	I/O	TTL	GPIO port E bit 1.
PE2	B11	I/O	TTL	GPIO port E bit 2.
PE3	A12	I/O	TTL	GPIO port E bit 3.
PE4	D1	I/O	TTL	GPIO port E bit 4.
PF0	M9	I/O	TTL	GPIO port F bit 0.
PF1	H12	I/O	TTL	GPIO port F bit 1.

Table 19-6. Signals by Signal Name *(continued)*

Pin Name	Pin Number	Pin Type	Buffer Type[a]	Description
PF2	J11	I/O	TTL	GPIO port F bit 2.
PF3	J12	I/O	TTL	GPIO port F bit 3.
PG0	K1	I/O	TTL	GPIO port G bit 0.
PG1	K2	I/O	TTL	GPIO port G bit 1.
PWM0	G1	O	TTL	PWM 0. This signal is controlled by PWM Generator 0.
PWM1	G2	O	TTL	PWM 1. This signal is controlled by PWM Generator 0.
RST	H11	I	TTL	System reset input.
RXIN	L7	I	Analog	RXIN of the Ethernet PHY.
RXIP	M7	I	Analog	RXIP of the Ethernet PHY.
SSI0Clk	M4	I/O	TTL	SSI module 0 clock.
SSI0Fss	L4	I/O	TTL	SSI module 0 frame.
SSI0Rx	L5	I	TTL	SSI module 0 receive.
SSI0Tx	M5	O	TTL	SSI module 0 transmit.
SWCLK	A9	I	TTL	JTAG/SWD CLK.
SWDIO	B9	I/O	TTL	JTAG TMS and SWDIO.
SWO	A10	O	TTL	JTAG TDO and SWO.
TCK	A9	I	TTL	JTAG/SWD CLK.
TDI	B8	I	TTL	JTAG TDI.
TDO	A10	O	TTL	JTAG TDO and SWO.
TMS	B9	I/O	TTL	JTAG TMS and SWDIO.
TRST	A8	I	TTL	JTAG TRST.
TXON	L8	O	Analog	TXON of the Ethernet PHY.
TXOP	M8	O	Analog	TXOP of the Ethernet PHY.
U0Rx	L3	I	TTL	UART module 0 receive. When in IrDA mode, this signal has IrDA modulation.
U0Tx	M3	O	TTL	UART module 0 transmit. When in IrDA mode, this signal has IrDA modulation.
U1Rx	H2	I	TTL	UART module 1 receive. When in IrDA mode, this signal has IrDA modulation.
U1Tx	H1	O	TTL	UART module 1 transmit. When in IrDA mode, this signal has IrDA modulation.
VCCPHY	C10 D10 D11	-	Power	VCC of the Ethernet PHY.
VDD	L12	-	Power	Positive supply for I/O and some logic.
VDD25	C3 D3 F3 G3	-	Power	Positive supply for most of the logic function, including the processor core and most peripherals.

Table 19-6. Signals by Signal Name *(continued)*

Pin Name	Pin Number	Pin Type	Buffer Type[a]	Description
VDD33	E10 G10 G11 G12 H10 K7 K8 K9	-	Power	Positive supply for I/O and some logic.
VDDA	C6 C7	-	Power	The positive supply (3.3 V) for the analog circuits (ADC, Analog Comparators, etc.). These are separated from VDD to minimize the electrical noise contained on VDD from affecting the analog functions. VDDA pins must be connected to 3.3 V, regardless of system implementation.
XTALNPHY	J1	O	TTL	Ethernet PHY XTALN 25-MHz oscillator crystal output. Leave unconnected when using a single-ended 25-MHz clock input connected to the XTALPPHY pin.
XTALPPHY	J2	I	TTL	Ethernet PHY XTALP 25-MHz oscillator crystal input or external clock reference input.

a. The TTL designation indicates the pin has TTL-compatible voltage levels.

Table 19-7. Signals by Function, Except for GPIO

Function	Pin Name	Pin Number	Pin Type	Buffer Type[a]	Description
ADC	ADC0	B1	I	Analog	Analog-to-digital converter input 0.
	ADC1	A1	I	Analog	Analog-to-digital converter input 1.
	ADC2	B3	I	Analog	Analog-to-digital converter input 2.
Analog Comparators	C0+	A7	I	Analog	Analog comparator 0 positive input.
	C0-	A6	I	Analog	Analog comparator 0 negative input.
	C0o	F1	O	TTL	Analog comparator 0 output.
	C1+	M1	I	Analog	Analog comparator 1 positive input.
	C1-	B7	I	Analog	Analog comparator 1 negative input.

Table 19-7. Signals by Function, Except for GPIO *(continued)*

Function	Pin Name	Pin Number	Pin Type	Buffer Type[a]	Description
Ethernet	ERBIAS	K3	I	Analog	12.4-kΩ resistor (1% precision) used internally for Ethernet PHY.
	GNDPHY	C8 C9 K4	-	Power	GND of the Ethernet PHY.
	LED0	J12	O	TTL	Ethernet LED 0.
	LED1	J11	O	TTL	Ethernet LED 1.
	MDIO	L9	I/O	TTL	MDIO of the Ethernet PHY.
	RXIN	L7	I	Analog	RXIN of the Ethernet PHY.
	RXIP	M7	I	Analog	RXIP of the Ethernet PHY.
	TXON	L8	O	Analog	TXON of the Ethernet PHY.
	TXOP	M8	O	Analog	TXOP of the Ethernet PHY.
	VCCPHY	C10 D10 D11	-	Power	VCC of the Ethernet PHY.
	XTALNPHY	J1	O	TTL	Ethernet PHY XTALN 25-MHz oscillator crystal output. Leave unconnected when using a single-ended 25-MHz clock input connected to the XTALPPHY pin.
	XTALPPHY	J2	I	TTL	Ethernet PHY XTALP 25-MHz oscillator crystal input or external clock reference input.
General-Purpose Timers	CCP0	E12	I/O	TTL	Capture/Compare/PWM 0.
	CCP1	L6	I/O	TTL	Capture/Compare/PWM 1.
	CCP2	D12	I/O	TTL	Capture/Compare/PWM 2.
	CCP3	M2	I/O	TTL	Capture/Compare/PWM 3.
I2C	I2C0SCL	C11	I/O	OD	I^2C module 0 clock.
	I2C0SDA	C12	I/O	OD	I^2C module 0 data.
JTAG/SWD/SWO	SWCLK	A9	I	TTL	JTAG/SWD CLK.
	SWDIO	B9	I/O	TTL	JTAG TMS and SWDIO.
	SWO	A10	O	TTL	JTAG TDO and SWO.
	TCK	A9	I	TTL	JTAG/SWD CLK.
	TDI	B8	I	TTL	JTAG TDI.
	TDO	A10	O	TTL	JTAG TDO and SWO.
	TMS	B9	I/O	TTL	JTAG TMS and SWDIO.
	$\overline{\text{TRST}}$	A8	I	TTL	JTAG $\overline{\text{TRST}}$.
PWM	Fault	F2	I	TTL	PWM Fault.
	PWM0	G1	O	TTL	PWM 0. This signal is controlled by PWM Generator 0.
	PWM1	G2	O	TTL	PWM 1. This signal is controlled by PWM Generator 0.

Table 19-7. Signals by Function, Except for GPIO *(continued)*

Function	Pin Name	Pin Number	Pin Type	Buffer Type[a]	Description
Power	GND	B6 C4 C5 F10 F11 F12 H3 J3 J10 K5 K6 K10 L10	-	Power	Ground reference for logic and I/O pins.
	GNDA	A5 B5	-	Power	The ground reference for the analog circuits (ADC, Analog Comparators, etc.). These are separated from GND to minimize the electrical noise contained on VDD from affecting the analog functions.
	LDO	E3	-	Power	Low drop-out regulator output voltage. This pin requires an external capacitor between the pin and GND of 1 µF or greater. When the on-chip LDO is used to provide power to the logic, the LDO pin must also be connected to the VDD25 pins at the board level in addition to the decoupling capacitor(s).
	VDD	L12	-	Power	Positive supply for I/O and some logic.
	VDD25	C3 D3 F3 G3	-	Power	Positive supply for most of the logic function, including the processor core and most peripherals.
	VDD33	E10 G10 G11 G12 H10 K7 K8 K9	-	Power	Positive supply for I/O and some logic.
	VDDA	C6 C7	-	Power	The positive supply (3.3 V) for the analog circuits (ADC, Analog Comparators, etc.). These are separated from VDD to minimize the electrical noise contained on VDD from affecting the analog functions. VDDA pins must be connected to 3.3 V, regardless of system implementation.
SSI	SSI0Clk	M4	I/O	TTL	SSI module 0 clock.
	SSI0Fss	L4	I/O	TTL	SSI module 0 frame.
	SSI0Rx	L5	I	TTL	SSI module 0 receive.
	SSI0Tx	M5	O	TTL	SSI module 0 transmit.

Table 19-7. Signals by Function, Except for GPIO *(continued)*

Function	Pin Name	Pin Number	Pin Type	Buffer Type[a]	Description
System Control & Clocks	CMOD0	E11	I	TTL	CPU Mode bit 0. Input must be set to logic 0 (grounded); other encodings reserved.
	CMOD1	B10	I	TTL	CPU Mode bit 1. Input must be set to logic 0 (grounded); other encodings reserved.
	OSC0	L11	I	Analog	Main oscillator crystal input or an external clock reference input.
	OSC1	M11	O	Analog	Main oscillator crystal output. Leave unconnected when using a single-ended clock source.
	RST	H11	I	TTL	System reset input.
UART	U0Rx	L3	I	TTL	UART module 0 receive. When in IrDA mode, this signal has IrDA modulation.
	U0Tx	M3	O	TTL	UART module 0 transmit. When in IrDA mode, this signal has IrDA modulation.
	U1Rx	H2	I	TTL	UART module 1 receive. When in IrDA mode, this signal has IrDA modulation.
	U1Tx	H1	O	TTL	UART module 1 transmit. When in IrDA mode, this signal has IrDA modulation.

a. The TTL designation indicates the pin has TTL-compatible voltage levels.

Table 19-8. GPIO Pins and Alternate Functions

IO	Pin Number	Multiplexed Function	Multiplexed Function
PA0	L3	U0Rx	
PA1	M3	U0Tx	
PA2	M4	SSI0Clk	
PA3	L4	SSI0Fss	
PA4	L5	SSI0Rx	
PA5	M5	SSI0Tx	
PA6	L6	CCP1	
PA7	M6		
PB0	E12	CCP0	
PB1	D12	CCP2	
PB2	C11	I2C0SCL	
PB3	C12	I2C0SDA	
PB4	A6	C0-	
PB5	B7	C1-	
PB6	A7	C0+	
PB7	A8	TRST	
PC0	A9	TCK	SWCLK
PC1	B9	TMS	SWDIO
PC2	B8	TDI	
PC3	A10	TDO	SWO
PC4	L1		
PC5	M1	C1+	
PC6	M2	CCP3	

Table 19-8. GPIO Pins and Alternate Functions *(continued)*

IO	Pin Number	Multiplexed Function	Multiplexed Function
PC7	L2		
PD0	G1	PWM0	
PD1	G2	PWM1	
PD2	H2	U1Rx	
PD3	H1	U1Tx	
PD4	E1		
PD5	E2		
PD6	F2	Fault	
PD7	F1	C0o	
PE0	A11		
PE1	B12		
PE2	B11		
PE3	A12		
PE4	D1		
PF0	M9		
PF1	H12		
PF2	J11	LED1	
PF3	J12	LED0	
PG0	K1		
PG1	K2		

20 Operating Characteristics

Table 20-1. Temperature Characteristics

Characteristic	Symbol	Value	Unit
Industrial operating temperature range	T_A	-40 to +85	°C
Extended operating temperature range	T_A	-40 to +105	°C
Unpowered storage temperature range	T_S	-65 to +150	°C

Table 20-2. Thermal Characteristics

Characteristic	Symbol	Value	Unit
Thermal resistance (junction to ambient)[a]	Θ_{JA}	34	°C/W
Average junction temperature[b]	T_J	$T_A + (P_{AVG} \bullet \Theta_{JA})$	°C

a. Junction to ambient thermal resistance θ_{JA} numbers are determined by a package simulator.

b. Power dissipation is a function of temperature.

Table 20-3. ESD Absolute Maximum Ratings[a]

Parameter Name	Min	Nom	Max	Unit
V_{ESDHBM}	-	-	2.0	kV
V_{ESDCDM}	-	-	1.0	kV
V_{ESDMM}	-	-	100	V

a. All Stellaris parts are ESD tested following the JEDEC standard.

21 Electrical Characteristics

21.1 DC Characteristics

21.1.1 Maximum Ratings

The maximum ratings are the limits to which the device can be subjected without permanently damaging the device.

Note: The device is not guaranteed to operate properly at the maximum ratings.

Table 21-1. Maximum Ratings

Characteristic [a]	Symbol	Value		Unit
		Min	Max	
I/O supply voltage (V_{DD})	V_{DD}	0	4	V
Core supply voltage (V_{DD25})	V_{DD25}	0	3	V
Analog supply voltage (V_{DDA})	V_{DDA}	0	4	V
Ethernet PHY supply voltage (V_{CCPHY})	V_{CCPHY}	0	4	V
Input voltage	V_{IN}	-0.3	5.5	V
Maximum current per output pins	I	-	25	mA

a. Voltages are measured with respect to GND.

Important: This device contains circuitry to protect the inputs against damage due to high-static voltages or electric fields; however, it is advised that normal precautions be taken to avoid application of any voltage higher than maximum-rated voltages to this high-impedance circuit. Reliability of operation is enhanced if unused inputs are connected to an appropriate logic voltage level (for example, either GND or V_{DD}).

21.1.2 Recommended DC Operating Conditions

For special high-current applications, the GPIO output buffers may be used with the following restrictions. With the GPIO pins configured as 8-mA output drivers, a total of four GPIO outputs may be used to sink current loads up to 18 mA each. At 18-mA sink current loading, the V_{OL} value is specified as 1.2 V. The high-current GPIO package pins must be selected such that there are only a maximum of two per side of the physical package or BGA pin group with the total number of high-current GPIO outputs not exceeding four for the entire package.

Table 21-2. Recommended DC Operating Conditions

Parameter	Parameter Name	Min	Nom	Max	Unit
V_{DD}	I/O supply voltage	3.0	3.3	3.6	V
V_{DD25}	Core supply voltage	2.25	2.5	2.75	V
V_{DDA}	Analog supply voltage	3.0	3.3	3.6	V
V_{CCPHY}	Ethernet PHY supply voltage	3.0	3.3	3.6	V
V_{IH}	High-level input voltage	2.0	-	5.0	V
V_{IL}	Low-level input voltage	-0.3	-	1.3	V
V_{OH} [a]	High-level output voltage	2.4	-	-	V
V_{OL} [a]	Low-level output voltage	-	-	0.4	V

Table 21-2. Recommended DC Operating Conditions *(continued)*

Parameter	Parameter Name	Min	Nom	Max	Unit
I_{OH}	High-level source current, V_{OH}=2.4 V				
	2-mA Drive	2.0	-	-	mA
	4-mA Drive	4.0	-	-	mA
	8-mA Drive	8.0	-	-	mA
I_{OL}	Low-level sink current, V_{OL}=0.4 V				
	2-mA Drive	2.0	-	-	mA
	4-mA Drive	4.0	-	-	mA
	8-mA Drive	8.0	-	-	mA

a. V_{OL} and V_{OH} shift to 1.2 V when using high-current GPIOs.

21.1.3 On-Chip Low Drop-Out (LDO) Regulator Characteristics

Table 21-3. LDO Regulator Characteristics

Parameter	Parameter Name	Min	Nom	Max	Unit
V_{LDOOUT}	Programmable internal (logic) power supply output value	2.25	2.5	2.75	V
	Output voltage accuracy	-	2%	-	%
t_{PON}	Power-on time	-	-	100	µs
t_{ON}	Time on	-	-	200	µs
t_{OFF}	Time off	-	-	100	µs
V_{STEP}	Step programming incremental voltage	-	50	-	mV
C_{LDO}	External filter capacitor size for internal power supply	1.0	-	3.0	µF

21.1.4 GPIO Module Characteristics

Table 21-4. GPIO Module DC Characteristics

Parameter	Parameter Name	Min	Nom	Max	Unit
R_{GPIOPU}	GPIO internal pull-up resistor	50	-	110	kΩ
R_{GPIOPD}	GPIO internal pull-down resistor	55	-	180	kΩ

21.1.5 Power Specifications

The power measurements specified in the tables that follow are run on the core processor using SRAM with the following specifications (except as noted):

- V_{DD} = 3.3 V
- V_{DD25} = 2.50 V
- V_{DDA} = 3.3 V
- V_{DDPHY} = 3.3 V
- Temperature = 25°C

- Clock Source (MOSC) =3.579545 MHz Crystal Oscillator
- Main oscillator (MOSC) = enabled
- Internal oscillator (IOSC) = disabled

Table 21-5. Detailed Power Specifications

Parameter	Parameter Name	Conditions	3.3 V V_{DD}, V_{DDA}, V_{DDPHY}		2.5 V V_{DD25}		Unit
			Nom	Max	Nom	Max	
I_{DD_RUN}	Run mode 1 (Flash loop)	V_{DD25} = 2.50 V Code= while(1){} executed in Flash Peripherals = All ON System Clock = 50 MHz (with PLL)	48	pending[a]	108	pending[a]	mA
	Run mode 2 (Flash loop)	V_{DD25} = 2.50 V Code= while(1){} executed in Flash Peripherals = All OFF System Clock = 50 MHz (with PLL)	5	pending[a]	52	pending[a]	mA
	Run mode 1 (SRAM loop)	V_{DD25} = 2.50 V Code= while(1){} executed in SRAM Peripherals = All ON System Clock = 50 MHz (with PLL)	48	pending[a]	100	pending[a]	mA
	Run mode 2 (SRAM loop)	V_{DD25} = 2.50 V Code= while(1){} executed in SRAM Peripherals = All OFF System Clock = 50 MHz (with PLL)	5	pending[a]	45	pending[a]	mA
I_{DD_SLEEP}	Sleep mode	V_{DD25} = 2.50 V Peripherals = All OFF System Clock = 50 MHz (with PLL)	5	pending[a]	16	pending[a]	mA
$I_{DD_DEEPSLEEP}$	Deep-Sleep mode	LDO = 2.25 V Peripherals = All OFF System Clock = IOSC30KHZ/64	4.6	pending[a]	0.21	pending[a]	mA

a. Pending characterization completion.

21.1.6 Flash Memory Characteristics

Table 21-6. Flash Memory Characteristics

Parameter	Parameter Name	Min	Nom	Max	Unit
PE_{CYC}	Number of guaranteed program/erase cycles before failure[a]	10,000	100,000	-	cycles
T_{RET}	Data retention at average operating temperature of 85˚C (industrial) or 105˚C (extended)	10	-	-	years
T_{PROG}	Word program time	20	-	-	µs
T_{ERASE}	Page erase time	20	-	-	ms

Table 21-6. Flash Memory Characteristics *(continued)*

Parameter	Parameter Name	Min	Nom	Max	Unit
T_{ME}	Mass erase time	-	-	250	ms

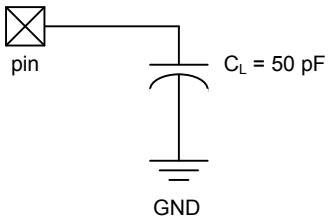
a. A program/erase cycle is defined as switching the bits from 1-> 0 -> 1.

21.2 AC Characteristics

21.2.1 Load Conditions

Unless otherwise specified, the following conditions are true for all timing measurements. Timing measurements are for 4-mA drive strength.

Figure 21-1. Load Conditions



21.2.2 Clocks

Table 21-7. Phase Locked Loop (PLL) Characteristics

Parameter	Parameter Name	Min	Nom	Max	Unit
$f_{ref_crystal}$	Crystal reference[a]	3.579545	-	8.192	MHz
f_{ref_ext}	External clock reference[a]	3.579545	-	8.192	MHz
f_{pll}	PLL frequency[b]	-	400	-	MHz
T_{READY}	PLL lock time	-	-	0.5	ms

a. The exact value is determined by the crystal value programmed into the XTAL field of the **Run-Mode Clock Configuration (RCC)** register.

b. PLL frequency is automatically calculated by the hardware based on the XTAL field of the **RCC** register.

Table 21-8 on page 530 shows the actual frequency of the PLL based on the crystal frequency used (defined by the XTAL field in the **RCC** register).

Table 21-8. Actual PLL Frequency

XTAL	Crystal Frequency (MHz)	PLL Frequency (MHz)	Error
0x4	3.5795	400.904	0.0023%
0x5	3.6864	398.1312	0.0047%
0x6	4.0	400	-
0x7	4.096	401.408	0.0035%
0x8	4.9152	398.1312	0.0047%
0x9	5.0	400	-
0xA	5.12	399.36	0.0016%
0xB	6.0	400	-
0xC	6.144	399.36	0.0016%

Table 21-8. Actual PLL Frequency *(continued)*

XTAL	Crystal Frequency (MHz)	PLL Frequency (MHz)	Error
0xD	7.3728	398.1312	0.0047%
0xE	8.0	400	0.0047%
0xF	8.192	398.6773333	0.0033%

Table 21-9. Clock Characteristics

Parameter	Parameter Name	Min	Nom	Max	Unit
f_{IOSC}	Internal 12 MHz oscillator frequency	8.4	12	15.6	MHz
$f_{IOSC30KHZ}$	Internal 30 KHz oscillator frequency	15	30	45	KHz
f_{MOSC}	Main oscillator frequency	1	-	8.192	MHz
t_{MOSC_per}	Main oscillator period	125	-	1000	ns
$f_{ref_crystal_bypass}$	Crystal reference using the main oscillator (PLL in BYPASS mode)[a]	1	-	8.192	MHz
$f_{ref_ext_bypass}$	External clock reference (PLL in BYPASS mode)[a]	0	-	50	MHz
f_{system_clock}	System clock	0	-	50	MHz

a. The ADC must be clocked from the PLL or directly from a 16-MHz clock source to operate properly.

Table 21-10. Crystal Characteristics

Parameter Name	Value				Units
Frequency	8	6	4	3.5	MHz
Frequency tolerance	±50	±50	±50	±50	ppm
Aging	±5	±5	±5	±5	ppm/yr
Oscillation mode	Parallel	Parallel	Parallel	Parallel	-
Temperature stability (-40°C to 85°C)	±25	±25	±25	±25	ppm
Temperature stability (-40°C to 105°C)	±25	±25	±25	±25	ppm
Motional capacitance (typ)	27.8	37.0	55.6	63.5	pF
Motional inductance (typ)	14.3	19.1	28.6	32.7	mH
Equivalent series resistance (max)	120	160	200	220	Ω
Shunt capacitance (max)	10	10	10	10	pF
Load capacitance (typ)	16	16	16	16	pF
Drive level (typ)	100	100	100	100	µW

21.2.2.1 System Clock Specifications with ADC Operation

Table 21-11. System Clock Characteristics with ADC Operation

Parameter	Parameter Name	Min	Nom	Max	Unit
f_{sysadc}	System clock frequency when the ADC module is operating (when PLL is bypassed)	16	-	-	MHz

21.2.3 JTAG and Boundary Scan

Table 21-12. JTAG Characteristics

Parameter No.	Parameter	Parameter Name	Min	Nom	Max	Unit
J1	f_{TCK}	TCK operational clock frequency	0	-	10	MHz
J2	t_{TCK}	TCK operational clock period	100	-	-	ns
J3	t_{TCK_LOW}	TCK clock Low time	-	t_{TCK}	-	ns
J4	t_{TCK_HIGH}	TCK clock High time	-	t_{TCK}	-	ns
J5	t_{TCK_R}	TCK rise time	0	-	10	ns
J6	t_{TCK_F}	TCK fall time	0	-	10	ns
J7	t_{TMS_SU}	TMS setup time to TCK rise	20	-	-	ns
J8	t_{TMS_HLD}	TMS hold time from TCK rise	20	-	-	ns
J9	t_{TDI_SU}	TDI setup time to TCK rise	25	-	-	ns
J10	t_{TDI_HLD}	TDI hold time from TCK rise	25	-	-	ns
J11 t_{TDO_ZDV}	TCK fall to Data Valid from High-Z	2-mA drive	-	23	35	ns
		4-mA drive		15	26	ns
		8-mA drive		14	25	ns
		8-mA drive with slew rate control		18	29	ns
J12 t_{TDO_DV}	TCK fall to Data Valid from Data Valid	2-mA drive	-	21	35	ns
		4-mA drive		14	25	ns
		8-mA drive		13	24	ns
		8-mA drive with slew rate control		18	28	ns
J13 t_{TDO_DVZ}	TCK fall to High-Z from Data Valid	2-mA drive	-	9	11	ns
		4-mA drive		7	9	ns
		8-mA drive		6	8	ns
		8-mA drive with slew rate control		7	9	ns
J14	t_{TRST}	$\overline{TRST}$ assertion time	100	-	-	ns
J15	t_{TRST_SU}	$\overline{TRST}$ setup time to TCK rise	10	-	-	ns

Figure 21-2. JTAG Test Clock Input Timing

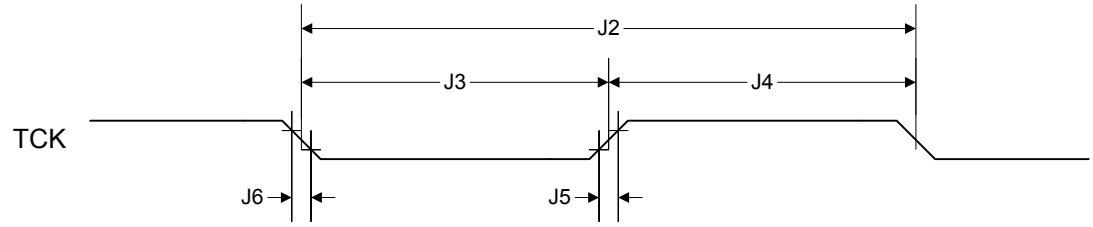


Figure 21-3. JTAG Test Access Port (TAP) Timing

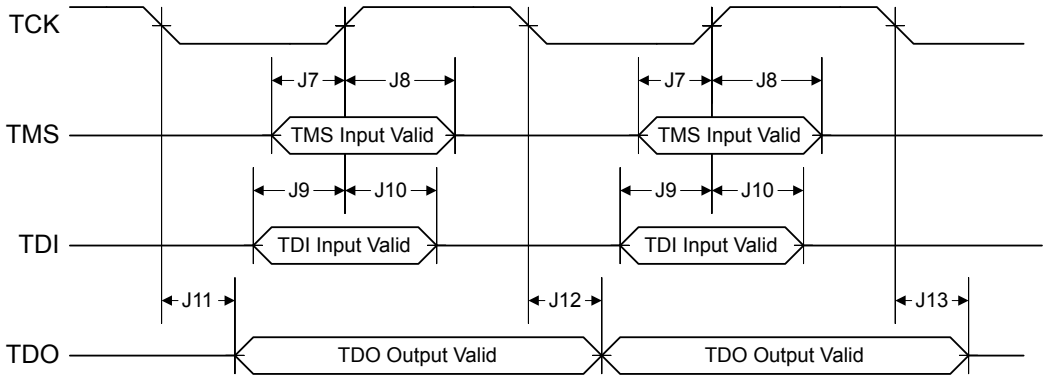
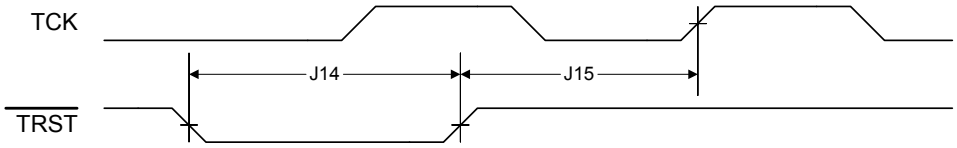


Figure 21-4. JTAG TRST Timing



21.2.4 Reset

Table 21-13. Reset Characteristics

Parameter No.	Parameter	Parameter Name	Min	Nom	Max	Unit
R1	V_{TH}	Reset threshold	-	2.0	-	V
R2	V_{BTH}	Brown-Out threshold	2.85	2.9	2.95	V
R3	T_{POR}	Power-On Reset timeout	-	10	-	ms
R4	T_{BOR}	Brown-Out timeout	-	500	-	µs
R5	T_{IRPOR}	Internal reset timeout after POR	6	-	11	ms
R6	T_{IRBOR}	Internal reset timeout after BOR[a]	0	-	1	µs
R7	T_{IRHWR}	Internal reset timeout after hardware reset (RST pin)	0	-	1	ms
R8	T_{IRSWR}	Internal reset timeout after software-initiated system reset [a]	2.5	-	20	µs
R9	T_{IRWDR}	Internal reset timeout after watchdog reset[a]	2.5	-	20	µs
R10	$T_{VDDRISE}$	Supply voltage (V_{DD}) rise time (0V-3.3V)	-	-	250	µs
R11	T_{MIN}	Minimum RST pulse width	2	-	-	µs

a. 20 * t_{MOSC_per}

Figure 21-5. External Reset Timing ($\overline{RST}$)

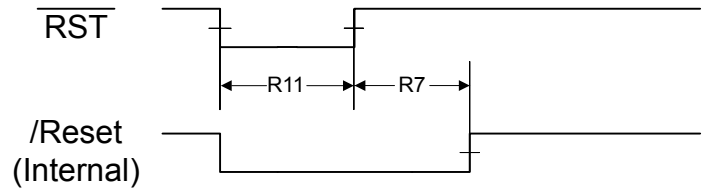


Figure 21-6. Power-On Reset Timing

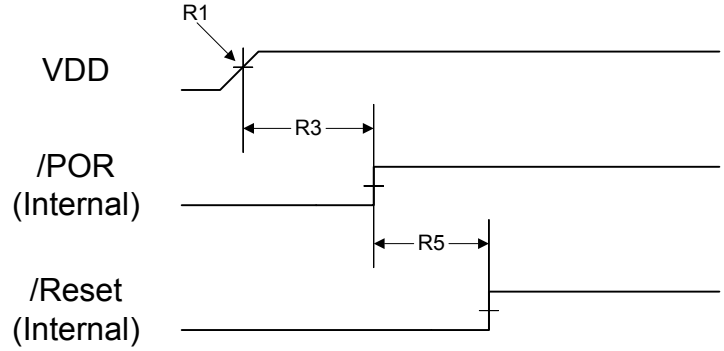


Figure 21-7. Brown-Out Reset Timing

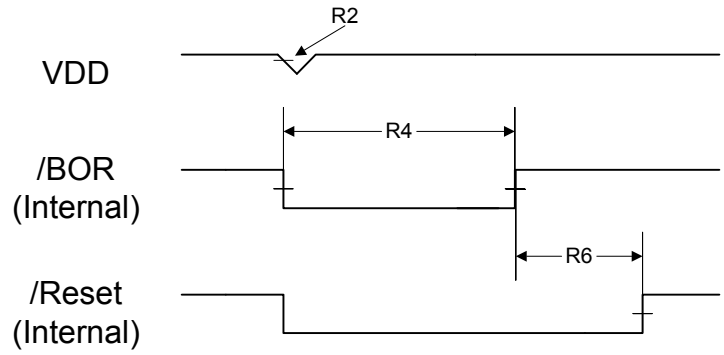


Figure 21-8. Software Reset Timing

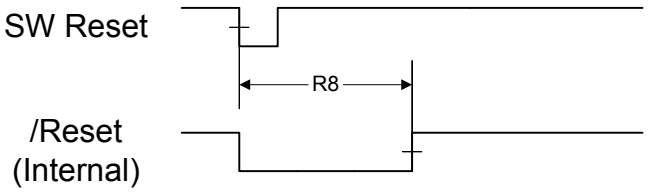
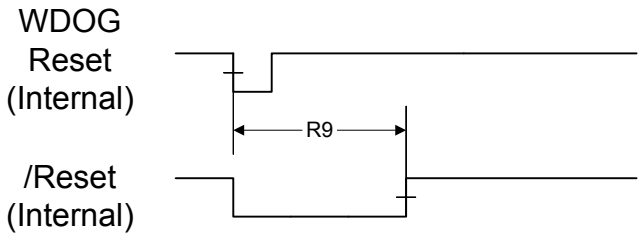


Figure 21-9. Watchdog Reset Timing



21.2.5 Sleep Modes

Table 21-14. Sleep Modes AC Characteristics[a]

Parameter No	Parameter	Parameter Name	Min	Nom	Max	Unit
D1	t_{WAKE_S}	Time to wake from interrupt in sleep or deep-sleep mode, not using the PLL	-	-	7	system clocks
D2	$t_{WAKE_PLL_S}$	Time to wake from interrupt in sleep or deep-sleep mode when using the PLL	-	-	T_{READY}	ms

a. Values in this table assume the IOSC is the clock source during sleep or deep-sleep mode.

21.2.6 General-Purpose I/O (GPIO)

Note: All GPIOs are 5 V-tolerant.

Table 21-15. GPIO Characteristics

Parameter	Parameter Name	Condition	Min	Nom	Max	Unit
t_{GPIOR}	GPIO Rise Time (from 20% to 80% of V_{DD})	2-mA drive	-	17	26	ns
		4-mA drive		9	13	ns
		8-mA drive		6	9	ns
		8-mA drive with slew rate control		10	12	ns
t_{GPIOF}	GPIO Fall Time (from 80% to 20% of V_{DD})	2-mA drive	-	17	25	ns
		4-mA drive		8	12	ns
		8-mA drive		6	10	ns
		8-mA drive with slew rate control		11	13	ns

21.2.7 Analog-to-Digital Converter

Table 21-16. ADC Characteristics[a]

Parameter	Parameter Name	Min	Nom	Max	Unit
V_{ADCIN}	Maximum single-ended, full-scale analog input voltage	-	-	3.0	V
	Minimum single-ended, full-scale analog input voltage	0.0	-	-	V
	Maximum differential, full-scale analog input voltage	-	-	1.5	V
	Minimum differential, full-scale analog input voltage	0.0	-	-	V
N	Resolution		10		bits
f_{ADC}	ADC internal clock frequency[b]	14	16	18	MHz
$t_{ADCCONV}$	Conversion time[c]				µs

Table 21-16. ADC Characteristics *(continued)*

Parameter	Parameter Name	Min	Nom	Max	Unit
$f_{ADCCONV}$	Conversion rate[c]				k samples/s
t_{LT}	Latency from trigger to start of conversion	-	2	-	system clocks
I_L	ADC input leakage	-	-	±3.0	µA
R_{ADC}	ADC equivalent resistance	-	-	10	kΩ
C_{ADC}	ADC equivalent capacitance	0.9	1.0	1.1	pF
E_L	Integral nonlinearity error	-	-	±1	LSB
E_D	Differential nonlinearity error	-	-	±1	LSB
E_O	Offset error	-	-	±1	LSB
E_G	Full-scale gain error	-	-	±3	LSB
E_{TS}	Temperature sensor accuracy	-	-	±5	°C

a. The ADC reference voltage is 3.0 V. This reference voltage is internally generated from the 3.3 VDDA supply by a band gap circuit.

b. The ADC must be clocked from the PLL or directly from an external clock source to operate properly.

c. The conversion time and rate scale from the specified number if the ADC internal clock frequency is any value other than 16 MHz.

Figure 21-10. ADC Input Equivalency Diagram

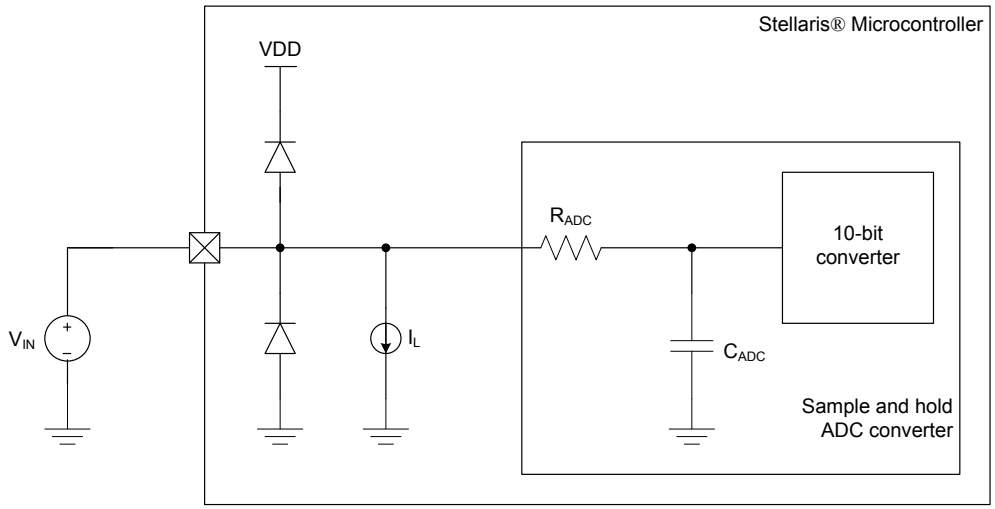


Table 21-17. ADC Module Internal Reference Characteristics

Parameter	Parameter Name	Min	Nom	Max	Unit
V_{REFI}	Internal voltage reference for ADC	-	3.0	-	V
E_{IR}	Internal voltage reference error	-	-	±2.5	%

21.2.8 Synchronous Serial Interface (SSI)

Table 21-18. SSI Characteristics

Parameter No.	Parameter	Parameter Name	Min	Nom	Max	Unit
S1	t_{clk_per}	SSIClk cycle time	2	-	65024	system clocks

Table 21-18. SSI Characteristics *(continued)*

Parameter No.	Parameter	Parameter Name	Min	Nom	Max	Unit
S2	t_{clk_high}	SSIClk high time	-	0.5	-	t clk_per
S3	t_{clk_low}	SSIClk low time	-	0.5	-	t clk_per
S4	t_{clkrf}	SSIClk rise/fall time	-	7.4	26	ns
S5	t_{DMd}	Data from master valid delay time	0	-	1	system clocks
S6	t_{DMs}	Data from master setup time	1	-	-	system clocks
S7	t_{DMh}	Data from master hold time	2	-	-	system clocks
S8	t_{DSs}	Data from slave setup time	1	-	-	system clocks
S9	t_{DSh}	Data from slave hold time	2	-	-	system clocks

Figure 21-11. SSI Timing for TI Frame Format (FRF=01), Single Transfer Timing Measurement

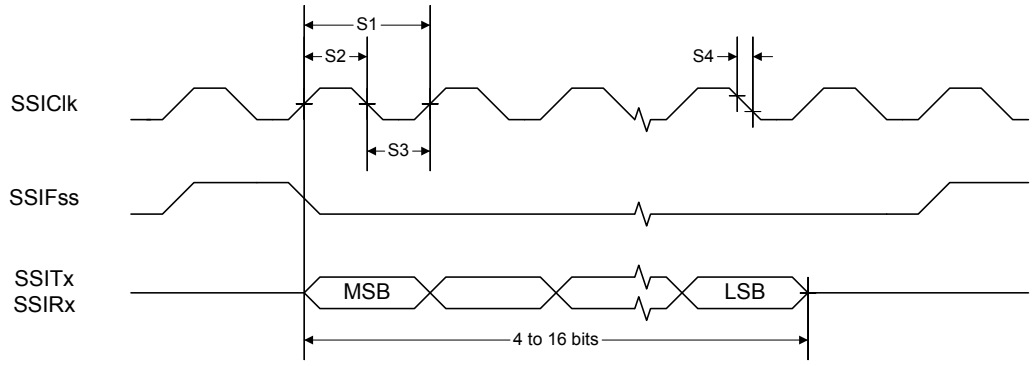


Figure 21-12. SSI Timing for MICROWIRE Frame Format (FRF=10), Single Transfer

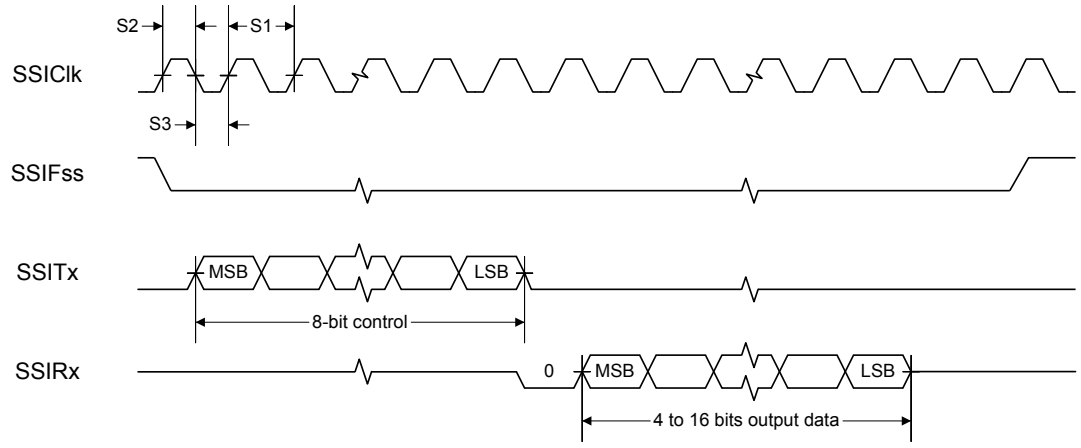
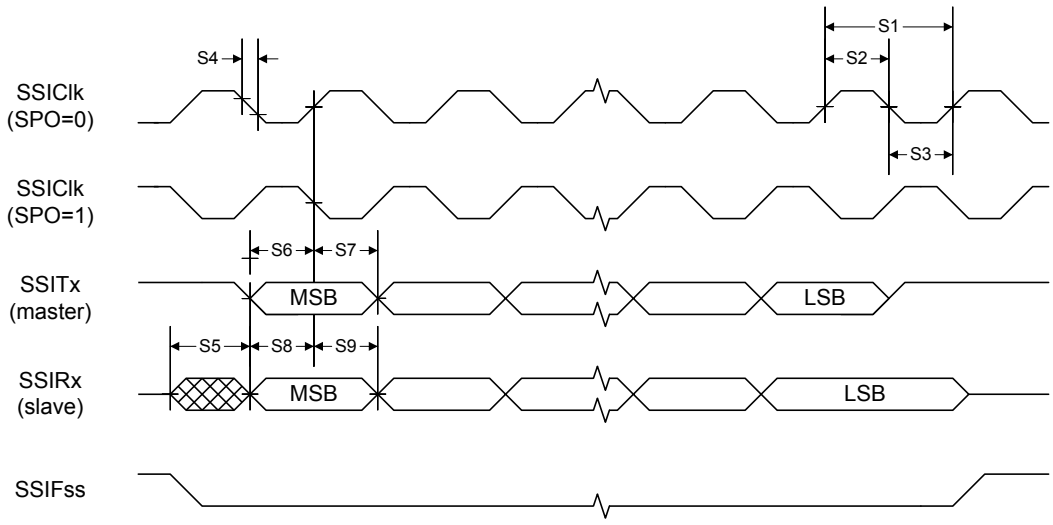


Figure 21-13. SSI Timing for SPI Frame Format (FRF=00), with SPH=1



21.2.9 Inter-Integrated Circuit (I^2C) Interface

Table 21-19. I^2C Characteristics

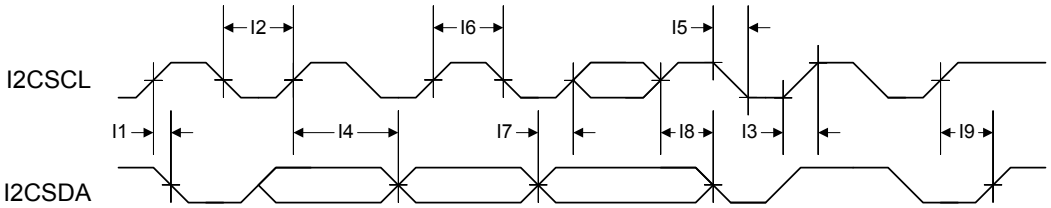
Parameter No.	Parameter	Parameter Name	Min	Nom	Max	Unit
I1[a]	t_{SCH}	Start condition hold time	36	-	-	system clocks
I2[a]	t_{LP}	Clock Low period	36	-	-	system clocks
I3[b]	t_{SRT}	I2CSCL/I2CSDA rise time (V_{IL} =0.5 V to V_{IH} =2.4 V)	-	-	(see note b)	ns
I4[a]	t_{DH}	Data hold time	2	-	-	system clocks
I5[c]	t_{SFT}	I2CSCL/I2CSDA fall time (V_{IH} =2.4 V to V_{IL} =0.5 V)	-	9	10	ns
I6[a]	t_{HT}	Clock High time	24	-	-	system clocks
I7[a]	t_{DS}	Data setup time	18	-	-	system clocks
I8[a]	t_{SCSR}	Start condition setup time (for repeated start condition only)	36	-	-	system clocks
I9[a]	t_{SCS}	Stop condition setup time	24	-	-	system clocks

a. Values depend on the value programmed into the TPR bit in the **I^2C Master Timer Period (I2CMTPR)** register; a TPR programmed for the maximum I2CSCL frequency (TPR=0x2) results in a minimum output timing as shown in the table above. The I^2C interface is designed to scale the actual data transition time to move it to the middle of the I2CSCL Low period. The actual position is affected by the value programmed into the TPR; however, the numbers given in the above values are minimum values.

b. Because I2CSCL and I2CSDA are open-drain-type outputs, which the controller can only actively drive Low, the time I2CSCL or I2CSDA takes to reach a high level depends on external signal capacitance and pull-up resistor values.

c. Specified at a nominal 50 pF load.

Figure 21-14. I²C Timing



21.2.10 Ethernet Controller

Table 21-20. 100BASE-TX Transmitter Characteristics[a]

Parameter Name	Min	Nom	Max	Unit
Peak output amplitude	950	-	1050	mVpk
Output amplitude symmetry	98	-	102	%
Output overshoot	-	-	5	%
Rise/Fall time	3	-	5	ns
Rise/Fall time imbalance	-	-	500	ps
Duty cycle distortion	-	-	-	ps
Jitter	-	-	1.4	ns

a. Measured at the line side of the transformer.

Table 21-21. 100BASE-TX Transmitter Characteristics (informative)[a]

Parameter Name	Min	Nom	Max	Unit
Return loss	16	-	-	dB
Open-circuit inductance	350	-	-	µH

a. The specifications in this table are included for information only. They are mainly a function of the external transformer and termination resistors used for measurements.

Table 21-22. 100BASE-TX Receiver Characteristics

Parameter Name	Min	Nom	Max	Unit
Signal detect assertion threshold	600	700	-	mVppd
Signal detect de-assertion threshold	350	425	-	mVppd
Differential input resistance	-	20	-	kΩ
Jitter tolerance (pk-pk)	4	-	-	ns
Baseline wander tracking	-75	-	+75	%
Signal detect assertion time	-	-	1000	µs
Signal detect de-assertion time	-	-	4	µs

Table 21-23. 10BASE-T Transmitter Characteristics[a]

Parameter Name	Min	Nom	Max	Unit
Peak differential output signal	2.2	-	2.8	V
Harmonic content	27	-	-	dB
Link pulse width	-	100	-	ns

Table 21-23. 10BASE-T Transmitter Characteristics *(continued)*

Parameter Name	Min	Nom	Max	Unit
Start-of-idle pulse width	-	300 350	-	ns

a. The Manchester-encoded data pulses, the link pulse and the start-of-idle pulse are tested against the templates and using the procedures found in Clause 14 of *IEEE 802.3*.

Table 21-24. 10BASE-T Transmitter Characteristics (informative)[a]

Parameter Name	Min	Nom	Max	Unit
Output return loss	15	-	-	dB
Output impedance balance	29-17log(f/10)	-	-	dB
Peak common-mode output voltage	-	-	50	mV
Common-mode rejection	-	-	100	mV
Common-mode rejection jitter	-	-	1	ns

a. The specifications in this table are included for information only. They are mainly a function of the external transformer and termination resistors used for measurements.

Table 21-25. 10BASE-T Receiver Characteristics

Parameter Name	Min	Nom	Max	Unit
DLL phase acquisition time	-	10	-	BT
Jitter tolerance (pk-pk)	30	-	-	ns
Input squelched threshold	500	600	700	mVppd
Input unsquelched threshold	275	350	425	mVppd
Differential input resistance	-	20	-	kΩ
Bit error ratio	-	10^{-10}	-	-
Common-mode rejection	25	-	-	V

Table 21-26. Isolation Transformers[a]

Name	Value	Condition
Turns ratio	1 CT : 1 CT	+/- 5%
Open-circuit inductance	350 uH (min)	@ 10 mV, 10 kHz
Leakage inductance	0.40 uH (max)	@ 1 MHz (min)
Inter-winding capacitance	25 pF (max)	
DC resistance	0.9 Ohm (max)	
Insertion loss	0.4 dB (typ)	0-65 MHz
HIPOT	1500	Vrms

a. Two simple 1:1 isolation transformers are required at the line interface. Transformers with integrated common-mode chokes are recommended for exceeding FCC requirements. This table gives the recommended line transformer characteristics.

Note: The 100Base-TX amplitude specifications assume a transformer loss of 0.4 dB. For the transmit line transformer with higher insertion losses, up to 1.2 dB of insertion loss can be compensated by selecting the appropriate setting in the Transmit Amplitude Selection (TX0) bits in the **MR19** register.

Table 21-27. Ethernet Reference Crystal[a]

Name	Value	Condition
Frequency	25.00000	MHz
Frequency tolerance	±50	PPM
Aging	±2	PPM/yr
Temperature stability (-40° to 85°)	±5	PPM
Temperature stability (-40° to 105°)	±5	PPM
Oscillation mode	Parallel resonance, fundamental mode	
Parameters at 25° C ±2° C; Drive level = 0.5 mW		
Drive level (typ)	50-100	µW
Shunt capacitance (max)	10	pF
Motional capacitance (min)	10	fF
Series resistance (max)	60	Ω
Spurious response (max)	> 5 dB below main within 500 kHz	

a. If the internal crystal oscillator is used, select a crystal that meets these specifications.

Figure 21-15. External XTLP Oscillator Characteristics

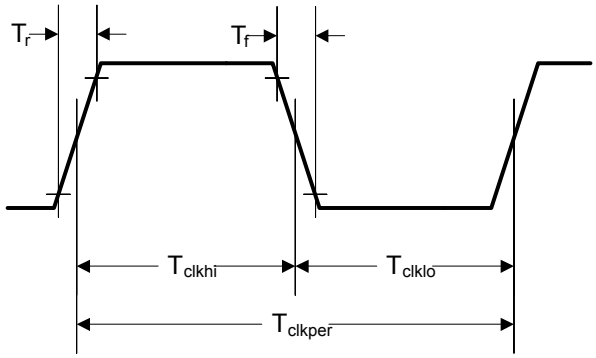


Table 21-28. External XTLP Oscillator Characteristics

Parameter Name	Symbol	Min	Nom	Max	Unit
XTLN Input Low Voltage	$XTLN_{ILV}$	-	-	0.8	-
XTLP Frequency[a]	$XTLP_f$	-	25.0	-	-
XTLP Period[b]	T_{clkper}	-	40	-	-
XTLP Duty Cycle	$XTLP_{DC}$	40 40	-	60 60	%
Rise/Fall Time	T_r , T_f	-	-	4.0	ns
Absolute Jitter	T_{JITTER}	-	-	0.1	ns

a. IEEE 802.3 frequency tolerance ±50 ppm.

b. IEEE 802.3 frequency tolerance ±50 ppm.

21.2.11 Analog Comparator

Table 21-29. Analog Comparator Characteristics

Parameter	Parameter Name	Min	Nom	Max	Unit
V_{OS}	Input offset voltage	-	±10	±25	mV
V_{CM}	Input common mode voltage range	0	-	V_{DD}-1.5	V
C_{MRR}	Common mode rejection ratio	50	-	-	dB
T_{RT}	Response time	-	-	1	µs
T_{MC}	Comparator mode change to Output Valid	-	-	10	µs

Table 21-30. Analog Comparator Voltage Reference Characteristics

Parameter	Parameter Name	Min	Nom	Max	Unit
R_{HR}	Resolution high range	-	V_{DD}/31	-	LSB
R_{LR}	Resolution low range	-	V_{DD}/23	-	LSB
A_{HR}	Absolute accuracy high range	-	-	±1/2	LSB
A_{LR}	Absolute accuracy low range	-	-	±1/4	LSB

A Serial Flash Loader

A.1 Serial Flash Loader

The Stellaris® serial flash loader is a preprogrammed flash-resident utility used to download code to the flash memory of a device without the use of a debug interface. The serial flash loader uses a simple packet interface to provide synchronous communication with the device. The flash loader runs off the crystal and does not enable the PLL, so its speed is determined by the crystal used. The two serial interfaces that can be used are the UART0 and SSI0 interfaces. For simplicity, both the data format and communication protocol are identical for both serial interfaces.

A.2 Interfaces

Once communication with the flash loader is established via one of the serial interfaces, that interface is used until the flash loader is reset or new code takes over. For example, once you start communicating using the SSI port, communications with the flash loader via the UART are disabled until the device is reset.

A.2.1 UART

The Universal Asynchronous Receivers/Transmitters (UART) communication uses a fixed serial format of 8 bits of data, no parity, and 1 stop bit. The baud rate used for communication is automatically detected by the flash loader and can be any valid baud rate supported by the host and the device. The auto detection sequence requires that the baud rate should be no more than 1/32 the crystal frequency of the board that is running the serial flash loader. This is actually the same as the hardware limitation for the maximum baud rate for any UART on a Stellaris® device which is calculated as follows:

```
Max Baud Rate = System Clock Frequency / 16
```

In order to determine the baud rate, the serial flash loader needs to determine the relationship between its own crystal frequency and the baud rate. This is enough information for the flash loader to configure its UART to the same baud rate as the host. This automatic baud-rate detection allows the host to use any valid baud rate that it wants to communicate with the device.

The method used to perform this automatic synchronization relies on the host sending the flash loader two bytes that are both 0x55. This generates a series of pulses to the flash loader that it can use to calculate the ratios needed to program the UART to match the host's baud rate. After the host sends the pattern, it attempts to read back one byte of data from the UART. The flash loader returns the value of 0xCC to indicate successful detection of the baud rate. If this byte is not received after at least twice the time required to transfer the two bytes, the host can resend another pattern of 0x55, 0x55, and wait for the 0xCC byte again until the flash loader acknowledges that it has received a synchronization pattern correctly. For example, the time to wait for data back from the flash loader should be calculated as at least 2*(20(bits/sync)/baud rate (bits/sec)). For a baud rate of 115200, this time is 2*(20/115200) or 0.35 ms.

A.2.2 SSI

The Synchronous Serial Interface (SSI) port also uses a fixed serial format for communications, with the framing defined as Motorola format with SPH set to 1 and SPO set to 1. See "Frame Formats" on page 333 in the SSI chapter for more information on formats for this transfer protocol. Like the UART, this interface has hardware requirements that limit the maximum speed that the SSI clock can run. This allows the SSI clock to be at most 1/12 the crystal frequency of the board running

the flash loader. Since the host device is the master, the SSI on the flash loader device does not need to determine the clock as it is provided directly by the host.

A.3 Packet Handling

All communications, with the exception of the UART auto-baud, are done via defined packets that are acknowledged (ACK) or not acknowledged (NAK) by the devices. The packets use the same format for receiving and sending packets, including the method used to acknowledge successful or unsuccessful reception of a packet.

A.3.1 Packet Format

All packets sent and received from the device use the following byte-packed format.

```
struct
{
 unsigned char ucSize;
 unsigned char ucCheckSum;
 unsigned char Data[];
};
```

ucSize	The first byte received holds the total size of the transfer including the size and checksum bytes.
ucChecksum	This holds a simple checksum of the bytes in the data buffer only. The algorithm is Data[0]+Data[1]+…+ Data[`ucSize`-3].
Data	This is the raw data intended for the device, which is formatted in some form of command interface. There should be `ucSize`–2 bytes of data provided in this buffer to or from the device.

A.3.2 Sending Packets

The actual bytes of the packet can be sent individually or all at once; the only limitation is that commands that cause flash memory access should limit the download sizes to prevent losing bytes during flash programming. This limitation is discussed further in the section that describes the serial flash loader command, COMMAND_SEND_DATA (see “COMMAND_SEND_DATA (0x24)” on page 546).

Once the packet has been formatted correctly by the host, it should be sent out over the UART or SSI interface. Then the host should poll the UART or SSI interface for the first non-zero data returned from the device. The first non-zero byte will either be an ACK (0xCC) or a NAK (0x33) byte from the device indicating the packet was received successfully (ACK) or unsuccessfully (NAK). This does not indicate that the actual contents of the command issued in the data portion of the packet were valid, just that the packet was received correctly.

A.3.3 Receiving Packets

The flash loader sends a packet of data in the same format that it receives a packet. The flash loader may transfer leading zero data before the first actual byte of data is sent out. The first non-zero byte is the size of the packet followed by a checksum byte, and finally followed by the data itself. There is no break in the data after the first non-zero byte is sent from the flash loader. Once the device communicating with the flash loader receives all the bytes, it must either ACK or NAK the packet to indicate that the transmission was successful. The appropriate response after sending a NAK to the flash loader is to resend the command that failed and request the data again. If needed, the host may send leading zeros before sending down the ACK/NAK signal to the flash loader, as the

flash loader only accepts the first non-zero data as a valid response. This zero padding is needed by the SSI interface in order to receive data to or from the flash loader.

A.4 Commands

The next section defines the list of commands that can be sent to the flash loader. The first byte of the data should always be one of the defined commands, followed by data or parameters as determined by the command that is sent.

A.4.1 COMMAND_PING (0X20)

This command simply accepts the command and sets the global status to success. The format of the packet is as follows:

```
Byte[0] = 0x03;
Byte[1] = checksum(Byte[2]);
Byte[2] = COMMAND_PING;
```

The ping command has 3 bytes and the value for COMMAND_PING is 0x20 and the checksum of one byte is that same byte, making Byte[1] also 0x20. Since the ping command has no real return status, the receipt of an ACK can be interpreted as a successful ping to the flash loader.

A.4.2 COMMAND_GET_STATUS (0x23)

This command returns the status of the last command that was issued. Typically, this command should be sent after every command to ensure that the previous command was successful or to properly respond to a failure. The command requires one byte in the data of the packet and should be followed by reading a packet with one byte of data that contains a status code. The last step is to ACK or NAK the received data so the flash loader knows that the data has been read.

```
Byte[0] = 0x03
Byte[1] = checksum(Byte[2])
Byte[2] = COMMAND_GET_STATUS
```

A.4.3 COMMAND_DOWNLOAD (0x21)

This command is sent to the flash loader to indicate where to store data and how many bytes will be sent by the COMMAND_SEND_DATA commands that follow. The command consists of two 32-bit values that are both transferred MSB first. The first 32-bit value is the address to start programming data into, while the second is the 32-bit size of the data that will be sent. This command also triggers an erase of the full area to be programmed so this command takes longer than other commands. This results in a longer time to receive the ACK/NAK back from the board. This command should be followed by a COMMAND_GET_STATUS to ensure that the Program Address and Program size are valid for the device running the flash loader.

The format of the packet to send this command is a follows:

```
Byte[0] = 11
Byte[1] = checksum(Bytes[2:10])
Byte[2] = COMMAND_DOWNLOAD
Byte[3] = Program Address [31:24]
Byte[4] = Program Address [23:16]
Byte[5] = Program Address [15:8]
Byte[6] = Program Address [7:0]
Byte[7] = Program Size [31:24]
```

```
Byte[8] = Program Size [23:16]
Byte[9] = Program Size [15:8]
Byte[10] = Program Size [7:0]
```

A.4.4 COMMAND_SEND_DATA (0x24)

This command should only follow a COMMAND_DOWNLOAD command or another COMMAND_SEND_DATA command if more data is needed. Consecutive send data commands automatically increment address and continue programming from the previous location. The caller should limit transfers of data to a maximum 8 bytes of packet data to allow the flash to program successfully and not overflow input buffers of the serial interfaces. The command terminates programming once the number of bytes indicated by the COMMAND_DOWNLOAD command has been received. Each time this function is called it should be followed by a COMMAND_GET_STATUS to ensure that the data was successfully programmed into the flash. If the flash loader sends a NAK to this command, the flash loader does not increment the current address to allow retransmission of the previous data.

```
Byte[0] = 11
Byte[1] = checksum(Bytes[2:10])
Byte[2] = COMMAND_SEND_DATA
Byte[3] = Data[0]
Byte[4] = Data[1]
Byte[5] = Data[2]
Byte[6] = Data[3]
Byte[7] = Data[4]
Byte[8] = Data[5]
Byte[9] = Data[6]
Byte[10] = Data[7]
```

A.4.5 COMMAND_RUN (0x22)

This command is used to tell the flash loader to execute from the address passed as the parameter in this command. This command consists of a single 32-bit value that is interpreted as the address to execute. The 32-bit value is transmitted MSB first and the flash loader responds with an ACK signal back to the host device before actually executing the code at the given address. This allows the host to know that the command was received successfully and the code is now running.

```
Byte[0] = 7
Byte[1] = checksum(Bytes[2:6])
Byte[2] = COMMAND_RUN
Byte[3] = Execute Address[31:24]
Byte[4] = Execute Address[23:16]
Byte[5] = Execute Address[15:8]
Byte[6] = Execute Address[7:0]
```

A.4.6 COMMAND_RESET (0x25)

This command is used to tell the flash loader device to reset. This is useful when downloading a new image that overwrote the flash loader and wants to start from a full reset. Unlike the COMMAND_RUN command, this allows the initial stack pointer to be read by the hardware and set up for the new code. It can also be used to reset the flash loader if a critical error occurs and the host device wants to restart communication with the flash loader.

```
Byte[0] = 3
Byte[1] = checksum(Byte[2])
Byte[2] = COMMAND_RESET
```

The flash loader responds with an ACK signal back to the host device before actually executing the software reset to the device running the flash loader. This allows the host to know that the command was received successfully and the part will be reset.

B Register Quick Reference

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
System Control Base 0x400F.E000															
DID0, type RO, offset 0x000, reset -															
	VER							CLASS							
MAJOR								MINOR							
PBORCTL, type R/W, offset 0x030, reset 0x0000.7FFD															
														BORIOR	
LDOPCTL, type R/W, offset 0x034, reset 0x0000.0000															
										VADJ					
RIS, type RO, offset 0x050, reset 0x0000.0000															
									PLLLRIS					BORRIS	
IMC, type R/W, offset 0x054, reset 0x0000.0000															
									PLLLIM					BORIM	
MISC, type R/W1C, offset 0x058, reset 0x0000.0000															
									PLLLMIS					BORMIS	
RESC, type R/W, offset 0x05C, reset -															
											SW	WDT	BOR	POR	EXT
RCC, type R/W, offset 0x060, reset 0x078E.3AD1															
				ACG	SYSDIV				USESYSDIV		USEPWMDIV	PWMDIV			
		PWRDN		BYPASS		XTAL				OSCSRC				IOSCDIS	MOSCDIS
PLLCFG, type RO, offset 0x064, reset -															
		F									R				
RCC2, type R/W, offset 0x070, reset 0x0780.2810															
USERCC2			SYSDIV2												
		PWRDN2		BYPASS2					OSCSRC2						
DSLPCLKCFG, type R/W, offset 0x144, reset 0x0780.0000															
			DSDIVORIDE												
									DSOSCSRC						
DID1, type RO, offset 0x004, reset -															
VER				FAM				PARTNO							
PINCOUNT								TEMP			PKG		ROHS	QUAL	
DC0, type RO, offset 0x008, reset 0x007F.002F															
SRAMSZ															
FLASHSZ															
DC1, type RO, offset 0x010, reset 0x0011.31BF															
											PWM				ADC
MINSYSDIV						MAXADCSPD		MPU		TEMPSNS	PLL	WDT	SWO	SWD	JTAG
DC2, type RO, offset 0x014, reset 0x0307.1013															
						COMP1	COMP0						TIMER2	TIMER1	TIMER0
			I2C0								SSI0			UART1	UART0
DC3, type RO, offset 0x018, reset 0x8F07.87C3															
32KHZ				CCP3	CCP2	CCP1	CCP0						ADC2	ADC1	ADC0
PWMFAULT					C1PLUS	C1MINUS	C0O	C0PLUS	C0MINUS					PWM1	PWM0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DC4, type RO, offset 0x01C, reset 0x5000.007F															
	EPHY0		EMAC0												
									GPIOG	GPIOF	GPIOE	GPIOD	GPIOC	GPIOB	GPIOA
RCGC0, type R/W, offset 0x100, reset 0x00000040															
											PWM				ADC
						MAXADCSPD						WDT			
SCGC0, type R/W, offset 0x110, reset 0x00000040															
											PWM				ADC
						MAXADCSPD						WDT			
DCGC0, type R/W, offset 0x120, reset 0x00000040															
											PWM				ADC
												WDT			
RCGC1, type R/W, offset 0x104, reset 0x00000000															
						COMP1	COMP0						TIMER2	TIMER1	TIMER0
			I2C0								SSI0			UART1	UART0
SCGC1, type R/W, offset 0x114, reset 0x00000000															
						COMP1	COMP0						TIMER2	TIMER1	TIMER0
			I2C0								SSI0			UART1	UART0
DCGC1, type R/W, offset 0x124, reset 0x00000000															
						COMP1	COMP0						TIMER2	TIMER1	TIMER0
			I2C0								SSI0			UART1	UART0
RCGC2, type R/W, offset 0x108, reset 0x00000000															
	EPHY0		EMAC0												
									GPIOG	GPIOF	GPIOE	GPIOD	GPIOC	GPIOB	GPIOA
SCGC2, type R/W, offset 0x118, reset 0x00000000															
	EPHY0		EMAC0												
									GPIOG	GPIOF	GPIOE	GPIOD	GPIOC	GPIOB	GPIOA
DCGC2, type R/W, offset 0x128, reset 0x00000000															
	EPHY0		EMAC0												
									GPIOG	GPIOF	GPIOE	GPIOD	GPIOC	GPIOB	GPIOA
SRCR0, type R/W, offset 0x040, reset 0x00000000															
											PWM				ADC
												WDT			
SRCR1, type R/W, offset 0x044, reset 0x00000000															
						COMP1	COMP0						TIMER2	TIMER1	TIMER0
			I2C0								SSI0			UART1	UART0
SRCR2, type R/W, offset 0x048, reset 0x00000000															
	EPHY0		EMAC0												
									GPIOG	GPIOF	GPIOE	GPIOD	GPIOC	GPIOB	GPIOA
Internal Memory **Flash Memory Control Registers (Flash Control Offset)** Base 0x400F.D000															
FMA, type R/W, offset 0x000, reset 0x0000.0000															
															OFFSET
OFFSET															
FMD, type R/W, offset 0x004, reset 0x0000.0000															
DATA															
DATA															
FMC, type R/W, offset 0x008, reset 0x0000.0000															
WRKEY															
												COMT	MERASE	ERASE	WRITE

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FCRIS, type RO, offset 0x00C, reset 0x0000.0000															
														PRIS	ARIS
FCIM, type R/W, offset 0x010, reset 0x0000.0000															
														PMASK	AMASK
FCMISC, type R/W1C, offset 0x014, reset 0x0000.0000															
														PMISC	AMISC

Internal Memory

Flash Memory Protection Registers (System Control Offset)

Base 0x400F.E000

USECRL, type R/W, offset 0x140, reset 0x31															
								USEC							
FMPRE0, type R/W, offset 0x130 and 0x200, reset 0xFFFF.FFFF															
READ_ENABLE															
READ_ENABLE															
FMPPE0, type R/W, offset 0x134 and 0x400, reset 0xFFFF.FFFF															
PROG_ENABLE															
PROG_ENABLE															
USER_DBG, type R/W, offset 0x1D0, reset 0xFFFF.FFFE															
NW	DATA														
DATA														DBG1	DBG0
USER_REG0, type R/W, offset 0x1E0, reset 0xFFFF.FFFF															
NW	DATA														
DATA															
USER_REG1, type R/W, offset 0x1E4, reset 0xFFFF.FFFF															
NW	DATA														
DATA															
FMPRE1, type R/W, offset 0x204, reset 0x0000.FFFF															
READ_ENABLE															
READ_ENABLE															
FMPRE2, type R/W, offset 0x208, reset 0x0000.0000															
READ_ENABLE															
READ_ENABLE															
FMPRE3, type R/W, offset 0x20C, reset 0x0000.0000															
READ_ENABLE															
READ_ENABLE															
FMPPE1, type R/W, offset 0x404, reset 0x0000.FFFF															
PROG_ENABLE															
PROG_ENABLE															
FMPPE2, type R/W, offset 0x408, reset 0x0000.0000															
PROG_ENABLE															
PROG_ENABLE															
FMPPE3, type R/W, offset 0x40C, reset 0x0000.0000															
PROG_ENABLE															
PROG_ENABLE															

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15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
General-Purpose Input/Outputs (GPIOs) GPIO Port A base: 0x4000.4000 GPIO Port B base: 0x4000.5000 GPIO Port C base: 0x4000.6000 GPIO Port D base: 0x4000.7000 GPIO Port E base: 0x4002.4000 GPIO Port F base: 0x4002.5000 GPIO Port G base: 0x4002.6000															
GPIODATA, type R/W, offset 0x000, reset 0x0000.0000															
								DATA							
GPIODIR, type R/W, offset 0x400, reset 0x0000.0000															
								DIR							
GPIOIS, type R/W, offset 0x404, reset 0x0000.0000															
								IS							
GPIOIBE, type R/W, offset 0x408, reset 0x0000.0000															
								IBE							
GPIOIEV, type R/W, offset 0x40C, reset 0x0000.0000															
								IEV							
GPIOIM, type R/W, offset 0x410, reset 0x0000.0000															
								IME							
GPIORIS, type RO, offset 0x414, reset 0x0000.0000															
								RIS							
GPIOMIS, type RO, offset 0x418, reset 0x0000.0000															
								MIS							
GPIOICR, type W1C, offset 0x41C, reset 0x0000.0000															
								IC							
GPIOAFSEL, type R/W, offset 0x420, reset -															
								AFSEL							
GPIODR2R, type R/W, offset 0x500, reset 0x0000.00FF															
								DRV2							
GPIODR4R, type R/W, offset 0x504, reset 0x0000.0000															
								DRV4							
GPIODR8R, type R/W, offset 0x508, reset 0x0000.0000															
								DRV8							
GPIOODR, type R/W, offset 0x50C, reset 0x0000.0000															
								ODE							
GPIOPUR, type R/W, offset 0x510, reset -															
								PUE							

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GPIOPDR, type R/W, offset 0x514, reset 0x0000.0000															
								PDE							
GPIOSLR, type R/W, offset 0x518, reset 0x0000.0000															
								SRL							
GPIODEN, type R/W, offset 0x51C, reset -															
								DEN							
GPIOLOCK, type R/W, offset 0x520, reset 0x0000.0001															
LOCK															
LOCK															
GPIOCR, type -, offset 0x524, reset -															
								CR							
GPIOPeriphID4, type RO, offset 0xFD0, reset 0x0000.0000															
								PID4							
GPIOPeriphID5, type RO, offset 0xFD4, reset 0x0000.0000															
								PID5							
GPIOPeriphID6, type RO, offset 0xFD8, reset 0x0000.0000															
								PID6							
GPIOPeriphID7, type RO, offset 0xFDC, reset 0x0000.0000															
								PID7							
GPIOPeriphID0, type RO, offset 0xFE0, reset 0x0000.0061															
								PID0							
GPIOPeriphID1, type RO, offset 0xFE4, reset 0x0000.0000															
								PID1							
GPIOPeriphID2, type RO, offset 0xFE8, reset 0x0000.0018															
								PID2							
GPIOPeriphID3, type RO, offset 0xFEC, reset 0x0000.0001															
								PID3							
GPIOPCellID0, type RO, offset 0xFF0, reset 0x0000.000D															
								CID0							
GPIOPCellID1, type RO, offset 0xFF4, reset 0x0000.00F0															
								CID1							
GPIOPCellID2, type RO, offset 0xFF8, reset 0x0000.0005															
								CID2							
GPIOPCellID3, type RO, offset 0xFFC, reset 0x0000.00B1															
								CID3							

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
General-Purpose Timers Timer0 base: 0x4003.0000 Timer1 base: 0x4003.1000 Timer2 base: 0x4003.2000															
GPTMCFG, type R/W, offset 0x000, reset 0x0000.0000															
													GPTMCFG		
GPTMTAMR, type R/W, offset 0x004, reset 0x0000.0000															
												TAAMS	TACMR	TAMR	
GPTMTBMR, type R/W, offset 0x008, reset 0x0000.0000															
												TBAMS	TBCMR	TBMR	
GPTMCTL, type R/W, offset 0x00C, reset 0x0000.0000															
	TBPWML	TBOTE		TBEVENT		TBSTALL	TBEN		TAPWML	TAOTE	RTCEN	TAEVENT		TASTALL	TAEN
GPTMIMR, type R/W, offset 0x018, reset 0x0000.0000															
					CBEIM	CBMIM	TBTOIM					RTCIM	CAEIM	CAMIM	TATOIM
GPTMRIS, type RO, offset 0x01C, reset 0x0000.0000															
					CBERIS	CBMRIS	TBTORIS					RTCRIS	CAERIS	CAMRIS	TATORIS
GPTMMIS, type RO, offset 0x020, reset 0x0000.0000															
					CBEMIS	CBMMIS	TBTOMIS					RTCMIS	CAEMIS	CAMMIS	TATOMIS
GPTMICR, type W1C, offset 0x024, reset 0x0000.0000															
					CBECINT	CBMCINT	TBTOCINT					RTCCINT	CAECINT	CAMCINT	TATOCINT
GPTMTAILR, type R/W, offset 0x028, reset 0xFFFF.FFFF															
TAILRH															
TAILRL															
GPTMTBILR, type R/W, offset 0x02C, reset 0x0000.FFFF															
TBILRL															
GPTMTAMATCHR, type R/W, offset 0x030, reset 0xFFFF.FFFF															
TAMRH															
TAMRL															
GPTMTBMATCHR, type R/W, offset 0x034, reset 0x0000.FFFF															
TBMRL															
GPTMTAPR, type R/W, offset 0x038, reset 0x0000.0000															
								TAPSR							
GPTMTBPR, type R/W, offset 0x03C, reset 0x0000.0000															
								TBPSR							
GPTMTAPMR, type R/W, offset 0x040, reset 0x0000.0000															
								TAPSMR							
GPTMTBPMR, type R/W, offset 0x044, reset 0x0000.0000															
								TBPSMR							

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GPTMTAR, type RO, offset 0x048, reset 0xFFFF.FFFF															
TARH															
TARL															
GPTMTBR, type RO, offset 0x04C, reset 0x0000.FFFF															
TBRL															
Watchdog Timer Base 0x4000.0000															
WDTLOAD, type R/W, offset 0x000, reset 0xFFFF.FFFF															
WDTLoad															
WDTLoad															
WDTVALUE, type RO, offset 0x004, reset 0xFFFF.FFFF															
WDTValue															
WDTValue															
WDTCTL, type R/W, offset 0x008, reset 0x0000.0000															
														RESEN	INTEN
WDTICR, type WO, offset 0x00C, reset -															
WDTIntClr															
WDTIntClr															
WDTRIS, type RO, offset 0x010, reset 0x0000.0000															
															WDTRIS
WDTMIS, type RO, offset 0x014, reset 0x0000.0000															
															WDTMIS
WDTTEST, type R/W, offset 0x418, reset 0x0000.0000															
							STALL								
WDTLOCK, type R/W, offset 0xC00, reset 0x0000.0000															
WDTLock															
WDTLock															
WDTPeriphID4, type RO, offset 0xFD0, reset 0x0000.0000															
								PID4							
WDTPeriphID5, type RO, offset 0xFD4, reset 0x0000.0000															
								PID5							
WDTPeriphID6, type RO, offset 0xFD8, reset 0x0000.0000															
								PID6							
WDTPeriphID7, type RO, offset 0xFDC, reset 0x0000.0000															
								PID7							
WDTPeriphID0, type RO, offset 0xFE0, reset 0x0000.0005															
								PID0							
WDTPeriphID1, type RO, offset 0xFE4, reset 0x0000.0018															
								PID1							
WDTPeriphID2, type RO, offset 0xFE8, reset 0x0000.0018															
								PID2							

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WDTPeriphID3, type RO, offset 0xFEC, reset 0x0000.0001															
								PID3							
WDTPCellID0, type RO, offset 0xFF0, reset 0x0000.000D															
								CID0							
WDTPCellID1, type RO, offset 0xFF4, reset 0x0000.00F0															
								CID1							
WDTPCellID2, type RO, offset 0xFF8, reset 0x0000.0005															
								CID2							
WDTPCellID3, type RO, offset 0xFFC, reset 0x0000.00B1															
								CID3							
Analog-to-Digital Converter (ADC) Base 0x4003.8000															
ADCACTSS, type R/W, offset 0x000, reset 0x0000.0000															
												ASEN3	ASEN2	ASEN1	ASEN0
ADCRIS, type RO, offset 0x004, reset 0x0000.0000															
												INR3	INR2	INR1	INR0
ADCIM, type R/W, offset 0x008, reset 0x0000.0000															
												MASK3	MASK2	MASK1	MASK0
ADCISC, type R/W1C, offset 0x00C, reset 0x0000.0000															
												IN3	IN2	IN1	IN0
ADCOSTAT, type R/W1C, offset 0x010, reset 0x0000.0000															
												OV3	OV2	OV1	OV0
ADCEMUX, type R/W, offset 0x014, reset 0x0000.0000															
EM3				EM2				EM1				EM0			
ADCUSTAT, type R/W1C, offset 0x018, reset 0x0000.0000															
												UV3	UV2	UV1	UV0
ADCSSPRI, type R/W, offset 0x020, reset 0x0000.3210															
		SS3				SS2				SS1				SS0	
ADCPSSI, type WO, offset 0x028, reset -															
												SS3	SS2	SS1	SS0
ADCSAC, type R/W, offset 0x030, reset 0x0000.0000															
													AVG		
ADCSSMUX0, type R/W, offset 0x040, reset 0x0000.0000															
		MUX7				MUX6				MUX5				MUX4	
		MUX3				MUX2				MUX1				MUX0	
ADCSSCTL0, type R/W, offset 0x044, reset 0x0000.0000															
TS7	IE7	END7	D7	TS6	IE6	END6	D6	TS5	IE5	END5	D5	TS4	IE4	END4	D4
TS3	IE3	END3	D3	TS2	IE2	END2	D2	TS1	IE1	END1	D1	TS0	IE0	END0	D0

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15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADCSSFIFO0, type RO, offset 0x048, reset -															
						DATA									
ADCSSFIFO1, type RO, offset 0x068, reset -															
						DATA									
ADCSSFIFO2, type RO, offset 0x088, reset -															
						DATA									
ADCSSFIFO3, type RO, offset 0x0A8, reset -															
						DATA									
ADCSSFSTAT0, type RO, offset 0x04C, reset 0x0000.0100															
			FULL				EMPTY			HPTR				TPTR	
ADCSSFSTAT1, type RO, offset 0x06C, reset 0x0000.0100															
			FULL				EMPTY			HPTR				TPTR	
ADCSSFSTAT2, type RO, offset 0x08C, reset 0x0000.0100															
			FULL				EMPTY			HPTR				TPTR	
ADCSSFSTAT3, type RO, offset 0x0AC, reset 0x0000.0100															
			FULL				EMPTY			HPTR				TPTR	
ADCSSMUX1, type R/W, offset 0x060, reset 0x0000.0000															
		MUX3				MUX2				MUX1				MUX0	
ADCSSMUX2, type R/W, offset 0x080, reset 0x0000.0000															
		MUX3				MUX2				MUX1				MUX0	
ADCSSCTL1, type R/W, offset 0x064, reset 0x0000.0000															
TS3	IE3	END3	D3	TS2	IE2	END2	D2	TS1	IE1	END1	D1	TS0	IE0	END0	D0
ADCSSCTL2, type R/W, offset 0x084, reset 0x0000.0000															
TS3	IE3	END3	D3	TS2	IE2	END2	D2	TS1	IE1	END1	D1	TS0	IE0	END0	D0
ADCSSMUX3, type R/W, offset 0x0A0, reset 0x0000.0000															
														MUX0	
ADCSSCTL3, type R/W, offset 0x0A4, reset 0x0000.0002															
												TS0	IE0	END0	D0
ADCTMLB, type R/W, offset 0x100, reset 0x0000.0000															
															LB
Universal Asynchronous Receivers/Transmitters (UARTs) UART0 base: 0x4000.C000 UART1 base: 0x4000.D000															
UARTDR, type R/W, offset 0x000, reset 0x0000.0000															
				OE	BE	PE	FE			DATA					

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
UARTRSR/UARTECR, type RO, offset 0x004, reset 0x0000.0000 (Reads)															
												OE	BE	PE	FE
UARTRSR/UARTECR, type WO, offset 0x004, reset 0x0000.0000 (Writes)															
								DATA							
UARTFR, type RO, offset 0x018, reset 0x0000.0090															
								TXFE	RXFF	TXFF	RXFE	BUSY			
UARTILPR, type R/W, offset 0x020, reset 0x0000.0000															
								ILPDVSR							
UARTIBRD, type R/W, offset 0x024, reset 0x0000.0000															
DIVINT															
UARTFBRD, type R/W, offset 0x028, reset 0x0000.0000															
										DIVFRAC					
UARTLCRH, type R/W, offset 0x02C, reset 0x0000.0000															
								SPS	WLEN		FEN	STP2	EPS	PEN	BRK
UARTCTL, type R/W, offset 0x030, reset 0x0000.0300															
						RXE	TXE	LBE					SIRLP	SIREN	UARTEN
UARTIFLS, type R/W, offset 0x034, reset 0x0000.0012															
										RXIFLSEL			TXIFLSEL		
UARTIM, type R/W, offset 0x038, reset 0x0000.0000															
					OEIM	BEIM	PEIM	FEIM	RTIM	TXIM	RXIM				
UARTRIS, type RO, offset 0x03C, reset 0x0000.000F															
					OERIS	BERIS	PERIS	FERIS	RTRIS	TXRIS	RXRIS				
UARTMIS, type RO, offset 0x040, reset 0x0000.0000															
					OEMIS	BEMIS	PEMIS	FEMIS	RTMIS	TXMIS	RXMIS				
UARTICR, type W1C, offset 0x044, reset 0x0000.0000															
					OEIC	BEIC	PEIC	FEIC	RTIC	TXIC	RXIC				
UARTPeriphID4, type RO, offset 0xFD0, reset 0x0000.0000															
								PID4							
UARTPeriphID5, type RO, offset 0xFD4, reset 0x0000.0000															
								PID5							
UARTPeriphID6, type RO, offset 0xFD8, reset 0x0000.0000															
								PID6							
UARTPeriphID7, type RO, offset 0xFDC, reset 0x0000.0000															
								PID7							

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
UARTPeriphID0, type RO, offset 0xFE0, reset 0x0000.0011															
								PID0							
UARTPeriphID1, type RO, offset 0xFE4, reset 0x0000.0000															
								PID1							
UARTPeriphID2, type RO, offset 0xFE8, reset 0x0000.0018															
								PID2							
UARTPeriphID3, type RO, offset 0xFEC, reset 0x0000.0001															
								PID3							
UARTPCellID0, type RO, offset 0xFF0, reset 0x0000.000D															
								CID0							
UARTPCellID1, type RO, offset 0xFF4, reset 0x0000.00F0															
								CID1							
UARTPCellID2, type RO, offset 0xFF8, reset 0x0000.0005															
								CID2							
UARTPCellID3, type RO, offset 0xFFC, reset 0x0000.00B1															
								CID3							
Synchronous Serial Interface (SSI) SSI0 base: 0x4000.8000															
SSICR0, type R/W, offset 0x000, reset 0x0000.0000															
SCR								SPH	SPO	FRF		DSS			
SSICR1, type R/W, offset 0x004, reset 0x0000.0000															
												SOD	MS	SSE	LBM
SSIDR, type R/W, offset 0x008, reset 0x0000.0000															
DATA															
SSISR, type RO, offset 0x00C, reset 0x0000.0003															
											BSY	RFF	RNE	TNF	TFE
SSICPSR, type R/W, offset 0x010, reset 0x0000.0000															
								CPSDVSR							
SSIIM, type R/W, offset 0x014, reset 0x0000.0000															
												TXIM	RXIM	RTIM	RORIM
SSIRIS, type RO, offset 0x018, reset 0x0000.0008															
												TXRIS	RXRIS	RTRIS	RORRIS
SSIMIS, type RO, offset 0x01C, reset 0x0000.0000															
												TXMIS	RXMIS	RTMIS	RORMIS
SSIICR, type W1C, offset 0x020, reset 0x0000.0000															
														RTIC	RORIC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SSIPeriphID4, type RO, offset 0xFD0, reset 0x0000.0000															
								PID4							
SSIPeriphID5, type RO, offset 0xFD4, reset 0x0000.0000															
								PID5							
SSIPeriphID6, type RO, offset 0xFD8, reset 0x0000.0000															
								PID6							
SSIPeriphID7, type RO, offset 0xFDC, reset 0x0000.0000															
								PID7							
SSIPeriphID0, type RO, offset 0xFE0, reset 0x0000.0022															
								PID0							
SSIPeriphID1, type RO, offset 0xFE4, reset 0x0000.0000															
								PID1							
SSIPeriphID2, type RO, offset 0xFE8, reset 0x0000.0018															
								PID2							
SSIPeriphID3, type RO, offset 0xFEC, reset 0x0000.0001															
								PID3							
SSIPCellID0, type RO, offset 0xFF0, reset 0x0000.000D															
								CID0							
SSIPCellID1, type RO, offset 0xFF4, reset 0x0000.00F0															
								CID1							
SSIPCellID2, type RO, offset 0xFF8, reset 0x0000.0005															
								CID2							
SSIPCellID3, type RO, offset 0xFFC, reset 0x0000.00B1															
								CID3							

Inter-Integrated Circuit (I^2C) Interface

I^2C Master

I2C Master 0 base: 0x4002.0000

I2CMSA, type R/W, offset 0x000, reset 0x0000.0000															
								SA							R/S
I2CMCS, type RO, offset 0x004, reset 0x0000.0000 (Reads)															
									BUSBSY	IDLE	ARBLST	DATACK	ADRACK	ERROR	BUSY
I2CMCS, type WO, offset 0x004, reset 0x0000.0000 (Writes)															
												ACK	STOP	START	RUN
I2CMDR, type R/W, offset 0x008, reset 0x0000.0000															
								DATA							

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
I2CMTPR, type R/W, offset 0x00C, reset 0x0000.0001															
									TPR						
I2CMIMR, type R/W, offset 0x010, reset 0x0000.0000															
															IM
I2CMRIS, type RO, offset 0x014, reset 0x0000.0000															
															RIS
I2CMMIS, type RO, offset 0x018, reset 0x0000.0000															
															MIS
I2CMICR, type WO, offset 0x01C, reset 0x0000.0000															
															IC
I2CMCR, type R/W, offset 0x020, reset 0x0000.0000															
										SFE	MFE				LPBK
Inter-Integrated Circuit (I^2C) Interface **I^2C Slave** I2C Slave 0 base: 0x4002.0800															
I2CSOAR, type R/W, offset 0x000, reset 0x0000.0000															
									OAR						
I2CSCSR, type RO, offset 0x004, reset 0x0000.0000 (Reads)															
													FBR	TREQ	RREQ
I2CSCSR, type WO, offset 0x004, reset 0x0000.0000 (Writes)															
															DA
I2CSDR, type R/W, offset 0x008, reset 0x0000.0000															
								DATA							
I2CSIMR, type R/W, offset 0x00C, reset 0x0000.0000															
															DATAIM
I2CSRIS, type RO, offset 0x010, reset 0x0000.0000															
															DATARIS
I2CSMIS, type RO, offset 0x014, reset 0x0000.0000															
															DATAMIS
I2CSICR, type WO, offset 0x018, reset 0x0000.0000															
															DATAIC
Ethernet Controller **Ethernet MAC** Base 0x4004.8000															
MACRIS/MACIACK, type RO, offset 0x000, reset 0x0000.0000 (Reads)															
									PHYINT	MDINT	RXER	FOV	TXEMP	TXER	RXINT

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MACRIS/MACIACK, type WO, offset 0x000, reset 0x0000.0000 (Writes)															
									PHYINT	MDINT	RXER	FOV	TXEMP	TXER	RXINT
MACIM, type R/W, offset 0x004, reset 0x0000.007F															
									PHYINTM	MDINTM	RXERM	FOVM	TXEMPM	TXERM	RXINTM
MACRCTL, type R/W, offset 0x008, reset 0x0000.0008															
											RSTFIFO	BADCRC	PRMS	AMUL	RXEN
MACTCTL, type R/W, offset 0x00C, reset 0x0000.0000															
											DUPLEX		CRC	PADEN	TXEN
MACDATA, type RO, offset 0x010, reset 0x0000.0000 (Reads)															
RXDATA															
RXDATA															
MACDATA, type WO, offset 0x010, reset 0x0000.0000 (Writes)															
TXDATA															
TXDATA															
MACIA0, type R/W, offset 0x014, reset 0x0000.0000															
MACOCT4								MACOCT3							
MACOCT2								MACOCT1							
MACIA1, type R/W, offset 0x018, reset 0x0000.0000															
MACOCT6								MACOCT5							
MACTHR, type R/W, offset 0x01C, reset 0x0000.003F															
										THRESH					
MACMCTL, type R/W, offset 0x020, reset 0x0000.0000															
								REGADR						WRITE	START
MACMDV, type R/W, offset 0x024, reset 0x0000.0080															
								DIV							
MACMTXD, type R/W, offset 0x02C, reset 0x0000.0000															
MDTX															
MACMRXD, type R/W, offset 0x030, reset 0x0000.0000															
MDRX															
MACNP, type RO, offset 0x034, reset 0x0000.0000															
										NPR					
MACTR, type R/W, offset 0x038, reset 0x0000.0000															
															NEWTX
Ethernet Controller **MII Management**															
MR0, type R/W, address 0x00, reset 0x3100															
RESET	LOOPBK	SPEEDSL	ANEGEN	PWRDN	ISO	RANEG	DUPLEX	COLT							
MR1, type RO, address 0x01, reset 0x7849															
	100X_F	100X_H	10T_F	10T_H					MFPS	ANEGC	RFAULT	ANEGA	LINK	JAB	EXTD

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MR2, type RO, address 0x02, reset 0x000E															
OUI[21:6]															
MR3, type RO, address 0x03, reset 0x7237															
OUI[5:0]						MN						RN			
MR4, type R/W, address 0x04, reset 0x01E1															
NP		RF					A3	A2	A1	A0	S				
MR5, type RO, address 0x05, reset 0x0000															
NP	ACK	RF	A[7:0]								S				
MR6, type RO, address 0x06, reset 0x0000															
											PDF	LPNPA		PRX	LPANEGA
MR16, type R/W, address 0x10, reset 0x0140															
RPTR	INPOL		TXHIM	SQEI	NL10					APOL	RVSPOL			PCSBP	RXCC
MR17, type R/W, address 0x11, reset 0x0000															
JABBER_IE	RXER_IE	PRX_IE	PDF_IE	LPACK_IE	LSCHG_IE	RFAULT_IE	ANEGCOMP_IE	JABBER_INT	RXER_INT	PRX_INT	PDF_INT	LPACK_INT	LSCHG_INT	RFAULT_INT	ANEGCOMP_INT
MR18, type RO, address 0x12, reset 0x0000															
			ANEGF	DPLX	RATE	RXSD	RX_LOCK								
MR19, type R/W, address 0x13, reset 0x4000															
TXO															
MR23, type R/W, address 0x17, reset 0x0010															
								LED1[3:0]				LED0[3:0]			
MR24, type R/W, address 0x18, reset 0x00C0															
								PD_MODE	AUTO_SW	MDIX	MDIX_CM	MDIX_SD			
Analog Comparators Base 0x4003.C000															
ACMIS, type R/W1C, offset 0x000, reset 0x0000.0000															
														IN1	IN0
ACRIS, type RO, offset 0x004, reset 0x0000.0000															
														IN1	IN0
ACINTEN, type R/W, offset 0x008, reset 0x0000.0000															
														IN1	IN0
ACREFCTL, type R/W, offset 0x010, reset 0x0000.0000															
						EN	RNG					VREF			
ACSTAT0, type RO, offset 0x020, reset 0x0000.0000															
														OVAL	
ACSTAT1, type RO, offset 0x040, reset 0x0000.0000															
														OVAL	
ACCTL0, type R/W, offset 0x024, reset 0x0000.0000															
				TOEN	ASRCP			TSLVAL	TSEN		ISLVAL	ISEN		CINV	
ACCTL1, type R/W, offset 0x044, reset 0x0000.0000															
				TOEN	ASRCP			TSLVAL	TSEN		ISLVAL	ISEN		CINV	

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Pulse Width Modulator (PWM) Base 0x4002.8000															
PWMCTL, type R/W, offset 0x000, reset 0x0000.0000															
															GlobalSync0
PWMSYNC, type R/W, offset 0x004, reset 0x0000.0000															
															Sync0
PWMENABLE, type R/W, offset 0x008, reset 0x0000.0000															
														PWM1En	PWM0En
PWMINVERT, type R/W, offset 0x00C, reset 0x0000.0000															
														PWM1Inv	PWM0Inv
PWMFAULT, type R/W, offset 0x010, reset 0x0000.0000															
														Fault1	Fault0
PWMINTEN, type R/W, offset 0x014, reset 0x0000.0000															
															IntFault
															IntPWM0
PWMRIS, type RO, offset 0x018, reset 0x0000.0000															
															IntFault
															IntPWM0
PWMISC, type R/W1C, offset 0x01C, reset 0x0000.0000															
															IntFault
															IntPWM0
PWMSTATUS, type RO, offset 0x020, reset 0x0000.0000															
															Fault
PWM0CTL, type R/W, offset 0x040, reset 0x0000.0000															
										CmpBUpd	CmpAUpd	LoadUpd	Debug	Mode	Enable
PWM0INTEN, type R/W, offset 0x044, reset 0x0000.0000															
		TrCmpBD	TrCmpBU	TrCmpAD	TrCmpAU	TrCntLoad	TrCntZero			IntCmpBD	IntCmpBU	IntCmpAD	IntCmpAU	IntCntLoad	IntCntZero
PWM0RIS, type RO, offset 0x048, reset 0x0000.0000															
										IntCmpBD	IntCmpBU	IntCmpAD	IntCmpAU	IntCntLoad	IntCntZero
PWM0ISC, type R/W1C, offset 0x04C, reset 0x0000.0000															
										IntCmpBD	IntCmpBU	IntCmpAD	IntCmpAU	IntCntLoad	IntCntZero
PWM0LOAD, type R/W, offset 0x050, reset 0x0000.0000															
Load															
PWM0COUNT, type RO, offset 0x054, reset 0x0000.0000															
Count															
PWM0CMPA, type R/W, offset 0x058, reset 0x0000.0000															
CompA															
PWM0CMPB, type R/W, offset 0x05C, reset 0x0000.0000															
CompB															

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PWM0GENA, type R/W, offset 0x060, reset 0x0000.0000															
				ActCmpBD		ActCmpBU		ActCmpAD		ActCmpAU		ActLoad		ActZero	
PWM0GENB, type R/W, offset 0x064, reset 0x0000.0000															
				ActCmpBD		ActCmpBU		ActCmpAD		ActCmpAU		ActLoad		ActZero	
PWM0DBCTL, type R/W, offset 0x068, reset 0x0000.0000															
															Enable
PWM0DBRISE, type R/W, offset 0x06C, reset 0x0000.0000															
				RiseDelay											
PWM0DBFALL, type R/W, offset 0x070, reset 0x0000.0000															
				FallDelay											

C Ordering and Contact Information

C.1 Ordering Information

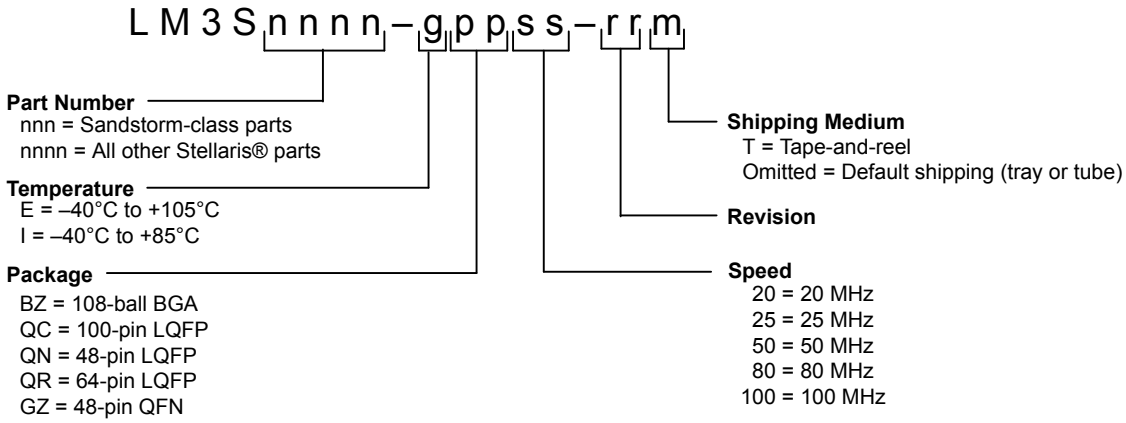


Table C-1. Part Ordering Information

Orderable Part Number	Description
LM3S6432-IBZ50-A2	Stellaris® LM3S6432 Microcontroller Industrial Temperature 108-ball BGA
LM3S6432-IBZ50-A2T	Stellaris® LM3S6432 Microcontroller Industrial Temperature 108-ball BGA Tape-and-reel
LM3S6432-EQC50-A2	Stellaris® LM3S6432 Microcontroller Extended Temperature 100-pin LQFP
LM3S6432-EQC50-A2T	Stellaris® LM3S6432 Microcontroller Extended Temperature 100-pin LQFP Tape-and-reel
LM3S6432-IQC50-A2	Stellaris® LM3S6432 Microcontroller Industrial Temperature 100-pin LQFP
LM3S6432-IQC50-A2T	Stellaris® LM3S6432 Microcontroller Industrial Temperature 100-pin LQFP Tape-and-reel

C.2 Part Markings

The Stellaris® microcontrollers are marked with an identifying number. This code contains the following information:

- The first line indicates the part number. In the example figure below, this is the LM3S6965.
- The first seven characters in the second line indicate the temperature, package, speed, and revision. In the example figure below, this is an Industrial temperature (I), 100-pin LQFP package (QC), 50-MHz (50), revision A2 (A2) device.
- The remaining characters contain internal tracking numbers.



C.3 Kits

The Stellaris® Family provides the hardware and software tools that engineers need to begin development quickly.

- Reference Design Kits accelerate product development by providing ready-to-run hardware and comprehensive documentation including hardware design files
- Evaluation Kits provide a low-cost and effective means of evaluating Stellaris® microcontrollers before purchase
- Development Kits provide you with all the tools you need to develop and prototype embedded applications right out of the box

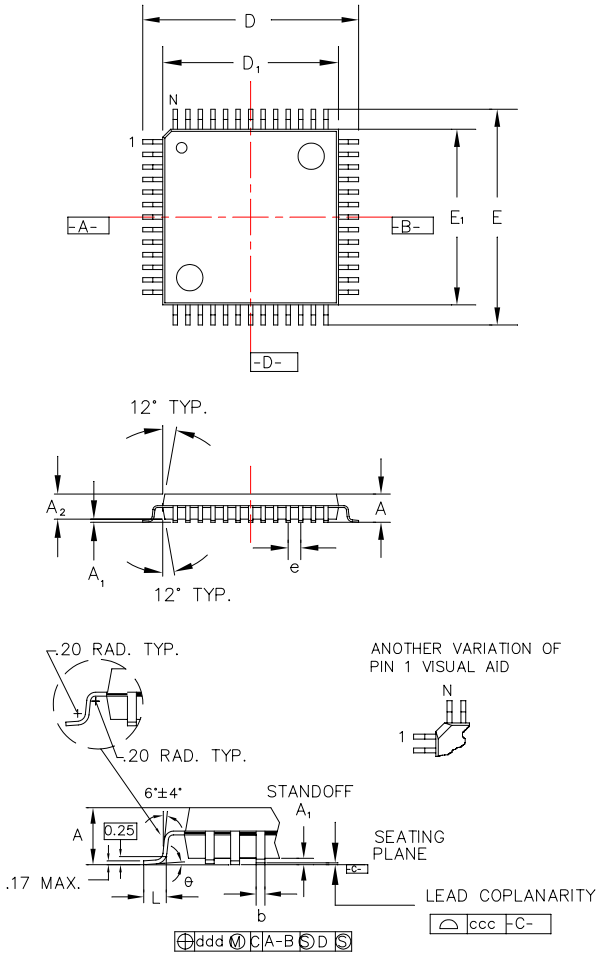
See the website at www.ti.com/stellaris for the latest tools available, or ask your distributor.

C.4 Support Information

For support on Stellaris® products, contact the TI Worldwide Product Information Center nearest you: http://www-k.ext.ti.com/sc/technical-support/product-information-centers.htm.

D Package Information

Figure D-1. 100-Pin LQFP Package

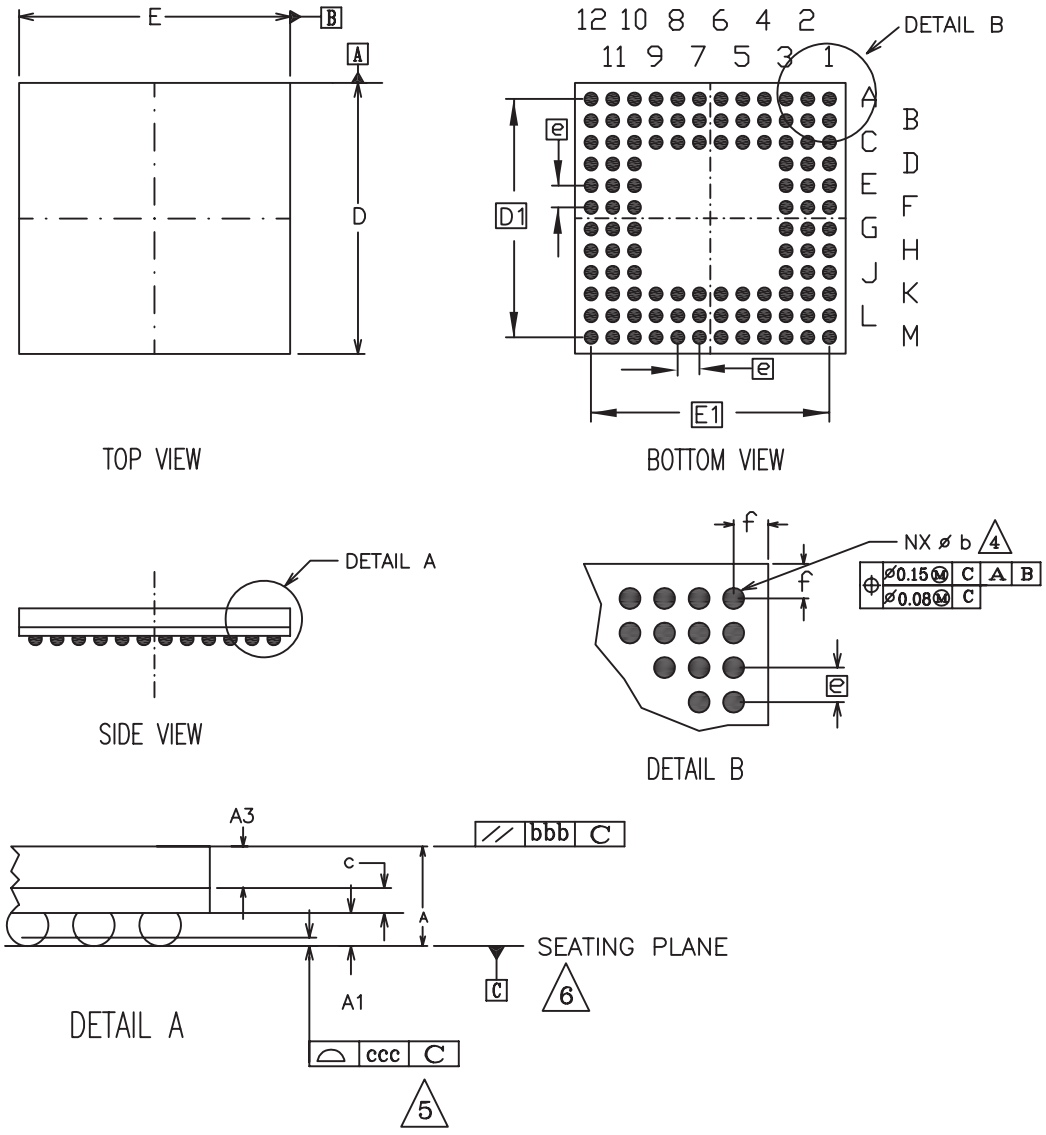


Note: The following notes apply to the package drawing.

1. All dimensions shown in mm.
2. Dimensions shown are nominal with tolerances indicated.
3. Foot length 'L' is measured at gage plane 0.25 mm above seating plane.

Body +2.00 mm Footprint, 1.4 mm package thickness		
Symbols	Leads	100L
A	Max.	1.60
A_1	-	0.05 Min./0.15 Max.
A_2	±0.05	1.40
D	±0.20	16.00
D_1	±0.05	14.00
E	±0.20	16.00
E_1	±0.05	14.00
L	+0.15/-0.10	0.60
e	Basic	0.50
b	+0.05	0.22
θ	-	0˚-7˚
ddd	Max.	0.08
ccc	Max.	0.08
JEDEC Reference Drawing		MS-026
Variation Designator		BED

Figure D-2. 108-Ball BGA Package



Note: The following notes apply to the package drawing.

1. ALL DIMENSIONS ARE IN MILLIMETERS.
2. 'e' REPRESENTS THE BASIC SOLDER BALL GRID PITCH.
3. 'M' REPRESENTS THE BASIC SOLDER BALL MATRIX SIZE.
   AND SYMBOL 'N' IS THE NUMBER OF BALLS AFTER DEPOPULATING.
4. 'b' IS MEASURABLE AT THE MAXIMUM SOLDER BALL DIAMETER AFTER REFLOW
   PARALLEL TO PRIMARY DAIUM [C] .
5. DIMENSION 'ccc' IS MEASURED PARALLEL TO PRIMARY DATUM [C].
6. PRIMARY DATUM [C] AND SEATING PLANE ARE DEFINED BY THE SPHERICAL
   CROWNS OF THE SOLDER BALLS.
7. PACKAGE SURFACE SHALL BE MATTE FINISH CHARMILLES 24 TO 27.
8. SUBSTRATE MATERIAL BASE IS BT RESIN.
9. THE OVERALL PACKAGE THICKNESS "A" ALREADY CONSIDERS COLLAPSE BALLS
10. DIMENSIONING AND TOLERANCING PER ASME Y14.5M 1994.
11. EXCEPT DIMENSION b.

Symbols	MIN	NOM	MAX
A	1.22	1.36	1.50
A1	0.29	0.34	0.39
A3	0.65	0.70	0.75
c	0.28	0.32	0.36
D	9.85	10.00	10.15
D1		8.80 BSC	
E	9.85	10.00	10.15
E1		8.80 BSC	
b	0.43	0.48	0.53
bbb		.20	
ddd		.12	
e		0.80 BSC	
f	-	0.60	-
M		12	
n		108	
REF: JEDEC MO-219F			